

Environment Notes for UPSC CSE GS3 by Pmfias.com

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Solved Environment Prelims Questions (2011-15)

Solved Environment Prelims Questions (2011-15)

- The questions from the previous 3-4 years' prelims papers help us understand the nature of questions asked in prelims and the trend the UPSC is following.
- Environment is the high priority topic for prelims and can be studied in relatively less hours compared to other subjects [Very High Cost to Benefit Ratio].

Here I have not given explanation to location based question [Location of Biosphere reserves, Tiger reserves, National Parks etc.].

I will come up with a separate notes on Location Based Question that are important for Prelims under Geography and Environment.

Most of the questions below are explained in detail under respective headings.

If you found any mistakes, inform me at poormansfriend2485@gmail.com or My FB

Page: <https://www.facebook.com/PoorMansFriend2485>

I advise you to first go through the full notes before trying to understand these question. Page | 3

2011-2012

Q1. Which one of the following is not a site for in-situ method of conservation of flora?

- Biosphere Reserves
- Botanical Garden
- National Park
- Wildlife Sanctuary

In situ conservation = on the site conservation without displacing the affected organism.

Ex situ conservation = conserving the organism in an artificial habitat by displacing it from its natural habitat.

Botanical Garden = Plants are bred in a protected environment far from their natural home, especially for research purposes. So its Ex situ conservation.

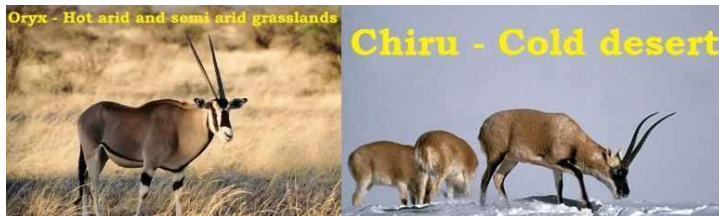
Rest all along with protected forests and reserved forests are In situ conservation methods.

Answer: b) Botanical Garden

Q2. What is the difference between the antelopes Oryx and Chiru?

- Oryx is adapted to live in hot and arid areas whereas Chiru is adapted to live in steppes and semi-desert areas of cold high mountains. .
- Oryx is poached for its antlers whereas Chiru is poached for its musk.
- Oryx exists in western India only whereas Chiru exists in north-east India only.
- None of the statements a, b, and c given above is correct.

They are both antelopes.

**Q4. Consider the following:**

1. Black-necked crane
2. Cheetah
3. Flying squirrel
4. Snow leopard

Which of the above are naturally found in India ?

- a. 1, 2 and 3 only
- b. 1, 3 and 4 only
- c. 2 and 4 only
- d. 1, 2, 3 and 4

Black-necked crane is commonly found in Tibetan and trans-Himalayan region. In winters they migrate to less colder regions of Indian Himalayas.

Cheetah is an extinct species. They have gone extinct during pre-independence era. Reason: They were hunted down by various Indian kings and British officers.

Flying Squirrels are found in many Indian forests.

Snow leopard is an 'endangered' specie found in the Himalayan ranges.

Answer: a)

Q3. Among the following States, which one has the most suitable climatic conditions for the cultivation of a large variety of orchids with minimum cost of production, and can develop an export oriented industry in this field ?

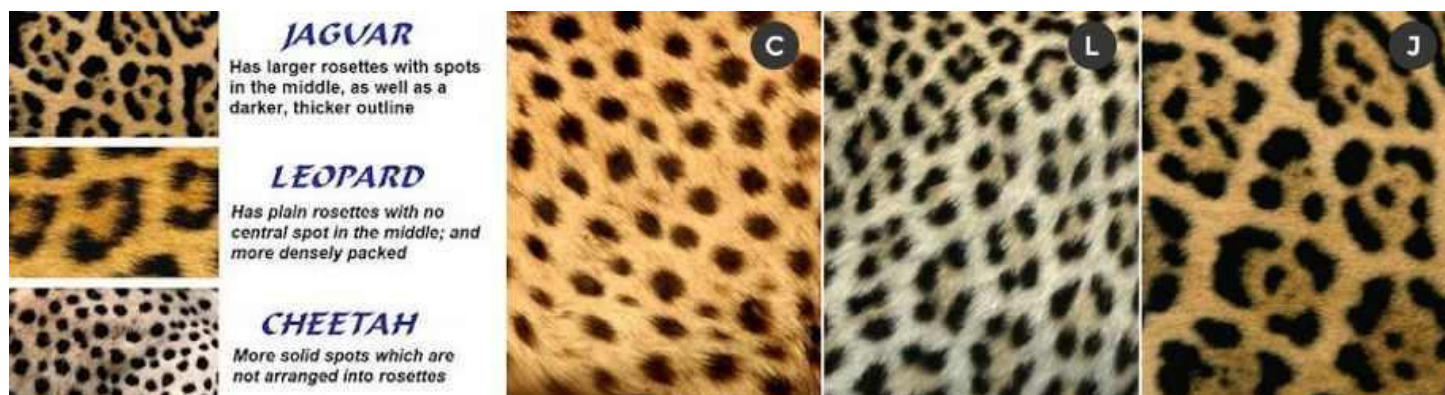
- a. Andhra Pradesh
- b. Arunachal Pradesh
- c. Madhya Pradesh
- d. Uttar Pradesh

Orchids are decorative flowering plants. They grow in regions with moderate climatic conditions [Sub-tropics with decent rainfall]

They are typical to North-Eastern states.

Great demand for these decorative flowering plants exists in South-East Asian region.

Answer: b)



Answer: b) 1, 3 and 4 only

Q5. A sandy and saline area is the natural habitat of an Indian animal species. The animal has no predators in that area but its existence is threatened due to the destruction of its habitat. Which one of the following could be that animal?

- a. Indian wild buffalo

- b. Indian wild ass
- c. Indian wild boar
- d. Indian Gazelle

Sandy saline area = Kutch region

Indian wild buffalo = Terai region

Indian wild boar = can survive in different types of habitat: grasslands,

taiga, tropical rainforests, but they prefer life in deciduous forests.

Chinkara (Indian gazelle) = Thar desert

Answer: b) Indian wild ass

Q6. Consider the following kinds of organisms

1. Bat
2. Bee
3. Bird

Which of the above is/are pollinating agent/agents?

- a. 1 and 2 only
- b. 2 only
- c. 1 and 3 only
- d. 1, 2 and 3

Pollination = transfer of pollen from stamen to pistil.

Bees are well known pollinators.

Birds help in pollination while trying to catch insects or trying to suck nectar.

Bats help in pollination while trying to catch insects. [Bats are pollinators – Mentioned several times in NCERT]

Answer: All

Q7. The 'Red Data Books' published by the International Union for Conservation of Nature and Natural Resources (IUCN) contain lists of

1. Endemic plant and animal species present in the biodiversity hotspots.
2. Threatened plant and animal species.
3. Protected sites for conservation of nature & natural resources in various countries.

Which of the statement given above is/are correct ?

- a. 1 & 3
- b. 2 only
- c. 2 & 3

d. 3 only

IUCN is an NGO. It publishes Red data book which contains a list of 'Threatened species' (vulnerable, endangered and critically endangered).

Answer: b) 2 only

Q8. Which one of the following groups of animals belongs to the category of endangered species?

- a. Great Indian Bustard, Musk Deer, Red Panda and Asiatic Wild Ass
- b. Kashmir Stag, Cheetal, Blue Bull and Great Indian Bustard
- c. Snow Leopard, Swamp Deer, Rhesus Monkey and Saras (Crane)
- d. Lion-tailed Macaque, Blue Bull, Hanuman Langur and Cheetal

As of 2016, the status of most of these animals has changed. Outdated question.

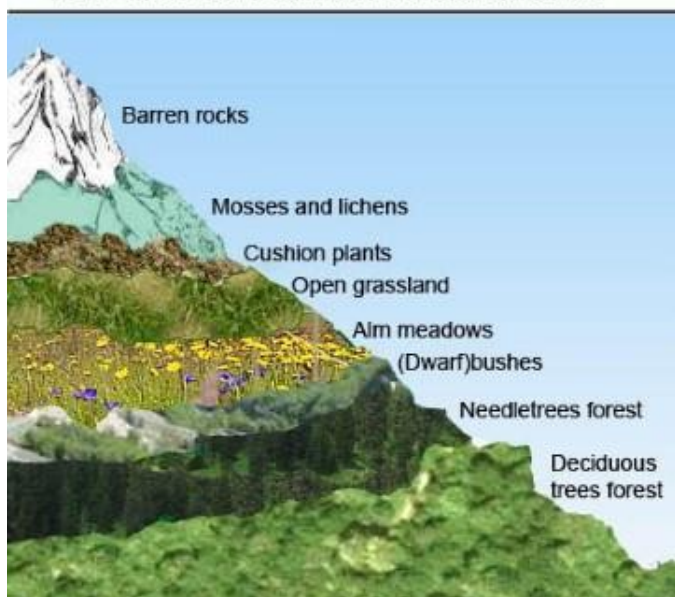
Q9. The Himalayan Range is very rich in species diversity. Which one among the following is the most appropriate reason for this phenomenon?

- a. It has a high rainfall that supports luxuriant vegetative growth.
- b. It is a confluence of different bio geographical zones.
- c. Exotic and invasive species have not been introduced in this region.
- d. It has less human interference.

Explanation:

Options c) and d) are wrong statements.

Options a) and b) are correct statements.

VEGETATION ZONATION IN MOUNTAINS

Greater Diversity is due to

1. various types of forests found along the Himalayas Ranges [Shiwaliks, Himachal and Himadri].
2. Vertical zonation of vegetation [chief characteristic of Himachal and Himadri].

Answer is b)

Q10. If a tropical rain forest is removed, it does not regenerate quickly as compared to a tropical deciduous forest. This is because

- a. the soil of rain forest is deficient in nutrients
- b. propagules of the trees in a rain forest have poor viability
- c. the rain forest species are slow-growing
- d. exotic species invade the fertile soil of rain forest.

Explanation:

1. the soil of rain forest is deficient in nutrients: Rainforest = Rainfall through the year. It rains almost every day =

The top soil is continuously washed away = nutrients are also washed away = **leaching of nutrients** = very little fertility remains in top soil = most of the seeds don't germinate for years = regeneration is very slow (it takes decades). But the layer below top soil (sub-soil) is very fertile. So plants grow very quickly once their roots reach the sub-soil and if they receive enough sunlight.

2. propagules of the trees in a rain forest have poor viability: Propagule = detachable structure that can give rise to a new plant, e.g. a bud, sucker, or spore [Asexual Reproduction in plants]. Seed bearing plants are more significant than Propagules in rainforest. So propagules don't really matter.
3. the rain forest species are slow-growing: The plant species in rainforests compete for sunlight. So they grow as rapidly as they can.
4. exotic species invade the **fertile** soil of rain forest: this statement is wrong as the rainforest soil is heavily leached. Hence they are not fertile. But exotic invasive species are a threat to rain forests (E.g. Most plantation crops like rubber, palm etc.).

Answer: a)

Q11. When the bark of a tree is removed in a circular fashion all around near its base, it generally dries up and dies because

- a. Water from soil cannot rise to aerial parts.
- b. Roots are starved of energy
- c. Tree is infected by soil microbes
- d. Roots do not receive oxygen for respiration

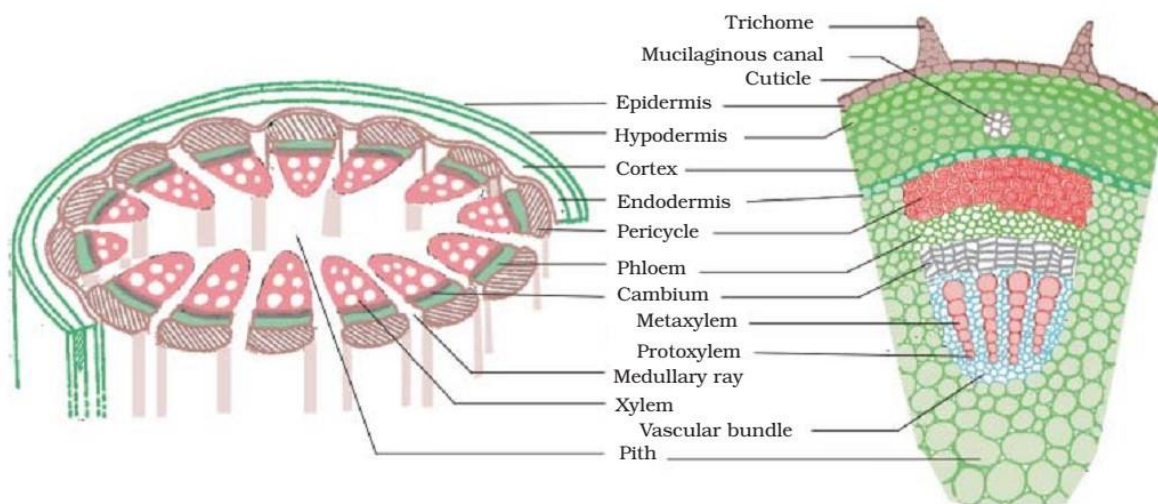


Fig. 6.3: Section of a stem

Answer: b) roots are starved of energy as the phloem (food carrying tissue) is cut off.

- c. 1, 3 and 4 only
d. 1, 2, 3 and 4

Q12. Consider the following:

1. Photosynthesis
2. Respiration
3. Decay of organic matter
4. Volcanic action

Which of the above add carbon dioxide to the carbon cycle on Earth ?

- a. 1 and 4 only
b. 2 and 3 only
c. 2,3 and 4 only
d. 1, 2, 3 and 4

Answer: c) Photosynthesis takes out CO₂ from carbon cycle. Rest all ads CO₂.

Q13. Consider the following statements: Chlorofluorocarbons, known as ozone-depleting substances, are used

1. in the production of plastic foams
2. in the production of tubeless tyres
3. in cleaning certain electronic components
4. as pressurizing agents in aerosol cans

Which of the statements given above is/are correct?

- a. 1, 2 and 3 only
b. 4 only

Answer: c) CFCs were used as refrigerants, pressurizing agents (foam and aerosol cans) and for cleaning electronic equipment.

Q14. Consider the following:

1. Carbon dioxide
2. Oxides of Nitrogen
3. Oxides of Sulphur

Which of the above is/are the emission/emissions from coal combustion at thermal power plants ?

- a. 1 only
b. 2 and 3 only
c. 1 and 3 only
d. 1, 2 and 3

Burning coal releases CO, CO₂, sulphur dioxide and oxides of nitrogen – air pollutants.

Answer: d) 1, 2 and 3.

Q13. Consider the following statements: Chlorofluorocarbons, known as ozone-depleting substances, are used

1. in the production of plastic foams
2. in the production of tubeless tyres
3. in cleaning certain electronic components
4. as pressurizing agents in aerosol cans

Which of the statements given above is/are correct?

- a. 1, 2 and 3 only
- b. 4 only
- c. 1, 3 and 4 only
- d. 1, 2, 3 and 4

Answer: c) CFCs were used as refrigerants, pressurizing agents (foam and aerosol cans) and for cleaning electronic equipment.

Q15. The increasing amount of carbon dioxide in the air is slowly raising the temperature of the atmosphere, because it absorbs

- a. the water vapour of the air and retains its heat.
- b. the ultraviolet part of the solar radiation.
- c. all the solar radiations.
- d. the infrared part of the solar radiation

Answer: d) the infrared part of the solar radiation (outgoing radiation).

Among GHGs, only water vapor has the ability to absorb both incoming (UV) and outgoing (infrared) radiation.

Q16. The formation of ozone hole in the Antarctic region has been a cause of concern. What could be the reason for the formation of this hole?

- a. Presence of prominent tropospheric turbulence; and inflow of chlorofluorocarbons
- b. Presence of prominent polar front and stratospheric clouds; and inflow of chloro fluorocarbons
- c. Absence of polar front and stratospheric clouds; and inflow of methane and chloro fluorocarbons.
- d. Increased temperature at polar region due to global warming

Answer: b) Presence of prominent polar front and stratospheric Clouds and inflow of chlorofluorocarbons. [Explained in the Post on Ozone Depletion]

Q17. The acidification of oceans is increasing. Why is this phenomenon a cause of concern?

1. The growth and survival of calcareous phytoplankton will be adversely affected.
2. The growth and survival of coral reefs will be adversely affected.
3. The survival of some animals that have phytoplanktonic larvae will be adversely affected.
4. The cloud seeding and formation of clouds will be adversely affected.

Which of statements given above is / are correct?

- a. 1, 2 and 3 only
- b. 2 only
- c. 1 and 3 only
- d. 1, 2, 3 and 4

Explanation:

Ocean acidification decreases the calcifying ability of corals, calcareous plankton, crustaceans etc. It also adversely affects cloud formation and cloud seeding. So Options 1, 2 and 4 are correct [Detailed explanation is given under "Ocean Acidification"].

Answer: d) 1, 2, 3 and 4

Q18. Regarding "carbon credits", which one of the following statements is not correct?

- a. The carbon credit system was ratified in conjunction with the Kyoto Protocol
- b. Carbon credits are awarded to countries or groups that have reduced greenhouse gases below their emission quota
- c. The goal of the carbon credit system is to limit the increase of carbon emission quota
- d. Carbon credits are traded at a price fixed from time to time by the United Nations Environment Programme.

Answer d) Carbon credit prices are traded on an exchange. So their prices are never fixed.

Q19. With reference to India, consider the following Central Acts:

1. Import and Export (Control) Act, 1947.
2. Mining and Mineral Development (Regulation) Act, 1957
3. Customs Act, 1962
4. Indian Forest Act, 1927

Which of above Acts have relevance to/bearing on the biodiversity conservation in the country?

- a. 1 and 3 only
- b. 2,3 and 4 only
- c. 1,2,3 and 4
- d. None of the above Acts

Answer: c) all

Central Acts and Rules having Relevance to Biodiversity Conservation

- Fisheries Act, 1897
- Destructive Insects and Pests Act, 1914
- **The Indian Forest Act, 1927**
- Agricultural Produce (Grading and Marketing) Act, 1937
- Indian Coffee Act, 1942
- Import and Export (Control) Act, 1947
- Rubber (Production and Marketing) Act, 1947
- Tea Act, 1953
- **Mining and Mineral Development (Regulation) Act, 1957**
- Prevention of Cruelty to Animals Act, 1960
- **Customs Act, 1962**
- Cardamom Act, 1965
- Seeds Act, 1966
- The Patents Act, 1970
- Wildlife (Protection) Act, 1972
- Marine Products Export Development Authority Act, 1972
- Water (Prevention and Control of Pollution) Act, 1974
- Tobacco Board Act, 1975
- Territorial Water, Continental Shelf, Exclusive Economic Zone and other Maritime Zones Act, 1976
- Water (Prevention and Control of Pollution) Cess Act, 1977

- Maritime Zones of India (Regulation and Fishing by Foreign Vessels) Act, 1980
- Forest (Conservation) Act, 1980
- Air (Prevention and Control of Pollution) Act, 1981
- Agricultural and Processed Food Products Export Development Authority Act, 1985/1986
- Environment (Protection) Act, 1986
- Spices Board Act, 1986
- National Dairy Development Board, 1987
- Rules for the manufacture, use/import/export and storage of hazardous microorganisms/ genetically engineered organisms or cells, 1989
- **Foreign Trade (Development and Regulation) Act, 1992 [Imports and Exports (Control) Act, 1947 got repealed with the enactment of Foreign Trade (Development And Regulation) Act, 1992.]**
- Protection of Plant Varieties and Farmers' Rights (PPVFR) Act, 2001
- Biological Diversity Act, 2002
- Plant Quarantine (Regulation of Import into India) Order, 2003
- Biological Diversity Rules, 2004
- The Food Safety and Standards Act, 2006
- Scheduled Tribes and Other Traditional Forest Dwellers (Recognition of Forest Rights) Act, 2006.

Q20. The National Green Tribunal Act, 2010 was enacted in consonance with which of the following provisions of the Constitution of India?

1. Right to healthy environment, construed as a part of part of Right to life under Article 21.
2. Provision of grants for raising the level of administration in the Scheduled Areas for the welfare of Scheduled Tribes under Article 275(1)

Which of the statements given above is/are correct ?

- a. 1 only
- b. 2 only
- c. Both 1 and 2

d. Neither 1 nor 2

Answer: a)

Q21. How does National Biodiversity Authority (NBA) help in protecting the Indian agriculture?

1. NBA checks the biopiracy and protects the indigenous and traditional genetic resources.
2. NBA directly monitors and supervises the scientific research on genetic modification of crop plants.
3. Application for intellectual Property Rights related to genetic / biological resources cannot be made without the approval of NBA.

Which of the statements given above is/are correct?

- a. 1 only
- b. 2 and 3 only
- c. 1 and 3 only
- d. 1, 2 and 3

The top biotech regulator in India for Genetically Modified Organisms is **Genetic Engineering Appraisal Committee (GEAC)**.

Answer: c) 1 and 3 only

Q22. Recently, 'oilzapper' was in the news. What is it?

- a. It is an eco-friendly technology for the remediation of oil sludge and oil spills.
- b. It is the latest technology developed for under- sea oil exploration.
- c. It is a genetically engineered high biofuel yielding maize variety.
- d. It is the latest technology to control the accidentally caused flames from oil wells.

Explanation:

Using bioremediation techniques, TERI has developed a mixture of bacteria called '**Oilzapper and Oilivorous-S**' which degrades the pollutants of oil-contaminated sites, leaving behind no harmful residues. This

technique is not only environment friendly, but also highly cost-effective.

Answer: a) remediation of oil sludge and oil spills

Q23. Government of India encourages the cultivation of 'sea buckthorn'. What is the importance of this plant ? Page | 10

1. It helps in controlling soil erosion and in preventing desertification.
2. It is a rich source of biodiesel.
3. It has nutritional value and is well-adapted to live in cold areas of high altitudes.
4. Its timber is of great commercial value.

Which of the statements given above is /are correct ?

- a. 2, 3 and 4 only
- b. **1 and 3 only**
- c. 1, 2, 3 and 4
- d. 1 only

Q24. Two important rivers - one with its source in Jharkhand (and known by a different name in Odisha), and another, with its source in Odisha - merge at a place only a short distance from the coast of Bay of Bengal before flowing into the sea. This is an important site of wildlife and biodiversity and a protected area. Which one of the following could be this ?

- a. Bhitarkanika
- b. Chandipur-on-sea
- c. Gopalpur-on-sea
- d. Simlipal

Simlipal National Park is far away from coast.

Chandipur-on-sea and Gopalpur-on-sea are tourist destinations with resorts. There is no significant wildlife or bio-diversity.

Answer: a) Bhitarkanika

Q25. A particular State in India has the following characteristics:

1. It is located on the same latitude which passes through northern Rajasthan.

2. It has over 80% of its area under forest cover.
3. Over 12% of forest cover constitutes **Protected Area Network** in this State.

Which one among the following States has all the above characteristics?

- a. **Arunachal Pradesh**
- b. Assam
- c. Himachal Pradesh
- d. Uttarakhand

2013

Q1. Consider the following pairs:

1. Nokrek Bio-sphere Reserve: Garo Hills
2. Logtak (Loktak) Lake : Barail Range
3. Namdapha National Park : Dafla Hills

Which of the above pairs is/are correctly matched?

- a. **1 only**
- b. 2 and 3 only
- c. 1, 2 and 3
- d. None

Q2. Consider the following pairs about parks and rivers passing through them:

1. Corbett National Park: Ganga
2. Kaziranga National Park: Manas
3. Silent Valley National Park: Kaveri

Which of the above pairs is/are correctly matched?

- a. 1 and 2
- b. 3 only
- c. 1 and 3
- d. **None**

Q3. In which of the following States is lion-tailed macaque found in its natural habitat?

1. Tamil Nadu
2. Kerala
3. Karnataka
4. Andhra Pradesh

Select the correct answer using the codes given below.

- a. 1, 2 and 3 only
- b. 2 only
- c. 1, 3 and 4 only
- d. 1, 2, 3 and 4

Lion-tailed macaque is found in Western Ghats. So AP is the odd option.

Answer: a) 1, 2 and 3 only

Q4. Which of the following is/are unique characteristic/characteristics of equatorial forests?

1. Presence of tall, closely set trees with crowns forming a continuous canopy
2. Coexistence of a large number of species
3. Presence of numerous varieties of epiphytes

Select the correct answer using the code given below:

- a. 1 only
- b. 2 and 3 only
- c. 1 and 3 only
- d. 1, 2 and 3

Equatorial Vegetation

- High temperature and abundant rainfall support a luxuriant **tropical rain forest**.
- In the Amazon lowlands, the forest is so dense that it is called '**selvas**'. [selvas: A dense tropical rainforest usually having a cloud cover (**dense canopy**)]
- All plants struggle upwards (most **epiphytes**) for sunlight resulting in a peculiar layer arrangement [Canopy].



Old World monkeys are located in Africa, central to southern Asia, Japan, and India.



Figure 4.20 Examples of Bony fishes :
(a) *Hippocampus* (b) *Catla*

Answer: a) 1, 2 and 3 only

Q7. Consider the following animals:

1. Sea cow
2. Sea horse
3. Sea lion

Which of the above is/are mammal/mammals?

- a. 1 only
- b. 1 and 3 only
- c. 2 and 3 only
- d. 1, 2 and 3

- Sea cow (Dugong) found in shallow Indian seas is a herbivorous mammal
- Sea lion is also a mammal.
- Sea horse is a Bony Fish.



Answer: Too easy. d) All

Q5. Consider the following fauna of India:

1. Gharial
2. Leatherback turtle
3. Swamp deer

Which of the above is/are endangered?

- a. 1 and 2 only
- b. 3 only
- c. 1, 2 and 3
- d. None

1 and 2 are 'Critically Endangered' 3 is 'Vulnerable'. Outdated question

Q6. Consider the following

1. Star tortoise
2. Monitor lizard
3. Pygmy hog
4. Spider monkey

Which of the above found in India?

- a. 1, 2 and 3 only
- b. 2 and 3 only
- c. 1 and 4 only
- d. 1, 2, 3 and 4

Google results don't show Spider Monkey in India.

Spider monkeys are New World monkeys. Monkeys and apes of the Old World.

The New World monkeys are found in Mexico, Central America, and South America, and the

Answer: b) 1 and 3 only

Q8. Mycorrhizal biotechnology has been used in rehabilitating degraded sites because mycorrhiza enables the plants to

1. resist drought and increase absorptive area
2. tolerate extremes of pH
3. Resist disease infestation

Select the correct answer using the codes given below:

- a. 1 only
- b. 2 and 3 only
- c. 1 and 3 only
- d. 1, 2 and 3

Answer: d) all [Explained under “Sustainable Agriculture”]

Q9. Consider the following organisms

1. Agaricus
2. Nostoc
3. Spirogyra

Which of the above is/are used as biofertilizer/biofertilizers

- a. 1 and 2
- b. 2 only
- c. 2 and 3
- d. 3 only

Answer: b) 2 only [Not sure]

Nostoc is a nitrogen fixing cyanobacteria.

Q10. Which of the following adds/add nitrogen to the soil?

1. Excretion of urea by animals
2. Burning of coal by man
3. Death of vegetation

Select the correct answer using the codes given below.

- a. 1 only
- b. 2 and 3 only
- c. 1 and 3 only

d. 1, 2 and 3

Explanation:

All the above three adds to nitrogen cycle.

Burning coal releases CO, CO₂, sulphur dioxide and oxides of nitrogen – air pollutants. Page | 13

Oxides of nitrogen fall on earth as acid rain. Acidic rain is a complex mixture of nitrous, nitric, sulfurous and sulfuric acids which all combine to lower the pH.

But, the question says “Which of the following adds/add nitrogen to the **soil**?”

1. Animal Waste like Urea, Uric acid and Death of vegetation add nitrogen in the form of nitrates directly into soil.
2. Coal combustion adds nitrogen to atmosphere and from there it falls back to earth in the form of acid rain and acid rain adds nitrogen to soil.

<http://www.ewh.com/how-does-5117705-burning-fuels-affect-nitrogen-cycle.html> says:

“The release of nitric oxides into the air in large quantities causes **smog and acid rain** that pollutes the atmosphere, soil and water and affects plants and animals. The increase in nitrogen and nitrous oxide is caused by automobiles, power plants and a wide variety of industries.

As nitrous oxides filter into soil, it loses nutrients like calcium and potassium, which are essential for maintaining a balance in plant ecosystems. With the loss of these compounds, soil fertility declines. Also, soils become significantly more acidic.”

And also <http://chemistry.elmhurst.edu/vchembo/ok/307nitrogen.html> says:

“The main component of the nitrogen cycle starts with the element nitrogen in the air. Two nitrogen oxides are found in the air as a result of interactions with oxygen. Nitrogen will only react with oxygen in the presence of high temperatures and pressures found

near **lightning bolts** and in combustion reactions in power plants or internal combustion engines. Nitric oxide, NO, and nitrogen dioxide, NO₂, are formed under these conditions. Eventually nitrogen dioxide may react with water in rain to form **nitric acid**, HNO₃. The nitrates thus formed may be utilized by plants as a nutrient (so, soil gets nitrogen from acid rain)."

So, soil gets nitrogen from acid rain.

From this, I feel Burning of coal by man adds nitrogen to soil but indirectly through acid rain.

The answer depends on how you interpret the question.

Answer: Official UPSC Key says answer is c) 1 and 3 only.

If you have anything more to add write it at poormansfriend2485@gmail.com

Q11. Contour bunding is a method of soil conservation used in

- desert margins, liable to strong wind action
- low flat plains, close to stream courses, liable to flooding
- scrublands, liable to spread of weed growth
- None of the above

Counter bunding is used along with terracing to protect sloped surfaces.

Answer: d) None

Q12. With reference to the usefulness of the by-products of sugar industry, which of the following statements is/are correct?

- Bagasse can be used as biomass fuel for the generation of energy.
- Molasses can be used as one of the feedstocks for the production of synthetic chemical fertilizers.
- Molasses can be used for the production of ethanol.

Select the correct answer using the codes given below.

- 1 only
- 2 and 3 only
- 1 and 3 only
- 1, 2 and 3

Answer: c) 1 and 3 only

Q13. Due to improper/indiscriminate disposal of old and used computers or their parts, which of the following are released into the environment as e-waste?

- Beryllium
- Cadmium
- Chromium
- Heptachlor
- Mercury
- Lead
- Plutonium

Select the correct answer using the codes given below.

- 1, 3, 4, 6 and 7 only
- 1, 2, 3, 5 and 6 only
- 2, 4, 5 and 7 only
- 1, 2, 3, 4, 5, 6 and 7

Explanation:

Heptachlor is a Chlorohydrocarbon (CHC) which is used as an insecticide.

Plutonium is a radioactive metal and hence not used in computers.

So, answer should not contain either 4) or 7).

Answer: b) 1, 2, 3, 5 and 6 only

Q14. Which of the following can be found as pollutants in the drinking water in some parts of India?

- Arsenic
- Sorbitol
- Fluoride
- Formaldehyde
- Uranium

Select the correct answer using the codes given below.

- 1 and 3 only
- 2, 4 and 5 only
- 1, 3 and 5 only
- 1, 2, 3, 4 and 5

Explanation:

Easiest option is 3) Fluoride. So, (b) can be eliminated.

Next easiest option is Arsenic (Heavy metal). Damn!

Nest easiest option is Uranium. Many columns are written on uranium mining and its ill-effects. So, (a) also can be elevated. (a) and (b) eliminated. (c) and (d) remains.

The knowledge of either "Sorbitol" or "Formaldehyde" will tell us the answer.

Formaldehyde is more known than sorbitol. Googling didn't give me any authentic source that called formaldehyde a water pollutant. It is used a preservative to preserve human organs. It finds some application in textile, resin and wood industry.

Answer: c) 1, 3 and 5 only

Sorbitol (glucitol) is a sugar alcohol with a sweet taste which the human body metabolizes slowly.

Q15. Which one among the following industries is the maximum consumer of water in India?

- Engineering
- Paper and pulp
- Textiles
- Thermal power

Table 12.4
Demand of Water in Industrial Areas in India

Sector	Average Water Consumption in Indian Industries	Globally Best
Thermal Power Plants	Average 80 cubic metre per megawatt (mwh)	Less than 10 cubic metre (M ³) per mwh
Textile Industry	200-250 M ³ /per tonne cotton cloth	Less than 100 M ³ /per tonne cotton cloth
Paper and Pulp	- Wood base mills: 150-200 M³ ton/per tonne - Waste paper based mills: 10-80 M³/per tonne	- Wood base mills: 50-75 M³ ton/per tonne - Waste paper based Mills: 10-25 M³/per tonne
Total Iron and Steel Plants	10-8 CM ³ per tonne of finished product (Average around 25 M ³)	5-10 M ³ per tonne of finished product. Best Practice less than 0.1 M ³ waste water per tonne finished products.
Distilleries	75-200 M ³ per tonne Alcohol Produced	Figures NA
Fertilizer Industry	- Nitrogenous Fertilizer Plant 5.0-20.0 M³ per tonne - Straight Phosphatic Plant 1.4-2.0 M³/per tonne - Complex Fertilizer 0.2-5.4 M³/per tonne	An effluent discharge of less than 1.5 M ³ tonne as P2 O5

Q16. Acid rain is caused by the pollution of environment by

- carbon dioxide and nitrogen
- carbon monoxide and carbon dioxide
- ozone and carbon dioxide
- nitrous oxide and sulphur dioxide

Explanation:

Carbon dioxide, carbon monoxide, oxides of nitrogen and oxides of sulphur in atmosphere can cause acid rains.

Both b) and d) are correct. But d) has more influence than b)

Answer: d) nitrous oxide and sulphur dioxide

Q17. Photochemical smog is a resultant of the reaction among

- NO₂, O₃ and peroxyacetyl nitrate in the presence of sunlight
- CO, O₂ and peroxyacetyl nitrate in the presence of sunlight
- CO, CO₂ and N₂ at low temperature
- High concentration of N₂, O₃ and CO in the evening

Explanation:

- When pollutants such as **nitrogen oxides** (primary pollutant) and **volatile organic compounds** (primary pollutant) react together in the presence of **SUNLIGHT, OZONE** (Secondary pollutant) and peroxyacetyl nitrate (PAN) (Secondary pollutant) are formed.

Reactions involved



- Photochemical (**summer smog**) is formed when the primary pollutant NO_2 reacts with secondary pollutants O_3 and peroxyacetyl nitrate in the presence of sunlight.

Answer: a) NO_2 , O_3 and peroxyacetyl nitrate in the presence of sunlight

Q18. With reference to food chains in ecosystems, consider the following statements:

- A food chain illustrates the order in which a chain of organisms feed upon each other.
- Food chains are found within the populations of a species.
- A food chain illustrates the numbers of each organism which are eaten by others.

Which of the statements given above is / are correct?

- 1 only
- 1 and 2 only
- 1, 2 and 3
- None

Explanation:

- A food chain illustrates the order in which a chain of organisms feed upon each other. (True)
- Food chains are found within the populations of a species. (Man won't eat man – so, false)
- A food chain illustrates the numbers of each organism which are eaten by others (food web illustrates the number not the food chain).

Answer: a)

Q19. Which one of the following terms describes not only the physical space occupied by an organism, but also its functional role in the community of organisms?

- Ecotone
- Ecological niche
- Habitat
- Home range

Answer: b) Niche

Explanation:

- Ecotone – zone of transition between two ecosystems. E.g. grasslands, mangroves etc.
- Habitat – surroundings in which an organism lives.
- Home Range - A home range is the area in which an animal lives and moves on a daily or periodic basis.

Q20. With reference to the food chains in ecosystems, which of the following kinds of organism is / are known as decomposer organism/organisms?

- Virus
- Fungi
- Bacteria

Select the correct answer using the codes given below.

- 1 only
- 2 and 3 only
- 1 and 3 only
- 1, 2 and 3

Explanation:

- Fungi and Bacteria are decomposers. They breakdown organic matter into simple inorganic substances.
- Virus represents dormant life. They are metabolically inactive as long as they are outside a host body. They are not decomposers. They invade host cells and use their nucleus (DNA machinery) to carry out their life processes.
- Protists are unicellular organisms, such as paramecium and euglena. They're not decomposers either.

Answer: b)

Q21. In the grasslands, trees do not replace the grasses as a part of an ecological succession because of

- a. insects and fungi
- b. limited sunlight and paucity of nutrients
- c. water limits and fire
- d. None of the above

Answer: c)**Explanation:**

- Grasses have this one good trick to monopolize a place. In dry season the grasses dry up and cause fires which destroy other plant species and their seeds.
- Also grasslands develop in regions with scanty rainfall where plant growth cannot be achieved.
- Though forests form the climax community in most of the ecosystems, but in grassland ecosystem [In the grasslands, trees do not replace the grasses as a part of an ecological succession] grasses form the climax community. Thanks to fire and lack of water.
- Grasslands are almost irreversible once deforestation in water scarce areas gives way to grasslands. [Ecological Damage]

Q22. Which one of the following is the correct sequence of ecosystems in the order of decreasing productivity?

- a. Oceans, lakes, grasslands, mangroves
- b. Mangroves, oceans, grasslands, lakes
- c. Mangroves, grasslands, lakes, oceans
- d. Oceans, mangroves, lakes, grasslands

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| 17**Explanation:**

Productivity = production/unit area/unit time

Production/unit area depends on number and diversity of producers.

From the topics discussed in this post, we can say that transitional zones (Ecotones) have very high productivity compared to normal ecosystems. [Tropical Rainforests is an exception as it has productivity comparable to wetlands because of its rich diversity of primary producers]

In the options, mangroves are transitional ecosystems while the rest are normal ecosystems.

Remember: Grasslands are not transitional all the time. E.g. Steppe. Non transitional grasslands have very low productivity because of very limited diversity of primary producers. [Grasslands become transitional only when they are narrow]

So the order of decreasing productivity will be like

Mangroves ,..... , ,

Oceans = very deep and hence productivity is limited to surface only (Below in the aphotic zone productivity is negligible. Aphotic zone in oceans is few kilometers).

Also surface water in oceans are very poor in nutrients. Nutrient rich cold water flows as a sub-surface flow lying in the aphotic zone. Sunlight and nutrients are far apart and hence primary productivity is very low except in regions where there is upwelling of nutrient rich cold water [Cold and Warm current mixing zones].

So open ocean ecosystem has the least productivity. [Desert ecosystem also has very low productivity, lesser than oceans].

So, the answer will look like Mangroves,, Oceans.

The only such option is c)

Answer: c) Mangroves, grasslands, lakes, oceans.

Lakes just like oceans have low productivity. But due to some plants in photic zone, lakes have productivity slightly greater than that of oceans.

2014

Q1. Which one of the following is the process involved in photosynthesis?

- Potential energy is released to form free energy
- Free energy is converted into potential energy and stored
- Food is oxidized to release carbon dioxide and water
- Oxygen is taken, and carbon dioxide and water vapour are given out

Answer: b) Sunlight (free energy) is converted into carbohydrates (potential energy) using water and carbon dioxide. Oxygen is released in the process.

Q2. Which of the following adds/add carbon dioxide to the carbon cycle on the planet Earth?

- Volcanic action
- Respiration
- Photosynthesis
- Decay of organic matter

Select the correct answer using the code given below.

- 1 and 3 only
- 2 only
- 1, 2 and 4 only**
- 1, 2, 3 and 4

Q3. In India, the problem of soil erosion is associated with which of the following?

- Terrace cultivation
- Deforestation
- Tropical climate

Select the correct answer using the code given below.

- 1 and 2 only
- 2 only
- 1 and 3 only
- 1, 2 and 3**

Terrace cultivation is a method to prevent soil erosion.

Soil erosion occurs in all climates.

Answer: b) 2 only

Q4. Lichens, which are capable of initiating ecological succession even on a bare rock, are actually a symbiotic association of

- algae and bacteria
- algae and fungi**
- bacteria and fungi
- fungi and mosses

More of a biology question. Fungus provides shelter, water and minerals to the algae and, in return, the alga provides food (photosynthesis).

Q5. Conserving Biodiversity

The most important strategy for the conservation of biodiversity together with traditional human life is the establishment of

- biosphere reserves
- botanical gardens
- national parks
- wildlife sanctuaries

Answer: a) biosphere reserves [Explained under "Biodiversity Conservation"]

Q6. With reference to 'Eco-Sensitive Zones', which of the following statements is/are correct?

1. Eco-Sensitive Zones are the areas that are declared under the Wildlife (Protection) Act, 1972.
2. The purpose of the declaration of Eco-Sensitive Zones is to prohibit all kinds of human activities, in those zones except agriculture.

Select the correct answer using the code given below.

- a. 1 only
- b. 2 only
- c. Both 1 and 2
- d. Neither 1 nor 2

Eco-Sensitive Zones are declared under **Environmental (Protection) Act, 1986**.

Eco-Sensitive Zones are transitional areas where human activity is regulated but not prohibited.

Answer: d) Neither

Q7. Consider the following pairs

1. Dampa Tiger Reserve : Mizoram
2. Gumti Wildlife Sanctuary : Sikkim
3. Saramati Peak : Nagaland

Which of the above pairs is /are correctly matched?

- a. 1 only
- b. 2 and 3 only
- c. 1 and 3 only
- d. 1, 2 and 3

Answer: c) 1 and 3 only

Q8. Consider the following pairs

Wetlands	Confluence of rivers
1. Harike Wetlands	Beas and Satluj/Sutlej
2. Keoladeo Park	Ghana National Banas and Chambal

3. Kolleru Lake
Confluence of Musi and Krishna

Which of the above pairs is/are correctly matched?

- a. 1 only
- b. 2 and 3 only
- c. 1 and 3 only
- d. 1, 2 and 3

Answer: a) 1 only

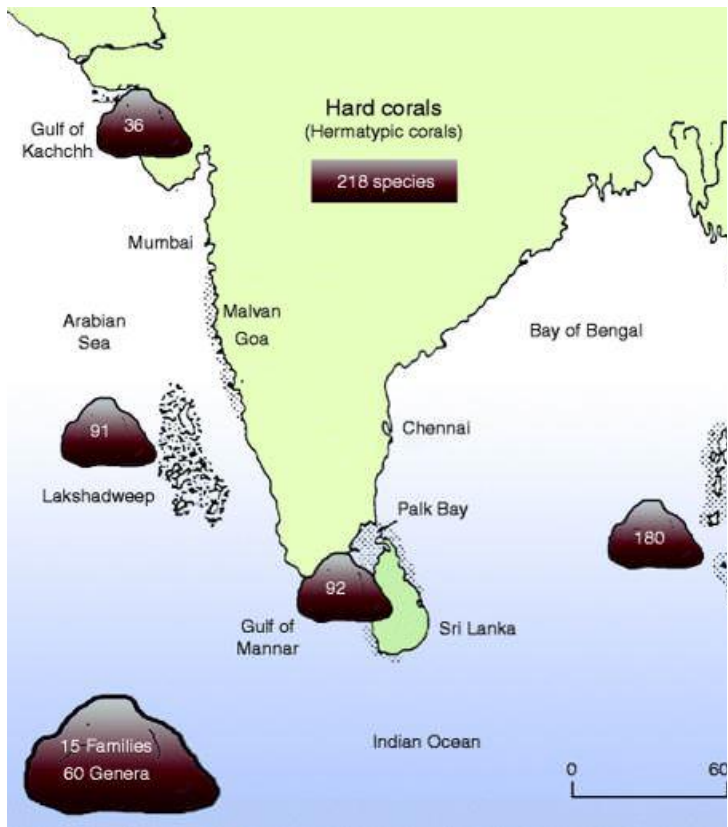
Q9. Which of the following have coral reefs?

1. Andaman and Nicobar Islands
2. Gulf of Kutch
3. Gulf of Mannar
4. Sunderbans

Select the correct answer using the code given below.

- a. 1, 2 and 3 only
- b. 2 and 4 only
- c. 1 and 3 only
- d. 1, 2, 3 and 4

Explanation:



The phenomenon of hibernation can be observed in which of the above kinds of animals?

- 1 and 2 only
- 2 only
- 1, 2 and 3
- Hibernation cannot be observed in any of the above

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<http://www.pmfias.com/classification-vertebrata-phylum-chordata/#warm-blooded-vs-cold-blooded-animals> [Hibernation also explained]

Answer: c) All

Q11. Among the following organisms, which one does not belong to the class of other three?

- Crab
- Mite
- Scorpion
- Spider

Insects, Arachnids and Crustaceans are subphylum of Arthropoda.

Mite, Scorpion and spider are Arachnids, while Crab is a Crustacean.

I missed this concept in biology. So I am updating here.

Arachnids

- Spiders, harvestmen, mites, ticks and other arachnids are members of the class Arachnida.

Coral reefs don't form in areas where there is significant intrusion of fresh water and cold water.

Most of the mighty Indian Rivers flow into Bay of Bengal and hence coral reefs are absent on the east coast of India.

Coral reefs are also absent on west coast of Africa (Benguela Current), South America (Peruvian or Humboldt Current), Europe (Canaries Current) and other continents due to cold currents.

Andaman and Nicobar Islands, Gulf of Kutch and Gulf of Mannar are regions where there is not significant freshwater intrusion. But Sunderbans are the mouth of Ganga-Brahmaputra river system and hence there is no coral reef formation there.

Answer: a) 1, 2 and 3 only

Q10. Consider the following:

- Bats
- Bears
- Rodents



Crustaceans

- Crustaceans make up a large group of arthropods that includes animals such as crabs, lobsters, crayfish and shrimp. They breathe with gills and have two pairs of antennae.



Insects

- In general, insects have three-part bodies, six jointed legs, compound eyes and two antennae.
- Bees, wasps, beetles, mosquitoes, flies, grasshoppers, ants, butterflies and moths, and dragonflies and damselflies are common types of insects.

Q12. Which one of the following is the correct sequence of a food chain?

- Diatoms-Crustaceans-Herrings
- Crustaceans-Diatoms-Herrings
- Diatoms-Herrings-Crustaceans
- Crustaceans-Herrings-Diatoms

Explanation:

- Food chain starts with a producer and ends with a top consumer.
- Phytoplankton** are the primary producers in the oceans. They include:

- diatoms (unicellular algae),**
- coccolithophores** (unicellular, eukaryotic protist),
- cyanobacteria (Bluegreen algae)**—synechococcus, prochlorococcus, nostoc, spirogyra etc..
- dinoflagellates (flagellated protists).**

So answer is between (a) and (c)

- Crustaceans form a very large group of arthropods which includes crabs, lobsters, crayfish, shrimp, krill and barnacles.
- Herrings is a fish.
- <http://octopus.gma.org/herring/biology/ecology/default.asp> says Herrings eat crustaceans.

Answer: (a). Tough question. There will be 3-5 questions like this one where you might have not even heard the name of the animal or plant. Don't worry.. rest 95 will be easy!

Q13. If you travel through the Himalayas, you are Likely to see which of the following plants naturally growing there?

- Oak
- Rhododendron
- Sandalwood

Select the correct answer using the code given below

- 1 and 2 only
- 3 only
- 1 and 3 only
- 1, 2 and 3

Explanation:

Oaks occur in Sub-tropical Broad-leaved Hill Forests, Montane Wet Temperate Forests and Himalayan Dry Temperate Forests [All these forest types are found in Himalayan regions]

Rhododendron occur in sub-alpine cold desert regions [Himalayan regions].

Sandalwood (Red Sanders) occur in Dry Deciduous forests [Ganga plains, South and Central India.].

Answer: a) 1 and 2 only

Q14. If you walk through countryside, you are likely to see some birds stalking alongside the cattle to seize the insects, disturbed by their movement through grasses, Which of the following is/are such bird/birds?

1. Painted Stork
2. Common Myna
3. Black-necked Crane

Select the correct answer using the code given below.

- a. 1 and 2
- b. **2 only**
- c. 2 and 3
- d. 3 only

Black-necked crane feeds on meadows in higher altitudes of Tibetan plateau.



Q15. Other than poaching, what are the possible reasons for the decline in the population of Ganges River Dolphins?

1. Construction of dams and barrages on rivers
2. Increase in the population of crocodiles in rivers
3. Getting trapped in fishing nets accidentally

4. Use of synthetic fertilizers and other agricultural chemicals in crop-fields in the vicinity of rivers

Select the correct answer using the code given below.

- a. 1 and 2 only
- b. 2 and 3 only
- c. **1, 3 and 4 only**
- d. 1, 2, 3 and 4

'Susu' or Ganges River Dolphin shares its habitat with crocodiles, freshwater turtles and wetland birds, many of which are fish eaters and are potential competitors with dolphins.

http://wwf.panda.org/what_we_do/endangered_species/cetaceans/about/river_dolphins/ganges_river_dolphin/

Q16. With reference to two non-conventional energy sources called 'coal bed methane' and 'shale gas', consider the following 'statements':

1. Coal bed methane is the pure methane gas extracted from coal seams, while shale gas is a mixture of propane and butane only that can be extracted from fine-grained sedimentary rocks.
2. In India abundant coal bed methane sources exist, but so far no shale gas sources have been found.

Which of the statements given above is/are correct?

- a. 1 only
- b. 2 only
- c. Both 1 and 2
- d. **Neither 1 nor 2**

CBM = Methane

Shale gas = Lot of **Methane** + Little **Ethane, Propane, & Butane** + very little **carbon dioxide, nitrogen, and hydrogen sulfide**.

Abundant shale reserves occur in India

[Explained in "Economic Geography Notes" <http://imojo.in/7rxalg>]

Q17. In India, cluster bean (Guar) is traditionally used as a vegetable or animal feed, but recently the cultivation of this has assumed significance. Which one of the following statements is correct in this context?

- The oil extracted from seeds is used in the manufacture of biodegradable plastics
- The gum made from its seeds is used in the extraction of shale gas**
- The leaf extract of this plant has the properties of anti-histamines
- It is a source of high quality biodiesel

Gaur is used to make water viscous before pumping into ground for hydraulic fracking used to extract shale gas.

[Explained in “Economic Geography Notes” <http://imojo.in/7rxal9>]

Q18. With reference to technologies for solar power production, consider the following statements:

- ‘Photovoltaics’ is a technology that generates electricity by direct conversion of light into electricity, while ‘Solar Thermal’ is a technology that utilizes the Sun’s rays to generate heat which is further used in electricity generation process.
- Photovoltaics generates Alternating Current (AC), while Solar Thermal generates Direct Current (DC).
- India has manufacturing base for Solar Thermal technology, but not for Photovoltaics.

Which of the statements given above is / are correct?

- 1 only
- 2 and 3 only
- 1, 2 and 3
- None

Explanation:

Photoelectric effect = When light strikes on a material, electrons are dislodged [photons dislodge electrons].

Photovoltaic = The dislodged electrons if channeled through a conductor will create electric current (voltage Or potential difference) = **Solar Panels**. [Electric current is nothing but movement of electrons from high potential to low potential area (more electrons to less electrons region)]

Solar thermal = converting light into heat = **solar cooker, solar water heater**.

Photovoltaics generate direct current (DC). [Rotating = AC, Stationary = DC. Electric generator, wind turbine generate AC while solar panels generate DC]

Solar thermal is mostly used for water heating purposes. Electricity can be generated by using hot water steam to rotate turbine = AC current.

In India both solar panels and solar cookers are manufactured. [Remember India – USA WTO ‘domestic content’ dispute?]

Answer: a) 1 only

Q19. There is some concern regarding the nanoparticles of some chemical elements that are used by the industry in the manufacture of various products. Why?

- They can accumulate in the environment, and contaminate water and soil.
- They can enter the food chains.
- They can trigger the production of free radicals.

Select the correct answer using the code given below.

- 1 and 2 only
- 3 only
- 1 and 3 only
- 1, 2 and 3**

Explained under “Air Pollution”

Q20. Which of the following are some important pollutants released by steel industry in India?

- Oxides of sulphur

2. Oxides of nitrogen
3. Carbon monoxide
4. Carbon dioxide

Select the correct answer using the code given below.

- a. 1, 3 and 4 only
- b. 2 and 3 only
- c. 1 and 4 only
- d. 1, 2, 3 and 4

Explanation:

Coke is used as fuel in blast furnace. Combustion of coke releases both CO and CO₂.

Oxides of sulphur and oxides of nitrogen are common pollutants released when any fossil fuel is burnt. (Coal and iron ore contains small amount of sulphur. Presence of sulphur makes iron weak)

Answer: d) All

Q21. Brominated flame retardants are used in many household products like mattresses and upholstery. Why is there some concern about their use?

1. They are highly resistant to degradation in the environment.
2. They are able to accumulate in humans and animals.

Select the correct answer using the code given below.

- a. 1 only
- b. 2 only
- c. Both 1 and 2
- d. Neither 1 nor 2

Q22. Consider the following countries:

1. Denmark
2. Japan
3. Russian Federation
4. United Kingdom
5. United States of America

Which of the above are the members of the 'Arctic Council'?

- a. 1, 2 and 3
- b. 2, 3 and 4
- c. 1, 4 and 5
- d. **1, 3 and 5**

- The Arctic Council consists of the eight Arctic States: **Canada, the Kingdom of Denmark (including Greenland and the Faroe Islands), Finland, Iceland, Norway, Russia, Sweden and the United States.**

Q23. With reference to 'Global Environment Facility', which of the following statements is/are correct?

- a. It serves as financial mechanism for 'Convention on Biological Diversity' and 'United Nations Framework Convention on Climate Change'.
- b. It undertakes scientific research on environmental issues at global level
- c. It is an agency under OECD to facilitate the transfer of technology and funds to underdeveloped countries with specific aim to protect their environment.
- d. Both (a) and (b)

Explanation:

GEF is an independent financial organization.

It may fund scientific research but it is not directly involved in scientific research.

IPCC takes care of the most of the research work.

Answer: a)

Q24. If a wetland of international importance is brought under the 'Montreux Record', what does it imply?

- a. Changes in ecological character have occurred, are occurring or are likely to occur in the wetland as a result of human interference.
- b. The country in which the wetland is located should enact a law to prohibit any human activity within five kilometers from the edge of the wetland
- c. The survival of the wetland depends on the cultural practices and traditions of

- certain communities living in its vicinity and therefore the cultural diversity therein should not be destroyed
- d. It is given the status of 'World Heritage Site'

- a. 1 and 2 only
b. 3 only,
c. 1 and 3 only
d. **1, 2. and 3**

World Heritage Convention explained in "Biodiversity", Rest two under "Fighting Climate Change-Biodiversity Protection" Page | 25

Q27. The scientific view is that the increase in global temperature should not exceed 2 °C above pre-industrial level. If the global temperature increases beyond 3°C above the pre-industrial level, what can be its possible impact/impacts on the world?

1. Terrestrial biosphere tends toward a net carbon source
2. Widespread coral mortality will occur.
3. All the global wetlands will permanently disappear.
4. Cultivation of cereals will not be possible anywhere in the world.

Select the correct answer using the code given below.

- a. 1 only
b. 1 and 2 only
c. 2, 3 and 4 only
d. 1, 2, 3 and 4

Explanation:

Taiga and temperate forests act as an important carbon sink. Global warming by 3°C will turn these forests into carbon source.

Corals are very sensitive to temperature changes. 3°C rise in global temperature will lead to widespread coral mortality.

3°C rise in global temperature will lead to submergence of many low lying coastal wetlands (not all) due to rise in sea levels. Inland wetlands like Keoladeo Ghana National Park will not be effected (it is 100s of kilometers away from coast).

Because of the slow inertia, long response time for parts of the climate system, it has been estimated that we are already committed to a sea-level rise of approximately 2.3 metres

The Montreux Record is a register of wetland sites on the List of Wetlands of International Importance where changes in ecological character have occurred, are occurring, or are likely to occur as a result of technological developments, pollution or other human interference. It is maintained as part of the Ramsar List.

Answer: a)

Q25. With reference to a conservation organization called Wetlands International', which of the following statements is/are correct?

1. It is an intergovernmental organization formed by the countries which are signatories to Ramsar Convention.
2. It works at the field level to develop and mobilize knowledge, and use the practical experience to advocate for better policies.

Select the correct answer using the code given below.

- a. 1 only
b. 2 only
c. Both 1 and 2
d. Neither 1 nor 2

Answer: b) 2 only. Wetlands International is an NGO

Q26. Consider the following international agreements:

1. The International Treaty on Plant Genetic Resources for Food and Agriculture
2. The United Nations Convention to Combat Desertification
3. The World Heritage Convention

Which of the above has/have a bearing on the biodiversity?

(7.5 ft) for each degree Celsius of temperature rise within the next 2,000 years.

<http://www.pnas.org/content/110/34/13745.abstract?sid=26fd1d37-7276-46e2-9192-0931e6ebf6ab>

Cultivation of cereals in tropics will take a hit. But in temperate regions their production increases in the short run.

Answer: b) 1 and 2 only

Q28. Consider the following statements:

1. Animal Welfare Board of India is established under the Environment (Protection) Act, 1986.
2. National Tiger Conservation Authority is a statutory body.
3. National Ganga River Basin Authority is chaired by the Prime Minister.

Which of the statements given above is/ are correct?

- a. 1 only
- b. 2 and 3 only
- c. 2 only
- d. 1, 2 and 3

Animal Welfare Board of India is established under Prevention of Cruelty to Animals Act.

Answer: b) 2 and 3 only

Q29. With reference to Bombay Natural History Society (BNHS), consider the following statements :

1. It is an autonomous organization under the Ministry of Environment and Forests.
2. It strives to conserve nature through action-based research, education and public awareness.
3. It organizes and conducts nature trails and camps for the general public.

Which of the statements given above is/are correct?

- a. 1 and 3 only
- b. 2 only

- c. 2 and 3 only
- d. 1, 2 and 3

- The Bombay Natural History Society, founded in 1883, is one of the largest non-governmental organizations in India engaged in **conservation and biodiversity research**.
- It supports many research efforts through grants and publishes the **Journal of the Bombay Natural History Society**.
- It organizes and conducts nature trails and camps for the general public.
- Many prominent naturalists, including the ornithologists Sálím Ali and S. Dillon Ripley, have been associated with it.

Answer: c) 2 and 3 only

Q30. Consider the following statements regarding 'Earth Hour'

1. It is an initiative of UNEP and UNESCO.
2. It is a movement in which the participants switch off the lights for one hour on a certain day every year.
3. It is a movement to raise the awareness about the climate change and the need to save the planet.

Which of the statements given above is / are correct?

- a. 1 and 3 only
- b. 2 only
- c. 2 and 3 only.
- d. 1, 2 and 3

Earth Hour

- Earth Hour is a worldwide movement for the planet organized by the **World Wide Fund for Nature (WWF)**.
- The event is held worldwide annually encouraging individuals, communities, households and businesses to turn off their non-essential lights for one hour, from 8:30 to 9:30 p.m. towards the end of March, as a symbol for their commitment to the planet.
- It was famously started as a lights-off event in Sydney, Australia in 2007.

- Earth Hour engages a massive mainstream community on a broad range of environmental issues.

Answer: c) 2 and 3 only

Q31. Every year, a month long ecologically important campaign/festival is held during which certain communities/ tribes plant saplings of fruit-bearing trees. Which of the following are such communities/tribes?

- Bhutia and Lepcha
- Gond and Korku**
- Irula and Toda
- Sahariya and Agariya

<http://www.thehindu.com/todays-paper/tp-miscellaneous/tp-others/their-own-green-revolution/article3823031.ece>

2015

Q1. Which of the following National Parks is unique in being a swamp with floating vegetation that supports a rich biodiversity?

- Bhitarkanika National Park
- Keibul Lamjao National Park**
- Keoladeo Ghana National park
- Sultanpur National park

Q2. What can be the impact of excessive/inappropriate use of nitrogenous fertilizers in agriculture?

- Proliferation of nitrogen-fixing microorganisms in soil can occur.
- Increase in the acidity of soil can take place
- Leaching of nitrate to the ground-water can occur.

Select the correct answer using the code given below.

- 1 and 3 only
- 2 only
- 2 and 3 only
- 1,2 and 3

Explanation:

Nitrification is important in agricultural systems, where fertilizer is often applied as ammonia. Conversion of this ammonia to nitrate **increases nitrogen leaching** because nitrate is more water-soluble than ammonia.

Agricultural fertilization and the use of nitrogen fixing plants also contribute to atmospheric NO_x, by promoting nitrogen fixation by microorganisms. Excess NO_x leads to acid rain. Acid rain lowers pH of soil (increase in acidity of soil)

The legume-rhizobium symbiosis is a classic example of mutualism—rhizobia supply ammonia or amino acids to the plant and in return receive organic acids as a carbon and energy source.

So, excessive/inappropriate use of nitrogenous fertilizers can make the plants independent of both symbiotic and free living nitrogen fixers. Fixers don't get the food from the plants due to broken relationship and other factors. So their population decreases.

Answer: c) 2 and 3 only

Q3. With reference to the International Union for Conservation of Nature and Natural Resources (IUCN) and the Convention on International Trade in Endangered Species of Wild Fauna and Flora (CITES), which of the following statements is/are correct?

- IUCN is an organ of the United Nations and CITES is an international agreement between governments
- IUCN runs thousands of field projects around the world to better manage natural environments.
- CITES is legally binding on the States that have joined it, but this Convention does not take the place of national laws.

Select the correct using the code given below.

- 1 only
- 2 and 3 only
- 1 and 3 only
- 1, 2 and 3

Explanation:

IUCN is an NGO. CITES is an international agreement between governments (multilateral treaty).

Answer: b) 2 and 3 only

Q4. With reference to 'fly ash' produced by the power plants using the coal as fuel, which of the following statements is/are correct?

1. Fly ash can be used in the production of bricks for building construction
2. Fly ash can be used as a replacement for some of the Portland cement contents of concrete
3. Fly ash is made up of silicon dioxide and calcium oxide only, and does not contain any toxic elements.

Select the correct answer using the code given below

- a. 1 and 2
- b. 2 only
- c. 1 and 3
- d. 3 only

Fly ash does contain heavy toxic elements like arsenic, cobalt, lead etc.

Answer: a) 1 and 2

Q5. With reference to 'dugong', a mammal found in India, which of the following statements is/are correct?

1. It is a herbivorous marine animal.
2. It is found along the entire coast of India
3. It is given legal protection under Schedule 1 of the Wildlife (Protection) Act, 1972.

Select the correct answer using the code given below.

- a. 1 and 2
- b. 2 only
- c. 1 and 3
- d. 3 only

Explanation:

- Dugong (sea cow) is an herbivorous animal. It eats sea grass and aquatic plants found in shallow oceans (At depths sea grass and aquatic plants don't grow due to absence of sunlight).
- Dugong 'vulnerable' herbivorous marine mammal.

Wildlife (Protection) Act, 1972

- Harming **endangered** (vulnerable, endangered, critically endangered) species listed in **Schedule 1** of the Act is prohibited throughout India.
- Hunting species, like those requiring special protection (Schedule II), big game (Schedule III), and small game (Schedule IV), is regulated through licensing.
- A few species classified as **vermin (Schedule V)**, may be hunted without restrictions.

Answer: c) 1 and 3

Q6. Which one of the following is the national aquatic animal of India?

- a. Saltwater crocodile
- b. Olive ridley turtle
- c. **Gangetic dolphin**
- d. Gharial

Q7. Which one of the following regions of India has a combination of mangrove forest, evergreen forest and deciduous forest?

- a. North Coastal Andhra Pradesh
- b. South-West Bengal
- c. Southern Saurashtra
- d. Andaman and Nicobar Islands

Explanation:

North Coastal Andhra Pradesh has deciduous and mangrove forests.

South West Bengal has mangrove, semi-evergreen and moist deciduous.

Southern Saurashtra has dry deciduous and tropical thorn.

Answer: d) Andaman and Nicobar Islands

Q8. Which one of the following is associated with the issue of control and phasing out of the use of ozone-depleting substances?

- Bretton Woods Conference
- Montreal Protocol
- Kyoto Protocol
- Nagoya Protocol

Explanation:

Bretton Woods Conference established the International Bank for Reconstruction and Development (IBRD) and the International Monetary Fund (IMF).

Montreal Protocol is an international treaty to protect the ozone layer by phasing out the production of ozone depleting substances. It is legally binding.

Kyoto Protocol implemented the objective of the UNFCCC to fight global warming by reducing greenhouse gas concentrations in the atmosphere to "a level that would prevent dangerous anthropogenic interference with the climate system". It is binding on the parties.

Nagoya Protocol is a supplementary agreement to the 1992 Convention on Biological Diversity (CBD) on "Access to Genetic Resources and the Fair and Equitable Sharing of Benefits Arising from their Utilization".

Answer: b)

Q9. What is Rio+20 Conference, often mentioned in the news?

- It is the United nations Conference on Sustainable Development
- It is a Ministerial Meeting of the World Trade Organization
- It is a Conference of the Inter-governmental Panel on Climate Change
- It is a Conference of the Member Countries of the Convention on Biological Diversity

Explanation:

- Earth Summit 1992 is also known as The **United Nations Conference on Environment and Development (UNCED)**.
- As a follow-up, the **World Summit on Sustainable Development (Rio+10)** was held in 2002 in Johannesburg, South Africa.
- In 2012, the **United Nations Conference on Sustainable Development** was also held in Rio, and is also commonly called **Rio+20** or Rio Earth Summit 2012.

Answer: a) United nations Conference on Sustainable Development

Q10. Which of the following statements regarding 'Green Climate Fund' is/are correct?

- It is intended to assist the developing countries in adaptation and mitigation practices to counter climate change.
- It is founded under the aegis of UNEP, OECD, Asian Development Bank and World Bank

Select the correct answer using the code given below.

- 1 only
- 2 only
- Both 1 and 2
- Neither 1 nor 2

Green Climate Fund

- The Green Climate Fund (GCF) is a fund within the framework of the UNFCCC.
- It is a mechanism to redistribute money from the developed to the developing world.
- GCF will help developing countries financially in adapting mitigation practices to counter climate change.

Answer: a) 1 only

Q11. Which one of the following National Parks has a climate that varies from

tropical to subtropical, temperate and arctic?

- Khangchendzonga National park
- Nandadevi National Park
- Neora Valley National Park
- Namdapha National park

Explanation:

Khangchendzonga National park and Nandadevi National Park lies in Greater Himalayas (Conifer to Alpine climate)

Neora Valley National Park lies near Darjeeling (Shiwaliks) [Sub-tropical broad leaved hill and Sub-tropical moist hill (pine) forests].

Namdapha National park lies in Purvanchal hills in Arunachal Pradesh.

In the Namdapha National Park, located in Arunachal Pradesh, the climate varies from tropical to subtropical, temperate and arctic. It is tropical and subtropical in the southern region while it is of arctic type in the northern part of the park.

Answer: d) Namdapha National park**Q12. 'BioCarbon Fund Initiative for Sustain-able Forest Landscapes' is managed by the**

- Asian Development Bank
- International Monetary Fund
- United Nations Environment Programme
- World Bank

Explanation:

The BioCarbon Fund Initiative for **Sustainable Forest Landscapes (ISFL)** is a multilateral fund, supported by donor governments and managed by the **World Bank**.

Answer: d) World Bank

Q13. The Genetic Engineering Appraisal Committee is constituted under the

- Food Safety and Standards Act, 2006

- Geographical Indications of Goods (Registration and Protection) Act, 1999
- Environment (Protection) Act, 1986
- Wildlife (Protection) Act, 1972

Explanation:

The Genetic Engineering Appraisal Committee (GEAC) is the apex body constituted in the Ministry of Environment and Forests under 'Rules for Manufacture, Use, Import, Export and Storage of Hazardous Microorganisms/Genetically Engineered Organisms or Cells 1989', under the Environment Protection Act, 1986.

Answer: c)

Q14. 'Basel III Accord' or simply 'Basel III', often seen in the news, seeks to

- develop national strategies for the conservation and sustainable use of biological diversity
- improve banking sector's ability to deal with financial and economic stress and improve risk management
- reduce the greenhouse gas emissions but places a heavier burden on developed countries
- transfer technology from developed Countries to poor countries to enable them to replace the use of chlorofluorocarbons in refrigeration with harmless chemicals

Basel III accord is about improving banking by dealing with financial and economic stress.

Answer: b) improve banking sector's ability

Q15. In a particular region in India, the local people train the roots of living tree into robust bridges across the streams. As the time passes these bridges become stronger. These unique 'Living Root Bridges' are found in

- Meghalaya
- Himachal Pradesh
- Jharkhand
- Tamil Nadu

Q16. Consider the following States:

1. Arunachal Pradesh
2. Himachal Pradesh
3. Mizoram

In which of the following states do “Tropical Wet Evergreen Forests” occur?

- a. 1 only
- b. 2 and 3 only
- c. 1 and 3 only
- d. 1, 2 and 3

Explanation:

Himachal Pradesh has many varieties of forests but not wet evergreen.

Purvanchal hills start at the syntaxial bend (Namcha Barwa) which is in Arunachal Pradesh. Wet Evergreen forests are scattered all along the Purvanchal hills.

Answer: c) 1 and 3 only

Q17. With reference to ‘Forest Carbon Partnership Facility’, which of the following statements is/are correct?

1. it is global partnership of governments, businesses, civil society and indigenous peoples
2. it provides financial aid to universities, individual scientists and institutions involved in scientific forestry research to develop eco-friendly and climate adaptation technologies for sustainable forest management
3. It assists the countries in their ‘REDD+ (Reducing Emission from Deforestation and Forest Degradation+)’ efforts by providing them with financial and technical assistance.

Select the correct answer using the code given below

- a. 1 only
- b. 2 and 3 only
- c. 1 and 3 only
- d. 1, 2 and 3

It provide financial incentives to countries (only) in their REDD+ efforts.

Answer: c) 1 and 3 only

Q18. With reference to an organization known as ‘Birdlife International’ which of the following statements is/are correct?

1. It is a Global Partnership of Conservation Organizations.
2. The concept of ‘biodiversity hotspots’ originated from this organization.
3. It identifies the sites known/referred to as ‘Important Bird and Biodiversity Areas’.

Select the correct answer using the code given below.

- a. 1 only
- b. 2 and 3 only
- c. 1 and 3 only
- d. 1, 2 and 3

Answer: c) 1 and 3 only

Q19. In India, in which one of the following types of forests is teak a dominant tree species?

- a. Tropical moist deciduous forest
- b. Tropical rain forest
- c. Tropical thorn scrub forest
- d. Temperate forest with grasslands

Answer: a)

Q20. Which one of the following is the best description of the term “ecosystem”?

- a. A community of organisms interacting with one another
- b. That part of the Earth which is inhabited by living organisms
- c. A community of organisms together with the environment in which they live.
- d. The flora and fauna of a geographical area.

Answer: c)

Explanation

1. A community of organisms interacting with one another – no specific term.

2. That part of the Earth which is inhabited by living organisms – biosphere.
3. A community of organisms together with the environment in which they live – ecosystem.
4. The flora and fauna of a geographical area – biodiversity.

Q21. With reference to bio-toilets used by the Indian Railways, consider the following statements:

1. The decomposition of human waste in the bio-toilets is initiated by a fungal inoculum.
2. Ammonia and water vapour are the only end products in this decomposition which are released into the atmosphere.

Which of the statements given above is/are correct?

- a. 1 only
- b. 2 only
- c. Both 1 and 2
- d. Neither 1 nor 2

Decomposition of human waste in bio-toilets is carried out by anaerobic bacteria.

The final waste is CO₂ and CH₄. [Explained in detail under “Water Pollution”]

Answer: d) Neither 1 nor 2

Q22. With reference to the Indian Renewable Energy Development Agency Limited (IREDA), which of the following statements is/are correct?

1. It is a Public Limited Government Company.
2. It is a Non – Banking Financial Company.

Select the correct answer using the code given below.

- a. 1 only
- b. 2 only
- c. Both 1 and 2
- d. Neither 1 or 2

IREDA is Public Limited Government Company established as a **Non-Banking Financial Institution** in 1987 engaged in promoting, developing and extending financial assistance for setting up projects relating to new and renewable sources of energy and energy efficiency/conservation with the motto: **“Energy For Ever”**. Page | 32

Answer: c) Both

Environment and Ecosystem - Components of an Ecosystem

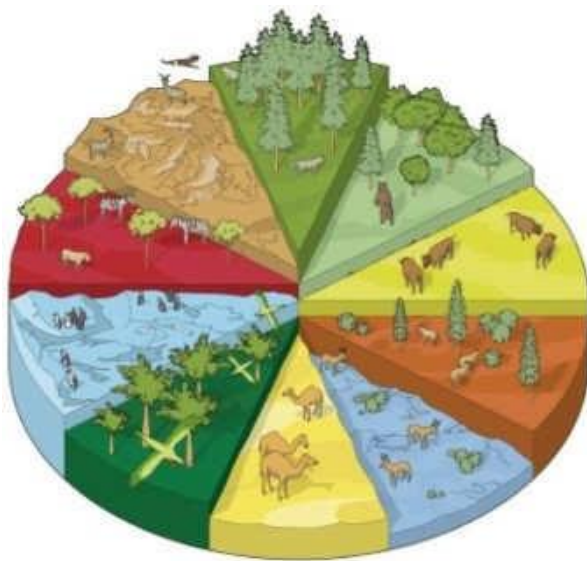
Environment

- Environment is the natural component in which biotic (living) and abiotic (nonliving) factors interact with each other. These interactions shape the habitat and ecosystem of an organism.
- In biological sense, environment constitute the physical (nutrients, water, air etc.) and biological factors (biomolecules, organisms) along with their chemical interactions (chemical cycles – carbon cycle, nitrogen cycle etc.) that affect an organism or a group of organisms.
- All organisms are dependent on the environment to carry out their natural life processes (birth to death) and to meet their physical requirements (food, energy, water, oxygen, shelter etc.).
- The environment is not static. Both biotic and abiotic factors are in a constant flux and keep changing continuously.

Habitat

- Habitat is the physical environment in which an organism lives (it corresponds to address of an organism).
- It is an ecological or environmental area inhabited by **particular species** of plants, animals, fungi, etc. **Many habitats make up the environment.**
- A single habitat may be common for more than one organism which have similar requirements.
- For example, a single aquatic habitat may support a fish, frog, crab, phytoplankton and many other kinds of organisms.

- The various species sharing a habitat thus have the same 'address'. Forest, river etc. are other examples of habitat.
- **All habitats are environments but all environments are not habitats.**



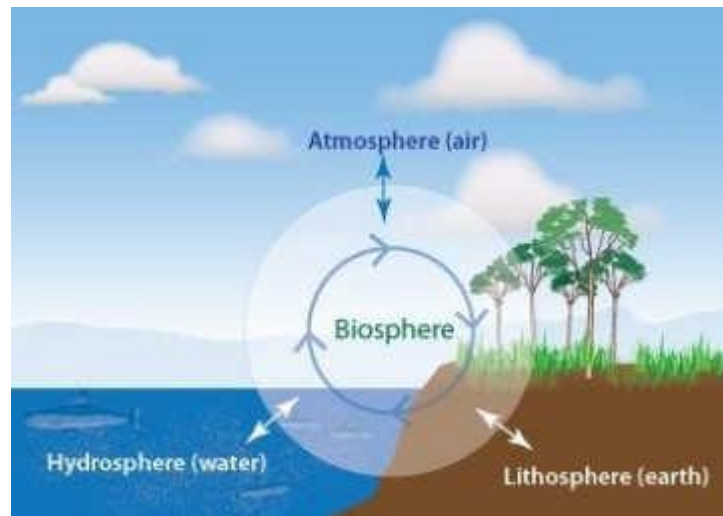
Difference between Habitat and Environment?

<http://www.differencebetween.com/difference-between-habitat-and-vs-environment/>

- A habitat always has **life** in it, whereas the environment does not necessarily have life in it.
- The habitat is a defined place or area of the environment according to the requirements of a particular life form. Therefore, a habitat is always an environment, but an environment is not always a habitat.
- A habitat is always a preference of **one species**, whereas an environment could be a preference of many species that could eventually become many habitats.
- Usually, the environment governs the properties of a habitat, but not vice versa.

Biosphere

- The biosphere is the **biological component** (supporting life) of earth which includes the lithosphere, hydrosphere and atmosphere.
- The biosphere includes all living organisms on earth, together with the dead organic matter produced by them.



Picture

Credits:

<https://briangrimmerblog.files.wordpress.com/2014/07/image.jpg>

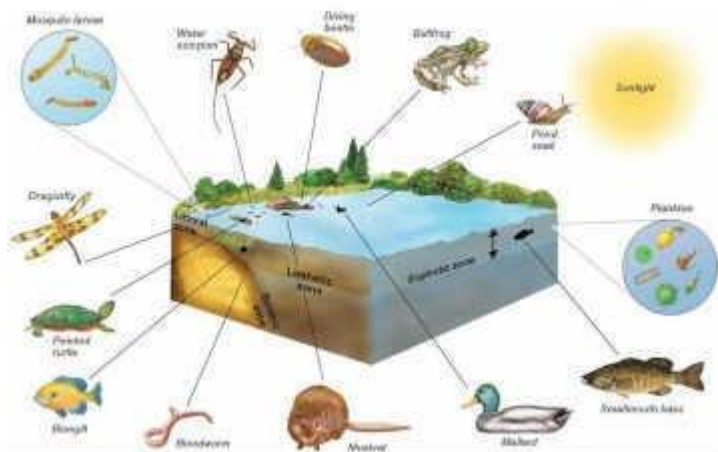
- Biosphere is absent at extremes of the North and South poles, the highest mountains and the deepest oceans, since existing hostile conditions there do not support life [Life is the characteristic feature of biosphere].
- Occasionally spores of fungi and bacteria do occur at great height beyond 8,000 metres, but they are metabolically inactive, and hence represent only dormant life.

Ecosystem

- An ecosystem can be visualised as a **functional unit of nature**, where living organisms [**producers, consumers, and decomposers**] interact among themselves and also with the surrounding physical environment.
- Ecosystem varies greatly in size from a small pond to a large forest or a sea.
- Forest, grassland and desert are some examples of terrestrial ecosystems; pond, lake, wetland, river and estuary are some examples of aquatic ecosystems. Crop fields and an aquarium may also be considered as man-made ecosystems.
- In the ecosystem, biotic and abiotic components are linked together through **nutrient cycles** and **energy flows**.
- An ecosystem can be of any size but usually encompasses **specific and limited species**. Eg: Aquatic Ecosystem. [This is

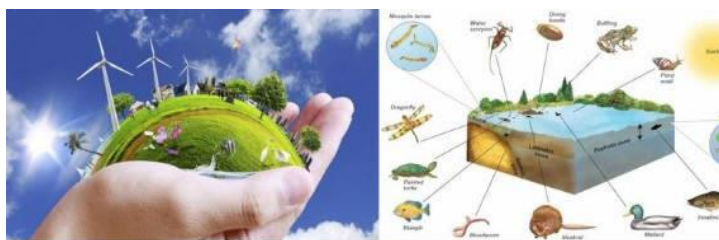
how ecosystem is different from Environment]

- Everything that lives in an ecosystem is dependent on the other species and elements that are also part of that **ecological community**. If one part of an ecosystem is damaged or disappears, it has an impact on everything else.



Aquatic Ecosystem

Difference Between Ecology, Environment & Ecosystem



Ecology is the study of interactions between organisms, organisms and the surroundings occurring within an ecosystem or environment.

- Ecology is the study of the ecosystems and the environment.
- Environment is a group of ecosystems.
- Ecosystem is a functional unit of environment (mostly biosphere).

Environment → Can be Almost Everything or a Small region.

Habitat → Area where an organism lives.

Biosphere → The region on earth that supports life.

Ecosystem → **Producers, Consumers, Decomposers and their relationships (tiny environment). It is the functional unit of the environment.**

Q0. Which one of the following is the best description of the term "ecosystem"?

Page | 34

- A community of organisms interacting with one another
- That part of the Earth which is inhabited by living organisms
- A community of organisms together with the environment in which they live.
- The flora and fauna of a geographical area.

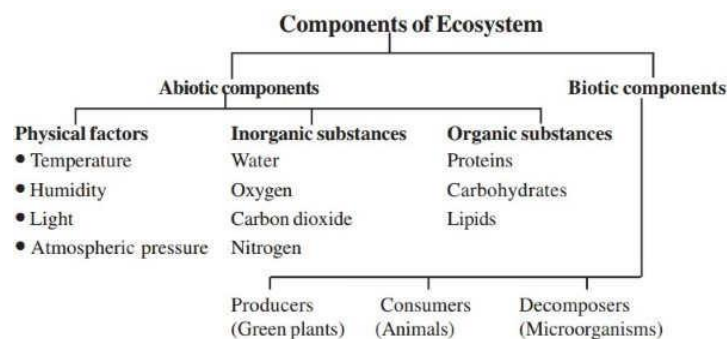
Answer: c)

Explanation

- A community of organisms interacting with one another – no specific term.
- That part of the Earth which is inhabited by living organisms – biosphere.
- A community of organisms together with the environment in which they live – ecosystem.
- The flora and fauna of a geographical area – biodiversity.

Components of an Ecosystem

- The components of the ecosystem are categorized into abiotic or non-living and biotic or living components. Both the components of ecosystem and environment are **same**.



Abiotic Components

- Abiotic components are the inorganic and non-living parts which act as major **limiting factors**.

Limiting factor

- Lot of factors determine the survival of an organism. One single factor can limit the range of an organism. This single factor is called as a limiting factor.
- For example, seeds don't germinate quickly in evergreen rain forests in spite of good rains and vegetation as the surface soil is heavily **leached** (nutrients washed away by running water). Here, poor soil is the limiting factor.
- Likewise, germinated saplings may not survive due to lack of light because of the dense canopy. Here, light [shade of the forest] is the limiting factor.

Q1. If a tropical rain forest is removed, it does not regenerate quickly as compared to a tropical deciduous forest. This is because [Prelims Question]

- the soil of rain forest is deficient in nutrients
- propagules of the trees in a rain forest have poor viability
- the rain forest species are slow-growing
- exotic species invade the fertile soil of rain forest.

Explanation:

- the soil of rain forest is deficient in nutrients: Rainforest = Rainfall through the year. It rains almost every day = The top soil is continuously washed away = nutrients are also washed away = **leaching of nutrients** = very little fertility remains in top soil = most of the seeds don't germinate for years = regeneration is very slow (it takes decades). But the layer below top soil (sub-soil) is very fertile. So plants grow very quickly once their roots reach the sub-soil and if they receive enough sunlight.
- propagules of the trees in a rain forest have poor viability: Propagule = detachable structure that can give rise to a new plant, e.g. a bud, sucker, or

spore [Asexual Reproduction in plants]. Seed bearing plants are more significant than Propagules in rainforest. So propagules don't really matter.

- the rain forest species are slow-growing: The plant species in rainforests compete for sunlight. So they grow as rapidly as they can.
- exotic species invade the **fertile** soil of rain forest: this statement is wrong as the rainforest soil is heavily leached. Hence they are not fertile. But exotic invasive species are a threat to rain forests (E.g. Most plantation crops like rubber, palm etc.).

Answer: a)

- Light:** The spectral quality of solar radiation is important for life. The UV component of the spectrum is harmful to many organisms.
- Rainfall:** Majority of biochemical reactions take place in an aqueous medium. These biochemical reactions are important for survival of an organism. So rainfall is an important limiting factor.
- Temperature:** Latitudinal insolation decides the temperature range of a region. A few organisms can tolerate and thrive in a wide range of temperatures (they are called **eurythermal**), but, a vast majority of them are restricted to a narrow range of temperatures (such organisms are called **stenothermal**).
- Atmosphere:** 21% oxygen helps in survival of many organisms, 78% nitrogen prevents spontaneous combustion and 0.038% carbon dioxide helps primary producers in the synthesis of carbohydrates.
 - Organic compounds:** Proteins, carbohydrates, lipids etc. are essential for energy transfer in the living world.
 - Inorganic compound:** Carbon, carbon dioxide, water, sulphur, nitrates, phosphates, and ions of various metals are essential for organisms to survive.
 - Altitude:** Vertical zonation of vegetation is caused due to altitude. Change in temperature with altitude is the limiting factor.
 - Buffering capacity of earth:** The most unique feature of the earth is its

buffering action due to which a neutral pH (pH- 7) is maintained in the soil and water bodies. The neutral pH is conducive for the survival and sustenance of living organisms.

- **Salinity:** Some organisms are tolerant of a wide range of salinities (**euryhaline**) but others are restricted to a narrow range (**stenohaline**). So salinity is a limiting factor.

Effect Of Abiotic Components On Terrestrial Primary Producers – Plants

- Plants are the reason that the other animals are able to survive on land. So the effect of abiotic factors on plants is crucial.

Light

- Extremely high intensity **favors root growth than shoot growth** which results in **increased transpiration, short stem, smaller thicker leaves**.
- On the other hand low intensity of light **retards growth**, flowering and fruiting.
- When the Intensity of light is less than the minimum, the plants ceases to grow due to **accumulation of CO₂** and finally dies.
- Out of 7 colours in the visible part of spectrum, only **red** and **blue** are effective in photosynthesis.
- Plant grown in blue light are small, red light results in elongation of cells results in etiolated plants. Plants grown in ultraviolet and violet light are dwarf.

Frost

- Frost results in freezing the soil moisture. The plants growing in such soil, get exposed to direct sun light in the morning, they are killed due to **increased transpiration** when their roots are unable to supply moisture. This is the main reason for innumerable death of sal seedlings.
- As a result of frost, water in the intercellular spaces of the plant gets frozen into ice which withdraws water from the interior of the cells. This results in **increasing concentration of salts and**

dehydration of cells. Thus coagulation and precipitation of the cell colloid results in death of plant. Also, frost leads to formation of canker.

Snow

- Snow acts as blanket, prevents further drop in temperature and **protects** seedlings from excessive cold and frost.
- Accumulation of snow on tree parts can break the branches or even uproot the tree.
- Snow shortens the period of vegetative growth.

Temperature

- High temperature results in death of plant due to coagulation of protoplasmic proteins [Some bacteria can survive high temperatures because of their protoplasmic proteins that don't coagulate at normally high temperatures].
- High temperature disturbs the balance between respiration and photo synthesis thereby causes depletion of food resulting in greater susceptibility to fungal and bacterial attack.
- It also results in desiccation of plant tissues and depletion of moisture.

Die back

- Refers the progressive dying usually backwards from the tip of any portion of plant.
- This is one of the adaptive mechanisms to avoid adverse conditions like drought.
- In this mechanism, the root remains alive for years together but the shoots dies.
- E.g. Sal, Red sanders, Silk cotton tree etc..

Biotic Components

Primary producers - Autotrophs (self-nourishing)

- Primary producers are basically green plants, certain bacteria and algae that carry out photosynthesis.
- In terrestrial ecosystem, grasses, plants and trees are the primary producers while

in aquatic ecosystem, microscopic algae [plankton] are the primary producers.

Consumers — Heterotrophs or Phagotrophs (other nourishing)

- Consumers are incapable of producing their own food. They depend on organic food derived from plants, animals or both.
- Consumers can be divided into two broad groups namely micro and macro consumers.

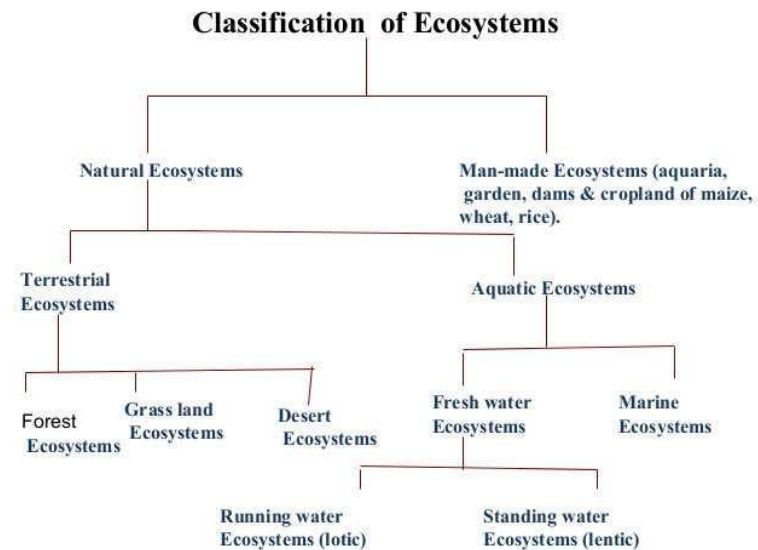
Macro consumers

- **Herbivores** are primary consumers which feed mainly on plants e.g. cow.
- Secondary consumers feed on primary consumers e.g. wolves, dogs, etc.
- **Carnivores** which feed on both primary and secondary consumers are called tertiary consumers e.g. lion which can eat wolves, snakes etc.
- **Omnivores** are organisms which consume both plants and animals e.g. man, bear, etc.

Micro consumers - Saprotrophs (decomposers or osmotrophs)

- They are bacteria and fungi which obtain energy and nutrients from dead organic substances (**detritus**) of plant and animals.
- The products of decomposition such as inorganic nutrients which are released in the ecosystem are reused by producers and thus recycled.
- Earthworm and certain soil organisms (such as nematodes, and arthropods) are detritus feeders and help in the decomposition of organic matter and are called **detritivores**.

Classification of Ecosystems



Ecology - Principles and Organizations

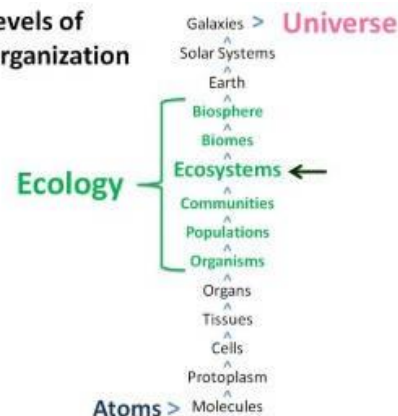
Levels of Organization

Ecologist study organisms ranging from the various levels of organization:

- Species
- Population
- Community
- Ecosystem
- Biome
- Biosphere



Levels of Organization



Ecology

- The term ecology was derived from two Greek words 'Oikos' meaning home and 'logos' meaning study.
- Ecology is the branch of biology concerned with the relations of organisms to one another (energy flow and mineral cycling) and to their physical surroundings (environment).
- Ecology encompasses study of individual, organisms, population, community, ecosystem, biome and biosphere which form the various levels of **ecological organization**.
- The Indian texts of **Vedas**, the **Samhitas**, the **Brahmanas** and the **Aranyakas-Upanishads** contain many references to ecological concepts. The Indian treatise on medicine, the **Caraka-Samhita** and the surgical text **Susruta-Samhita**, show that

people during this period had a good understanding of plant and animal ecology.

Levels Of Organizations In Ecology

- Ecology not only deals with the study of the relationship of individual organisms with their environment, but also with the study of populations, communities, ecosystems, biomes, and biosphere as a whole.

- **Individual:** Organism is an individual living being that has the ability to act or function independently. It may be any organism.
- **Species:** Species are a group of living organisms consisting of similar individuals capable of exchanging genes or of **interbreeding**, considered as the basic unit of taxonomy and denoted by a Latin binomial, e.g. Homo sapiens.
- **Population:** Population is a community of interbreeding organisms [**same species**], occupying a defined area during a specific time.
- Population growth rate is the percentage variation between the number of individuals in a population at two different times. It can be positive due to **birth** and/or **immigration** or negative due to **death** and/or **emigration**.
- The number of individuals per unit area at a given time is termed as population density.
- In case of large, mobile animals like tigers, leopards, lions, deer etc., the density may be determined by counting individual animals directly or by the **pugmarks** (foot imprints) left by the animals in a defined area.
- Pugmarks of each individual animals are unique and different from one others. Study of pug marks can provide the following information reliably if analyzed skillfully:
 1. Presence of **different species** in the area of study.
 2. Identification of **individual animals**.

3. Population of large cats (tigers, lions etc.).
4. **Sex ratio** and age (young or adult) of large cats. [Even sex of tigers can be determined using pugmarks. http://assets.wwfindia.org/downloads/reading_pugmarks.pdf also says the same]

- **Community:** Communities in most instances are named after the **dominant plant form** (species). For example: A grassland community is dominated by grasses, though it may contain herbs, shrubs, and trees, along with associated insects and animals of different species. A community is not fixed or rigid.
- On the basis of size and degree of relative independence communities may be divided into two types: Major Communities and Minor Communities.
 1. **Major Communities:** These are large sized and relatively independent. They depend only on the sun's energy from outside. Eg: Tropical evergreen forests.
 2. **Minor Communities:** These are dependent on neighboring communities and are often called **societies**. They are secondary aggregations within a major community. Eg: A mat of lichen on a cow dung pad.
- **Ecosystem:** An ecosystem is a community of organisms interacting with each other and with their environment such that energy is exchanged and system-level processes, such as the cycling of elements, emerge.
- **Biome:** Biome is a large naturally occurring community of flora and fauna occupying a major habitat. E.g. Rainforest biome or tundra biome.
- Plants and animals in a biome have common characteristics due to similar climates and can be found over a range of continents.
- Biomes are distinct from habitats, because any biome can comprise a variety of habitats (habitat: the natural home or environment of an organism).
- **Biosphere:** The biosphere is the biological component of earth which includes the lithosphere, hydrosphere and atmosphere.

The biosphere includes all living organisms on earth, together with the dead organic matter produced by them.

Principles of Ecology

- On planet Earth, life exists not just in a few favourable habitats but even in extreme and harsh habitats – scorching Rajasthan desert, perpetually rain-soaked Meghalaya forests, deep ocean trenches, torrential streams, permafrost polar regions, high mountain tops, boiling thermal springs, and stinking compost pits, to name a few. Even our intestine is a unique habitat for hundreds of species of microbes. How is this possible?

Adaptation

- Each organism is adapted to its particular environment. An adaptation is thus, “the appearance or behavior or structure or mode of life of an organism that allows it to survive in a particular environment”. E.g. Neck of a giraffe.
- Adaptation is any attribute of the organism (morphological – when trees grew higher, the giraffes neck got longer; physiological – in the absence of an external source of water, the kangaroo rat in North American deserts is capable of meeting all its water requirements through its internal fat oxidation; behavioral – animals migrating temporarily to a less stressful habitat) that enables the organism to survive and reproduce in its habitat.
- We need to breathe faster when we are on high mountains. After some days, our body adjusts to the changed conditions on the high mountain.
- Such small changes that take place in the body of a single organism over short periods, to overcome small problems due to changes in the surroundings, are called acclimatization.

Examples of Adaptation

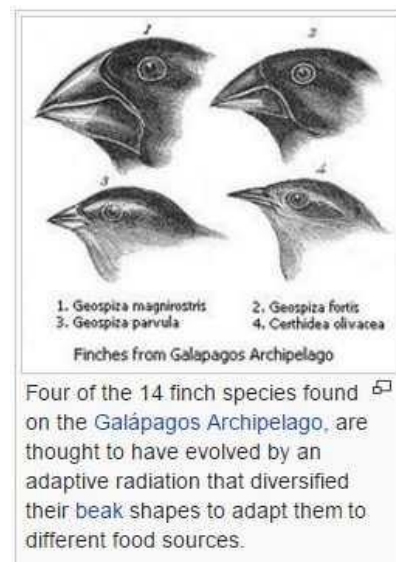
- Many adaptations have evolved over a long evolutionary time and are genetically fixed.

- In the absence of an external source of water, the **kangaroo rat** in North American deserts is capable of meeting all its water requirements through its internal **fat oxidation** (in which **water is a byproduct**).
- It also has the ability to **concentrate its urine** so that minimal volume of water is used to remove excretory products.
- Many desert plants have a thick **cuticle** on their leaf surfaces and have their stomata arranged in deep pits to minimise water loss through transpiration.
- They also have a special photosynthetic pathway (CAM) that enables their stomata to remain **closed during day time**.
- Some desert plants like **Opuntia**, have **no leaves** - they are reduced to **spines** and the photosynthetic function is taken over by the **flattened stems** [less leaves = less area available for transpiration].
- Mammals from colder climates generally have **shorter ears and limbs to minimise heat loss**. (This is called the Allen's Rule.) Guess why an elephant has a huge ear.
- In the polar seas aquatic mammals like seals have a thick layer of fat (blubber) below their skin that acts as an insulator and reduces loss of body heat.
- Some organisms possess adaptations that are physiological which allow them to respond quickly to a stressful situation. If you had ever been to any high altitude place (>3,500m) you must have experienced what is called **altitude sickness**.
- Its symptoms include nausea, fatigue and heart palpitations. This is because in the low atmospheric pressure of high altitudes, the body does not get enough oxygen. But, gradually you get **acclimatized** and stop experiencing altitude sickness.
- How did your body solve this problem? The body compensates low oxygen availability by **increasing red blood cell production, decreasing the binding capacity of hemoglobin** and by **increasing breathing rate**.
- Archaeobacteria** flourish in hot springs and deep sea hydrothermal vents where

temperatures far exceed 100 degree C. How is this possible?

- A **hyper thermophile** is an organism that thrives in extremely hot environments — from 60 °C. Hyper thermophiles are a subset of **extremophiles** (can tolerate temperatures of around 100 °C).
- They are able to survive because of the **protein structure** and the nature of the **cell membrane**. The protein molecules in the hyper thermophiles exhibit hyper thermo stability—that is, they can maintain structural stability (and therefore function) at high temperatures. The cell membrane contains **high levels of saturated fatty acids** to retain shape at high temperatures.
- Desert lizards lack the physiological ability that mammals have to deal with the high temperatures of their habitat, but manage to keep their body temperature fairly constant by behavioral means.
- They bask in the sun and absorb heat when their body temperature drops below the comfort zone, but move into shade when the ambient temperature starts increasing.
- Some species are capable of burrowing into the soil to hide and escape from the above-ground heat.

- In evolutionary biology, adaptive radiation is a process in which organisms diversify rapidly from an ancestral species into a multitude of new forms, particularly when a change in the environment makes new resources available, creates new challenges, or opens new environmental niches.



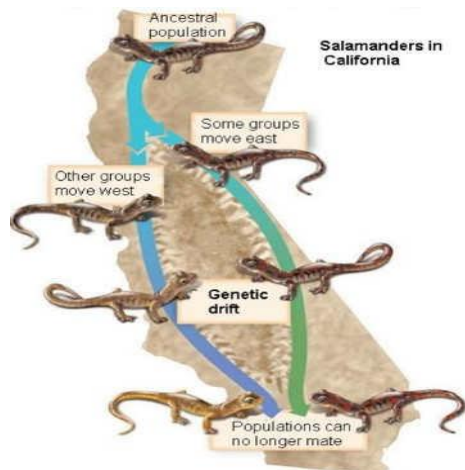
Speciation

- Speciation is the process by which new species are formed and evolution is the mechanism by which speciation is brought about.
- A species comprises of many populations. Often different populations of a species remain isolated due to some **geographic barrier** such as mountain, ocean, river, etc.
- The most common way a population undergoes speciation is by geographic isolation (**Allopatric speciation or geographic speciation**).

Variation

- Variations are induced by changes in genetic makeup due to **addition or deletion of certain genes**.
- Mutations, change in climate, geographical barriers etc. induce variations over a period of time.
- Species are generally composed of a number of distinct populations which freely interbreed even though they appear to be different in appearance [E.g. American man and Chinese women can interbreed. They sub species under Homo Sapiens].
- Difference in colour of skin, type of hair; curly or straight, eye colour, blood type among different ethnic groups represent variation within human species.

Adaptive radiation



- After a long period of time, the sub-populations become very different and get isolated, reproductively, i.e. they no longer interbreed.
- Later even when the barrier is removed the sub-populations are unable to interbreed and thus subsequently the sub-populations become two different species.

Mutation

- Mutation (a change in genetic material that results from an **error in replication of DNA**) causes new genes to arise in a population.
- Further, in a sexually reproducing population, **meiosis** and fertilization produce new combination of genes every generation, which is termed **recombination**.
- Thus members of the same species show 'variation' and are not exactly identical. Variations are heritable.

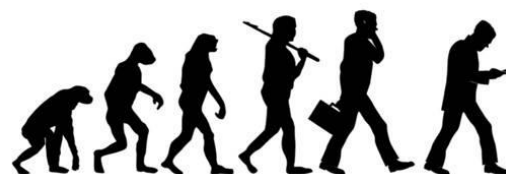
Natural Selection

- Natural Selection is the mechanism proposed by **Darwin** and Wallace. Natural selection is the process by which species **adapt** to their environment.
- It is an evolutionary force that selects among variations i.e. genes that help the organism to better adapt to its environment. Such genes are reproduced more in a population due to natural selection.

- Those offsprings which are suited to their immediate environment have a better chance of surviving, reaching reproductive age and passing on the suitable adaptations to their progeny.

Evolution

- Evolution is the change which gives rise to new species. It happens in order to make the organism better suit to the present environment.



- Climate change, competition, adaptability, need, changing environment etc. are the major forces behind evolution.
- **Mutation** and **recombination** are sources of 'variation' or differences in the genetic makeup or gene pool of a species. The variations over a period of time lead to the creation of a new species.
- Evolution involves the processes of natural selection, adaptation, variation etc.. Evolution leads to speciation or formation of new species.
- A valid theory of evolution was propounded by Charles Darwin and Alfred Wallace in 1859. This theory has been extended in the light of progress in genetics and is known as **Neo-Darwinism**.

Extinction

- Ever since life evolved on earth, new species better suited or adapted to the environment have appeared and older less successful forms have died or become extinct.
- The primary reason for these extinctions is environmental change or biological competition.
- Extinction occurs when species cannot evolve fast enough to cope with the

changes taking place in their environment.

- Extinction may take place due to catastrophic natural phenomena such as tsunamis, volcanoes etc.
- In recent time, human activities such as deportation, over exploitation, environmental pollution and environmental change are other factors responsible for extinction.

Q2. Which of the following are true?

1. The presence of specific features or certain habits, which enable a plant or an animal to live in its surroundings, is called Evolution.
2. The surroundings where an organism lives is called its habitat.
3. Small changes that take place in the body of a single organism over short periods, to overcome small problems due to changes in the surroundings, is called acclimatization
4. Gradual changes in an organism to survive in an environment is called Adaptation

Codes:

1. All
2. 2,3 only
3. 1,2,4 only
4. 1,2,3 only

Answer: b)

Explanation:

1. The presence of specific features or certain habits, which enable a plant or an animal to live in its surroundings, is called **Adaptation** and not evolution. E.g. Hibernation.
2. The surroundings where an organism lives is called its habitat. (True)
3. Small changes that take place in the body of a single organism over short periods, to overcome small problems due to changes in the surroundings, is called **acclimatization** (True). E.g. Soldiers undergo rigorous acclimatization training before they can

serve in harsh climatic regions like Siachen Glacier.

4. Gradual changes in an organism to survive in an environment is called Evolution and not Adaptation. E.g. The evolution of Giraffes neck over a period of time.

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Q3. Choose the incorrect pairs

Characteristic feature Vegetation

1. Sloping branches and needle like leaves - Desert vegetation
2. Deep roots - Taiga vegetation
3. Waxy stem, thick leaves or no leaves - Tundra vegetation
4. Canopy - Tropical vegetation

Codes:

- a) All
- b) 4 only
- c) 1,2,3 only
- d) 2,3 only

Answer: c) – incorrect pairs

Explanation:

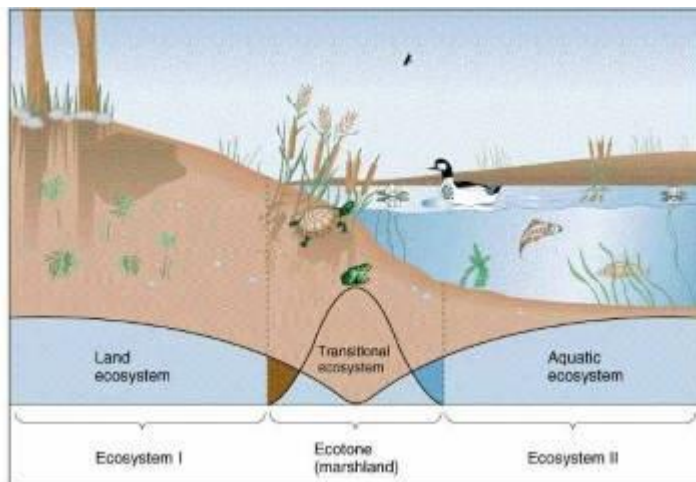
- Sloping branches (prevent accumulation of snow) and needle like leaves (reduce transpiration) – Taiga vegetation.
- Deep roots – Desert vegetation
- Waxy stem, thick leaves or no leaves – Desert vegetation
- Canopy – characteristic feature of tropical forests – rainforests, deciduous forests etc.

Ecotone - Edge Effect | Ecological Niche

Ecotone

- An ecotone is a **zone of junction** or a **transition area** between two biomes [diverse ecosystems]. It is where two communities meet and integrate.
- For e.g. the **mangrove forests** represent an ecotone between marine and terrestrial ecosystem. Other examples are **grassland** (between forest and desert), **estuary** (between fresh water and

salt water) and river bank or marsh land (between dry and wet).



Characteristics of Ecotone

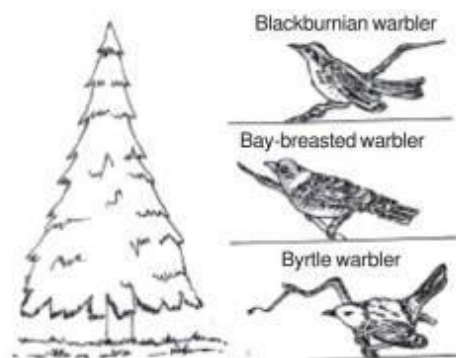
- It may be narrow (between grassland and forest) or wide (between forest and desert).
- As it is a zone of transition, it has conditions intermediate to the adjacent ecosystems. Hence it is a **zone of tension**.
- Usually, the number and the population density of the species of an outgoing community decreases as we move away from community or ecosystem.
- A well-developed ecotones contain some organisms which are entirely different from that of the adjoining communities.

Edge Effect – Edge Species

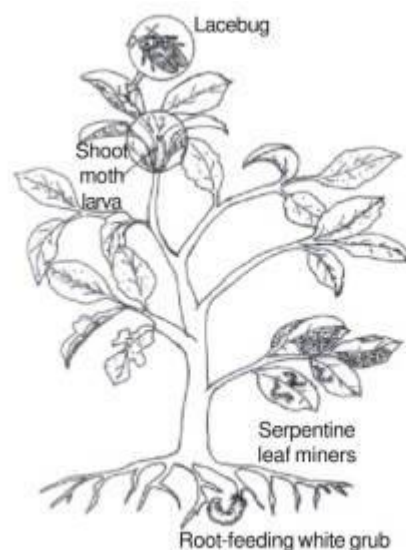
- In ecology, edge effects refer to the changes in population or community structures that occur at the boundary of two habitats (ecotone).
- Sometimes the number of species and the population density of some of the species in the ecotone is much greater than either community. This is called **edge effect**.
- The organisms which occur primarily or most abundantly in this zone are known as **edge species**.
- In the terrestrial ecosystems edge effect is especially applicable to **birds**. For example the density of birds is greater in the mixed habitat of the ecotone between the forest and the desert.

Ecological Niche

- Niche refers to the **unique functional role and position of a species** in its habitat or ecosystem.
- In nature, many species occupy the same habitat but they perform different functions.
- The functional characteristics of a species in its habitat is referred to as “niche” in that common habitat.
- Habitat of a species is like its ‘address’ (i.e. where it lives) whereas niche can be thought of as its “profession” (i.e. activities and responses specific to the species).
- A niche is unique for a species while many species share the habitat. No two species in a habitat can have the same niche. This is because of the **competition** with one another until one is displaced.



The three species of warbler birds search for insects as food in the forest at different levels in the tree and so occupy different niches



Different species of insects feeding on different parts of the same plant

- For example, a large number of different species of insects may be pests of the same

plant but they can co-exist as they feed on different parts of the same plant.

- A species' niche includes all of its interactions with the biotic and abiotic factors of its environment [habitat niche - where it lives, food niche - what it eats or decomposes & what species it competes with, reproductive niche - how and when it reproduces, physical & chemical niche - temperature, land shape, land slope, humidity & other requirement].
- An ecological niche describes how an organism or population responds to the distribution of resources and competitors (for example, by growing when resources are abundant, and when predators, parasites and pathogens are scarce) and how it in turn alters those same factors (for example, limiting access to resources by other organisms, acting as a food source for predators and a consumer of prey).
- Niche plays an important role in **conservation of organisms**. If we have to conserve species in its native habitat we should have knowledge about the niche requirements of the species and should ensure that all requirements of its niche are fulfilled.

Q4. Which one of the following terms describes not only the physical space occupied by an organism, but also its functional role in the community of organisms?

1. Ecotone
2. Ecological niche
3. Habitat
4. Home range

Answer: b) Niche

Explanation:

- Ecotone – zone of transition between two ecosystems. E.g. grasslands, mangroves etc.
- Habitat – surroundings in which an organism lives (home).
- Home Range – A home range is the area in which an animal lives and moves on a daily or periodic basis (a little bigger than habitat – home → office → home).

Ecological succession - Primary & Secondary Succession | Homeostasis

Functions Of Ecosystem

- The function of an ecosystem include
 1. **Ecological succession or ecosystem development**
 2. **Homeostasis (or cybernetic) or feedback control mechanisms**
 3. **Energy flow through food chain [Next Post]**
 4. **Nutrient cycling (biogeochemical cycles) [Later Post]**

Each will be discussed in detail in subsequent posts.

Ecological Succession

- Biotic communities are dynamic in nature and change over a period of time. The process by which communities of plant and animal species in an area are replaced or changed into another over a period of time is known as ecological succession.
- Succession is a universal process of directional change in vegetation, on an ecological time scale.
- Succession occurs when a series of communities replace one another due to large scale destruction (natural or manmade). This process continues with one community replacing another, until a stable, mature community develops.
- Succession is a progressive series of changes which leads to the establishment of a relatively **stable climax community**.
- The first plant to colonize an area is called the **pioneer community**. The final stage of succession is called the **climax community**. The stage leading to the climax community are called **successional stages** or **seres**. Each transitional (temporary) community that is formed and replaced during succession is called a stage in succession or a **seral community**.
- Succession is characterized by the following: increased productivity, the shift of nutrients from the reservoirs, increased diversity of organisms with increased niche

development, and a gradual increase in the complexity of food webs.

- Succession would occur faster in area existing in the **middle of the large continent**. This is because, here seeds of plants belonging to the different seres would reach much faster, establish and ultimately result in climax community.
- The terminal (final) stage of succession forms the community which is called as **climax community**. A climax

community is stable, mature, more complex and long lasting.

- The entire sequence of communities in a given area, succeeding each other, during the course of succession is termed **sere**. Succession that occurs on land where moisture content is low for e.g. on bare rock is known as **xerarch**. Succession that takes place in a water body, like ponds or lake is called **hydrarch**.

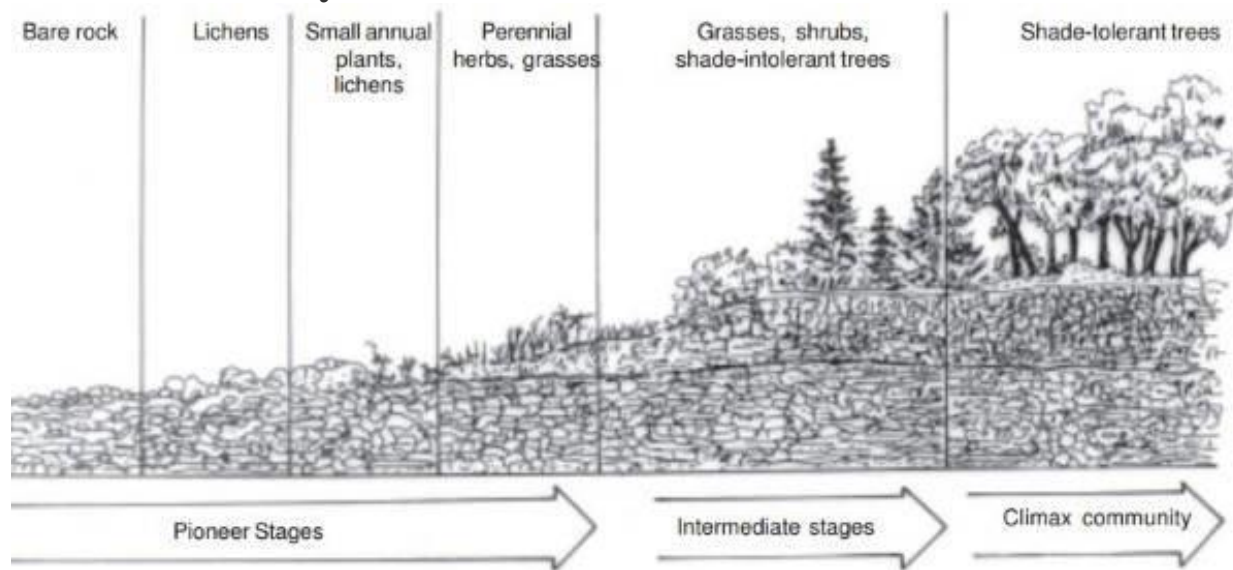


Fig 4.15: The orderly sequence of primary succession

Primary Succession

- Primary succession takes place on over a **bare or unoccupied areas** such as rocks outcrop, newly formed deltas and sand dunes, emerging volcano islands and lava flows as well as glacial moraines (muddy area exposed by a retreating glacier) where no community has existed previously.
- In primary succession on a terrestrial site the new site is first colonized by a few hardy pioneer species that are often **microbes, lichens and mosses**. The pioneers over a few generations alter the habitat conditions by their growth and development.

3. bacteria and fungi
4. fungi and mosses

More of a biology question. Fungus provides shelter, water and minerals to the algae and, in return, the alga provides food (photosynthesis).

- These new conditions may be conducive to the establishment of additional organisms that may subsequently arrive at the site. The pioneers through their death and decay leave patches of organic matter in which small animals can live.
- The organic matter produced by these pioneer species produces organic acids during decomposition that dissolve and etch the substratum releasing nutrients to the substratum. Organic debris accumulates in pockets and crevices, providing soil in which seeds can become lodged and grow.

Q5. Lichens, which are capable of initiating ecological succession even on a bare rock, are actually a symbiotic association of

1. algae and bacteria
2. **algae and fungi**

- As the community of organisms continues to develop, it becomes more diverse and competition increases, but at the same time new niche opportunities develop.
- The pioneer species disappear as the habitat conditions change and invasion of new species progresses, leading to the replacement of the preceding community.

Secondary Succession

- Secondary succession occurs when plants recognize an area in which the climax community has been disturbed.
- Secondary succession is the sequential development of biotic communities after the **complete or partial destruction of the existing community**.

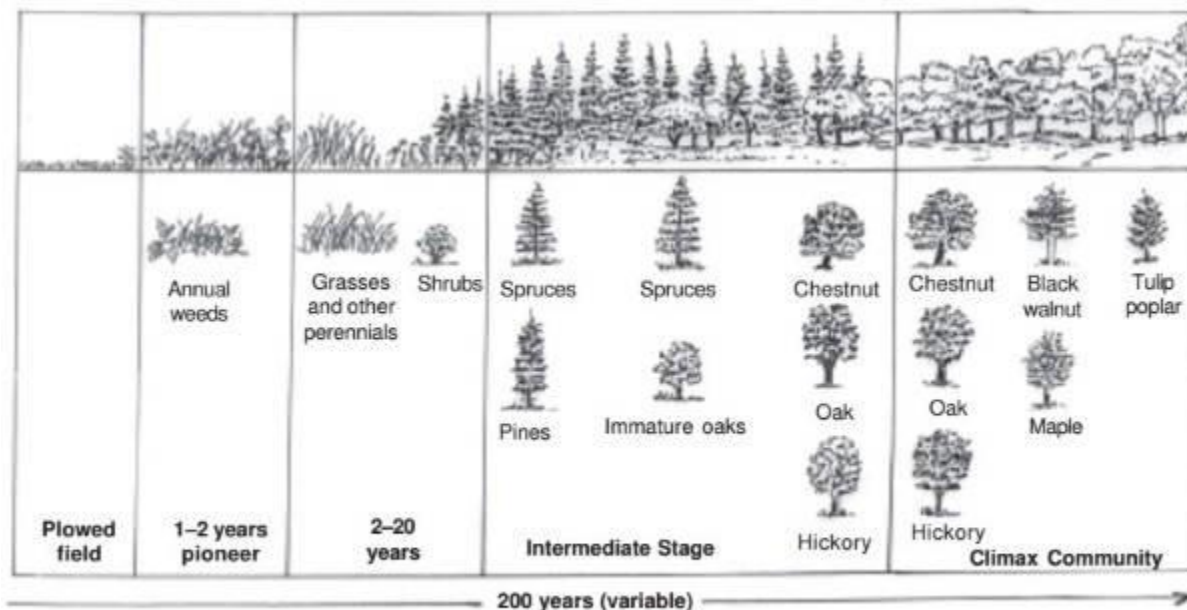


Fig. 4.16: Secondary succession on land

- A mature or intermediate community may be destroyed by natural events such as floods, droughts, fires, or storms or by human interventions such as deforestation, agriculture, overgrazing, etc.
- This abandoned farmland is first invaded by hardy species of grasses that can survive in bare, sunbaked soil. These grasses may be soon joined by tall grasses and herbaceous plants. These dominate the ecosystem for some years along with mice, rabbits, insects and seed-eating birds.
- Eventually, some trees come up in this area, seeds of which may be brought by wind or animals. And over the years, a forest community develops. Thus an abandoned farmland over a period becomes dominated by trees and is transformed into a forest.

- Unlike in the primary succession, the secondary succession starts on a **well-developed soil** already formed at the site. Thus secondary succession is relatively **faster** as compared to primary succession which may often require hundreds of years.

Autogenic and Allogenic Succession

- When succession is brought about by living inhabitants of that community itself, the process is called autogenic succession, while change brought about by outside forces is known as allogenic succession.
- Autogenic succession (caused by plants themselves) is succession driven by the biotic components of an ecosystem. In contrast, allogenic succession (caused by drought, fire, flooding etc.) is driven by the abiotic components of the ecosystem.

Difference Between Primary and Secondary Succession

Q6. In the grasslands, trees do not replace the grasses as a part of an ecological succession because of

1. insects and fungi
2. limited sunlight and paucity of nutrients
3. water limits and fire
4. None of the above

Answer: c)

Explanation:

- Grasses have this one good trick to monopolize a place. In dry season the grasses dry up and cause fires which destroy other plant species and their seeds.
- Also grasslands develop in regions with scanty rainfall where plant growth cannot be achieved.
- Though forests form the climax community in most of the ecosystems, but in grassland ecosystem [In the grasslands, trees do not replace the grasses as a part of an ecological succession] grasses form the climax community. Thanks to fire and lack of water.
- Grasslands are almost irreversible once deforestation in water scarce areas gives way to grasslands. [Ecological Damage]

Autotrophic and Heterotrophic succession

- Succession in which, initially the green plants are much greater in quantity is known as autotrophic succession; and the ones in which the heterotrophs are greater in quantity is known as heterotrophic succession.

Succession in Plants

- Based on the nature of the habitat - whether it is water (or very wet areas) or it is on very dry areas - succession of plants is called **hydrarch or xerarch**, respectively.
- Hydrarch succession takes place in wetter areas and the successional series progress from hydric to the mesic (intermediate) conditions.

- As against this, xerarch succession takes place in dry areas and the series progress from xeric to mesic conditions.
- Hence, both hydrarch and xerarch successions lead to **medium water conditions (mesic)** - neither too dry (xeric) nor too wet (hydric). With time the xerophytic habitat gets converted into a **Mesophytic one**.

Succession in Water

- In primary succession in water, the pioneers are the small phytoplanktons, they are replaced with time by free-floating angiosperms, then by rooted hydrophytes, sedges, grasses and finally the trees. The climax again would be a forest. With time the water body is converted into land.
- Another important fact is to understand that all succession whether taking place in water or on land, proceeds to a similar climax community - the **mesic**.

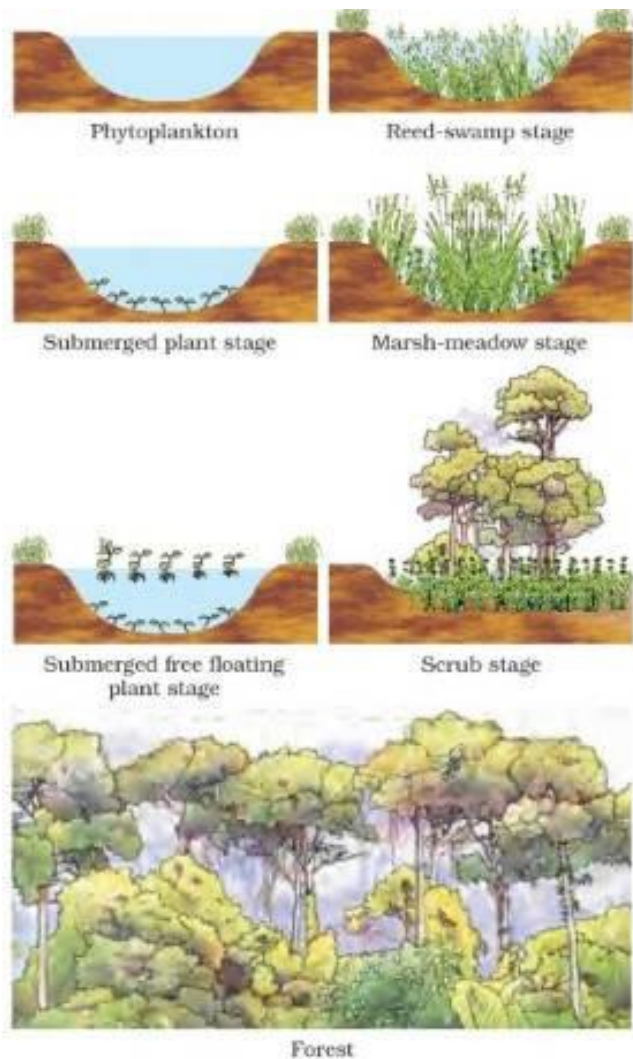


Figure 14.5 Diagrammatic representation of primary succession

- Some organisms are able to maintain homeostasis by physiological (sometimes behavioral – migrating to tree shade) means which ensures constant body temperature, constant osmotic concentration, etc.
- All birds and mammals, and a very few lower vertebrate and invertebrate species are indeed capable of such regulation (thermoregulation and osmoregulation).

[\[http://www.pmfias.com/classification-vertebrata-phylum-chordata/#warm-blooded-vs-cold-blooded-animals\]](http://www.pmfias.com/classification-vertebrata-phylum-chordata/#warm-blooded-vs-cold-blooded-animals)

- The ‘success’ of mammals is largely due to their ability to maintain a constant body temperature and thrive whether they live in Antarctica or in the Sahara desert.
- The mechanisms used by most mammals to regulate their body temperature are similar to the ones that we humans use. We maintain a constant body temperature of 37 °C.
- In summer, when outside temperature is more than our body temperature, we sweat profusely. The resulting evaporative cooling brings down the body temperature. In winter when the temperature is much lower than 37 °C, we start to shiver, a kind of exercise which produces heat and raises the body temperature.
- Plants, on the other hand, do not have such mechanisms to maintain internal temperatures.

Conform

- An overwhelming majority (99 per cent) of animals and nearly all plants cannot maintain a constant internal environment. Their body temperature changes with the ambient temperature.
- In aquatic animals, the osmotic concentration of the body fluids change with that of the ambient water osmotic concentration. These animals and plants are simply **conformers**.
- Considering the benefits of a constant internal environment to the organism, we must ask why these conformers had not evolved to become regulators.
- Thermoregulation is energetically expensive for many organisms. This is

Homeostasis

- Homeostasis is the maintenance of a **stable equilibrium**, especially through physiological (through bodily part functions. E.g. Cooling your body through sweating) processes.
- Organisms try to maintain the constancy of its internal environment (a process called homeostasis) despite varying external environmental conditions that tend to upset its homeostasis.
- For humans, it could be achieved at home by using an air conditioner in summer and heater in winter.
- Here the person’s homeostasis is accomplished, not through physiological, but artificial means. How do other living organisms cope with the situation?

Regulate

particularly true for small animals like shrews and humming birds.

- Heat loss or heat gain is a function of surface area. Since small animals have a larger surface area relative to their volume, they tend to lose body heat very fast when it is cold outside; then they have to expend much energy to generate body heat [lot of food goes into heat generation] through metabolism. This is the main reason why very small animals are rarely found in polar regions.
- During the course of evolution, the costs and benefits of maintaining a constant internal environment are taken into consideration.
- Some species have evolved the ability to regulate, but only over a limited range of environmental conditions, beyond which they simply conform.
- If the stressful external conditions are localised or remain only for a short duration, the organism has two other alternatives.

Migrate

- The organism can move away temporarily from the stressful habitat to a more hospitable area and return when stressful period is over
- Every winter the famous Keolado National Park (Bhartpur) in Rajasthan host thousands of migratory birds coming from Siberia and other extremely cold northern regions.

Suspend

- In bacteria, fungi and lower plants, various kinds of thick-walled spores are formed which help them to survive unfavorable conditions - these germinate on availability of suitable environment.
- In higher plants, seeds and some other vegetative reproductive structures serve as means to tide over periods of stress besides helping in dispersal - they germinate to form new plants under favourable moisture and temperature conditions. They do so by reducing their metabolic activity and going into a state of 'dormancy'.
- In animals, the organism, if unable to migrate, might avoid the stress by escaping

in time. The familiar case of bears going into **hibernation** during winter is an example of escape in time.

[\[http://www.pmfias.com/classification-vertebrata-phylum-chordata/#hibernation\]](http://www.pmfias.com/classification-vertebrata-phylum-chordata/#hibernation)

- Some snails and fish go into **aestivation** to avoid summer-related problems-heat and desiccation.
- Under unfavorable conditions many zooplankton species in lakes and ponds are known to enter **diapause**, a stage of suspended development.

Homeostasis in Ecosystem

- Ecosystems are capable of maintaining their state of equilibrium. They can regulate their own species structure and functional processes. This capacity of ecosystem of self-regulation is known as homeostasis.
- In ecology the term applies to the tendency for a biological systems to resist changes.
- For example, in a pond ecosystem, if the population of zooplankton increased, they would consume large number of the phytoplankton and as a result food would become scarce for zooplankton as well as other small fishes.
- When the number zooplankton is reduced because of starvation, phytoplankton population start increasing. After some time the population size of zooplankton also increases and this process continues at all the trophic levels of the food chain.

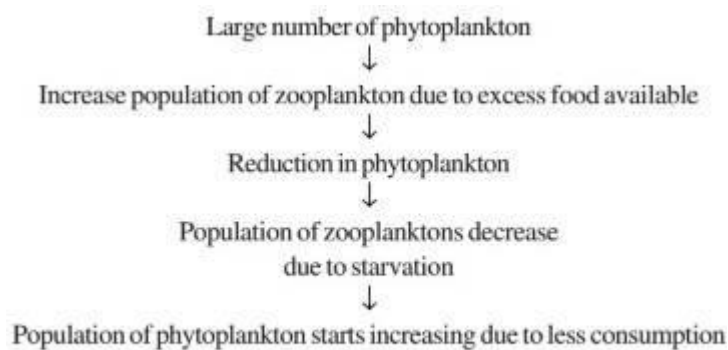


Fig. 5.9: Homeostasis in ecosystem

- Note that in a homeostatic system, negative feedback mechanism [here its

scarcity of food] is responsible for maintaining stability in an ecosystem.

- However, homeostatic capacity of ecosystems is not unlimited as well as not everything in an ecosystem is always well regulated. Humans are the greatest source of disturbance to ecosystems.

Trophic Levels - Food Chain - Food Web - Biotic Interaction

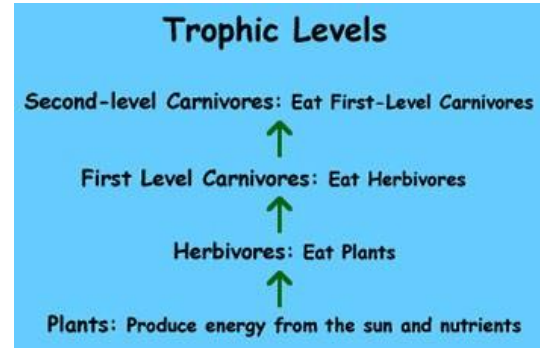
Functions Of Ecosystem

- The functions of an ecosystem include
 1. [Ecological succession or ecosystem development \[Previous Post\]](#)
 2. [Homeostasis \(or cybernetic\) or feedback control mechanisms \[Previous Post\]](#)
 3. [Energy flow through food chain \[This post\]](#)
 4. [Nutrient cycling \(biogeochemical cycles\) \[Next post\]](#)

Energy Flow Through an Ecosystem – Trophic Levels

(Trophe = Nourishment)

- The flow of energy from producer to top consumers is called energy flow which is **unidirectional**.
- To understand the energy flow through the ecosystem we need to study about the trophic levels [trophic level interaction].
- Trophic level is the representation of energy flow in an ecosystem. The trophic level of an organism is the position it occupies in a food chain.
- Trophic level interaction deals with how the members of an ecosystem are connected based on nutritional needs.



Trophic levels

Autotrophs	Green plants (producers)
Heterotrophs	Herbivore (primary consumers)
Heterotrophs	Carnivores (secondary consumers)
Heterotrophs	Carnivore (tertiary consumers)
Heterotrophs	Top carnivores (Quaternary consumers)

- Energy flows through the trophic levels from producers to subsequent trophic levels is unidirectional.
- Energy level **decreases** from the first trophic level upwards due to loss of energy in the form of heat at each trophic level.
- This energy loss at each trophic level is quite significant. Hence there are usually not more than four-five trophic levels [beyond this the energy available is negligible to support an organism].
- The trophic level interaction involves three concepts namely

1. **Food Chain**
2. **Food Web**
3. **Ecological Pyramids**

Food Chain

- Transfer of food energy from green plants (producers) through a series of organisms with repeated eating and being eaten link is called a food chain. E.g. Grasses → Grasshopper → Frog → Snake → Hawk/Eagle.
- Each step in the food chain is called trophic level. A food chain starts with producers and ends with top carnivores.
- The trophic level of an organism is the position it occupies in a food chain.

Types of Food Chains

1. Grazing food chain
2. Detritus food chain

Grazing food chain

- The consumers which start the food chain, utilizing the plant or plant part as their food, constitute the grazing food chain. This food chain begins from green plants at the base and the primary consumer is herbivore.
- For example, In terrestrial ecosystem, grass is eaten by caterpillar, which is eaten by lizard and lizard is eaten by snake.
- In Aquatic ecosystem phytoplankton (primary producers) are eaten by zoo planktons which are eaten by fishes and fishes are eaten by pelicans.

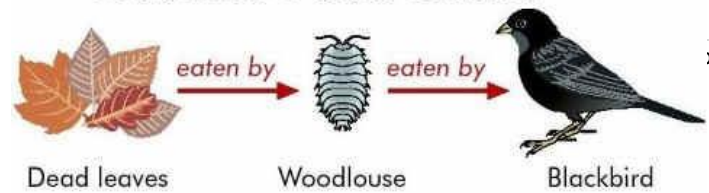


Detritus food chain

- This type of food chain starts from dead organic matter of decaying animals and plant bodies.
- Dead organic matter or detritus feeding organisms are called **detrivores or decomposer**. The detrivores are eaten by predators.
- The two food chains are linked. The initial energy source for detritus food chain is the waste materials and dead organic matter from the grazing food chain.
- In an **aquatic ecosystem, grazing food chain** is the major conduit for energy flow. As against this, in a terrestrial ecosystem, a much larger fraction of energy flows through the detritus food

chain than through the grazing food chain.

Detritus Food Chain



- Bacterial and fungal enzymes degrade detritus into simpler inorganic substances. This process is called as **catabolism**.
- Humification and mineralization occur during decomposition in the soil.
- **Humification** leads to accumulation of a dark coloured amorphous substance called humus that is highly resistant to microbial action and undergoes decomposition at an extremely slow rate.
- Being colloidal in nature, humus serves as a reservoir of nutrients. The humus is further degraded by some microbes and release of inorganic nutrients occur by the process known as **mineralization**.
- Warm and moist environment favor decomposition whereas low temperature and anaerobiosis inhibit decomposition resulting in buildup of organic materials.

Q1. With reference to the food chains in ecosystems, which of the following kinds of organism is / are known as decomposer organism/organisms?

1. Virus
2. Fungi
3. Bacteria

Select the correct answer using the codes given below.

- a. 1 only
- b. 2 and 3 only
- c. 1 and 3 only
- d. 1, 2 and 3

Explanation:

- Fungi and Bacteria are decomposers. They breakdown organic matter into simple

inorganic substances.

- Virus represents dormant life. They are metabolically inactive as long as they are outside a host body. They are not decomposers. They invade host cells and use their nucleus (DNA machinery) to carry out their life processes.
- Protists are unicellular organisms, such as paramecium and euglena. They're not decomposers either.

Answer: b)

Q2. Which one of the following is the correct sequence of a food chain?

1. Diatoms-Crustaceans-Herrings
2. Crustaceans-Diatoms-Herrings
3. Diatoms-Herrings-Crustaceans
4. Crustaceans-Herrings-Diatoms

Explanation:

- Food chain starts with a producer and ends with a top consumer.
- **Phytoplankton** are the primary producers in the oceans. They include:

1. **diatoms (unicellular algae),**
2. **coccolithophores** (unicellular, eukaryotic protist),
3. **cyanobacteria (Bluegreen algae)**—synechococcus, prochlorococcus, nostoc, spirogyra etc..
4. **dinoflagellates (flagellated protists).**

So answer is between (a) and (c)

- Crustaceans form a very large group of arthropods which includes crabs, lobsters, crayfish, shrimp, krill and barnacles.
- Herrings is a fish.
- <http://octopus.gma.org/herring/biology/ecology/default.asp> says Herrings eat crustaceans.

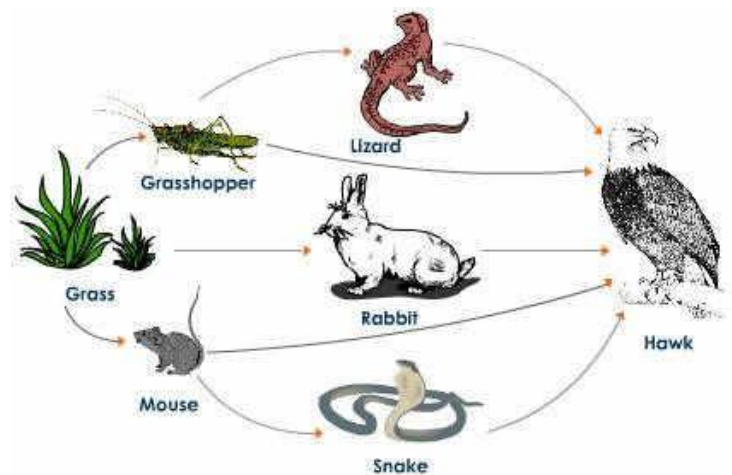
Answer: (a). Tough question. There will be 3-5 questions like this one where you might have not even heard the name of the animal or plant. Don't worry.. rest 95 will be easy!

Food Web

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- Multiple interlinked food chains make a food web. Food web represents all the possible paths of energy flow in an ecosystem.
- If any of the intermediate food chain is removed, the succeeding links of the chain will be affected largely.
- The food web provides more than one alternative for food to most of the organisms in an ecosystem and therefore increases their chance of survival.
- Also food availability and preferences of food of the organisms may shift seasonally e.g. we eat watermelon in summer and peaches in the winter. Thus there are interconnected networks of feeding relationships that take the form of food webs.



A Food Web in a Grassland Ecosystem With Five Possible Food Chains

Type of interaction	Species 1	Species 2	Effects of interaction
Negative Interactions			
Amensalism	-	0	one species is inhibited while the other species is unaffected
Predation	+	-	Predator—prey relationship: one species (predator) benefits while the second species (prey) is harmed and inhibited.
Parasitism	+	-	Beneficial to one species (parasite) and harmful to the other species (host).
Competition	-	-	Adversely affects both species
Positive Associations			
Commensalism	+	0	One species (the commensal) benefits, while the other species (the host) is neither harmed nor inhibited
Mutualism	+	+	Interaction is favourable to both species
Neutral Interactions			
Neutralism	0	0	Neither species affects the other

Q3. With reference to food chains in ecosystems, consider the following statements:

1. A food chain illustrates the order in which a chain of organisms feed upon each other.
2. Food chains are found within the populations of a species.
3. A food chain illustrates the numbers of each organism which are eaten by others.

Which of the statements given above is / are correct?

1. 1 only
2. 1 and 2 only
3. 1, 2 and 3
4. None

Explanation:

1. A food chain illustrates the order in which a chain of organisms feed upon each other. (True)
2. Food chains are found within the populations of a species. (Man won't eat man – so, false)
3. A food chain illustrates the numbers of each organism which are eaten by others (food web illustrates the number not the food chain).

Answer: a)

Biotic Interaction

- The interaction that occurs among different individuals of the same species is called intraspecific interaction while the interaction among individuals of different species in a community is termed as interspecific interaction.
- Specific terms are applied to interspecific interactions depending upon whether the interaction is beneficial, harmful or neutral to individuals of the species.
- Some types of interactions listed by the effects they have on each partner. '0' is no effect, '-' is detrimental and '+' is beneficial.

Possible biological interactions between two species.

+ = beneficial; - = harmful, 0 = unaffected or neutral.

Amensalism

- One species harms or restricts the other species without itself being adversely affected or harmed by the presence of the other species.
- Organisms that secrete **antibiotics** and the species that get inhibited by the antibiotics are examples of amensalism.

- For example the bread mould fungi *Pencillium* produce penicillin an antibiotic substance which inhibits the growth of a variety of bacteria.
- A large tree shades a small plant, retarding the growth of the small plant. The small plant has no effect on the large tree.

Predation

- Predators like leopards, tigers and cheetahs use speed, teeth and claws to hunt and kill their prey.
- They keep prey populations under control. But for predators, prey species could achieve very high population densities and cause ecosystem instability.
- When certain exotic species are introduced into a geographical area, they become invasive and start spreading fast because the invaded land does not have its natural predators.
- Predators also help in maintaining species diversity in a community, by reducing the intensity of competition among competing prey species.
- A wide variety of chemical substances that we extract from plants on a commercial scale (nicotine, caffeine, quinine, strychnine, opium, etc.) are produced by plants actually as defences against grazers and browsers.

Parasitism

- In this type of interaction, one species is harmed and the other benefits.
- Parasitism involves parasite usually a small size organism **living in or on** another living species called the host from which the parasite gets its nourishment and often shelter.
- Many organisms like animal, bacteria and viruses are parasites of plants and animals.
- Plants like **dodder** (**Cuscuta**) and **mistletoe** (**Loranthus**) are parasites that live on flowering plants.
- Tap worm, round worm, malarial parasite, many bacteria, fungi, and viruses are common parasites of humans.
- Parasites that feed on the external surface of the host organism are called **ectoparasites**. E.g. lice on humans.

Many marine fish are infested with ectoparasitic **copepods**.

- The female mosquito is not considered a parasite, although it needs our blood for reproduction. Why? Because it doesn't live on the host.
- In contrast, **endoparasites** are those that live inside the host body at different sites (liver, kidney, lungs, red blood cells, etc.).
- **Brood parasitism** in birds is a fascinating example of parasitism in which the parasitic bird lays its eggs in the nest of its host. E.g. cuckoo (koel).

Competition

- This is an interaction between two populations in which both species are harmed to some extent.
- Competition occurs when two populations or species, both need a vital resource that is in short supply.
- The vital resource could be food, water, shelter, nesting site, mates or space.
- Such competition can be:
 1. interspecific competition-occurring between individuals of two different species occurring in a habitat and
 2. intraspecific competition-occurs between individuals of same species.
- Intraspecific competition occurs between members of the same species and so it is very intense.

Commensalism

- In this relationship one of the species benefits while the other is neither harmed nor benefited.
- Some species obtain the benefit of shelter or transport from another species. For example **sucker fish**, **remora** often attaches to a shark by means of its sucker which is present on the top side of its head. This helps the remora get protection, a free ride as well as meal from the left over of the shark's meal. The shark does not however get any benefit nor is it adversely affected by this association.

- Another example of commensalisms is the relationship between **trees** and **epiphytic plants**.
- Epiphytes live on the surface of other plants like ferns, mosses and orchids and use the surface of trees for support and for **obtaining sunlight and moisture**. The tree gets no benefit from this relationship nor are they harmed.
- Cow dung provides food and shelter to dung beetles. The beetles have no effect on the cows.
- Another example of commensalism is the interaction between **sea anemone** that has stinging tentacles and the **clown fish** that lives among them.
- The fish gets protection from predators which stay away from the stinging tentacles. The anemone does not appear to derive any benefit by hosting the clown fish.

Mutualism

- This is a close association between two species in which both the species benefit. For example the **sea anemone**, a **cnidarian** gets attached to the shell of **hermit crabs** for benefit of transport and obtaining new food while the anemone provides camouflage and protection by means of its stinging cells to the hermit crab.
- However, some mutualisms are so intimate that the interacting species can no longer live without each other as they depend totally on each other to survive. Such close associations are called **symbiosis (symbiosis is intense mutualism – E.g. coral and zooxanthellae)**.
- An example of such close mutualistic association is that of **termite and their intestinal flagellates**. Termites can eat wood but have no enzymes to digest it. However, their intestine contains certain flagellate protists (protozoans) that have the necessary enzymes to digest the cellulose of the wood eaten by termites and convert it into sugar.
- The flagellates use some of this sugar for their own metabolism while enough is left for the termite. Both termite and flagellates cannot survive without each other.

- Another familiar example of symbiosis is seen in pollination of flowers where flowering plants are cross pollinated by the bees which benefit by getting nectar from the plants and both cannot survive without the other.
- Example: in pollination mutualisms, the pollinator gets food (pollen, nectar), and the plant has its pollen transferred to other flowers for cross-fertilization (reproduction).
- Lichens represent an intimate mutualistic relationship between a fungus and photosynthesizing algae or cyanobacteria.
- Similarly, the **mycorrhizae** are associations between **fungi and the roots of higher plants**. The fungi help the plant in the absorption of essential nutrients from the soil while the plant in turn provides the fungi with energy-yielding carbohydrates.

Neutralism

- Neutralism describes the relationship between two species which do interact but do not affect each other.
- True neutralism is extremely unlikely and impossible to prove.

Ecological Pyramids - Pyramid of numbers, biomass & energy

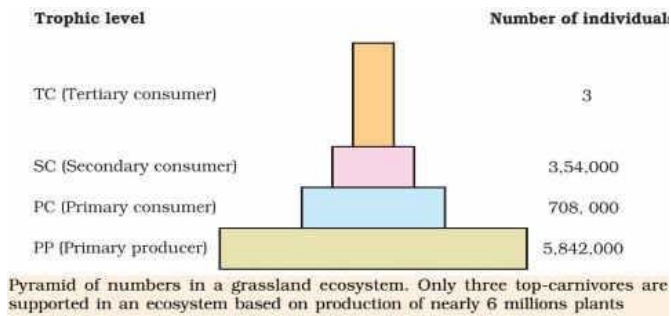
Ecological Pyramids

- The pyramidal representation of trophic levels of different organisms based on their ecological position [producer to final consumer] is called as an ecological pyramid.
- The food producer forms the base of the pyramid and the top carnivore forms the tip. Other consumer trophic levels are in between.
- The pyramid consists of a number of horizontal bars depicting specific trophic levels. The length of each bar represents the total number of individuals or biomass or energy at each trophic level in an ecosystem.
- The ecological pyramids are of three categories.

1. Pyramid of numbers,

2. **Pyramid of biomass**, and
3. **Pyramid of energy or productivity**.

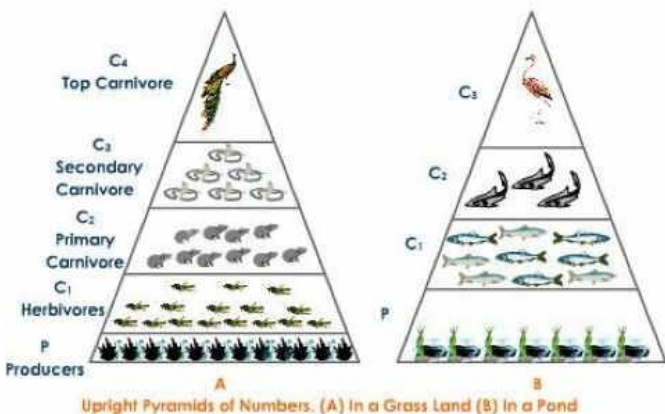
Pyramid of Numbers



- Pyramid of numbers represents the **total number of individuals of different species** (population) at each trophic level.
- Depending upon the size, the pyramid of numbers may not always be upright, and may even be completely inverted.
- It is very difficult to count all the organisms, in a pyramid of numbers and so the pyramid of number does not completely define the trophic structure for an ecosystem.

Pyramid of numbers – upright

- In this pyramid, the number of individuals is decreased from lower level to higher trophic level.



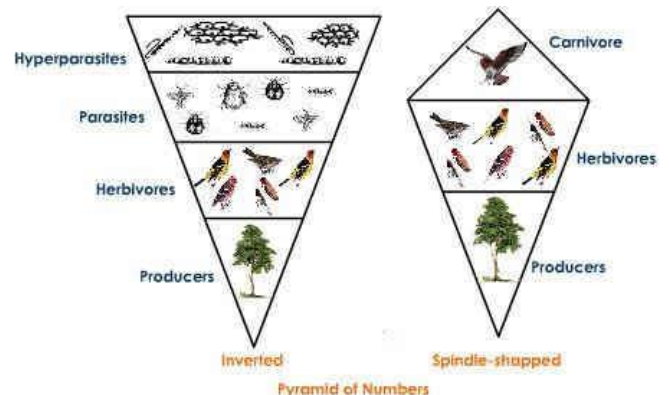
All Picture Credits:
<http://image.tutorvista.com/> + NCERT

- This type of pyramid can be seen in **grassland ecosystem** and **pond ecosystem**.

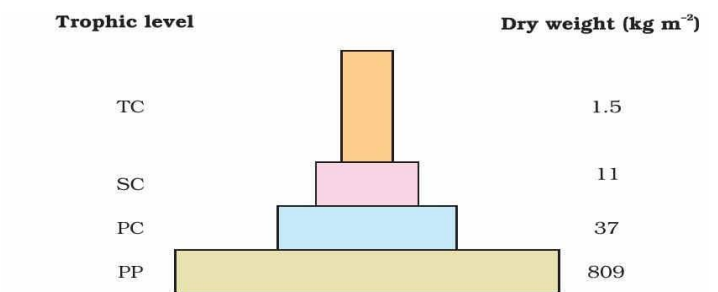
- The grasses occupy the lowest trophic level (base) because of their abundance.
- The next higher trophic level is primary consumer - herbivore (example - grasshopper).
- The individual number of grasshopper is less than that of grass. The next energy level is primary carnivore (example - rat).
- The number of rats are less than grasshopper, because, they feed on grasshopper. The next higher trophic level is secondary carnivore (example - snakes). They feed on rats.
- The next higher trophic level is the top carnivore. (Ex: Hawk).
- With each higher trophic level, the number of individual decreases.

Pyramid of numbers - inverted

- In this pyramid, the number of individuals is increased from lower level to higher trophic level. E.g. Tree ecosystem.



Pyramid of Biomass



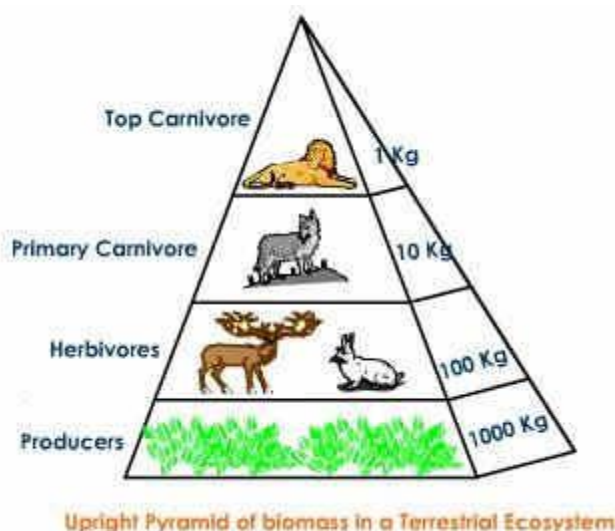
Pyramid of biomass shows a sharp decrease in biomass at higher trophic levels

- Pyramid of biomass is usually determined by collecting all organisms occupying each trophic level separately and measuring their **dry weight**.

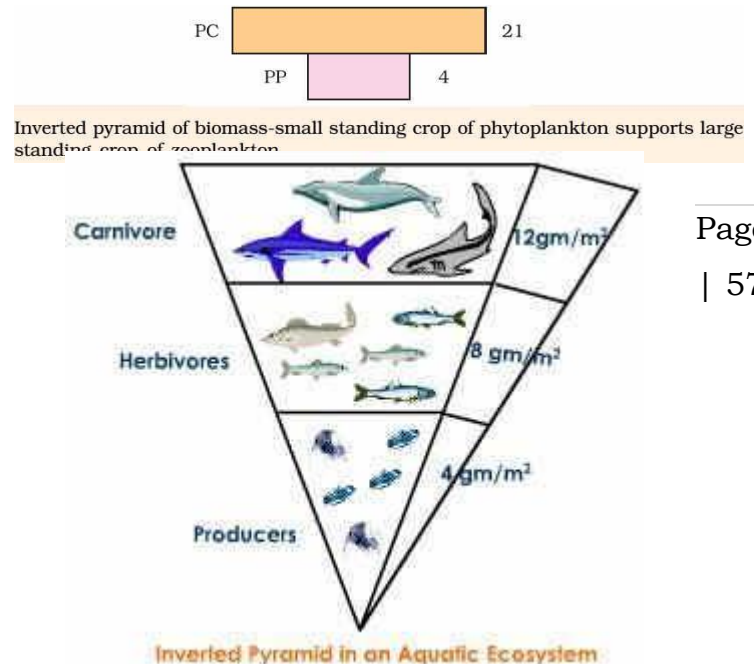
- This overcomes the size difference problem because all kinds of organisms at a trophic level are weighed. Biomass is measured in g/m^2 .
- The biomass of a species is expressed in terms of fresh or dry weight. Measurement of biomass in terms of dry weight is more accurate.
- Each trophic level has a certain mass of living material at a particular time called as the **standing crop**.
- The standing crop is measured as the mass of living organisms (biomass) or the number in a unit area.

Pyramid of Biomass – Upright

- For most ecosystems on land, the pyramid of biomass has a large base of primary producers with a smaller trophic level perched on top.
- The biomass of producers (autotrophs) is at the maximum. The biomass of next trophic level i.e. primary consumers is less than the producers. The biomass of next higher trophic level i.e. secondary consumers is less than the primary consumers. The top, high trophic level has very less amount of biomass.



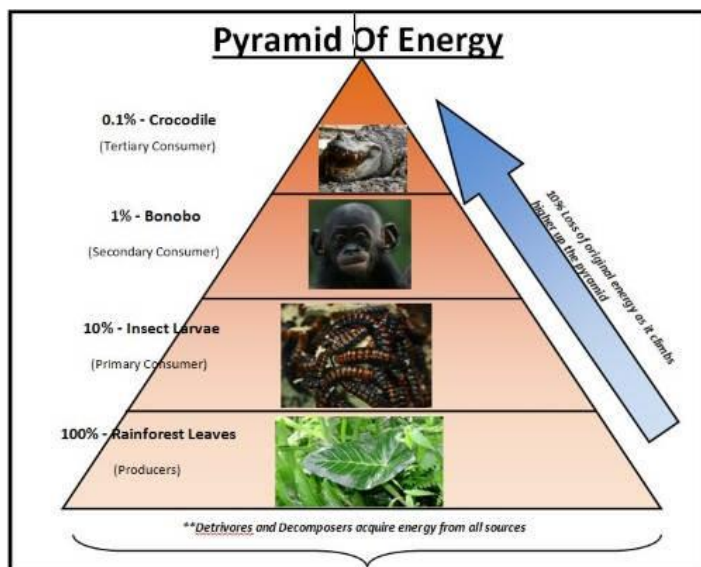
Pyramid of Biomass – Inverted



- In contrast, in many **aquatic ecosystems**, the pyramid of biomass may assume an inverted form. [Pyramid of numbers for aquatic ecosystem is upright]
- This is because the producers are tiny phytoplankton that grow and reproduce rapidly.
- Here, the pyramid of biomass has a small base, with the consumer biomass at any instant actually exceeding the producer biomass and the pyramid assumes inverted shape.

Pyramid of Energy

- To compare the functional roles of the trophic levels in an ecosystem, an energy pyramid is **most suitable**.
- An energy pyramid represents the amount of energy at each trophic level and loss of energy at each transfer to another trophic level. Hence the pyramid is **always upward**, with a large energy base at the bottom.



energy follows 10 per cent law – only 10 per cent of the energy is transferred to each trophic level from the lower trophic level. [above figure]

- The decreases at each subsequent trophic level is due to two reasons:

- At each trophic a part of the available **energy is lost in respiration or used up in metabolism.**
- A part of energy is lost at each **transformation**, i.e. when it moves from lower to higher trophic level as heat.

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Limitations of Ecological Pyramids

- It does not take into account the **same species belonging to two or more trophic levels.**
- It assumes a simple food chain, something that almost never exists in nature; it **does not accommodate a food web.**
- Moreover, saprophytes (plant, fungus, or microorganism that lives on decaying matter) are not given any place in ecological pyramids even though they play a vital role in the ecosystem.

Pollutants And Trophic Level – Biomagnification

Pollutants And Trophic Level

Chlorinated Hydrocarbons (CHC) or Organochloride

- Pollutants, especially non-degradable ones move through the various trophic levels in an ecosystem. Non-degradable pollutants mean materials, which cannot be metabolized by the living organisms. Example: **Chlorinated Hydrocarbons.**
- Chlorinated Hydrocarbons** or **Organochloride** or **CHC** are hydrocarbons whose some or most hydrogen atoms have been replaced by **chlorine** E.g. **DDT, endosulfan etc.**
- A variety of simple chlorinated hydrocarbons including **dichloromethane, chloroform, and carbon tetrachloride.**

Applications of CHC

Pic Credits:
<https://endangeredspeciesbiomesprojects.wikispaces.com>

- Suppose an ecosystem receives 1000 calories of light energy in a given day. Most of the energy is not absorbed; some is reflected back to space; of the energy absorbed only a small portion is utilized by green plants, out of which the plant uses up some for respiration and of the 1000 calories, therefore only 100 calories are stored as energy rich materials.
- Now suppose an animal, say a deer, eats the plant containing 100 calorie of food energy. The deer uses some of it for its own metabolism and stores only 10 calorie as food energy. A lion that eats the deer gets an even smaller amount of energy. Thus usable energy decreases from sunlight to producer to herbivore to carnivore. Therefore, the energy pyramid will always be upright.
- Energy pyramid** concept helps to explain the phenomenon of **biological magnification** - the tendency for toxic substances to increase in concentration progressively with higher trophic levels.

Ecological Efficiency

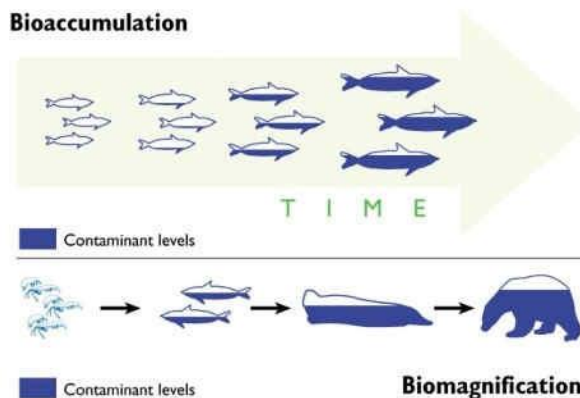
- Ecological efficiency describes the efficiency with which energy is transferred from one trophic level to the next.
- The number of trophic levels in the grazing food chain is restricted as the transfer of

- Production of vinyl chloride almost all of which was converted into polyvinylchloride (PVC) [PVC pipes].
- Chloroform, dichloromethane, dichloroethene, and trichloroethane are useful solvents. These solvents are immiscible with water and effective in cleaning applications such as degreasing and dry cleaning.
- Pesticides and insecticides such as DDT, heptachlor, and endosulfan are CHCs.

Effects of CHC

- Dioxins (highly toxic organic compound produced as a by-product in some manufacturing processes), produced when organic matter is burned in the presence of chlorine, and some insecticides, such as DDT, are persistent organic pollutants.
- DDT, which was widely used to control insects in the mid-20th century, accumulates in food chains, and causes reproductive problems (e.g., eggshell thinning) in certain bird species.
- DDT residues continue to be found in humans and mammals across the planet many years after production and use have been limited.
- In Arctic areas, particularly high levels are found in marine mammals. These chemicals concentrate in mammals, and are even found in human breast milk.
- In some species of marine mammals, particularly those that produce milk with a high fat content, males typically have far higher levels, as females reduce their concentration by transfer to their offspring through lactation.
- Endosulfan became a highly controversial agrichemical due to its acute toxicity, potential for bioaccumulation, and role as an **endocrine disruptor** (enhances the effect of estrogens causing reproductive and developmental damage in both animals and humans).
- Because of its threats to human health and the environment, a global ban on the manufacture and use of endosulfan was negotiated under the **Stockholm Convention** in April 2011.
- Movement of these pollutants involves two main processes:

1. **Bioaccumulation.**
2. **Biomagnification.**

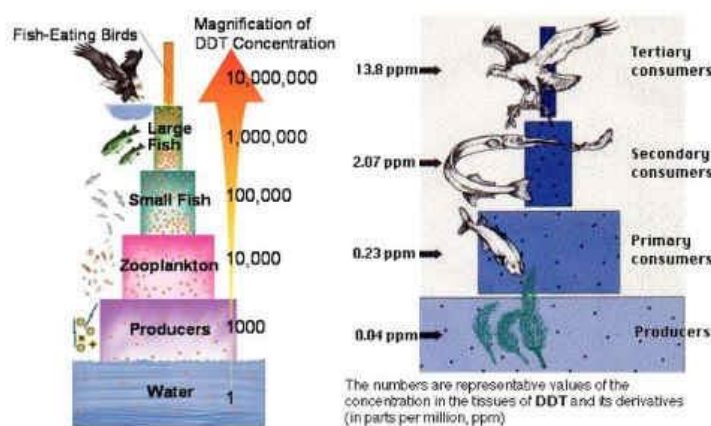


Bioaccumulation

- It refers to how pollutants enter a food chain.
- In bioaccumulation there is an increase in concentration of a pollutant from the environment to the first organism in a food chain.

Biomagnification

- Biomagnification refers to the tendency of pollutants to concentrate as they move from one trophic level to the next.
- Thus in biomagnification there is an increase in concentration of a pollutant from one link in a food chain to another.
- In order for biomagnification to occur, the pollutant must be: long-lived, mobile, soluble in fats, biologically active. E.g. DDT.



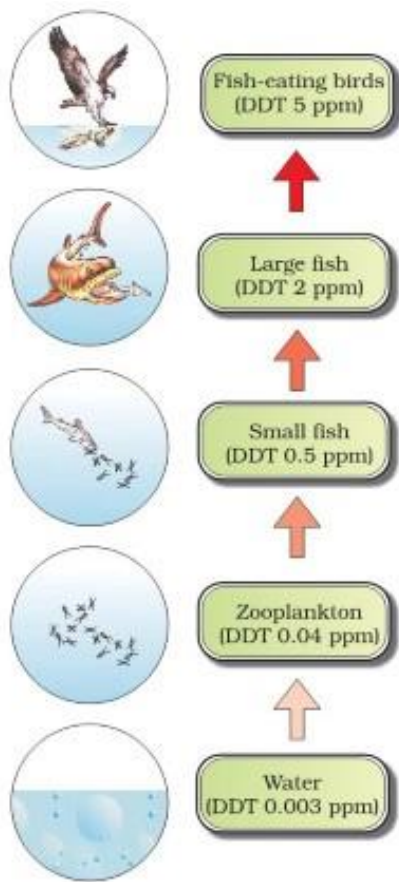


Figure 16.5 Biomagnification of DDT in an aquatic food chain

- If a pollutant is short-lived, it will be broken down before it can become dangerous.
- If it is not mobile, it will stay in one place and is unlikely to be taken up by organisms.
- If the pollutant is soluble in water, it will be excreted by the organism. Pollutants that **dissolve in fats**, however, may be retained for a long time.
- It is traditional to measure the amount of pollutants in fatty tissues of organisms such as fish.
- In mammals, we often test the milk produced by females, since the milk has a lot of fat in it and is often more susceptible to damage from toxins (poisons).

Nutrient Cycles | Biogeo Chemical Cycles

Functions Of An Ecosystem

- The functions of an ecosystem include

1. [Ecological succession or ecosystem development \[Previous Posts\]](#)
2. [Homeostasis \(or cybernetic\) or feedback control mechanisms \[Previous Posts\]](#)
3. [Energy flow through food chain \[Previous Post\]](#)
4. [Nutrient cycling \(biogeochemical cycles\) \[This post\]](#)

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Biogeo Chemical Cycling or Nutrient Cycling

- Energy flow and nutrient circulation are the major functions of the ecosystem. We have studied about energy flow through trophic levels in the previous posts.
- Energy is lost as heat forever in terms of the usefulness of the system. On the other hand, nutrients of food matter never get used up. They can be recycled again and again indefinitely.
- **Carbon, hydrogen, oxygen, nitrogen and phosphorus** as elements and compounds make up 97% of the mass of our bodies and are more than 95% of the mass of all living organisms.
- In addition to these, about 15 to 25 other elements are needed in some form for the survival and good health of plants and animals.
- These elements or mineral nutrients are always in circulation moving from non-living to living and then back to the non-living components of the ecosystem in a more or less circular fashion. This circular fashion is known as **biogeochemical cycling** (bio for living; geo for atmosphere).
- Among the most important nutrient cycles are the carbon nutrient cycle and the nitrogen nutrient cycle.
- There are many other nutrient cycles that are important in ecology, including a large number of trace mineral nutrient cycles.

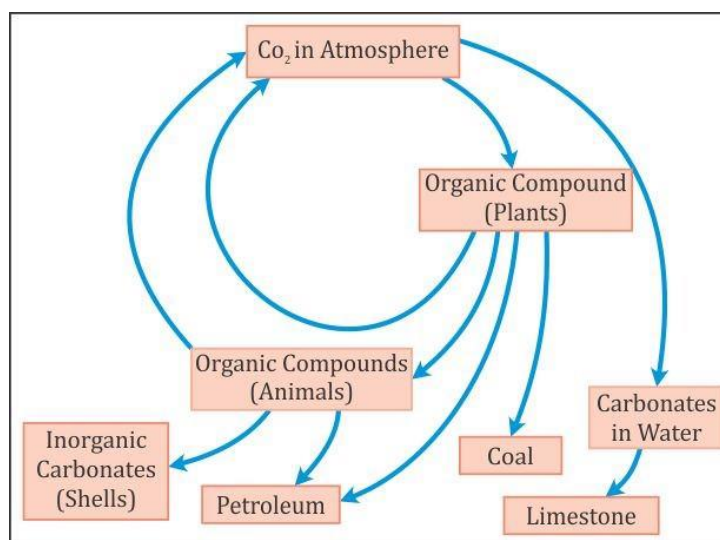
Nutrient Cycles

- Based on the replacement period, a nutrient cycle is referred to as Perfect or Imperfect cycle.
1. A perfect nutrient cycle is one in which nutrients are replaced as fast as they

are utilized. Most **gaseous cycles** are generally considered as perfect cycles.

2. In contrast **sedimentary cycles** are considered relatively imperfect, as some nutrients are lost from the cycle and get locked into sediments and so become unavailable for immediate cycling.
- Based on the nature of the reservoir, there are two types of cycles namely-
 1. **Gaseous Cycle** — where the reservoir is the **atmosphere or the hydrosphere** — water cycle, carbon cycle, nitrogen cycle, etc. and
 2. **Sedimentary Cycle** — where the reservoir is the **earth's crust** [elements mostly found in earth's crust] — phosphorous cycle, sulphur cycle, calcium cycle, magnesium cycle etc.

Carbon Cycle [Gaseous Cycle]



- Carbon is a minor constituent of the atmosphere as compared to oxygen and nitrogen.
- However, without carbon dioxide life could not exist because it is vital for the production of carbohydrates through photosynthesis by plants.
- It is the element that anchors all organic substances from coal and oil to DNA (deoxyribonucleic acid: the compound that carries genetic information).
- Carbon is present in the atmosphere, mainly in the form of carbon dioxide (CO_2).

- Carbon cycle involves a continuous exchange of carbon between the atmosphere and organisms.
- Carbon from the atmosphere moves to green plants by the process of **photosynthesis**, and then to animals.
- By process of **respiration** and **decomposition** of dead organic matter it returns back to atmosphere. It is usually a short term cycle.
- Some carbon also enters a long term cycle. It accumulates as un-decomposed organic matter in the peaty layers of marshy soil or as insoluble carbonates in bottom sediments of aquatic systems which take a long time to be released.
- In deep oceans such carbon can remain buried for millions of years till geological movement may lift these rocks above sea level. These rocks may be exposed to erosion, releasing their carbon dioxide, carbonates and bicarbonates into streams and rivers.
- Fossil fuels such as coals, oil and natural gas etc. are organic compounds that were buried before they could be decomposed and were subsequently transformed by time and geological processes into fossil fuels. When they are burned the carbon stored in them is released back into the atmosphere as carbon-dioxide.

https://upload.wikimedia.org/wikipedia/commons/d/d5/Carbon_cycle.jpg

Q1. Consider the following:

1. Photosynthesis
2. Respiration
3. Decay of organic matter
4. Volcanic action

Which of the above add carbon dioxide to the carbon cycle on Earth?

- a) 1 and 4 only
- b) 2 and 3 only
- c) 2, 3 and 4 only
- d) 1, 2, 3 and 4

Answer: c) Photosynthesis takes out CO_2 from carbon cycle. Rest all adds CO_2 .

Q2. Which one of the following is the process involved in photosynthesis?

- a) Potential energy is released to form free energy
- b) Free energy is converted into potential energy and stored
- c) Food is oxidized to release carbon dioxide and water
- d) Oxygen is taken, and carbon dioxide and water vapour are given out

Answer: b) Sunlight (free energy) is converted into carbohydrates (potential energy) using water and carbon dioxide. Oxygen is released in the process.

Nitrogen Cycle [Gaseous Cycle]

- Apart from carbon, hydrogen and oxygen, nitrogen is the most prevalent element in living organisms.
- Nitrogen is a constituent of **amino acids, proteins, hormones, chlorophylls** and many of the **vitamins**. [All of these explained under Biology (<http://imojo.in/76hnkb>)]
- Plants compete with microbes for the limited nitrogen that is available in soil. Thus, nitrogen is a **limiting nutrient** for both natural and agricultural ecosystems.
- Nitrogen exists as two nitrogen atoms (N_2) joined by a very strong **triple covalent bond** ($N \equiv N$).
- In nature, **lightning** and **ultraviolet radiation** provide enough energy to convert nitrogen to nitrogen oxides (NO , NO_2 , N_2O).
- Industrial combustions, forest fires, automobile exhausts and power-generating stations are also sources of atmospheric nitrogen oxides.

Nitrogen Fixing – Nitrogen to Ammonia (N_2 to NH_3)

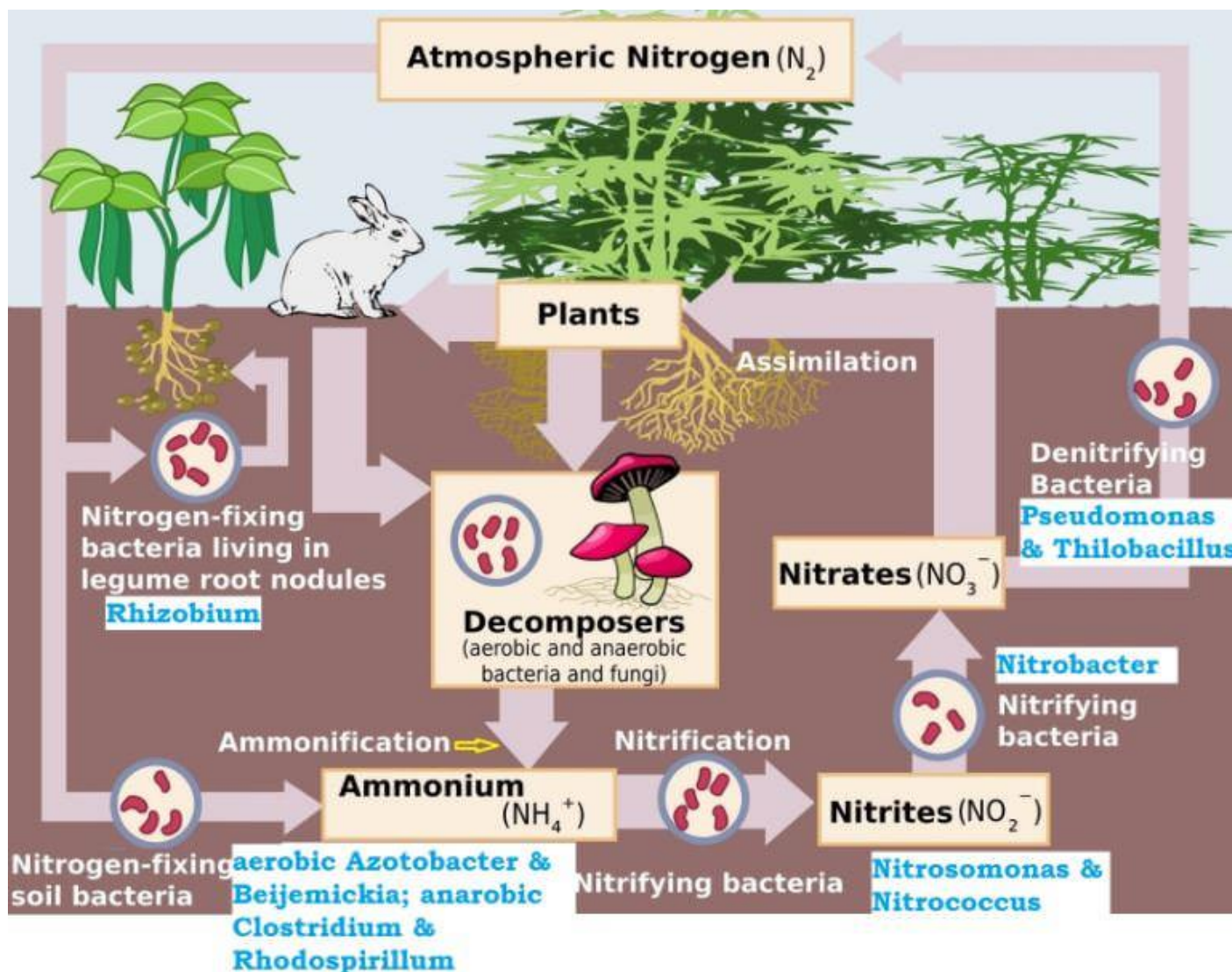
- There is an inexhaustible supply of nitrogen in the atmosphere but the elemental form cannot be used directly by most of the living organisms.
- Nitrogen needs to be '**fixed**', that is, **converted to ammonia, nitrites or nitrates**, before it can be taken up by plants.
- Nitrogen fixation on earth is accomplished in three different ways:

1. By microorganisms (bacteria and blue-green algae),
2. By man using industrial processes (fertilizer factories) and
3. To a limited extent by atmospheric phenomenon such as thunder and lightning.

- Certain microorganisms are capable of fixing atmospheric nitrogen into **ammonia (NH_3)** and **ammonium ions (NH_4^+)**.

Ammonia is a molecule consisting of nitrogen and hydrogen having molecular NH_3 , while ammonium (NH_4^+) is an ion of ammonia that is formed by accepting hydrogen ion.

- The enzyme, **nitrogenase** which is capable of **nitrogen reduction** is present exclusively in **prokaryotes**. Such microbes are called **N_2 -fixers**. These include:
 1. free living nitrogen fixing bacteria (non-symbiotic nitrogen fixing bacteria or nitrogen fixing soil bacteria) (e.g. aerobic **Azotobacter** and **Beijerinckia**; anaerobic **Clostridium** and **Rhodospirillum**),
 2. symbiotic nitrogen fixing bacteria (e.g. **Rhizobium**) living in association with leguminous plants and non-leguminous root nodule plants and
 3. some cyanobacteria (major source of nitrogen fixation in oceans) (**blue green algae**. E.g. **Nostoc**, **Anabaena**, **Spirulina** etc.).



Leguminous: denoting plants of the pea family (Leguminosae), typically having seeds in pods, distinctive flowers, and root nodules containing nitrogen-fixing bacteria.

Nitrification – Ammonia to Nitrates

- Ammonium ions can be directly taken up as a source of nitrogen by some plants.
- Others absorb **nitrates** which are obtained by oxidizing ammonia and ammonium ions.
- Ammonia and ammonium ions are oxidized to **nitrites or nitrates** by two groups of specialized bacteria.

1. **Ammonium** ions are first oxidized to **nitrite** by the bacteria **Nitrosomonas** and/or **Nitrococcus**.

2. The nitrite is further oxidized to **nitrate** with the help of the bacterium **Nitrobacter**.

- These steps are called **nitrification**. These nitrifying bacteria are **chemoautotrophs**.
- The nitrate thus formed is absorbed by plants and is transported to the leaves.
- In leaves, it is reduced to form ammonia that finally forms the amine group of **amino acids**, which are the building blocks of proteins. These then go through higher trophic levels of the ecosystem.

Nitrification is important in agricultural systems, where fertilizer is often applied as ammonia. Conversion of this ammonia to nitrate **increases nitrogen leaching** because nitrate is more water-soluble than ammonia.

Nitrification also plays an important role in the removal of nitrogen from municipal wastewater. The conventional removal is **nitrification**, followed by **denitrification**.

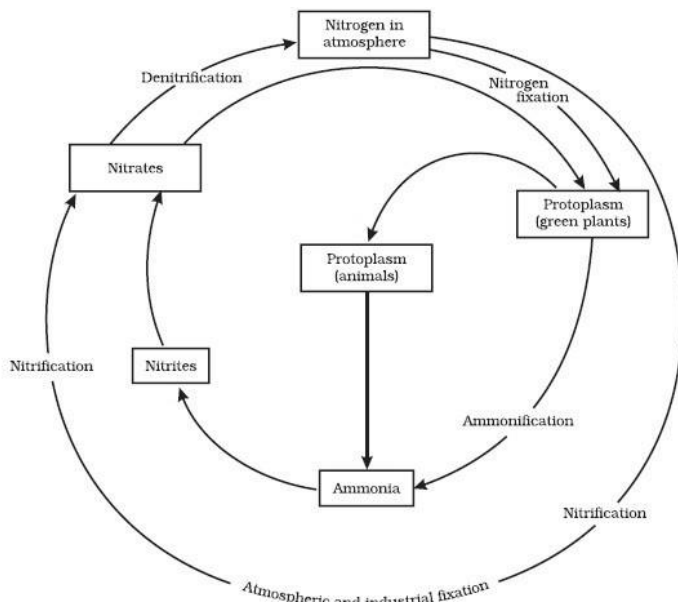


Fig.14.6: Nitrogen-cycle in nature

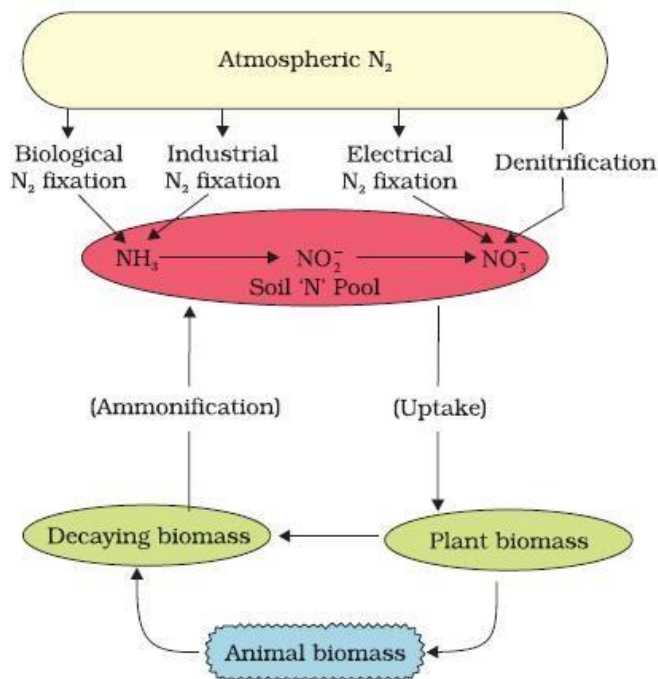


Figure 12.3 The nitrogen cycle showing relationship between the three main nitrogen pools - atmospheric, soil, and biomass

Ammonification – Urea, Uric Acid to Ammonia

- Living organisms produce nitrogenous waste products such as urea and uric acid (organic nitrogen).
- These waste products as well as dead remains of organisms are converted back into inorganic ammonia and ammonium ions by the bacteria. This process is called ammonification.
- Some of this ammonia volatilizes and re-enters the atmosphere but most of it is converted into **nitrate** by soil bacteria.

Denitrification – Nitrate to Nitrogen

- Nitrate present in the soil is **reduced to nitrogen** by the process of **denitrification**.
- In the soil as well as oceans there are special denitrifying bacteria (**Pseudomonas** and **Thiobacillus**), which convert the nitrates/nitrites to elemental nitrogen. This nitrogen escapes into the atmosphere, thus completing the cycle.

Step 1: N_2 Fixing è Nitrogen → Ammonia or Ammonium Ions

Step 2: Nitrification è Ammonia or Ammonium Ions → Nitrite → Nitrate

Step 3: Ammonification è Dead Matter + Animal Waste (Urea, Uric Acid) → Ammonia or Ammonium Ions.

Dead Matter + Animal Waste (Urea, Uric Acid) → Ammonia or Ammonium Ions [most of it escapes into atmosphere. Rest is Nitrified (**Step 2**) to nitrates]

Nitrate [some of it is available for plants. Rest is Denitrified (**Step 4**)]

Step 4: Denitrification è Nitrate → Nitrogen.

- The amount of Nitrogen fixed by man through industrial process has far exceeded the amount fixed by the Natural Cycle.
- As a result Nitrogen has become a pollutant which can disrupt the balance of nitrogen. It may lead to **Acid rain**,

Eutrophication and Harmful Blooms.**Algal****Q3. Which of the following adds/add nitrogen to the soil?**

1. Excretion of urea by animals
2. Burning of coal by man
3. Death of vegetation

Select the correct answer using the codes given below.

1. 1 only
2. 2 and 3 only
3. 1 and 3 only
4. 1, 2 and 3

Explanation:

All the above three adds to nitrogen cycle.

Burning coal releases CO, CO₂, sulphur dioxide and oxides of nitrogen – air pollutants.

Oxides of nitrogen fall on earth as acid rain. Acidic rain is a complex mixture of nitrous, nitric, sulfurous and sulfuric acids which all combine to lower the pH.

But, the question says “Which of the following adds/add nitrogen to the **soil**?”

1. Animal Waste like Urea, Uric acid and Death of vegetation add nitrogen in the form of nitrates directly into soil.
2. Coal combustion adds nitrogen to atmosphere and from there it falls back to earth in the form of acid rain and acid rain adds nitrogen to soil.

<http://www.ehow.com/how-does-5117705-burning-fuels-affect-nitrogen-cycle.html> says:

“The release of nitric oxides into the air in large quantities causes **smog and acid rain** that pollutes the atmosphere, soil and water and affects plants and animals. The increase in nitrogen and nitrous oxide is caused by automobiles, power plants and a wide variety of industries.

As nitrous oxides filter into soil, it loses nutrients like calcium and potassium, which are essential for maintaining a balance in plant ecosystems. With the loss of these compounds, soil fertility declines. Also, soils become significantly more acidic.”

And

also <http://chemistry.elmhurst.edu/vchembook/307nitrogen.html> says:

“The main component of the nitrogen cycle starts with the element nitrogen in the air. Two nitrogen oxides are found in the air as a result of interactions with oxygen. Nitrogen will only react with oxygen in the presence of high temperatures and pressures found near **lightning bolts** and in combustion reactions in power plants or internal combustion engines. Nitric oxide, NO, and nitrogen dioxide, NO₂, are formed under these conditions. Eventually nitrogen dioxide may react with water in rain to form **nitric acid**, HNO₃. The nitrates thus formed may be utilized by plants as a nutrient (so, soil gets nitrogen from acid rain).”

So, soil gets nitrogen from acid rain.

From this, I feel Burning of coal by man adds nitrogen to soil but indirectly through acid rain.

The answer depends on how you interpret the question.

Answer: Official UPSC Key says answer is c) 1 and 3 only.

If you have anything more to add write it at poormansfriend2485@gmail.com

Q4. Consider the following:

1. Carbon dioxide
2. Oxides of Nitrogen
3. Oxides of Sulphur

Which of the above is/are the emission/emissions from coal combustion at thermal power plants ?

- a) 1 only
- b) 2 and 3 only

- c) 1 and 3 only
- d) 1, 2 and 3

Burning coal releases CO, CO₂, sulphur dioxide and oxides of nitrogen – air pollutants.

Answer: d) 1, 2 and 3.

Q5. What can be the impact of excessive/inappropriate use of nitrogenous fertilizers in agriculture?

- Proliferation of nitrogen-fixing microorganisms in soil can occur.
- Increase in the acidity of soil can take place
- Leaching of nitrate to the ground-water can occur.

Select the correct answer using the code given below.

- 1. 1 and 3 only
- 2. 2 only
- 3. 2 and 3 only
- 4. 1,2 and 3

Explanation:

Nitrification is important in agricultural systems, where fertilizer is often applied as ammonia. Conversion of this ammonia to nitrate **increases nitrogen leaching** because nitrate is more water-soluble than ammonia.

Agricultural fertilization and the use of nitrogen fixing plants also contribute to atmospheric NO_x, by promoting nitrogen fixation by microorganisms. Excess NO_x leads to acid rain. Acid rain lowers pH of soil (increase in acidity of soil)

The legume–rhizobium symbiosis is a classic example of mutualism—rhizobia supply ammonia or amino acids to the plant and in return receive organic acids as a carbon and energy source.

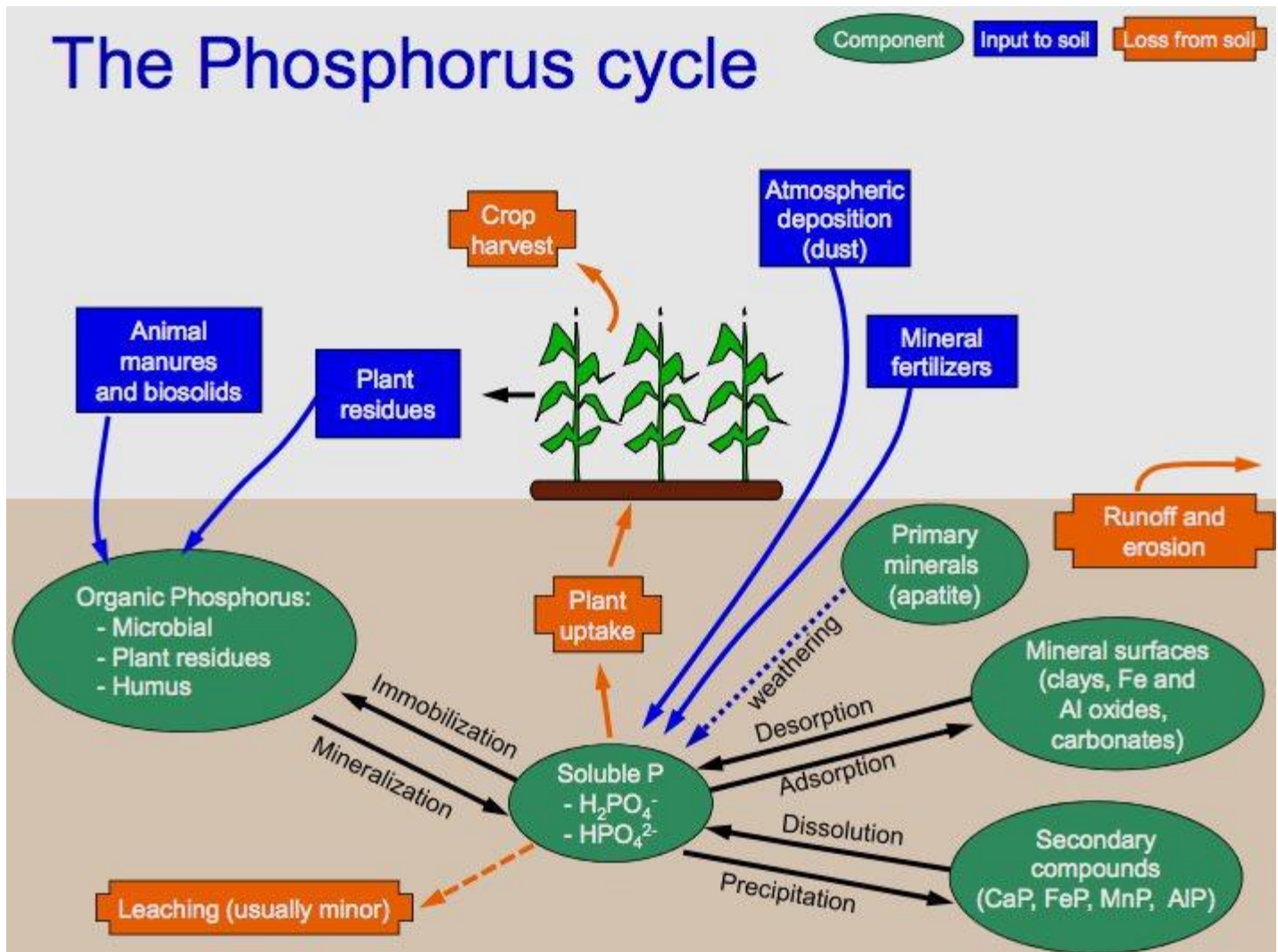
So, excessive/inappropriate use of nitrogenous fertilizers can make the plants independent of both symbiotic and free living nitrogen fixers. Fixers don't get the food from the plants due to broken relationship and other factors. So their population decreases.

Answer: c) 2 and 3 only

Phosphorus Cycle [Sedimentary cycle]

- Phosphorus plays a central role in aquatic ecosystems and water quality.
- Unlike carbon and nitrogen, which come primarily from the atmosphere, phosphorus occurs in large amounts as a mineral in phosphate rocks and enters the cycle from erosion and mining activities.
- This is the nutrient considered to be the main cause of excessive growth of rooted and free-floating microscopic plants (phytoplankton) in lakes [**Eutrophication**].
- The main storage for phosphorus is in the earth's crust. On land phosphorus is usually found in the form of **phosphates**.
- By the process of weathering and erosion phosphates enter rivers and streams that transport them to the ocean.
- In the ocean phosphorus accumulates on continental shelves in the form of insoluble deposits.
- After millions of years, the crustal plates rise from the sea floor and expose the phosphates on land.
- After more time, weathering will release them from rock and the cycle's geochemical phase begins again.

The Phosphorus cycle

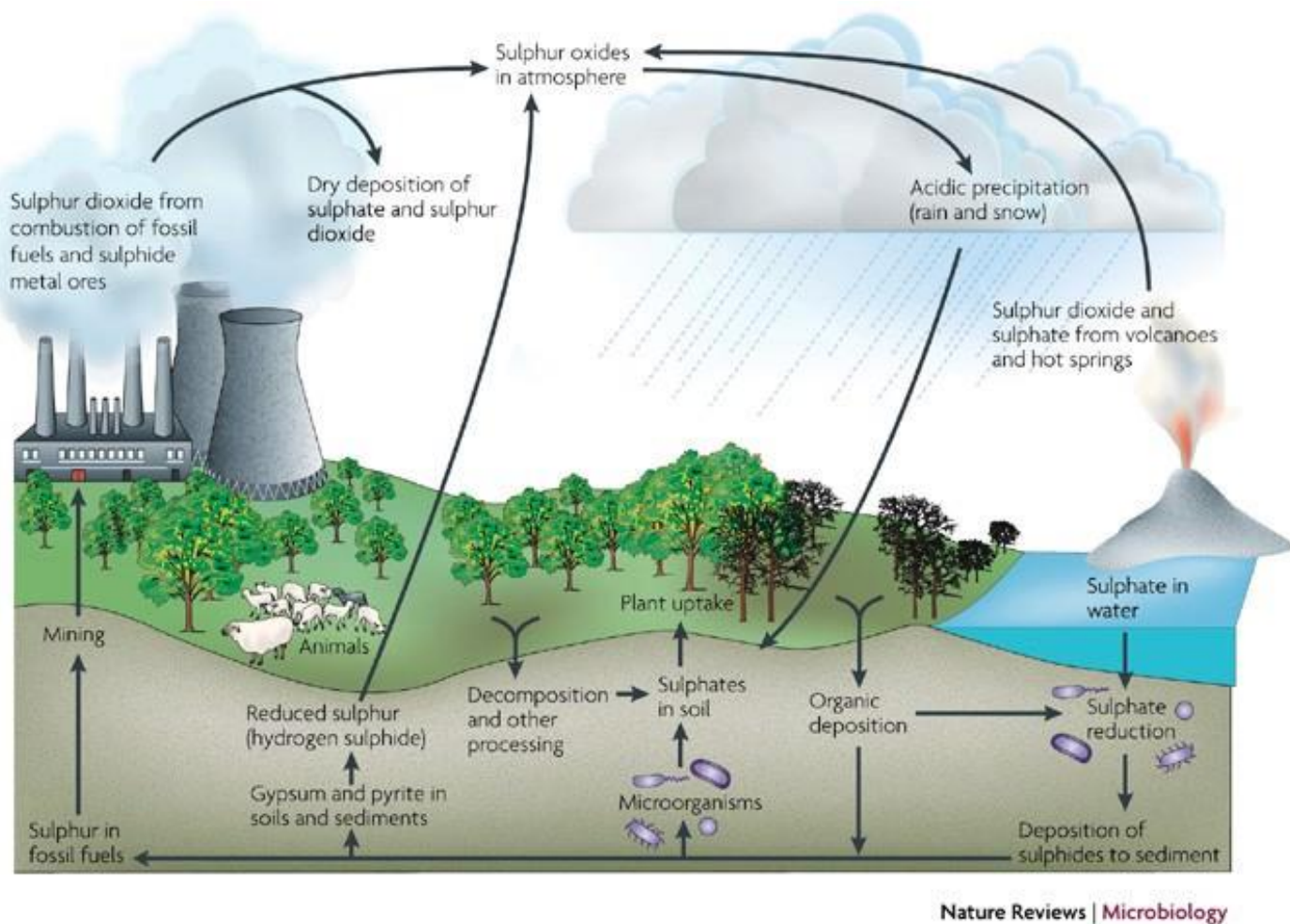


Sulphur Cycle [Sedimentary cycle]

- The sulphur reservoir is in the soil and sediments where it is locked in organic (**coal, oil and peat**) and inorganic deposits (**pyrite rock and sulphur rock**) in the form of **sulphates, sulphides** and **organic sulphur**.
- It is released by weathering of rocks, erosional runoff and decomposition of organic matter and is carried to terrestrial and aquatic ecosystems in salt solution.
- The sulphur cycle is mostly sedimentary except two of its compounds – **hydrogen sulphide (H_2S)** and **sulphur dioxide (SO_2)** which add a **gaseous component**.
- Sulphur enters the atmosphere from several sources like **volcanic eruptions, combustion of fossil fuels (coal, diesel**

etc.), from **surface of ocean** and from **gases released by decomposition**. Atmospheric hydrogen sulphide also gets oxidized into sulphur dioxide.

- Atmospheric sulphur dioxide is carried back to the earth after being dissolved in rainwater as weak **sulphuric acid**.
- Whatever the source, sulphur in the form of **sulphates** is taken up by plants and incorporated through a series of metabolic processes into **sulphur bearing amino acid** which is incorporated in the proteins of autotroph tissues. It then passes through the grazing food chain.
- Sulphur bound in living organism is carried back to the soil, to the bottom of ponds and lakes and seas through **excretion and decomposition** of dead organic material.



Biomes - Terrestrial Ecosystems

Natural Ecosystem

- A natural ecosystem is an assemblage of plants and animals which functions as a unit and is capable of maintaining its identity. There are two main categories of ecosystems.

Terrestrial ecosystem

- Ecosystems found on land e.g. forest, grasslands, deserts, tundra.

Tundra

- Arctic and Alpine Tundra Biome

Forest

- Taiga or Boreal Biome [Coniferous forests]
- Temperate Deciduous Biome [North Western Europe]

- Temperate Rainforest Biome
- Sub-Tropical Deciduous Biome in Eastern China, South Eastern USA
- Temperate Deciduous Biome [Mediterranean Region]
- Tropical Deciduous Biome [Monsoon Climate]
- Savanna or Tropical Wet and Dry Biome
- Tropical Rain Forest Biome

Grassland

- Steppe or Temperate Grassland Biome
- Savanna or Tropical Wet and Dry Biome [Tropical Grasslands]

Desert

- Tropical and Mid Latitude Desert Biome

Aquatic ecosystem

- Plants and animal community found in water bodies. These can be further classified into two sub groups.
- Fresh water ecosystems, such as rivers, lakes and ponds.
- Marine ecosystems, such as oceans, estuary and mangroves.

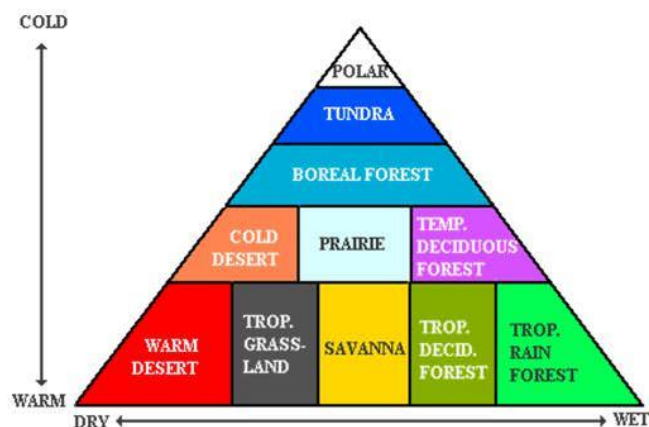
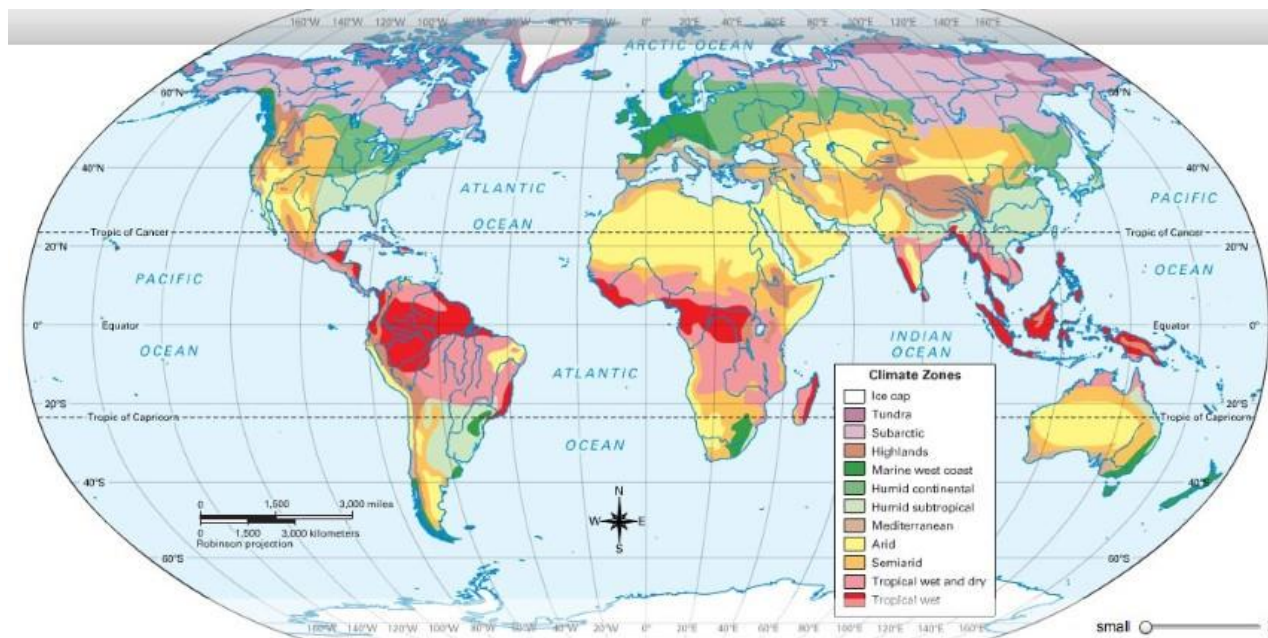
Biomes or Terrestrial Ecosystems

- The **terrestrial** part of the biosphere is divisible into enormous regions called biomes, which are characterized, by

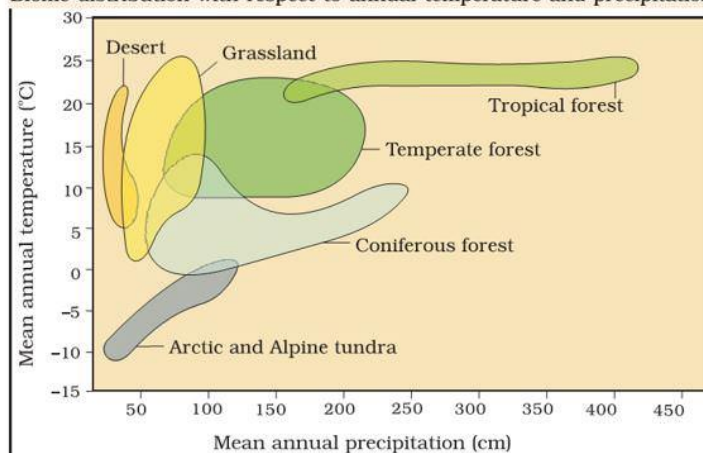
distinct climate [precipitation and temperature mainly], vegetation, animal life and general soil type.

- [They are nothing but '[Climatic Regions](http://www.pmfias.com/climatic-regions-upsc-ias-geography/)' <http://www.pmfias.com/climatic-regions-upsc-ias-geography/> explained under Geography Section]
- No two biomes are alike. The climate determines the boundaries of a biome and abundance of plants and animals found in each one of them. The most important climatic factors are temperature and precipitation.

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Biome distribution with respect to annual temperature and precipitation



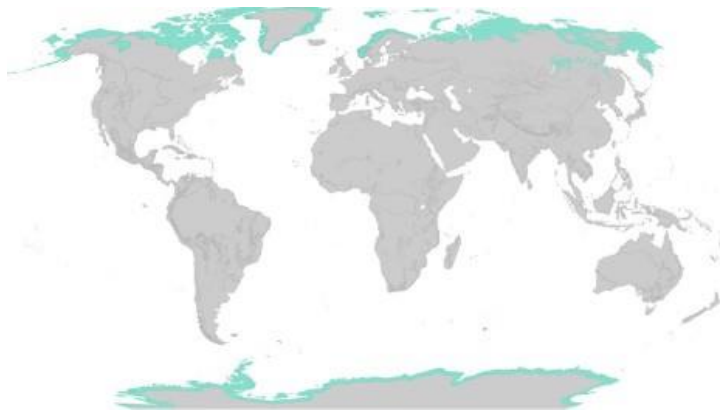
Tundra Biome

- There are two types of tundra – arctic and alpine.

Distribution

- Arctic tundra extends as a continuous belt below the polar ice cap and above the tree line (taiga) in the northern hemisphere.

- It occupies the northern fringe of Canada, Alaska, European Russia, Siberia and island group of Arctic Ocean.
- On the south pole, tundra is very small since most of it is covered by ocean.
- Alpine tundra occurs at high mountains above the tree line. E.g. High ranges of Himalayas, Andes, Alps etc.



Temperature

- The tundra climate is characterized by a very **low mean annual temperature**.
- In mid-winter temperatures are as low as 40 – 50 °C below freezing.

Precipitation

- Precipitation is mainly in the form of snow and sleet.

Natural Vegetation

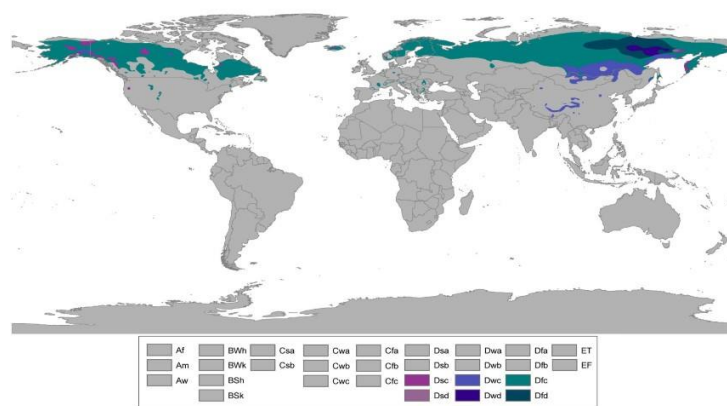
- There are **no trees** in the tundra (Ground is frozen). Lowest form of vegetation like **mosses, lichens** are found here and there.
- Coastal lowlands support hardy grasses and the reindeer moss which provide the only pasturage for **reindeers**.
- In the brief summer, berry-bearing bushes and Arctic flowers bloom.
- In the summer, birds migrate north to prey on the numerous **insects** which emerge when the snow thaws.
- Insects have short life cycles which are completed during favourable period of the year.
- Animals like the **reindeer, arctic fox, wolves, musk-ox, polar bear,**

lemming, arctic hare, arctic willow live in tundra region.

- Reptiles and amphibians are almost absent.
- Most of the animals have long life e.g. arctic willow has a life span of 150 to 300 years.
- They are protected from chillness by the presence of **thick cuticle** and **epidermal hair** or fur.
- Mammals of the tundra region have **large body size and small tail and ear to avoid the loss of heat from the surface** [less surface area = less heat loss = less food required to produce heat].

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Taiga or Boreal Biome



Temperature

- Summers are brief and warm reaching 20-25 °C whereas winters are long and brutally cold – 30-40 °C below freezing.

Precipitation

- Typical annual precipitation ranges from 38 cm to 63 cm.
- It is quite **well distributed throughout the year**, with a **summer maxima**.
- In winter the precipitation is in the form of snow.

Soil

- Boreal forest soils are characterized by thin **podzols** and are rather poor.
- This is because the weathering of rocks proceeds slowly in cold environments and because the litter derived from conifer needle (leaf) is decomposed very slowly and

is not rich in nutrients. Moreover conifers don't shed their leaves frequently.

- Most podzols are poor soils for agriculture due to the **sandy** portion, resulting in a **low level of moisture and nutrients**.
- Some are sandy and excessively drained. Others have shallow rooting zones and poor drainage due to subsoil cementation.
- A **low pH (acidic)** further compounds issues, along with phosphate deficiencies and aluminium toxicity.
- The low pH (acidic) factor is due to **excessive leaching of alkaline oriented cations** which if present would neutralize the organic acids of the accumulating litter.

Natural Vegetation

- The predominant vegetation is **evergreen coniferous forest**.
- Conifers are evergreen plant species such as Spruce, fir and pine trees, etc.
- The conifers require little moisture are best suited to this type of sub-Arctic climate.
- The productivity and community stability of a boreal forest are **lower** than those of any other forest ecosystem.
- Animals found in this region include Siberian tiger, wolverine, lynx, wolf, bear, red fox, squirrel, and amphibians like Hyla, Rana, etc.

Characteristics of Coniferous forests

- Unlike the equatorial rain forests, Coniferous forests are of **moderate density** and are more uniform. The trees in coniferous forests grow straight and tall.
- Almost all conifers are **evergreen**. There is no annual replacement of new leaves as in deciduous trees.
- The same leaf remains on the tree for as long as **five years**. Food is stored in the **trunks**, and the **bark is thick** to protect the trunk from excessive cold.
- Conifers are conical in shape. Their conical shape and sloping branches prevent snow accumulation. It also offers little grip to the winds.
- **Transpiration** can be quite rapid in the warm summer. So, leaves are small, thick, **leathery** and needle-shaped to **check excessive transpiration**.

- The soils of the coniferous forests are They are excessively **leached** and very **acidic**.
- Humus content is also low as the evergreen leaves barely fall and the rate of decomposition is slow.
- Under-growth is negligible because of the poor soil conditions.
- Absence of direct sunlight and the short duration of summer are other contributory factors.
- Coniferous forests are also found in regions with high elevation [Example: The forests just below the snowline in Himalayas].
- But on very steep slopes where soils are immature or non-existent, even the conifer cannot survive [Example: Southern slopes of Greater Himalayas].

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Temperate Deciduous Biome [North Western Europe]



- Moderately warm summers and fairly mild winters.

Temperature

- The mean annual temperatures are usually between 5° C and 15° C.
- Winters are **abnormally mild**. This is because of the warming effect brought by **warm North Atlantic Drift**. [Eastern Australian warm current in case of New Zealand]

Precipitation

- **Rainfall occurs throughout the year with winter maxima.**
- Adequate rainfall throughout the year.

Seasons

- As in other temperate regions there are **four distinct seasons**.
- Winter is the season of cloudy skies, foggy and misty mornings, and many rainy days from the passing depressions. (Trees shed their leaves in winter to prevent snow accumulation and protect themselves from severe cold)
- Spring is the driest and the **most refreshing** season when people emerge from the depressing winter to see everything becoming green again.
- This is followed by the long, sunny summer.
- Next is the autumn with the roar of gusty winds; and the cycle repeats itself.
- This type of climate with its four distinct seasons is something that is **conspicuously absent in the tropics**. [Rainforest == Only Rainy season, Tropical Monsoon == Summer, Winter and Rainy, Tropical Savanna == Summer (rains) and Winter]

Natural Vegetation

- Soils of temperate forests are podzolic and fairly deep.
- The natural vegetation of this climatic type is **deciduous forest**.
- The trees shed their leaves in the **cold season**.
- This is an adaptation for protecting themselves against the winter snow and frost.
- Shedding begins in **autumn, the 'fall' season**. Growth begins in spring.
- Some of the common species include **oak, elm, ash, birch, beech, and poplar**.
- In the wetter areas grow willows (Light weight cricket bats are made from willows. In India willows are found in Kashmir).
- Most animals are the familiar vertebrates and invertebrates.

Temperate Rainforest Biome

- Temperate rain forests receive an annual precipitation of 200 cm, mostly due to on shore westerlies.
- Precipitation occurs in the form of fog, rain as well as snow. Fog is quite common and is an important source of water.

Distribution

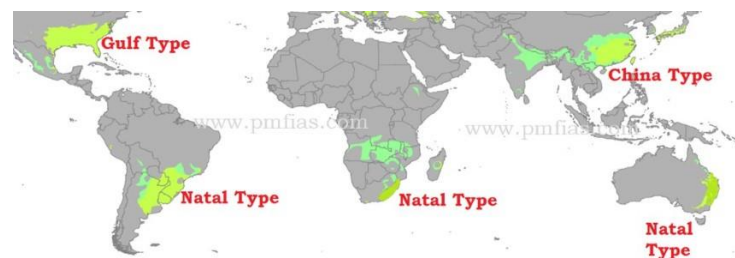
- This is a small biome in terms of area covered. The main stretch of this habitat is along the northwestern coast of North America from northern California through southern Alaska. There are also small areas in southern Chile, New Zealand, Australia and a few other places around the world.



Natural Vegetation

- Big coniferous trees dominate this habitat, including **Douglas fir, Western red cedar, Mountain hemlock, Western hemlock, Sitka spruce and Lodgepole pine**.
- In addition to the trees, mosses and lichens are very common, often growing as epiphytes.
- Grizzly Bears are the common mammals found in Alaska.

Sub-Tropical Deciduous Biome in Eastern China, South Eastern USA



Climate

- Characterized by a **warm moist summer** and a **cool, dry winter** (one exception: winters are also moist in Natal Type).

Temperature

- The mean monthly temperature varies between 4° C and 25° C and is strongly modified by **maritime influence**.

Precipitation

- Rainfall is more than moderate, anything from **60 cm to 150 cm**.
- There is the **fairly uniform distribution of rainfall throughout the year**.

Natural Vegetation

- Supports a luxuriant vegetation.
- The lowlands carry both evergreen broad-leaved forests and deciduous trees [**hardwood**].
- On the highlands, are various species of conifers such as pines and cypresses which are important
- Perennial plant growth is not checked by either a dry season or a cold season.

Steppe or Temperate Grassland Biome



Name of the Temperate Grassland	Region
Pustaz	Hungary and surrounding regions
Prairies	North America [between the foothills of the Rockies and the Great Lakes]
Pampas	Argentina and

	Uruguay [Rain-shadow effect]
Bush-veld (more tropical)	Northern South Africa
High Veld (more temperate)	Southern South Africa
Downs	Australia: Murray-Darling basin of southern Australia
Canterbury	New Zealand

Temperature

- Climate is continental with **extremes of temperature**.
- Temperatures vary greatly between summer and winter.

Precipitation

- The average rainfall may be taken as about 45 cm, but this varies according to location from 25 cm to 75 cm.

Natural Vegetation of Steppe Climate

Grasses

- Greatest difference from the tropical savanna is that steppes are practically **treeless** and the **grasses are much shorter**.
- Grasses are fresh and **nutritious**. This is typical of the grass of the wheat-lands in North America, the **rich black earth or chernozem areas of Russian Ukraine** and the better watered areas of the Asiatic Steppes.

Trees

- Polewards, an increase in precipitation gives rise to a transitional zone of wooded steppes where some **conifers** gradually appear.

Animals

- Does not have much animal diversity.
- Horses** are common in Asian Steppes.

Major Grasslands of the World

Savanna

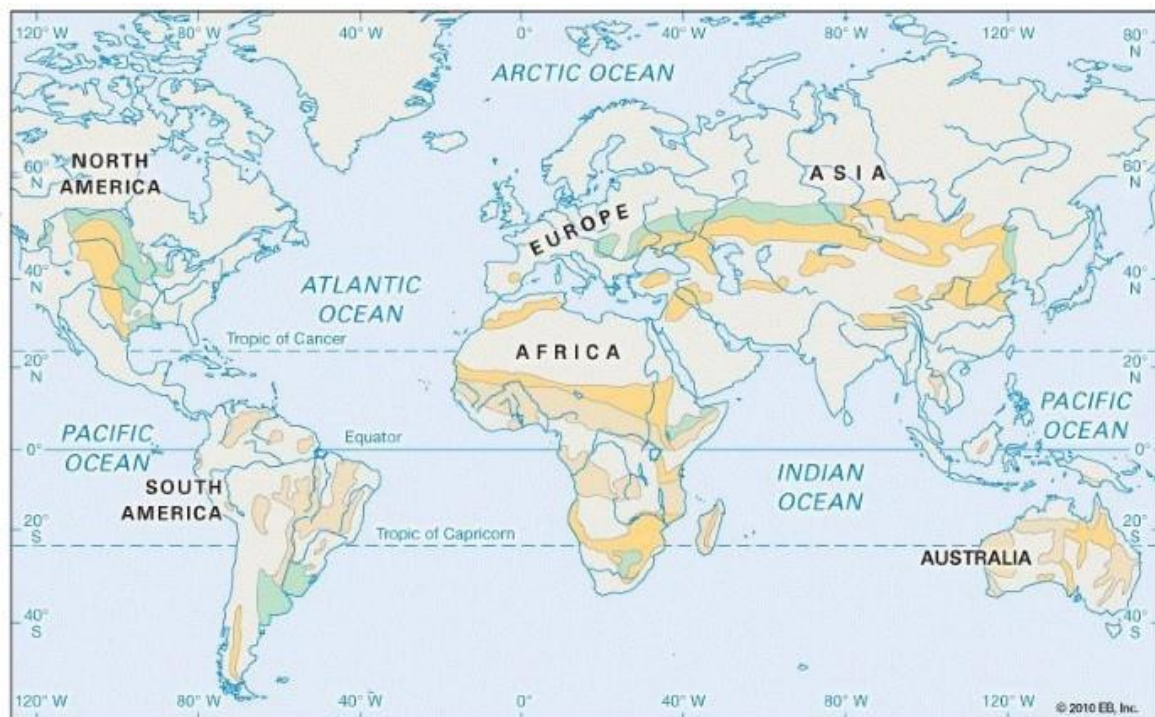
1. Llanos of the Orinoco in Venezuela and Colombia
2. Campos of Brazil
3. Sudan in Africa
4. South African veld
5. Australia

Prairie

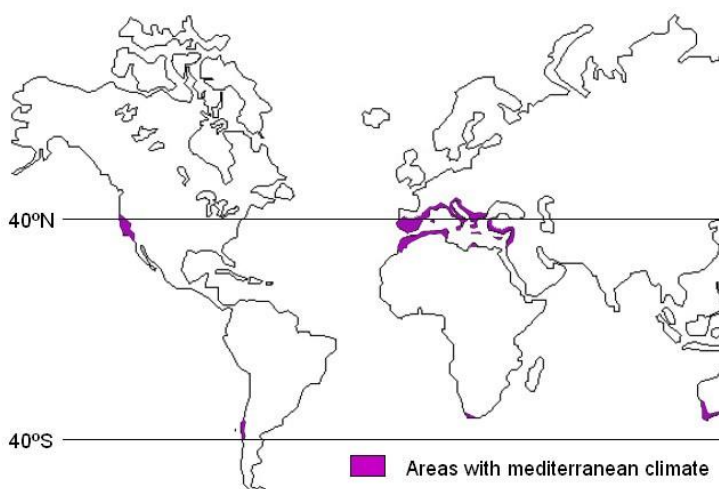
1. Midwestern United States and Canada
2. Pampa of Argentina, Uruguay, and southeastern Brazil
3. Plains of Hungary, Romania, and historic Yugoslavia
4. Black Earth Belt of Russia
5. Manchurian Plain

Steppe

1. Great Plains of North America
2. Kyrgyz Steppe
3. Australia
4. Sudan in Africa



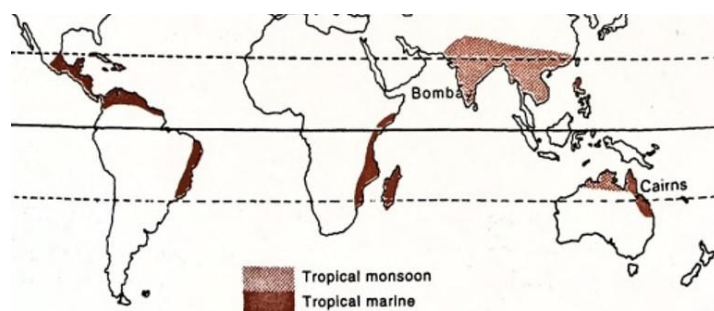
Temperate Deciduous Biome [Mediterranean Region]



- Parts of the world that have Mediterranean type of climate are characterized by **warm, dry summers and cool, moist winters**.
- Trees with **small broad leaves** are widely spaced and **never very tall**.
- Regions with adequate rainfall are inhabited by low broad leaved evergreen trees [mostly evergreen oaks].
- Fire is an important hazardous factor in this ecosystem and the adaptation of the plants enable them to **regenerate quickly** after being burnt.

- Plants are in a continuous struggle against heat, dry air, excessive evaporation and prolonged droughts.
- They are, in short **xerophytic [drought tolerant]**, a word used to describe the drought-resistant plants in an environment deficient in moisture.

Tropical Deciduous Biome [Monsoon Climate]



- Unlike equatorial wet climate, monsoon climate is characterized by **distinct wet and dry seasons** associated with **seasonal reversal of winds**.
- **Floods** in wet season and **droughts** in dry season are common.
- Usually there are three seasons namely **summer, winter and rainy**

Temperature

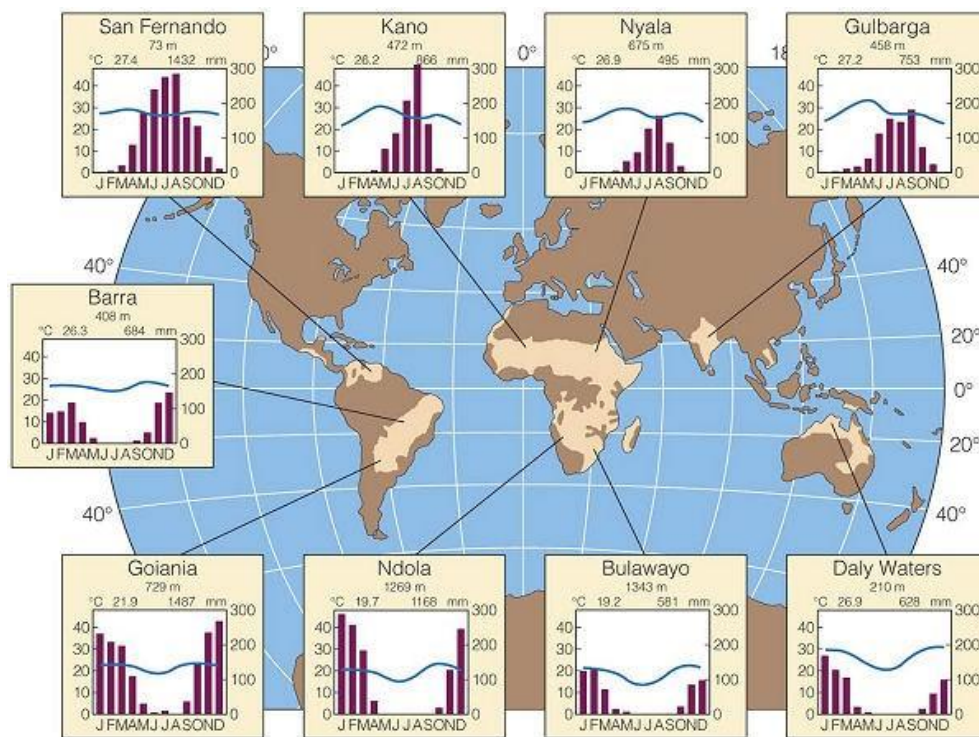
- Monthly mean temperatures **above 18 °C**.
- Temperatures range from 30-45° C in summer.
- In winters, temperature range is 15-30° C with mean temperature around 20-25° C.

Precipitation

- Annual mean rainfall ranges from 200-250 cm. In some regions it is around 350 cm.
- Places like **Cherrapunji** & **Mawsynram** receive an annual rainfall of about **1000 cm**.

Tropical Monsoon Forests

- Also known as **drought-deciduous forest; dry forest; dry-deciduous forest; tropical deciduous forest**.
- **Teak, neem, bamboos, sal, shisham, sandalwood, khair, mulberry** are some of the important species found here.



Savanna or Tropical Wet and Dry Biome

- This type of biome has **alternate wet and dry seasons** similar to monsoon climate but has **considerably less annual rainfall**.
- Also, there is **no distinct rainy season** like in monsoon climate. [Only two

seasons – winter and summer. **Rains occur in summer**].

- Floods and droughts are common.
- Vegetation, wildlife and human life are quite different from monsoon climate regions.

Rainfall

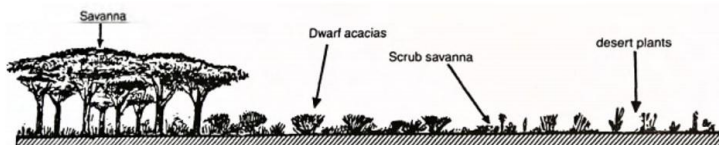
- Mean annual rainfall ranges from **80 – 160 cm** [Rainfall decreases with distance from equator].

Temperature

- Mean annual temperature is **greater than 18° C**.
- The monthly temperature hovers between 20° C and 32° C for lowland stations.

Natural Vegetation of Savanna Climate

- The savanna landscape is typified by **tall grass and short trees**.
 - The grasslands are also called as **'bush-veld'**.
 - The trees are **deciduous**, shedding their leaves in the cool, dry season to prevent excessive loss of water through transpiration, e.g. acacias.
 - Trees usually have **broad trunks**, with water-storing devices to survive through the prolonged drought.
 - Many trees are umbrella shaped, exposing only a narrow edge to the strong winds.
 - In true savanna lands, the grass is **tall and coarse**, growing 6 to 12 feet high. The **elephant grass** may attain a height of even 15 feet.
- Grasses appear greenish and well-nourished in the rainy season but turns yellow and dies down in the dry season that follows.
- As the rainfall diminishes towards the deserts the savanna merges into thorny scrub.

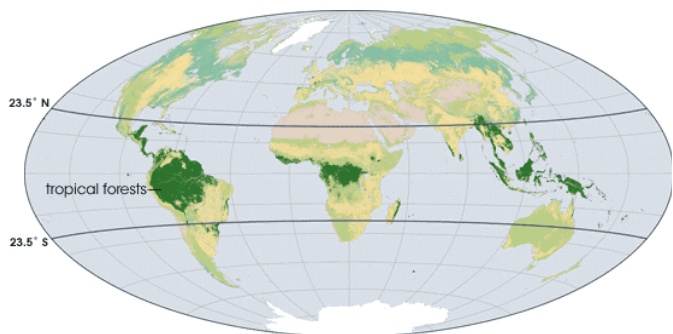


Animal Life of the Savanna

- There are two main groups of animals in the savanna, the grass-eating herbivorous animals and the fleshing-eating carnivorous animals.
- The herbivorous include the zebra, antelope, giraffe, deer, gazelle, elephant etc. [most of the National geographic and Animal Planet documentaries on wild animals are shot in savanna regions] and carnivorous animals include the lion, tiger, leopard, hyena, panther, jaguar, jackal etc..
- Species of reptiles and mammals including crocodiles, alligators, giant lizards live together with the larger rhinoceros and hippopotamus in rivers and marshy lakes.

Tropical Rain Forest Biome

- Also known as 'The Hot, Wet Equatorial Climate', 'Equatorial Rainforest Climate'.
- The regions are generally referred as 'Equatorial Rainforests', 'Equatorial Evergreen Forests', 'Tropical Moist Broadleaf Forest', 'Lowland Equatorial Evergreen Rainforest'.



Temperature

- Temperature is **uniform** throughout the year.
- The mean monthly temperatures are always around **27° C** with very little variation.
- **There is no winter.** [Typical to Equatorial Rainforest Climate]

Precipitation

- Precipitation is heavy and **well distributed throughout the year.**
- Annual average is always above **150 cm.** In some regions the annual average may be as high as 250 – 300 cm.

Equatorial Vegetation

- High temperature and abundant rainfall support a luxuriant **tropical rain forest.**
- In the Amazon lowlands, the forest is so dense that it is called '**selvas**'. [selvas: A dense tropical rainforest usually having a cloud cover (**dense canopy**)]
- Unlike the temperate regions, the **growing season here is all the year round**-seeding, flowering, fruiting and decaying do not take place in a seasonal pattern.
- The equatorial vegetation comprises a multitude of evergreen trees that yield **tropical hardwood, e.g. mahogany, ebony, dyewoods etc.**
- In the coastal areas and brackish swamps, **mangrove forests**
- All plants struggle upwards (most **epiphytes**) for sunlight resulting in a peculiar layer arrangement [Canopy].

Epiphyte (Commensalism – Epiphyte benefits without troubling the host): An epiphyte is a plant that grows harmlessly upon another plant (such as a tree) and derives its moisture and nutrients from the air, rain, and sometimes from debris accumulating around it.

Q1. Which of the following is/are unique characteristic/characteristics of equatorial forests?

1. Presence of tall, closely set trees with crowns forming a continuous canopy
2. Coexistence of a large number of species
3. Presence of numerous varieties of epiphytes

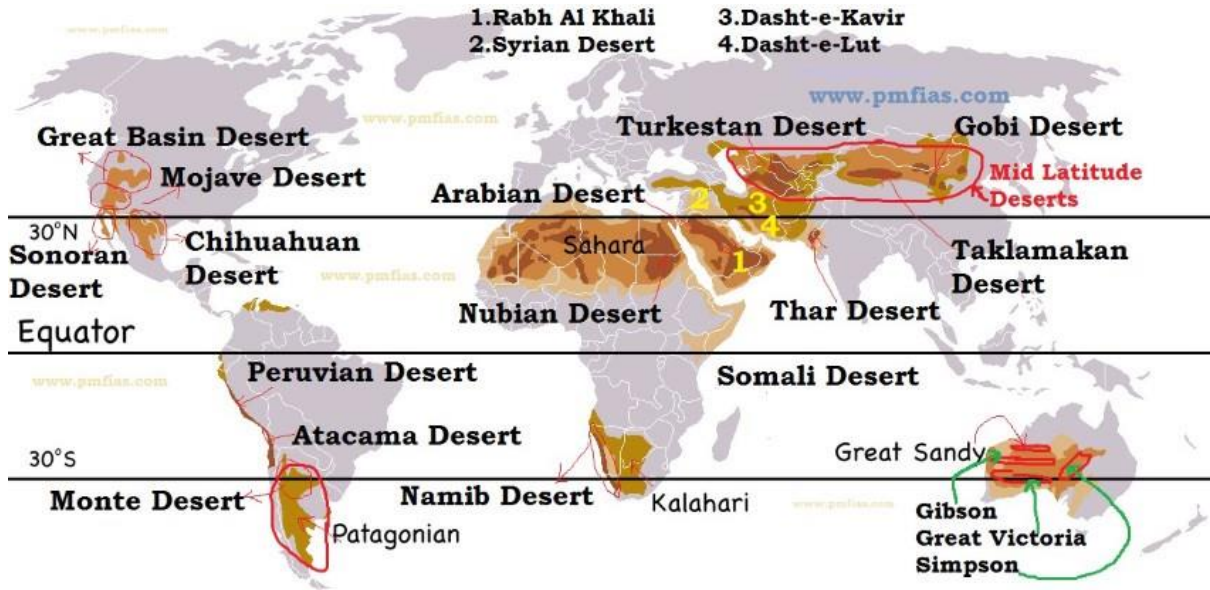
Select the correct answer using the code given below:

1. 1 only
2. 2 and 3 only

3. 1 and 3 only
4. 1, 2 and 3

Answer: Too easy. d) All

Desert Biome



Hot Deserts

- They include the biggest **Sahara Desert** (3.5 million square miles), **Great Australian Desert**, **Arabian Desert**, **Iranian Desert**, **Thar Desert**, **Kalahari** and **Namib Deserts**.
- In North America, the desert extends from Mexico into U.S.A. and is called by different names at different places, e.g. the **Mohave**, **Sonoran**, **Californian** and **Mexican Deserts**.
- In South America, the **Atacama** or **Peruvian Desert** is the **driest** of all deserts with less than 2 cm of rainfall annually.

Mid-Latitude Deserts

- The temperate deserts are rainless because of either **continentality** or **rain-shadow effect**. [**Gobi desert** is formed due to **continentality** and **Patagonian desert** due to **rain-shadow effect**]
- Amongst the mid-latitude deserts, many are found on plateau and are at a considerable distance from the sea. These are **Ladakh**, **The Kyzyl Kum**, **Turkestan**, **Taklimakan** and **Gobi deserts of Central**

- Deserts are regions where **evaporation exceeds precipitation**.
- There are mainly two types – hot like the **hot deserts** of the Saharan type and temperate as are the **mid-latitude deserts** like the Gobi.

Asia, drier portions of the Great Basin Desert of the western United States and Patagonian Deserts of Argentina etc..

- The Patagonian Desert is more due to its rain-shadow position on the leeward side of the lofty Andes than to continentality.

Rainfall (Both Hot and Cold deserts)

- Deserts, whether hot or mid-latitude have an annual precipitation of **less than 25 cm**.

Temperature of Hot deserts

- There is no cold season in the hot deserts and the average summer temperature is high around 30°C.
- The highest temperature recorded is **77° C** in 1922 at **A1 Azizia, Libya**.

Desert Vegetation

- The predominant vegetation of both hot and mid-latitude deserts is **xerophytic** or drought-resistant.

- This includes the cacti, thorny bushes, long-rooted wiry grasses and scattered dwarf acacias.
- Trees are rare except where there is abundant ground water to support clusters of **date palms**.
- Most desert shrubs have long roots and are well spaced out to gather moisture, and search for ground water. Plants have few or no leaves and the foliage is either **waxy, leathery, hairy or needle-shaped** to reduce the loss of water through transpiration.
- The seeds of many species of grasses and herbs have **thick, tough skins** to protect them while they lie dormant.

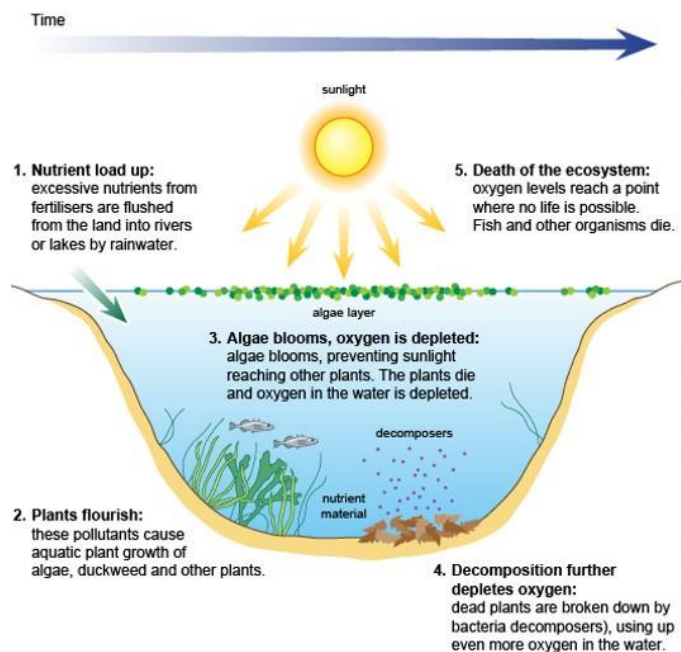
Eutrophication – Algal Bloom

Eutrophication – Algal Bloom

- Eutrophic water body: it is a body of water rich in nutrients and so supporting a dense plant population, the decomposition of which kills animal life by depriving it of oxygen.
- Eutrophication is the response to the addition of nutrients such as **nitrates** and **phosphates** naturally or artificially, fertilizing the aquatic ecosystem.
- **Algal blooms** are the consequence of Eutrophication.
- Eutrophication occurs naturally due to deposition of nutrients [such as in depositional environments] carried by flood waters. It takes over centuries for eutrophication to occur naturally.
- Similar nutrient enrichment of lakes at an accelerated rate is caused by human activities [discharge of wastewaters or agricultural runoff, Combustion of fossil fuel (produces gases —nitrogen oxides), growing urban population in the coastal areas) and the consequent phenomenon is known as '**cultural eutrophication**'. It takes only decades.
- Phytoplankton (algae and blue-green bacteria) thrive on the excess nutrients and their population explosion covers

almost entire surface layer. This condition is known as **algal bloom**.

- Oxygen in aquatic ecosystem is replenished by photosynthetic aquatic plants. Algal Blooms restrict the penetration of sunlight resulting in **death of aquatic plants**, and hence restricts the replenishment of oxygen.
- The oxygen level is already depleted due to the population explosion of phytoplankton.
- Phytoplankton are **photosynthetic during day time** adding oxygen to aquatic ecosystem. But **during nights, they consume far more oxygen** as they respire aggressively. i.e. Algal blooms accentuate the rate of oxygen depletion as the population of phytoplankton is very high.
- The primary consumers like small fish are killed due to oxygen deprivation caused by algal blooms.
- Death of primary consumers adversely affects the food chain and leads to the destruction of higher life forms.
- Further, more **oxygen is taken up by microorganisms during the decomposition process** of dead algae, plants and fishes. Due to reduced oxygen level, the remaining fishes and other aquatic organisms also die. All this



eventually leads to degradation of aquatic ecosystem.

- The new anaerobic conditions [absence of oxygen] created promote growth of bacteria such as **Clostridium botulinum** which

produces **toxins** deadly to aquatic organisms, birds and mammals.

Effects of Eutrophication

- **Loss of fresh water lakes:** Eutrophication eventually creates detritus layer in lakes and produces successively **shallower** depth of surface water. Eventually the water body is reduced into marsh whose plant community is **transformed** from an aquatic environment to recognizable **terrestrial** [Lakes are one of the major sources of fresh water]
- **New species invasion:** Eutrophication may cause the ecosystem competitive by transforming the normal limiting nutrient to abundant level. This cause shifting in species composition of ecosystem.
- **Toxicity:** Some algal blooms when died or eaten, release **neuro** & **hepatotoxins** which can kill aquatic organism & pose threat to humans. E.g. **Shellfish poisoning**.
- **Loss of coral reefs:** Occurs due to decrease in water transparency (increased turbidity).
- Affects navigation due to increased turbidity; creates colour (yellow, green, red), smell and water treatment problems; increases biomass of inedible toxic phytoplankton, benthic and epiphytic algae and bloom of gelatinous zooplankton.

Mitigation of Eutrophication

Checking water pollution is the ultimate solution to eutrophication.

- Treating Industrial effluents domestic sewage to remove nutrient rich sludge through waste water processing.
- Riparian buffer: Interfaces between a flowing body of water and land created near the waterways, farms, roads, etc. in an attempt to filter pollution. Sediments and nutrients are deposited in the buffer zones instead of deposition in water [Wetlands, estuaries are natural riparian buffers].

- Increase in efficiency of nitrogen & phosphorous fertilizers and using them in adequate levels.
- Nitrogen testing & modeling: N-Testing is a technique to find the optimum amount of fertilizer required for crop plants. It will reduce the amount of nitrogen lost to the surrounding area.
- Encouraging organic farming.
- Reduction in nitrogen emission from vehicles and power plants.

Harmful Algal Blooms

- Algae or phytoplankton are microscopic organisms that can be found naturally in coastal waters.
- They are major producers of oxygen and food for many of the animals that live in these waters.
- When environmental conditions are favorable for their development, these cells may multiply rapidly and form high numbers of cells and this is called an algal bloom.
- Water temperature has also been related to the occurrence of algal blooms, with unusually **warm water being conducive to blooms**.
- A bloom often results in a color change in the water. Algal blooms can be any color, but the most common ones are red or brown. These blooms are commonly referred to as **red or brown tides**.
- Most algal blooms are not harmful but some produce toxins and do affect fish, birds, marine mammals and humans. The toxins may also make the surrounding air difficult to breathe. These are known as Harmful Algal Blooms (HABs).
- Harmful Algal Blooms are considered an environmental hazard because these events can make people sick when contaminated **shellfish or finfish** are eaten, or when people breathe aerosolized HAB toxins near the beach.
- HAB events adversely affect commercial and recreational fishing, tourism, and valued habitats, creating a significant impact on local economies and the livelihood of coastal residents.

Red Tide

-
- A diagram of an ecosystem showing various organisms and their interactions. The central part is a cross-section of a body of water with a shoreline. The water is divided into three zones: the **Littoral zone** near the shore, the **Limnetic zone** in the middle, and the **Profundic zone** at the bottom. The shoreline is labeled **Terrestrial zone**. Various organisms are shown and labeled: **Biological factors** (a group of organisms), **Water scorpion**, **Living bubble**, **Bufo** (a frog), **Pond snail**, **Dragonfly**, **Plankton** (a group of small organisms), **Perched turtle**, **Shrimp**, **Skud**, **Noton** (a water bug), **Duck**, and **Brookstick fish**. The **Terrestrial zone** is also labeled. The diagram illustrates the complex interactions between these organisms and their environment.

Aquatic Organisms

The aquatic organisms are classified on the basis of their zone of occurrence.

- **Neuston:** These organisms live at the air-water interface e.g. floating plants.
- **Periphyton:** These are organisms which remain attached to stems and leaves of rooted plants or substances emerging above the bottom mud such as sessile algae.
- **Plankton:** Microscopic floating organisms such as algae, diatoms, protozoans and larval forms are called plankton. This group includes both microscopic plants like algae (phytoplankton) and animals like crustaceans and protozoans (zooplankton).
- The locomotory power of the planktons is limited so that their distribution is controlled, largely, by currents in the aquatic ecosystems.
- **Nekton:** This group contains powerful swimmers that can overcome the water currents.
- The animals range in size from the swimming insects to the largest blue whale.
- **Benthos:** The benthic organisms are those found living in the **bottom** of the water mass.

- Sunlight and oxygen are the most important limiting factors of the aquatic ecosystems.

Sunlight

- Sunlight penetration rapidly diminishes as it passes down the column of water.
- The depth to which light penetrates a lake determines the extent of plant distribution.
- Suspended particulate matters such as clay, silt, phytoplankton, etc. make the water turbid.
- Turbidity limits the extent of light penetration and the photosynthetic activity in a significant way.
- Based on light penetration and plant distribution they are classified as photic and aphotic zones.

Photic zone

- Photic (or "euphotic") zone is the portion that extends from the lake surface down to where the light level is 1% of that at the surface. The depth of this zone depends on the transparency of water.
- It is the upper layer of the aquatic ecosystems within which photosynthetic activity is confined. Both photosynthesis and respiration activity takes place.

Aphotic zone

- The lower layers of the aquatic ecosystems, where light penetration and plant growth are restricted forms the aphotic zone (**profundal zone**). Only respiration activity takes place in this zone.
- Aphotic zone extends from the end of the photic zones to bottom of the lake.

Dissolved oxygen

- In fresh water the average concentration of dissolved oxygen is 10 parts per million or 10 ppm by weight. This is 150 times lower than the concentration of oxygen in an equivalent volume of air.
- Oxygen enters the aquatic ecosystem through the air water interface and by the photosynthetic activities of aquatic plants.
- Dissolved oxygen escapes the water body through air-water interface and through respiration of organisms (fish, decomposers, zooplanktons, etc.).
- The amount of dissolved oxygen retained in water is also influenced by

temperature. **Oxygen is less soluble in warm water. Warm water also enhances decomposer activity.** Therefore, increasing the temperature of a water body increases the rate at which oxygen is depleted from water.

- When the dissolved oxygen level falls below 3-5 ppm, many aquatic organisms are likely to die.

Winterkill

- An ice layer on the top of a water body can effectively cut off light. Photosynthesis stops but respiration continues in such water body.
- If the water body is shallow, the oxygen gets depleted and the fish die. This condition is known as winterkill.

Temperature

- Since water temperatures are less subject to change, the aquatic organisms have **narrow temperature tolerance limit**.
- As a result, even small changes in water temperature are a great threat to the survival of aquatic organisms when compared to the changes in air temperatures in the terrestrial organisms.

Lake Ecology

- Any body of standing water, generally large enough in area and depth is known as lake.
- The largest lake in the world is lake Superior in North America. Lake Baikal in Siberia is the deepest. Chilka lake of Orissa is largest lake in India.
- Three main zones can be differentiated in a lake:-
 1. Peripheral zone (littoral zone) with shallow water.
 2. Open water beyond the littoral zone where water is quite deep.
 3. Benthic zone (bottom) or the floor of the lake.

Ageing of Lakes

- Lakes receive their water from surface runoff (sometimes also groundwater discharge) and along with it various chemical substances and mineral matter eroded from the land.
- Over periods spanning millennia, ageing occurs as the lakes accumulate mineral and organic matter and gradually, get filled up.
- The nutrient-enrichment of the lakes promotes the growth of algae, aquatic plants and various fauna. This process is known as natural '**eutrophication**'.
- Similar nutrient enrichment of lakes at an accelerated rate is caused by human activities (discharge of wastewaters or agricultural runoff) and the consequent ageing phenomenon is known as '**cultural eutrophication**'.
- On the basis of their nutrient content, lakes are categorized as **Oligotrophic** (very low nutrients), **Mesotrophic** (moderate nutrients) and **Eutrophic** (highly nutrient rich).
- Vast majority of lakes in India are either eutrophic or mesotrophic because of the nutrients derived from their surroundings or organic wastes entering them.

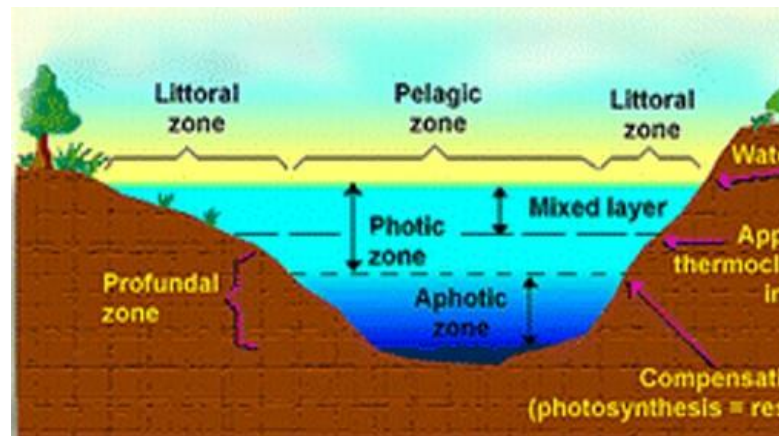
Parameter	Oligotrophic	Eutrophic
Oxygen in the bottom layer	Present	Absent
Depth	Tend to be deeper	Tend to be shallower
Water quality for domestic & industrial uses	Good	Poor
Number of plant and animal species	Many	Fewer

Wetlands - Wetland Ecosystem

Wetland Ecosystem

- Wetlands are areas of marsh or peatland with water that is static or flowing, fresh, brackish or saline, including areas of marine water the depth of which at low tide does not exceed 6 m.
- Wetlands are **transition zones** between terrestrial and aquatic ecosystems. E.g. Mangroves, lake littorals (marginal areas between highest and lowest water level of

the lakes), floodplains (areas lying adjacent to the river channels beyond the natural levees and periodically flooded during high discharge in the river) and other marshy or swampy areas.



- These habitats experience periodic flooding from adjacent deep water habitats and therefore supports plants and animals specifically adapted to such shallow flooding or water logging.
- Waterlogged soil**, adapted plant life (**hydrophytes**) and **hydric soils** (not enough O₂) are the chief characteristics of wetlands.
- India has totally 27,403 wetlands, of which 23,444 are inland wetlands and 3,959 are coastal wetlands.
- Wetlands occupy 18.4% of the country's area of which 70% are under **paddy cultivation**.
- Natural wetlands in India range from high altitude wetlands in Himalayas; flood plains of the major river systems; saline and temporary wetlands of the arid and semi-arid regions; coastal wetlands such as **lagoons, backwaters, estuaries, mangroves**, swamps and coral reefs, and so on.

Distinction from Lakes

- Lakes are generally less important when compared to wetland from the viewpoint of ecosystem and biodiversity conservation.
- There is no clear distinction between lakes and wetlands. Wetlands are **shallow** water bodies whereas lakes can be deep or shallow.

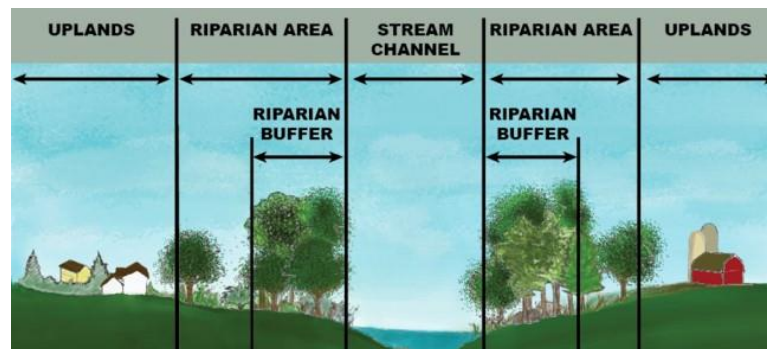
- National Lake Conservation Programme (NLCP) considers lakes as standing water bodies which have a minimum water depth of 3 m, generally cover a water spread of more than ten hectares, and have **no or very little aquatic vegetation**.
- Wetlands (generally less than 3 m deep over most of their area) are usually **rich in nutrients** (derived from surroundings and their sediments) and have **abundant growth of aquatic macrophytes**.
- They support **high densities and diverse fauna**, particularly birds, fish and macro invertebrates, and therefore, have high value for biodiversity conservation.
- Excessive growth of macrophytes (both submerged and free-floating; macrophytes –aquatic plant large enough to be seen by the naked eye) in wetlands affects the water quality adversely and interfere with the utilization of the water body.
- However, marginal aquatic vegetation is desirable as it **checks erosion, serves habitat for wildlife and helps improve water quality**.

Characteristic	Lake	Wetland (shallow lake)
Origin	Largest are due to tectonic forces: Fluvial, Geomorphic, increase in water table, etc.	Mostly Fluvial, Residual lakes
Water turnover	Permanent	Permanent or Temporary
Water level changes	Relatively small	Relatively Large
Thermal stratification	Yes	No
Vertical mixing	Thermally regulated	Wind regulated
Dominant Producer	Phytoplankton	Macrophytes
Food chain	Grazing Pathway	Detritus Pathway
Productivity	Low	High
Trophic status	Oligotrophic	Mostly Eutrophic
Functions-	Less	Significant

Flood control	Significant	
Waste treatment	No	Yes

Importance of Wetlands

- Wetlands are indispensable for the countless benefits or “ecosystem services” that they provide humanity, ranging from freshwater supply, food and building materials, and biodiversity, to flood control, groundwater recharge, and climate change mitigation.
- Wetlands are habitat to aquatic flora and fauna, numerous species of native and migratory birds.
- Wetlands are an important resource for sustainable tourism.
- They carry out water purification, filtration of sediments and nutrients from surface water.
- They help in nutrients recycling, ground water recharging and stabilization of local climate.
- Play an important role in flood mitigation by controlling rate of runoff.
- Buffer (act as riparian buffer) shorelines against erosion and pollutants.
- They act as genetic reservoir for various species of plants (especially rice).



Reasons for depletion

- Excessive pollutants (Industrial effluents, domestic waste, agricultural runoff etc.) are dumped into wetlands beyond the recycling capacity.
- Habitat destruction and deforestation creates ecological imbalance by altering the population of wetland species.
- Conversion of wetlands for agriculture and encroachment by public and mafia.

- Over fishing and fish farming (Aqua culture).
- Overgrazing in marshy soils.
- Removal of sand from beds near seas makes the wetland vulnerable to wave action and tidal bore.

Mitigation

- Demarcation of wetlands using latest technology, proper enforcement of laws and stringent punishments for violators.
- Preventing unsustainable aquaculture and cultivation of shellfish.
- Treating industrial effluents and water from farm lands before discharging into wetlands.
- Utilizing wetlands on a sustainable basis by giving enough time for natural regeneration. Artificial regeneration for quick recovery.
- Afforestation, weed control, preventing invasive species is the key to wetland conservation.
- Preventive measures to stop the introduction of exotic invasive species like water hyacinth.
- Soil conservation measures & afforestation. Preventing grazing in peripherals of wetlands.
- Wildlife conservation, sustainable tourism, eco-tourism and sensitizing local populace.
- Eutrophication abatement by processing nutrient rich discharge into the water body.
- Involving local population in the conservation of wetlands.

Ramsar Sites in India - Ramsar Convention on Wetlands

Measures to Protect Wetlands

- Ramsar Convention on Wetlands
- Ramsar Sites in India
- Wetlands International
- National Wetlands Conservation Programme (NWCP)

Ramsar Convention on Wetlands

- International treaty for “**the conservation and sustainable use of wetlands**”.

- It is also known as the Convention on Wetlands.
- It is named after the city of **Ramsar in Iran**.
- The Convention was signed on 2nd of February, 1971.
- The 2nd of February each year is World Wetlands Day.
- Number of parties to the convention (COP) is 169.
- At the center of the Ramsar philosophy is the “**wise use**” of wetlands.
- **Wise use:** maintenance of ecological character within the context of sustainable development.

Need for Such Convention

- Wetlands are indispensable for the countless benefits or “**ecosystem services**” that they provide humanity, ranging from freshwater supply, food and building materials, and biodiversity, to flood control, groundwater recharge, and climate change mitigation.
- 64% of the world's wetlands have disappeared in the last century.

What is wetland

- The Convention uses a broad definition of wetlands. It includes all lakes and rivers, underground aquifers, swamps and marshes, wet grasslands, peatland, oases, estuaries, deltas and tidal flats, mangroves and other coastal areas, coral reefs, and all human-made sites such as fish ponds, rice paddies, reservoirs and salt pans.

COP

- Conference of the Parties (COP) is the Convention's governing body consisting of all governments that have ratified the treaty.
- Every three years, representatives of the Contracting Parties meet as the Conference of the Contracting Parties (COP)
- COP is the policy-making organ of the Convention which adopts decisions (Resolutions and Recommendations) to administer the work of the Convention.

- The most recent **COP12** was held in **Punta del Este, Uruguay** in 2015. **COP13** will take place in **Dubai, United Arab Emirates**, in 2018.

Under the Convention, the Contracting Parties commit to:

- Work towards the **wise use** of all their wetlands;
- Designate suitable wetlands for the List of Wetlands of International Importance (the "Ramsar List") and ensure their effective management;
- Cooperate internationally on trans boundary wetlands, shared wetland systems and shared species.

Ramsar Site

- At the time of joining the Convention, each Contracting Party undertakes to designate at least one wetland site for inclusion in the List of Wetlands of International Importance.
- The inclusion of a "Ramsar Site" in the List embodies the government's commitment to take the steps necessary to ensure that its ecological character is maintained.
- The List of Wetlands of International Importance included 2,231 Ramsar Sites in March 2016.
- The country with the highest number of Sites is the United Kingdom with 170
- The country with the greatest area of listed wetlands is Bolivia.

The Montreux Record

- The Montreux Record is a register of wetland sites on the List of Wetlands of International Importance where changes in ecological character have occurred, are occurring, or are likely to occur as a result of technological developments, pollution or other human interference. It is maintained as part of the Ramsar List.

Q1. If a wetland of international importance is brought under the 'Montreux Record', what does it imply?

- a. Changes in ecological character have occurred, are occurring or are likely to occur in the wetland as a result of human interference.
- b. The country in which the wetland is located should enact a law to prohibit any human activity within five kilo meters from the edge of the wetland
- c. The survival of the wetland depends on the cultural practices and traditions of certain communities living in its vicinity and therefore the cultural diversity therein should not be destroyed
- d. It is given the status of 'World Heritage Site'

Answer: a)

International Organization Partners

- The Ramsar Convention works closely with six other organizations known as International Organization Partners (IOPs). These are:
 1. **Birdlife International**
 2. **International Union for Conservation of Nature (IUCN)**
 3. **International Water Management Institute (IWMI)**
 4. **Wetlands International**
 5. **WWF**
 6. **International Wildfowl & Wetlands Trust (WWT)**

Other Partners

- Convention on Biological Diversity (CBD)
- Convention to Combat Desertification (UNCCD),
- Convention on the Conservation of Migratory Species of Wild Animals
- Convention on Migratory Species (CMS),
- World Heritage Convention (WHC) and
- Convention on International Trade in Endangered Species (CITES).

Project funding is done by various groups like multilateral development banks, bilateral donors, UN agencies such as UNEP, UNDP, Non-governmental organizations etc.

Transboundary Ramsar Sites

- Contracting Parties are designating their new and existing Ramsar Sites as Transboundary Ramsar Sites.
- These are ecologically coherent, shared wetlands extending across national borders, which are managed collaboratively.

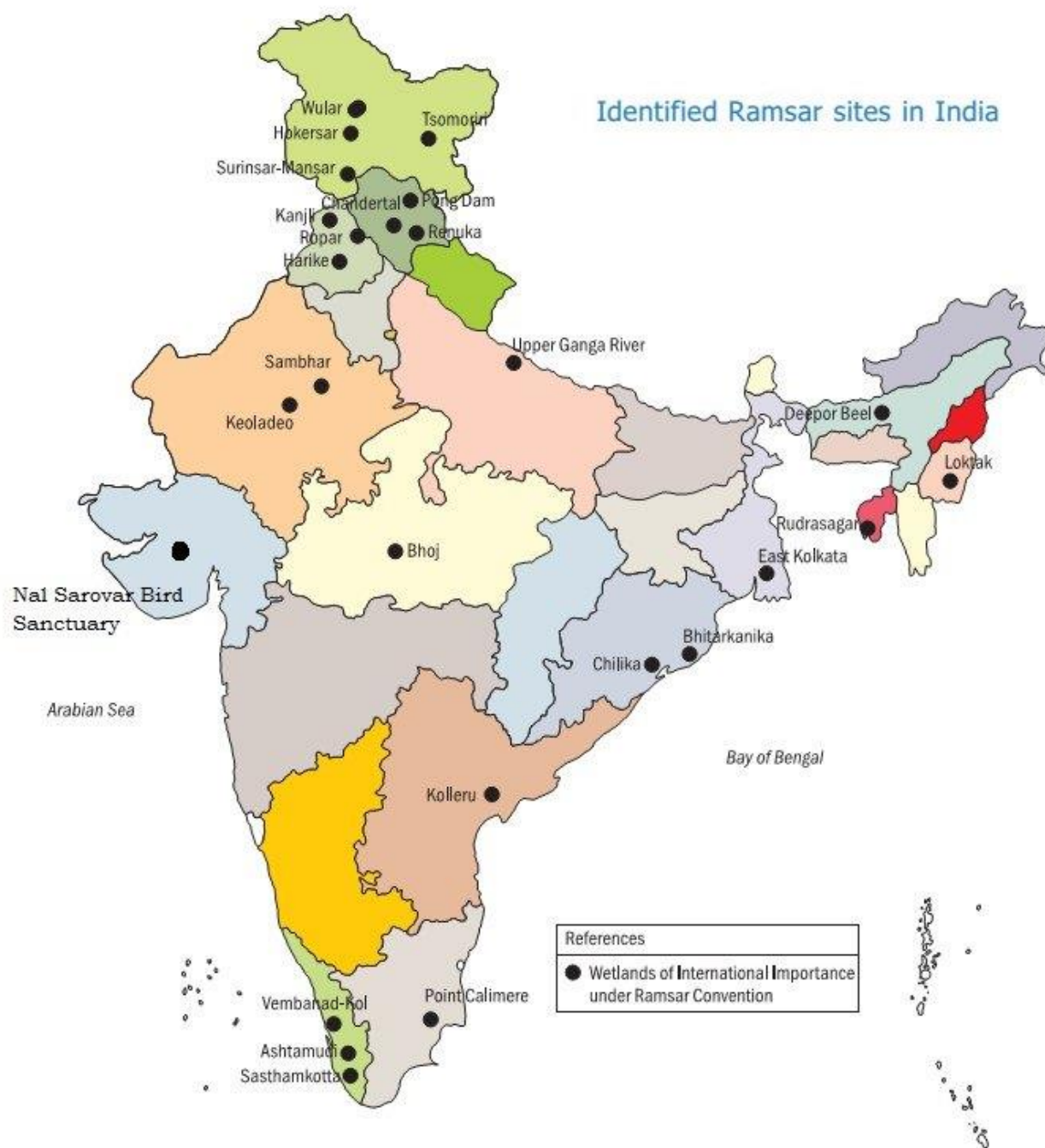
Criteria for Identification of Wetlands under Ramsar Convention

- If it contains a representative, rare, or unique example of a natural or near-natural wetland type.
- if it supports vulnerable, endangered, or critically endangered species; or threatened ecological communities.
- If it supports populations of plant and/or animal species important for maintaining the biological diversity of a particular biogeographic region.

- If it supports plant and/or animal species at a critical stage in their life cycles, or provides refuge during adverse conditions.
- If it regularly supports 20,000 or more water birds.
- If it regularly supports 1% of the individuals in a population of one species or subspecies of water birds.
- If it supports a significant proportion of indigenous fish subspecies
- If it is an important source of food for fishes, spawning ground, nursery and/or migration path.
- If it is an important source of food and water resource, increased possibilities for recreation and eco-tourism, etc.

Ramsar Sites in India

- At present, there are 26 Ramsar Sites in India.



	Wetland	State and Area in km ²	
1	Ashtamudi Wetland	Kerala 614	- A natural backwater in Kollam district. - River Kallada and Pallichal drains into it. - It forms an estuary with Sea at Neendakara which is a famous fishing harbour in Kerala. - National Waterway 3 passes through it.
2	Bhitarkanika Mangroves	Orissa 650	- The core area of Bhitarkanika Wildlife Sanctuary was declared Bhitarkanika National Park. - Bhitarkanika Wildlife Sanctuary includes Gahirmatha Marine Wildlife Sanctuary.

			- Bhitarkanika Mangroves, a part of Bhitarkanika Wildlife Sanctuary were designated a Ramsar Wetland of International Importance in 2002. - It is famous for its salt water crocodiles and Olive ridley sea turtle.
3	Bhoj Wetland	Madhya Pradesh 32	- The Bhoj Wetland consists of two lakes located in the city of Bhopal. - The two lakes are the Bhojtal and the Lower Lake. - It is a manmade reservoir. - A total of more than 20,000 birds are observed annually. - The largest bird of India, the sarus crane (<i>Grus antigone</i>) is found here.
4	Chandra Taal	Himachal Pradesh 0.49	
5	Chilika Lake	Orissa 1165	- Chilka Lake is a brackish water lagoon at the mouth of the Daya River - It is spread over the Puri, Khurda and Ganjam districts of Odisha state on the east coast of India. - It is the largest coastal lagoon in India and the second largest lagoon in the world. - The lagoon hosts - over 160 species of birds in the peak migratory season. Birds from as far as the Caspian Sea, Lake Baikal, Aral Sea and other remote parts of Russia, Kirghiz steppes of Mongolia, Central and southeast Asia, Ladakh and Himalayas come here. - In 1981, Chilika Lake was designated the first Indian wetland of international importance under the Ramsar Convention. - Birds: White bellied sea eagles, greylag geese, purple moorhen, jacana, flamingos, egrets, gray and purple herons, Indian roller, storks, white ibis, spoonbills, brahminy ducks, shovellers, pintails, and more. - Nalbana Island is the core area of the Ramsar designated wetlands of Chilika Lake. - Nalbana was notified in 1987 and declared a bird sanctuary in 1973 under the Wildlife Protection Act. - The Irrawaddy dolphin (<i>Orcaella brevirostris</i>) is the flagship species of Chilika lake. - Chilka is home to the only known

			population of Irrawaddy dolphins in India. It is classified as critically endangered, in five of the six other places it is known to live.
6	Deepor Beel	Assam 40	
7	East Calcutta Wetlands	West Bengal 125	- The wetland forms an urban facility for treating the city's waste water - The wetland provides about 150 tons of fresh vegetables daily, as well as some 10,500 tons of table fish per year.
8	Harike Wetland	Punjab 41	An important site for breeding, wintering and staging birds, supporting over 200,000 Anatidae (ducks, geese, swans, etc.) during migration.
9	Hokera Wetland	Jammu and Kashmir 13.75	- Hokera wetland is only 10 km away from Srinagar. 68 waterfowl species like Large Egret, Great Crested Grebe, Little Cormorant, Common Shelduck, Tufted Duck and endangered White-eyed Pochard, coming from Siberia, China, Central Asia, and Northern Europe.
10	Kanjli Wetland	Punjab 1.83	
11	Keoladeo National Park	Rajasthan 28.73	- A complex of ten artificial, seasonal lagoons, varying in size, situated in a densely populated region. - Placed on the Montreux Record in 1990 due to "water shortage and an unbalanced grazing regime". - Additionally, the invasive growth of the grass <i>Paspalum distichum</i> has changed the ecological character of large areas of the site, reducing its suitability for certain waterbird species, notably the Siberian crane.
12	Kolleru Lake	Andhra Pradesh 901	Bird sanctuary
13	Loktak Lake	Manipur 266	- Loktak Lake is the largest freshwater lake in the north-eastern region. - Keibul Lamjao the only floating national park in the world floats over it.

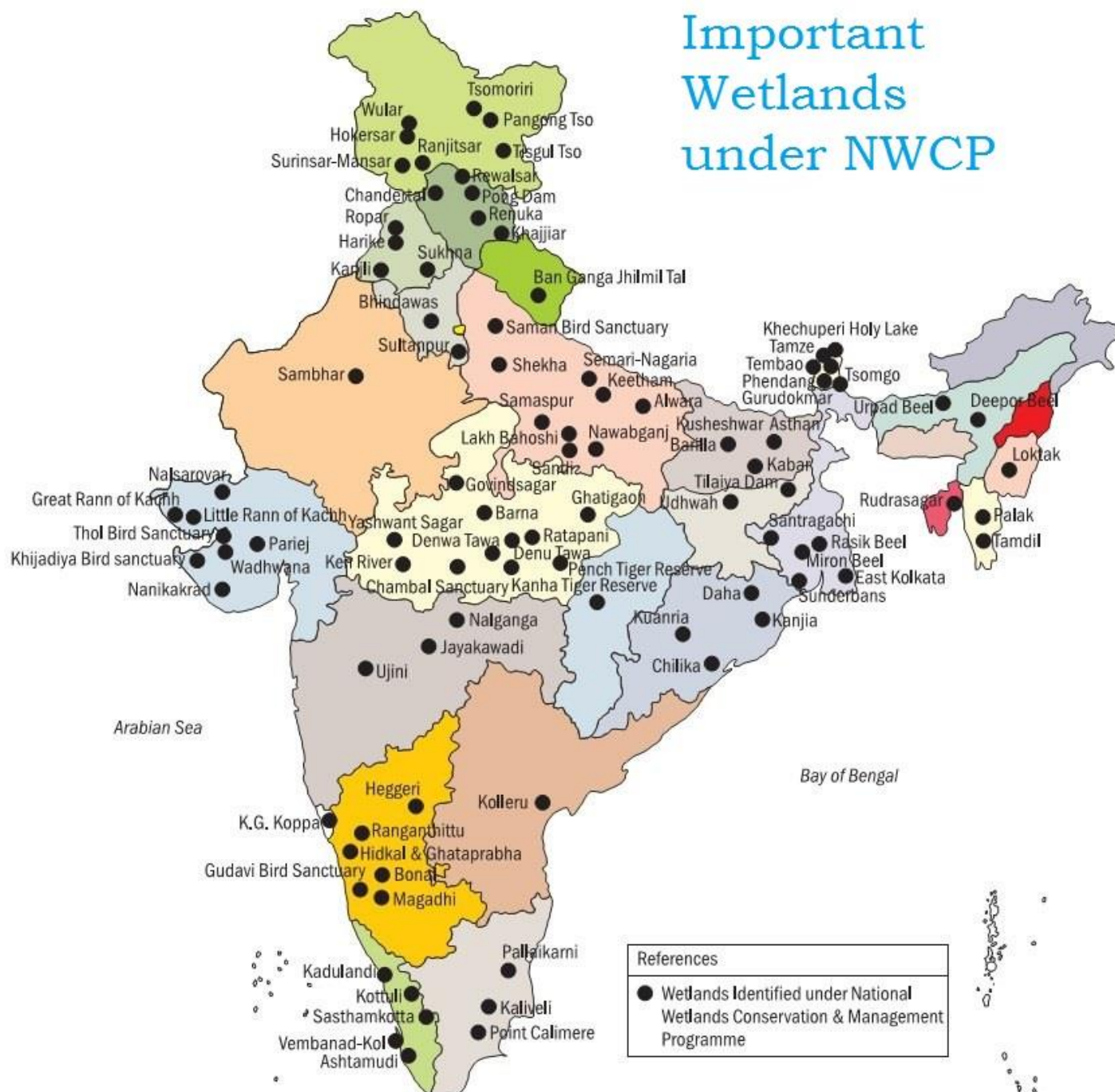
14	Nalsarovar Bird Sanctuary [Declared as Ramsar site in 2012. Most recent Ramsar Site in India]	Gujarat 123	- A natural freshwater lake (a relict sea) that is the largest natural wetland in the Thar Desert Biogeographic Province and represents a dynamic environment with salinity and depth varying depending on rainfall. - Besides a few mammalian species including the endangered wild ass and the black buck, its migratory bird population includes rosy pelicans, flamingoes, white storks, brahminy ducks and herons. Thousands of migratory waterfowl flock to this sanctuary just after the Indian monsoon season. - The wetland is also a lifeline for a satellite population of the endangered Indian Wild Ass (<i>Equus hemionus khur</i>) which uses this area in the dry season.
15	Point Calimere Wildlife and Bird Sanctuary	Tamil Nadu 385	- One of the last remnants of Dry Evergreen Forests. - Habitat: Dry Evergreen Forests, Mangrove & Wetlands. - Winter Migrants : Spoon Billed Sandpiper, Greater Flamingos.
16	Pong Dam Lake	Himachal Pradesh 156	A water storage reservoir created in 1975 on the Beas River in the low foothills of the Himalaya on the northern edge of the Indo-Gangetic plain.
17	Renuka Lake	Himachal Pradesh 0.2	- A natural wetland with freshwater springs and inland subterranean karst formations, fed by a small stream flowing from the lower Himalayan out to the Giri river. - There are 103 species of birds of which 66 are residents, e.g. Crimson-breasted barbet, Mayna, Bulbul, Pheasants, Egrets, Herons, Mallards and Lapwing. - Among ungulates Sambhar, Barking deer and Ghorals are also abundant in the area. - The lake has high religious significance and is named after the mother of Hindu sage Parshuram, and is thus visited by thousands of pilgrims and tourists.
18	Ropar_Wetland	Punjab 13.65	A humanmade wetland of lake and river formed by the 1952 construction of a barrage for diversion of water from the Sutlej River for drinking and irrigation supplies.

			- The site is an important breeding place for the nationally protected Smooth Indian Otter, Hog Deer, Sambar, and several reptiles, and the endangered Indian Pangolin (<i>Manis crassicaudata</i>) is thought to be present. - Some 35 species of fish play an important role in the food chain, and about 150 species of local and migratory birds are supported.
19	Rudrasagar Lake	Tripura 2.4	
20	Sambhar Lake	Rajasthan 240	- The Sambhar Salt Lake, India's largest inland salt lake. - Sambhar has been designated as a Ramsar site (recognized wetland of international importance) because the wetland is a key wintering area for tens of thousands of flamingos and other birds that migrate from northern Asia. - The specialized algae and bacteria growing in the lake provide striking water colours and support the lake ecology that, in turn, sustains the migrating waterfowl. There is other wildlife in the nearby forests, where Nilgai move freely along with deer and foxes.
21	Sasthamkotta Lake	Kerala 3.73	It is the largest freshwater lake in Kerala, situated in Kollam district.
22	Surinsar-Mansar Lakes	Jammu and Kashmir 3.5	
23	Tsomoriri	Jammu and Kashmir 120	- A freshwater to brackish lake lying at 4,595m above sea level. The site is said to represent the only breeding ground outside of China for one of the most endangered cranes, the Black-necked crane (<i>Grus nigricollis</i>), and the only breeding ground for Bar-headed geese in India. - The Great Tibetan Sheep or Argali (<i>Ovis ammon hodgsoni</i>) and Tibetan Wild Ass (<i>Equus kiang</i>) are endemic to the Tibetan plateau, of which the Changthang is the westernmost part.
24	Upper Ganga River (Brijghat to Narora Stretch)	Uttar Pradesh 266	The river provides habitat for IUCN Red listed Ganges River Dolphin, Gharial, Crocodile, 6 species of turtles, otters, 82

			species of fish and more than hundred species of birds.
25	Vembanad-Kol Wetland	Kerala 1512	• Largest lake of Kerala. • Famous tourist locations like Alappuzha and Kumarakom, known for house boats falls here. • River mouths of Pamba-Achenkovil rivers in Vembanad forms one of the unique wetland topography of Kerala, the Kuttanad. It is below sea level and is famous for exotic fish varieties and Paddy fields that are below sea level.
26	Wular Lake	Jammu and Kashmir 189	• The largest freshwater lake in India.

Important Wetlands under NWCP

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Wetlands International

- Wetlands International is a global organization (NGO) that works to sustain and restore wetlands and their resources for people and biodiversity.
- It is an independent, not-for-profit organization, supported by government and NGO membership from around the world.
- Wetlands International's work ranges from research, advocacy and engagement with

governments, corporate and international policy fora and conventions.

Q1. With reference to a conservation organization called 'Wetlands International', which of the following statements is/are correct?

- It is an intergovernmental organization formed by the countries which are signatories to Ramsar Convention.

2. It works at the field level to develop and mobilize knowledge, and use the practical experience to advocate for better policies.

Select the correct answer using the code given below.

- 1 only
- 2 only
- Both 1 and 2
- Neither 1 nor 2

Answer: b) 2 only. Wetlands International is an NGO

National Wetlands Conservation Programme (NWCP)

- NWCP was implemented in the year 1985-86.
- Under the programme, 115 wetlands have been identified by the MoEF which require urgent conservation and management interventions.
- Criteria for identification of wetlands of national importance under NWCP are same as those prescribed under the 'Ramsar Convention on Wetlands'.

Aim

- Conservation of wetlands to prevent their further degradation and ensuring their wise use for the benefit of local communities and overall conservation of biodiversity.

Objectives

- to lay down policy guidelines for conservation and management of wetlands.
- to provide financial assistance for undertaking intensive conservation measures.
- to monitor implementation of the programme.
- to prepare an inventory of Indian wetlands.

Federal management

- The **Central Government** is responsible for overall coordination of wetland conservation programmes.

- It also provides guidelines, financial & technical assistance to state govt.
- Since the land resources belong to them, the State Governments/UT Administration are responsible for **management of wetlands**.

Mangrove and Estuarine Ecosystems

Estuarine Ecosystem

- An estuary is a place where a river or a stream opens into the sea (mouth of the river).
- It is a partially enclosed coastal area of brackish water (salinity varies between 0-35 ppt) with one or more rivers or streams flowing into it, and with a free connection to the open sea.
- At the estuaries, fresh water carrying fertile silt and runoff from the land mixes with the salty sea water.
- Estuaries form a **transition zone (ecotone)** between river environments and maritime environments.
- Examples of estuaries are **river mouths, coastal bays, tidal marshes, lagoons and deltas**.



- Estuaries are formed due to rise in sea level, movement of sand and sandbars, glacial processes and tectonic processes.
- All the plants and animals in the estuaries are subjected to variations in salinity to which they are adapted (osmoregulation).
- Estuaries are greatly influenced by tidal action. They are periodically washed by sea water once or twice a day based on the number of tides.

- In some narrow estuaries, **tidal bores** are significant. Tidal bores cause great damage to the estuarine ecology.

Importance of Estuaries

- They are the **most productive** (more productive than wetlands) water bodies in the world because of the mixing of fresh water and saline water zone where marine organisms of both the ecosystems meet.

Ecotone regions (transitional zones) like **mangroves, wetlands, estuaries, grasslands** etc. have far greater productivity compared to natural ecosystems like forest ecosystem, ocean ecosystem, pond ecosystem, riverine ecosystem, desert ecosystem etc.. This is because of the wide ranging species from the adjacent ecosystems being present in the ecotone.

- Also, an estuary has **very little wave action**, so it provides a calm refuge from the open sea and hence becomes ideal for the survival of numerous aquatic species.
- Estuaries are most heavily populated areas throughout the world, with about 60% of the world's population living along estuaries and the coast.
- The vast mangrove forests on the seaward side of an estuary act as barrier for the coastal habitat to check the wind speed during cyclones and high velocity landward winds.
- Mangroves act as a filter trapping suspended mud and sand carried by rivers which leads to delta formations around estuaries.
- Precipitation of clay and alluvium particles in estuarine region is high because of the exposure to saline water (saline water precipitates fine alluvium).
- Estuaries store and recycle Nutrients, traps sediment and forms a buffer between coastal catchments and the marine environment.
- They also absorb, trap and detoxify pollutants, acting as a natural water filter.
- Estuaries with their wetlands, creeks, lagoons, mangroves and sea-grass beds are rich in natural resources including fisheries.
- They are deep and well protected from marine transgressions and hence they are

ideal locations for the construction of **ports and harbours**.

- The banks of estuarine channels form a favored location for human settlements, which use the estuaries for fishing and commerce, but nowadays also for dumping civic and industrial waste.

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Differences between Lagoon and Estuary

- A lagoon is a stretch of salt water separated from the sea by a low sandbank or coral reef.
- Backwaters in Kerala are mostly lagoons where sea water flows inwards through a small inlet that is open towards the sea.
- The main difference between lagoons and estuaries is in the flow dynamics of the water bodies: in estuaries, the water flows fast and strong, while in lagoons the water is more shallow and flows sluggishly.
- Estuaries are usually deeper than lagoons. Also, lagoons mostly don't have any fresh water source while the estuaries have at least one. Lagoons more saline than estuaries.
- Lagoons are formed due to fall in sea levels (coastline of emergence. E.g. Kerala Coast) whereas estuaries are mostly formed due to rise in sea levels (coastline of submergence. E.g. Konkan coast)

Estuarine Vegetation

- Only certain types of plants and animals specially adapted to the "brackish" estuarine waters flourish in the estuaries.
- Factors influencing the growth and distribution of organism in an estuary are its salinity and the amount of flooding.
- Estuaries support diverse habitats, such as mangroves, salt marshes, sea-grass, mudflats etc.
- Estuaries are very dynamic and productive ecosystems since the river flow, tidal range and sediment distribution is continuously changing in them.
- In general the phytoplanktons of estuaries are diatoms, dinoflagellates, green algae, blue-green algae.
- Towards the sea coast of the estuaries there are large algae and sea grasses. Near

the mouth of the rivers and deltas there are mangrove forests.

- Estuaries are homes to all kind of terrestrial or land-based plants and animals, such as wood storks, pelicans, coniferous and deciduous trees and butterflies.
- Estuaries are also homes to unique aquatic plants and animals, such as sea turtles, sea lions, sea catfish, saltworts, eelgrass, saltgrasses, cordgrasses, sea grass, sedge, bulrush etc..

India Estuarine Ecosystem

- The Country has 14 major, 44 medium and 162 minor rivers drains into the sea through various estuaries.
- Major estuaries occur in the Bay of Bengal. Many estuaries are locations of some of the major seaports.
- Most of the India's major estuaries occur on the east coast. In contrast, the estuaries on the west coast are **smaller** (In environmental studies, deltas are considered as sub sections of estuaries).
- Two typical examples of estuaries on the west coast are the **Mandovi** and **Zuari estuaries**.

Issues of Indian Estuarine Ecosystem

- Modifications of the estuarine catchments results in changes in water flow in various estuaries, either far in excess or much lower than required (Eg: Hooghly, Narmada, Krishna, Godavari, Pulicat etc.)
- Pollution through industries and combined city sewage discharge.
- Recreational boating and fishing.
- Navigation, dredging and shipping (e.g. Hooghly)
- Expansion of urban and rural settlements, mining & industries, agriculture and dumping of solid wastes
- Over exploitation of target fish stock due to increased demand.
- Reclaiming the fringed areas for intensive aquaculture in pens.
- Obstructing the migratory routes of fish and prawn recruitment (e.g., Chilka, Pulicat)
- Polluting the environment through feeding of stocked fish and prawn in pens (Chilka)

- Destruction of biodiversity through prawn seed collection and operation of small-meshed nets (e.g., Hooghly, Chilka, Pulicat)
- Submergence of catchment areas due to rise in water level.

Mangroves

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- Mangroves represent a characteristic littoral (near the sea shore) forest ecosystem.
- These are **mostly evergreen** forests that grow in sheltered low lying coasts, estuaries, mudflats, tidal creeks backwaters (coastal waters held back on land), marshes and lagoons of tropical and subtropical regions.
- Mangroves grow below the high water level of spring tides. The best locations are where abundant silt is brought down by rivers or on the backshore of accreting sandy beaches.
- Mangroves are highly productive ecosystems and the trees may vary in height from 8 to 20 m. They protect the shoreline from the effect of cyclones and tsunamis.
- They are breeding and spawning ground for many commercially important fishes.
- Since mangroves are located between the land and sea they represent the best example of ecotone.
- Mangroves are shrubs or small trees that grow in coastal saline or brackish water.
- Mangroves are salt tolerant trees, also called **halophytes**, and are adapted to life in harsh coastal conditions.
- Mangrove vegetation facilitates **more water loss**. Leaves are thick and contain salt secreting glands. Some block absorption of salt at their roots itself.
- They contain a complex salt filtration system and complex root system to cope with salt water immersion and wave action.
- They are adapted to the **low oxygen (anoxic)** conditions of waterlogged mud.
- They produces **pneumatophores (blind roots)** to overcome respiration problem in the anaerobic soil conditions.
- Mangroves occur worldwide in the tropics and subtropics, mainly between latitudes 25° N and 25° S.
- They require high solar radiation to filter saline water through their roots. This

explains why mangroves are confined to only tropical and sub-tropical coastal waters.

- Mangroves occur in a variety of configurations. Some species (e.g. **Rhizophora**) send arching **prop roots** down into the water. While other (e.g. **Avicennia**) send **vertical "Pneumatophores"** or air roots up from the mud.



Prop roots and pneumatophores



Stilt roots

- Adventitious roots which emerged from the main trunk of a tree above ground level are called **stilt roots**.
- Mangroves exhibit **Viviparity** mode of reproduction. i.e. seeds germinate in the tree itself (before falling to the ground).
- This is an adaptive mechanism to overcome the problem of germination in saline water.

Mangroves in India



- The mangroves of Sundarbans are the largest single block of tidal holophytic mangroves of the world.
- The major species of this dense mangrove forest include *Herritiera fomes*, *Rhizophora* spp., *Bruguiera* spp., *Ceriops decandra*, *Sonneratia* spp. and *Avicennia* spp., *Nypa fruticans* are found along the creeks.
- This mangrove forest is famous for the Royal Bengal Tiger and crocodiles. Mangrove areas here are being cleared for agricultural use.
- The mangroves of Bhitarkanika (Orissa), which is the second largest in the Indian sub-continent, harbour high concentration of typical mangrove species and high genetic diversity.
- Mangrove swamps occur in profusion in the intertidal mudflats on both side of the creeks in the Godavari-Krishna deltaic regions of Andhra Pradesh.
- Mangroves of Pichavaram and Vedaranyam are degraded mainly due to construction of aquaculture ponds and salt pans.
- On the west coast of India, mangroves, mostly scrubby and degraded occur along the intertidal region of estuaries and

creeks in Maharashtra, Goa and Karnataka.

- The mangrove vegetation in the coastal zone of Kerala is very sparse and thin.
- In Gujarat (north-west coast) mangroves *Avicennia marina*, *Avicennia officinalis* and *Rhizophora mucronata* are found mainly in Gulf of Kachchh and the Kori creek.
- Mangroves are of scrubby type with stunted growth, forming narrow, discontinuous patches on soft clayey mud.
- The condition of the mangroves is improving especially in the Kori creek region, which is a paleodelta of the Indus river.
- In size, mangroves range from bushy stands of dwarf mangroves found in Gulf of Kuchchh, to taller stands found in the Sunderbans.
- On the Andaman & Nicobar Islands, the small tidal estuaries, neritic inlets and the lagoons support a dense and diverse undisturbed mangrove flora.

Importance of Mangroves

- Mangrove plants have (additional) special roots such as prop roots, pneumatophores which help to impede water flow and thereby enhance the deposition of sediment in areas (where it is already occurring), stabilize the coastal shores, provide breeding ground for fishes.
- Mangroves moderate monsoonal tidal floods and reduce inundation of coastal lowlands.
- They prevent coastal soil erosion.
- They protect coastal lands from tsunami, hurricanes and floods.
- Mangroves enhance natural recycling of nutrients.
- Mangrove supports numerous flora, avifauna and wild life.
- Provide a safe and favorable environment for breeding, spawning, rearing of several fishes.
- They supply woods, fire wood, medicinal plants and edible plants to local people.
- They provide numerous employment opportunities to local communities and augment their livelihood.

Threats

- They are destroyed for conversion of area for agricultural purpose, fuel, fodder and, salinization, mining, oil spills, aquaculture (shrimp farming), use of chemical pesticides & fertilizers, industrial purposes.

Q2. Which one of the following is the correct sequence of ecosystems in the order of decreasing productivity?

- Oceans, lakes, grasslands, mangroves
- Mangroves, oceans, grasslands, lakes
- Mangroves, grasslands, lakes, oceans
- Oceans, mangroves, lakes, grasslands

Explanation:

Productivity = production/unit area/unit time

Production/unit area depends on number and diversity of producers.

From the topics discussed in this post, we can say that transitional zones (Ecotones) have very high productivity compared to normal ecosystems. [Tropical Rainforests is an exception as it has productivity comparable to wetlands because of its rich diversity of primary producers]

In the options, mangroves are transitional ecosystems while the rest are normal ecosystems.

Remember: Grasslands are not transitional all the time. E.g. Steppe. Non transitional grasslands have very low productivity because of very limited diversity of primary producers. [Grasslands become transitional only when they are narrow]

So the order of decreasing productivity will be like

Mangroves ,..... , ,

Oceans = very deep and hence productivity is limited to surface only (Below in the aphotic zone productivity is negligible. Aphotic zone in oceans is few kilometers).

Also surface water in oceans are very poor in nutrients. Nutrient rich cold water flows as a

sub-surface flow lying in the aphotic zone. Sunlight and nutrients are far apart and hence primary productivity is very low except in regions where there is upwelling of nutrient rich cold water [Cold and Warm current mixing zones].

So open ocean ecosystem has the least productivity. [Desert ecosystem also has very low productivity, lesser than oceans].

So, the answer will look like Mangroves,, Oceans.

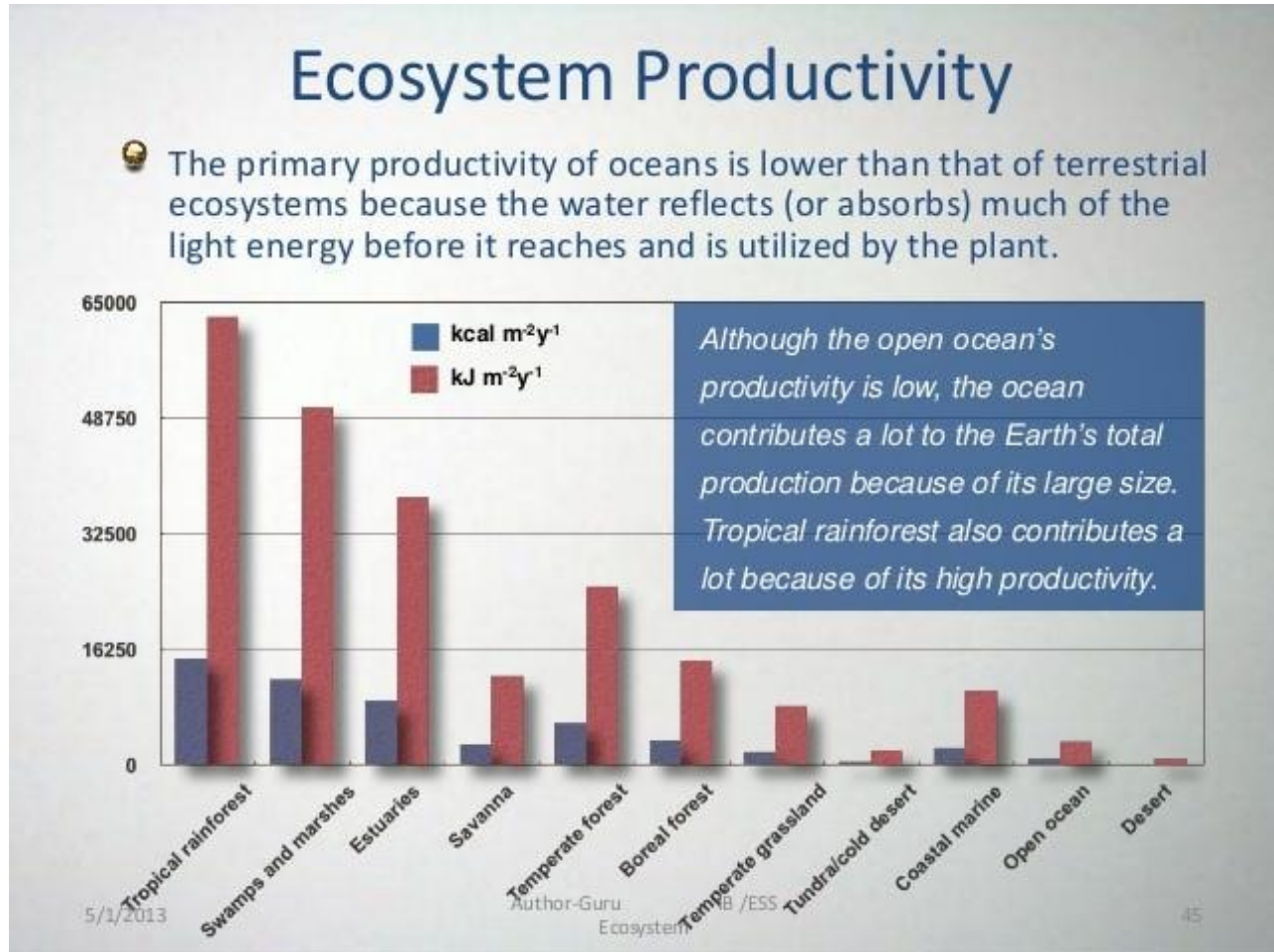
The only such option is c)

Answer: c) Mangroves, grasslands, lakes, oceans.

Lakes just like oceans have low productivity. But due to some plants in photic zone, lakes have productivity slightly greater than that of oceans.

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Environmental Degradation & Himalayan Environment

Environmental Degradation

- Environmental degradation is making the environment unfit or less suitable for the survival of different life forms thereby causing immense ecological damage.
- Population explosion, urbanization and associated increase in human needs and

comforts have resulted in rapid industrialization.

- Rapid industrialization in turn led to overexploitation of natural resources.
- The consequences of such exploitation are evident in the form of soil erosion, desertification, loss of biodiversity, pollution of land, air and water bodies.

Major Causes of Environmental Degradation

- **Air/water/soil pollution:** Gaseous pollutants: Oxides of carbon, nitrogen and sulphur. [All types of pollution are explained in detail in future posts]
- Particulate matter: Fine metal dust, fly ash, soot, cotton dust and radioactive substances.
- Burning of plastics: Emit poly chlorinated biphenyles (PCBs).
- Industrial and domestic effluents: Pathogens, heavy metals in water bodies and soil.
- Industrial catastrophes: Accidental release of some poisonous gases like **phosgene** (COCl₂) and **methyl isocyanate** (as it happened in Bhopal) were fatal.
- Secondary air pollutants formed from complex reactions between primary pollutants, such as smog and acid rain.
- **Habitat fragmentation and destruction:** Industrialization and associated infrastructure development has led to the destruction of habitats of many species.
- **Hazardous waste/Toxic substances:** Use of various type of chemicals today have serious health implications. Incidence of cancer, genetic mutations and damage to nervous, immune and hormonal systems.
- **Invasive populations:** Introduction of new alien species or non-native species reduces the populations growth of native species.
- **Stress due to over harvesting:** Overgrazing by livestock results in soil erosion and loss of productivity. Similarly over harvesting of edible fishes reduces population and may become completely extinct if the over fishing continues for long.
- **Global warming:** Increasing use of fossil fuels is a leading cause of increasing levels of CO₂ and other greenhouse gases in the atmosphere.
- Over exploitation of natural resources: Rapidly growing population results in over exploitation of resources. Over exploitation and introduction of a new or genetically modified species reduce the productivity of natural ecosystems.
- Deforestation, over grazing, intensive cultivation, over irrigation etc. results in the loss of top soil and fertility of the land. Prolonged degradation of land leads to desertification.

Consequences of Environmental Degradation

- **Increased sensitivity to diseases:** Cultivated species of plants, fishes and other domesticated animals have become increasingly sensitive to pest and diseases.
- **Genetic resistance:** An increased use of insecticides, pesticides and antibiotics has speeded up directional natural selection and caused genetic resistance in pathogens.
- **Effect on nutrient recycling:** Use of fertilizers in agricultural fields interferes with the natural biogeochemical cycles.
- **Loss of biodiversity:** Habitat destruction and habitat fragmentation are threatening the survival of many endangered species.
- **Climate change/global warming:** Air pollution is making the plant inhospitable due to events like global warming, acid rain, ocean acidification etc.
- **Melting of ice caps and glaciers/Sea level change:** Global warming is posing serious threat to Himalayan and polar ecosystems by melting snow at a faster rate. Rise in sea level in future is a serious threat to costal landforms.
- **Depletion of water resources:** Water pollution has created tremendous scarcity of quality water resources.
- **Desertification:** Lot of arable land is converted to waste land due to erosion, salinity and loss of fertility.
- **Change in weather patterns:** The incidence of extreme weather events and change in weather patterns have become common due to environmental degradation.

Human Modified Ecosystems and Environmental Degradation

- Some examples of human modified ecosystems are: agro-ecosystems, plantation forests, rural and urban ecosystems, aquaculture etc.

Characteristics of human modified ecosystems

- Highly unstable, unsustainable and have a devastating effect on the environment.

- Highly simplified with a very low species diversity and poor nutrient cycling.
- Food chains are simple, small and undergo frequent changes.
- Depend on human (anthropogenic) support for survival.
- Species are highly susceptible to epidemic diseases.
- Highly subjected to soil erosion and soil degradation.
- Regeneration and recovery of the environment is very slow. Some adverse effects are irreversible for a very long time.

Agro ecosystems

- Agro ecosystems are large areas where commercial crops are cultivated.
- Crops are mostly cultivated as **monoculture** (growing only one type of crop) on the entire field.

Characteristics of agro-ecosystems

- Attract weeds and susceptible to plant diseases.
- Soil are poor, deficient in nutrients, require supplement of chemical or fertilizers.
- Need artificial irrigation and water management.

Disadvantages of agro-ecosystem

- Large scale monoculture of agricultural crops results in severe loss of native biodiversity including genetic diversity of crop plants.
- High yielding varieties of crop plants are more susceptible to disease. E.g. smut of sugarcane, maize and sorghum and rust of wheat and bajra.
- Protecting crops from pests and diseases requires large scale use of pesticides and chemicals which pollute the environment.
- Deplete ground water in many areas due to well irrigation.
- Run off water from agricultural field laden with fertilizers and pesticides pollute river, lakes and ponds (Eutrophication).

Plantation Forest

- It is a manmade ecosystem consisting of individuals of a particular tree species.

- The aim is to grow fast growing trees which are commercially valuable. E.g. Palm, rubber, eucalyptus etc.
- Trees planted on barren land, private land, village panchayat land, roadsides, canal banks, along with railway line and on land not suitable for agriculture.

Characteristics of plantation forests

- Plantation forests are generally monoculture, like oil palm plantation, rubber plantation, coffee plantation, **Jatropha curcare** (biodiesel).
- Plantation forests have trees of approximately same age.
- Plantation forests are highly susceptible to pathogens.

Economic importance

- Tree plantation are raised for fruits, oil, rubber, coffee, timber, fire wood, pulp wood for making rayon and paper industries.
- Trees are also planted to serve as wind breaks or shelter belts.
- Tree plantations are also raised for controlling soil erosion and for increasing soil fertility.

Aquaculture

- Aquaculture is the artificial cultivation of aquatic plants or animals.
- It is primarily carried out for cultivating certain commercially important edible species of fresh and marine water fishes, molluscs, crustaceans and aquatic plants.
- Fisheries include the extraction of food from the sea and the fresh water whereas aquaculture is rearing of the aquatic organisms in artificially made water bodies e.g. culture of fish like carps, tilapia (also known as aquatic chicken.).

There are two types of aquaculture

- **Fish farming** is cultivation of fish in a controlled environment often a coastal or inland pond, lake, reservoir or rice field (paddy) and harvesting when they reach the desired size.

- **Fish ranching** is a practice of keeping fishes in captivity for the first few years in floating cages in coastal lagoons and releasing them from captivity into water bodies. Adults are harvested when they return for spawning to the lagoons. E.g. Salmon and Hilsa which migrate to rivers to spawn are cultivated by fish ranching method.

Merits of aquaculture

- Ecological efficiency is high. 2 kg. of grains are required to add 1 kg live weight.
- High yield in small volume of water.
- Improved qualities of fish obtained by selection and breeding and genetic engineering.
- Aquaculture reduces over harvesting of fisheries.
- High profit.

Demerits of aquaculture

- Large inputs of feed, water and land are required.
- Loss of native aquatic biodiversity. As it replaced by monoculture of a commercially important fish species.
- Produces large amounts of fish wastes that pollute water bodies.
- Destroys mangrove forests or coastal vegetation.
- Aquaculture fishes are very sensitive to pesticide runoff from croplands.
- In aquaculture ponds high population density is maintained that makes them highly vulnerable to diseases leading to total collapse of the crop.
- Aquaculture tanks or reservoirs are often get contaminated after a few years.

Dams, Reservoirs And Diversions

- A dam is a structure built in order to store river or tidal water. Dams, reservoirs and diversions capture and store runoff water and release it as needed. They are used for:
- controlling or moderating floods,
- producing hydroelectric power, and
- supply water for irrigation, industry and other uses to rural, suburban and urban areas. Support recreational activities such as swimming and boating.

Advantages of dams

- Water released from dams is used to generate electricity.
- Hydroelectricity n reduces dependence on coal and hence reduces CO2 emission.
- Reduces downstream flooding.
- Reduces river silting below the dam.
- Supply irrigation water for croplands.

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Disadvantages of dams

- Permanently submerge large areas of forests and crop lands.
- Displace large number of native people.
- Increase water pollution on account of reduced water flow.
- Reduce nutrients replenishment of downstream flood plains.
- Disrupt spawning and migration of some fish species.
- High costs and long gestation period.

Large reservoirs increase the risk of earthquakes (Killari Reservoir is speculated to have Induced Latur Earthquake of 1993 that has killed more than 9,000 people).

Environmental Issues in Himalayas

Importance of Himalayan Region

- The Indian Himalayan Region has wide ranging ecological and socioeconomic significance.

Ecological Services

- Natural water reservoir that feeds several perennial rivers.
- With wide ranging natural vegetation, it acts as a giant '**carbon sink**'.
- It is one of the most important regions of rich biodiversity, especially the eastern Himalayan region which is a global biodiversity hotspot. It is storehouse of endemic species.

Role in Indian climate

- The Himalayan region acts as a barrier to the south-west monsoon winds causing copious rainfall (orographic rainfall) in the foothill and the Ganga-Brahmaputra plain region. Without out Himalayas, this region would have been a desert.
- It stops the cold waves from central Asia, China and Siberian Russia from entering India.

Economic importance

- Himalayan region is ideal for hydroelectric power generation.
- With the most beautiful landscapes on earth, the Himalayan region is a great tourist destination.
- Himalayas are well known for their cultural tourism.

Degradation of Himalayan Environment

- The Himalayan region, with its young and unstable topography, is one of the most fragile ecosystems on earth.
- It has recently gained notice for its 'vulnerability' due to various unsustainable economic and developmental activities such as unchecked tourism and mountaineering, hydroelectric power projects, indiscriminate urbanization etc..

Biotic pressure

- Population explosion has resulted in high biotic pressure by putting strain on natural resources.
- Many species are now threatened due to habitat destruction and habitat fragmentation caused by anthropogenic activities.
- Man-animal conflict: snow Leopards are hunted to save livestock.

Pollution and solid waste

- Air pollution due to increase in vehicle density; water pollution and soil pollution due to domestic sewage are making the region more vulnerable.
- Indiscriminate solid waste dumping as a result of urbanization and tourism development have resulted in

accumulation of debris along the water courses which cause flash floods in wet season.

Hydroelectric power projects

- Hydroelectric power projects are a serious cause of concern. Their construction has weakened the structural stability of the steep slopes causing landslides.
- Tunnel drilling has created new unnatural water courses that cause more soil erosion and makes the stratum highly unstable increasing the risk of landslides.
- Change of river courses causes accumulation of debris in new sites which increase the chances of flash floods in the wet season.

Urbanization to meet tourism demands

- Thriving tourist industry has created great demand for lodging facilities. Buildings are constructed indiscriminately by flouting rules. General non-compliance with prescribed norms and guidelines has become very common.
- The natural sites for these illegal constructions are river banks, dried and old river courses and moderately sloped regions.
- During wet seasons, landslides become common along sloped regions that are affected by construction. The debris from the construction sites block the river courses causing flash floods. All these cause a great damage to the fragile ecology.

Infrastructure development

- Growing population, thriving tourism, military (security) needs have led to rapid construction of roads, and railways on the Himalayan slopes.
- Instability caused by infrastructure development is evident in numerous landslides that occur in along the slopes in the region.

Cultural/Pilgrimage Tourism

- The Himalayas is known to be a home of saints, destination of pilgrimage since time immemorial.
- For example, Badrinath, Kedarnath, Gangotri-Yamunotri and Hemkund Sahib in Uttarakhand, Manimahesh, Jwala Devi, Chintpurni, Naina Devi in Himachal Pradesh and Vaishnav Devi and Amarnath in Jammu & Kashmir, Khecheopalri and other sacred lakes in Sikkim are particularly important destinations.
- Unfortunately, most of these places lack adequate facilities of transport, accommodation, waste disposal and other amenities for the ever growing number of pilgrims that visit them every year.
- Also, there is a gross lack of regulatory mechanism for infrastructure creation, management, and for controlling the tourist inflow in such sites.
- As a result the sensitive ecosystems and cultural values of these areas are facing pressures far beyond their carrying capacities.

Deforestation

- Deforestation is rampant all over India. In spite of their fragile ecosystem, Himalayas are no exception.
- Trees are felled for their timber which find many domestic and industrial applications.
- Deforestation leads to soil erosion and slope instability. Landslides become common.

Climate change/global warming

- Climate change has evolved as the single biggest threat to the Himalayan ecosystem. Melting of glaciers, ice caps and bursting of lakes create havoc downstream.
- The receding snowline has threatened the local biodiversity. Many species acclimatized to the cold environment are moving further up the slopes. They lose their normal prey and hence struggle to survive. Their natural habitat turns hostile leading to extinction of many critical species.

Initiatives to save Himalayan Environment

Ban on Plastic in HP

- State government enacted the Himachal Pradesh Non-Biodegradable Garbage (Control) Act, 1995 in order to prevent throwing or depositing non-biodegradable garbage in public drains, roads.
- It has then increased the minimum thickness of plastic carry bags to 70 microns of virgin material, which exceeded the 20 micron thickness recommended by Central Rules.
- Further, the state Government has taken a Cabinet decision to ban plastics altogether in the entire state since 2009.

Participatory Conservation of Lakes in the Region

- The Nainital Lake is the sole source of drinking water for Nainital town, an important tourist destination in Uttarakhand state.
- Increasing inflow of tourists, urban waste making its way into the lake is adversely affecting water quality.
- To conserve the water body, the residents have switched on to a scientific garbage disposal system - under the project name 'Mission Butterfly' by Nainital Lake Conservation Project.
- The sweepers, on a small monthly charge, collect waste from each household and directly transfer it to the compost pits.
- Apart from the residents, schools and hotel owners have extended full cooperation to the authorities, to save its precious eco-system.
- In addition, they are able to generate income and employment by converting it to manure.

Assam Hill Land and Ecological Sites Act, 2006

- The Assam Hill Land and Ecological Sites (Protection and Management) Act, 2006 to prevent indiscriminate cutting of trees and filling up of water bodies in urban areas.

- Under the Act, the state government can bring any hill under its purview for protection.

Sikkim's Ecotourism Policy

- "Sikkim - the Ultimate Tourist Destination" is the policy motto of the state.
- The state is employing a system of environmental fees, and permits for entries, and stay time restrictions in some environmentally sensitive high altitude/pristine areas.
- Operationalization of tourism in various modes, such as village tourism, nature tourism, wildlife tourism, trekking/adventure tourism, and cultural tourism in the state and institutionalization of tourism management at the community level.
- Promotion and use of local art & craft, cuisines, etc., along with organizing tourism fairs and festivals.
- Imparting training in tourism related service industries.
- The efforts made by Sikkim can be a basis of responsible tourism in other Himalayan states.

Measures That Can Be Taken to Protect Himalayan Ecosystem

Harnessing Religious Sentiments for Conservation

- Encouraging devotees and tribals to undertake participatory plantation for rehabilitation of degraded areas.
- Promoting the concept of eco-cultural landscapes. Both landscapes are highly evolved with high level of economic and ecological efficiencies.
- Involving them in maintenance and strengthening of sacred groves/landscape (e.g., Sacred Groves of Meghalaya: The tribal communities - Khasis, Garos, and Jaintias, have a tradition of environmental conservation based on religious beliefs and customary law and are protected from any product extraction.
- Himalayan Homestays: The Himalayan Homestays programme fosters

conservation-based community managed tourism development in remote settlements, by gradually building local capacity and ownership.

Promoting Ecotourism and Regulation of Commercial Tourism

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- Eco-tourism villages, parks, sanctuaries and other areas should be identified to establish a primary base for ecotourism.
- Village communities, especially youths, and rural women should be involved in Ecotourism.
- Restrictions on the entry of vehicles and visitors per day/ per group should be imposed in sensitive ecological sites.
- Local art, crafts, cuisines, and dishes should be promoted and made an integral part of tourist experience in order to ensure economic benefits to the locals and their cultural integrity/ entity is not lost.
- Best practices on commercial trekking should be imposed on a mandatory basis
- Creation of log/bamboo huts should be promoted in busy mountain areas.

Soil Erosion - Wind & Water Erosion | Desertification

Soil erosion

- Soil erosion is the loosening and displacement of topsoil particles from the land.

Pace Of Soil Erosion

- Soil erosion in nature may be
- a slow process (or geological erosion) or
- a fast process promoted by deforestation, floods, tornadoes or other human activities. These two processes are explained below:

Water Erosion

- Running water is one of the main agents, which carries away soil particles.
- Soil erosion by water occurs by means of raindrops, waves or ice.

- Soil erosion by water is termed differently according to the intensity and nature of erosion.
- (i) Raindrop erosion (ii) Sheet erosion (iii) Rill erosion (iv) Stream bank erosion (v) Erosion due to landslides (vi) Coastal erosion.

Raindrop erosion

- Raindrops falling on land surface cause detachment of the soil particles. The loose soil particles are washed away by flowing water.
- An average size of raindrop is approximately 5 mm in diameter falling through the air hits the soil at a velocity of 32 km/hr.
- Larger raindrops and gusts of wind hit the soil surface even at higher velocities.
- Raindrops behave like tiny bombs when falling on exposed soil, displace soil particles and destroy soil structure.
- Presence of vegetation on land prevents raindrops from falling directly on the soil thus erosion of soil in areas covered by vegetation is prevented.
- With continued rainfall the displaced soil particles fill in the spaces between soil particles and so prevent water from seeping into the soil.
- After some time this result in accumulation of water called 'ponding' on the land. This water begins to flow. This flowing water is called runoff. As the water moves it erodes the soil surface.

Sheet erosion

- The detachment and transportation of soil particles by flowing rainwater is called sheet or wash off erosion. This is very slow process and often remain unnoticed.

Rill erosion

- In rill erosion finger like rills appear on the cultivated land after it has undergone sheet erosion. These rills are usually smoothened out every year while forming. Each year the rills slowly increase in number become wider and deeper. When

rills increase in size they are called gullies. Ravines are deep gullies.

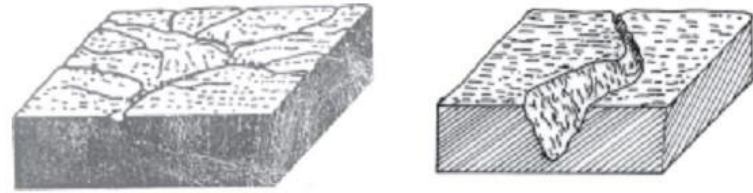


Fig. 17.5: (a) Rill erosion, (b) Gully erosion

Stream bank erosion

- The erosion of soil from the banks (shores) of the streams or rivers due to the flowing water is called bank erosion.
- In certain areas where river changes its course, the river banks get eroded at a rapid rate. Stream bank erosion damages the adjoining agricultural lands, highways and bridges.



Landslide

- Sudden mass movement of soil is called landslide. Landslides occur due to instability or loss of balance of land mass with respect to gravity.
- Loss in balance occurred mainly due to excessive water or moisture in the earth mass.
- Gravity acts on such an unstable landmass and causes the large chunks of surface materials such as soil and rocks slide down rapidly.

Coastal erosion

- Coastal erosion of soil occurs along sea shores. It is caused by the wave action of the sea and the inward movement of the sea into the land.

Consequences of water erosion

- The fine particles of the topsoil which contain the bulk of nutrients and organic matter needed by the plants are lost from soil erosion. Erosion removes the most fertile part of soil. The less fertile subsoil is left.
- Erosion may result in removal of seeds or seedlings so that the soil becomes bare. Bare soil is more vulnerable to erosion both by wind and water.
- Removal of seeds and seedlings reduces the ability of soil to store water.
- Sheet, rill, gully and stream bank erosion also cause siltation of rivers, streams and fields. Deposition of silt results in damage of crops and pastures, and sedimentation of water bodies like streams, dams, reservoirs etc.
- Sedimentation of water bodies deteriorate water quality and damage aquatic habitats and organisms.
- Gully erosion also results in loss of large volumes of soil. Wider deep gullies sometimes reach 30 m and thus severely limit land use.
- Large gullies disrupt normal farm operation.
- Stream bank erosion not only causes loss of land, but also changes the course of a river or stream.
- Stream banks erosion also damage public roads.
- Mass movement of land or landslides also inhibits farm production and land use.
- It also causes mortality in animals and humans.
- Coastal erosion causes the adjoining land to become covered by sand.

Prevention of water erosion

- It is essential to retain vegetation cover that soil is not exposed to rain.
- Plants intercept rainfall and protect soil from direct impact of raindrops.
- Cattle grazing should be controlled.
- Crop rotation and keeping the land fallow (not planting anything in the soil for some time) should be adopted.
- Vegetation and soil management should be improved in order to increase soil organic matter.

- To prevent stream bank erosion runoff water should be stored in the catchment for as possible by maintaining vegetation cover and as by constructing dams for storing water.
- For prevention or reduction of coastal erosion, protective vegetation along the beaches should be re-established.
- The best method of controlling coastal dune erosion is not to disturb the dunes and the coastal system.
- Further, construction of buildings and other development should be located behind the dune system.

Wind erosion

- Soil erosion by wind is more common in areas where the natural vegetation has been destroyed. Such conditions occur mainly in arid and dry areas along the sandy shores of oceans, lakes and rivers.
- The loose soil particles are blown and transported from wind by following three ways: (i) Siltation: blown by wind in a series of short bounces. (ii) Suspension: transported over long distances in the form of suspended particles. (iii) Surface creep: transported at ground level by high velocity winds.

Consequences of wind erosion

- Wind erosion removes the finer soil material including organic matter, clay and silt, in a suspension (colloidal) form and leaving behind coarser, less fertile material.
- Productive capacity of the soil is lost as most of the plant nutrients which remain attached smaller colloidal soil fraction are lost.
- Wind erosion also damages roads and fertile agricultural fields by depositing large quantities of air blown soil particles.

Prevention of wind erosion

- The vegetation cover over sandy soils should be kept above 30%. Access of wind to the soil should be controlled by leaving the stubble or mulch on the soil. (Stubble is the remains of crop left after harvesting).

- Wind speed can be broken or controlled by planting trees in form of a shelter belt.
- The practice of leaving the land fallow (i.e. not planting anything in the field) and use of machinery should be modified. This can be done by using direct- drilling techniques (ploughing the field) and by using direct-drilling techniques.
- Over grazing by cattle should be avoided.
- Deforestation includes cutting and felling of trees, removal of forest litter.
- Browsing and trampling by livestock, forest fires, also leads to cause deforestation etc.
- Deforestation leads to erosion. Deforestation further leads to land degradation, nutrient and the disruption of the delicate soil plant relationship.

Farming

- Crops are grown, harvested, land is ploughed and is exposed to wind. All this prevents replenishment of moisture.
- Agriculture also causes the worst type of soil erosion on farmland in the form of wash-off or sheet erosion.
- On the arid and semiarid areas, sand blows and sand shifts act in a similar fashion as sheet erosion does, where water is the chief agent. Consequently, a creeping effect of desertification sets in and the fertility of the land is lost progressively.

The following agricultural practices can lead to accelerated soil erosion

- **Tilling or ploughing** increases the chances of erosion because it disturbs the natural soil surface and protective vegetation.
- **Continuous cropping:** Continuous cropping of the same land and extending of cultivation of marginal and sub-marginal lands encourages soil erosion.
- **Cultivation on mountain slopes:** Cultivation on mountain slopes without appropriate land treatment measures such as bounding, terracing and trenching

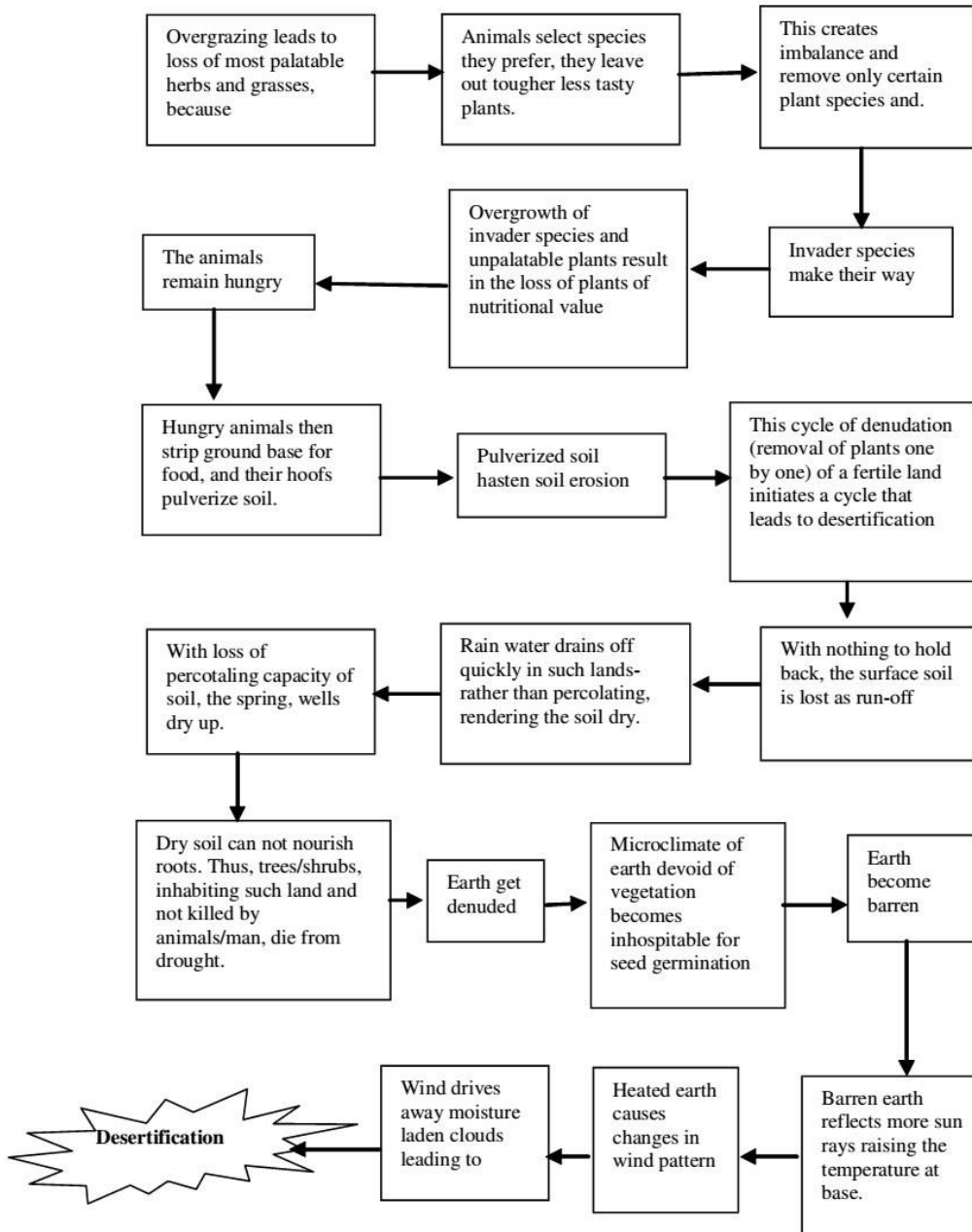


Fig 14.4: Factors causing desertification

Soil Erosion Caused By Human Activities

Deforestation

cause soil erosion and loss of soil nutrients.

- **Monoculture:** Monoculture refers to the practice of planting of the same variety of crop in the field.
- Monoculture practices can lead to soil erosion in three ways.

1. A monoculture crop is harvested all at one time, which leaves the entire fields bare exposing it to both water and wind.
2. Without vegetation natural rainfall is not retained by the soil and flows rapidly over the surface rather than into the ground. It also carries away the top soil which results in soil erosion and degradation.
3. In the event any disease or pest invades the field, the entire crop is usually wiped out leaving the bare soil susceptible to water and wind.

Overgrazing

- It means too many animals are allowed to feed on a piece of grassland. Trampling and grazing by cattle destroys the vegetation of the area.
- In the absence of adequate vegetative cover the land becomes highly susceptible to both wind and water erosion.

Economic activities

- Soil erosion also occurs due to economic activities. The extraction of useful natural resources such as metals, minerals and fossil fuels etc., from the land causes serious disturbance to the land leading to soil erosion and drastic changes in the landscape.

Developmental activities

- Soil erosion may also occur because of various developmental activities such as housing, transport, communication, recreation, etc.
- Building construction also promotes soil erosion because accelerated soil erosion takes place during construction of houses, roads, rail tracks etc.
- The construction of such facilities causes massive disturbance to land, resulting in

soil erosion and disruption of natural drainage system.

Soil degradation and soil conservation are already explained under “**Indian Geography**”

Desertification

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- Desertification is diminution of the biological potential of the land which ultimately leads to the formation of desert.
- The land that has lost its productivity (ability to grow plants) is called a desert.
- A desert landscape supports a very limited growth of sparse vegetation and stunted growth of plants.
- Substantial part of earth's terrestrial area is facing desertification due to overexploitation and mismanage of land resources for human activities. Some of the principal causes, which promote desertification, are: over cultivation, overgrazing, deforestation, and salt accumulation due to irrigation.

Over Cultivation

- Every cycle of cultivation is preceded by ploughing to remove weeds. The ploughed land turns soil upside down thus exposing rich sub-soil to wind and water erosion.
- Such land may remain barren for most part of the year and in turn lose more soil due to erosion. Such erosion is most pronounced on slopes.
- Moreover, in regions where rainfall is low, the soil is often dry and is more susceptible to erosion. Ploughed soil loses more water by evaporation.

Overgrazing

- Deserts receive less rainfall. Deserts have sparse vegetation mostly consisting of grasses and herbs less and best used for grazing.
- Overgrazing by goats, domestic cattle remove the protective vegetation and expose the soil. Further the movement of grazing animals loosen the soil surface by their hoofs.
- Unprotected loose soil becomes highly susceptible to erosion by wind and water.

Such conditions leads to progressive desertification.

Deforestation

- Forests and vegetation prevent soil erosion and to hold water in soil. Plant roots absorb and recycle nutrients released from the decaying organic matter.
- Forests are often cleared to agriculture, timber, construction wood, firewood, raw material for paper etc. All this leads to barrenness of the land leading to desertification.

Salting due to Irrigation

- With demand for more land for agriculture, crops are grown in areas that have little access to natural water bodies.
- The water is supplied to these growing areas by artificial means and improved irrigation methods. Such water brings salts dissolved in it.
- Even the best quality of irrigation water contain 200-500 ppm of salts. Water used for irrigation is lost from agriculture field through evaporation and transpiration by crop plant. The water gets evaporated but the dissolved salt keeps on accumulating which makes the soils more salty.
- Saline accumulation retards plant growth. Land devoid of plant cover easily becomes desertified.
- Accumulation of excessive salt in soil or salinization makes the soil unfit for agriculture.

Air Pollution - Major Air Pollutants - Prevention & Control

Pollution

- Pollution may be defined as addition of undesirable material into the environment as a result of human activities. The agents which cause environmental pollution are called pollutants.
- A pollutant may be defined as a physical, chemical or biological substance released into the environment which is directly or indirectly harmful to humans and other living organisms.

- Pollution may be of the following types: Air pollution, Noise pollution, Water pollution, Soil pollution, Thermal pollution and Radiation pollution.
- In order to control environmental pollution, the Government of India has passed the **Environment (Protection) Act, 1986** to protect and improve the quality of our environment (air, water and soil).

Air Pollution

- Air pollution may be defined as the presence of any solid, liquid or gaseous substance including **noise** and **radioactive radiation** in the atmosphere in such concentration that may be directly and/or indirectly injurious to humans or other living organisms, property or interferes with the normal environmental processes.
- An ever increasing use of fossil fuels in power plants, industries, transportation, mining, construction of buildings, stone quarries had led to air pollution.
- Fossil fuels contain small amounts of **nitrogen** and **sulphur**. Burning of fossil fuels like coal (thermal power plants) and petroleum (petroleum refineries) release different **oxides of nitrogen and sulphur** into the atmosphere.
- These gases react with the water vapour present in the atmosphere to form sulphuric acid and nitric acid. The acids drop down with rain, making the rain acidic. This is called **acid rain**.
- Acid rain corrodes the marble monuments like Taj Mahal. This phenomenon is called as **"Marble cancer"**.
- Other kinds of pollutants are **chlorofluorocarbons (CFCs)** which are used in **refrigerators, air conditioners and as pressurizing agents in aerosol sprays**. CFCs damage the ozone layer of the atmosphere.
- The combustion of fossil fuels also increases the amount of suspended particles in air. These suspended particles could be unburnt carbon particles or substances called hydrocarbons.
- Presence of high levels of all these pollutants cause visibility to be lowered, especially in cold weather when water also condenses out of air. This is known as **smog** and is a visible indication of air pollution.

Classification of Pollutants

According to the form in which they persist after release into the environment.

- **Primary pollutants:** These persist in the form in which they are added to the environment e.g. DDT, plastic.
- **Secondary Pollutants:** These are formed by interaction among the primary pollutants. For example, **peroxyacetyl nitrate (PAN)** is formed by the interaction of **nitrogen oxides** and **hydrocarbons**.

According to their existence in nature

- **Quantitative Pollutants:** These occur in nature and become pollutant when their concentration reaches beyond a threshold level. E.g. carbon dioxide, nitrogen oxide.
- **Qualitative Pollutants:** These do not occur in nature and are man-made. E.g. fungicides, herbicides, DDT etc.

Particulate pollutants

Particulate matter suspended in air are dust and soot released from the industrial chimneys. Their size ranges from 0.001 to 500 micrometers (μm) in diameter.

- Particles less than 10 μm float and move freely with the air current. Particles which are more than 10 μm in diameter settle down. Particles less than 0.02 μm form persistent aerosols.
- Major source of SPM (suspended particulate matter) are vehicles, power plants, construction activities, oil refinery, railway yard, market place, industries, etc.
- According to Central Pollution Control Board (CPCB), particulate size 2.5 μm or less in diameter (**PM 2.5**) are responsible for causing the greatest harm to human health.
- These fine particulates can be inhaled deep into the lungs and can cause breathing and respiratory symptoms, irritation, inflammations and **pneumoconiosis** – a disease of the lungs due to inhalation of dust, characterized by inflammation, coughing, and fibrosis..

Fly ash

- Fly ash is ejected mostly by thermal power plants as byproducts of coal burning operations.
- Fly ash pollutes air and water and may cause heavy metal pollution in water bodies.
- Fly ash affects vegetation as a result of its direct deposition on leaf surfaces or indirectly through its deposition on soil.
- Fly ash in the air slowly settles on leaves and crops in fields in areas near to thermal power plants and lowers the plant yield.
- Fly ash is now being used for making **bricks** and as a **land fill**

Composition

- Fly ash particles are oxide rich and consist of silica, alumina, oxides of iron, calcium, and magnesium and toxic heavy metals like **lead, arsenic, cobalt, and copper**.
- Major oxides are present are **Aluminium silicate** (in large amounts), **silicon dioxide (SiO₂)** and **calcium oxide (CaO)**.

Advantages

- Cement can be replaced by fly ash up to 35%, thus reducing the cost of construction, making roads, etc.
- Fly ash bricks are light in weight and offer high strength and durability.
- Fly ash is a better fill material for road embankments and in concrete roads.
- Fly ash can be used in **reclamation of wastelands**.
- Abandoned mines can be filled up with fly ash.
- Fly ash can increase the crop yield and it also enhances water holding capacity of the land .

Policy measures of MoEF

- The Ministry of Environment and Forests has made it mandatory to use Fly Ash based products in all construction projects, road embankment works and low lying land filling works within 100 kms radius of Thermal Power Station and mine filling activities within 50 kms radius of Thermal Power Station.

Q1. With reference to 'fly ash' produced by the power plants using the coal as fuel, which of the following statements is/are correct?

1. Fly ash can be used in the production of bricks for building construction
2. Fly ash can be used as a replacement for some of the Portland cement contents of concrete
3. Fly ash is made up of silicon dioxide and calcium oxide only, and does not contain any toxic elements.

Select the correct answer using the code given below

- a. 1 and 2
- b. 2 only
- c. 1 and 3
- d. 3 only

Fly ash does contain heavy toxic elements like arsenic, cobalt, lead etc.

Answer: a) 1 and 2

Lead

- It is present in petrol, diesel, lead batteries, paints, hair dye products, etc.
- It can cause **nervous system damage** and digestive problems and, in some cases, cause cancer. Lead affects children in particular.
- **Tetraethyl lead (TEL)** is used as an **anti-knock agent** in petrol for smooth and easy running of vehicles.
- The lead particles coming out from the exhaust pipes of vehicles is mixed with air. If inhaled it produces injurious effects on **kidney and liver** and interferes with development of **red blood cells**.
- Lead mixed with water and food can create cumulative poisoning. It has long term effects on children as it **lowers intelligence**.

Metallic Oxides

- Oxides of iron, aluminum, manganese, magnesium, zinc and other metals have adverse effect due to deposition of dust on plants during mining operations and metallurgical processes.
- They create physiological, biochemical and developmental disorders in plants and also contribute towards **reproductive failure in plants**.

Nanoparticles – NPs

Source:

<http://www.ncbi.nlm.nih.gov/pmc/articles/PMC3388445/>

- Nanoparticles are particle with dimensions comparable to $1/10^9$ of a meter [1 divided by 100 crores].
- Major natural processes that release NPs in the atmosphere are forest fires, volcanic eruptions, weathering, dust storms from desert etc.
- Naturally occurring NPs are quite heterogeneous in size and can be transported over thousands of kilometres and remain suspended in the air for several days.
- Nanotechnology has a global socioeconomic value, with applications ranging from electronics to biomedical uses (delivering drugs to target sites).
- Man-made NPs are unknowingly or purposely released in the environment during various industrial and mechanical processes.

Effects of Nanoparticles on the environment

- After release in the environment, NPs will accumulate in various environmental matrices such as air, water, soil and sediments including wastewater sludge.
- NPs in the environment influences dust cloud formation, environmental hydroxyl radical concentration, ozone depletion, or stratospheric temperature change.

Effect of NNPs on dust cloud formation

- NNPs are thought to play an important role in dust-clouds formation after being released into the environment as they coagulate and form dust cloud.
- Dust cloud formation decreases sun light intensity.

Asian brown clouds impact on Himalayan glaciers

- Asian brown clouds carry large amounts of soot and black carbon (NPs) which are deposited on the glaciers.
- This could lead to higher absorption of the sun's heat and potentially contributing to the increased melting of glaciers.

NPs and ozone depletion

- The nanoparticles have greater chemical reactivity. They can result in increased production of reactive oxygen species (ROS), including free radicals like $\text{Cl}^\cdot$.
- Radicals like $\text{Cl}^\cdot$ destroy ozone. [Explained in "Ozone Depletion"]

In chemistry, a radical (a free radical) is an atom, molecule, or ion that has unpaired valence electrons.

Effect of NPs on stratospheric temperature

- NPs in the troposphere interact with molecular hydrogen accidentally released from hydrogen fuel cells and other sources.
- Molecular hydrogen along with NPs moves up to the stratosphere, resulting in the abundance of water vapour in the stratosphere.
- This will cause stratospheric cooling due to formation Stratospheric clouds (mostly ice crystals).
- Stratospheric clouds destroys ozone. [Explained in "Ozone Depletion"]

There is some concern regarding the nanoparticles of some chemical elements that are used by the industry in the manufacture of various products. Why?

1. They can accumulate in the environment, and contaminate water and soil.
2. They can enter the food chains.
3. They can trigger the production of free radicals.

Select the correct answer using the code given below.

- a. 1 and 2 only
- b. 3 only
- c. 1 and 3 only
- d. **1, 2 and 3**

Major Gaseous Air Pollutants, Their Sources & Effects

Carbon monoxide (CO)

- Carbon monoxide (CO) is a colorless, odorless, tasteless and highly toxic gas that is slightly

less dense than air. It is **short-lived** (stay only few months) in the atmosphere.

- Carbon monoxide is produced from the exhaust of internal combustion engines and from incomplete combustion of various other fuels. Iron smelting also produce carbon monoxide as a byproduct.
- It forms when there is not enough oxygen to produce carbon dioxide (CO_2).
- In the presence of oxygen, carbon monoxide burns with a blue flame, producing carbon dioxide.
- Worldwide, the largest source of carbon monoxide is natural in origin, due to photochemical reactions in the troposphere.
- Other natural sources of CO include volcanoes, forest fires, and other forms of combustion.

Health Effects

- Carbon monoxide poisoning is the most common type of fatal air poisoning.
- It is toxic to hemoglobin animals (including humans) when encountered in concentrations above about 35 ppm.
- It is also produced in normal animal metabolism in low quantities.
- It combines with hemoglobin to produce **carboxyhemoglobin**, which usurps the space in hemoglobin that normally carries oxygen.

Environmental Effects

- In the atmosphere, it is spatially variable and short lived, having a role in the formation of **ground-level ozone** (tropospheric ozone) and can elevate concentrations of **methane**.
- Carbon monoxide reacts with hydroxyl radical ($-\text{OH}$) to produce peroxy radical. Peroxy radical reacts with nitrogen oxide (NO) to form nitrogen dioxide (NO_2) and hydroxyl radical. **NO_2 gives O_3 via photolysis** (separation of molecules by the action of light).

Carbon dioxide (CO_2)

- Colorless and odorless gas vital to life on Earth. It is heavier than air.
- Natural sources include volcanoes, hot springs and geysers, and it is freed from carbonate rocks by dissolution in water and acids.

- Because carbon dioxide is soluble in water, it occurs naturally in groundwater, rivers and lakes, in ice caps and glaciers and also in seawater.

Effects on Health

- CO₂ is an asphyxiant gas (asphyxia → a condition arising when the body is deprived of oxygen, causing unconsciousness or death.).
- Concentrations of 7% may cause suffocation, even in the presence of sufficient oxygen, manifesting as dizziness, headache, and unconsciousness.

Effects on Environment

- Carbon dioxide is an important greenhouse gas. Burning of carbon-based fuels since the industrial revolution has led to **global warming**.
- It is also a major cause of **ocean acidification** because it dissolves in water to form **carbonic acid**.

Chlorofluorocarbons (CFCs)

- Chlorofluorocarbons (CFCs) are used in refrigerators, air conditioners and aerosol sprays.
- Since the late 1970s, the use of CFCs has been heavily regulated because of their destructive effects on the ozone layer. (Explained in a separate post on Ozone Depletion)
- The **Montreal Protocol** on Substances that Deplete the Ozone Layer (a protocol to the **Vienna Convention for the Protection of the Ozone Layer**) is an **international treaty** designed to protect the ozone layer by phasing out the production of numerous substances including CFCs which are responsible for ozone depletion.

Ozone (O₃)

- It occurs naturally in the stratosphere. Here it absorbs harmful ultraviolet rays of the sun.
- However, at the ground level, it is a pollutant (**Greenhouse gas**) with highly toxic effects.
- Vehicles and industries are the major source of ground-level ozone emissions.
- **Carbon monoxide, Nitrogen dioxide** play a major role in converting O₂ to O₃.

- Ozone makes our eyes itchy, and watery. It lowers our resistance to cold and pneumonia.

Nitrogen oxide (NO_x)

- NO_x is a generic term for the various nitrogen oxides produced during combustion.
- They are produced mainly in **internal combustion engines** and **coal burning power plants**. They are produced naturally by **lightening**.

[Oxygen and nitrogen do not react at ambient temperatures. But at high temperatures they produce various oxides of nitrogen. Such temperatures arise inside an internal combustion engine or a power station boiler]

- **Agricultural fertilization** and the **use of nitrogen fixing plants** also contribute to atmospheric NO_x, by promoting nitrogen fixation by microorganisms.
- **NO_x (contributes to global cooling)** should not be confused with **nitrous oxide (N₂O)**, which is a **greenhouse gas** and has many uses as an oxidizer.

Effects on Health and Environment

- They are believed to aggravate **asthmatic conditions** and create many respiratory health issues, especially in children.
- The reduction of NO_x emissions is one of the most important technical challenges facing **biodiesel**.
- NO_x gases react to form **smog** and **acid rain** as well as being central to the formation of **tropospheric ozone**.
- When NO_x and volatile organic compounds (VOCs) react in the presence of sunlight, they form **photochemical smog**.
- Mono-nitrogen oxides eventually form nitric acid when dissolved in atmospheric moisture, forming a component of acid rain.
- NO_x emissions cause **global cooling** through the formation of -OH radicals that **destroy methane molecules**, countering the effect of greenhouse gases.

Sulphur dioxide (SO₂)

- It is a toxic gas with a pungent, irritating smell. It contributes to **acid rain**

- It is released naturally by volcanic activity. It is abundantly available in the atmosphere of **Venus**.
- Sulfur dioxide is primarily produced for sulfuric acid manufacture.
- Inhaling sulfur dioxide is associated with increased respiratory symptoms and disease, difficulty in breathing, and premature death. It also weakens the functioning of certain nerves.

It is also produced by

- **burning coal in thermal power plants** and **diesel fuels**.
- some industrial processes, such as production of paper and smelting of metals.
- reactions involving Hydrogen Sulphide (H_2S) and oxygen.
- The roasting of sulfide ores such as pyrite, sphalerite, and cinnabar (mercury sulfide). Pollutants

Volatile organic compounds (VOCs)

- Volatile Organic Compounds (VOCs) are a large group of **carbon-based** chemicals that easily **evaporate** at room temperature.
- For example, **formaldehyde**, which evaporates from paint, has a boiling point of only $-19^\circ C$. Formaldehyde causes irritation to the eyes and nose and allergies.
- The main indoor sources are perfumes, hair sprays, furniture polish, glues, air fresheners, moth repellents, wood preservatives, and other products.
- Health effect - irritation of the eye, nose and throat, headaches, nausea and loss of coordination.
- Long term - suspected to damage the liver and other parts of the body.

Benzene and Ethylene

- Benzene is a natural constituent of crude oil and is one of the elementary petrochemicals.
- Because benzene has a **high octane number**, it is an important component of gasoline.
- Benzene increases the **risk of cancer** and other illnesses. Benzene is a notorious cause of **bone marrow failure**.
- Ethylene is widely used in the chemical industry. Much of this production goes toward **polyethylene**, a widely used plastic

containing polymer chains of ethylene units in various chain lengths.

- Ethylene is also an important **natural plant hormone**, used in agriculture to force the **ripening of fruits**.
- Ethylene is of low toxicity to humans and exposure to excess ethylene cause adverse health effects like headache, drowsiness, dizziness and unconsciousness.
- Ethylene is not but **Ethylene oxide is a carcinogen**.

Tobacco Smoke

- Tobacco smoke generates a wide range of harmful chemicals and is **carcinogenic** (cancer causing).
- Health effect - burning eyes, nose, and throat irritation to cancer, bronchitis, severe asthma, and a decrease in lung function.

Biological pollutants

- It includes pollen from plants, mite, and hair from pets, fungi, parasites, and some bacteria.
- Most of them are allergens and can cause asthma, hay fever, and other allergic diseases.

Asbestos

- Asbestos refers to a set of six naturally occurring silicate **fibrous minerals** — chrysotile, crocidolite, amosite, anthophyllite, tremolite, and actinolite.
- It is now known that prolonged inhalation of asbestos fibers can cause serious and fatal illnesses including **lung cancer**, **mesothelioma**, and **asbestosis** (a type of pneumoconiosis).

Radon

- It is a gas that is emitted naturally by the **soil**. Due to modern houses having poor ventilation, it is confined inside the house and causes lung cancers.

Smog - Photochemical smog - Sulfurous smog

Prevention and Control of air Pollution

Indoor Air Pollution

- Poor ventilation due to faulty design of buildings leads to pollution of the confined space.
- Paints, carpets, furniture, etc. in rooms may give out volatile organic compounds (VOCs).
- Use of disinfectants, fumigants, etc. may release hazardous gases.
- In hospitals, pathogens present in waste remain in the air in the form of spores. This can result in hospital acquired infections and is an occupational health hazard.
- In congested areas, slums and rural areas burning of firewood and biomass results in lot of smoke.
- Children and ladies exposed to smoke may suffer from acute respiratory problems.

Prevention and control of indoor air pollution

- Use of wood and dung cakes should be replaced by cleaner fuels such as biogas, kerosene, LPG or electricity. But supply of electricity is limited. Similarly kerosene is also limited. The use of solar cookers must be encouraged.
- Old stoves must be replaced with improved stoves with high thermal efficiency and reduced emission of pollutants including smoke.
- The house designs should incorporate a well-ventilated kitchen.
- Those species of trees such as baval (Acacia nilotica) which are least smoky should be planted and used. Charcoal is a comparatively cleaner fuel.
- Indoor pollution due to decay of exposed kitchen waste can be reduced by covering the waste properly.
- Segregation of waste, pretreatment at source, sterilization of rooms will help in checking indoor air pollution.

Control of Industrial Pollution

- Industrial pollution can be greatly reduced by:
 1. use of cleaner fuels such as liquefied natural gas (LNG) in power plants, fertilizer plants etc. which is cheaper in addition to being environmentally friendly.
 2. employing environment friendly industrial processes so that emission of pollutants and hazardous waste is minimized.

3. installing devices which reduce release of pollutants.

- Devices like filters, electrostatic precipitators, inertial collectors, scrubbers, gravel bed filters or dry scrubbers are described below:

Filters

- Filters remove particulate matter from the gas stream. The medium of a filter may be made of fibrous materials like cloth, granular material like sand, a rigid material like screen, or any mat like felt pad.
- Baghouse filtration system is the most common one and is made of cotton or synthetic fibres (for low temperatures) or glass cloth fabrics (for higher temperature up to 290° C).

Electrostatic precipitators (ESP)

- There are several ways of removing particulate matter; the most widely used method is electrostatic precipitation, which can remove over 99 per cent particulate matter present in the exhaust from a thermal power plant.
- The emanating dust is charged with ions and the ionized particulate matter is collected on an oppositely charged surface.
- An electrostatic precipitator has electrode wires that are maintained at several thousand volts, which produce a **corona** that releases **electrons**.
- These electrons attach to dust particles giving them a net negative charge. The collecting plates are grounded (relatively positive charge) and attract the charged dust particles.
- The velocity of air between the plates must be low enough to allow the dust to fall.
- The particles are removed from the collection surface by occasional shaking or by rapping the surface.
- ESPs are used in boilers, furnaces, and many other units of thermal power plants, cement factories, steel plants, etc.

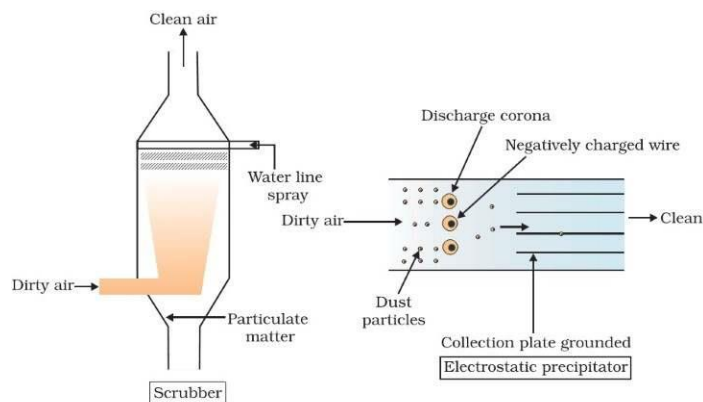


Figure 16.1 Electrostatic precipitator

Inertial collectors

- It works on the principle that inertia of SPM (suspended particulate matter) in a gas is higher than its solvent and as inertia is a function of the mass of the particulate matter, this device collects heavier particles more efficiently (centrifugation is the technique). 'Cyclone' is a common inertial collector used in gas cleaning plants.

Scrubbers

- Scrubbers are wet collectors. They remove **aerosols** from a stream of gas either by collecting wet particles on a surface followed by their removal, or else the particles are wetted by a scrubbing liquid.
- The particles get trapped as they travel from supporting gaseous medium across the interface to the liquid scrubbing medium. (this is just like mucus in trachea trapping dust)
- A scrubber can remove gases like **sulphur dioxide**.

Catalytic converter

- Proper maintenance of automobiles along with use of lead-free petrol or diesel can reduce the pollutants they emit.
- Catalytic converters, having expensive metals namely **platinum-palladium** and **rhodium** as the catalysts, are fitted into automobiles for reducing emission of poisonous gases.
- As the exhaust passes through the catalytic converter, unburnt hydrocarbons are converted into carbon dioxide and water, and carbon monoxide and nitric oxide are changed to carbon dioxide and nitrogen gas, respectively.

- Motor vehicles equipped with catalytic converter should use **unleaded petrol** because lead in the petrol inactivates the catalyst.

Apart from the use of above mentioned devices, other control measures are -

- increasing the height of chimneys.
- closing industries which pollute the environment.
- shifting of polluting industries away from cities and heavily populated areas.
- development and maintenance of green belt of adequate width.

Steps Taken to Control Vehicular Pollution

- The emission standards for automobiles have been set which if followed will reduce the pollution. Standards have been set for the durability of catalytic converters which reduce vehicular emission.
- In cities like Delhi, motor vehicles need to obtain Pollution Under Control (PUC) certificate at regular intervals. This ensures that levels of pollutants emitted from vehicle exhaust are not beyond the prescribed legal limits.
- The price of diesel is much cheaper than petrol which promotes use of diesel. To reduce emission of sulphur dioxide, **sulphur content in diesel has been reduced** to 0.05%.
- Earlier lead in the form of **tetraethyl lead** was added in the petrol to **raise octane level** for **smooth running** of engines. Addition of lead in petrol has been banned to prevent emission of lead particles with the vehicular emission.
- Usage of alternate fuels like CNG in public transport vehicles is made mandatory in cities like Delhi. All the buses of Delhi were converted to run on CNG by the end of 2002.

CNG burns most efficiently, unlike petrol or diesel, in the automobiles and very little of it is left unburnt. Moreover, CNG is cheaper than petrol or diesel, cannot be siphoned off by thieves and adulterated like petrol or diesel. The main problem with switching over to CNG is the difficulty of laying down pipelines to deliver CNG through distribution points/pumps and ensuring uninterrupted

supply.

- Simultaneously parallel steps taken in Delhi for reducing vehicular pollution include phasing out of old vehicles, use of unleaded petrol, use of low-sulphur petrol and diesel, use of catalytic converters in vehicles, application of stringent pollution- level norms for vehicles, etc.
- The Government of India through a new auto fuel policy has laid out a roadmap to cut down vehicular pollution in Indian cities.
- More stringent norms for fuels means steadily reducing the sulphur and aromatics content in petrol and diesel fuels.
- The goal, according to the roadmap, is to **reduce sulphur** to 50 ppm in petrol and diesel and **bring down the level of aromatic hydrocarbons** to 35 per cent.
- The Bharat Stage II (equivalent to Euro-II norms), which is currently in place in Delhi, Mumbai, Kolkata, Chennai, Bangalore, and other major cities was made applicable to all automobiles throughout the country.
- All automobiles in major cities were expected to meet the Euro III emission specifications by 2005 and Euro-IV norms by 2010.
- The rest of the country was expected to meet Euro-III emission norm by 2010.
- According to an estimate, a substantial fall in CO₂ and SO₂ level has been found in Delhi between 1997 and 2005. But still their levels are not under permissible levels.
- The recent odd-even formula is expected to improve the worsening air quality in Delhi.
- In India, the **Air (Prevention and Control of Pollution) Act** came into force in 1981, but was amended in 1987 to include **noise** as an air pollutant. Noise is undesired high level of sound.

Government Initiative

National Air Quality Monitoring Programme

- Central Pollution Control Board (CPCB) has been executing a nationwide programme of ambient air quality monitoring known as National Air Quality Monitoring Programme (NAMP).

The National Air Quality Monitoring Programme (NAMP) is undertaken

- to determine status and trends of ambient air quality;
- to ascertain the compliance of NAAQS;
- to identify non-attainment cities;
- to understand the natural process of cleaning in the atmosphere; and
- to undertake preventive and corrective measures.

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National Ambient Air Quality Standards (NAAQS)

- The NAAQS have been revisited and revised in November 2009 for 12 pollutants, which include
- **sulphur dioxide (SO₂),**
- **nitrogen dioxide (NO₂),**
- **particulate matter having micron (PM₁₀),**
- **particulate matter having size less than 2.5 micron (PM_{2.5}),**
- **ozone,**
- **lead,**
- **carbon monoxide (CO),**
- **arsenic,**
- **nickel,**
- **benzene,**
- **ammonia, and**
- **benzopyrene.**

National Air Quality Index (AQI)

- Launched by the Environment Ministry in April 2015.
- Initiative under 'Swachh Bharat'.

AQI

- It helps the common man to judge the air quality within his vicinity.
- Index constituted as a part of Government's mission to improve the culture of cleanliness.

Old vs new

- While the earlier measuring index was limited to three indicators, the current measurement index had been made quite comprehensive by the addition of more parameters.

Previously who measured Air pollution

- Central Pollution Control Board along with State Pollution Control Boards has been operating **National Air Monitoring Program (NAMP)**.

Why is AQI necessary

- Quality of data from some cities remains weak and the standards set for pollutants fall short of World Health Organization recommendations.
- The pollution related analysis using vast number of complex parameters was complicated for the common man to understand.

Categories of air pollution under AQI

- There are six AQI categories, namely Good, Satisfactory, Moderately polluted, Poor, Very Poor, and Severe.

Pollutants considered

- The proposed AQI will consider eight pollutants (PM₁₀, PM_{2.5}, NO₂, SO₂, CO, O₃, NH₃, and Pb).

Alternatives

- Government is disincentivising use of private vehicles through **congestion charging**.
- The National Green Tribunal has ordered that diesel vehicles over 10 years old not ply on Delhi roads.
- Odd even formula.
- Making city roads friendly to bicycle users.
- Promoting rooftop solar power as an alternative to coal power.

Photochemical smog - Sulfurous smog

In this Post: Smog - Primary and secondary pollutants; Types of Smog: Sulfurous smog [London Smog] and Photochemical smog [Los Angeles Smog]; Effects of Smog; Question UPSC Mains 2015: Mumbai, Delhi and Kolkata are the three mega cities of the country but the air pollution is much more serious problem in Delhi as compared to the other two. Why is this so? [200 words]

Previous post: Must read post: [Condensation – Forms of Condensation – Types of Clouds](#)

Smog



http://www.youtube.com/watch?v=yoO_xDA m2mI

- Smog = **smoke + fog (smoky fog)** caused by the burning of large amounts of **coal, vehicular emission and industrial fumes** (Primary pollutants).
- Smog contains soot particulates like **smoke, sulphur dioxide, nitrogen dioxide** and other components.
- At least two distinct types of smog are recognized: **sulfurous smog** and **photochemical smog**.

Primary and secondary pollutants

- A primary pollutant is an air pollutant emitted directly from a source.
- A secondary pollutant is not directly emitted as such, but forms when other pollutants (primary pollutants) react in the atmosphere.
- Examples of a secondary pollutant include **ozone**, which is formed when
 - hydrocarbons (HC) and nitrogen oxides (NO_x) combine in the presence of sunlight;
 - NO combines with oxygen in the air; and
 - acid rain, which is formed when sulfur dioxide or nitrogen oxides react with water.

Sulfurous smog

- Sulfurous smog is also called "**London smog**," (first formed in London).
- Sulfurous smog results from a high concentration of **SULFUR OXIDES** in the air and is caused by the use of **sulfur-bearing fossil fuels, particularly coal** (Coal was the main source of power in London during

nineteenth century. The effects of coal burning were observed in early twentieth century).

- This type of smog is aggravated by **dampness** and a **high concentration of suspended particulate matter** in the air.



Photochemical smog

- Photochemical smog is also known as “**Los Angeles smog**”.
- Photochemical smog occurs most prominently in urban areas that have large numbers of automobiles (**Nitrogen oxides** are the primary emissions).
- Photochemical (**summer smog**) forms when pollutants such as **nitrogen oxides** (primary pollutant) and **organic compounds** (primary pollutants) react together in the presence of **SUNLIGHT**. A gas called **OZONE** (Secondary pollutant) is formed.

Nitrogen

Dioxide + Sunlight + Hydrocarbons = Ozone (Ozone in stratosphere it is beneficial, but near the earth's surface it results in global warming as it is a greenhouse gas)

- The resulting smog causes a light brownish coloration of the atmosphere, reduced visibility, plant damage, irritation of the eyes, and respiratory distress.



Reactions involved

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Atmospheric oxidant production:

1. $\text{NO} + \text{VOC} \longrightarrow \text{NO}_2$ (nitrogen dioxide)
2. $\text{NO}_2 + \text{UV} \longrightarrow \text{NO} + \text{O}$ (nitric oxide + atomic oxygen)
3. $\text{O} + \text{O}_2 \longrightarrow \text{O}_3$ (ozone)
4. $\text{NO}_2 + \text{VOC} \longrightarrow \text{PAN, etc.}$ (peroxyacetyl nitrate)

Net results:



Toxic Chemical	Sources www.pmfias.com	Environmental Effects
Nitrogen Oxides (NO and NO₂)	☐ combustion of oil, coal, gas ☐ bacterial action in soil ☐ forest fires, volcanic action ☐ lightning	☐ decreased visibility due to yellowish color of NO ₂ ☐ NO ₂ can suppress plant growth
Volatile Organic Compounds (VOCs)	☐ evaporation of fuels ☐ incomplete combustion of fossil fuels www.pmfias.com	☐ eye irritation ☐ respiratory irritation ☐ some are carcinogenic ☐ decreased visibility due to blue-brown haze
Ozone (O₃)	☐ formed from photolysis of NO ₂ ☐ sometimes results from stratospheric ozone intrusions	☐ decreased crop yields ☐ retards plant growth ☐ damages plastics ☐ breaks down rubber www.pmfias.com
Peroxyacetyl Nitrates (PAN)	☐ formed by the reaction of NO ₂ with VOCs	☐ eye irritation ☐ high toxicity to plants ☐ damaging to proteins

Haze

Haze is traditionally an atmospheric phenomenon where dust, smoke and other dry particles obscure the clarity of the sky (No condensation. Smog is similar to haze but there is condensation in smog). Sources for haze particles include farming (ploughing in dry weather), traffic, industry, and wildfires.



- Smog is a combination of airborne particulate matter, like soot, and invisible toxic gases including **ozone (O₃)**, **carbon monoxide (CO)**, **sulfur dioxide (SO₂)**, which are **carcinogens (cancer causing agents)**.
- Temperature inversions are accentuated and **precipitation is reduced**.
- Smog related Haze lowers visibility.

Question: UPSC Mains 2015

Mumbai, Delhi and Kolkata are the three mega cities of the country but the air pollution is much more serious problem in Delhi as compared to the other two. Why is this so? [200 words]

- In spite of similar urbanization, air pollution is much more severe in Delhi compared to that in Mumbai and Kolkata. This is because of

Polluting Industry in close vicinity in Delhi.

- Delhi and its immediate neighborhood is the hot bed of polluting industries which are primarily coal fueled. Burning coal releases Oxides of sulphur which forms sulphurous smog. This type of smog is more pronounced in Delhi than in the other two cities due to geography and climate.

Vehicular Emissions

Effects of Smog

- The **atmospheric pollution** levels of Los Angeles, Beijing, Delhi, Mexico City and other cities are increased by **inversion** that traps pollution close to the ground.

Temperature Inversion – Types – Effects on Weather

- It is usually highly toxic to humans and can cause severe sickness, shortened life or death.

- All the three cities contribute nearly equal vehicular emissions rich in CO₂ and NO₂. NO₂ results in photochemical smog. Here again, Delhi is worst hit due to its geography and climate.

Geography and Climate

- This the **most detrimental factor**. Delhi is a **continental city** while the other two are coastal. Land and Sea Breezes in Mumbai and Kolkata carry pollutants away from the city. There is no such advantage to Delhi as it is land locked.
- Also, the **duration of monsoon winds** is short in Delhi compared to the other two.
- Delhi faces severe cold wave in winter compared to the other two. Cold climate here creates temperature inversion which traps the pollutants, mainly smog, for a longer duration.

Farm Straw Burning

- Delhi is at the heart of major agricultural region. Burning of farm straw in the surrounding regions also adds to Delhi's pollution levels.

226 words. Cutting those 26 words will be detrimental for success.

Ozone Depletion | Ozone Hole - Causes, Effects

A layer of ozone preset in the atmosphere (stratosphere) protects the biosphere from the harmful effects of UV radiation by absorbing them.

Ozone depletion or ozone hole refer to the damage suffered by the ozone layer due to natural and anthropogenic causes.

Each of the factors/causes responsible for ozone depletion are explained below.

Ozone Hole – Ozone Depletion

- Polar vortex and ozone depletion are two distinct but related phenomena.
- Ozone gas is continuously formed by the action of UV rays on molecular oxygen in the stratosphere. Also, ozone is simultaneously

degraded into molecular oxygen in the stratosphere.

- There should be a balance between production and degradation of ozone in the stratosphere so that there is a continuous layer of ozone.
- Of late, the balance has been disrupted due to enhancement of ozone degradation by chlorofluorocarbons (CFCs) [chlorofluorocarbons (CFCs) are **halocarbons**]. There is a steady decline of about **4%** in the total volume of ozone in Earth's stratosphere.
- Much larger decrease in stratospheric ozone is observed around **Earth's polar regions**.

Halocarbon == a compound in which the hydrogen of a hydrocarbon is replaced by halogens like chlorine, bromine, iodine etc.

Halogen == group of reactive non-metallic elements like fluorine, chlorine, bromine, iodine, and astatine.

The thickness of the ozone in a column of air from the ground to the top of the atmosphere is measured in terms of **Dobson units (DU)**.

The ozone measurement instruments and techniques are varied. Some of them are the **Dobson spectrophotometer** and the **filter ozonometer** called M83.

Halogen atoms like chlorine destroy ozone

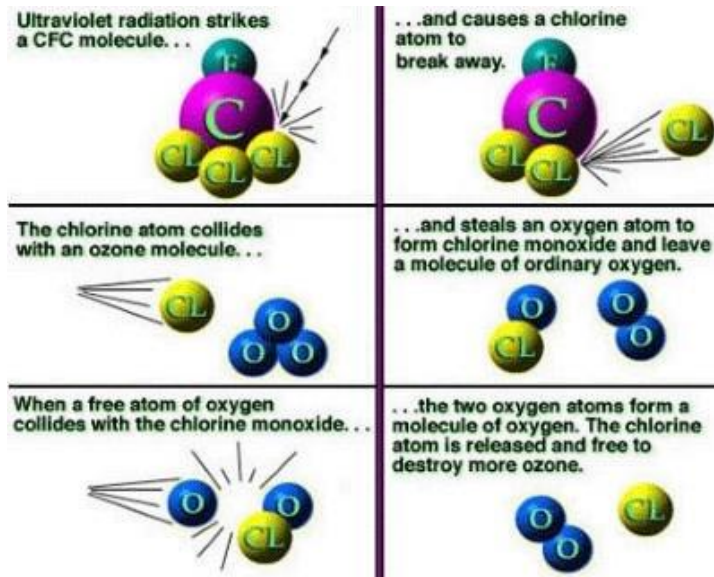
- **Photodissociation** (under the influence of sunlight) of **ozone-depleting substances (ODS)** like
 1. halocarbon refrigerants (CFCs),
 2. halocarbon solvents (Methyl Chloroform, carbon tetrachloride),
 3. propellants, and foam-blowing agents (CFCs, HCFCs, carbon tetrachloride and trichloroethane, freons, halons [used in firefighting])

creates **free chlorine atoms** that destroy ozone.

- **Bromine** containing compounds called **halons** and **HBFCs**, i.e. **hydrobromofluorocarbons** [both used in **fire extinguishers** and **methyl bromide** (a widely

used pesticide)] release bromine atoms similar to CFCs that release chlorine atoms.

- Each bromine atom destroys **hundred times** of more ozone molecules than what a chlorine atom does.



Q1. Consider the following statements: Chlorofluorocarbons, known as ozone-depleting substances, are used

- in the production of plastic foams
- in the production of tubeless tyres
- in cleaning certain electronic components
- as pressurizing agents in aerosol cans

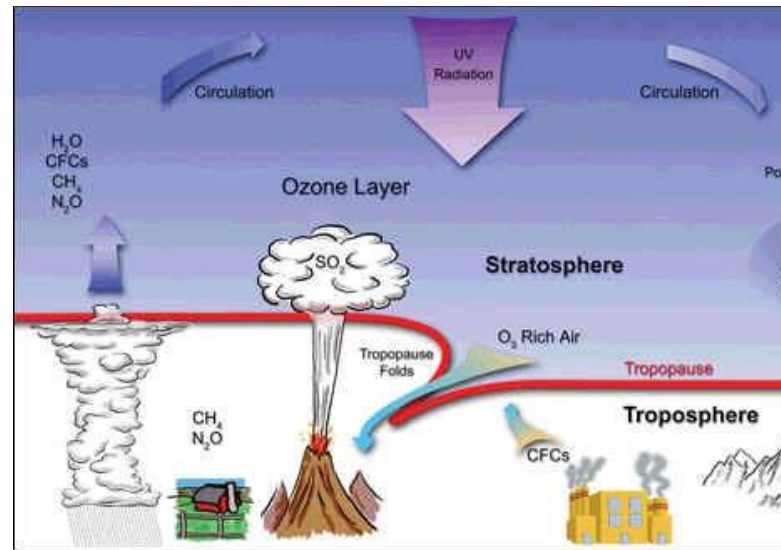
Which of the statements given above is/are correct?

- 1, 2 and 3 only
- 4 only
- 1, 3 and 4 only
- 1, 2, 3 and 4

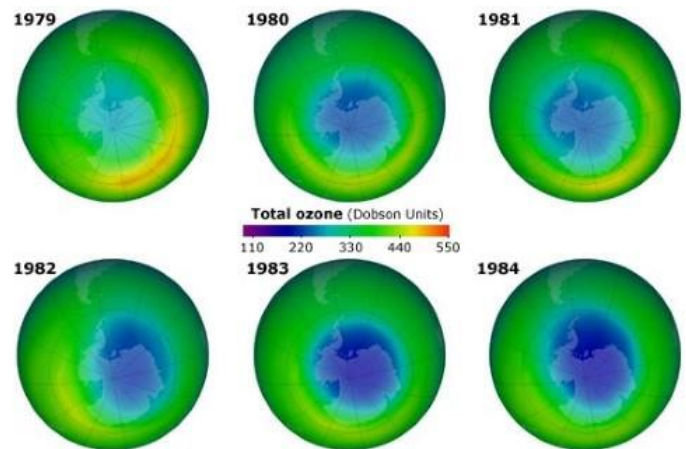
Answer: c) CFCs were used as refrigerants, pressurizing agents (foam and aerosol cans) and for cleaning electronic equipment.

But how does a chlorine atom reach to such high levels of atmosphere?

- Methane (CH₄), chlorofluorocarbons (CFCs), nitrous oxide (N₂O) and water are injected into the stratosphere through towering **tropical cumulus clouds**.
- These compounds are broken down by the ultraviolet radiation in the stratosphere.



The ozone depletion is significant near poles (especially south pole) than at any other place. This is because of high altitude polar clouds called **Polar Stratospheric Clouds** formed during **Polar Vortex** events.



Polar Vortex

- Polar vortex (circumpolar vortex) is a **polar cyclone**.
- A polar vortex is a large pocket of very cold air, typically the coldest air in the Northern Hemisphere, which sits over the polar region during the winter season.
- Polar vortex is closely associated with **jet streams [Rossby waves]**.
- It is formed mainly in winter and gets **weaker in summer**.
- It surrounds **polar highs** and lie within the polar front (boundary separating the temperate and polar air masses).



Polar Stratospheric Clouds (PSCs)

- Extend from 12 km – 22 km above the surface.
- They are nacreous clouds.

Nacreous clouds

Nacreous clouds, sometimes called mother-of-pearl clouds, are rare clouds.

They are mostly visible within two hours after sunset or before dawn.

They form in frigid regions of the lower stratosphere, some 15 - 25 km (9 -16 mile) high and well above tropospheric clouds.

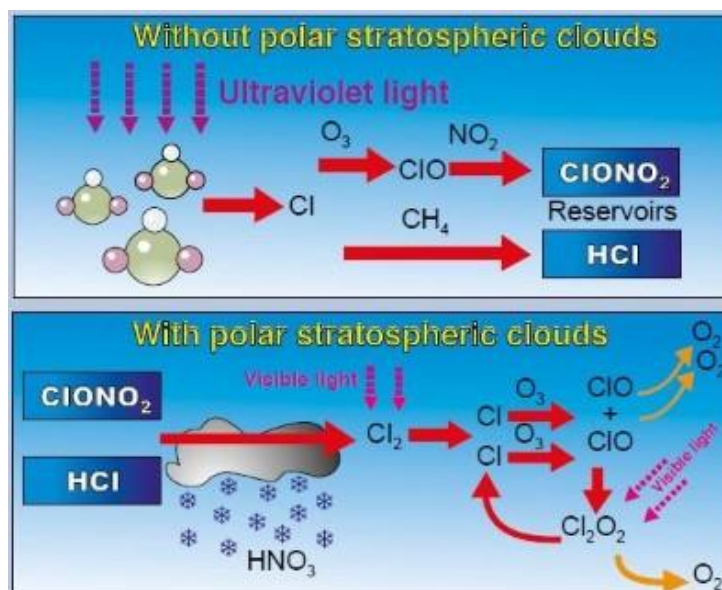
They are bright even after sunset and before dawn because at those heights there is still sunlight.

They are seen mostly during winter at high latitudes like Scandinavia, Iceland, Alaska and Northern Canada. Sometimes, however, they occur as far south as England.

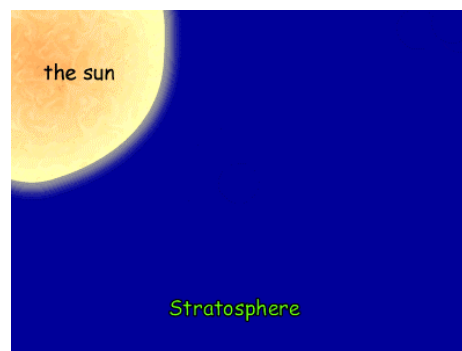


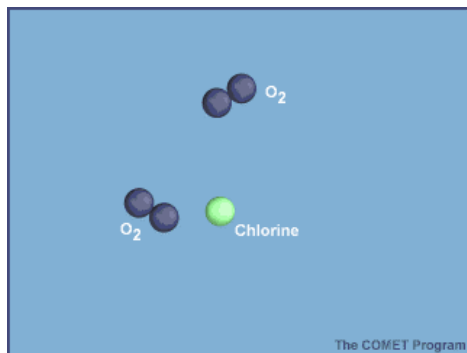
- PSCs or nacreous clouds contain **water, nitric acid and/or sulfuric acid**.
- They are formed mainly during the event of **polar vortex in winter**; more intense at **south pole**.
- The Cl-catalyzed ozone depletion is dramatically enhanced in the presence of polar stratospheric clouds (PSCs) [Finally this how polar vortex contributes to ozone depletion].
- A number of naturally occurring substances like **Hydrogen oxide (HOx), Methane (CH₄), Hydrogen gas (H₂), Nitrogen oxides (NO_x)** aid the process of ozone depletion.

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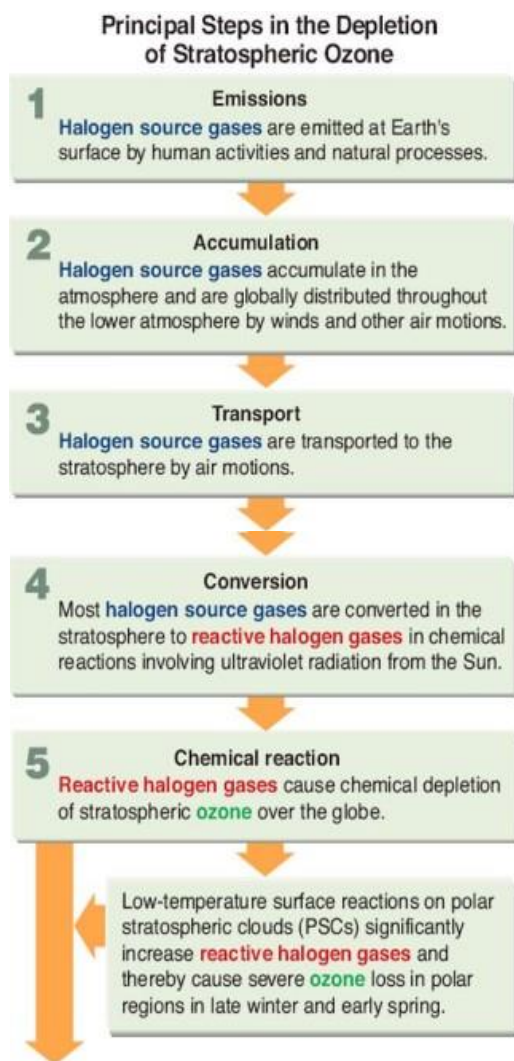


- PSCs convert "reservoir" compounds into reactive free radicals (**Cl and ClO**).
- These free radicals deplete ozone as shown in the animation below.
- So PSC accelerate ozone depletion.**





Gif Images: View in power point in full screen mode.



Q2. The formation of ozone hole in the Antarctic region has been a cause of concern. What could be the reason for ozone depletion at poles?

- a. Presence of prominent tropospheric turbulence; and inflow of chlorofluorocarbons

- b. Presence of prominent polar front and stratospheric Clouds and inflow of chlorofluorocarbons
- c. Absence of polar front and stratospheric clouds; and inflow of methane and chlorofluorocarbons
- d. Increased temperature at polar region due to global warming

Answer: b) Presence of prominent polar front and stratospheric clouds and inflow of chlorofluorocarbons

Harmful Effects of Ozone Depletion

Effects on Humans

- UV rays are highly injurious to living organisms since DNA and proteins of living organisms preferentially absorb UV rays, and its high energy breaks the chemical bonds within these molecules.
- UV radiation of wavelengths shorter than UV-B, are almost completely absorbed by Earth's atmosphere, given that the ozone layer is intact.
- But if UV-B manages to reach the troposphere due to ozone depletion, DNA and mutation may occur. It causes aging of skin, damage to skin cells and various types of skin cancers.
- In human eye, cornea absorbs UV-B radiation, and a high dose of UV-B causes inflammation of cornea, called snow-blindness cataract, etc. Such exposure may permanently damage the cornea.

In short

- Increased susceptibility to skin-cancer
- Increase cataract (a medical condition in which the lens of the eye becomes progressively opaque, resulting in blurred vision)
- Damage DNA (DNA mutations)
- Damage to cornea and causes retinal diseases
- Suppresses human immune systems

Effects on terrestrial plants

Psychological and developmental processes of plants are affected by UV-B radiation.

- Inhibits photosynthesis
- Inhibits metabolism

- Represses growth
- Destroys cells
- Causes DNA mutations
- Decline in forest productivity

Effects on aquatic ecosystems

- Exposure to solar UV-B radiation has been shown to affect both orientation mechanisms and motility in phytoplankton, resulting in reduced survival rates for plankton population adversely affecting marine food chain.
- Marine/freshwater organisms are very sensitive to UV-rays. E.g. Corals

Effects on air quality

- Increase in UV-B radiation result in higher photo dissociation rates of key trace gases that control the chemical reactivity of the troposphere.
- This can increase both production and destruction of ozone (O₃) and related oxidants such as hydrogen peroxide (H₂O₂), which are known to have adverse effects on human health, terrestrial plants, and outdoor materials.
- Changes in the atmospheric concentrations of the hydroxyl radical (OH) may change the atmospheric lifetimes of climatically important gases such as methane (CH₄) and the CFC substitutes.
- Increased tropospheric reactivity could also lead to increased production of particulates such as cloud condensation nuclei, from the oxidation and subsequent nucleation of sulphur.

Effects on materials

- Accelerate breakdown of paints
- Accelerate breakdown of plastics
- Affect temperature gradient levels in the atmosphere
- Affect atmospheric circulation pattern, climatic changes.

Measures to Prevent Ozone (O₃) Layer Depletion

Monitoring of ozone layer is taken up by

1. World Meteorological Organization (WMO)
2. World Weather Watch (WWW)

3. Integrated Global Ocean Services Systems (IGOSS)
4. Global Climate Observing System (GCOS)

CFC substitutes

- Further, use of HCFCs (Hydrochloric fluorocarbons) as a substitute for CFCs is being recommended on temporary basis because HCFCs are relatively less damaging to ozone layer as compared to CFCs, but they are not completely ozone safe.

International Efforts

- Recognizing the deleterious effects of ozone depletion, an international treaty, known as the Montreal Protocol, was signed at Montreal (Canada) in 1987 (effective in 1989) to control the emission of ozone depleting substances.

Vienna Convention for the Protection of the Ozone Layer

- Multilateral Environmental Agreement.
- It was agreed upon at the Vienna Conference of 1985 and entered into force in 1988.
- It acts as a framework for the international efforts to protect the ozone layer.
- However, it does not include legally binding reduction goals for the use of CFCs, the main chemical agents causing ozone depletion. These are laid out in the accompanying Montreal Protocol.

Montreal Protocol

- The Montreal Protocol on Substances that Deplete the Ozone Layer (a protocol to the Vienna Convention for the Protection of the Ozone Layer) is an international treaty designed to protect the ozone layer by **phasing out** the production of numerous substances that are responsible for ozone depletion.
- It was agreed in 1987, and entered into force in 1989, followed by a first meeting in Helsinki, May 1989. Since then, it has undergone eight revisions.
- As a result of the international agreement, the ozone hole in Antarctica is slowly **recovering**.
- Climate projections indicate that the ozone layer will return to 1980 levels between 2050 and 2070.

- It is the single most successful international agreement to date.

The two ozone treaties (Vienna Convention and Montreal Protocol) have been ratified by 197 parties [196 UN states + European Union] making them the first universally ratified treaties in United Nations history.

UN Framework Convention on Climate Change is also ratified by 197 parties.

Q3. Which one of the following is associated with the issue of control and phasing out of the use of ozone-depleting substances?

- Bretton Woods Conference
- Montreal Protocol
- Kyoto Protocol
- Nagoya Protocol

Bretton Woods Conference established the International Bank for Reconstruction and Development (IBRD) and the International Monetary Fund (IMF).

Montreal Protocol is an international treaty to protect the ozone layer by phasing out the production of ozone depleting substances. It is legally binding.

Kyoto Protocol implemented the objective of the UNFCCC to fight global warming by reducing greenhouse gas concentrations in the atmosphere to "a level that would prevent dangerous anthropogenic interference with the climate system". It is binding on the parties.

Nagoya Protocol is a supplementary agreement to the 1992 Convention on Biological Diversity (CBD) on "Access to Genetic Resources and the Fair and Equitable Sharing of Benefits Arising from their Utilization".

Answer: b)

Acid Rain – Causes, Effects | Ocean Acidification

Acid Rain – Acidification

Page

- Acid rain refers to any precipitation (rain, fog, mist, snow) that is more acidic than normal (pH of less than 5.6. [pH below 7 = acidic]).
- Acid rain is caused by atmospheric pollution from acidic gases such as **sulphur dioxide** and **oxides of nitrogen** emitted from burning of fossil fuels.
- It is also recognized that acidic smog, fog, mist, move out of the atmosphere and settle on dust particles which in turn accumulate on vegetation as acid depositions. When rain falls, the acid from these depositions leak and form acid dews.

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The pH scale

- The pH scale is a measure of how acidic or basic (alkaline) a solution is.
- It ranges from 0 to 14. A pH of 7 is neutral.
- A pH less than 7 is acidic, and a pH greater than 7 is basic.
- It is based on **hydrogen ion concentration** in an aqueous solution.
- pH values decreases as hydrogen ion levels increases.
- A solution with pH 4 is ten times more acidic than solution with pH 5, and a hundred times more acidic than solution with pH 6.
- Whilst the pH range is usually given as 0 to 14, lower and higher values are theoretically possible.

Acidic gases and their emission sources

Acidic gases	Source
CO ₂ (Carbon dioxide)	Fossil fuel burning, industrial process, respiration.
CH ₄ (Methane)	Paddy fields, wetlands, gas drilling, landfills, decomposition of animals wastes and carcasses.
CO (Carbon monoxide)	Biomass burning, Industrial sources: smelting of iron ore, Biogenesis, Plant isoprene's.
SO _x (Sulphur oxides)	Fossil fuel burning, power plants, smelting of metal sulfide ores, industrial sources, industrial production of sulfuric acid in metallurgical, chemical and fertilizer industries volcanoes, seas and oceans, decomposition of organic matter.

NO. (nitrogen oxides)	Fossil fuel burning, lightening, biomass burning, forest fires, oceans, power plants.
Formic acid (HCOOH)	Biomass burning due to forest fires causes emission of formic acid (HCOOH) and formaldehyde (HCHO) into the atmosphere. Large fraction formaldehyde gets photo — oxidation and forms formic acid in the atmosphere.
Carbonic acid (H ₂ CO ₃)	Carbon monoxide and carbon dioxide dissolve in water (water vapor) to form carbonic acid.

Q1. Acid rain is caused by the pollution of environment by

- a) carbon dioxide and nitrogen
- b) carbon monoxide and carbon dioxide
- c) ozone and carbon dioxide
- d) nitrous oxide and sulphur dioxide

Explanation:

Carbon dioxide, carbon monoxide, oxides of nitrogen and oxides of sulphur in atmosphere can cause acid rains.

Both b) and d) are correct. But d) has more influence than b)

Answer: d) nitrous oxide and sulphur dioxide

Types Of Acid Deposition

"Acid rain" is a broad term referring to a mixture of wet and dry deposition (form of deposition material) from the atmosphere.

Wet Deposition

- If the acid chemicals in the air are blown into areas where the weather is wet, the acids can fall to the ground in the form of rain, snow, fog, or mist.
- As this acidic water flows over and through the ground, it affects a variety of plants and animals.

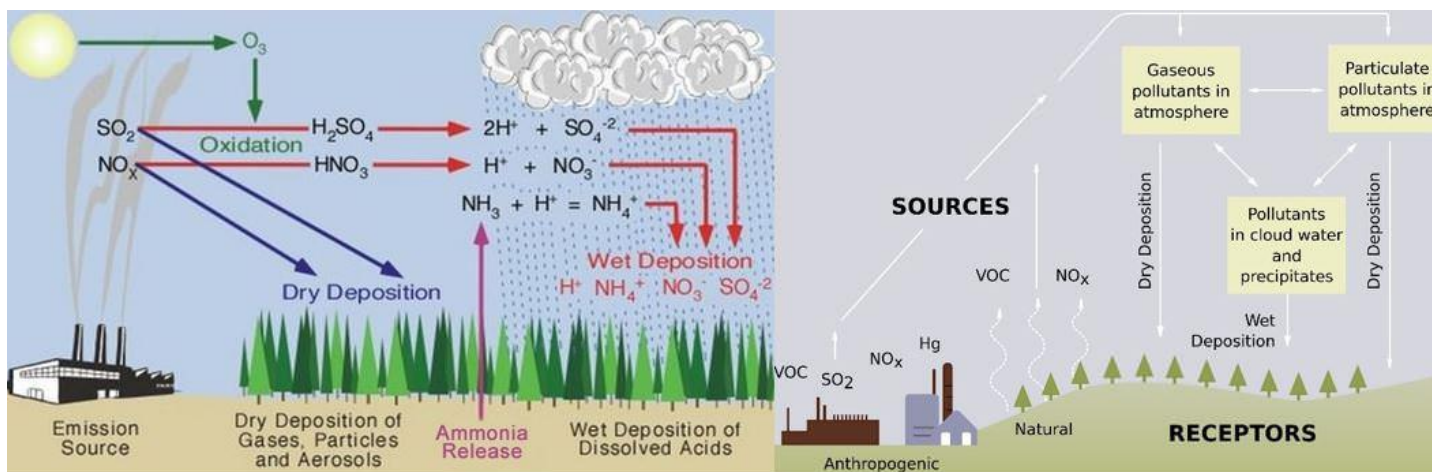
Dry Deposition

- In areas where the weather is dry, the acid chemicals may become incorporated into dust or smoke and fall to the ground through dry deposition, sticking to the ground, buildings, vegetation, cars, etc.
- Dry deposited gases and particles can be washed from these surfaces by rainstorms, through runoff. This runoff water makes the resulting mixture more acidic.
- About half of the acidity in the atmosphere falls back to earth through **dry deposition**.

Chemistry of Acid Rain

Six basic steps are involved in the formation of acid rain:

- The atmosphere receives oxides of sulfur and nitrogen from natural and man-made sources.
- Some of these oxides fall back directly to the ground as dry deposition, either close to the place of origin or some distance away.
- Sunlight** stimulates the formation of **photo-oxidants** (such as ozone) in the atmosphere.
- These photo-oxidants interact with the oxides of sulfur and nitrogen and other gases (like NH₃) to produce H₂SO₄ (sulphuric acid) and HNO₃ (nitric acid) by oxidation.
- Acid rain containing ions of **sulfate**, **nitrate**, **ammonium** and **hydrogen** falls as wet deposition.



Harmful effects of acid rain

- Acid precipitation affects both aquatic and terrestrial organisms. It also damages buildings and monuments.

Effects on humans

- Acid rain affects human health in a number of ways.
- The obvious ones are bad smells, reduced visibility; irritation of the skin, eyes and the respiratory tract.
- Some direct effects include chronic bronchitis, pulmonary emphysema and cancer.
- Some indirect effects include food poisoning via drinking water and food.
- An increase in the levels of toxic heavy-metals like manganese, copper, cadmium and aluminium also contribute to the detrimental effects on human health.

Effects on soil

- The exchange between hydrogen ions and the nutrient cations like potassium and

magnesium in the soil cause **leaching** of the nutrients, making the soil infertile. This is accompanied by a decrease in the respiration of soil organisms.

- An increase in ammonia in the soil due to a decrease in other nutrients **decrease the rate of decomposition**. The nitrate level of the soil is also found to decrease.
- The impact of acid rain on soil is less in India; because Indian soils are mostly **alkaline, with good buffering ability**.

Effects on aquatic life

- The pH of the medium is very important for metabolic processes of aquatic organisms.
- Eggs or sperms of fish, frogs and other aquatic organisms are sensitive to pH changes.
- Acid rain kills their gametes affecting the life cycles and productivity. Death or their inability to increase in numbers causes severe ecosystem imbalances.
- Acidic lake waters may kill bacteria/microbes/planktons and the acidic lakes become unproductive and lifeless. Such acidic and lifeless ponds/lakes adversely affect fisheries and livelihood.

Naturally acidic lakes	Anthropogenically acidified lakes
Brown to yellow colour caused by humic substances	Very clear water caused by reduced primary productivity
Concentrations of dissolved organic carbon are high while transparency is low	Dissolved organic carbon concentrations are low. Whereas the transparency is high.
Low pH but well buffered.	Poorly buffered
Abound with aquatic life.	Some of the more sensitive taxa, such as blue-green algae, some bacteria, snails, mussels crustaceans, mayflies and fish either decrease or /are eliminated.

Effect on terrestrial life

- Acid rain damage **cuticle** of plant leaves resulting etiolation (pale and weak) of foliage. This in turn reduces photosynthesis.

- Reduced photosynthesis accompanied by leaf fall reduces plant and crop productivity.
- Acidic medium promotes **leaching** of heavy metals like **aluminum, lead** and **mercury**. Such metals when percolate into ground water affect soil micro flora/fauna.
- Absorption of these toxic metal ions by plants/microorganisms affect their metabolism.
- Acid rain can directly affect the eggs and tadpoles of frogs and salamanders that breed in small forest ponds.
- It has been postulated that acid rain can indirectly affect wildlife by allowing metals bound on soils and sediments to be released into the aquatic environment, where toxic substances may be ingested by animals, like birds, that feed in such an environment.
- Other indirect effects of acid rain on wildlife are loss or alteration of food and habitat resources.

Effects on micro organisms

pH determines the proliferation of any microbial species in a particular environment and the rate at which it can produce.

The optimum pH of most bacteria and protozoa is near neutrality; most fungi prefer

Material	Type of Impact	Principal Air Pollutants
Metals	Corrosion, tarnishing	Sulphur Oxides and other acid gases
Paints and organic coatings	Surface erosion, discoloration, soiling	Sulphur dioxides, hydrogen sulphide
Textiles	Fading, colour change	Nitrogen oxides, ozone
Leather	Weakening, powdered surface	Sulphur oxides
Rubber	Cracking	Ozone

Trigger Effect of Acid Rain on Pollutants

- A low pH of the rainwater and subsequent increased acidity in the environment can trigger off or aggravate the effects of certain harmful pollutants.

Mercury

- Methyl mercury and related short chain alkyl mercurial compounds are most dangerous to humans, as they accumulate in edible fish tissue.
- Although acid deposition may not increase the production of methyl mercury, it may increase the partitioning of methyl mercury into the water column.

an acidic environment, most blue-green bacteria prefer an alkaline environment.

- So after a long run of acid rain, microbial species in the soil and water shift from bacteria-bound to fungi-bound and cause an imbalance in the micro flora.
- This causes a delay in the decomposition of soil organic material, and an increase in fungal disease in aquatic life and forests.

Effects on forests

- Acid rains kill vegetation and causes severe damage to the forest landscape.

Effect on buildings, monuments and materials

- Many old, historic, ancient buildings and works of art/textile etc. are adversely affected by acid rain.
- Limestone and marble are destroyed by acid rain. Smoke and soot cover such objects. They slowly dissolve/flake away the surfaces because of acid fumes in the air. Many buildings/monuments such as Taj Mahal in Agra have suffered from acid rain.

- The use of lime has helped in reducing the mercury levels in fish.

Aluminium

- Acidified waters are known to leach substantial amounts of aluminium from watersheds.
- Even at relatively low levels, aluminium has been implicated in dialysis dementia, a disorder of the central nervous system, which may be toxic to individuals with impaired kidney function.

Cadmium

- Cadmium can enter the drinking water supply through corrosion of galvanized pipe or from

the copper-zinc solder used in the distribution systems.

- A decrease in water pH from 6.5 to 4.5 can result in a fivefold increase in cadmium and could cause renal tubular damage.

Lead

- Foetuses and infants are highly susceptible to drinking water lead contamination.
- High blood lead levels in children (>30 mug/ Ml) are believed to induce biochemical and neurophysiological dysfunction.
- However, lower than normal blood levels of lead can cause mental deficiencies and behavioural problems.

Asbestos

- Asbestos in natural rock can be released by acidic waters.

Acid Rain Areas

- They are concentrated in the industrialized belt of the northern hemisphere.
- They are often upland and/or mountainous areas, which are well-watered by rain and snow.
- Due to the abundance of water, they possess numerous lakes and streams and also have more land covered with vegetation. Being upland, they often have thin soils and glaciated bedrock.
- Many parts of Scandinavia, Canada, the North and Northeast United States and Northern Europe (particularly West Germany and upland Britain) share these features.
- Across the Atlantic there are number of acid rain hot spots including Nova Scotia, Southern Ontario and Quebec in Canada, the Adirondack Mountains in New York, Great Smoky mountains, parts of Wisconsin, Minnesota, and the Colorado Rockies of the US.

In India

- In India, the first report of acid rain came from Bombay in 1974. Instances of acid rain are being reported from metropolitan cities.
- In India, the annual SO₂ emission has almost doubled in the last decade due to increased fossil fuel consumption.

- Lowering of soil pH is reported from north-eastern India, coastal Karnataka and Kerala, parts of Orissa, West Bengal and Bihar.

Indicators

- **Lichens** serve as good bio-indicators for air pollution.
- In the variety of pH around 6.0, several animals, those are important food items for fish decline. These include the freshwater shrimp, crayfish, snails and some small mussels.

Acid Rain Control

- Any procedure that shall reduce, minimize, or halt emission of sulphur and nitrogen oxides into the atmosphere shall control acid rain.
- Use of low sulphur fuel or natural gas or washed coal (chemical washing of pulverized coal) in thermal plants can reduce incidences of acid rain.
- Buffering - the practice of adding a neutralizing agent to the acidified water to increase the pH is one of the important control measures. Usually lime in the form of calcium oxide and calcium carbonate is used.

Ocean Acidification

- Ocean acidification has been called the “evil twin of global warming” and “the other CO₂ problem”.
- Ocean acidification is the ongoing decrease in the pH of the Earth's oceans, caused by the uptake of carbon dioxide (CO₂) from the atmosphere.
- An estimated 30–40% of the carbon dioxide from human activity released into the atmosphere dissolves into oceans, rivers and lakes.
- To achieve chemical equilibrium, some of it reacts with the water to form **carbonic acid**.
- Some of these extra carbonic acid molecules react with a water molecule to give a bicarbonate ion and a **hydronium ion**, thus increasing ocean acidity (H⁺ ion concentration).
- Checking CO and CO₂ emissions and controlling pollution are the only means to reduce ocean acidification.

Other contributors

- Acid rain can have a pH between 1 and 6 and has impact on surface ocean chemistry. It has major effect on ocean acidification locally and regionally but very small globally.
- Eutrophication leads to large plankton blooms, and when these blooms collapse and sink to the sea bed the subsequent respiration of bacteria decomposing the algae leads to a decrease in sea water oxygen and an increase in CO₂ (a decline in pH).

Effects of Ocean Acidification

- Oceans are an important reservoir for CO₂, absorbing a significant quantity of it (one-third) produced by anthropogenic activities and effectively buffering climate change.
 - The uptake of atmospheric carbon dioxide is occurring at a rate exceeding the natural buffering capacity of the oceans creating new problems in the form of ocean acidification.
1. Increasing acidity depresses metabolic rates and immune responses in some organisms.
 2. Other chemical reactions are triggered which result in a net decrease in the amount of carbonate ions available. This makes it more difficult for marine calcifying organisms, such as coral (calcareous corals) and some plankton (calcareous plankton), to form biogenic calcium carbonate.
 3. Commercial fisheries are threatened because acidification harms calcifying organisms which form the base of the Arctic food webs.
 4. Acidification could damage the Arctic tourism economy and affect the way of life of indigenous peoples. A major pillar of Arctic tourism is the sport fishing and hunting industry.
 5. Increasing acidity accentuates coral bleaching as corals are very sensitive to changes in water composition.

How ocean acidification effects calcifying ability

- Seawater absorbs CO₂ to produce carbonic acid, bicarbonate and carbonate ions.
- The carbonate ions are essential to the calcification process that allows certain marine organisms to build their calcium carbonate shells and skeletons (e.g. hard tropical corals, cold water corals, molluscs,

crustaceans, sea urchins, certain types of plankton, lobsters, etc.).

- However, increase in atmospheric CO₂ levels lead to decrease in pH level, increase in the concentration of carbonic acid and bicarbonate ions, causing a decrease in the concentration of **carbonate ions**.
- Thus carbonate ions are less available and calcification is therefore harder to achieve, and may be prevented altogether.

Saturation horizons

- Deep, cold ocean waters are naturally under saturated with carbonate ions causing the shells of most calcifying organisms to dissolve.
- Surface waters are oversaturated with carbonate ions and do not readily dissolve shells of calcifying organisms.
- The saturation horizon is the level below which calcium carbonate minerals undergo dissolution.
- Those organisms that can survive below the saturation horizon do so due to special mechanisms to protect their calcium carbonate from dissolving.
- As ocean acidification causes this horizon to rise vertically (upwelling) in the water column so more and more calcifying organisms will be exposed to under saturated water and thus vulnerable to dissolution of their shells and skeletons.

Impact of Ocean Acidification on Cloud Formation

- The majority of sulfur in the atmosphere is emitted from the ocean, often in the form of dimethylsulfide (DMS) produced by phytoplankton.
- Some of DMS produced by phytoplankton enters the atmosphere and reacts to make sulphuric acid, which clumps into aerosols, or microscopic airborne particles.
- **Aerosols seed the formation of clouds**, which help cool the Earth by reflecting sunlight.
- **But, in acidified ocean water, phytoplankton produce less DMS. This reduction of sulfur may lead to decreased cloud formation, raising global temperatures.**

Artificial Cloud seeding

- Cloud seeding is the process of spreading either **dry ice**, or more commonly, **silver iodide aerosols**, into the upper part of clouds to try to stimulate the precipitation process and form rain.
- Since most rainfall starts through the growth of ice crystals from super-cooled cloud droplets in the upper parts of clouds, the **silver iodide** particles are meant to encourage the growth of new ice particles.

Q2. The acidification of oceans is increasing. Why is this phenomenon a cause of concern?

1. The growth and survival of calcareous phytoplankton will be adversely affected.
2. The growth and survival of coral reefs will be adversely affected.
3. The survival of some animals that have phytoplanktonic larvae will be adversely affected.
4. The cloud seeding and formation of clouds will be adversely affected.

Which of statements given above is / are correct?

- a. 1, 2 and 3 only
- b. 2 only
- c. 1 and 3 only
- d. 1, 2, 3 and 4

Explanation:

We have already learnt that ocean acidification decreases the calcifying ability of corals, calcareous plankton, crustaceans etc. It also adversely affects cloud formation and cloud seeding. So Options 1, 2 and 4 are correct.

Answer: d) 1, 2, 3 and 4

Water Pollution - Causes, Control Measures

Water Pollution

- Water pollution is the addition/presence of undesirable substances to/in water such as organic, inorganic, biological, radiological,

heat, which degrades the quality of water so that it becomes unfit for use'.

- Water pollution is caused by a variety of human activities such as industrial, agricultural and domestic.
- Natural sources of pollution of water are soil erosion, leaching of minerals from rocks and decaying of organic matter.

Point and non-point sources of pollution

- Rivers, lakes, seas, oceans, estuaries and ground water sources may be polluted by point or non-point sources.
- When pollutants are discharged from a specific location such as a drain pipe carrying industrial effluents discharged directly into a water body it represents point source pollution.
- In contrast non-point sources include discharge of pollutants from diffused sources or from a larger area such as run off from agricultural fields, grazing lands, construction sites, abandoned mines and pits, roads and streets.

Causes of Water Pollution

Sewage Water

- Sewage water include discharges from houses, commercial and industrial establishments connected to public sewerage system.
- The sewage contains human and animal excreta, food residues, cleaning agents, detergents and other wastes.
- Domestic and hospital sewage contain many undesirable pathogenic microorganisms, and its disposal into a water without proper treatment.

Putrescibility is the process of decomposition of organic matter present in water by microorganisms using oxygen.

Dissolved Oxygen (DO) – Biological Oxygen Demand (BOD) – Chemical oxygen demand (COD)

DO

- Presence of organic and inorganic wastes in water decreases the dissolved Oxygen (DO) content of the water.
- Water having DO content below 8.0 mg/L may be considered as contaminated. Water having DO content below 4.0 mg/L is considered to be highly polluted.
- DO content of water is important for the survival of aquatic organisms. A number of factors like surface turbulence, photosynthetic activity, O₂ consumption by organisms and decomposition of organic matter are the factors which determine the amount of DO present in water.
- The higher amounts of waste increases the rates of decomposition and O₂ consumption, thereby decreases the DO content of water.

BOD

- The demand for O₂ is directly related to increasing input of organic wastes and is expressed as biological oxygen demand (BOD) of water.
- Water pollution by organic wastes is measured in terms of Biochemical Oxygen Demand (BOD).
- BOD is the amount of dissolved oxygen needed by bacteria in decomposing the organic wastes present in water. It is expressed in milligrams of oxygen per litre of water.

- The higher value of BOD indicates low DO content of water. Since BOD is **limited to biodegradable materials** Therefore, it is not a reliable method of measuring pollution load in water.

COD

- Chemical oxygen demand (COD) is a slightly better mode used to measure pollution load in water.
- COD measures the amount of oxygen in parts per million required to oxidize **organic (biodegradable and non-biodegradable)** and **oxidizable inorganic** compounds in the water sample.

Industrial Wastes

- The industries discharge several inorganic and organic pollutants, which may prove highly toxic to the living beings.
- Discharge of waste water from industries like petroleum, paper manufacturing, metal extraction and processing, chemical manufacturing, etc., that often contain toxic substances, notably, heavy metals (defined as elements with density > 5 g/cm³ such as **mercury, cadmium, copper, lead, arsenic**) and a variety of organic compounds.

Type of Industry	Inorganic pollutants	Organic pollutant
Mining	Chlorides, ferrous sulphate, sulphuric acid, hydrogen sulphide, ferric hydroxide and heavy metals .	
Iron and Steel	Iron cyanide, thiocyanates, sulphides, oxides of copper, chromium, cadmium, and mercury.	Oil, phenol and naphtha
Chemical Plants	Various acids and alkalies, chlorides, sulphates, nitrates of metals, phosphorus, fluorine, silica and suspended particles.	Aromatic compounds solvents, organic acids, nitro compound dyes, etc.
Pharmaceutical		Proteins, carbohydrates, organic solvent intermediate products, drugs and antibiotics
Soap and Detergent	Tertiary ammonium compounds, alkalies (e.g. lime or caustic soda)	Flats and fatty acids, glycerol, polyphosphates, sulphonated hydrocarbons.
Food processing		pathogens
Paper and Pulp	Sulphides, bleaching liquors.	organic acids.

Agricultural sources

- Agricultural runoff contains dissolved salts such as **nitrates, phosphates, ammonia** and other nutrients, and toxic metal ions and organic compounds.
- Fertilizers contain major plant nutrients such as **nitrogen, phosphorus** and **potassium**.
- Excess fertilizers may reach the ground water by leaching or may be mixed with surface water of rivers, lakes and ponds by runoff and drainage.
- Pesticides include insecticides, fungicides, herbicides, nematicides, rodenticides and soil fumigants. They contain a wide range of chemicals such as **chlorinated hydrocarbons (CHCs. E.g. DDT, Endosulfan etc.), organophosphates, metallic salts, carbonates, thiocarbonates, derivatives of acetic acid** Many of the pesticides are non-degradable and their residues have long life.
- The animal excreta such as dung, wastes from poultry farms, piggeries and slaughter houses etc. reach the water through runoff and surface leaching during rainy season.

Thermal and Radiation Pollution

- Power plants – thermal and nuclear, chemical and other industries use lot of water for cooling purposes and the used hot water is discharged into rivers, streams or oceans.
- Discharge of hot water may increase the temperature of the receiving water by 10 to 15 °C above the ambient water temperature. This is thermal pollution.
- Increase in water temperature **decreases dissolved oxygen** in water which adversely affects aquatic life.
- Unlike terrestrial organisms, aquatic organisms are adapted to a uniform steady temperature of environment. Sudden rise in temperature kills fishes and other aquatic animals.
- Discharge of hot water in water body affects feeding in fishes, increases their metabolism and affects their growth. Their swimming efficiency declines. Running away from predators or chasing prey becomes difficult. Their resistance to diseases and parasites decreases.
- One of the best methods of reducing thermal pollution is to store the hot water in cooling

ponds, allow the water to cool before releasing into any receiving water body

- Nuclear accidents near water bodies or during natural calamities like tsunami and earthquakes poses the risk of radiation leakage (radiation exposure) into water bodies. E.g. Fukushima Daiichi nuclear disaster.

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Radiation exposure causes **mutations in DNA** of marine organisms. If those mutations are not repaired, the cell may turn cancerous. 135

Radioactive iodine tends to be absorbed by the thyroid gland and can cause thyroid cancer.

Oil Spills

- Oil spills are most glaring of all oceanic pollution.
- The most common cause of oil spill is leakage during marine transport and leakage from underground storage tanks. Oil spill could occur during offshore oil production as well.

Impact of oil spill on marine life

- Oil being lighter than water covers the water surface as a thin film cutting off oxygen to floating plants and other producers.
- Within hours of oil spill, the fishes, shellfish, plankton die due to suffocation and metabolic disorders.
- Birds and sea mammals that consume dead fishes and plankton die due to poisoning. Death of these organisms severely damages marine ecosystems.

Impact of oil spills on terrestrial life

- Bays, estuaries, shores, reefs, beaches particularly near large coastal cities or at the mouth of rivers are relatively more susceptible to the hazards of oil spills.
- A number of coastal activities, especially recreational such as bathing, boating, angling, diving, rafting are affected. As a result tourism and hotel business in the coastal areas suffers seriously.

Invasive species

- Plants of **water hyacinth** are the world's most problematic aquatic weed, also called '**Terror of Bengal**'.



- They grow abundantly in eutrophic water bodies, and lead to an imbalance in the ecosystem dynamics of the water body.
- They cause havoc by their excessive growth leading to stagnation of polluted water.

Underground water pollution

- In India at many places, the ground water is threatened with contamination due to seepage from industrial and municipal wastes and effluents, sewage channels and agricultural runoff.
- Pollutants like fluorides, **uranium**, heavy metals and nutrients like nitrates and phosphates are common in many parts of India.

Q1. Which of the following can be found as pollutants in the drinking water in some parts of India?

- Arsenic
- Sorbitol
- Fluoride
- Formaldehyde
- Uranium

Select the correct answer using the codes given below.

- 1 and 3 only
- 2, 4 and 5 only
- 1, 3 and 5 only
- 1, 2, 3, 4 and 5

Easiest option is 3) Fluoride. So, (b) can be eliminated.

Next easiest option is Arsenic (Heavy metal). Damn!

Nest easiest option is Uranium. Many columns are written on uranium mining and its ill-effects. So, (a) also can be elevated. (a) and (b) eliminated. (c) and (d) remains.

The knowledge of either "Sorbitol" or "Formaldehyde" will tell us the answer.

Formaldehyde is more known than sorbitol. Googling didn't give me any authentic source that called formaldehyde a water pollutant. It is used a preservative to preserve human organs. It finds some application in textile, resign and wood industry.

Answer: c) 1, 3 and 5 only

Sorbitol (glucitol) is a sugar alcohol with a sweet taste which the human body metabolizes slowly.

Marine pollution

- Oceans are the ultimate sink of all natural and manmade pollutants.
- Rivers discharge their pollutants into the sea.
- The sewerage and garbage of coastal cities are also dumped into the sea.
- The other sources of oceanic pollution are navigational discharge of oil, grease, detergents, sewage, garbage and radioactive wastes, off shore oil mining, oil spills.

Ground Water

Titbit: 22 March is celebrated as the world water day.

General Assembly of the United Nations proclaimed the period 2005 – 2015 as the International Decade for action on "Water for life"

Natural stores of water in the global hydrological cycle

Stores	Percentage (%)
Oceans	97.71
Ice caps	1.9
Ground water	0.5
Soil moisture	0.01

Lakes and rivers	0.009
Atmosphere	0.0001

- The moisture in the soil indicates the presence of water underground.
- If we dig deeper and deeper, we would reach a level where all the space between particles of soil and gaps between rocks are filled with water. The upper limit of this layer is called the water table.
- The water table may be at a depth of less than a metre or may be several metres below the ground. The water found below the water table is called groundwater.
- The process of seeping of water into the ground is called infiltration.
- At places the groundwater is stored between layers of hard rock below the water table. This is known as an aquifer.
- The rainwater can be used to recharge the groundwater. This is referred to as water harvesting.

Mahatma Gandhi said: “No one need to wait for anyone else to adopt a humane and enlightened course of action.”

Groundwater Contaminants and Their Effects

Nitrates

- Dissolved nitrates commonly contaminate groundwater.
- Excess nitrate in drinking water reacts with hemoglobin to form non-functional **methaemoglobin**, and impairs oxygen transport. This condition is called **methaemoglobinemia** or **blue baby syndrome**.

Methemoglobin is a form of the oxygen-carrying metalloprotein hemoglobin. Methemoglobin cannot bind oxygen, unlike oxyhemoglobin.

- High level of nitrates may form carcinogens and can accelerate eutrophication in surface waters.

Pathogens

- Poor hygiene of wells may cause pathogenic contamination. Water seepage from solid

waste dumps and municipal drains may also cause pathogenic contamination.

Trace metals

- Include **lead, mercury, cadmium, copper, chromium** and **nickel**. These metals can be toxic and carcinogenic.

Arsenic

- Seepage of industrial and mine discharges, fly ash ponds of thermal power plants can lead to metals in groundwater.
- In India and Bangladesh [Ganges Delta], millions of people are exposed to groundwater contaminated with high levels of arsenic, a highly toxic and dangerous pollutant.
- Chronic exposure to arsenic causes **black foot disease**. It also causes diarrhoea, peripheral neuritis, hyperkeratosis and also lung and skin cancer.

Organic compounds

- Seepage of agricultural runoff loaded with organic compounds like pesticides and may cause pesticide pollution of ground water.

Fluoride

- Excess fluoride in drinking water causes **neuromuscular disorders, gastro-intestinal problems, teeth deformity, hardening of bones and stiff and painful joints (skeletal fluorosis)**.
- Fluorosis is a common problem in several states of the country due to intake of high fluoride content water.
- Fluorides cause dental fluorosis, stiffness of joints (particularly spinal cord) causing humped back.
- Pain in bones and joint and outward bending of legs from the knees is called Knock-Knee syndrome.
- High concentration of fluoride ions is present in drinking water in 13 states of India. The maximum level of fluoride, which the human body can tolerate is 1.5 parts per million (mg/L of water). Long term ingestion of fluoride ions causes fluorosis.

Major Water Issues Of India

Water scarcity

- Due to un-even distribution of rainfall in time and space and ever-increasing demand of water for agricultural, industrial and domestic activities, the water resources are over-exploited. This is resulting in shrinking or even drying up of many water bodies for considerable periods in a year.
- Reducing demands by optimum use, minimization of wastage, efforts to reduce the percolation and evaporation losses, conservation efforts in domestic uses, groundwater recharging, rain water harvesting, afforestation, recycling and reuse are important to combat this problem.

Pathogenic pollution

- Water borne diseases are the most important water quality issues in India. This is mainly due to inadequate arrangements for transport and treatment of wastewaters.

Oxygen depletion

Eutrophication [oxygen depletion due to algal blooms] is a common problem in most of the India lakes and rivers due to discharge of untreated sewage and industrial effluents.

Salinity

- There are number of cases where salinity is increasing in both surface water and groundwater.
- The increase in groundwater salinity is mainly due to increased irrigation activities or sea water intrusion in coastal areas.

Toxic pollution

- Due to discharge of toxic effluents from many industries and increased use of chemicals in agriculture and their subsequent contribution to the water bodies, many water bodies in the country are polluted due to presence of toxic substances.

Ecological health

- A large number of areas in our aquatic environment support rare species of aquatic and amphibious plants and animals and are,

therefore, ecologically very sensitive. They need special protection.

Water Pollution Control Measures

- Realizing the importance of maintaining the cleanliness of the water bodies, the Government of India has passed the **Water (Prevention and Control of Pollution) Act, 1974** to safeguard our water resources.
- An ambitious plan to save the river, called the **Ganga Action Plan** was launched in 1985. It aimed to reduce the pollution levels in the river. However, the increasing population and industrialization have already damaged this mighty river beyond repair.
- In India, the **Central Pollution Control Board (CPCB)**, an apex body in the field of water quality management, has developed a concept of "designated best use".
- Accordingly the water body is designated as A, B, C, D, E on the basis of
 - **pH,**
 - **dissolved oxygen, mg/l**
 - **BOD, (200C) mg/l**
 - **total coliform (MPN/100ml)**
 - **free ammonia mg/l,**
 - **electrical conductivity etc.**
- The CPCB, in collaboration with the concerned State Pollution Control Boards, has classified all the water bodies including coastal waters in the country according to their "designated best uses".
- This classification helps the water quality managers and planners to set water quality targets and identify needs and priority for water quality restoration programmes for various water bodies in the country.
- The famous **Ganga Action Plan** and subsequently the **National River Action Plan** are results of such exercise.
- **Riparian buffers:** A riparian buffer is a vegetated area (a "buffer strip") near a stream, usually forested, which helps shade and partially protect a stream from the impact of adjacent land uses. It plays a key role in increasing water quality in associated streams, rivers, and lakes, thus providing environmental benefits.
- Treatment of sewage water and the industrial effluents before releasing it into water bodies. Hot water should be cooled before release from the power plants.
- Excessive use of fertilizers and pesticides should be avoided. Organic farming and

efficient use of animal residues as fertilizers can replace chemical fertilizers.

- **Water hyacinth** (an **aquatic weed, invasive specie**) can purify water by taking some toxic materials and a number of heavy metals from water.
- Oil spills in water can be cleaned with the help of **bregoli** — a by-product of paper industry resembling saw dust, oil zapper, microorganisms.
- It has been suggested that we should plant **eucalyptus** trees all along sewage ponds. These trees absorb all surplus wastewater rapidly and release pure water vapor into the atmosphere.

Bioremediation

- Bioremediation is the use of microorganisms (bacteria and fungi) to degrade the environmental contaminants into less toxic forms.
- The microorganisms may be indigenous to a contaminated area or they may be isolated from elsewhere and brought to the contaminated site.
- The process of bioremediation can be monitored indirectly by measuring the **Oxidation Reduction Potential or redox** in soil and groundwater, together with pH, temperature, oxygen content, electron acceptor/donor concentrations, and concentration of breakdown products (e.g. carbon dioxide)

In situ bioremediation

- In situ — It involves treatment of the contaminated material at the site.
- **Bioventing**: supply of air and nutrients through wells to contaminated soil to stimulate the growth of indigenous bacteria. It is used for simple hydrocarbons and can be used where the contamination is deep under the surface.
- **Biosparging**: Injection of air under pressure below the water table to increase groundwater oxygen concentrations and enhance the rate of biological degradation of contaminants by naturally occurring bacteria
- **Bioaugmentation**: Microorganisms are imported to a contaminated site to enhance degradation process.

Using bioremediation techniques, TERI has developed a mixture of bacteria called '**Oilzapper and Oilivorous-S**' which degrades the pollutants of oil-contaminated sites, leaving behind no harmful residues. This technique is not only environment friendly, but also highly cost-effective.

Q2. Recently, 'oilzapper' was in the news. What is it?

1. It is an eco-friendly technology for the remediation of oil sludge and oil spills.
2. It is the latest technology developed for under-sea oil exploration.
3. It is a genetically engineered high biofuel yielding maize variety.
4. It is the latest technology to control the accidentally caused flames from oil wells.

Answer: a) Current Affairs based question. Keep track of latest developments.

Ex situ bioremediation

- Ex situ — involves the removal of the contaminated material to be treated elsewhere.
- **Land farming**: contaminated soil is excavated and spread over a prepared bed and periodically tilled until pollutants are degraded. The goal is to stimulate indigenous biodegradative microorganisms and facilitate their aerobic degradation of contaminants.
- **Biopiles**: it is a hybrid of land farming and composting. Essentially, engineered cells are constructed as aerated composted piles. Typically used for treatment of surface contamination with petroleum hydrocarbons.
- **Bioreactors**: it involves the processing of contaminated solid material (soil, sediment, sludge) or water through an engineered containment system.
- **Composting**: Composting is nature's process of recycling decomposed organic materials into a rich soil known as compost.

Advantages and Disadvantages of bioremediation

Advantages of bioremediation

- Useful for the complete destruction of a wide variety of contaminants.

- The complete destruction of target pollutants is possible.
- Less expensive.
- Environment friendly.

Disadvantages of bioremediation

- Bioremediation is limited to those compounds that are **biodegradable**. Not all compounds are susceptible to rapid and complete degradation.
- Biological processes are often highly specific.
- It is difficult to extrapolate from bench and pilot-scale studies to full-scale field operations.
- Bioremediation often takes longer time than other treatment process.

Phytoremediation

- Phytoremediation is use of **plants** to remove contaminants from soil and water. Natural phytoremediation is carried out by mangroves, estuarine vegetation and other wetland vegetation.

Phytoextraction/phytoaccumulation: plants accumulate contaminants into the roots and aboveground shoots or leaves.

- **Phytotransformation/phytodegradation:** uptake of organic contaminants from soil, and their transformation to more stable, less toxic, less mobile form.

- **Phytostabilization:** plants reduce the mobility and migration of contaminated soil. Leachable constituents are adsorbed and bound into the plant structure.

- **Rhizodegradation:** breakdown of contaminants through the activity existing in the rhizosphere (region of soil in the vicinity of plant roots). This activity is due to the presence of proteins and enzymes produced by the plants or by soil organisms such as bacteria, yeast, and fungi.

- **Rhizofiltration:** water remediation technique that involves the uptake of contaminants by plant roots. Rhizofiltration is used to reduce contamination in natural wetlands and estuary areas (E.g. Mangroves).

- **Mycoremediation:** fungi are used to decontaminate the area.

- **Mycofiltration:** using fungal mycelia to filter toxic waste and microorganisms.

- The bacterium **Deinococcus radiodurans** has been used to detoxify toluene and ionic

mercury which are released from radioactive nuclear waste.

Effects of Water Pollution on Health and Environment

Effects of Water Pollution

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Effects of Water Pollution on Human Health

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- Domestic and hospital sewage contain many undesirable pathogenic microorganisms, and its disposal into a water without proper treatment may cause outbreak of serious diseases, such as, amoebiasis dysentery, typhoid, jaundice, cholera, etc.
- Metals like **lead, zinc, arsenic, copper, mercury** and **cadmium** in industrial waste waters adversely affect humans and other animals.
- **Arsenic pollution** of ground water has been reported from West Bengal, Orissa, Bihar, Western U.P. Consumption of such arsenic polluted water leads to accumulation of arsenic in the body parts like blood, nails and hairs causing skin lesions, rough skin, dry and thickening of skin and ultimately **skin cancer**.
- Mercury compounds in waste water are converted by bacterial action into extremely toxic **methyl mercury**, which can cause numbness of limbs, lips and tongue, deafness, blurring of vision and mental derangement.
- Pollution of water bodies by mercury causes **Minamata** (neurological syndrome) disease in humans and **dropsy** in fishes.
- Lead causes **lead poisoning** (Lead interferes with a variety of body processes and is toxic to many organs and tissues). The compounds of lead cause anaemia, headache, loss of muscle power and bluish line around the gum.
- Cadmium poisoning causes cancer of lungs and liver and **Itai - Itai** disease (a painful disease of bones and joints, causes softening of the bones and kidney failure) etc.
- Water contaminated with cadmium can cause itai itai disease also called ouch-ouch disease (a painful disease of bones and joints) and cancer of lungs and liver.

Water Borne Diseases

Disease	Causative organism	Mode of spread	Symptoms
Typhoid	Salmonella typhi	Contaminated food, water, milk, unwashed raw vegetables and flies	Continuous fever which increases day by day Temperature higher in evening than morning, body ache, headache and constipation. Haemorrhage from an ulceration in small intestine
Cholera	Vibrio cholerae	Water or food contaminated by bacteria from stools of cholera patient	Painless diarrhoea, vomiting, 30-40 stools per day which soon becomes typically watery and colourless with flakes of mucous floating in them
Bacterial Diarrhoea	Shigella spp.	Contaminated food, water and by direct personal contact	Diarrhoea, with blood and dysentery mucous in the stools along with severe gripping pain in the abdomen. Stools not too frequent (4-10 per day), faecal matter scanty. Patient looks ill
Leptospirosis	Leptospira	Rodents primary hosts-carry organisms in kidneys. Infection by wading or swimming in water contaminated with rodent urine	Fever, pain in legs, nausea, vomiting are common, congestion of the conjunctival blood vessels around corneas of the eyes
Viral diseases			
Infective Hepatitis	Hepatitis virus	Food and water contaminated with virus in stools	Loss of appetite, nausea, vomiting and diarrhoea, accompanied with fever. Urine dark coloured. Eye and skin appear yellow
Protozoan diseases			
Amoebic dysentery	Entamoeba histolytica	Ingestion of cysts in food and water	Abdominal discomfort and diarrhoea, with or without blood or mucous in stools, fever, chills and gripping pain in abdomen
Diarrhoea	Giardia (=Lamblia) intestinalis	Food or water contaminated with faeces having cysts	Intestinal disorders leading to epigastric pain, abdominal discomfort, loss of appetite, headache and loose bowels
Bilharzia	Schistosoma spp	Cercaria larvae of flukes in water penetrate skin of persons wading in water	Allergy-like itch, rash, aches, fever, eosinophilia etc. When infection becomes heavy, eggs may block arterioles of lungs cardio-pulmonary water causing schistosomiasis and may lead to congestive heart failure.
Guinea worm	Dracunculus medinensis	Unfiltered water	Blister near the ankle, causing allergy and aches

Vector borne diseases related with water

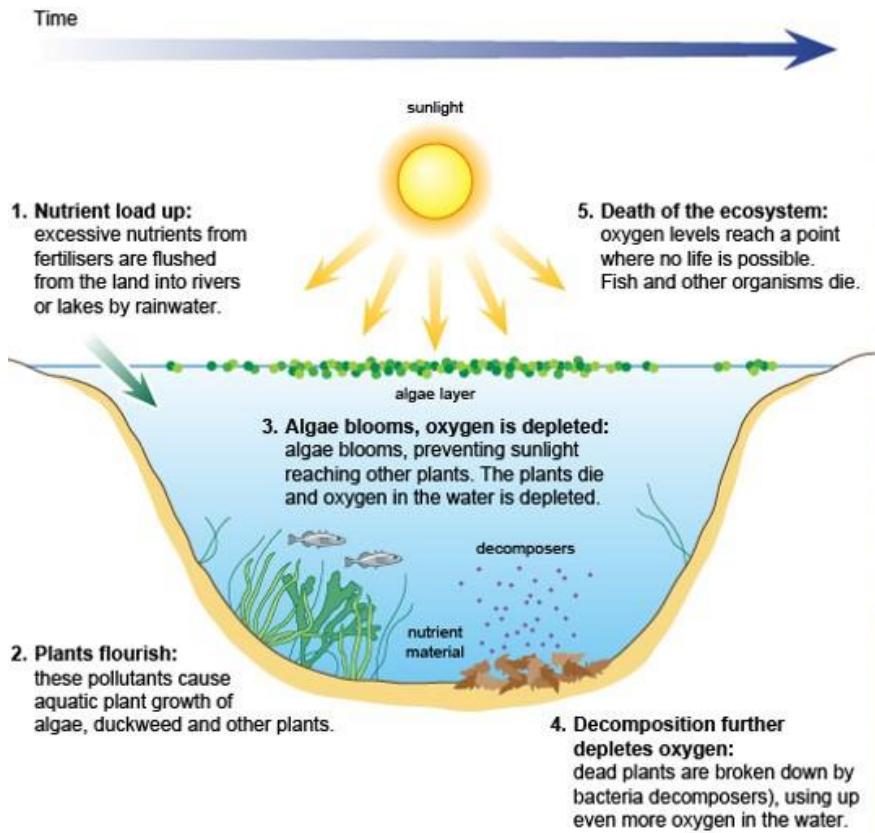
Disease	Causative organisms	Vector	Hosts	Symptoms
Malaria	Plasmodium sp	Female Anopheles (primary or final hosts)	Man (intermediate hosts)	Shivering, chills and sweating. As chills subside body temperature rises as high as 106° F. When temperature comes down patient sweats profusely and becomes comfortable until next attack which takes place at regular intervals
Filaria (Elephantiasis)	Wuchereria (=filaria)	Culex fatigans	Man (final hosts)	Enlargement of limbs and scrotum
Dengue	Barbo-virus	Aedes aegypti	Man (reservoir)	Sudden onset of moderately high fever, excruciating joint pain, intense pain behind eyes, a second rise in temp following brief remission, reduction in neutrophilic white blood cells

Effects of Water Pollution on Environment

- Micro-organisms involved in biodegradation of organic matter in sewage waste consume lot of oxygen, and make water oxygen deficient killing fish and other aquatic creatures.

- Presence of large amounts of nutrients in water results in **algal bloom** [excessive growth of planktonic (free-floating) algae [Harmful

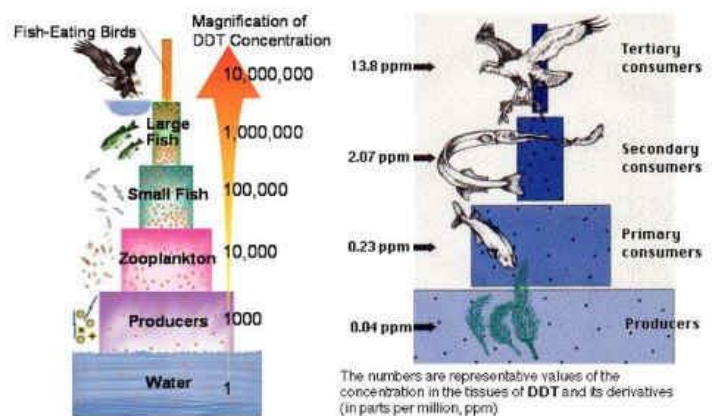
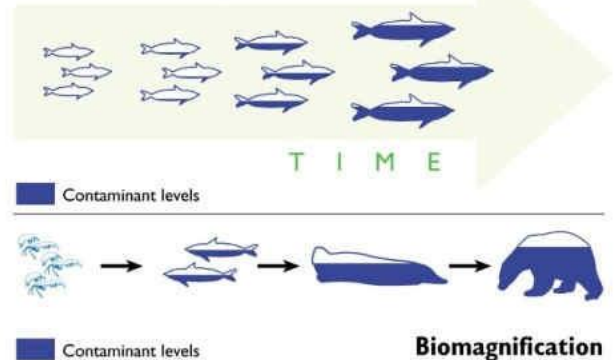
Algal Blooms and **Eutrophication** are explained in the previous post]. This leads to **ageing of lakes**.



A few toxic substances, often present in industrial waste waters, can undergo biological magnification (Biomagnification) in the aquatic food chain. This phenomenon is well-known for **mercury** and **DDT**.

- High concentrations of DDT disturb calcium metabolism in birds, which causes thinning of eggshell and their premature breaking, eventually causing decline in bird populations.
- Thermal wastewater eliminates or reduces the number of organisms sensitive to high temperature, and may enhance the growth of plants and fish in extremely cold areas but, only after causing damage to the indigenous flora and fauna.
- Aquatic organisms take up pesticides from water which get into the food chain and move up the food chain. At higher trophic level they get concentrated and may reach the upper end of the food chain [Biomagnification explained in 'Trophic Levels'].

Bioaccumulation



Effects of Water Pollution on Aquatic Ecosystem

- Polluted water reduces Dissolved Oxygen (DO) content, thereby, eliminates sensitive organisms like plankton, molluscs and fish etc.
- However a few tolerant species like **Tubifex (annelid worm)** and some insect larvae may survive in highly polluted water with low DO content. Such species are recognized as **indicator species** for polluted water.
- **Biocides, polychlorinated biphenyls (PCBs)** and **heavy metals** directly eliminate sensitive aquatic organisms.
- Hot waters discharged from industries, when added to water bodies, lowers its DO content.

Water Conservation and Management

Water Conservation and Management

Primary source of water in India is south-west and north-east monsoons. Monsoon, however, is erratic and amount of rain fall is highly variable in different parts of our country. Hence, surface runoff needs be conserved.

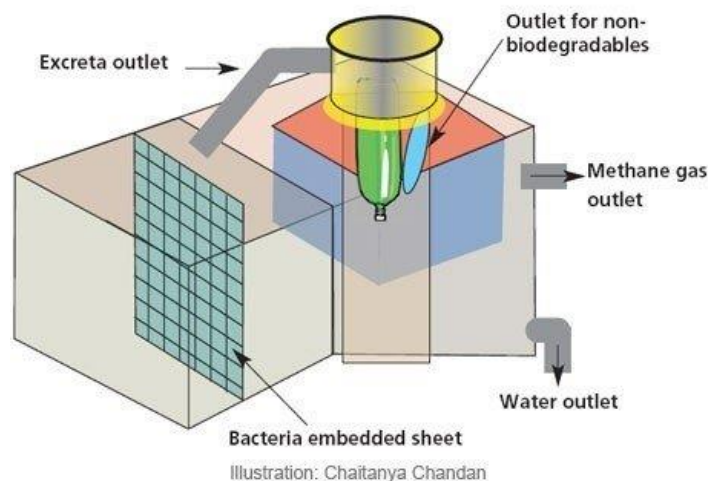
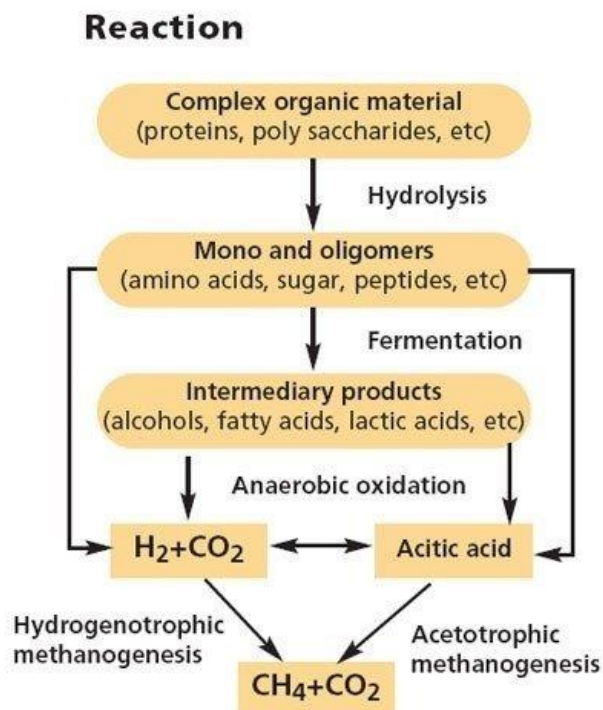
EcoSan toilets

Can you imagine the amount of water that one can save if one didn't have to flush the toilet? Well, this is already a reality. Ecological sanitation is a sustainable system for handling human excreta, using dry composting toilets.

- This is a practical, hygienic, efficient and cost-effective solution to human waste disposal.
- With this composting method, human excreta can be recycled into a resource (as natural fertilizer), which reduces the need for chemical fertilizers. There are working 'EcoSan' toilets in many areas of Kerala and Sri Lanka.

HOW BIO-TOILETS WORK

Bio-toilets have a colony of anaerobic bacteria that converts human waste into water and small amounts of gases. The gases are released into the atmosphere and the water is discharged after chlorination on the tracks



Bio-Toilets

Designed by

- Railways along with **DRDO**.

Why Bio Toilets in Rail?

- Direct discharge of human waste from the existing toilet system in trains causes

corrosion of the tracks, costing crores to replace the rail tracks.

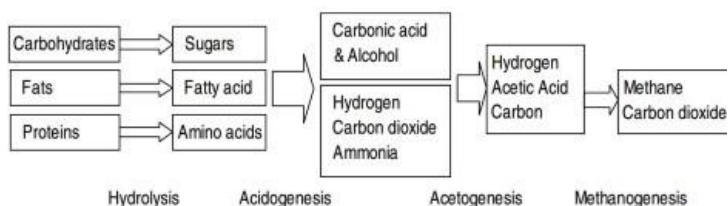
- The bio-toilets are fitted underneath the lavatories and the human waste discharged into them is acted upon by a particular kind of **bacteria** that converts it into non-corrosive **neutral water**.

Terms associated with Bio-Toilets

- Bio-digesters:** The term bio digester is used for the shells made up of steel for the **anaerobic** digestion of human waste.
- Bio tank:** The term bio tank is used for the tanks made up of concrete for the anaerobic digestion of human waste.
- Aerobic Bacteria:** Aerobic bacteria are those which flourish in the presence of free dissolved oxygen in the waste water and consume organic matter for their food, and thereby oxidizing it to stable end products.
- Anaerobic Bacteria:** Anaerobic bacteria flourish in the absence of free dissolved oxygen, and survive by utilizing the bounded molecular oxygen in compounds like nitrates (NO₃) and sulphates (SO₄) etc. thereby reducing them to stable end products along with evolution of foul smelling gases like **H₂S (hydrogen sulphide), CH₄ (methane)**
- Facultative Bacteria:** Facultative bacteria can operate either as aerobically or as anaerobically.
- Anaerobic Microbial inoculums:** mixture of different types of bacteria (hydrolytic, Acidogenic, acetogenic and methanogenic groups) responsible for breakdown of complex polymers into simple sugars which are further broken down into low chain fatty acids and finally into **biogas**.

Anaerobic biodegradation system

- Anaerobic digestion is a collection of processes by which microorganisms break down biodegradable material in the absence of oxygen.



The final waste are Methane and Carbon Dioxide.

Aerobic biodegradation Vs Anaerobic biodegradation

Aerobic biodegradation

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- Forced aeration is essential which is energy intensive.
- Incomplete aeration (partial aerobic condition) leads to foul smell.
- Less effective pathogen inactivation.
- Cannot tolerate detergents
- Generate large amount of sludge.
- Repeated addition of bacteria/enzyme is required for the process.
- Maintenance & recurring cost is high.

Anaerobic biodegradation

- No aeration is required.
- Complete anaerobic conditions.
- More than 99% pathogen inactivation.
- Anaerobes can even degrade detergents/phenyl
- Sludge generation is very less.
- One time bacterial inoculation is enough.
- Minimal maintenance and no recurring cost.

With reference to bio-toilets used by the Indian Railways, consider the following statements:

- The decomposition of human waste in the bio-toilets is initiated by a fungal inoculum.
- Ammonia and water vapour are the only end products in this decomposition which are released into the atmosphere.

Which of the statements given above is/are correct?

- 1 only
- 2 only
- Both 1 and 2
- Neither 1 nor 2

Decomposition of human waste in bio-toilets is carried out by anaerobic bacteria.

The final waste is CO₂ and CH₄.

Answer: d) Neither 1 nor 2

Contour farming

- Contour farming is an example of harvesting technique involving water and moisture control at a very simple level.
- It often consists of rows of rocks placed along the contour of steps. Runoff captured by these barriers also allows for retention of soil, thereby serving as erosion control measure on gentle slopes.
- This technique is especially suitable for areas having rainfall of considerable intensity, spread over large part i.e. in Himalayan area, north east states and Andaman and Nicobar islands.
- In areas where rainfall is scanty and for a short duration, it is worth attempting these techniques, which will induce surface runoff, which can then be stored.

Ground water conservation

Artificial recharge

- Increasing the surface area for percolation, percolation tank construction etc. are some artificial recharge methods.

Catchment area protection (CAP)

- It helps in withholding runoff water albeit temporarily by a check bund constructed across the streams in hilly terrains to delay the run off so that greater time is available for water to seep underground.
- Such methods are in use in north-east states, in hilly areas of tribal belts. This technique also helps in soil conservation. Afforestation in the catchment area is also adopted for water and soil conservation.

Inter-basin transfer of water

- Western and peninsular regions have comparatively low water resources/cultivable land ratio. Northern and eastern region which are drained by Ganga and Brahmaputra have substantial water resources.
- Hence, the scheme of diverting water from region with surplus water to water deficit region can be adopted.
- Ganga-Cauvery link would enable the transfer of vast quantities of Ganga basin flood water

running out to sea, to west and south west India.

- The transfer of the surplus Ganga water would make up for the periodical shortage in Son, Narmada, Godavari, Krishna and Cauvery.

Adoption of drip sprinkler irrigation

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- Surface irrigation methods leads to water loss due to evaporation and percolation.
- Drip irrigation is an efficient method of irrigation in which a limited area near the plant is irrigated by dripping water. This method is particularly useful in row crop.
- Similarly sprinkler method is also suitable for such water scarce areas. About 80% water consumption can be reduced by this method, whereas the drip irrigation can reduce water consumption by 50 to 70 %.

Management of growing pattern of crops

- In water scarce areas, the crop selection should be based on efficiency of the crop to utilize the water. Some of the plants suitable for water scarce areas are:
 1. plants with shorter growth period;
 2. high yielding plants that require no increase in water supply;
 3. plants with deep and well trenched roots and
 4. plants which cannot tolerate surface irrigation.

Selection of crop varieties

- Crop performance and yield are the results of genotype expression as modulated by continuous interactions with the environment.
- Generally, the new varieties of crop do not require more water than the older ones. However, they require timely supply of water because their productivity is high.
- Frequent light irrigation is more conducive than heavy irrigation at large intervals for obtaining high yields.

Nutritional management

- Potassium plays a major role under stress conditions. It improves the tissue water

potential by osmoregulation, ultimately increasing the water use efficiency.

- Experiments conducted at the Water Technology Centre, Coimbatore, indicated that foliar application of 0.5% potassium chloride can reduce the moisture stress in soyabean, sorghum and groundnut.

Role of antitranspirants

- Application of antitranspirants reduces transpiration maintaining thereby the tissue water potential. Plants then take up less water from soil.
- Antitranspirants can prolong the irrigation intervals by slowing down soil water depletion. Application of Kaolin (3%) and lime wash (2%) was found to maintain the water balance of plant and resulted in normal yield of sorghum under moisture stress conditions.
- Certain growth regulators reduce the plants susceptibility to water stress. Application of **cycoel**, a growth retardant increases the ability to withstand drought.
- Cycoel application also reduces production of **gibberellic acid** which leads to closing of stomata. Transpiration loss of water gets reduced.

Reducing evapotranspiration

- Evapotranspiration losses can be reduced by reducing the evaporation from soil surface and transpiration from the plants, in arid zones, considerable amount of water is lost in evaporation from soil surface.
- This can be prevented by placing water tight moisture barriers or water tight mulches on the soil surface. Non-porous materials like papers, asphalt, plastic foils or metal foils can also be used for preventing evaporation losses.
- Transpiration losses can be reduced by reducing air movement over a crop by putting wind breaks and evolving such types of crops which possess xerophytic adaptations.

Recycling of water

- The wastewater from industrial or domestic sources can be used after proper treatment, for irrigation, recharging ground water, and even for industrial or municipal use. If agricultural lands are available close to cities,

municipal waste water can be easily used for irrigation.

Reuse of wastewater

- Wastewater contains lots of nutrients. Its use for irrigation saves these nutrients. It improves the productivity of crops and soil fertility.
- Wastewater is a resource rather than a waste since it contains appreciable amount of nitrogen, phosphorus and potash.
- Stabilization ponds can be used for fish aquaculture. The effluent can also be used for cultivation of short-term and long term, ornamental, commercial and fodder crops.

The potential applications of reusing of treated wastewater are in the following fields or areas:

- Agricultural use through irrigation of crops as well as for improving river amenity;
- Industrial cooling especially in large industrial enterprises;
- Reuse in municipal public areas such as watering lawns, parks, play grounds and trees;
- Flushing toilets in hotels and residential districts;
- Reuse of the treated wastewater for urban landscape purposes.
- Treated waste water can also be used for groundwater recharging.

Grey water reuse

- Grey water is defined as untreated household wastewater, which has not come into contact with toilet waste. It can originate from the shower, bath, bathroom, washing basin, clothes washing machine and laundry trough. Grey water can be used in agriculture and many industries.

Reduce the loss of water due to evaporation

- The methods that reduce evaporation from water bodies are - installing wind breaks, reducing energy available for evaporation, constructing artificial aquifers, minimizing exposed surface through reservoir regulation, reducing ratio of area/volume of water bodies,

locating reservoirs at higher altitudes and applying monomolecular films.

- There are numerous methods to reduce losses due to evaporation and to improve soil moisture. Some of them are listed below:
- Mulching i.e. the application of organic or inorganic materials such as plant debris, compost, etc., slows down the surface run-off, improves soil moisture, reduces evaporation losses and improves soil fertility.
- Soil covered by crops, slow down run-off and minimize evaporation losses, hence, fields should not be left bare for long periods of time.
- Ploughing helps to move the soil around. As a consequence it retains more water thereby reducing evaporation.
- Shelter belt of trees and bushes along the edge of agricultural fields slow down the wind speed and reduce evaporation and erosion.
- Planting of trees, grass, and bushes breaks the force of rain and helps rainwater penetrate the soil.
- Fog and dew contain substantial amounts of water that can be used directly by adapted plant species. Artificial surfaces such as netting-surface traps or polythene sheets can be exposed to fog and dew; the resulting water can be used for crops.
- Contour farming is adopted in hilly areas and in lowland areas for paddy fields. Farmers recognize the efficiency of contour based systems for conserving soil and water.
- Salt-resistant varieties of crops have been also developed recently. Because these grow in saline areas, overall agricultural productivity is increased without making additional demands on fresh water sources. Thus, this is a good water conservation strategy.
- Desalination technologies such as distillation, electro-dialysis and reverse osmosis are available.

Water Treatment for Domestic Use

- Substances that are removed during the process of drinking water treatment include suspended solids, bacteria, algae, viruses, fungi, and minerals such as iron and manganese.
- The processes involved in removing the contaminants include physical processes such as settling and filtration, chemical processes such as disinfection and coagulation and

biological processes such as slow sand filtration.

Coagulation / Flocculation

- **Aluminium sulphate (alum)** is the most common coagulant used for water purification. Other chemicals, such as ferric sulphate or sodium aluminate, may also be used.
- During coagulation, liquid aluminium sulfate (alum) is added to untreated water.
- This causes the tiny particles of dirt in the water to stick together or coagulate.
- Next, groups of dirt particles stick together to form larger particles called flocs.
- Flocs are easier to remove by settling or filtration.

Sedimentation

- As the water and the floc particles progress through the treatment process, they move into sedimentation basins where the water moves slowly, causing the heavy floc particles to settle to the bottom.
- Floc which collects on the bottom of the basin is called sludge, and is piped to drying lagoons.
- In Direct Filtration, the sedimentation step is not included, and the floc is removed by filtration only.

Filtration

- Water flows through a filter designed to remove particles in the water. The filters are made of layers of sand and gravel, and in some cases, crushed anthracite.
- Filtration collects the suspended impurities in water and enhances the effectiveness of disinfection. The filters are routinely cleaned by backwashing.

Disinfection

- Water is disinfected before it enters the distribution system to ensure that any disease-causing bacteria, viruses, and parasites are destroyed.
- Chlorine is used because it is a very effective disinfectant, and residual concentrations can be maintained to guard against possible biological contamination in the water distribution system.

- The addition of chlorine or chlorine compounds to drinking water is called chlorination.
- Chlorine can combine with certain naturally occurring organic compounds in water to produce **chloroform** and other potentially harmful byproducts. The risk of this is very small, however, when chlorine is applied after coagulation, sedimentation, and filtration.
- Ozone gas may also be used for disinfection of drinking water. However, since ozone is unstable, it cannot be stored and must be produced on-site, making the process more expensive than chlorination.
- Ozone has the advantage of not causing taste or odour problems. It also leaves no residue in the disinfected water.
- The lack of an ozone residue, however, makes it difficult to monitor its continued effectiveness as water flows through the distribution system.

Sludge Drying

- Solids that are collected and settled out of the water by sedimentation and filtration are removed to drying lagoons.

Fluoridation

- Water fluoridation is the treatment of community water supplies for the purpose of adjusting the concentration of the free fluoride ion to the optimum level sufficient to reduce dental caries.
- Fluoride is generally present in all natural water. Its concentration up to certain level is not harmful. Beyond that level, the bones start disintegrating. This disease is called **fluorosis**.
- We have fluoride problem in many parts of our country. Bureau of Indian Standards prescribes 1.0 mg/l as desirable and 1.5 mg/l as maximum permissible limit for drinking water.
- Defluoridation at domestic level can be carried out by mixing water for treatment with adequate amount of aluminum sulphate (alum) solution, lime or sodium carbonate and bleaching powder depending upon its alkalinity (concentration of bicarbonates and carbonates in water) and fluoride contents.

pH Correction

- Lime is added to the filtered water to adjust the pH and stabilize the naturally soft water in order to minimise corrosion in the distribution system, and within customers' plumbing.

Titbit: National Environmental Engineering Research Institute (NEERI) is at Nagpur.

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Removal of iron

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- In many parts of our country we have problem of excess iron in drinking water especially in North-East regions. Iron causes bad taste and odour to the drinking water. Bureau of Indian Standards prescribes desirable limit for iron as 0.3 mg/l.
- A major part of iron is oxidized. Then the water is made to react with oxidizing media (lime stone). By aeration and further oxidation the dissolved iron is converted to insoluble ferric hydroxide. The insoluble iron can thus be easily removed through filtration.

Removal of arsenic

- Arsenic is found in ground water in some parts of West Bengal. Arsenic is highly toxic in nature. It may cause a number of skin disorders or even cancer. Bureau of Indian Standards prescribes desirable limit for arsenic as 0.05 mg/l.
- Removal of arsenic is essential.
- Bleaching powder and alum are used for removal of arsenic.

Watershed Management

- Watershed is an area that contribute water to a stream or a water body through run-off or underground path.
- That is the region from which surface water draws into a river, a lake, wet land or other body of water is called its watershed or drainage basin.
- Watershed management is a technique for conservation of water and soil in a watershed.
- The presence of water in soil is essential for the growth of plants and vegetation. Forests and their associated soils and litter layers are excellent filters as well as sponges, and water that passes through this system is relatively pure.
- Various kinds of forest disturbances can speed up the movement of water from the

system and in effect, reduce the filtering action.

- In mountainous terrain the forests play a prominent role in prevention of soil erosion.
- Erosion threat can be tackled by the maintenance of continual cover. Ideally, this is achieved by single stem harvesting; only one tree is felled at any one point, and the small gap so created is soon sealed by the outward growth of its neighbors.
- Despite the uncertain balance of water gain and loss, forests offer the most desirable cover for water management strategies.
- In contrast to the rapid flows of short duration characteristics of sparsely vegetated land water yields are gradual, reliable and uniform in forests. Deforested land sheds water swiftly, causing sudden rises in the rivers below.
- Over a large river system, such as that of the Ganga and the Yamuna, forests are a definite advantage since they lessen the risk of floods. They also provide conditions more favourable to fishing and navigation than does un-forested land.
- All natural streams contain varying amounts of dissolved and suspended matter, although streams contain varying amounts of dissolved and suspended matter, although streams issuing from undisturbed watershed are ordinarily of high quality.
- Waters from forested areas are not only low in foreign substances, but they also are relatively high in oxygen and low in unwanted chemicals.
- The belief that forests increase rainfall has not been substantiated by scientific inquiry. Local effects can, however, prove substantial, particularly in semiarid regions where every millimeter of rain counts.
- The air above a forest, as contrasted with grassland, remains relatively cool and humid on hot days, so that showers are more frequent.
- Many areas in India used to get significant rainfall when they were forested are now facing severe draught due to denudation (example Rajasthan desert).

Individual And Community Role

- Building check dams on seasonal rivers.
- Micro-watershed management project.
- Constructing percolation tanks in every residential and industrial complex.

- Roof top rain water collection for drinking purposes. [2/3rds of water is wasted in RO – reverse osmosis filtration]

Government's efforts on water conservation

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- National Water Policy 2002 strongly | emphasize conservation of water.
- Construction of large number of dams on various river systems.
- Interlinking of rivers (proposed).
- Promotion of bunds at village level.
- Promotion of rain water harvesting.
- Promotion of reuse and recycling of wastewater.
- Steps to protect water quality.
- Drought-proofing the future.

Permanent conservation measures may include:

- Subsidizing use of water-efficient faucets, toilets and showerheads
- Public education and voluntary use reduction.
- Billing practices that impose higher rates for higher amounts of water use
- Building codes that require water-efficient fixtures or appliances
- Leak detection surveys and meter testing, repair and replacement
- Reduction in use and increase in recycling of industrial water

Temporary cutbacks may include:

- Reduction of system-wide operating pressure
- Water use bans, restrictions, and rationing
- Strengthening of local or municipal bodies could help addressing the issue of water shortage and its management in cities.

Q3. Which one among the following industries is the maximum consumer of water in India?

- Engineering
- Paper and pulp
- Textiles
- Thermal power**

Water consumption by various industries

1. Thermal power

2. Textile industry
3. Paper and Pulp
4. Iron and Steel industry
5. Fertilizer Industry

Radioactive Pollution - Ionizing and Non-Ionizing Radiation Effects

Radioactive Pollution

Sources

Artificial Sources of Radioactive pollution

- Accidents in nuclear power plants and nuclear waste.
- Nuclear weapon testing and explosion (Nuclear fallout). The fall Out contains radioactive substances such as **strontium-90, cesium-137, iodine-131**, etc.
- Uranium mining and mining of other radioactive material like **thorium** Uranium contamination is well observed in India.
- Radiation therapy and direct exposures to radiation for diagnostic purposes (e.g. X-rays), chemotherapy etc.
- The slow nuclear radiations can emanate from a variety of sources viz. nuclear reactors, laboratories, hospitals, and direct exposures to X-rays etc.

Natural Sources

- They include cosmic rays from space and terrestrial radiations from radio-nuclides present in earth's crust such as **radium-224, uranium-238, thorium-232, potassium-40, carbon-14**, etc.
- Some species of animals and plants preferentially accumulate specific radioactive materials. For example, oysters deposit ^{65}Zn , fish accumulate ^{55}Fe , marine animals selectively deposit ^{90}Sr .

Effects of Radioactive pollution

- The use of nuclear energy has two very serious inherent problems.
- 1. accidental leakage, as occurred in the **Three Mile Island, Chernobyl and Fukushima incidents** and
- 2. safe disposal of radioactive wastes. It has been recommended that storage of nuclear waste,

after sufficient pre-treatment, should be done in suitably shielded containers buried within the rocks, about 500 m deep below the earth's surface. However, this method of disposal is meeting stiff opposition from the public.

- The quick devastating and immediate effects of nuclear radiations are well known as witnessed following Hiroshima and Nagasaki in Japan during world war II.
- Continued small dose exposure to nuclear radiation can cause **childhood leukemia, miscarriage, underweight babies, infant deaths, increased susceptibility to AIDS and other immune disorders and increased criminalities.**
- Underground bomb testing releases radiations in very small doses of radicals that pollutes water and soil.
- This radioactive water is taken by plants through roots. The radioactivity enters food chain when such plants are eaten by animals and humans. Such radioactivity has been detected even in the milk.
- Radiation, that is given off by nuclear waste is extremely damaging to biological organisms, because it causes mutations to occur at a very high rate.
- At high doses, nuclear radiation is lethal but at lower doses, it creates various disorders, the most frequent of all being cancer.

Ionizing and Non-Ionizing Radiation

- Radioactivity is a phenomenon of spontaneous emission of **proton (alpha-particles), electrons (beta-particles)** and **gamma rays (short wave electromagnetic waves)** due to **disintegration of atomic nuclei** of some elements. These cause radioactive pollution.
- Radiations can be categorized into two groups namely the non-ionizing radiations and the ionizing radiations.

Non-ionizing radiations

- Non-ionizing radiations are constituted by the electromagnetic waves at the **longer wavelength** of the spectrum ranging from near **infra-red rays to radio waves** [include **higher wavelength ultraviolet rays, microwaves**].
- These waves have energies enough to excite the atoms and molecules of the medium

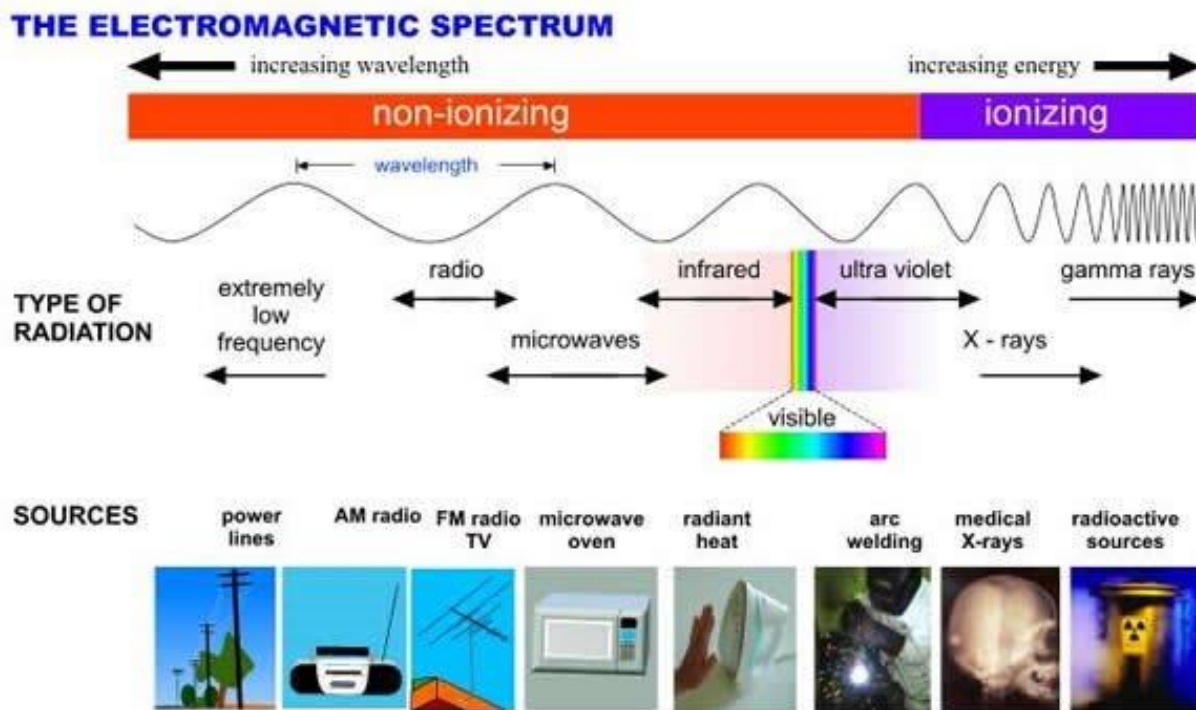
through which they pass, causing them to vibrate faster but **not strong enough to ionize them**.

- In a microwave oven the radiation causes water molecules in the cooking medium to vibrate faster and thus raising its temperature.

- They may damage eyes which may be caused by **reflections from coastal sand**, snow (**snow blindness**) directly looking towards sun during eclipse.
- They injure the cells of skin and blood capillaries producing blisters and reddening called sunburns.

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Ionizing radiations

- Ionizing radiations cause **ionization** (one or more electrons are peeled out from the outer shells of an atom) of atoms and molecules of the medium through which they pass.

Ionization is the process by which an atom or a molecule acquires a negative or positive charge by gaining or losing electrons to form ions, often in conjunction with other chemical changes.

- Electromagnetic radiations such as **short wavelength ultra violet radiations (UV)**, **X-rays** and **gamma rays** and energetic particles produced in nuclear processes, electrically charged particles like **alpha and beta particles** produced in radioactive decay and **neutrons** produced in nuclear fission, are highly damaging to living organisms.
- Electrically charged particles produced in the nuclear processes can have sufficient energy to knock electrons out of the atoms or

molecules of the medium, thereby producing ions.

- The ions produced in water molecules, for example, can induce reactions that can **break bonds** in proteins and other important molecules.
- An example of this would be when a gamma ray passes through a cell, the water molecules near the DNA might be ionized and the ions might react with the DNA causing it to break.
- They can also cause chemical changes by breaking the chemical bonds, which can damage living tissues.
- Short range effects include burns, impaired metabolism, dead tissues and death of the organisms.
- Long range effects are **mutations** increased incidence of **tumors and cancer**, shortening of life-span and developmental changes.

Non-ionizing radiations affect only those components which absorb them and have low penetrability.

Ionizing radiations have high penetration power and cause breakage of macro molecules.

Biological Damage Due to Ionizing Radiations

- The biological damage resulting from ionizing radiations is generally termed as **radiation damage**.
- Large amounts of radiation can kill cells that can dramatically affect the exposed organism as well as possibly its offspring.
- Affected cells can mutate and result in cancer. A large enough dose of radiation can kill the organism.
- Radiation damage can be divided into two types: (a) **somatic damage** (also called **radiation sickness**) and (b) **genetic damage**.
- Somatic damage refers to damage to cells that are not associated with reproduction.
- Effects of somatic radiation damage include reddening of the skin, loss of hair, ulceration, **fibrosis of the lungs**, the formation of holes in tissue, a **reduction of white blood cells**, and the induction of **cataract in the eyes**. This damage can also result in cancer and death.
- Genetic damage refers to damage to cells associated with reproduction. This damage can subsequently cause genetic damage from **gene mutation** resulting in abnormalities. Genetic damages are passed on to next generation.

Radiation dose

- The biological damage caused by the radiation is determined by the intensity of radiation and duration of the exposure.
- It depends on the amount of energy deposited by the radiation in the biological system.
- For example, alpha particles (protons) do much more damage per unit energy deposited than do beta particles (electrons).
- A traditional unit of human-equivalent dose is the **rem**, which stands for radiation equivalent in man.
- At low doses, such as what we receive every day from background radiation (<1 m rem), the cells repair the damage rapidly.
- At higher doses (up to 100 rem), the cells might not be able to repair the damage, and

the cells may either be changed permanently or die.

- Cells changed permanently may go on to produce abnormal cells when they divide and may become cancerous.
- At even higher doses, the cells cannot be replaced fast enough and tissues fail to function. An example of this would be "**radiation sickness**." This is a condition that results after high doses is given to the whole body (>100 rem).

Damage due to radiation particles

- Alpha particles, can be blocked by a piece of paper and human skin.
- Beta particles can penetrate through skin, while can be blocked by some pieces of glass and metal.
- Gamma rays can penetrate easily to human skin and damage cells on its way through, reaching far, and can only be blocked by a very thick, strong, massive piece of concrete.

Half-Life - Period of Radioactivity

- Each radioactive material has a constant decay rate. Half-life is the time needed for half of its atoms to decay.
- Half-life of a radio nuclide refers to its period of radioactivity. The half-life may vary from a fraction of a second to thousands of years.
- The radio nuclides with **long half-time** are the chief source of environmental radioactive pollution.

Accidents at nuclear power plants

- Nuclear fission in the reactor core produces lot of heat which if not controlled can lead to a meltdown of fuel rods in the reactor core.
- If a meltdown happens by accident, it will release large quantities of highly dangerous radioactive materials in the environment with disastrous consequences to the humans, animals and plants.
- To prevent this type of accidents and reactor blow up, the reactors are designed to have a number of safety features. Inspire of these safety measures three disasters in the nuclear power plants are noteworthy – **Three Mile Island' in Middletown (U.S.A.) in 1979, Chernobyl (U.S.S.R.) in 1986 and**

Fukushima Daiichi nuclear disaster in 2011.

- In the first two cases a series of mishaps and errors resulted in over heating of the reactor core and lot of radiation was released into the environment.
- The leakage from Three Mile Island reactor was apparently low and no one was injured immediately. However, in case of Chernobyl the leakage was very heavy causing death of some workers and radiation spread over large areas scattered all over Europe.
- The latest one – Fukushima Daiichi nuclear accident was triggered by an earthquake.
- Other important nuclear power plant disasters include Chalk river, Canada, Windscale Plutonium Production Center, U.K and Monju, Japan.
- Accidents with nuclear submarines and nuclear warships is a possibility.

Safe Disposal Of Nuclear Wastes

- Radioactive wastes are of two types
 1. low level radioactive wastes (LLW) which include civilian applications of radionuclides in medicine, research and industry, materials from decommissioned reactors, protection clothing worn by persons working with radioactive materials or working in nuclear establishments.
 2. High level radioactive wastes (HLW) results from spent nuclear fuel rods and obsolete nuclear weapons.

Some proposed methods of disposing nuclear waste are:

- bury it deep underground in insulated containers. This is a strategy being pursued in United States.
- shoot it into the space or into the sun. The cost would be very high and a launch accident should be disastrous.
- bury it under the ice sheet of Antarctica or Greenland ice cap. The ice could be destabilized by heat from the waste. The method has been prohibited by international law.
- dump it into deep oceans by keeping the waste into glass and steel cases. But the containers might leak and contaminate the ocean.

- change it into harmless or less harmful isotopes. Currently no method is known to do that and the method would be too costly.
- presently waste fuel rods are being stored in special storage ponds at reactor sites or sent to reprocessing plants. Even though reprocessing is more expensive but some countries use reprocessing as an alternative to waste storage.

Preventive/Control Measures

- Prevention is the best control measure as there is no cure available for radiation damage.
- All safety measures should be strictly enforced. UN should have more powers to perform safety checks in various nuclear establishments across the world.
- Worldwide monitoring of radiation leakage should be a priority.
- Proper technologies should be developed to prevent contamination of water and soil by radioactive waste and radioactive materials.
- More avenues for safe disposal of radioactive must be worked out.
- Regular monitoring through frequent sampling and quantitative analysis in domestic nuclear establishments.
- Appropriate steps should be taken to protect from occupational exposure.
- Gradually decreasing the share of nuclear power is a necessity.
- World must unite to ban production and use of nuclear weapons.

Impact Of Radiation From Mobile Phone Towers

- The radiation that comes from mobile tower radiation is non-ionizing radiation.

Health Impacts

- Every antenna on cell phone tower radiates electro-magnetic radiation (power).
- One cell phone tower is being used by a number of operators, more the number of antennas more is the power intensity in the nearby area.
- The power level near towers is higher and reduces as we move away.
- EMR may cause cellular and psychological changes in human beings due to thermal

effects that are generated due to absorption of microwave radiation.

- The exposure can lead to genetic defects, effects on reproduction and development, Central Nervous System behavior etc.
- EMR can also cause non thermal effects which are caused by radio frequency fields at levels too low to produce significant heating and are due to movement of calcium and other ions across cell membranes.
- Such exposure is known to be responsible for fatigue, nausea, irritability, headaches, loss of appetite and other psychological disorders.
- The current exposure safety standards are purely based on the thermal effects considering few evidences from exposure to non-thermal effects.

Impact on birds

- The surface area of bird is relatively larger than their body weight in comparison to human body so they absorb more radiation.
- Also the fluid content in the body of the bird is less due to small body weight so it gets heated up very fast.
- Magnetic field from the towers disturbs birds' navigation skills hence when birds are exposed to EMR they disorient and begin to fly in all directions.
- A large number of birds die each year from collisions with telecommunication masts.

What are the responsibilities of Stakeholders?

MoEF

- The MoEF has to notify the impacts of communication towers on wildlife and human health to the concerned agencies for regulating the norms for notification of standards for safe limit of EMR.

State/Local Bodies:

- Regular monitoring and auditing in urban localities/educational/hospital/industrial/residential/recreational premises including the Protected Areas and ecologically sensitive areas.
- Carry out an 'Ecological Impact Assessment' before giving permission for construction of

towers in wildlife and ecologically important areas.

State Environment and Forest Department

- State Environment and Forest Department are entrusted with the task of providing regular awareness among the people about the norms on cell phone towers and dangers of EMR from them.

Department of Telecommunications

- Avoid overlapping of high radiation fields. New towers should not be permitted within a radius of one kilometer of the existing tower.
- The location and frequencies of cell phone towers and other towers emitting EMR should be made available in the public domain GIS mapping of all the cell phone towers to be maintained to monitor the population of bird and bees in and around the wildlife protected area and the mobile towers.
- Need to refine the Indian standard on safe limits of exposure to EMR, keeping in view the available literature on impacts on various life forms.
- To undertake Precautionary approaches to minimize the exposure levels and adopt stricter norms perennial, that live longer than agricultural crops.
- Any study conducted on impact of EMF radiation on wildlife needs to be shared to facilitate appropriate policy formulations.

Soil Pollution

- Soil pollution is defined as the 'addition of substances to the soil, which adversely affect physical, chemical and biological properties of soil and reduces its productivity.
- It is build-up of persistent toxic compounds, chemicals, salts, radioactive materials, or disease causing agents in soil which have adverse effects on plant growth, human and animal health.

Causes and Sources of Soil Pollution

Plastic bags

- They accumulate in soil and prevents germination of seeds. They stay in soil for

centuries without decomposing (non-biodegradable).

- Burning of plastic in garbage dumps release highly toxic and poisonous gases like **carbon monoxide, carbon dioxide, phosgene, dioxins** and other poisonous **chlorinated compounds**.
- Toxic solid residue left after burning remains in soil. The harmful gases enters soils through chemical cycles.

Industrial sources

- They includes **fly ash, metallic residues, mercury, lead, copper, zinc, cadmium, cynides, thiocynates, chromates, acids, alkalies, organic substances, nuclear wastes**
- Large number of industrial chemicals, dyes, acids, etc. find their way into the soil and are known to create many health hazards including cancer.

Pesticides

- Pesticides are chemicals that include insecticides, fungicides, algicides, rodenticides, weedicides sprayed in order to improve productivity of agriculture, forestry and horticulture.
- **Chlorohydrocarbons (CHCs) like DDT, endosulfan, heptachlor** accumulate in soil and cause biomagnification. Some of these pesticides like DDT and endosulfan are banned by most of the countries.

Fertilizers and manures

- Excessive use of chemical fertilizers **reduces the population of soil borne organisms** and the crumb structure of the soil, productivity of the soil and increases salt content of the soil.

Discarded materials

- It includes concrete, asphalt, rungs, leather, cans, plastics, glass, discarded food, paper and carcasses.

Radioactive wastes

- Radioactive elements from mining and nuclear power plants, find their way into water and then into the soil.

Other pollutants

- Many air pollutants (acid rain) and water pollutants ultimately become part of the soil and the soil also receives some toxic chemicals during weathering of certain rocks.

Effects of soil pollution

Agriculture

- Reduced soil fertility due to increase in alkalinity, salinity or pH.
- Reduced crop yield due to reduced fertility.
- Reduced nitrogen fixation due to the reduced number of nitrogen fixers.
- Increased erosion due to loss of forests and other vegetation.
- Run off due to deforestation cause loss of soil and nutrients.
- Deposition of silt in tanks and reservoirs due to soil erosion.

Health

- Health effects are similar to effects of water pollution.

Environment

- Ecological imbalance.
- Foul smell and release of gases.
- Waste management problems.

Control measures

More or less same as for water pollution

- Indiscriminate disposal of solid waste should be avoided.
- To control soil pollution, it is essential to stop the use of plastic bags and instead use bags of degradable materials like paper and cloth.
- Sewage should be treated properly before using as fertilizer and as landfills.
- The organic matter from domestic, agricultural and other waste should be segregated and subjected to vermicomposting which generates useful manure as a byproduct.
- The industrial wastes prior to disposal should be properly treated for removing hazardous materials.

- Biomedical waste should be separately collected and incinerated in proper incinerators.
- Use of bio pesticides, bio fertilizers. Organic farming.
- Four R's: Refuse, Reduce, Reuse, and Recycle.
- Afforestation and Reforestation.
- Solid waste treatment.
- Reduction of waste from construction areas.

Noise Pollution

- Noise by definition is "sound without value" or "any noise that is unwanted by the recipient".
- Noise level is measured in terms of decibels (dB). An increase of about 10 dB is approximately double the increase in loudness.
- H.O. (World Health Organization) has prescribed optimum noise level as 45 dB by day and 35 dB by night. Anything above 80 dB is hazardous.

Effects of noise pollution

- Noise pollution leads to irritation, increased blood pressure, loss of temper, mental depression and annoyance, decrease in work efficiency, loss of hearing which may be first temporary but can become permanent in the noise stress continues.

Prevention and control of noise pollution

- Road traffic noise can be reduced by better designing and proper maintenance of vehicles.
- Noise abatement measures include creating noise mounds, noise attenuation walls and well maintained roads and smooth surfacing of roads.
- Retrofitting of locomotives, continuously welded rail track, use of electric locomotives or deployment of quieter rolling stock will reduce noises emanating from trains.
- Air traffic noise can be reduced by appropriate insulation and introduction of noise regulations for takeoff and landing of aircrafts at the airport.
- Industrial noises can be reduced by sound proofing equipment like generators and areas producing lot of noise.
- Power tools, very loud music and land movers, public functions using loudspeakers, etc.

should not be permitted at night. Use of horns, alarms, refrigeration units, etc. is to be restricted. Use of fire crackers which are noisy and cause air pollution should be restricted.

- A **green belt of trees** is an efficient noise absorber.

Ambient Noise Level Monitoring

- Noise Pollution (Control and Regulation) Rules, 2000 define ambient noise levels for various areas as follows:

Category of Area/Zone	Limits in dB(A) Leq	
	Day Time	Night Time
	6 a.m. to 10 p.m.	10 p.m to 6 a.m
Industrial Area	75	70
Commercial Area	65	55
Residential Area	55	45
Silence Zone	50	40

- The Government of India on Mar 2011 launched a Real time Ambient Noise Monitoring Network.
- Under this network, in phase-I, five Remote Noise Monitoring Terminals each have been installed in different noise zones in seven metros (Delhi, Hyderabad, Kolkata, Mumbai, Bangalore, Chennai and Lucknow).
- In Phase II another 35 monitoring stations will be installed in the same seven cities. Phase III will cover installing 90 stations in 18 other cities.
- Phase-III cities are Kanpur, Pune, Surat, Ahmedabad, Nagpur, Jaipur, Indore, Bhopal, Ludhiana, Guwahati, Dehradun, Thiruvananthapuram, Bhubaneswar, Patna, Gandhinagar, Ranchi, Amritsar and Raipur.
- **Silence Zone** is an area comprising not less than 100 metres around hospitals, educational institutions, courts, religious places or any other area declared as such by a competent authority.

Solid Hazardous E - Waste & Heavy Metal Toxicity

Solid Wastes

- Solid wastes or municipal solid wastes generally comprise paper, food wastes,

plastics, glass, metals, rubber, leather, textile, etc.

- Open-burning reduces the volume of the wastes, although it is generally not burnt to completion and open dumps often serve as the breeding ground for rats and flies.
- Sanitary landfills were adopted as the substitute for open-burning dumps. In a sanitary landfill, wastes are dumped in a depression or trench after compaction, and covered with dirt every day.
- Landfills are also not really much of a solution since the amount of garbage generation specially in the metros has increased so much that these sites are getting filled too.
- Also there is danger of seepage of chemicals, etc. from these landfills polluting the underground water resources.

Effects of Plastic Waste

- Conventional plastics, right from their manufacture to their disposal are a major problem to the environment.
- The land gets littered by plastic bag garbage and becomes ugly and unhygienic.
- Conventional plastics have been associated with **reproductive problems** in both humans and wildlife.
- **Dioxin** (highly carcinogenic and toxic) byproduct of the manufacturing process is one of the chemicals believed to be passed on through **breast milk** to the nursing infant.
- Burning of plastics, especially **PVC** releases dioxin and also **furan** into the atmosphere.

Dioxins are environmental pollutants. They belong to the so-called “dirty dozen” - a group of dangerous chemicals known as **persistent organic pollutants** (POPs).

The name "dioxins" is often used for the family of structurally and chemically related **polychlorinated dibenzo para dioxins (PCDDs)** and **polychlorinated dibenzofurans (PCDFs)**.

- Plastic bags can also contaminate foodstuffs due to leaching of toxic dyes and transfer of pathogens.
- Careless disposal of plastic bags chokes drains, blocks the porosity of the soil and causes problems for groundwater recharge.
- Plastic disturbs the soil microbe activity. The terrestrial and aquatic animals misunderstand

plastic garbage as food items, swallow them and die.

- Plastic bags deteriorates soil fertility as it forms part of manure and remains in the soil for years.

Designing eco-friendly, biodegradable plastics is the need of the hour.

Remedy for Plastic Waste

- **Polyblend** is a fine powder of recycled and modified plastic waste. This mixture is mixed with the **bitumen** that is used to lay roads.
- Blends of Polyblend and bitumen, when used to lay roads, enhanced the bitumen's water repellent properties, and helped to increase road life by a factor of three.

Industrial solid waste

- Thermal power plants producing coal ash/fly ash;
- The integrated iron and steel mills producing blast furnace slag;
- Non-ferrous industries like aluminium, copper and zinc producing **red mud** and tailings;
- Sugar industries generating **press mud**;
- Pulp and paper industries producing **lime mud**;
- Fertilizer and allied industries producing **gypsum**;

Recoverable Matter

- Pulp and paper: Ligno-sulphate, sodium salts.
- Textile: Caustic soda.
- Distillery: Potassium salts, yeast Fertilizer (phosphatic) Calcium sulphate, fluoride.
- Coke oven: Ammonia, ammonium sulphate, tar, naphthalene, phenol.

Concept Of Cleaner Technologies

- One way to emulate nature is to recycle and reuse the chemicals used in industries instead of dumping them into the environment.
- Industries may interact in such a way that they establish a “resource exchange” programme in which waste of one industry or manufacturer is utilized as raw material by another-industry- similar to food web in nature.

- Use of CNG by automobiles instead of petrol, as an automobile fuel, is an example of cleaner technology which has reduced pollution of the environment.
- Instead of throw away economy which creates huge amount of waste, the manufacturers can make more money if their product is redesigned so that it uses minimum amount of raw materials lasts longer, easy to maintain, repair, remanufacture, reuse or recycle.

Hazardous Waste

- Any substance that is present in the environment or released into the environment causing substantial damage to public health and welfare of the environment is called hazardous substance.
- Any hazardous substance could exhibit any one or more of the following

characteristics: **toxicity**, **ignitability**, **corrosivity** or **reactivity** (explosive). Thus, any waste that contains hazardous or very hazardous substance is called hazardous waste.

- Hazardous wastes can originate from various sources such as: house-hold, local areas, urban, industry, agriculture, construction activity, hospitals and laboratories, power plants and other sources.
- The hazardous waste when disposed of release a number of environmentally unfriendly substance(s).
- Hospitals generate hazardous wastes that contain disinfectants and other harmful chemicals, and also pathogenic micro-organisms. Such wastes also require careful treatment and disposal. The use of **incinerators** (destroy, especially waste material, by burning) is crucial to disposal of hospital waste.

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Source	Disposed/used as	Polluting agent	Effect
Industrial waste	Incineration of waste	Toxic fumes e.g. Chlorine, polyvinylchloride	Chlorine could cause acid rain
	Incomplete combustion	Dioxins/organochlorides	Carcinogenic
	Release into water bodies	Chlorophenol, fluorine compounds, aldehydes , SO ₂ , CO	Cause environmental pollution
	Plastic	Polythene, poly propylene, polyesters etc. on burning release gases	Toxic, ecological pollution
Nuclear waste	Hospitals Laboratories	Slow/sustained in medical/agriculture use	Health hazard, carcinogenic, mutation
Agricultural waste	Forms of Nitrogen wastes	Manure/Dung rich in NO ₃ /NO ₂ ²	Accumulate in vegetables, cause methanoglobinemia cyanosis
		Nitrosamines/ NO ₃ /NO ₁	Carcinogenic contribute to acid rain
		N ₂ O – Nitrous Oxide	Greenhouse effect [NO _x – NO, NO ₂ are global coolers , N ₂ O is greenhouse gas]
		NH ₃ ⁺ (from livestock breeding)	Affect aquatic life; stimulate fungal growth; epiphytes; cause weathering of forests
	Phosphates		Eutrophication of aquatic environment
	Phytosanitary product	Insecticides/pesticides/fungicides/herbicides	Enter soil as run off, polluter water table affect aquatic life, carcinogenic, renal failure
	Methane	Ruminating cattle, fermentation of organic matter	Powerful greenhouse effect

Stockholm Convention on Persistent Organic Pollutants

- Stockholm Convention is an international environmental treaty
- Came into effective in 2004
- Aims to eliminate or restrict the production and use of persistent organic pollutants (POPs).
- POPs are defined as "chemical substances that persist in the environment, bio-accumulate through the food web, and pose a risk of causing adverse effects to human health and the environment".

Important Listed substances

- Aldrin:** Used as a local ectoparasiticide and insecticide
- Heptachlor:** Uses as a termiticide (including in the structure of houses and underground), for organic treatment and in underground cable boxes
- Hexachlorobenzene:** Use as a chemical intermediate and a solvent for pesticides
- Endrin:** Endrin has been used primarily as an agricultural insecticide on tobacco, apple trees, cotton, sugar cane, rice, cereal, and grains.
- Polychlorinated biphenyl:** PCB's commercial utility was based largely on their chemical stability, including low flammability, and physical properties, including electrical insulating properties. They are highly toxic.
- DDT:** DDT is the best-known of several chlorine-containing pesticides used in the 1940s and 1950s.

Basel Convention

- Basel Convention on the Control of Transboundary Movements of Hazardous Wastes and Their Disposal.

- International treaty that was designed to reduce the movements of hazardous waste between nations.
- Main goal is to prevent transfer of hazardous waste from developed to less developed countries (LDCs).
- It **does not** address the movement of **radioactive waste**.
- 182 states and the European Union are parties to the Convention
- Location è Basel, Switzerland

Rotterdam Convention

- Rotterdam Convention on the Prior Informed Consent Procedure for Certain Hazardous Chemicals and Pesticides in International Trade.
- Multilateral treaty to promote shared responsibilities in relation to importation of hazardous chemicals.
- The convention promotes open exchange of information and calls on exporters of hazardous chemicals to use proper labeling, safe handling, and inform purchasers of any known restrictions or bans.
- Signatory nations can decide whether to allow or ban the importation of chemicals listed in the treaty..

Electronic waste | E - WASTE

- The discarded and end-of-life electronic products ranging from computers, equipment, home appliances, audio and video products and all of their peripherals are popularly known as Electronic waste (E-waste).
- E-waste is not hazardous if it is stocked in safe storage or recycled by scientific methods or transported from one place to the other in parts or in totality in the formal sector. The e-waste can, however, be considered hazardous if recycled by primitive methods.

Source and health effects

Particulars	Source	Health Effects
Lead	Used in glass panels and gaskets in computer monitors	Lead causes damage to the central and peripheral nervous systems, blood systems, kidney and reproductive system in humans. It also effects the endocrine system, and impedes brain development among children.
	Solder in printed circuit boards and other Components	
		Lead tends to accumulate in the environment and has high acute and chronic effects on plants, animals and microorganisms.

Cadmium	Occurs in SMD chip resistors, infra-red detectors, and semiconductor chips Some older cathode ray tubes contain cadmium	Toxic cadmium compounds accumulate in the human body, especially the kidneys.
Mercury	It is estimated that 22 % of the yearly world consumption of mercury is used in electrical and electronic equipment Mercury is used in thermostats, sensors, relays, switches, medical equipment, lamps, mobile phones and in batteries Mercury, used in flat panel displays, will likely increase as their use replaces cathode ray tubes	Mercury can cause damage to organs including the brain and kidneys, as well as the foetus. The developing foetus is highly vulnerable to mercury exposure. When inorganic mercury spreads out in the water, it is transformed to methylated mercury which bio-accumulates in living organisms and concentrates through the food chain, particularly via fish.
Hexavalent Chromium/ Chromium VI 29	Chromium VI is used as corrosion protector of untreated and galvanized steel plates and as a decorative or hardener for steel housings Plastics (including PVC): Dioxin is released when PVC is burned. The largest volume of plastics (26%) used in electronics has been PVC. PVC elements are found in cabling and computer housings. Many computer moldings are now made with the somewhat more benign ABS plastics	Chromium VI can cause damage to DNA and is extremely toxic in the environment.
Brominated flame retardants (BFRs):	BFRs are used in the plastic housings of electronic equipment and in circuit boards to prevent flammability	
Barium	Barium is a soft silvery-white metal that is used in computers in the front panel of a CRT, to protect users from radiation	Studies have shown that short-term exposure to barium causes brain swelling, muscle weakness, damage to the heart, liver, and spleen.
Beryllium	Beryllium is commonly found on motherboards and finger clips It is used as a copper- beryllium alloy to strengthen connectors and tiny plugs while maintaining electrical conductivity	Exposure to beryllium can cause lung cancer. Beryllium also causes a skin disease that is characterized by poor wound healing and wart like bumps. Studies have shown that people can develop beryllium disease many years following the last exposure.
Toners	Found in the plastic printer cartridge containing black and color toners.	Inhalation is the primary exposure pathway, and acute exposure may lead to respiratory tract irritation. Carbon black has been classified as a class 2B carcinogen, possibly carcinogenic to humans. Reports indicate that colour toners (cyan, magenta and yellow) contain heavy metals.
Phosphor and additives	Phosphor is an inorganic chemical compound that is applied as a coat on the interior of the CRT faceplate.	The phosphor coating on cathode ray tubes contains heavy metals, such as cadmium , and other rare earth metals, for example, zinc , vanadium as additives. These metals and their compounds are very toxic. This is a serious hazard posed for those who dismantle CRTs by hand.

Q1: Due to improper/indiscriminate disposal of old and used computers or their parts, which of the following are released into the environment as e-waste?

1. Beryllium
2. Cadmium
3. Chromium
4. Heptachlor
5. Mercury
6. Lead
7. Plutonium

Select the correct answer using the codes given below.

1. 1, 3, 4, 6 and 7 only
2. 1, 2, 3, 5 and 6 only

3. 2, 4, 5 and 7 only
4. 1, 2, 3, 4, 5, 6 and 7

Explanation:

Heptachlor is a Chlorohydrocarbon (CHC) which is used as an insecticide.

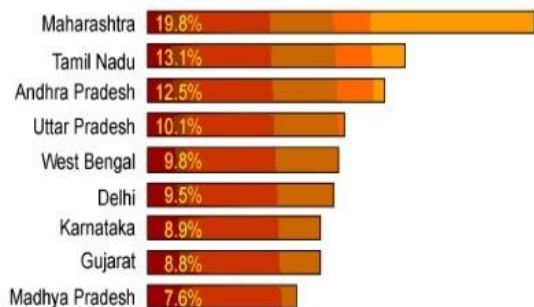
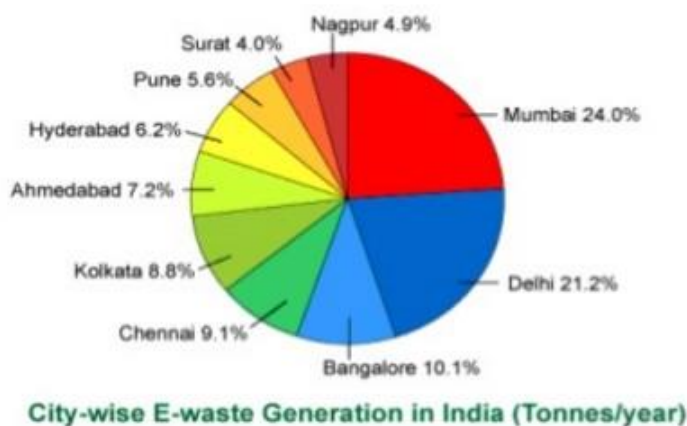
Plutonium is a radioactive metal and hence not used in computers.

So, answer should not contain either 4) or 7).

Answer: b) 1, 2, 3, 5 and 6 only

E — Waste in India

- India generates about 18.5 lakh metric tonnes (MT) of electronic waste every year, with Mumbai and Delhi-NCR accounting for the biggest chunk. The figure is likely to reach up to 30 lakh MT per year by 2018.
- Bangalore, Chennai, Kolkata, Ahmadabad, Hyderabad, Pune, Surat and Nagpur are other important cities generating substantial amount of e-waste.
- Among the eight largest e-waste generating states, **Maharashtra** ranks first followed by **Tamil Nadu** (2nd), **Andhra Pradesh** (3rd), Uttar Pradesh (4th), Delhi (5th), Gujarat (6th), Karnataka (7th) and West Bengal (8th).



State-wise E-waste Generation in India (Tonnes/year)

- Over half of the e-wastes generated in the developed world are exported to developing countries, mainly to China, India and Pakistan, where metals like copper, iron, silicon, nickel and gold are recovered during recycling process.
- Unlike developed countries, which have specifically built facilities for recycling of e-wastes, recycling in developing countries often involves manual participation thus exposing workers to toxic substances present in e-wastes.

Heavy Metal Toxicity And Methods Of Their Prevention

- Toxic metals are dispersed in the environment through metal smelting industrial emissions, burning of organic wastes, automobiles and coal based power generation.
- Heavy metals can be carried to places far away from their source of origin by winds when they are emitted in gaseous form or in form of fine particulates.
- Rain ultimately washes the air having metallic pollutants and brings them to the land and to water bodies.
- Heavy metals may endanger public health after being incorporated in food chain.
- Heavy metals cannot be destroyed by biological degradation.
- Incidence of heavy metal accumulation in fish, oysters, mussels, sediments and other components of aquatic ecosystems have been reported from all over the world.
- The heavy metals often encountered in the environment include lead, mercury, arsenic, chromium. These are known to cause toxic effects in living organisms.

Lead

- Lead enters the atmosphere from automobile exhaust. **Tetraethyl lead (TEL)** was added to petrol as an **anti-knock agent** for smooth running of automobile engines.
- TEL has now been replaced by other anti-knock compounds to prevent emission of lead by automobiles. Lead in petrol is being phased out by introduction of lead free petrol.
- Many industrial processes use lead and it is often released as a pollutant. Battery scrap also contain lead. It can get mixed up with water and food and create cumulative poisoning.
- Lead can cause irreversible behavioral disturbances, neurological damage and other developmental problems in young children and babies. It is a carcinogen of the lungs and kidneys.

Mercury

- In Japan, mass mercury poisoning (**Minamata disease**) was observed in 1960s, caused by eating fish from Minamata Bay which were contaminated with **methyl mercury**.
- Largest source of mercury pollution is through aquatic animals such as fish which accumulate mercury as methyl mercury.
- Mercury kills cells in the body and damages organs which come in contact with mercury and thus impairs their functioning.
- Inhalation of mercury vapours is more dangerous than its ingestion.
- Chronic exposure causes lesions in the mouth and skin and neurological problems.
- Typical symptoms of mercury poisoning are irritability, excitability, loss of memory, insomnia, tremor and gingivitis.
- Exposure to mercury can be prevented by taking care that mercury is not released in the environment as well as by replacing mercury by other materials.
- Mercury thermometers used earlier are getting replaced by mercury free thermometer.

Arsenic

- Arsenic is associated with **copper, iron and silver ores**.
- Arsenic is also emitted from **fossil fuel burning**.
- Liquid effluents from **fertilizer plants** also contain arsenic.

- Ground water contamination with arsenic is very common in areas where it is present.
- Chronic arsenic poisoning causes melanosis and keratosis (dark spots on the upper chest, back and arms are known as melanosis. The next stage is keratosis in which palms become hard) and leads to loss of appetite, weight, diarrhoea, gastrointestinal disturbances and skin cancer.
- Surface waters are generally free from arsenic pollution and should be preferred for drinking and cooking.
- Alternatively the tube well/ hand pump water should be purified to remove arsenic before consumption. Techniques for removing arsenic from water are available.

Cadmium

- Mining especially of **zinc** and metallurgical operations, electroplating industries, etc. release cadmium in the environment.
- It may enter the human body by inhalation or from aquatic sources including fish, etc.
- It may cause hypertension, liver cirrhosis, brittle bones, kidney damage and lung cancer.
- **Itai-itai** disease first reported from Japan in 1965 was attributed to cadmium contamination in water and rice caused by discharge of effluents from a zinc smelter into a river.

Other Heavy Metals

- Metals such as **zinc, chromium, antimony** and **tin** enter food from cheap cooking utensils.
- Preserved foods stored in tin cans also cause contamination by tin.
- Zinc is a skin irritant and affects pulmonary system.
- Problems of heavy metal toxicity can be prevented by avoiding the use of utensils made from materials containing these heavy metals or use of drinking water and consuming fish having these heavy metals.

Occupational Health Hazards

Black lung disease

- In coal mining areas coal dust is the main air pollutant. The deposits of coal dust makes

miners lungs look black instead of a healthy pink and hence the name black lung disease.

- Black lung disease is the common name for **pneumoconiosis (CWP) or anthracosis**, a lung disease of older workers in the coal industry, caused by inhalation over many years, of small amounts of coal dust.
- The particles of fine coal dust accumulate in lungs. Eventually this build-up causes thickening and scarring making the lungs less efficient in supplying oxygen to the blood.
- In some cases a progressive massive fibrosis develops, in which damage continues in the upper parts of the lungs even after exposure to dust has ended.
- X-rays can detect black lung disease before it causes any symptoms.

Noise

- Workers in mining, manufacturing and construction industries are exposed to high levels of noise which is a very important stress factor.
- Sound levels higher than 80 to 90 dB for more than eight hours are harmful to human ear. Some of the adverse effects of sound are –

Psychological

- Noise leads to emotional disturbances such as annoyance, disturbed sleep, lack of concentration and reduced efficiency.

Auditory effects

- Auditory fatigue – Occurs when noise level is in the range of 85 to 90 dB e.g. noise of a food blender.
- Deafness or impaired hearing – It may be temporary or permanent. Temporary hearing loss occurs on continuous exposure to noise as in case of telephone operators.
- Repeated or continuous exposure to noise more than 90 dB may result in permanent loss of hearing.

Non-auditory effects

- Interference with speech and communication.
- Annoyance: Most people are annoyed by noise and some may become neurotic. Neurotic people lose their temper quickly and become irritable.

- Efficiency: High level of noise at the work place reduces working efficiency. Quiet environment helps in increasing efficiency.
- General change in the body: Exposure to noise increases blood pressure, pulse rate, breathing and sweating or headache.

Chemicals and Biological Agents

- Workers in many industries are exposed to chemicals which are hazardous and may be even carcinogenic such as in textiles, cement and construction industries.
- Substances such as **benzene, chromium, nitrosamines** and **asbestos** may cause cancers of lung, bladder, skin, mesothelium, liver, etc.
- Occupational asthma is caused due to exposure to organic dusts, microorganisms, bacteria, fungi and moulds and several chemicals.
- **Silicosis** first reported from Kolar gold mines in 1947 is a common disease among miners, pottery and ceramic industry workers.
- **Pneumoconiosis** and **byssinosis** are common among mica and textile industry workers respectively.

Treatment and disposal of solid waste

Open dumps

- Open dumps refer to uncovered areas that are used to dump solid waste of all kinds.
- The waste is untreated, uncovered, and not segregated. It is the breeding ground for flies, rats, and other insects that spread disease.
- The rainwater runoff from these dumps contaminates nearby land and water thereby spreading disease. Treatment by open dumps is to be phased out.

Landfills

- It is a pit that is dug in the ground. The garbage is dumped and the pit is covered with soil everyday thus preventing the breeding of flies and rats.
- After the landfill is full, the area is covered with a thick layer of mud and the site can thereafter be developed as a parking lot or a park.
- Problems - All types of waste are dumped in landfills and when water seeps through them

it gets contaminated and in turn pollutes the surrounding area. This contamination of groundwater and soil through landfills is known as leaching.

Sanitary landfills

- Sanitary landfill is more hygienic and built in a methodical manner to solve the problem of leaching.
- These are lined with materials that are impermeable such as plastics and clay, and are also built over impermeable soil. Constructing sanitary landfills is very costly.

Incineration plants

- The process of burning waste in large furnaces at high temperature is known as incineration.
- In these plants the recyclable material is segregated and the rest of the material is burnt and ash is produced.
- Burning garbage is not a clean process as it produces tonnes of toxic ash and pollutes the air and water.
- A large amount of the waste that is burnt here can be recovered and recycled. In fact, at present, incineration is kept as the last resort and is used mainly for treating the infectious waste.

Pyrolysis

- It is a process of combustion in **absence of oxygen** or the material burnt under controlled atmosphere of oxygen. It is an alternative to incineration.
- The gas and liquid thus obtained can be used as fuels.
- Pyrolysis of carbonaceous wastes like firewood, coconut, palm waste, corn combs, cashew shell, rice husk paddy straw and saw dust, yields **charcoal** along with products like **tar, methyl alcohol, acetic acid, acetone** and **fuel gas**.

Composting

- Composting is a biological process in which micro-organisms, mainly fungi and bacteria, decompose degradable organic waste into humus like substance in the presence of oxygen.

- This finished product, which looks like soil, is high in carbon and nitrogen and is an excellent medium for growing plants.
- It increases the soil's ability to hold water and makes the soil easier to cultivate. It helps the soil retain more plant nutrients.
- It recycles the nutrients and returns them back to soil as nutrients.
- Apart from being clean, cheap, and safe, composting can significantly reduce the amount of disposable garbage.

Vermiculture

- It is also known as earthworm farming. In this method, Earth worms are added to the compost. These worms break the waste and the added excreta of the worms makes the compost very rich in nutrients.
- Four R's – Reduce, Reuse, Recycle and Recover.

Waste Minimization Circles (WMC)

- WMC helps Small and Medium Industrial Clusters in waste minimization in their industrial plants.
- This is assisted by the **World Bank** with the Ministry of Environment and Forests acting as the nodal ministry.
- The project is being implemented with the assistance of **National Productivity Council (NPC), New Delhi**.
- The initiative also aims to realize the objectives of the Policy Statement for Abatement of Pollution (1992), which states that the government should educate citizens about environmental risks, the economic and health dangers of resource degradation and the real economic cost of natural resources.
- The policy also recognizes that citizens and non-governmental organizations play a role in environmental monitoring, therefore, enabling them to supplement the regulatory system and recognizing their expertise where such exists and where their commitments and vigilance would be cost effective.

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Environmental Impact Assessment (EIA)

Environmental Impact Assessment

- Development projects in the past were undertaken without any consideration to their environmental consequences. As a result the whole environment got polluted and degraded.
- In view of the colossal damage to the environment, governments and public are now concerned about the environmental impacts of developmental activities. So, to assess the environmental impacts, the mechanism of EIA was introduced.
- EIA is a tool to anticipate the likely environmental impacts that may arise out of the proposed developmental activities and suggest mitigation measures and strategies. EIA was introduced in India in 1978, with respect to river valley projects. Later the EIA legislation was enhanced to include other developmental sections since 1941.
- EIA comes under **Notification on Environmental Impact Assessment (EIA) of developmental projects 1994** under the provisions of **Environment (Protection) Act, 1986**.
- Besides EIA, the Government of India under Environment (Protection) Act 1986 issued a number of other notifications, which are related to environmental impact assessment.
- EIA is now mandatory for **30 categories of projects**, and these projects get **Environmental Clearance (EC)** only after the EIA requirements are fulfilled.
- Environmental clearance or the 'go ahead' signal is granted by the **Impact Assessment Agency** in the Ministry of Environment and Forests, Government of India.

All projects that require clearance from central government can be broadly categorized into the following

1. **Industries**
2. **Mining**

3. **Thermal power plants**
4. **River valley projects**
5. **Infrastructure and CRZ (Coastal Regulation Zone)**
6. **Nuclear power projects.**

Individual projects that need require clearance from central government

- Nuclear power and related projects such as heavy water plants, nuclear fuel complex, rare earths.
- River valley projects including hydel power, major irrigation and their combination including flood control.
- Ports, harbours, airports (except minor ports and harbours).
- Petroleum refineries including crude and product pipelines.
- Chemical fertilizers (nitrogenous and phosphatic other than single superphosphate).
- Pesticides (technical).
- Petrochemical complexes (both olefinic and aromatic) and petrochemical intermediates such as DMT, Caprolactam, LAB etc., and production of basic plastics such as LDPE, HDPE, PP, PVC.
- Bulk drugs and pharmaceuticals
- Exploration for oil and gas and their production, transportation and storage
- Synthetic rubber
- Asbestos and asbestos products
- Hydrocyanic acid and its derivatives
- Primary metallurgical industries (such as production of iron and steel, aluminium, copper, zinc, lead, and ferro-alloys)
- Chlor alkali industry
- Integrated paint complex including manufacture of resins and basic raw materials required in the manufacture of paints
- Viscose staple fibre and filament yarn
- Storage batteries integrated with manufacture of oxides of lead and lead antimony alloy
- All tourism projects between 200m-500 metres of High Water Line and at locations with an elevation of more than 1000 metres with investment of more than Rs. 5 crore
- Thermal power plants
- Mining projects (with lease more than 5 hectares)
- Highway projects except projects relating to improvement work including widening and strengthening of roads with marginal land acquisition along the existing alignments provided it **does not pass through**

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ecologically sensitive areas such as National Parks, Sanctuaries, Tiger Reserves, Reserve Forests

- Tarred roads in the Himalayas and forest areas
- Distilleries
- Raw skins and hide
- Pulp, paper and newsprint
- Dyes
- Cement
- Foundries (Individual)
- Electroplating
- Meta aminophenol

The important aspects of EIA are

1. risk assessment,
2. environmental management and
3. post product monitoring.

EIA is to

1. serve as a primary environmental tool with clear provisions.
2. apply consistently to all proposals with potential environmental impacts.
3. use scientific practice and suggest strategies for mitigation.
4. address all possible factors such as short term, long term, small scale and large scale effects.
5. consider sustainable aspects such as capacity for assimilation, carrying capacity, biodiversity protection.
6. lay down a flexible approach for **public involvement**.
7. have in built mechanism of follow up and feedback.
8. include mechanisms for monitoring, auditing and evaluation.

In order to carry out an environmental impact assessment, the following are essential:

1. assessment of existing environmental status.
2. assessment of various factors of ecosystem (air, water, land, biological).
3. analysis of adverse environmental impacts of the proposed project to be started.
4. impact on people in the neighborhood.

Benefits of EIA

- EIA provides a cost effective method to eliminate or minimize the adverse impact of developmental projects.
- EIA enables the decision makers to analyses the effect of developmental activities on the environment well before the developmental project is implemented.
- EIA encourages the adaptation of mitigation strategies in the developmental plan.
- EIA makes sure that the developmental plan is environmentally sound and within limits of the capacity of assimilation and regeneration of the ecosystem.
- EIA links environment with development. The goal is to ensure **environmentally safe and sustainable development**.

Environmental Components Of EIA

- The EIA process looks into the following components of the environment.

Air environment

- Quality of ambient air present and predicted.
- Meteorological data: Wind speed, direction, humidity etc.
- Quantity of emission likely from project.
- Impact of the emission on the area.
- Pollution control desires/air quality standards.

Noise

- Levels of noise present and predicted
- Strategies for reducing noise pollution.

Water environment

- Existing ground and surface water resources, their quality and quantity within the zone.
- Impact of proposed project on water resources.

Biological environment

- Flora and fauna in impact zone.
- Potential damage (likely) due to project, due to effluents, emissions and landscaping.
- Biological stress (prediction).

Land environment

- Study of soil characteristics, land use, and drainage pattern, and the likely adverse impact of the project.
- Impact on historical monuments and heritage site.
- Assessment of expected economic benefits arising out of the project have to be compared to the all the above mentioned factors. Thus we can say that environmental concerns have to be made a part of the decision to set up a project.

EIA Process And Procedures

Steps in Preparation of EIA report

- Collection of baseline data from primary and secondary sources;
- Prediction of impacts based on past experience and mathematical modelling;
- Evolution of impacts versus evaluation of net cost benefit;
- Preparation of **environmental management plans** to reduce the impacts to the minimum;
- Quantitative estimation of financial cost of monitoring plan and the mitigation measures.

Environment Management Plan

- Delineation of mitigation measures including prevention and control for each environmental component and rehabilitation and resettlement plan.

Steps in EIA process

EIA involves the steps mentioned below. However, EIA process is cyclical with interaction between the various steps.

- **Screening:** The project plan is screened for scale of investment, location and type of development and if the project needs statutory clearance.
- **Scoping:** The project's potential impacts, zone of impacts, mitigation possibilities and need for monitoring.
- **Collection of baseline data:** Baseline data is the environmental status of study area.
- **Impact prediction:** Positive and negative, reversible and irreversible and temporary and permanent impacts need to be predicted which presupposes a good understanding of the project by the assessment agency.

- **Mitigation measures and EIA report:** The EIA report should include the actions and steps for preventing, minimizing or by passing the impacts or else the level of compensation for probable environmental damage or loss.
- **Public hearing:** On completion of the EIA report, public and environmental groups living close to project site may be informed and consulted.
- **Decision making:** Impact Assessment (IA) Authority along with the experts consult the project-in-charge along with consultant to take the final decision, keeping mind EIA and EMP (Environment Management Plan).
- **Monitoring and implementation of environmental management plan:** The various phases of implementation of the project are monitored.
- **Assessment of Alternatives, Delineation of Mitigation Measures and Environmental Impact Assessment Report:** For every project, possible alternatives should be identified and environmental attributes compared. Alternatives should cover both project location and process technologies.
- Once alternatives have been reviewed, a mitigation plan should be drawn up for the selected option and is supplemented with an Environmental Management Plan (EMP) to guide the proponent towards environmental improvements.
- **Risk assessment:** Inventory analysis and hazard probability and index also form part of EIA procedures.

Composition of the expert committees for EIA

- The Committees will consist of experts in the following disciplines:
- Eco-system management
- Air/water pollution control
- Water resource management
- Flora/fauna conservation and management
- Land use planning
- Social Sciences/Rehabilitation
- Project appraisal
- Ecology
- Environmental Health
- Subject Area Specialists
- **Representatives of NGOs/persons concerned with environmental issues**
- The Chairman will be an outstanding and experienced ecologist or environmentalist or

technical professional with wide managerial experience in the relevant development.

- The representative of Impact Assessment Agency will act as a Member-Secretary.
- Chairman and members will serve in their individual capacities except those specifically nominated as representatives.
- The membership of a committee **shall not exceed 15 members**.

Environmental Appraisal Procedure In India

- An Appraisal Committee constituted by the Ministry of Environment and Forests to first scrutinized a project based on the data presented by the project authorities.
- If necessary, the Ministry of Environment and Forests may also hold consultations with the investors and experts on specific issues as and when necessary.
- After considering all the facets of a projects, environmental clearance is accorded subject to implementation of the stipulated environmental safeguards.
- In case of projects where the project proponents have submitted complete information, a decision is taken within **90 days**.
- The six regional offices of the Ministry functioning at **Shillong, Bhubaneswar, Chandigarh, Bangalore, Lucknow and Bhopal** undertake monitoring of cleared projects.
- The primary objectives of this procedure is to ensure adequacy of the suggested safeguards and also to undertake mid-course corrections if required.
- Sometimes one or more natural resources becomes limiting resource in a given region and that restrict the scopes of development projects.

EIA of Coasts

- **Coastal Zone Management Plans (CZMPs)** are prepared by coastal states or Union Territories as per rules set by CRZ notification 1991.
- CZMPs are prepared based on identification and categorization of coastal areas for different activities and then submitted to the MoEF for approval.
- The ministry then forms a task force for examining their plans.

Environmental Clearance/Rejection Letter

Single window clearance

Environmental clearance + Forestry clearance.

- When a project requires both environmental clearance as well as approval under the Forest (Conservation) Act, 1980, proposals for both are required to be given simultaneously to the concerned divisions of the Ministry.
- The processing is done simultaneously for clearance or rejection. If the project does not involve diversion of forestland, the case is processed only for environmental clearance.

Time frame

- Once all the requisite documents and data from the project authorities are received and public hearings (where required) have been held, assessment and evaluation of the project from the environment angle is completed within **90 days** and the decision of the ministry shall be conveyed within 30 days thereafter [**120 days for final decision**].

Post project monitoring

- Whenever a project is given environment clearance, a set of conditions are stipulated by the Appraisal Committee on a case to case basis, which have to be complied with by the project proponent.
- The project authorities are required to submit a half-yearly compliance report to the Ministry about the compliance of conditions stipulated.
- Cases of non-compliance of the recommendations and conditions by cleared projects/ units are brought to the notice of the Ministry, which may then initiate action against the project authorities.

The Main Participants Of EIA

EIA applies to public and private sections. The six main players are:

1. Those who propose the project
2. The environmental consultant who prepare EIA on behalf of project proponent.
3. Pollution Control Board (State or National).

4. Public has the right to express their opinion.
5. The Impact Assessment Agency.
6. Regional center of the Ministry of Environment and Forest.

Salient Features of 2006 Amendment to EIA Notification

- Environment Impact Assessment Notification of 2006 has **decentralized** the environmental clearance projects by categorizing the developmental projects in two categories, i.e., Category A (national level appraisal) and Category B (state level appraisal).
- 'Category A' projects are appraised at national level by Impact Assessment Agency (IAA) and the Expert Appraisal Committee (EAC) and Category B projects are appraised at state level.
- State Level Environment Impact Assessment Authority (SEIAA) and State Level Expert Appraisal Committee (SEAC) are constituted to provide clearance to Category B process.

After 2006 Amendment the EIA cycle comprises of four stages

1. Screening
2. Scoping
3. Public hearing
4. Appraisal

- Category A projects require mandatory environmental clearance and thus they do not undergo the screening process.
- Category B projects undergoes screening process and they are classified into two types.

1. Category B, projects (Mandatory requires EIA).
2. Category B2 projects (Do not require EIA).

- Thus Category A projects and Category B, projects undergo the complete EIA process whereas Category B2 projects are excluded from complete EIA process.

Procedure For Public Hearing

Notice of Public Hearing

- Whoever applies for environmental clearance of projects, should request the concerned State Pollution Control Board to initiate a public hearing.

- The State Pollution Control Board issues a notice for environmental public hearing which will be published in at least two newspapers widely circulated in the region around the project, one of which will be in the vernacular language of the locality concerned.
- State Pollution Control Board mentions the date, time and place of public hearing.
- Suggestions, views, comments and objections of the public will be invited within thirty days from the date of publication of the notification.
- All persons including the residents, environmental groups and others located at the project site/sites of displacement/sites likely to be affected can participate in the public hearing. They can also make oral/written suggestions to the State Pollution Control Board.

Composition of public hearing panel

The composition of Public Hearing Panel may consist of the following, namely:

- Representative of State Pollution Control Board;
- District Collector or his nominee;
- Representative of State Government dealing with the subject;
- Representative of Department of the State Government dealing with Environment;
- Not more than three representatives of the local bodies such as Municipalities or panchayats;
- Not more than three senior citizens of the area nominated by the District Collector.

Shortcomings of Environmental Impact Assessment

Applicability

- There are several projects with significant environmental impacts that are exempted from the notification either because they are not listed in schedule I, or their investments are less than what is provided for in the notification.

Composition of expert committees and standards

- It is being found that the team formed for conducting EIA studies is lacking the expertise in various fields such as

environmentalists, wild life experts, Anthropologists and Social Scientists (to study the social impact of the project).

- For example for the preparation of EIA report of the proposed oil exploration in coast of Orissa by the reliance group has been given to the life science Dept of Berhampur university which has no expertise on the study of turtles and its life cycle.

Public hearing

- Public comments are not taken into account at the early stage, which often leads to conflict at the later stage of project clearance.
- A number of projects with significant environmental and social impacts have been excluded from the mandatory public hearing process.
- The documents which the public are entitled to are seldom available on time.
- The data collectors do not pay respect to the indigenous knowledge of local people.

Quality of EIA

- One of the biggest concerns with the environmental clearance process is related to the quality of EIA report that are being carried out.
- The reports are generally incomplete and provided with false data.
- Many EIA report are based on single season data.
- The EIA document in itself is so bulky and technical, which makes it very difficult to decipher so as to aid in the decision making process.

Lack of Credibility

- It is the responsibility of the **project proponent** to commission the preparation of the EIA for its project.
- The EIA is actually funded by an agency or individual whose primary interest is to procure clearance for the project proposed.
- There is little chance that the final assessment presented is un biased, even if the consultant may provide an unbiased assessment that is critical of the proposed project.
- There are so many cases of fraudulent EIA studies where erroneous data has been used, same facts used for two totally different places etc.

- There is no accreditation of EIA consultants, therefore any such consultant with a track record of fraudulent cases cannot be held liable for discrepancies.
- It is hard to imagine any consultant after being paid lakh of rupees, preparing a report for the project proponents, indicating that the project is not viable.

Case Study

- The MoEF constituted the Western Ghats Experts Ecology Panel (WGEEP) in 2010 under the Chairmanship of Prof. Madhav Gadgil.
- The Panel submitted its report in 2011 but it was not made public immediately due to its stringent assessment of the condition of Western Ghats.
- The report suggested many radical changes that needs to be brought to conserve Western Ghats. The recommendation if implemented would adversely affect mining mafia, sand mafia and local encroachers.
- Under pressure from various stakeholders, MoEF set up the High Level Working Group (HLWG) under the Chairmanship of Dr. K. Kasturirangan to study recommendations of WGEEP.
- The very constitution of the HLWG raised suspicions that this has been formed to dilute the recommendations of the WGEEP.
- The HLWG had indeed diluted many recommendations of WGEEP to satisfy the interests of various mafia.

Monitoring, compliance and institutional arrangements

- Often, and more so for strategic industries such as nuclear energy projected, the EMPs are kept **confidential** for political and administrative reasons.
- Details regarding the effectiveness and implementation of mitigation measures are often not provided.
- Emergency preparedness plans are not discussed in sufficient details and the information not disseminated to the communities.

Recommendations to improve EIA process

- **Independent** EIA Authority.

- Sector wide EIAs needed.
- Creation of a centralized baseline data bank.
- Dissemination of all information related to projects from notification to clearance to local communities and general public.

Applicability

- All those projects where there is likely to be a significant alternation of ecosystems need to go through the process of environmental clearance, without exception.
- No industrial developmental activity should be permitted in ecologically sensitive areas.

Public hearing

- Public hearings should be applicable to all hitherto exempt categories of projects which have environmental impacts.

Quality

- The focus of EIA needs to shift from utilization and exploitation of natural resources to conservation of natural resources.
- At present EIA reports are extremely weak when it comes to assessment of biological diversity of a project area and the consequent impacts on it. This gap needs to be plugged.
- All EIA reports should clearly state what are the adverse impacts that a proposed projects will have. This should be a separate chapter and not hidden within technical details.
- The sub components or subsidiary reports of EIA reports (e.g. Assessments of Biodiversity impacts done by a sub consultant) should be made publicly accessible as standalone reports with the EIA. This should be available on the websites of the MOEF.
- EIAs should be based on full studies carried out over at least one year. Single season data on environmental parameters like biodiversity, as is being done for several rapid assessments is not adequate to gain understanding of the full impact of the proposed project.
- It is critical that the preparation of an EIA is completely independent of the project proponent.
- State and central governments should maintain a list of credible, independent and competent agencies that can carry out EIAs. similarly the EIA consultant those are making false reports should be black listed.

- A national level accreditation to environment consultancy should be adopted

Grant of clearance

- The notification needs to make it clear that the provision for site clearance does not imply any commitment on the part of the impact Assessment agency to grant full environmental clearance.
- The prior informed consent of local communities and urban wards or residents association needs to be made mandatory before the grant of environmental clearance. The consent should be from the full general body.
- The language used for specifying conditions of clearance must be clear and specific.

Composition of expert committees

- The present executive committees should be replaced by expert's people from various stakeholder groups, who are reputed in environmental and other relevant fields.
- The process of selection of those committees should be open and transparent. The minutes, decisions and advice by these committee should be open to public.

Monitoring, compliance and institutional arrangements

- The EIA notification needs to build within it an automatic withdrawal of clearance if the conditions of clearance are being violated, and introduce more stringent punishment for noncompliance. At present the EIA notification limits itself to the stage when environmental clearance is granted.
- The MOEF should set up more regional offices with advisory Expert committees, each with smaller areas of jurisdiction, to effectively monitor the compliance of clearance conditions.

Redressal

- The composition of the NGT needs to be changed to include more judicials from the field of environment.
- Citizen should be able to access the authority for redressal of all violation of the EIA notification as well as issues relating to non-compliance.

Capacity building

- NGOs, civil society groups and local communities need to build their capacities to use the EIA notification towards better decision making on projects.

Biodiversity - Biodiversity of India

Biodiversity

- Biodiversity is the variety of plant and animal life in the world or in a particular habitat.

Measurement of Biodiversity

- Biodiversity is measured by two major components:
 1. species richness, and
 2. species evenness.

Species richness

- It is the measure of number of species found in a community

Alpha diversity

- It refers to the diversity within a particular area or ecosystem, and is usually expressed by the number of species (i.e., species richness) in that ecosystem.

Beta diversity

- It is a comparison of diversity between ecosystems, usually measured as the change in amount of species between the ecosystems.

Gamma diversity

- It is a measure of the overall diversity for the different ecosystems within a region.

Species evenness

- It measures the proportion of species at a given site, e.g. low evenness indicates that a few species dominate the site.

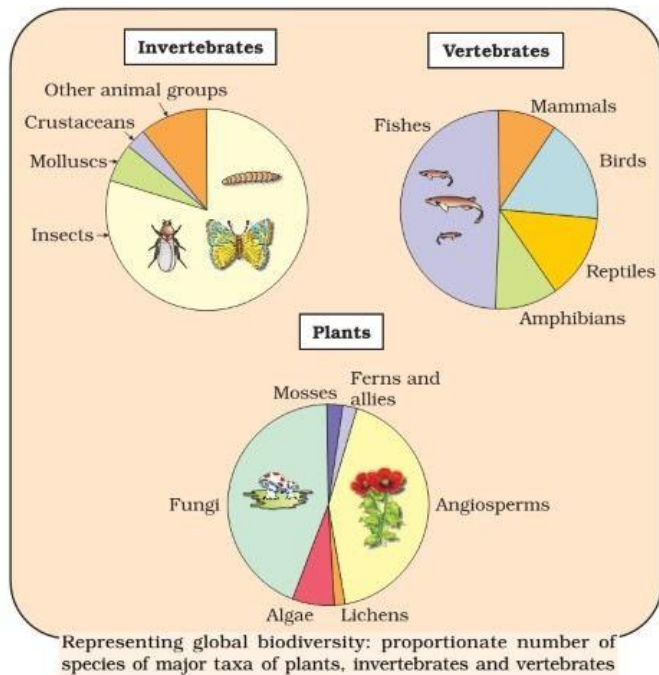
Genetic diversity

- Genetic diversity is the total number of genetic characteristics in the genetic makeup of a species.
- A single species might show high diversity at the genetic level [E.g. Man: Chinese, Indian American, African etc.]. India has more than 50,000 genetically different strains of rice, and 1,000 varieties of mango.
- Genetic diversity allows species to adapt to changing environments. This diversity aims to ensure that some species survive drastic changes and thus carry on desirable genes.

Species diversity

- It is the ratio of one species population over total number of organisms across all species in the given biome. 'Zero' would be infinite diversity, and 'one' represents only one species present.
- Species diversity is a measure of the diversity within an ecological community that incorporates both **species richness** (the number of species in a community) and the **evenness of species**.
- For example, the Western Ghats have a greater amphibian species diversity than the Eastern Ghats. There are more than 200000 species in India of which several are confined to India (endemic).
- **Endemism** is the ecological state of a species being unique to a defined geographic location, such as an island, nation, country or other defined zone, or habitat type; organisms that are indigenous to a place are not endemic to it if they are also found elsewhere. A particular type of animal or plant may be endemic to a zone, a state or a country. The extreme opposite of endemism is **cosmopolitan distribution**.
- Species differ from one another, markedly in their genetic makeup, do not inter-breed in nature. Closely-related species however have in common much of their hereditary characteristics. For instance, about 98.4 per cent of the genes of humans and chimpanzees are the same.
- According to the IUCN (2004), the total number of plant and animal species described so far is slightly more than 1.5 million, but we have no clear idea of how many species are yet to be discovered and described. A large proportion of the species waiting to be discovered are in the tropics. Estimate place the global species diversity at about 7 million.

- More than 70 per cent of all the species recorded are animals, while plants (including algae, fungi, bryophytes, gymnosperms and angiosperms) comprise no more than 22 per cent of the total.
- Among animals, insects are the most species-rich taxonomic group, making up more than 70 per cent of the total. That means, out of every 10 animals on this planet, 7 are insects.



- The number of fungi species in the world is more than the combined total of the species of fishes, amphibians, reptiles and mammals.
- It should be noted that these estimates do not give any figures for prokaryotes. Biologists are not sure about how many prokaryotic species there might be.
- In general, species diversity decreases as we move away from the equator towards the poles. With very few exceptions, tropics (latitudinal range of 23.5° N to 23.5° S) harbour more species than temperate or polar areas.
- India, with much of its land area in the tropical latitudes, has more than 1,200 species of birds.
- The largely tropical Amazonian rain forest in South America has the greatest biodiversity on earth- it is home to more than 40,000 species of plants, 3,000 of fishes, 1,300 of birds, 427 of mammals, 427 of amphibians, 378 of reptiles and of more than 1,25,000 invertebrates.

Why tropics have greater biological diversity?

- Speciation is generally a function of time, unlike temperate regions subjected to frequent glaciations in the past, tropical latitudes have remained relatively undisturbed for millions of years and thus, had a long evolutionary time for species diversification.
- Tropical environments, unlike temperate ones, are less seasonal, relatively more constant and predictable. Such constant environments promote niche specialization and lead to a greater species diversity.
- There is more solar energy available in the tropics, which contributes to higher productivity; this in turn might contribute indirectly to greater diversity.

The importance of Species Diversity to the Ecosystem

- For many decades, ecologists believed that communities with more species, generally, tend to be more stable than those with less species.
- What exactly is stability for a biological community? A stable community should not show too much variation in productivity from year to year; it must be either resistant or resilient to occasional disturbances (natural or man-made), and it must also be resistant to invasions by alien species.
- Although, we may not understand completely how species richness contributes to the well-being of an ecosystem, we know enough to realize that rich biodiversity is not only essential for ecosystem health but imperative for the very survival of the human race on this planet.

Bioprospecting: nations endowed with rich biodiversity explore molecular, genetic and species-level diversity to derive products of economic importance.

Keystone species and Foundation Species

Keystone species is a species whose addition to or loss from an ecosystem leads to major changes in occurrence of at least one other species.

Certain species in an ecosystem is considered more important in determining the presence of

many other species in that ecosystem. All **top predators** (Tiger, Lion, Crocodile, Elephant) are considered as keystone species because it regulates all other animal population indirectly. Hence top predators are given much consideration in conservation.

If keystone species is lost, it will result in the degradation of whole ecosystem. For example certain plant species (ebony tree, Indian-laurel) exclusively depends upon bats for its pollination. If the bat population is reduced then regeneration of particular plants becomes more difficult.

Foundation species is a dominant primary producer in an ecosystem both in terms of abundance and influence. Example: kelp in kelp forests and corals in coral reefs.

Flagship species

A flagship species is a species chosen to represent an environmental cause, such as an ecosystem in need of conservation.

These species are chosen for their vulnerability, attractiveness or distinctiveness in order to engender support and acknowledgement from the public at large.

Example: Indian tiger, African elephant, giant panda of China, mountain gorilla of Central Africa, orangutan of Southeast Asia and the leatherback sea turtle.

Ecological diversity

- Ecological diversity refers to the different types of habitats. A habitat is the cumulative factor of the climate, vegetation and geography of a region.
- It includes various biological zones, like lake, desert, coast, estuaries, wetlands, mangroves, coral reefs etc.
- At the ecosystem level, India, for instance, with its deserts, rain forests, mangroves, coral reefs, wetlands, estuaries, and alpine meadows has a greater ecosystem diversity than a Scandinavian country like Norway.

Services provided by Biodiversity

- Ecosystem services.
- Protection of water resources.
- Soils formation and protection.
- Nutrient storage and recycling.
- Pollution breakdown and absorption.

- Contribution to climate stability.
- Maintenance of ecosystems.
- Recovery from unpredictable events.

Biological services

- Food and shelter.
- Medicinal resources.
- Wood products.
- Ornamental plants.
- Breeding stocks, population reservoirs.
- Future resources.
- Diversity in genes, species and ecosystems.

Social services

- Research, education and monitoring.
- Recreation and tourism.
- Cultural values.

Biodiversity of India

- India is a recognized as one of the **mega-diverse countries**, rich in biodiversity and associated traditional knowledge.
- India has 23.39% of its geographical area under forest and tree cover.
- With just 2.4% of the land area, India accounts for nearly **7%** of the recorded species even while supporting almost 18% of human population.
- In terms of species richness, India ranks **seventh** in mammals, **ninth** in birds and **fifth** in reptiles.
- In terms of endemism of vertebrate groups, India's position is **tenth in birds** with 69 species, **fifth in reptiles** with 156 species and **seventh in amphibians** with 110 species.
- India's share of crops is 44% as compared to the world average of 11%.

India Represents

- Two 'Realms'
- Five Biomes
- Ten Bio-geographic Zones
- Twenty five Bio-geographic provinces

Realms

- Biogeographic realms are large spatial regions within which ecosystems share a broadly similar biota.

- Realm is a continent or sub-continent sized area with unifying features of geography and fauna & flora.
- The Indian region is composed of two realms. They are:

1. the Himalayan region represented by Palearctic Realm and
2. the rest of the sub-continent represented by Malayan Realm

- In world Eight terrestrial biogeographic realms are typically recognized. They are

1. Nearctic Realm
2. Palaearctic Realm
3. Afrotropical Realm
4. Indomalayan Realm
5. Oceania Realm
6. Australian Realm
7. Antarctic Realm
8. Neotropical Realm

Biomes of India

The term biome means the main groups of plants and animals living in areas of certain climate patterns.

- It includes the way in which animals, vegetation and soil interact together. The plants and animals of that area have adapted to that environment.

The five biomes of India are:

1. Tropical Humid Forests
2. Tropical Dry or Deciduous Forests (including Monsoon Forests)
3. Warm deserts and semi-deserts
4. Coniferous forests and
5. Alpine meadows.

Bio-geographic Zones

- Biogeography deals with the geographical distribution of plants and animals.
- Biogeographic zones were used as a basis for planning wildlife protected areas in India.
- There are 10 biogeographic zones which are distinguished clearly in India. They are as follows:

Trans-Himalayas

- An extension of the Tibetan plateau, harboring high-altitude cold desert in Laddakh (J&K) and Lahaul Spiti (H.P) comprising 5.7 % of the country's landmass.

Himalayas

- The entire mountain chain running from north-western to northeastern India, comprising a diverse range of biotic provinces and biomes, 7.2 % of the country's landmass.

Desert

- The extremely arid area west of the Aravalli hill range, comprising both the salty desert of Gujarat and the sand desert of Rajasthan. 6.9% of the country's landmass.

Semi-arid

- The zone between the desert and the Deccan plateau, including the Aravalli hill range. 15.6 % of the country's landmass.

Western Ghats

- The hill ranges and plains running along the western coastline, south of the Tapti river, covering an extremely diverse range of biotic provinces and biomes. 5.8% of the country's landmass.

Deccan Peninsula

- The largest of the zones, covering much of the southern and southcentral plateau with a predominantly deciduous vegetation. 4.3 % of the country's landmass.

Gangetic plain

- Defined by the Ganges river system, these plains are relatively homogenous. 11% of the country's landmass.

North-east India

- The plains and non-Himalayan hill ranges of northeastern India, with a wide variation of vegetation. 5.2% of the country's landmass.

Islands

- The Andaman and Nicobar Islands in the Bay of Bengal, with a highly diverse set of biomes. 0.03% of the country's landmass.

two; Lakshadweep islands are included in this with the percent area being negligible.

Bio-geographic provinces

Coasts

- A large coastline distributed both to the west and east, with distinct differences between the

- Bio-geographic Province is a ecosystematic or biotic subdivision of realms. India is divided into 25 bio geographic zones.

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Biogeographic Zones (10)	Biogeographic Provinces (25)
Trans Himalaya	1A: Himalaya – Ladakh Mountains 1B: Himalaya – Tibetan Plateau 1C: Trans – Himalaya Sikkim
The Himalaya	2A: Himalaya - North West Himalaya 2B: Himalaya - West Himalaya 2C: Himalaya - Central Himalaya 2D: Himalaya - East Himalaya
The Indian Desert	3A: Desert - Thar 3B: Desert - Katchchh
The Semi-Arid	4A: Semi - Arid - Punjab Plains 4B: Semi - Arid - Gujarat Rajputana
The Western Ghats	5A: Western Ghats - Malabar Plains 5B: Western Ghats - Western Ghats Mountains
The Deccan Peninsula	6A: Deccan Peninsular - Central Highlands 6B: Deccan Peninsular - Chotta Nagpur 6C: Deccan Peninsular - Eastern Highlands 6D: Deccan Peninsular - Central Plateau 6E: Deccan Peninsular - Deccan South
The Gangetic Plains	7A: Gangetic Plain - Upper Gangetic Plains 7B: Gangetic Plain - Lower Gangetic Plains
The Coasts	8A: Coasts - West Coast 8B: Coasts - East Coast 8C: Coasts - Lakshadweep
Northeast India	9A: North - East - Brahmaputra Valley 9B: North - East - North East Hills
Islands	10A: Islands - Andaman

Wildlife Diversity Of India

Himalayan mountain system

- The west Himalayas have low rainfall, heavy snowfall (temperate conditions), whereas in east Himalayas, there is heavy rainfall, snowfall only at very high altitudes, whereas at lower altitudes conditions are similar to the tropical rain forests. Flora and fauna of both Himalayas differ.

Himalayan foothills

- Flora: Natural monsoon evergreen and semi-evergreen forests; dominant species are **sal**, silk-cotton trees, giant bamboos; tall grassy meadow with savannahs in terai.
- Fauna: Includes big mammals of like elephant, sambar, swamp deer, cheetal, hog deer, barking deer, wild boar tiger, panther, hyena, black bear, sloth bear, Great Indian one-horned rhinoceros, wild buffalo, Gangetic gharial, golden langur.

[Some of these which are threatened are explained in the next post: Red Data Book]

Western Himalayas (High altitude region)

- Flora: Natural monsoon evergreen and semi-evergreen forests; **rhododendrons**; dwarf hill bamboo and birch forests mixed with alpine pastures.
- Fauna: Tibetan wild ass (kiang) [Don't confuse this with Asiatic wild ass which is found in Kutch region], wild goats (thar, ibex) and blue sheep; antelopes (Chiru and Tibetan gazelle), deers (hangul of Kashmir stag and shou or Sikkim stag, musk deer); golden eagle, snow cocks, snow partridges; snow leopard, black and brown bears; birds like Griffon vultures.

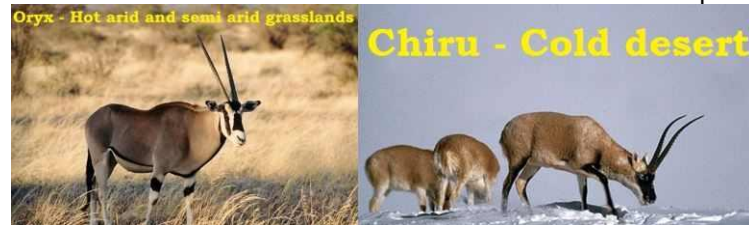
What is the difference between the antelopes Oryx and Chiru?

- Oryx is adapted to live in hot and arid areas whereas Chiru is adapted to live in steppes and semi-desert areas of cold high mountains.
- Oryx is poached for its antlers whereas Chiru is poached for its musk.

- Oryx exists in western India only whereas Chiru exists in north-east India only.
- None of the statements a, b, and c given above is correct.

They are both antelopes.

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Answer: a)

Eastern Himalayas

- Flora: **Oaks**, magnolias, laurels and birches covered with moss and ferns; coniferous forests of pine, fir, yew and junipers with undergrowth of scrubby **rhododendrons** and dwarf bamboos; lichens, mosses, orchids, and other epiphytes dominant (due to high humidity and high rainfall).
- Fauna: **Red panda**, hog badgers, forest badgers, crestless porcupines, takins etc.

Peninsular - Indian sub-region

- It has two zones.
- peninsular India and its extension into the drainage basin of the Ganges river system, and
 - desert region of Rajasthan-the Thar of Indian desert region.

Peninsular India

- It is home to tropical moist deciduous to tropical dry deciduous and scrub vegetation depending upon the variation in rainfall and humidity.
- Flora: **Sal** in north and east extensions (higher rainfall) and **teak** in southern plateau are dominant trees.
- West Ghats have evergreen vegetation (flora and fauna similar to evergreen rainforests of north eastern of India. In dry areas of Rajasthan and Aravalli hills, trees are scattered and thorny scrub species

predominate. The forests give way to more open savannah habit.

- Fauna: Elephant, wild boar, deers (cheetal or axis deer), hog deer swamp deer or barasinga, sambar, muntjak or barking deer, antelopes (four-horned antelope, nilgiri, blackbuck, chinkara gazelle), wild dog or dhole, tiger, leopard, cheetah, lion, wild pig, monkey, striped hyena, jackal, gaur.

Indian desert

- Thar desert of Rajasthan has unique flora and fauna.
- Flora: Thorny trees with reduced leaves; cacti, other succulents are the main plants.
- Fauna: Animals are mostly burrowing ones. Among mammals rodents are the largest group. The Indian desert gerbils are mouse like, rodents, other animals are, Asiatic wild ass, black buck, desert cat, caracal, red fox; reptiles (snakes, lizards and tortoise) well represented. Desert lizards include agamids, and geckos. Among birds the most discussed is **Great Indian bustard**.

Tropical rain forest region

- Distributed in areas of western Ghats and north east India.
- Flora: Extensive grass lands interspersed with densely forested gorges of evergreen vegetation known as sholas occur in the Nilgiris (an offshoot of Western Ghats). **Sholas** also occur in Anamalai and Palani hills.
- The rain forests of the Western Ghats have dense and lofty trees with much species diversity. Mosses, ferns, epiphytes, orchids, lianas and vines, herbs, shrubs make diverse habitat. **Ebony** trees predominate in these forests.
- Fauna: It is very rich with all kinds of animals. There are wild elephants, gaur and other larger animals. Most species are tree dwellers. The most prominent are hoolock gibbon (only ape found in India), golden langur, capped langur or leaf monkey, Assam macaque and the pig-tailed macaque, lion-tailed macaque, Nilgiri langur slender loris, bats, gaint squirrel, civets, flying squirrels, Nilgiri mongoose, spiny mouse.

Andaman and Nicobar Islands

- Flora: These are home for tropical rain forests. Mangroves are distributed in the coastal areas.
- Fauna: Among mammals, bats and rats; Andaman pig, crab-eating macaque, palm civet and deers (spotted deer, barking deer, hog deer, sambar).
- Among marine mammals there are dugong, false killer whale, dolphin.
- Among birds are rare one is Narcondum hornbill, white-bellied sea-eagle.
- Salt-water crocodile, a number of marine turtles, coconut crab, lizards (the largest being water monitor), 40 species of snakes including cobra, viper, voral and sea snake, python, etc. are present.

Mangrove swamps of Sunderbans

- Sunderbans are delta of the Ganges where both the Brahmaputra and Ganges join and drain into the Bay of Bengal.
- Flora: Various species of mangroves.
- Fauna. In the higher regions of mangroves, there are spotted deer, pigs, monitor lizard, monkeys. The most interesting animal of Sunderbans is the Royal Bengal Tiger.

Loss of Biodiversity - Man Animal Conflict - Culling

Loss of Biodiversity

- The IUCN Red List (2004) documents the extinction of 784 species in the last 500 years. Some examples of recent extinctions include the three subspecies (Bali, Javan, Caspian) of tiger.
- During the long period (> 3 billion years) since the origin and diversification of life on earth there were five episodes of mass extinction of species.
- Sixth Extinction, presently is in progress with current species extinction rates estimated to be 100 to 1,000 times faster than in the pre-human times.
- Ecologists warn that if the present trends continue, nearly half of all the species on earth might be wiped out within the next 100 years.
- In general, loss of biodiversity in a region may lead to
- decline in plant production,

- lowered resistance to environmental perturbations such as drought and
- increased variability in certain ecosystem processes such as plant productivity, water use, and pest and disease cycles.

Cause for the loss of biodiversity

- There are four major causes 'The Evil Quartet' are:

Habitat loss and fragmentation

- This is the most important cause driving animals and plants to extinction.
- Due to the growing human population, wetlands are being made dry through landfills, as the demand for land increases.
- Natural forests are cleared for industry, agriculture, dams, habitation, recreational sports, etc.
- The most dramatic examples of habitat loss come from tropical rain forests.
- Once covering more than 14 per cent of the earth's land surface, these rain forests now cover no more than 6 per cent. They are being destroyed fast.
- The Amazon rain forest (it is so huge that it is called the 'lungs of the planet') harboring probably millions of species is being cut and cleared for cultivating soya beans or for conversion to grasslands for raising beef cattle.
- Besides total loss, the degradation of many habitats by pollution also threatens the survival of many species.
- For example, pollution that leads to greenhouse effect results in global warming. All those species that are slow to adjust to the changed environment are eventually lost.
- A vast array of factors causing environmental degradation may result in the loss of biodiversity.
- Some of these factors are: global warming, increased CO₂ concentration in atmosphere, nuclear radiation; UV-exposure; oil spills, etc.
- When large habitats are broken up into small fragments due to various human activities, mammals and birds requiring large territories and certain animals with migratory habits are badly affected, leading to population declines.

Over-exploitation

- Humans have always depended on nature for food and shelter, but when 'need' turns to 'greed', it leads to over-exploitation of natural resources.
- In the last 500 years, many species extinctions (Steller's sea cow, passenger pigeon) were due to over exploitation by humans.
- Presently many marine fish populations around the world are over harvested, endangering the continued existence of some commercially important species.
- Whales for oil, fish for food, trees for wood, plants for medicines etc. are being removed by humans at higher rates than they can be replaced.
- Excessive cutting of trees, overgrazing, collection of fire-wood, hunting of wild animals for skin (for example tigers from reserve forests of India), ivory etc. all result in gradual loss of species.

Alien species invasions

- When alien species are introduced unintentionally or deliberately for whatever purpose, some of them turn invasive, and cause decline or extinction of indigenous species.
- The **Nile perch** introduced into Lake Victoria in east Africa led eventually to the extinction of an ecologically unique assemblage of more than 200 species of cichlid fish in the lake.
- You must be familiar with the environmental damage caused and threat posed to our native species by invasive weed species like **carrot grass (Parthenium)**, **Argemone**, **Lantana** and **water hyacinth (Eicchornia)**.
- The recent illegal introduction of the **African catfish** *Clarias gariepinus* for aquaculture purposes is posing a threat to the indigenous catfishes in our rivers.
- The colonization of tropical Pacific Islands by humans is said to have led to the extinction of more than 2,000 species of native birds.

Co-extinctions

- When a species becomes extinct, the plant and animal species associated with it in an obligatory way also become extinct.
- When a host fish species becomes extinct, its unique assemblage of parasites also meets the same fate.

- Another example is the case of a coevolved plant-pollinator mutualism where extinction of one invariably leads to the extinction of the other.

Poaching

- Large mammals such as the tiger, rhinoceros and the elephant once faced the distinct possibility of complete extinction due to rampant hunting and poaching.
- Global warming (Climate change), natural calamities are other reasons for loss of biodiversity.

Diseases that spread from cattle to wild animals

Diseases	Causative organism	Animal susceptible
Tuberculosis	Mycobacterium spp	Deer, cat, primates, elephant
Anthrax	Bacillus anthracis	Gaur, chital, wild pig, barking deer
Rabies	Rabies virus	Tiger, lion, bear, mongoose, squirrel
Foot and mouth disease	FMD virus	Gaur, nilgai, chital, sambar, yak, mithun
Rinderpest	Microbillivirus	Deer, wild pig, wild buffalo
Trypanosomiasis	Trypanosoma virus	Tiger, elephant, sambar, macaque
Toxoplasmosis	Toxoplasma gondii	Rhesus macaque, civet cat

Man - Animal Conflict

- It refers to the interaction between wild animals and people and the resultant negative impact on people or their resources, or wild animals or their habitat.
- It occurs when wildlife needs overlap with those of human populations, creating costs to residents and wild animals.

Causes

- Human population growth and encroachment into forest lands.
- Land use transformation – industrialization, infrastructure development, commercial farming etc.

- Species habitat loss, degradation and fragmentation due to above mentioned reasons.
- Increasing livestock populations and competitive exclusion of wild herbivores.
- Growing interest in ecotourism and increasing access to nature reserves.
- Abundant distribution of prey in the form of livestock on the periphery of forest lands.
- Increasing wildlife population as a result of conservation programmes.
- Climatic factors – climate change induced habitat destruction.
- Stochastic events (e.g. fire, floods etc.)

Impacts

- Crop damage and damage to property – elephants damage crops and villages.
- Livestock depredation – Himalayan snow leopard preys on goats in the Himalayan region. Farmers trap and kill snow leopards to save their livestock.
- Injuries and deaths – Man eater tigers are reported to have injured and killed villagers living on the periphery.
- Injuries to wildlife – leopards and other wild animals are hacked to death by mobs.

Preventive and Mitigation strategies

- Artificial and natural barriers (physical and biological) – very expensive.
- Guarding – very expensive.
- Alternative high-cost livestock husbandry practices
- Relocation: voluntary human population resettlement.
- Waste management systems that restrict wildlife access to refuse.
- Community based natural resource management schemes (CBNRMS)

Culling of animals – Conservation or Biodiversity loss?

Natural culling

- Culling means 'Selection'.
- In the wild, it is the process of weeding out of the weak.
- A population boom makes individuals compete for food and safety, and the weaker ones lose

out, leaving a smaller population of more able individuals.

- Culling naturally occurs by starvation, disease and predation. It is **nature's way of controlling population**.

Culling by humans – controlled culling

- In the post-conservation era, human intervention became necessary for the management of wildlife populations through controlled hunting, which is now referred to as culling.

Why did controlled culling become necessary?

- Man-Animal conflict – too many wild animals compete with humans for resources.
- Threat to life and livelihood (crop, property damage) makes culling necessary.
- Loss of forestland to mines, industry, agriculture, etc. is the primary reason behind man-animal conflict.
- Crop-raiding by smaller herbivores due to a population boom & animals raiding nutrient

rich crops like wheat and maize are other major reasons for man-animal conflict.

Reasons for wild herbivore population boom

- bans or restrictions on hunting,
- loss of natural predators,
- availability of non-forest food sources (cropland, garbage dumps), etc.

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The practice of Culling worldwide

- In the US, some areas require seasonal culling to ease pressure on livestock feed.
- In parts of Africa, culling has been used for commercial harvesting.
- Australia culls feral cats to protect native species.
- Australia also culls kangaroos.

Animal welfare activists vs. Pro culling lobby

Culling lobby: Affected farmers and government administration.



Arguments by animal welfare activists against culling	Arguments by Culling lobby
Ethical grounds: if we don't consider culling overpopulated humans, we do not have the right to cull other species.	They support practical and realistic approach.
Animal welfare activists believe that every individual animal is ethically indispensable, even at the cost of putting entire species at risk.	Conservationists are concerned about the integrity of the ecosystem and the future of entire species.
Culling is against animal conservation.	Culling is for conservation By law, wildlife are protected because they are too few and require protection. But when certain pockets see a population boom in herbivores, farmers cannot wait for predators to reoccupy such areas. The absence of lawful intervention often

Culling drives are often not well monitored frequently leads to free-for-all shooting sprees.

Culling can increase peoples apathy (lack of interest or enthusiasm) for conservation of other forms of life.



[Some people recruited to cull animals post their pictures along with their prey on social media. This is totally unprofessional as it glorifies killing an animal.]

Culling creates a conducive atmosphere for the poaching mafia to move in.

They argue that man-animal conflict can be controlled through non-invasive means, including

fencing crop fields,
planting chilli around cropland,
creating buffer areas between croplands and the forest's edge,
selecting non-edible crops,
providing adequate and regular compensation for crop loss, etc.

triggers retaliation by illegal means. In anger, farmers may indiscriminately target wildlife, including those that may not be causing any problem.

Can be monitored easily using strict guidelines and law enforcement.

No concrete evidence to prove that culling increases insensitivity among people.

[But practically it does. Lot of people enjoy hunting animals and others might take inspiration from culling]

It is specious (misleading) to claim that time-bound, limited permissions to cull create an atmosphere for poaching.

This is because most of the cropland areas may not host commercially lucrative wildlife species.

Fencing merely shifts conflict to the next accessible cropland.

Fencing is expensive and its maintenance is not effective.

Even selection of crops that traditionally repelled animals does not seem to work any longer. For example, farmers in Sirmour, Himachal Pradesh, now complain that monkeys raid garlic fields that they avoided until recently.

Other measures are mostly long-term and can keep a check on conflict only in situations where crop damage is still reasonable.

But in pockets where the situation is already out of hand, the only option is often to reduce the number of habitual crop-raiders.

Source: <http://goo.gl/Hm46Sm>

Invasive Alien Species

- Purposely or accidentally, people often bring non-native species into new areas where the species have few or no natural predators to keep their populations in check.
- Aliens are species that occur outside their natural range. Alien species that threaten native plants and animals or other aspects of biodiversity are called alien invasive species. They occur in all groups of plants and animals, as competitors, predators, pathogens and parasites, and they have invaded almost every type of native ecosystem.
- Biological invasion by alien species is recognized as one of the major threats to native species and ecosystems. The effects on biodiversity are enormous and often irreversible.

Invasion and Species Richness?

- The invasions potentially lead to an increase in species richness, as invasive species are added to the existing species pool.
- But it also leads to extinction of native species, resulting in decrease of species richness.
- The negative interactions are primarily the competition with natives for food and sustenance, which may not allow coexistence and also by predation.

Effects

- Loss of Biodiversity
- Decline of Native Species (Endemics).
- Habitat Loss
- Introduced pathogens reduce crop and stock yields
- Degradation of marine and freshwater ecosystems
- This biological invasion constitutes the greatest threat to biodiversity, and it has already had devastating consequences for the planet and challenges for the conservation managers.
- About 235 invasive alien species (both flora and fauna) are reported in India by Global Invasive Alien Species Database.

Some Invasive fauna in India are

- A new invasive gall forming insect of Eucalyptus in Southern India.

- Leptocybe invasa - a new insect pest detected from few pockets of coastal Tamil Nadu and it has spread to peninsular India.
- It is a tiny wasp that forms leaf and stem galls in Eucalyptus.
- Crazy ant
- Giant African snail
- Myna
- Gold Fish
- Pigeon
- Donkey
- House Gecko
- Tilapia

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Some Invasive Alien Flora Of India

Needle Bush

- Nativity: Trop. South America
- Distribution in India: Throughout
- A shrub or small tree.
- Remarks: Occasional in thorny scrub and dry degraded forests and often creates close thickets.

Black Wattle

- Nativity: South East Australia
- Distribution in India: Western Ghats
- Remarks: Introduced for afforestation in Western Ghats. Regenerates rapidly after fire and forms dense thickets. It is distributed in forests and grazing lands in high altitude areas.

Goat weed

- Nativity: Trop. America
- Distribution in India: Throughout
- Remarks: Aggressive colonizer. Troublesome weed in gardens, cultivated fields and forests.

Alternanthera paronychioides

- Nativity: Trop. America
- Distribution in India: Throughout
- Remarks: Occasional weed along edges of tanks, ditches and in marshy lands.

Prickly Poppy

- Nativity: Trop. Central & South America
- Distribution in India: Throughout

- Remarks: Aggressive colonizer. Common winter season weed in cultivated fields, scrub lands and fringes of forests.

Blumea eriantha

- Nativity: Trop. America
- Distribution in India: Throughout
- Remarks: Aggressive colonizer. Abundant along railway tracks, road sides and degraded forest lands.

Palmyra, Toddy Palm

- Nativity: Trop. Africa
- Distribution in India: Throughout
- Remarks: Aggressive colonizer. Cultivated and self-sown, occasionally found to be gregarious nearby cultivated fields, scrub lands and waste lands.

Calotropis / Madar, Swallo Wort

- Nativity: Trop. Africa
- Distribution in India: Throughout
- Remarks: Aggressive colonizer. Common in cultivated fields, scrub lands and waste lands.

Datura, Mad Plant, Thorn Apple

- Nativity: Trop. America
- Distribution in India: Throughout
- Remarks: Aggressive colonizer. Occasional weed on disturbed ground.

Water Hyacinth

- Nativity: Trop. America
- Distribution in India: Throughout
- Remarks: Aggressive colonizer. Abundant in still or slow floating waters. Nuisance for aquatic ecosystems.

Impatiens, Balsam

- Nativity: Trop. America
- Distribution in India: Throughout
- Remarks: Aggressive colonizer. Common along streams of moist forests and occasionally along railway tracks; also runs wild in gardens.

Ipomoea / the pink morning glory

- Nativity: Trop. America
- Distribution in India: Throughout
- Remarks: Aggressive colonizer. Common weed of marshy lands and along the edges of tanks and ditches.

Lantana camara / Lantana, Wild Sage

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- Nativity: Trop. America
- Distribution in India: Throughout
- Remarks: Aggressive colonizer. Common weed of forests, plantations, habitation, waste lands and scrub lands.

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Black Mimosa

- Nativity: Trop. North America
- Distribution in India: Himalaya, Western Ghats
- Remarks: Aggressive colonizer. It invades water courses and seasonally flooded wetlands.

Touch-Me-Not, Sleeping Grass

- Nativity: Brazil
- Distribution in India: Throughout
- Remarks: Aggressive colonizer. Common weed of cultivated fields, scrub lands and degraded forests.

4 'O' clock plant.

- Nativity: Peru
- Distribution in India: Throughout
- Remarks: Aggressive colonizer. Runs wild in gardens and near habitation.

Parthenium / Congress grass, Parthenium

- Nativity: Trop. North America
- Distribution in India: Throughout
- Remarks: Aggressive colonizer. Common weed of cultivated fields, forests, overgrazed pastures, waste lands and gardens.

Prosopis juliflora / Mesquite

- Mesquite
- Nativity: Mexico
- Distribution in India: Throughout

- Remarks: Aggressive colonizer. Common weed of waste lands, scrub lands and degraded forests.

Townsend grass

- Nativity: Trop. W. Asia
- Distribution in India: Throughout
- Remarks: Very common along streams and banks of rivers.

Species Extinction

Extinction is caused through various processes:

- Deterministic processes that have a cause and effect. E.g. glaciations, human interference such as deforestation.
- Stochastic processes (chance and random events) that effect the survival and reproduction of individuals. E.g. unexpected changes of weather patterns, decreased food supply, disease, increase of competitors, predators or parasites, etc. that may act independently or add to deterministic effects.
- The impact of these processes will of course depend on the size and degree of genetic diversity and resilience of populations.
- Traits that adversely affect or increase a species vulnerability to extinction due to habitat fragmentation have been identified. These are:

1. rarity or low abundance
2. poor dispersal ability
3. high trophic status - as animals occupying a higher trophic level (i.e. the position of a species in a food chain) usually have smaller populations than those at lower levels (e.g. carnivores are fewer in number than herbivores)
4. low adult survival rates

Natural extinctions

Have been caused due to several factors:

- continent drifting,
- climate change,
- tectonic activity,
- increased volcanic activity,
- The late Ordovician global glaciations (439 Mya).

- The late Cretaceous extinction assumed to be associated with an extra-terrestrial impact.
- Extinction in vascular plants has been more gradual compared with the loss of animals. It is believed that extinction among this group was due more to competitive displacement by more advanced plant forms, or due to a gradual climate change, than due to any sudden catastrophic event.

Artificial Extinction

- Even though species extinction is a natural process which can happen without the intervention of humans, extinctions caused by humans is now happening over and above the reasonable estimate of natural extinction rates.
- Species are threatened with extinction by the intervention of humans due to:
- Direct causes - such as hunting, collection or capture and persecution
- Indirect causes - such as habitat loss, modification and fragmentation and the introduction of invasive species.

Biodiversity Conservation - In Situ & Ex Situ Conservation

Biodiversity Conservation

- When we conserve and protect the whole ecosystem, its biodiversity at all levels is protected. E.g. we save the entire forest to save the tiger. This approach is called in **situ (on site) conservation**.
- However, when there are situations where an animal or plant is endangered or threatened and needs urgent measures to save it from extinction, **ex situ (off site) conservation** is the desirable approach.

Benefits of Biodiversity conservation

- Conservation of biological diversity leads to conservation of essential ecological diversity to preserve the continuity of food chains.
- The genetic diversity of plants and animals is preserved.
- It ensures the sustainable utilization of life support systems on earth.
- It provides a vast knowledge of potential use to the community.

- A reservoir of wild animals and plants is preserved, thus enabling them to be introduced, if need be, in the surrounding areas.
- Biodiversity conservation assures sustainable utilization of potential resources.

In situ conservation

- Faced with the conflict between development and conservation, many nations find it unrealistic and economically not feasible to conserve all their biological wealth.
- On a global basis, this problem has been addressed by eminent conservationists. They identified for maximum protection certain **'biodiversity hotspots'** regions with **very high levels of species richness** and **high degree of endemism** (that is, species confined to that region and not found anywhere else).
- Initially 25 biodiversity hotspots were identified but subsequently nine more have been added to the list, bringing the total number of biodiversity hotspots in the world to 34.
- These hotspots are also regions of accelerated habitat loss. Three of these hotspots - Western Ghats and Sri Lanka, Indo-Burma and Eastern Himalayas cover our country's exceptionally high biodiversity regions.
- Although all the biodiversity hotspots put together cover less than 2 percent of the earth's land area, the number of species they collectively harbour is extremely high and strict protection of these hotspots could reduce the ongoing mass extinctions by almost 30 per cent.
- In India, ecologically unique and biodiversity-rich regions are legally protected as **biosphere reserves, national parks, sanctuaries, reserved forests, protected forests** and **nature reserves**.
- India now has 14 biosphere reserves, 90 national parks and 448 wildlife sanctuaries.
- **Plantation, cultivation, grazing, felling trees, hunting and poaching are prohibited in biosphere reserves, national parks and sanctuaries.**

Protected Area Network in India

- **National Board for Wildlife (NBWL)**, chaired by the **Prime Minister of India** provides for

policy framework for wildlife conservation in the country.

- The **National Wildlife Action Plan** (2002-2016) was adopted in 2002, emphasizing the people's participation and their support for wildlife conservation.

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Reserved & Protected Forests

- As of present, reserved forests and protected forests differ in one important way:
 1. Rights to all activities like hunting, grazing, etc. in reserved forests are banned unless specific orders are issued otherwise.
 2. In protected areas, rights to activities like hunting and grazing are sometimes given to communities living on the fringes of the forest, who sustain their livelihood partially or wholly from forest resources or products.
- The first reserve forest in India was **Satpura National Park** in Madhya Pradesh.
- Typically, reserved forests are often upgraded to the status of wildlife sanctuaries, which in turn may be upgraded to the status of national parks, with each category receiving a higher degree of protection and government funding.

Wildlife Sanctuaries or wildlife refuges

- Wildlife Sanctuaries or wildlife refuges are home to various endangered species.
- They are safe from hunting, predation or competition.
- They are safeguarded from extinction in their natural habitat.
- Certain rights of people living inside the Sanctuary could be permitted.
- Grazing, firewood collection by tribals is allowed but strictly regulated.
- Settlements not allowed (few exceptions: tribal settlements do exist constant; efforts are made to relocate them).
- A Sanctuary can be promoted to a National Park.
- There are more than 500 wildlife sanctuaries in India.

National Park

- National parks are areas reserved for wild life where they can freely use the habitats and natural resources.
- The difference between a Sanctuary and a National Park mainly lies in the vesting of rights of people living inside.
- Unlike a Sanctuary, where certain rights can be allowed, in a National Park, no rights are allowed.
- No grazing of any livestock shall also be permitted inside a National Park while in a Sanctuary, the Chief Wildlife Warden may regulate, control or prohibit it.

Eco-Sensitive Zones

- The **National Wildlife Action Plan (2002–2016)** of MoEFCC stipulated that state governments should declare land falling within 10 km of the boundaries of national parks and wildlife sanctuaries as eco fragile zones or ESZs under the **Environmental (Protection) Act, 1986**.
- The purpose of the ESZ was to provide more protection to the parks by acting as a shock absorber or transition zone.
- Eco-Sensitive Zones would minimise forest depletion and man-animal conflict.
- The protected areas are based on the core and buffer model of management.
- The core area has the legal status of being a national park.
- The buffer area, however, does not have legal status of being a national park and could be a reserved forest, wildlife sanctuary or tiger reserve.

With reference to 'Eco-Sensitive Zones', which of the following statements is/are correct?

- Eco-Sensitive Zones are the areas that are declared under the Wildlife (Protection) Act, 1972.
- The purpose of the declaration of Eco-Sensitive Zones is to prohibit all kinds of human activities, in those zones except agriculture.

Select the correct answer using the code given below.

1. 1 only
2. 2 only
3. Both 1 and 2

4. Neither 1 nor 2

Answer: d) Neither

Biosphere Reserve

- Large areas of protected land for conservation of wild life, plant and animal resources and traditional life of the **tribals** living in the area.
- May have one more national parks or wildlife sanctuaries in it.

Core area

- Comprises a strictly protected ecosystem for conserving ecosystems, species and genetic variation. In core or natural zone **human activity is not allowed**.

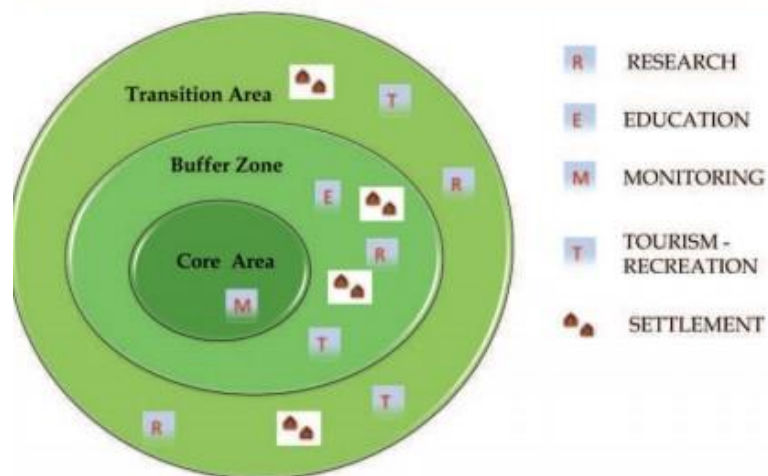
Buffer zone

- Used for scientific **research, monitoring, training and education**.

Transition area

- Ecologically sustainable human settlements and economic activities (tourism) are permitted.
- With the cooperation of reserve management and local people several human activities like settlements, cropping, recreation, and forestry are carried out without disturbing the environment.

BIOSPHERE RESERVE ZONING



Credits: <http://cdn.intechopen.com/>

Biosphere Reserves in India

Name	Location	State	Type	Key fauna
Nilgiri Biosphere Reserve	Waynad, Nagarhole, Bandipur and Mudumalai, Nilambur, Silent Valley and Anaimalai Hills	Tamil Nadu, Kerala and Karnataka	Western Ghats	Nilgiri tahr, lion-tailed macaque
Nanda Devi National Park & Biosphere Reserve	Parts of Chamoli District, Pithoragarh District & Bageshwar District	Uttarakhand	Western Himalayas	Snow Leopard, Himalayan Black Bear
Gulf of Mannar	Indian part of Gulf of Mannar extending from Rameswaram island in the North to Kanyakumari in the South of Tamil Nadu and Sri Lanka	Tamil Nadu	Coasts	Dugong or sea cow
Nokrek	Part of Garo Hills	Meghalaya	East Himalayas	Red panda
Sundarbans	Part of delta of Ganges and Brahmaputra river system	West Bengal	Gangetic Delta	Royal Bengal tiger
Manas	Part of Kokrajhar, Bongaigaon, Barpeta, Nalbari, Kamrup and Darrang Districts	Assam	East Himalayas	Golden langur, red panda
Simlipal	Part of Mayurbhanj district	Odisha	Deccan Peninsula	Gaur, royal Bengal tiger, elephant
Dihang-Dibang	Part of Siang and Dibang Valley	Arunachal Pradesh	Eastern Himalaya	
Pachmarhi Biosphere Reserve	Parts of Betul District, Hoshangabad District and Chhindwara District	Madhya Pradesh	Semi-Arid	Giant squirrel, flying squirrel
Achanakmar-Amarkantak Biosphere Reserve	Part of Annupur, Dindori and Bilaspur districts	Madhya Pradesh, Chhattisgarh	Maikala Hills	Four-horned antelope, Indian wild dog, Sarus crane, White-rumped vulture, Philautus sanctisilvaticus (Sacred grove bush frog)
Great Rann of Kutch	Part of Kutch, Rajkot, Surendranagar and Patan Districts	Gujarat	Desert	Indian wild ass
Cold Desert	Pin Valley National Park and surroundings; Chandratal and Sarchu & Kibber Wildlife Sanctuary	Himachal Pradesh	Western Himalayas	Snow leopard
Khangchendzonga	Parts of Kangchenjunga	Sikkim	East Himalayas	Snow leopard, red panda
Agasthyamalai Biosphere Reserve	Neyyar, Peppara and Shendurjun Wildlife Sanctuary and their adjoining areas	Kerala, Tamil Nadu	Western Ghats	Nilgiri tahr, elephants
Great Nicobar Biosphere Reserve	Southern most islands of Andaman and Nicobar Islands	Andaman and Nicobar Islands	Islands	Saltwater crocodile
Dibru-Saikhowa	Part of Dibrugarh and Tinsukia districts	Assam	East Himalayas	Golden langur
Seshachalam Hills	Seshachalam Hill Ranges covering parts of Chittoor and Kadapa districts	Andhra Pradesh	Eastern Ghats	
Panna	Part of Panna District and Chhatarpur District	Madhya Pradesh	Catchment Area of the Ken River	Tiger, chital, chinkara, sambhar and sloth bear

Conserving Biodiversity

The most important strategy for the conservation of biodiversity together with traditional human life is the establishment of

1. biosphere reserves
2. botanical gardens
3. national parks
4. wildlife sanctuaries

Answer: a)

Man and Biosphere Programme (MAB programme)

- It was first started by UNESCO in 1971.
- Later introduced in India in 1986.

Aim

1. Studying the effects of human interference and pollution on the biotic and abiotic components of ecosystems.
2. Conservation the ecosystems for the present as well as future.

The main objects of MAB programme are to:

1. Conserve representative samples of ecosystem.
2. Provide long term in situ conservation of genetic diversity.
3. Provide opportunities for education and training.
4. Provide appropriate sustainable managements of the living resources.
5. Promote international co-operation.

Tiger Reserves

- Same as sanctuaries. But they are monitored by NTCA under Project tiger.
- The various tiger reserves were created in the country based on 'core-buffer' strategy.

Core area

- The core areas are freed of all human activities.
- It has the legal status of a national park or wildlife sanctuary.
- Collection of minor forest produce, grazing, and other human disturbances are not allowed.

Buffer areas

- Twin objectives:

1. providing habitat supplement to spillover population of wild animals from core area.
2. provide site specific co-developmental inputs to surrounding villages for relieving their impact on core area.

- Collection of minor forest produce and grazing by tribals is allowed on a sustainable basis.
- The Forest Rights Act passed by the Indian government in 2006 recognizes the rights of some forest dwelling communities in forest areas.

Conservation Reserves

- Conservation Reserves can be declared by the State Governments in any area owned by the Government, particularly the areas adjacent to National Parks and Sanctuaries and those areas which link one Protected Area with another.
- Such declaration should be made after having consultations with the local communities.
- The rights of people living inside a Conservation Reserve are not affected.

Community Reserves

- Community Reserves can be declared by the State Government in any private or community land, not comprised within a National Park, Sanctuary or a Conservation Reserve, where an individual or a community has volunteered to conserve wildlife and its habitat.
- As in the case of a Conservation Reserve, the rights of people living inside a Community Reserve are not affected.

Sacred Groves

- India has a history of religious/cultural traditions that emphasised protection of nature.
- In many cultures, tracts of forest were set aside, and all the trees and wildlife within were venerated and given total protection.
- Such sacred groves are found in **Khasi and Jaintia Hills in Meghalaya, Aravalli Hills of Rajasthan, Western Ghat regions of Karnataka and Maharashtra and the Sarguja, Chanda and Bastar areas of Madhya Pradesh.**

- In Meghalaya, the sacred groves are the last refuges for a large number of rare and threatened plants.

Ex situ Conservation

- In this approach, threatened animals and plants are taken out from their natural habitat and placed in special setting where they can be protected and given special care.
- **Zoological parks, botanical gardens, wildlife safari parks and seed banks** serve this purpose.
- There are many animals that have become extinct in the wild but continue to be maintained in zoological parks.
- In recent years ex situ conservation has advanced beyond keeping threatened species. Now **gametes** of threatened species can be preserved in viable and fertile condition for long periods using cryopreservation techniques, eggs can be fertilized in vitro, and plants can be propagated using tissue culture methods.
- Seeds of different genetic strains of commercially important plants can be kept for long periods in **seed banks**.
- The national gene bank at **National Bureau Of Plant Genetic Resources (NBPGR), Delhi** is primarily responsible for conservation of unique accessions on long-term basis, as base collections for posterity, predominantly in the form of seeds.

Botanical garden

- Botanical garden refers to the scientifically planned collection of living trees, shrubs, herbs, climbers and other plants from various parts of the globe.

Purpose of botanical gardens

- To study the taxonomy as well as growth of plants.
- To study the introduction and acclimatization process of exotic plants.
- It acts as a germplasm collection.
- It helps development of new hybrids.
- It augments conserving rare and threatened species.
- It facilitates training of staff.
- It acts as a source of recreation.

Zoo

- Zoo is an establishment, whether stationary or mobile, where captive animals are kept for exhibition to the public and includes a circus and rescue centers but does not include an establishment of a licensed dealer in captive animals.
- The initial purpose of zoos was entertainment, over the decades, zoos have got transformed into centers for wildlife conservation and environmental education.
- Apart from saving individual animals, zoos have a role to play in species conservation too (through captive breeding).
- Zoos provide an opportunity to open up a whole new world, and this could be used in sensitizing visitors regarding the value and need for conservation of wildlife.

Which one of the following is not a site for in-situ method of conservation of flora?

1. Biosphere Reserves
2. Botanical Garden
3. National Park
4. Wildlife Sanctuary

In situ conservation = on the site conservation without displacing the affected organism.

Ex situ conservation = conserving the organism in an artificial habitat by displacing it from its natural habitat.

Botanical Garden = Plants are bred in a protected environment far from their natural home, especially for research purposes. So its Ex situ conservation.

Rest all along with protected forests and reserved forests are In situ conservation methods.

Answer: b) Botanical Garden

Constraints in biodiversity conservation

- Low priority for conservation of living natural resources.
- Exploitation of living natural resources for monetary gain.
- Values and knowledge about the species and ecosystem inadequately known.

- Unplanned urbanization and uncontrolled industrialization.

Historic Citizen Movements to Conserve Biodiversity

Chipko Movement

- It is a social-ecological movement that practiced the Gandhian methods of satyagraha and nonviolent resistance, through the act of hugging trees to protect them from falling.
- The modern Chipko movement started in the early 1970s in the **Garhwal Himalayas** of Uttarakhand, with growing awareness towards rapid deforestation.
- The landmark event in this struggle took place on March 26, 1974, when a group of peasant women in Reni village, Hemwalghati, in Chamoli district, Uttarakhand, India, acted to prevent the cutting of trees and **reclaim their traditional forest rights** that were threatened by the contractor system of the state Forest Department. Their actions inspired hundreds of such actions at the grassroots level throughout the region.
- By the 1980s the movement had spread throughout India and led to formulation of people-sensitive forest policies, which put a stop to the open felling of trees in regions as far reaching as Vindhyas and the Western Ghats.
- The first recorded event of Chipko however, took place in village Khejarli, Jodhpur district, in 1730 AD, when 363 Bishnois, led by Amrita Devi sacrificed their lives while protecting green Khejri trees, considered sacred by the community, by hugging them, and braved the axes of loggers sent by the local ruler, today it is seen an inspiration and a precursor for Chipko movement of Garhwal.

Appiko Movement

- Appiko movement was a revolutionary movement based on environmental conservation in India.
- The Chipko movement in Uttarakhand in the Himalayas inspired the villagers of the district of Karnataka province in southern India to launch a similar movement to save their forests.

- In September 1983, men, women and children of Salkani 'hugged the trees' in Kalase forest. (The local term for 'hugging' in Kannada is appiko.)
- Appiko movement gave birth to a new awareness all over southern India.

Biodiversity Hot Spots Across India and World

Biodiversity Hot Spots

- Hot spots are the richest and most threatened reservoirs of plant and animal life of the earth. They have **maximum number of endemic species**.
- 25 terrestrial hot spots have been identified for the conservation of biodiversity. They occupy 1.4% of the earth's surface and 20% of worlds the human population lives in these areas. Western Ghats and Eastern Himalayas (Includes some parts of Indo-Burma or Purvanchal Hills) are two hot spots of India.

To qualify as a hot spot, a region must meet two strict criteria:

1. **Species endemism:** the region must contain at least 1,500 species of vascular plants (> 0.5% of the world's total) as endemics, and
 2. **Degree of threat:** the region has to have lost at least 70% of its original habitat.
- Each biodiversity hot spot represents a remarkable universe of extraordinary floral and faunal endemism struggling to survive in rapidly shrinking ecosystems.
 - Over 50 percent of the world's plant species and 42 percent of all terrestrial vertebrate species are endemic to the 34 biodiversity hot spots.
 - Some hot spots are much richer than others in terms of their numbers of endemics. Five key factors have been taken into consideration and those biodiversity hot spot that tops the list with respect to these five factors are considered as hottest hot spots.

Factors

1. Endemic plants,
2. Endemic vertebrates,
3. Endemic plants/area ratio (species per 100km²),

- Endemic vertebrates/area ratio (species per 100km²) and
- Remaining primary vegetation as % of original extent.

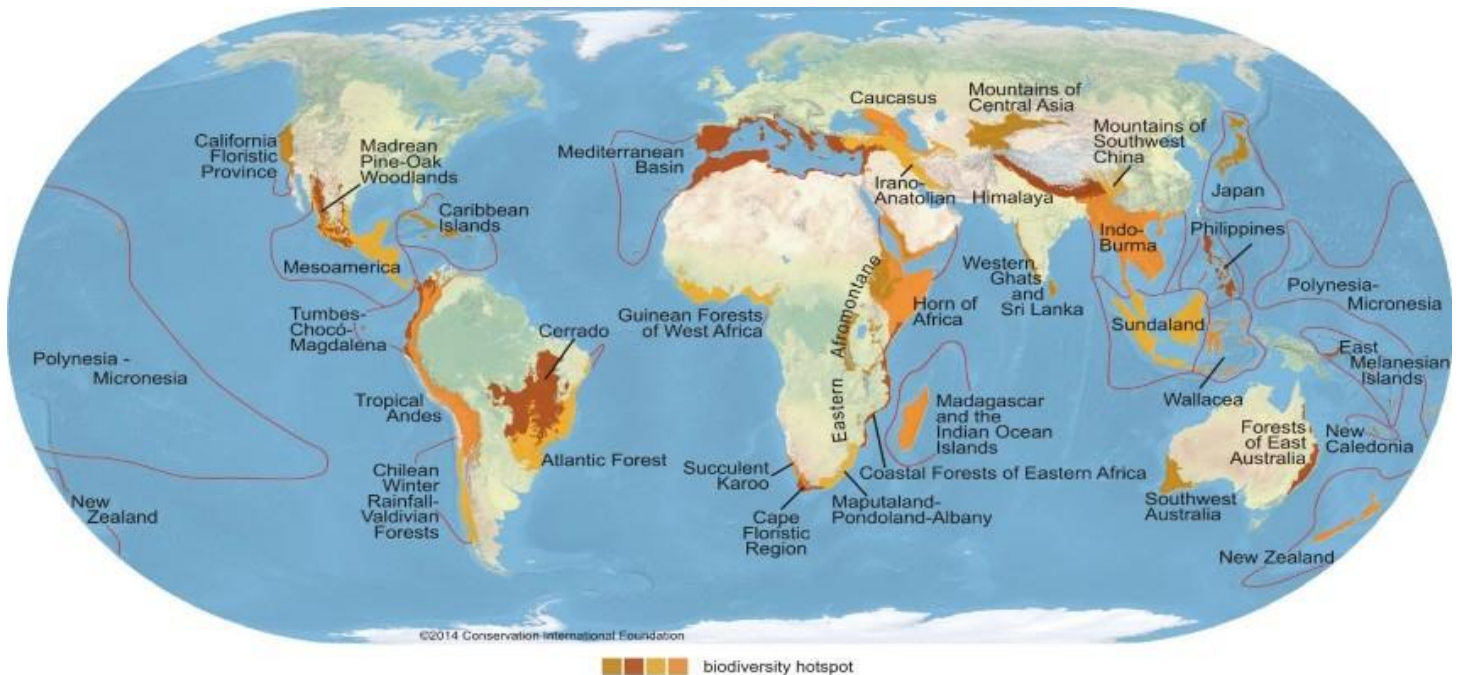
Biodiversity Hotspots Across the World

The eight hottest hot spots in terms of the above five factors are:

- Madagascar

- Philippines
- Sundaland [South East Asia]
- Brazil's Atlantic Forest
- Caribbean
- Indo-Burma
- Western Ghats and Sri Lanka
- Eastern Arc and Coastal Forests of Tanzania/Kenya

These eight 'hottest hot spots', appear at least three times in the top ten listings for each factor.



Conservation International (conservation.org) defines 35 biodiversity hotspots — extraordinary places that harbor vast numbers of plant and animal species found nowhere else. All are heavily threatened by habitat loss and degradation, making their conservation crucial to protecting nature for the benefit of all life on Earth.

Indian Biodiversity Hot Spots

There are 3 biodiversity hot spots present in India. They are:

- The Eastern Himalayas [Arunachal Pradesh, Bhutan, Eastern Nepal]**
- Indo-Burma and [Purvanchal Hills, Arakan Yoma, Eastern Bangladesh]**
- The Western Ghats and Sri Lanka**

Indo-Burma Region

- The Indo-Burma region encompasses several countries.
- It is spread out from Eastern Bangladesh to Malaysia and includes North-Eastern India south of Brahmaputra river, Myanmar, the southern part of China's Yunnan province,

Lao People's Democratic Republic, Cambodia, Vietnam and Thailand.

- The Indo-Burma region is spread over 2 million sq. km of tropical Asia.
- Since this hotspot is spread over such a large area and across several major landforms, there is a wide diversity of climate and habitat patterns in this region.

Biodiversity of Indo-Burma Region

- Much of this region has been deteriorating rapidly in the past few decades.
- This region is home to several primate species such as monkeys, langurs and gibbons with populations numbering only in the hundreds.
- Many of the species, especially some freshwater turtle species, are endemic.
- Almost 1,300 bird species exist in this region including the threatened white-eared night-

heron [Endangered], the grey-crowned crocias [Endangered], and the orange-necked partridge [Near Threatened].

- It is estimated that there are about 13,500 plant species in this hotspot, with over half of them endemic. Ginger, for example, is native to this region.

The Eastern Himalayas

- The Eastern Himalayas is the region encompassing Bhutan, northeastern India, and southern, central, and eastern Nepal.
- The abrupt rise of the Himalayan Mountains from less than 500 meters to more than 8,000 meters results in a diversity of ecosystems that range from alluvial grasslands and subtropical broad leaf forests along the foothills to temperate broad leaf forests in the mid hills, mixed conifer and conifer forests in the higher hills, and alpine meadows above the tree line.

Biodiversity of the Eastern Himalayas

- The Eastern Himalayan hotspot has nearly 163 globally threatened species (both flora and fauna) including the One-horned Rhinoceros [Vulnerable], the Wild Asian Water buffalo [Endangered].
- There are an estimated 10,000 species of plants in the Himalayas, of which one-third are endemic and found nowhere else in the world.
- A few threatened endemic bird species such as the Himalayan Quail, Cheer pheasant, Western tragopan are found here, along with some of Asia's largest and most endangered birds such as the Himalayan vulture and White-bellied heron.
- Mammals like the Golden langur, The Himalayan tahr, the pygmy hog, Lang-urs, Asiatic wild dogs, sloth bears, Gaurs, Muntjac, Sambar, Snow leopard, Black bear, Blue sheep, Takin, the Gangetic dolphin, wild water buffalo, swamp deer call the Himalayan ranged their home.

Western Ghats and Sri Lanka

- Western Ghats, also known as the "Sahyadri Hills" encompasses the mountain forests in the southwestern parts of India and highlands of southwestern Sri Lanka.

- The entire extent of hotspot was originally about 1,82,500 square kms, but due to tremendous population pressure, now only 12,445 square Km or 6.8% is in pristine condition.
- The wide variation of rainfall patterns in the Western Ghats, coupled with the region's complex geography, produces a great variety of vegetation types.
- These include scrub forests in the low-lying rainshadow areas and the plains, deciduous and tropical rainforests up to about 1,500 meters, and a unique mosaic of montane forests and rolling grasslands above 1,500 meters.
- In Sri Lanka diversity includes dry evergreen forests to dipterocarpus dominated rainforests to tropical montane cloud forest.
- The important populations include Asian elephant, Niligiri tahr, Indian tigers, lion tailed macaque [All Endangered], Indian Giant squirrel [Least Concern], etc.

World Heritage Sites

- World Heritage Sites means "Sites any of various areas or objects inscribed on the United Nations Educational, Scientific, and Cultural Organization (UNESCO) World Heritage List".
- The sites are designated as having outstanding universal value under the Convention concerning the Protection of the World Cultural and Natural Heritage.
- This Convention, which was adopted by the UNESCO in 1972 (and enforced in 1975) provides a framework for international cooperation in preserving and protecting cultural treasures and natural areas throughout the world. The first list of World Heritage state was published in 1978.
- The convention defines the kind of sites which can be considered for inscription of the World heritage list (ancient monuments, museums, **biodiversity and geological heritage**), and sets out the duties of the State Parties in identifying potential sites and their role in protecting them.

"Natural heritage sites are restricted to those natural areas that

- furnish outstanding examples of the Earth's record of life or its geologic processes.

- provide excellent examples of ongoing ecological and biological evolutionary processes.
- contain natural phenomena that are rare, unique, superlative, or of outstanding beauty

or

- furnish habitats or rare endangered animals or plants or are sites of exceptional biodiversity”.
- There are ten criteria for cultural heritage and natural heritage.
- Nominated sites must be of "outstanding universal value" and meet at least one of the ten criteria. The criteria are given below.

Criteria

1. to represent a masterpiece of human creative genius;
 2. to exhibit an important interchange of human values, over a span of time or within a cultural area of the world;
 3. to bear a unique or at least exceptional testimony to a cultural tradition or to a civilization which is living or which has disappeared;
 4. to be an outstanding example of a type of building, architectural or technological ensemble or landscape which illustrates a significant stage(s) in human history;
 5. to be an outstanding example of a traditional human settlement, land-use, or sea-use,;
 6. to contain superlative natural phenomena or areas of exceptional natural beauty and aesthetic importance;
 7. to be outstanding examples representing major stages of Earth's history;
 8. to contain the most important and significant natural habitats for insists conservation of biological diversity, including those containing threatened species of outstanding universal value from the point of view of science or conservation.
- The UNESCO funds numerous efforts to preserve and restore World Heritage Sites in developing nations.
 - It maintains also a List of World Heritage Sites in developing nations.
 - It maintains also a List of World Heritage in danger facing threat of pollution and other natural hazards.

- Sites subject to unusual levels of pollution, natural hazards, or other problems may be placed for restoration. Such designated sites facilitate promotion of tourism.

International Year of Biodiversity

Page

- The United Nations declared 2010 to be the International Year of Biodiversity. It is a celebration of life on earth and of the value of biodiversity for our lives.

Slogan

"Biodiversity is variety of life on earth

Biodiversity is life.

Biodiversity is our life".

IUCN Red List India - Red Data List - Red Book

International Union for Conservation of Nature

- (IUCN) is an international organization (**NGO**) working in the field of nature conservation and sustainable use of natural resources.
- It is involved in data gathering and analysis, research, field projects, advocacy, lobbying and education.
- The organization is best known for compiling and publishing the IUCN Red List, which assesses the conservation status of species worldwide.
- Its headquarters are in Gland, Switzerland.

BirdLife International

- BirdLife International is the world's largest nature conservation Partnership.
- Together they are 120 BirdLife Partners worldwide.
- BirdLife International strives to conserve **birds**, their habitats and global biodiversity, working with people towards sustainability in the use of natural resources.
- BirdLife International is the official **Red List** authority for birds, for the International Union for Conservation of Nature.
- It identifies the sites known/referred to as **'Important Bird and Biodiversity Areas'**.

With reference to an organization known as 'Birdlife International' which of the following statements is/are correct?

- It is a Global Partnership of Conservation Organizations.
- The concept of 'biodiversity hotspots' originated from this organization.
- It identifies the sites known/referred to as 'Important Bird and Biodiversity Areas'.

Select the correct answer using the code given below.

1. 1 only
2. 2 and 3 only
3. 1 and 3 only
4. 1, 2 and 3

BirdLife International has nothing to do with the concept "biodiversity hotspots".

Answer: c) 1 and 3 only

CITES

CITES (the Convention on International Trade in Endangered Species of Wild Fauna and Flora) is also known as the **Washington Convention**

- It is a **multilateral treaty** drafted as a result of a resolution adopted in 1963 at a meeting of members of the International Union for Conservation of Nature (**IUCN**).
- CITES entered into force in 1975.
- Its aim is to ensure that international trade in specimens of wild animals and plants does not threaten the survival of the species in the wild, and it accords varying degrees of protection to more than 35,000 species of animals and plants.
- Although CITES is **legally binding** on the Parties – in other words they have to implement the Convention – it **does not take the place of national laws**.
- Rather it provides a framework to be respected by each Party, which has to adopt its own domestic legislation to ensure that CITES is implemented at the national level.
- Parties = 181
- Similar treaty: Monitoring the Illegal Killing of Elephants – MIKE

With reference to the International Union for Conservation of Nature and Natural

Resources (IUCN) and the Convention on International Trade in Endangered Species of Wild Fauna and Flora (CITES), which of the following statements is/are correct?

- IUCN is an organ of the United Nations and CITES is an international agreement between governments
- IUCN runs thousands of field projects around the world to better manage natural environments.
- CITES is legally binding on the States that have joined it, but this Convention does not take the place of national laws.

Select the correct using the code given below.

1. 1 only
2. 2 and 3 only
3. 1 and 3 only
4. 1, 2 and 3

IUCN is an NGO. CITES is an international agreement between governments (multilateral treaty).

Answer: b) 2 and 3 only

TRAFFIC

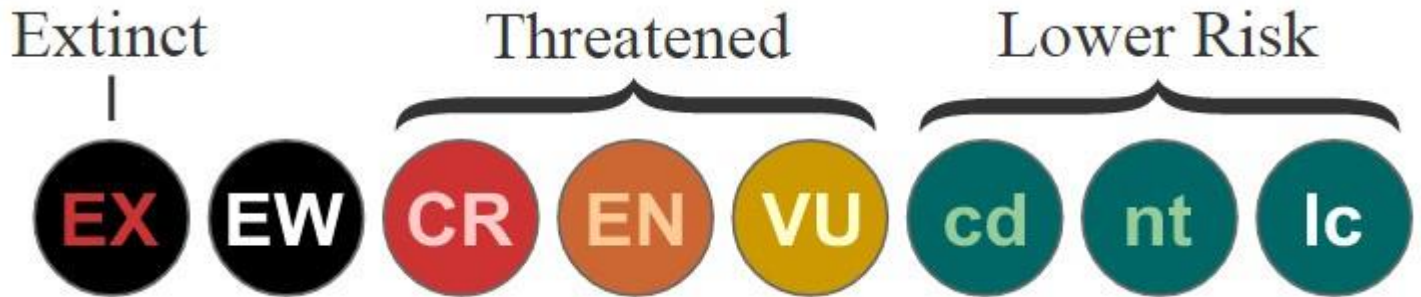
- The Wildlife Trade Monitoring Network (TRAFFIC)
- TRAFFIC is a **non-governmental organization** working globally on trade in wild animals and plants in the context of both biodiversity conservation and sustainable development.
- TRAFFIC is a joint programme of **World Wide Fund for Nature (WWF)** and **IUCN**.
- Traffic is complimentary to Convention on International Trade in Endangered Species of Wild Fauna and Flora (CITES).
- The programme was founded in 1976, with headquarters now located in Cambridge, United Kingdom.
- TRAFFIC's mission is to ensure that trade in wild plants and animals is not a threat to the conservation of nature.
- It investigates and analyses wildlife trade trends, patterns, impacts and drivers to provide the leading knowledge base on trade in wild animals and plants.

IUCN Red List or Red Data List or Red Book

- The IUCN Red List of **Threatened Species**, founded in 1964, is the world's most

comprehensive inventory of the global conservation status of biological species.

- When discussing the IUCN Red List, the official term "**threatened**" is a grouping of three categories: **Critically Endangered, Endangered, and Vulnerable**.



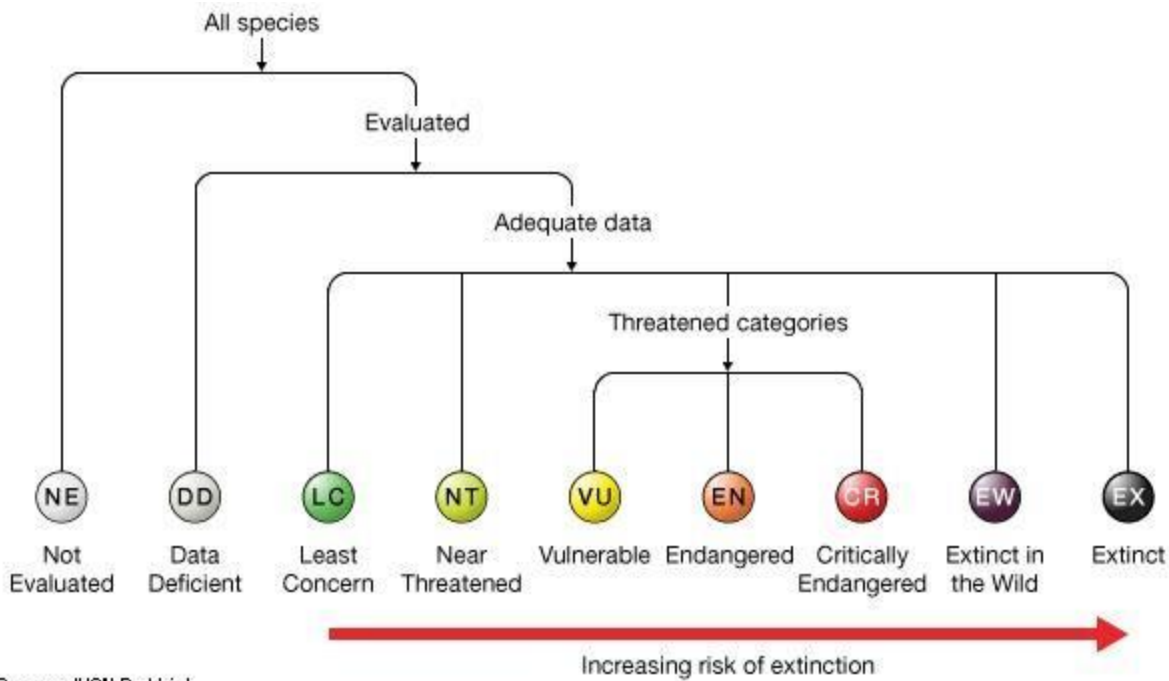
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- The pink pages in this publication include the critically endangered species. As the status of the species changes, new pages are sent to the subscribers.
- Green pages are used for those species that were formerly endangered, but have now recovered to a point where they are no longer threatened.

- With passing time, the number of pink pages continue to increase. There are pitifully few green pages.

Species are classified by the IUCN Red List into nine groups



Source : IUCN Red List

Picture Credits: <http://media-2.web.britannica.com/>

- Extinct (EX)** – No known individuals remaining.
- Extinct in the wild (EW)** – Known only to survive in captivity, or as a naturalized population outside its historic range.

- Critically endangered (CR)** – Extremely high risk of extinction in the wild.

Criteria

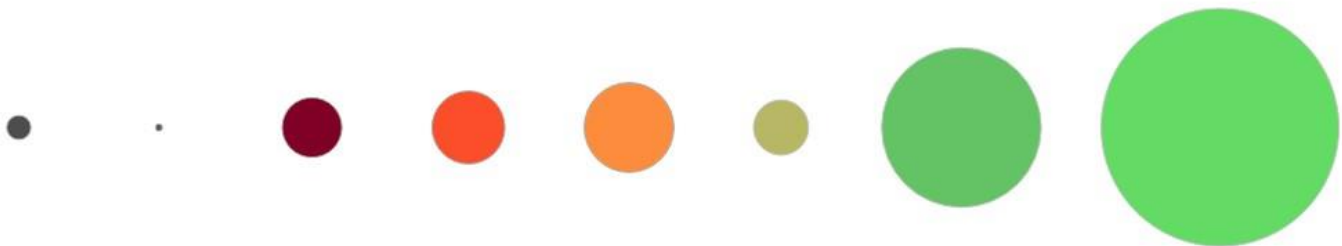
- reduction in population (> 90% over the last 10 years),

- population size (number less than 50 mature individuals),
- quantitative analysis showing the probability of extinction in wild in at least 50% in their 10 years) and
- it is therefore considered to be facing an extremely high risk of extinction in the wild.
- **Endangered (EN)** – High risk of extinction in the wild.
- **Vulnerable (VU)** – High risk of endangerment in the wild.

- **Near threatened (NT)** – Likely to become endangered in the near future.
- **Least concern (LC)** – Lowest risk. Does not qualify for a more at-risk category. Widespread and abundant taxa are included in this category.
- **Data deficient (DD)** – Not enough data to make an assessment of its risk of extinction.
- **Not evaluated (NE)** – Has not yet been evaluated against the criteria.

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Extinct	Extinct in Wild	Critically Endangered	Endangered	Vulnerable	Near Threatened	Least Concern	All Species
868	78	5176	7705	11654	4406	36952	82065



Pic Credits: <https://upload.wikimedia.org/>

The 'Red Data Books' published by the International Union for Conservation of Nature and Natural Resources (IUCN) contain lists of

- Endemic plant and animal species present in the biodiversity hotspots.
- Threatened plant and animal species.
- Protected sites for conservation of nature & natural resources in various countries.

Which of the statement given above is/are correct ?

1. 1 & 3
2. 2 only
3. 2 & 3
4. 3 only

IUCN is an NGO. It publishes Red data book which contains a list of 'Threatened species' (vulnerable, endangered and critically endangered).

Answer: b) 2 only

2015 IUCN Red List India [As of April 24, 2016]

- India has 988 species on IUCN 'Red List' as of 2015.
- India has added 15 more species to the "Red List" of threatened species [973 in 2014].
- The list contains **critically endangered, endangered and vulnerable species.**

'Critically Endangered' Mammals

Himalayan Brown/Red Bear (Ursus arctos isabellinus)



- Distribution: Nepal, Tibet, north India, and north Pakistan.
- Threats: loss of suitable habitat and persecution by humans.

Pygmy Hog (*Porcula salvania*)



- Is the world's smallest wild pig. This species constructs a nest throughout the year.
- It is one of the most useful indicators of the management status of grassland habitats.
- The grasslands where the pygmy hog resides are crucial for the survival of other threatened species such as Indian Rhinoceros, Swamp Deer, Wild Buffalo, Hispid Hare, Bengal Florican and Swamp Francolin.
- Habitat: Relatively undisturbed, tall 'terai' grasslands.
- Distribution: Previously spread across India, Nepal, and Bhutan. Now only found in Assam (Manas Wildlife Sanctuary and its buffer reserves).
- Pygmy hog-sucking Louse (*Haematopinus oliveri*), a parasite that feeds only on Pygmy Hogs will also fall in the same risk category of critically endangered as its survival is linked to that of the host species.
- Threats: The main threats are loss and degradation of grasslands, dry-season burning, livestock grazing and afforestation of grasslands. Hunting is also a threat.

Andaman White-toothed Shrew (*Crocidura andamanensis*)



- Distribution: It is endemic to the South Andaman Island of India.
- They are usually active by twilight or in the night.

- Threats: Habitat loss due to selective logging, natural disasters such as tsunami and drastic weather change are thought to contribute to current population declines.

Kondana Rat (*Millardia kondana*)



- Distribution: Known only from the small Sinhagarh Plateau (about one km²), near Pune in Maharashtra.
- It is a nocturnal burrowing rodent that is found only in India.
- Habitat: Tropical and subtropical dry deciduous forests and tropical scrub.
- Threats: Major threats are habitat loss, overgrazing of vegetation and disturbance from tourism and recreational activities.

Large Rock Rat or Elvira Rat (*Cremnomys elvira*)



- It is a medium sized, nocturnal and burrowing rodent endemic to India.
- Habitat: Tropical dry deciduous shrub land forest, seen in rocky areas.
- Habitat/distribution: Known only from Eastern Ghats of Tamil Nadu. Recorded from an elevation of about 600 m above mean sea level.
- Threats: Major threats are habitat loss, conversion of forests and fuel wood collection.

Namdapha Flying Squirrel (*Biswamoyopterus biswasi*)



- It is a unique (the only one in its genus) flying squirrel that is restricted to a single valley in the Namdapha N.P. (or) W.L.S. in Arunachal Pradesh.
- Habitat: Tropical forest.
- Habitat/distribution: Found only in Namdapha Tiger Reserve in Arunachal Pradesh.
- Threats: Hunted for food.

Malabar Civet (*Viverra civettina*)



- It is endemic to India and was first reported from Travancore, Kerala.
- It is nocturnal in nature and found exclusively in the Western Ghats.
- Habitat: Wooded plains and hill slopes of evergreen rainforests.
- Habitat/distribution: Western Ghats.
- Threats: Deforestation and commercial plantations are major threats.

Sumatran Rhinoceros (*Dicerorhinus sumatrensis*)



- It is the smallest and most endangered of the five rhinoceros species.
- It is now thought to be regionally extinct in India, though it once occurred in the foothills of the Himalayas and north-east India.
- The Javan Rhinoceros (*Rhinoceros sondaicus*) is also believed to be extinct in India and only a small number survive in Java and Vietnam.
- Threat: Poaching, loss of habitat.

Kashmir stag/hangul (*Cervus elaphus hanglu*)



- It subspecies of Red Deer which is native to India.
- Habitat: Dense riverine forests, high valleys, and mountains of the Kashmir valley and northern Chamba in Himachal Pradesh.
- State animal of Jammu and Kashmir.
- Threat: habitat destruction, over-grazing by domestic livestock, and poaching.

'Endangered' Mammals

Red Panda



- Status: Moved from Vulnerable to Endangered recently.
- Red panda is endemic to the temperate forests of the Himalayas.
- Habitat: Sikkim and Assam, northern Arunachal Pradesh.
- Threats: habitat loss and fragmentation, poaching, and inbreeding depression.

Wild ass/ khur (*Equus hemionus khur*)



- Distribution: Mostly occurs in Rann of Kutch region.
- Population steadily increasing.
- Today, its last refuge lies in the Indian Wild Ass Sanctuary, Little Rann of Kutch.
- Threat – Diseases, habitat degradation due to salt activities, Invasive species *Prosopis juliflora* shrub, and encroachment and grazing by the Maldhari.

A sandy and saline area is the natural habitat of an Indian animal species. The animal has no predators in that area but its existence is threatened due to the destruction of its habitat. Which one of the following could be that animal?

1. Indian wild buffalo
2. Indian wild ass
3. Indian wild boar
4. Indian Gazelle

Sandy saline area = Kutch region

Indian wild buffalo = Terai region

Indian wild boar = can survive in different types of habitat: grasslands, taiga, tropical rainforests, but they prefer life in deciduous forests.

Chinkara (Indian gazelle) = Thar desert

Answer: b) Indian wild ass

Dhole/ Asiatic wild dog or Indian wild dog (*Cuon alpinus*)



- Distribution: They occur in most of India south of the Ganges, particularly in the Central Indian Highlands and the Western and Eastern Ghats of the southern states.
- In north-east India, they inhabit Arunachal Pradesh, Assam, Meghalaya, and West Bengal.
- Threats: Habitat loss, depletion of its prey base, competition from other predators, persecution and possibly diseases from domestic and feral dogs.

Eld's deer/ thamin or brow-antlered deer (*Panolia eldii*)



- Distribution: Keibul Lamjao National Park (KLNP), Manipur.

- Threats: Overgrazing, loss of grassland habitat.

Golden langur (*Trachypithecus geei*)



- Primate, is an Old World monkey
- Distribution - small region of western Assam and in the neighboring foothills of the Black Mountains of Bhutan.
- Threats: Deforestation, human encroachments.

Himalayan / White-bellied Musk Deer



- Habitat: Kashmir, Kumaon and Sikkim.
- Threat: poaching & illegal trade for its musk. Only males produce the musk.

Hispid hare/ Assam rabbit (*Caprolagus hispidus*)



- Habitat: Southern foothills of the central Himalayas.
- Threats: The habitat of hispid hares is highly fragmented due to increasing agriculture, flood control, and human development.

Hog deer



- Habitat: Terai region and grasslands in northern India.
- Threats: Hunting, habitat loss, habitat fragmentation, human encroachments.

Lion-tailed macaque/ wanderoo (*Macaca silenus*)



- Endemic to the Western Ghats.
- Avoid human presence and they do not live, feed or travel through plantations.
- Habitat: Evergreen forests in the Western Ghats range.
- Threat: Habitat fragmentation due to spread of agriculture and tea, coffee, teak and cinchona, construction of water reservoirs and human settlements to support such activities.

In which of the following States is lion-tailed macaque found in its natural habitat?

1. Tamil Nadu
2. Kerala
3. Karnataka
4. Andhra Pradesh

Select the correct answer using the codes given below.

1. 1, 2 and 3 only
2. 2 only
3. 1, 3 and 4 only
4. 1, 2, 3 and 4

Lion-tailed macaque is found in Western Ghats. So AP is the odd option.

Answer: a) 1, 2 and 3 only

Chita / Tibetan Antelope



- Status – Moved from Vulnerable to Endangered recently.
- Habitat : Tibet cold desert.
- Threat : The chiru is threatened by hunting for its fine wool which is used to make the shahtoosh scarves, meat, magnificent horns.

Nilgiri langur/ Nilgiri leaf monkey (*Trachypithecus johnii*)



- Habitat: Hilly areas of Western Ghats in Tamil Nadu and Kerala.
- Threats: Habitat degradation, development activities, introduction of exotic tree species.

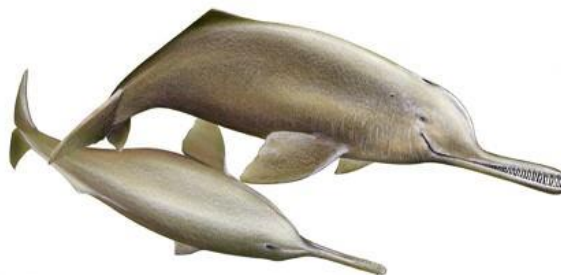
Nilgiri tahr



- The Nilgiri tahr is the largest of the three tahr species, inhabit montane grasslands of Western Ghats. It is the state animal of Tamil Nadu.
- Threats: Habitat loss, overgrazing, illegal hunting.

'Endangered' Marine Mammals

Freshwater / river dolphin



- Distribution: India, Bangladesh, Nepal and Pakistan which is split into two subspecies, the Ganges river dolphin and Indus river dolphin.
- Threats: Unintentional killing through entanglement in fishing gear; habitat loss and degradation – water development projects (barrages, high dams, and embankments), pollution – industrial waste and pesticides, municipal sewage discharge and noise from vessel traffic.

Ganges river dolphin

- Habitat: Ganges and Brahmaputra Rivers and their tributaries in Bangladesh, India and Nepal.

- The Ganges river dolphin has been recognized by the government of India as its **National Aquatic Animal**.

Indus river dolphin

- Habitat: Indus River in Pakistan and its Beas and Sutlej tributaries.

Other than poaching, what are the possible reasons for the decline in the population of Ganges River Dolphins?

- Construction of dams and barrages on rivers
- Increase in the population of crocodiles in rivers
- Getting trapped in fishing nets accidentally
- Use of synthetic fertilizers and other agricultural chemicals in crop-fields in the vicinity of rivers

Select the correct answer using the code given below.

- 1 and 2 only
- 2 and 3 only
- 1, 3 and 4 only**
- 1, 2, 3 and 4

'Susu' shares its habitat with crocodiles, freshwater turtles and wetland birds, many of which are fish eaters and are potential competitors with dolphins.

<http://goo.gl/7xvkDi>

Crocodile species in India

- The Mugger (or marsh) crocodile, gharial and the estuarine (coast) crocodile.
- Gharials (Critically Endangered)** once thrived in all the major river systems of the Indian Subcontinent. Today, they are extinct in the Indus River, in the Brahmaputra of Bhutan and Bangladesh, and in the Irrawaddy River. Their distribution is now limited to only 2% of their former range.
- The **mugger (Vulnerable)** crocodile is a freshwater species found sparsely in various lakes, rivers and marshes in the Indian subcontinent.
- Apart from the eastern coast of India, the **saltwater crocodile (Least concern)** is extremely rare on the Indian subcontinent.

- A huge population is present within the **Bhitarkanika Wildlife Sanctuary of Odisha** and they are known to be present in smaller numbers throughout the Indian and Bangladeshi portions of the Sunderbans.

Which one of the following is the national aquatic animal of India?

- Saltwater crocodile
- Olive ridley turtle
- Gangetic dolphin**
- Gharial

'Vulnerable' Mammals

Great Indian one horn Rhinoceros



- Habitat: Found only in the tall grasslands and forests in the foothills of the Himalayas [Terai region].
- National Parks: Kaziranga National Park, Pabitora wildlife sanctuary, Manas National Park, Assam.
- Status in the Wild: Moved from Endangered to Vulnerable.
- Threat – Oriental belief that its horn, among other parts, has medicinal properties, habitat loss, habitat fragmentation.

Gaur/Indian Bison



- The gaur (Bos gaurus), also called Indian bison, is a large bovine native to South Asia and Southeast Asia.

- Gaur are largely confined to evergreen forests or semi-evergreen and moist deciduous forests, but also occur in deciduous forest areas at the periphery of their range.
- The domesticated form of the gaur, *Bos frontalis*, is called gayal or mithun.
- Threats: Habitat loss, habitat fragmentation.



Four-horned antelope, Chousingha



- Endemic to the Western Ghats. Inhabits areas that are far from human disturbance.
- Threat: habitat loss and fragmentation, hunting for its fur.
- Only species of marten found in southern India.

Barasingha or swamp deer (*Rucervus duvaucelii*)



- Habitat: Isolated localities in northern and central India, and southwestern Nepal.
- Threats: Hunting for horns, habitat fragmentation and habitat loss.

Oriental small-clawed otter/ Asian small-clawed otter (*Aonyx cinerea*)



- Semiaquatic mammals which feed on fish and shellfish, and also other invertebrates, amphibians, birds and small mammals.
- It is a smallest otter species in the world.
- Habitat: It lives in mangrove swamps and freshwater wetlands.

The four-horned antelope must drink water regularly in order to survive.
Distribution: Presently it is confined to the Indian subcontinent. Scattered between the foothills of the Himalayas in the north to the Deccan Plateau in the south. Gir National Park has 1000 of these animals.
Threats: Loss of its natural habitat due to agricultural expansion. Four-horned skull and horns have made it a popular target for hunters.

Takin



- Distribution: Mountainous regions in the Himalayan Mountains and western China.
- Threats: Largely due to overhunting and the destruction of their natural habitat, takin are considered Endangered in China and Vulnerable per the IUCN.

Nilgiri marten

- Threat: habitat loss, pollution and hunting.

Clouded leopard (*Neofelis nebulosa*)



- Habitat: Himalayan foothills through mainland Southeast Asia into China. They occur in northern West Bengal, Sikkim, Arunachal Pradesh, Manipur, Meghalaya, Mizoram, Nagaland and Tripura.
- Threat: deforestation and poaching.

Asian black bear/ moon bear or white-chested bear (*Ursus thibetanus*)



- Habitat: Seen across much of the Himalayas, Korea, northeastern China, the Russian far east and the Honshu and Shikoku islands of Japan.
- Threats: deforestation and active hunting for its body parts.

‘Vulnerable’ Herbivorous Marine Mammals

- Include dugong and manatees and they inhabit swamps, rivers, estuaries, marine wetlands, and coastal marine waters.

Dugong/Sea Cow



- Threat: hunting (meat and oil), habitat degradation, and fishing-related fatalities.

Manatees Species

- Dugong belongs to manatees species.
- Habitat: India seas [near shore waters of Gulf of Mannar, Gulf of Kachchh and Andaman and Nicobar Islands], Caribbean Sea, Gulf of Mexico, the Amazon Basin, and West Africa
- Threat: Coastal development, red tide, hunting.

With reference to ‘dugong’, a mammal found in India, which of the following statements is/are correct?

- It is a herbivorous marine animal.
- It is found along the entire coast of India
- It is given legal protection under Schedule 1 of the Wildlife (Protection) Act, 1972.

Select the correct answer using the code given below.

1. 1 and 2
2. 2 only
3. 1 and 3
4. 3 only

Explanation:

- Dugong is an herbivorous animal. It eats sea grass and aquatic plants found in shallow oceans (At depths sea grass and aquatic plants don't grow due to absence of sunlight).

Wildlife (Protection) Act, 1972

- Harming **endangered** (vulnerable, endangered, critically endangered) species listed in **Schedule 1** of the Act is prohibited throughout India.
- Hunting species, like those requiring special protection (Schedule II), big game (Schedule III), and small game (Schedule IV), is regulated through licensing.
- A few species classified as **vermin (Schedule V)**, may be hunted without restrictions.

Answer: c) 1 and 3

‘Near Threatened’ Mammals

Marbled cat (*Pardofelis marmorata*)



- Habitat: northern India and Nepal, through south-eastern Asia to Borneo and Sumatra. In India - Sikkim, Darjeeling, moist tropical forest.
- Threats: hunting, habitat destruction for marbled cat and its prey.

Blackbuck (*Salman Khan*)



- Distribution: In the Indian subcontinent, the blackbuck can also be found in deserts (in the northwestern region), coastal areas, mountains (in the northern-northeastern region) Habitat: Grass land.

- Threat: excessive hunting for meat and sporting trophies, as well as habitat loss.

Himalayan tahr



- Habitat: Himalayas.
- Threats: The major threats in China are uncontrolled hunting and deforestation. In India, Himalayan tahr is sometimes hunted for meat, and there is apparently significant competition with livestock for summer grazing in some areas.

Markhor (*Capra falconeri*)



- The markhor is the national animal of Pakistan.
- Habitat: Mountains of central Asia. In India - some parts of Jammu and Kashmir .
- Status: Moved from endangered to Near Threatened in 2015
- Threats: Hunting (both for meat and for its twisted horns), armed conflict and habitat loss.

‘Not Evaluated’ Mammals

Himalayan wolf



- Conservation Status is 'Not Evaluated'. Several biologists feel that it needs be in the 'Critically Endangered List'.
- Distribution – Trans-Himalayan region of Himachal Pradesh, Jammu and Kashmir.
- Threats – Climate Change, Prey by humans to protect their cattle.

'Critically Endangered' Birds

The Jerdon's Courser (*Rhinoptilus bitorquatus*)



- It is a nocturnal bird found only in the northern part of the state of Andhra Pradesh in peninsular India [Sri Lankamaleswara Wildlife Sanctuary].
- Habitat: Undisturbed scrub jungle with open areas.
- Distribution: Jerdon's Courser is endemic to Andhra Pradesh.
- Threats: Clearing of scrub jungle, creation of new pastures, growing of dry land crops, Illegal trapping of birds, plantations of exotic trees, quarrying and the construction of the River Canals.

Forest Owlet (*Heteroglaux blewitti*)



- Habitat: Dry deciduous forest.
- Habitat: South Madhya Pradesh, in north-west Maharashtra and north-central Maharashtra.
- Threats: Logging operations, burning and cutting of trees damage roosting and nesting trees of the Forest Owlet.

White-bellied Heron (*Ardea insignis*)



- Distribution: Extremely rare bird found in five or six sites in Assam and Arunachal Pradesh, one or two sites in Bhutan, and a few in Myanmar.
- Habitat: Rivers with sand or gravel bars or inland lakes.
- Threats: Loss and degradation of lowland forests and wetlands through direct exploitation and disturbance by humans.

Bengal Florican (*Houbaropsis bengalensis*)



- A rare bustard species that is very well known for its mating dance.
- Habitat: Grasslands occasionally interspersed with scrublands.
- Distribution: Native to only 3 countries in the world - Cambodia, India and Nepal. In India, it occurs in 3 states, namely Uttar Pradesh, Assam and Arunachal Pradesh.
- Threats: Ongoing conversion of the bird's grassland habitat for various purposes including agriculture is mainly responsible for its population decline.

Himalayan Quail (*Ophrysia superciliosa*)



- Habitat: Tall grass and scrub on steep hillsides.
- Distribution: Western Himalayas.
- Threats: Indiscriminate hunting during the colonial period along with habitat modification.

Pink-headed Duck (*Rhodonessa caryophyllacea*)



- Males have a deep pink head and neck from which the bird derives its name.
- Habitat: Overgrown still-water pools, marshes and swamps in lowland forests and tall grasslands.
- Distribution: Recorded in India, Bangladesh and Myanmar. Maximum records are from north-east India.
- Threats: Wetland degradation and loss of habitat, along with hunting are the main causes of its decline.

Sociable Lapwing (*Vanellus gregarius*)



- It is a winter migrant to India.
- Habitat: Fallow fields and scrub desert.
- Distribution: central Asia, Asia Minor, Russia, Egypt, India, Pakistan. In India, habitat / distribution is restricted to the north and north-west of the country.
- Threats: Conversion of habitat to arable land, illegal hunting and proximity to human settlements.

Spoon Billed Sandpiper (*Eurynorhynchus pygmeus*)



- India is home to some of the last existing wintering grounds of this species.
- Habitat: Coastal areas with sparse vegetation. No breeding records further inland than 7 km from the seashore.
- Distribution: Has been recorded in West Bengal, Orissa, Kerala and Tamil Nadu.
- Threats: Habitat degradation and land reclamation. Human disturbance also leads to high incidence of nest desertion.

Siberian Crane (*Grus leucogeranus*)



- It is a large, strikingly majestic migratory bird that breeds and winters in wetlands.
- They are known to winter at Keoladeo National Park, Rajasthan.
- Habitat: Wetland areas.
- Located distribution: Keoladeo National Park in Rajasthan.
- Threats: Pesticide pollution, wetland drainage, development of prime habitat into agricultural fields, and to some extent, hunting.

‘Critically Endangered’ Reptiles

Gharial (*Gavialis gangeticus*)



- It is the most uniquely evolved crocodilian in the world, a specialized, river-dwelling, fish-eater.
- Habitat: Clean rivers with sand banks.
- Distribution: Only viable population in the National Chambal Sanctuary, spread across three states of Uttar Pradesh, Rajasthan and Madhya Pradesh in India.
- Small non-breeding populations exist in Son, Gandak, Hoogly and Ghagra rivers. Now extinct in Myanmar, Pakistan, Bhutan and Bangladesh.
- Threats: The combined effects of dams, barrages, artificial embankments, change in river course, pollution, sand-mining, riparian agriculture and ingress of domestic and feral livestock caused irreversible loss of riverine habitat and consequently of the gharial.

Hawksbill Turtle (*Eretmochelys imbricata*)



- The species is migratory in nature and nesting occurs in about 70 countries across the world. Maturation is slow and is estimated between 25 — 40 years.
- Habitat: Nesting occurs on insular, sandy beaches.
- Distribution: In India they are found in the Andaman and Nicobar Islands, the coast of Tamil Nadu and Orissa.

- Threats: Turtle shell trade, egg collection, slaughter for meat, oil pollution and destruction of nesting and foraging habitats.

Leatherback Turtle (*Dermochelys coriacea*)



- It is the largest of the living sea turtles, weighing as much as 900 kg.
- Jellyfish is their primary food.
- Habitat: Tropical and subtropical oceans.
- Distribution: Found in tropical and temperate waters of the Atlantic, Pacific, and into Indian Oceans.
- Threats: High sea fishing operations, harvesting of eggs, destruction of nests by wild predators and domesticated species such as cats, dogs and pigs. Artificial lighting

disorients hatchlings and adults and causes them to migrate inland rather than towards the sea. Threats to habitat include construction, mining and plantation of exotics.

Consider the following fauna of India:

- Gharial
- Leatherback turtle
- Swamp deer

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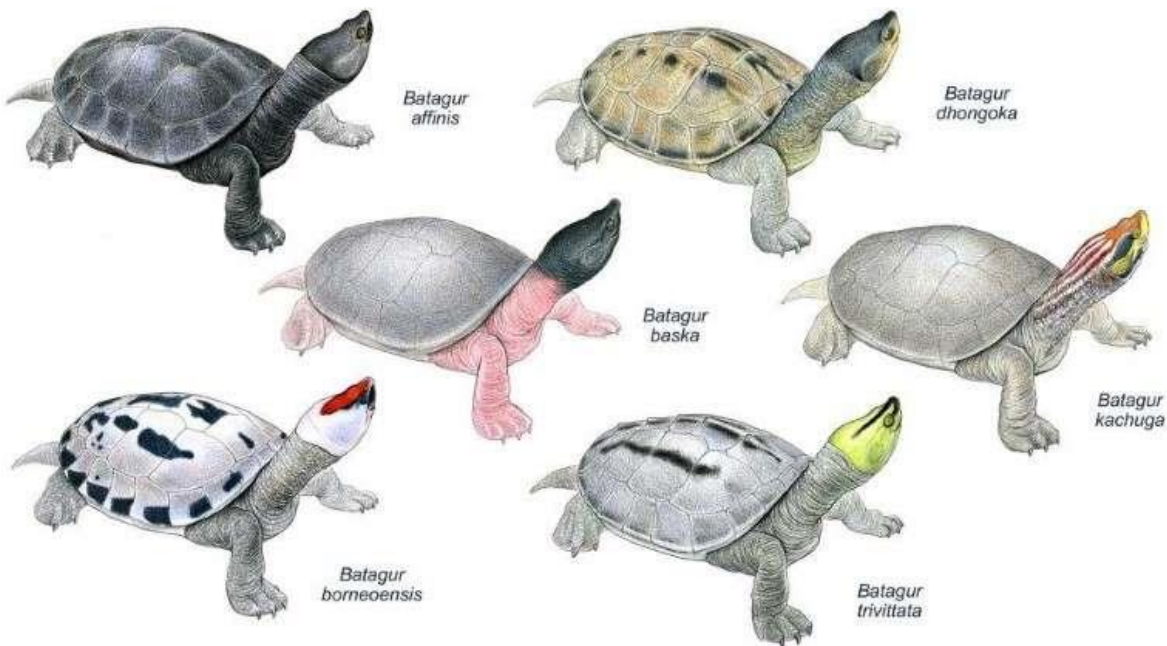
Which of the above is/are endangered?

- 1 and 2 only
- 3 only
- 1, 2 and 3
- None

1 and 2 are 'Critically Endangered' 3 is 'Vulnerable'.

Answer: a) 1 and 2 only

Four-toed River Terrapin or River Terrapin (*Batagur baska*)



- Habitat: Freshwater rivers and lakes.
- Distribution: Bangladesh, Cambodia, India, Indonesia and Malaysia.

- Threats: Use of flesh for medicinal purposes, demand for eggs, which are considered a delicacy

Red-crowned Roofed Turtle or the Bengal Roof Turtle (*Batagur kachuga*)

- Habitat: Deep, flowing rivers but with terrestrial nest sites.
- Distribution: Found in India, Bangladesh and Nepal. In India it resides basically in the watershed of the Ganga.
- Threats: Water development projects, water pollution, human disturbance and poaching for the illegal wildlife market.

'Near Threatened' Reptiles

Sispara day gecko (*Cnemaspis sisparensis*)



- Distribution: Endemic to Western Ghats, and found in Sispara, Nilgiris, Kavalai near Cochin.
- Threats: Habitat conversion and modification.

Vulnerable Reptiles

Olive ridley sea turtle



- Also known as the Pacific ridley sea turtle.
- Distribution: found in warm and tropical waters, primarily in the Pacific and Indian Oceans.

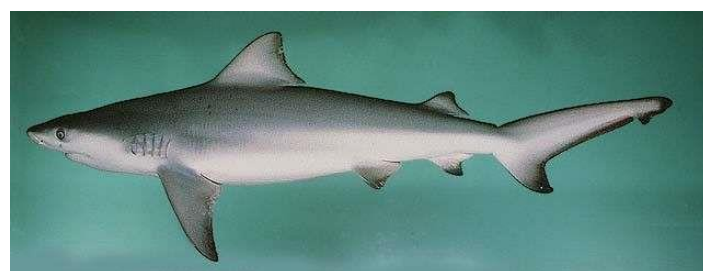
- Olive ridley turtles are best known for their behavior of synchronized nesting in mass numbers
- In the Indian Ocean, the majority of olive ridleys nest near **Gahirmatha in Odisha**. The coast of Odisha in India is the largest mass nesting site for the olive ridleys.
- Threats: unsustainable egg collection, slaughtering nesting females on the beach, and direct harvesting adults at sea for commercial sale of both the meat and hides.
- Coastal development, natural disasters, climate change, and other sources of beach erosion have also been cited as potential threats to nesting grounds.

What's the difference between turtles and tortoises?

Turtle	Tortoise
Lives in water	Lives on land
Come to land to lay eggs	Everything happens on land
Good swimmers	Bad swimmers
Carnivores (flesh eating)	Herbivores (plant eating)
Terrapins are almost a combination of turtles and tortoises. They live in water, mostly small dams or ponds, however they can also live on land.	

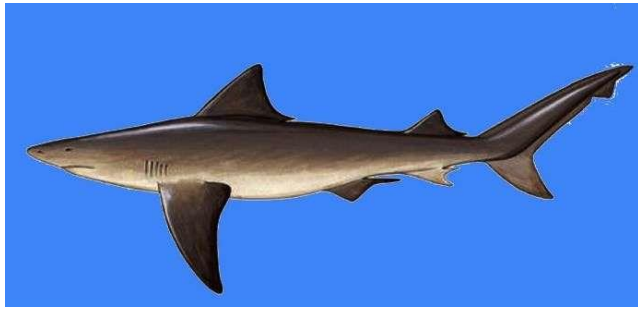
'Critically Endangered' Fish

The Pondicherry Shark (*Carcharhinus hemiodon*)



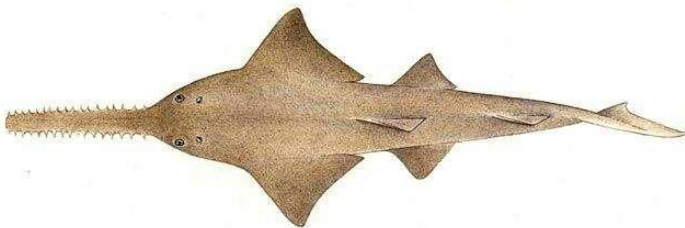
- Distribution: Indian Ocean - from Gulf of Oman to Pakistan, India and Sri Lanka.
- In scattered localities spanning India to New Guinea. Also been recorded at the mouth of the Hooghly river.
- Threats: Large, expanding, and unregulated commercial fisheries in inshore localities and habitats.

The Ganges Shark (*Glyphis gangeticus*)



- It occurs in the turbid waters of the Ganga river and the Bay of Bengal.
- The small eyes suggest that it is adapted to living in turbid water, while the slender teeth of the species suggests that it is primarily a fish-eater.
- Distribution: It occurs in India and possibly in Pakistan. The Ganga river system and Hooghly river mouth are its known habitats.
- Threats: Major fisheries targeting sharks. Other probable threats include overfishing, pollution, increasing river use and construction of dams and barrages. A few jaws of the species were found to have been traded in the international market during recent years, which testifies that the species is not extinct.

Large-tooth Sawfish (*Pristis microdon*)



- Distribution: Western part of the Indo-Pacific (East Africa to New Guinea, Philippines and Vietnam to Australia).
- In India, it is known to enter the Mahanadi river, up to 64 km inland, and also is very common in the estuaries of the Ganga and Brahmaputra.
- Threats: The principal threat to all sawfish are fisheries. Their long tooth-studded saw, makes them extraordinarily vulnerable to entanglement in any sort of net gear.
- When sawfish are caught in by catch, they often end up being traded because of the very high value of their products (meat is high quality and fins and saws extremely valuable in international trade).

- Major habitat changes include construction of dams over rivers, siltation, pollution from industries and mining operations.

Long-comb Sawfish or Narrow-snout Sawfish (*Pristis zijsron*)



- This species was reported as frequently found in shallow water. It inhabits muddy bottoms and also enters estuaries.
- Distribution: Indo-Pacific region including Australia, Cambodia, China, India, Indonesia and Malaysia.
- Threats: This species has been damaged intensively, both as a target species and as incidental by catch in commercial, sport or shark-control net fisheries, as well as for aquarium display. As a result, it has become severely depleted in recent decades, and now appears to have been extirpated from many parts of its range.

'Endangered' Fishes

Knife-tooth Sawfish (*Anoxypristis cuspidata*)



- Distribution: Widespread in western part of the Indo-Pacific region, including Red Sea.
- Threats: The principal threat to all sawfish are fisheries. Their long tooth-studded saw, makes them extraordinarily vulnerable to entanglement in any sort of net gear.
- When sawfish are caught in by catch, they often end up being traded because of the very high value of their products (meat is high quality and fins and saws extremely valuable in international trade).

‘Critically Endangered’ Spiders

Rameshwaram Ornamental or Rameshwaram Parachute Spider (Poecilotheria hanumavilasumica)



- It was recently described in 2004, and is only found in India.
- The species is semi-social, which means they live partly in groups.
- Habitat: Arboreal and tend to live in hiding.
- Distribution: Endemic to India. Spread along the coastal savannah, tropical lowland rain forests and montane forests up to an altitude of 2000 m above mean sea level.
- Threats: Major threats causing the disappearance of this species is habitat alteration and degradation.

Gooty Tarantula, Metallic Tarantula or Peacock Tarantula (Poecilotheria metallica)



- It was first found in Gooty (Ooty/Udagamandalam).
- Habitat: Wooded mountain area.
- Distribution: Endemic to South India.
- Threats: They are one of the most expensive spiders in the illegal pet trade. Large areas where the species occurs have been deforested, or subjected to habitat degradation due to local fuel wood collection, leading to decline in its population.

‘Critically Endangered’ Corals

Fire corals (Millepora boschmai)



- They are more closely related to jellyfish than corals.
- Distribution: Indonesia, Gulf of Chiriqui, Panama Pacific Province. Possibly extinct from Australia, India, Indonesia, Malaysia, Panama, Singapore and Thailand.
- Threats: Collected for decoration and jewellery trade. This group is also sensitive to temperature rise, and is thought to have completely disappeared from the majority of marine areas possibly because of growing global warming related bleaching effects.

In News

Flying squirrel



- Flying squirrels are mammals too, but they don't really fly.
- They jump from high in a trees glide through the air like a kite.
- Flying squirrels are a tribe of 44 species of squirrels.
- Their conservation status varies from Near Threatened to Endangered.
- Indian giant flying squirrel is included under 'Least Concerned'.

Consider the following:

1. Black-necked crane
2. Cheetah
3. Flying squirrel
4. Snow leopard

Which of the above are naturally found in India ?

- a. 1, 2 and 3 only
- b. 1, 3 and 4 only
- c. 2 and 4 only
- d. 1, 2, 3 and 4

Black-necked crane is commonly found in Tibetan and trans-Himalayan region. In winters they migrate to less colder regions of Indian Himalayas.

Cheetah is an extinct species. They have gone extinct during pre-independence era. Reason: They were hunted down by various Indian kings and British officers.

Flying Squirrels are found in many Indian forests.

Snow leopard is an 'endangered' specie found in the Himalayan ranges.

Answer: b) 1, 3 and 4 only

Guadalupe Fur Seal

- Distribution: west coast of California and off the Guadalupe islands of Mexico
- Moved from the "near threatened" to the "least concern" category in 2015.

Iberian Lynx

- Moved from the "critically endangered" to the "endangered" category.
- Found in Iberian Peninsula [Western Europe].

Lion

- Placed in Vulnerable category.
- Threats: Trade in bones is the major reason for their dwindling numbers.

Tiger and Bengal Tiger

- Placed in Endangered category.
- 2015: International Union for Conservation of Nature (IUCN), said that tiger numbers have grown to 3,890—a marked increase from estimates in 2010 that put the number at "as few as 3,200."
- The general observation is that the population of tigers in India has increased while that of the rest of the world has decreased.

Black-Necked Crane (Grus nigricollis)

- Black-necked crane
- Status: Vulnerable.

- The threat to the future of a vulnerable bird species has halted the Rs. 6,400-crore hydro power project in Tawang, Arunachal Pradesh.
- The species breeds on the Tibetan plateau and migrates to Tawang for the winter.
- The bird, most commonly found in China, is legally protected in Bhutan and India and is considered sacred to certain Buddhist traditions.

Which one of the following groups of animals belongs to the category of endangered species?

1. Great Indian Bustard, Musk Deer, Red Panda and Asiatic Wild Ass
2. Kashmir Stag, Cheetal, Blue Bull and Great Indian Bustard
3. Snow Leopard, Swamp Deer, Rhesus Monkey and Saras (Crane)
4. Lion-tailed Macaque, Blue Bull, Hanuman Langur and Cheetal

As of 2016, the status of many of these animals has changed. Outdated question.

Migratory Birds

- Migration refers to the cyclical seasonal movement of birds from one place to other. The distance of migration ranged from short distance to thousands of kilometers. But at the end of period, birds will eventually return to the original place.

Reasons for migration

- To avoid adverse factors (extreme climatic condition)
- To manage food shortage
- To manage shortage of water
- To have a better breeding conditions
- Less competition for safe nesting places

Migratory birds of India

Winter birds	Summer birds
Siberian Cranes, Greater Flamingo, Common Teal, Yellow Wagtail, White Wagtail, Northern Shoveler, Rosy Pelican, Wood Sandpiper, Spotted Sandpiper, Eurasian	Asian Koel, Black Crowned Night Heron, Eurasian Golden Oriole, Comb Duck, Blue Cheeked Bee Eater, Blue Tailed

Pigeon, Black Tailed Godwit, Redshank, Bluethroat, Long Billed Pipit.	Bee-Eater, Spotted Starling, Cuckoos.
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Steps Taken by the Government for Wildlife Protection

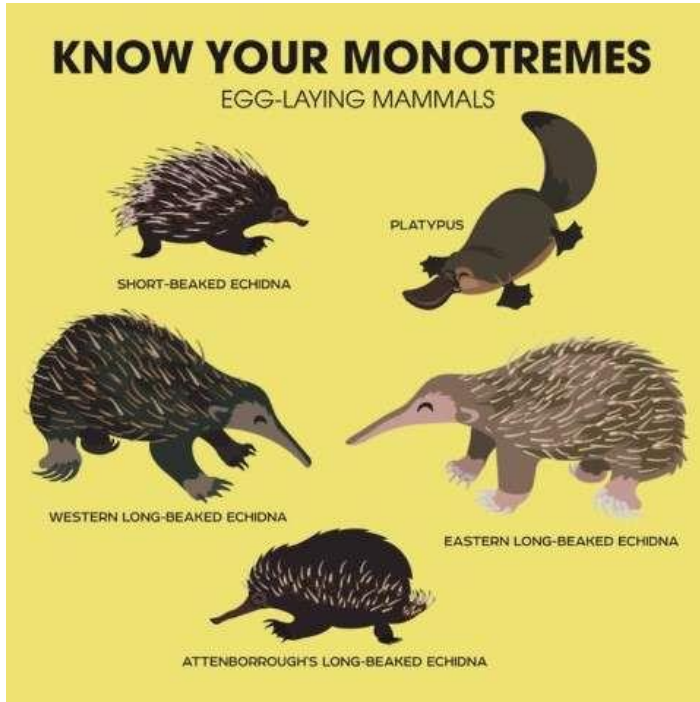
- All the states, excluding Jammu and Kashmir (it has its own Act), adopted Wildlife (Protection) Act in 1972 which prohibits any type of trade for endangered and rare species.
- The Central Government provides every type of financial assistance to the State Governments for the conservation and protection of the endangered species.
- National ban was imposed on tiger hunting in 1970 and Wildlife Protection Act became effective in 1972.
- Innumerable numbers of National Parks, Wildlife Sanctuaries, Parks etc. have been established by the government.
- In 1992, Central Zoo Authority (CZA) was initiated for the supervision of the management of zoological parks in the country.
- In 1996, the Wildlife Advisory Committee and Wildlife Institute were established for seeking advice on different features of Wildlife conservation and matters related to it.
- India is part of five main International Conventions which are associated to wildlife conservation. They are
- Convention on International Trade in Endangered Species (CITES),
- Coalition Against Wildlife Trafficking (CAWT),
- International Whaling Commission (IWC),
- United Nations Educational, Scientific and Cultural Organization – World Heritage Committee (UNESCO – WHC) and
- Convention on Migratory Species (CHS).

Misc

Egg Laying Mammals

- The unique feature of **monotremes**, a sub division of mammal, is that monotremes lay eggs rather than giving birth to their young.
- There are only five living Monotreme/egg laying Mammals species: they are – the **duck-billed platypus** and four species of spiny anteaters (also known as **echidna**).

- All of them are found only in **Australia** and **New Guinea**.
- In echidnas, the egg is carried in a pouch on the female's belly until the young hatches, at which point the barely-developed young must find a mammary gland and latch onto it for nourishment.



Marsupials

- Marsupials are the group of mammals commonly thought of as pouched mammals (like the wallaby and kangaroo).
- Marsupial mammals have placenta but it is very short-lived and does not make as much of a contribution to fetal nourishment.
- They give birth very early and the young animal, essentially a helpless embryo, climbs from the mother's birth canal to the nipples.
- There it grabs on with its mouth and continues to develop, often for weeks or months depending on the species.

List of Marsupials

- Kola
- Kangaroo
- Wallaby
- Wombats
- Opossum
- Tasmanian devils
- Mursupial Mole (4 foot)
- Bandi coot

- Tasmanian Wolf /Tiger



Insectivorous Plants of India

Insectivorous Plants

- These plants are specialized in trapping insects and are popularly known as insectivorous plants.
- They are very different from normal plants in their mode of nutrition. They, however, never prey upon humans or large animals.
- Insectivorous plants can broadly be divided into active and passive types based on their method of trapping their prey.
- The active ones can close their leaf traps the moment insects land on them.
- The passive plants have a 'pitfall' mechanism, having some kind of jar or pitcher-like structure into which the insect slips and falls, to eventually be digested.
- The insectivorous plants often have several attractions such as brilliant colors, sweet secretions and other curios to lure their innocent victims.

Why do they hunt despite having normal roots and photosynthetic leaves?

- These plants are usually associated with **rain-washed, nutrient-poor soils, or wet and acidic areas** that are ill-drained.
- Such wetlands are **acidic due to anaerobic conditions**, which cause partial decomposition of organic matter releasing acidic compounds into the surroundings.

- As a result, most microorganisms necessary for complete decomposition of organic matter cannot survive in such poorly oxygenated conditions.
- Normal plants find it difficult to survive in such nutrient poor habitats. The hunter plants are successful in such places because they supplement their photosynthetic food production by trapping insects and digesting their nitrogen rich bodies.

Insectivorous plants of India

Insectivorous plants of India belong mainly to three families:

1. **Droseraceae (3 species),**
2. **Nepenthaceae (1 species) and**
3. **Lentibulariaceae (36 species).**

Drosera and Aldrovanda

Drosera and Aldrovanda belong to family Droseraceae.

Drosera or Sundew inhabit wet infertile soils or marshy places.

Aldrovanda is a free-floating, rootless aquatic plant, the only species found in India, occurs in the salt marshes of Sunderbans, south of Calcutta. It also grows in fresh water bodies like ponds, tanks and lakes.

Insect trapping mechanism of Drosera



- The tentacles on the leaves secrete a sticky fluid that shines in the sun like dew-drops. Therefore the Drosera are commonly known as 'sundews'.

- When an insect lured by these glistening drops alights on the leaf surface it gets stuck in this fluid and are absorbed and digested.

Insect trapping mechanism of Aldrovanda



- On the leaf midrib are found some sensitive trigger hairs. The two halves of the leaf blade of Aldrovanda close along the midrib the moment an insect comes into contact with the leaf, trapping the victim inside.

Pitcher Plants Family: Nepenthaceae

- Pitcher plants belong to family nepenthaceae. The members of the family are commonly known as 'pitcher plants' because their leaves bear jar-like structures.
- Distribution: Confined to the high rainfall hills and plateaus of north-eastern region, at altitudes ranging from 100 —1500 m, particularly in Garo, Khasi and Jaintia hills of Meghalaya.

Insect trapping mechanism of pitcher plant



- Nepenthes conforms to the pitfall type of trap. A honey like substance is secreted from glands at the entrance of the pitcher. Once the insect enters into the pitcher, it falls down because of the slipperiness.
- The inner wall, towards its lower half, bears numerous glands, which secrete a proteolytic enzyme. This enzyme digests the body of the trapped insects and nutrients are absorbed.

Utricularia and Pinguicula

- Utricularia and Pinguicula belong to family Lentibulariaceae.

Utricularia or Bladderworts



The Bladderworts generally inhabit freshwater wetlands and waterlogged areas. Some species are associated with moist moss covered rock surfaces, and damp soils during rains.

Insect trapping: Utricularia in its bladders mouth, has sensitive bristles or hairs. When an insect happens to contact these hairs the door opens, carrying the insect into the bladder along with a little current of water. The door is shut when water fills the bladder, The enzymes produced by the inner wall of the bladder digest the insect.

Pinguicula or Butterwort



- It grows in the alpine heights of Himalayas, from Kashmir to Sikkim, along stream-sides in cool boggy places.
- Insect trapping mechanism: In Pinguicula, an entire leaf works as trap. When an insect lands on the leaf surface, it gets stuck in the sticky exudate. the leaf margins roll up thus trapping the victim.

Medicinal Properties of Insectivorous Plants

- Drosera are capable of curdling milk, its bruised leaves are applied on blisters and used for dyeing silk.
- Nepenthes in local medicine to treat cholera patients, the liquid inside the pitcher is useful for urinary troubles, it is also used as eye drops.
- Utricularia is useful against cough, for dressing of wounds, as a remedy for urinary disease.

Conservation Status of Indian Insectivores

- In India, species like Drosera peltata, Aldrovanda vesiculosa and Nepenthes khasiana have been included in the Red Data Book as **endangered plants**.

Threats

- Gardening trading for medicinal properties is one of the main causes for their decline.
- Habitat destruction is also rampant, the wetlands harboring such plants being the main casualties during the expansion of urban and rural habitation.
- Pollution caused by effluents containing detergents, fertilizers, pesticides, sewage etc. into the wetlands is yet another major cause for their decline (Since insectivorous plants do not tolerate high nutrient levels).
- Moreover, polluted water bodies are dominated by prolific water weeds which cause elimination of the delicate insectivorous plants.

Diclofenac - Indian Vulture Crisis

Indian Vulture Crisis

- India is most favourable region for Vultures: Hindus do not eat cows, which they consider sacred, and when a cow dies, it is left to be fed on by vultures. India has a high species diversity and hence vultures get lot of food.
- Nine species of vulture can be found living in India. But today, most are in danger of extinction due to a veterinary drug called diclofenac (vultures do not have a particular enzyme to break down diclofenac).

Diclofenac

- Diclofenac is a common anti-inflammatory drug administered to livestock and is used to treat the symptoms of inflammation, fevers and/or pain associated with disease or wounds.
- Diclofenac leads to renal failure in vultures damaging their excretory system [direct inhibition of uric acid secretion in vultures].
- Gyps species were the most affected by diclofenac.
- The population of the White-rumped vulture (*Gyps bengalensis*) fell 99.7% between 1993 and 2002.
- The populations of the Indian vulture (*Gyps indicus*) and the slender-billed vulture (*Gyps tenuirostris*) fell 97.4%.
- The percentages differ slightly because the white-rumped vulture is more sensitive to diclofenac than the other two species, but all three were in danger of extinction.
- Two other species of Gyps, the Himalayan vulture (*Gyps himalayensis*) and the Eurasian griffon (*Gyps fulvus*) were less affected because they come to India only in winters
- They are exclusively mountain-dwelling and hence less vulnerable to diclofenac contamination.
- Vulture populations have continued to decline in India at a rate of between 20% and 40% each year since 2007.

Consequences of Depopulation of Vultures

- Vultures previously played an important role in public sanitation in India and their disappearance has resulted in an explosion of rats and wild dogs and the spread of diseases resulting in an estimated costs of up to ₹1700 billion (US\$25 billion) (as of 2015).

- The carcasses formerly eaten by vultures rot in village fields leading to contaminated drinking water.
- These newly abundant scavengers are not as efficient as vultures. A vulture's metabolism is a true "dead-end" for pathogens, but dogs and rats become carriers of the pathogens.
- The mammals also carry diseases from rotting carcasses such as rabies, anthrax, plague etc. and are indirectly responsible for thousands of human deaths.

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Diclofenac Alternative

- The drug was banned for veterinary use in India in March, 2006; Nepal and Pakistan followed suit shortly thereafter.
- A replacement drug was quickly developed and proposed after tests on vultures in captivity: meloxicam.
- Meloxicam affects cattle the same way as diclofenac, but is harmless for vultures.
- Diclofenac for human use was still being diverted into veterinary uses through black markets in certain parts of India.

Conservation Status of India Vultures [As of April-2016]

Critically Endangered

- White-Rumped Vulture (*Gyps bengalensis*)
- White-Backed Vulture (*Gyps africanus*)
- Ruppell's Vulture (*Gyps rueppellii*)
- Indian Vulture (*Gyps indicus*)
- Slender-Billed Vulture (*Gyps tenuirostris*)

Endangered

- Cape Vulture (*Gyps coprotheres*)

Near Threatened

- Himalayan Vulture (*Gyps himalayensis*)

Least Concern

- Griffon Vulture (*Gyps fulvus*)

Critically Endangered

- White-Rumped Vulture (*Gyps bengalensis*)



- White-Backed Vulture (*Gyps africanus*)



Endangered

- Cape Vulture (*Gyps coprotheres*)



- Ruppell's Vulture (*Gyps rueppellii*)



Near Threatened

- Himalayan Vulture (*Gyps himalayensis*)



- Indian Vulture (*Gyps indicus*)



Least Concern

- Griffon Vulture (*Gyps fulvus*)



- Slender-Billed Vulture (*Gyps tenuirostris*)



Vulture Safety Zones

- The concept of a VSZ is unique for the Asian continent but similar VSZ are in operation in both Europe and Africa.
- Aim of developing VSZs is to establish targeted awareness activities surrounding 150 km radius of vultures' colonies so that no diclofenac or the veterinary toxic drugs are found in cattle carcasses, the main food of vultures (to provide safe food).
- The VSZ is spread around in several hundred kilometers covering the Jim Corbett in Uttarakhand, Dudhwa and Kartamaghat forest reserves in UP which is adjoining the Indo-Nepal border. Nepal has already set up VSZ on the Indian borders.

VSZ's provide

- A safe source of food that is free of contamination from veterinary drugs, poisons and other agricultural chemicals.
- A place where vultures can feed free from human disturbances.
- Supplement the ever decreasing food base for vultures.
- Extra food close to breeding colonies, this helps to increase their breeding success by improving the survival chances of the young vultures.
- Additional food, such as bone fragments, which can be supplied to breeding birds.
- An economical and practical way of disposing of old and unproductive cattle.
- Help to reduce the risk of spreading diseases.
- A place for scientists to study the biology and ecology of these threatened species.
- An opportunity to raise public awareness on vulture conservation and to raise funds.
- An excellent opportunity for eco-tourists to observe these magnificent birds.

Zones

- The zone between Uttarakhand to Nepal, which spans from Corbett to Katriya Ghat, a Tarai belt, covering 30,000 square kilometers will be earmarked as Vulture Safe zone. Slender-billed vulture and white-backed vulture are found in this area, which is marshy grassland, savannas and forests.
- Similarly, a belt between Dibrugarh in Assam to North Lakhimpur in Arunachal Pradesh will

also be conserved as a vulture safe zone where slender-billed and white-backed species of vultures are found.

- The third zone would be in central India, covering Chhatisgarh, where white-backed and long-billed vultures are found.

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What have to be done?

- Diclofenac free zones' (DFZ) meant the complete removal of diclofenac in the identified vulture safe zones i.e. places where vulture colonies have been identified.

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Vulture Restaurants

- At this restaurant, tables are reserved only for the unique and rare vultures by Maharashtra and Punjab forest departments.

Aim

- Conserve the fast dwindling vulture population.
- As uncontaminated food shortage is one of the reasons for vultures' decline, these scavengers will be fed by serving diclofenac free carcasses of cattle through restaurants.
- VR includes involvement of local communities in in-situ conservation, is having dual benefits to vultures and to our society.
- People inform the forest department in case of the death of an animal in their village and the department tests the dead animal for the presence of diclofenac.
- In their absence the department pays monetary benefits to the owner of the animal and informer, transports it to the vulture restaurant.
- Apart from this, whenever a vulture nesting is found, conservation measures like providing safe food near nesting trees, constant protection from all sorts of disturbances, etc., are put in place without delay.

Benefits

- Conservation of vulture from extinction
- Community participation in conservation
- Economic incentive to local cattle breeders
- Phasing out the use of diclofenac

Greenhouse Effect & Global Warming | Carbon Sink

Major Global Environmental Issues

- Increased human activity, urbanization, industrialization have led to rapid deterioration of the environment. This has severely affected the life supporting system.

The important global environmental issues are:

- [biodiversity loss](#)
- [desertification](#)
- [depletion of ozone layer](#)
- [acid rain](#)
- [oil spills](#)
- [dumping of hazardous wastes](#)
- Climate change due to greenhouse effect and global warming

Climate Change

- Climate is the long-term average of a region's weather events. The Earth's climate is not static. Over the billions of years of earth's existence, it has changed many times in response to natural causes like sun spot, ice age glaciations, etc.
- "Climate change" means a change of climate which is attributed directly or indirectly to human activity that alters the composition of the global atmosphere and which is in addition to natural climate variability observed over comparable time periods (100 years).
- Climate change is usually measured in major shifts in temperature, rainfall, snow, and wind patterns lasting decades or more.
- Humans are creating climate change by burning large amounts of fossil fuels (coal, oil, natural gas), deforestation (when forests are cut down or burned, they can no longer store carbon, and the carbon is released to the atmosphere).

Greenhouse Effect

- A greenhouse is a structure with walls and roof made chiefly of transparent material, such as glass, in which plants requiring regulated climatic conditions are grown.
- In a greenhouse, the incident solar radiation (the visible and adjacent portions of the infrared and ultraviolet ranges of the spectrum) passes through the glass roof and walls and is absorbed by the floor, earth, and

contents, which become warmer and re-emit the energy as **longer-wavelength infrared radiation** (heat radiation).

- Glass and other materials used for greenhouse walls **do not transmit infrared radiation**, so the infrared cannot escape via radiative transfer.
- As the structure is not open to the atmosphere, heat also cannot escape via convection, so the temperature inside the greenhouse rises. This is known as the 'greenhouse effect'.

Importance of Natural Greenhouse Effect

- The green-house effect is a natural phenomenon and has been occurring for millions of years on the earth.
- Life on the earth has been possible because of this natural greenhouse effect which is due to **water vapour** and small particles of water present in the atmosphere.
- Together, these produce more than 95 percent of total greenhouse warming.
- Average global temperatures is maintained at about 15°C due to natural greenhouse effect.
- Without this phenomenon, average global temperatures might have been around -17°C and at such low temperature life would not be able to exist.

Greenhouse Effect And Global Warming Due to Greenhouse Gases

- Atmospheric gases like **carbondioxide, methane, nitrous oxide (N₂O), water vapour, and chlorofluorocarbons** are capable of **trapping the out-going infrared radiation** from the earth's surface. Thus, the temperature of the global atmosphere is increased.

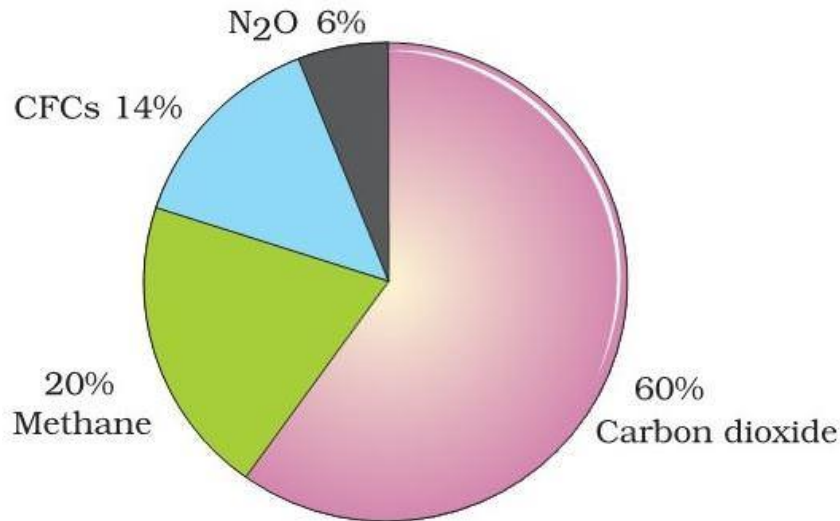
[Oxides of Nitrogen with general formula **NO_x** – **NO, NO₂** – Nitrogen oxide, Nitrogen dioxide etc. are **global cooling gasses** while **Nitrous oxide (N₂O)** is a **greenhouse gas**]

- As this phenomenon of increase in temperature is observed in green houses, in the botanical gardens these gases are known as greenhouse gases and the heating effect is known as greenhouse effect.

- If greenhouse gases are not checked, by the turn of the century the temperature may rise by **5°C**.
- Scientists believe that this rise in temperature will lead to deleterious changes in the environment and resulting in odd climatic changes (e.g. El Nino effect), thus leading to increased melting of polar ice caps as well as of other places like the Himalayan snow caps.

- Over many years, this will result in a **rise in sea level** that can submerge many coastal areas and lead to loss of coastal areas and ecosystems like swamps and marshes (most important ecosystems from the point of ecological services), etc.

Greenhouse Gases



Relative contribution of various greenhouse gases to total global warming

Gas	Sources and Causes
Carbon dioxide (CO ₂)	Burning of fossil fuels, deforestation
Chlorofluorocarbons (CFCs)	Refrigeration, solvents, insulation foams, aero propellants, industrial and commercial uses
Methane (CH ₄)	Growing paddy, excreta of cattle and other livestock, termites, burning of fossil fuel, wood, landfills, wetlands, fertilizer factories.
Nitrogen oxides (N ₂ O)	Burning of fossil fuels, fertilizers; burning of wood and crop residue.
Carbon Monoxide (CO)	Iron ore smelting, burning of fossil fuels, burning e-waste.

Carbon dioxide

- Carbon dioxide is meteorologically a very important gas as it is **transparent to the incoming solar radiation** but **opaque to the outgoing terrestrial radiation**.
- It absorbs a part of terrestrial radiation and reflects back some part of it towards the earth's surface. It is largely responsible for the greenhouse effect.
- Its concentration is greater close to the earth's surface as it is denser than air.

Ozone

- Ozone is another important greenhouse gas. But it is in very small proportions at the surface.
- Most of it is confined to the **stratosphere** where it absorbs the harmful UV radiation.
- At ground level, pollutants like NO₂ react with volatile organic compounds in the presence of sunlight to produce ozone (tropospheric ozone).

Water vapour

- Water vapour is also a variable gas in the atmosphere, which decreases with altitude.

Water vapour also decreases from the equator towards the poles.

- In the warm and wet tropics, it may account for four per cent of the air by volume, while in the dry and cold areas of desert and polar regions, it may be less than one per cent of the air.
- One unique feature about this greenhouse gas is that it absorbs **both** incoming (a part of incoming) and outgoing solar radiation.

Methane

- Methane is the most important greenhouse gas after carbon dioxide.
- It is produced from **decomposition of animal wastes and biological matter**.
- The emission of this gas can be restricted by using animal wastes and biological matter to produce gobar gas (methane).

Nitrous Oxide (N₂O)

- NO_x and N₂O are Different. N₂O or **Nitrous Oxide** is a **greenhouse gas**.
- NO_x[NO and NO₂ (nitric oxide or nitrogen oxide and nitrogen dioxide)] emissions cause **global cooling** through the formation of (OH) radicals that **destroy methane** molecules, countering the effect of greenhouse gases.

Carbon Monoxide

- Carbon monoxide is a short-lived greenhouse gas (it is less denser than air).
- It has an indirect radiative forcing effect by elevating concentrations of **methane** and tropospheric **ozone** through chemical reactions with other atmospheric constituents (e.g., the hydroxyl radical, OH.) that would otherwise destroy them.
- Through natural processes in the atmosphere, it is eventually oxidized to carbon dioxide.

Fluorinated gases

Chlorofluorocarbons (CFCs)

- CFCs were phased out via the Montreal Protocol due to their part in ozone depletion.
- This anthropogenic compound is also a greenhouse gas, with a **much higher potential** to enhance the greenhouse effect than CO₂.

Hydrofluorocarbons

- Hydrofluorocarbons are used as refrigerants, aerosol propellants, solvents, and fire retardants.
- These chemicals were developed as a replacement for chlorofluorocarbons (CFCs).
- Unfortunately, HFCs are potent greenhouse gases with long atmospheric lifetimes.

Perfluorocarbons

- Perfluorocarbons are compounds produced as a by-product in aluminum production and the manufacturing of semiconductors.
- Like HFCs, PFCs generally have long atmospheric lifetimes and high global warming potential.

Sulfur hexafluoride

- Sulfur hexafluoride is also a greenhouse gas.
- Sulfur hexafluoride is used in magnesium processing and semiconductor manufacturing, as well as a tracer gas for leak detection.
- Sulfur hexafluoride is used in electrical transmission equipment, including circuit breakers.

Black Carbon or Soot

- Black carbon (BC) is a solid particle or aerosol, (though not a gas) contributes to warming of the atmosphere. Black carbon, commonly known as soot
- Soot is a form of particulate air pollutant, produced from incomplete combustion.
- Black carbon warms the Earth by absorbing heat in the atmosphere and by reducing albedo, (the ability to reflect sunlight) when deposited on snow and ice.
- BC is the strongest absorber of sunlight and heats the air directly.
- In addition, it darkens snow packs and glaciers through deposition and leads to melting of ice and snow.
- Regionally, BC disrupts cloudiness and monsoon rainfall.
- Black carbon stays in the atmosphere for only several days to weeks. Thus the effects of BC on the atmospheric warming and glacier retreat disappear within months of reducing emissions.

Brown Carbon

- Brown carbon is a ubiquitous and unidentified component of organic aerosol.
- Biomass burning (possibly domestic wood burning) is shown to be a major source of brown carbon
- "Brown carbon" is generally referred for greenhouse gases and "black carbon" for particles resulting from impure combustion, such as soot and dust.

The increasing amount of carbon dioxide in the air is slowly raising the temperature of the atmosphere, because it absorbs

1. the water vapour of the air and retains its heat.
2. the ultraviolet part of the solar radiation.
3. all the solar radiations.
4. the infrared part of the solar radiation

Answer: d) the infrared part of the solar radiation (outgoing radiation).

Among GHGs, only water vapor has the ability to absorb both incoming (UV) and outgoing (infrared) radiation.

Global Warming Potential (GWP) & Lifetime of Green House Gases

Gas	GWP (100-year)	Lifetime (years)
Carbon di oxide	1	50-200
Methane	21	12
Nitrous oxide	310	120
Hydrofluorocarbons (HFCs)	140 -11,700	1-270
Perfluorocarbons (PFCs)	6,500-9,200	800-50,000
Sulfur hexafluoride (SF6)	23,900	3,200

Global Warming – Impacts

Melting of the ice caps

- Melting of the ice caps and glaciers will lead to rise in sea level. Thermal expansion also contributes to sea level rise.
- Fertile agricultural lands on the coast will be submerged and saline water intrusions will degrade the neighboring land. Ground water in such regions will become useless.
- Populous cities lying on the coasts will be submerged under the sea.
- The Himalayan glaciers are a source of fresh water for perennial rivers, in particular the Indus, Ganga, and Brahmaputra river systems.
- There is evidence that some Himalayan glaciers have retreated significantly since the 19th century.
- Flooding in Himalayas and Ganga plains in wet season and drought in dry season will severely affect the country.
- As a result of thawing of snow, the amount of arable land in high-latitude region is likely to increase by reduction of the amount of frozen lands.
- At the same time arable land along the coast lines are bound to be reduced as a result of rising sea level and saline water inundations.

Extreme Climatic Events

- Increased likelihood of extreme events such as heat wave, flooding, hurricanes, etc. will offset all the economic advancements made.
- Changes in rainfall patterns (E.g. Chennai floods) will severely impact agriculture.

Environmental Degradation

- Reduced hydroelectric power generation due to abnormal behavior of glaciers will further increase dependence on fossil fuels.
- Widespread vanishing of animal populations due to habitat loss will add more species to the 'threatened' and 'extinct' list.

Rising Health Related Issues

- Spread of diseases (like malaria, etc.) in tropics will put more pressure on the health care sector.
- It is anticipated that there will be an increase in the number of deaths due to greater frequency and severity of heat waves and other extreme weather events.
- Lack of freshwater during droughts and contamination of freshwater supplies during floods compromise hygiene, thus increasing rates diseases like cholera, diarrhoea etc..

Biodiversity Loss

- Loss of Plankton due to warming of seas will adversely affects marine food chain.
- Bleaching of Coral Reefs [Rain forests of the ocean] will cause great loss of marine biodiversity.
- Rising temperature would increase fertilizer requirement for the same production targets and result in higher GHG emissions, ammonia volatilization and cost of crop production.
- Rising temperatures will further affect the physical, chemical and biological properties of fresh water lakes and rivers, with adverse impacts on many individual fresh water species.

No Food Security

- Climate Change affects crops by impacting irrigation, insolation as well as the prevalence of pests.
- Increased frequencies of droughts, floods, storms and cyclones are likely to increase agricultural production variability.
- Moderate warming (increase of 1 to 3°C in mean temperature) is expected to benefit crop yields in temperate regions, while in lower latitudes the crops will take a hit. However, the natural calamities due to global warming can offset the benefits in temperature regions.
- In coastal areas, sea level rise will exacerbate water resource constraints due to increased salinization of groundwater supplies.

Deterioration of Carbon sinks

- High latitude forests store more carbon than tropical rainforests.
- One third of the world's soil-bound carbon is in taiga and tundra areas.
- When the permafrost melts due to global warming, it releases carbon in the form of **carbon dioxide** and **methane** (Both are GHGs).
- In the 1970s the tundra was a carbon sink, but today, it is a carbon source, thanks to global warming. [global warming leads to more global warming]

Some methods to reduce CO2 in atmosphere

Clean coal technology

Current Scenario

- Half of the world's electricity is generated by burning coal.
- Coal will remain a dominant energy source for years to come.
- Since CO₂ contributes to global warming, reducing its release into the atmosphere has become a major international concern.
- CO₂ and CO (carbon monoxide) are the major greenhouse gas which are released during burning of coal.
- Along with the above gases, nitrogen oxides (destroys ozone) and sulphur oxides (acid rains) are also released.

Clean coal technology

- Clean coal technology seeks to reduce harsh environmental effects by using multiple technologies to clean coal and contain its emissions.
- Some clean coal technologies purify the coal before it burns.
- One type of coal preparation, coal washing, removes unwanted minerals by mixing crushed coal with a liquid and allowing the impurities to separate and settle.
- Other systems control the coal burn to minimize emissions of sulfur dioxide, nitrogen oxides and particulates.
- Electrostatic precipitators remove particulates that aggravate asthma and cause respiratory ailments by charging particles with an electrical field and then capturing them on collection plates.
- Gasification avoids burning coal altogether. With gasification, steam and hot pressurized air or oxygen combine with coal in a reaction that forces carbon molecules apart. The resulting syngas, a mixture of carbon monoxide and hydrogen, is then cleaned and burned in a gas turbine to make electricity.
- Wet scrubbers, or flue gas desulfurization systems, remove sulfur dioxide, a major cause of acid rain, by spraying flue gas with limestone and water.
- Low-NO_x (nitrogen oxide) burners reduce the creation of nitrogen oxides, a cause of ground-level ozone, by restricting oxygen and manipulating the combustion process.

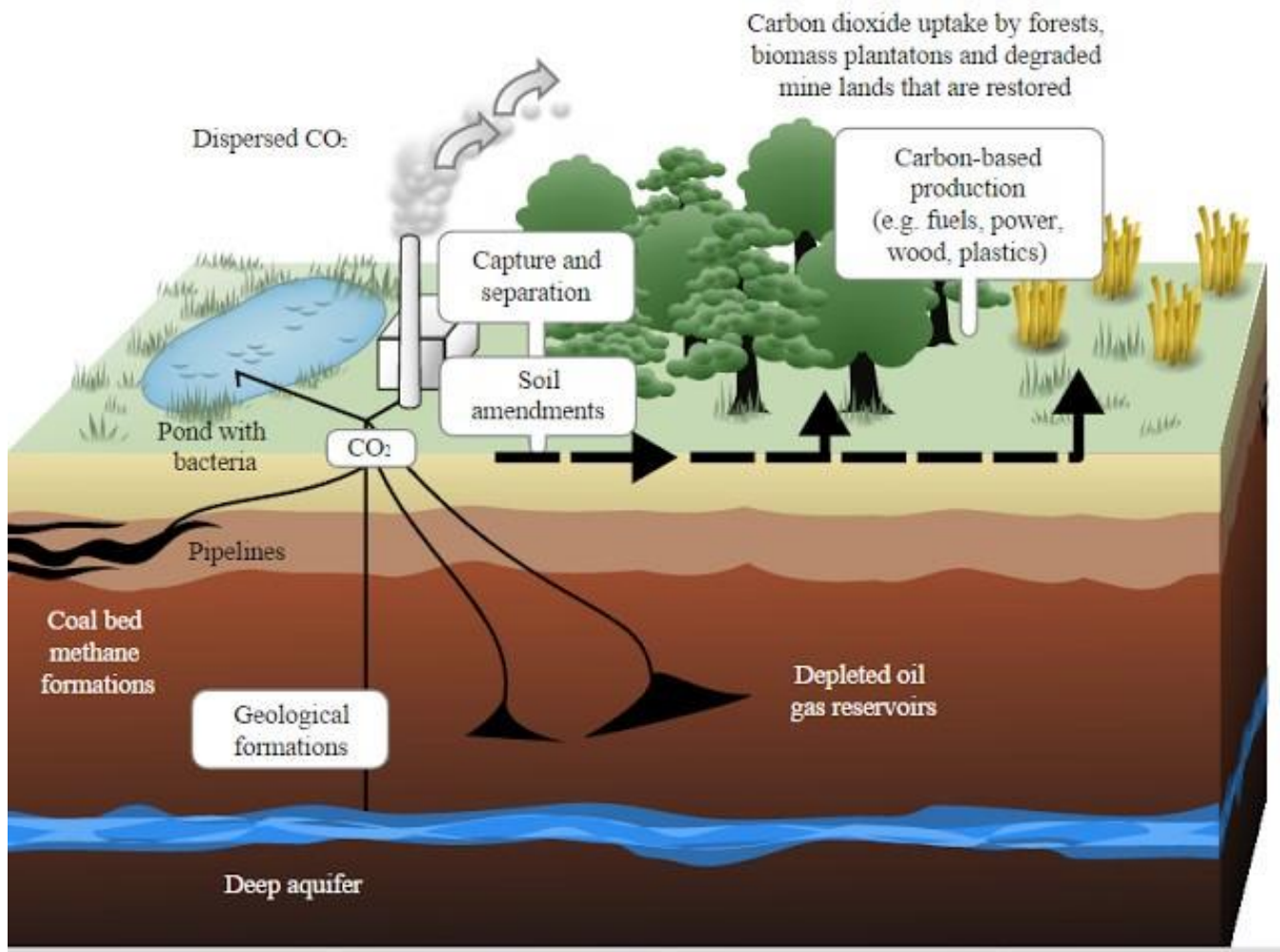
India's coal

- Coal mined in India is a poor quality coal with less carbon, high ash (hard to dispose) and high moisture content (more gases; less fuel efficiency) [India's coal is not Carboniferous Coal].

- To improve efficiency and reduce adverse effects, India should do away with its present sub-critical coal power plants and build more super-critical and ultra-super-critical ones (15-20% increase in efficiency).
- Should employ clean coal technology.

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Carbon capture and storage

- Carbon capture and storage catches and sequesters (hide) carbon dioxide (CO₂) from stationary sources like power plants.
- Capture: Flue-gas separation removes CO₂ and condenses it into a concentrated CO₂ stream.
- After capture, secure containers sequester the collected CO₂ to prevent or stall its reentry into the atmosphere.
- The two storage options are geologic and oceanic (must hide the CO₂ until peak emissions subside hundreds of years from now).
- Geologic storage involves injecting CO₂ into the earth.

- Depleted oil or gas fields and deep saline aquifers safely store CO₂ while coal seams absorb it.
- Ocean storage, a technology still in its early stages, involves injecting liquid CO₂ into waters 500 to 3,000 meters deep, where it dissolves under pressure. However, this method would slightly decrease pH and potentially harm marine habitats.

Carbon Sink

- A carbon sink is a natural or artificial reservoir that accumulates and stores some carbon-containing chemical compound for an indefinite period.

- The process by which carbon sinks remove carbon dioxide (CO₂) from the atmosphere is known as carbon sequestration.

Carbon Sink vs Carbon Source

- A carbon sink is anything that absorbs more carbon than it releases, whilst a carbon source is anything that releases more carbon than it absorbs.
- Forests, soils, oceans and the atmosphere all store carbon and this carbon moves between them in a continuous cycle.
- This constant movement of carbon means that forests act as sources or sinks at different times.

Carbon sequestration

- Carbon sequestration is the process of capture and long-term storage of atmospheric carbon dioxide (CO₂)
- It has been proposed as a way to slow the atmospheric and marine accumulation of greenhouse gases
- Carbon dioxide is naturally captured from the atmosphere through biological, chemical, or physical processes.

Forests as carbon Sinks

- Forests are carbon stores, and they are **carbon dioxide sinks** when they are increasing in density or area.
- In Canada's boreal forests as much as 80% of the total carbon is stored in the soils as dead organic matter.
- Tropical forests absorb about 18% of all carbon dioxide added by fossil fuels.
- In the context of climate change, the most important carbon stores are fossil fuel deposits as they have the unique benefit of being buried deep inside the earth, naturally separated from the carbon cycling in the atmosphere.
- This separation ends when humans burn coal, oil and natural gas, turning fossil carbon stores into atmospheric carbon.
- This release of carbon from fossil fuel has caused greenhouse gas (GHG) concentrations in the atmosphere to soar to levels more than 30 per cent higher than at the beginning of the industrial revolution.

- Because of this increase in atmospheric carbon, a lot of emphasis and hope has been put into the ability of trees, other plants and the soil to temporarily sink the carbon that fossil fuel burning releases into the atmosphere.
- Indeed, the Kyoto Protocol, the international communities' main instrument for halting global warming suggests that the absorption of carbon dioxide by trees and the soil is just as valid a means to achieve emission reduction commitments as cutting carbon dioxide emissions from fossil fuels.

The scientific view is that the increase in global temperature should not exceed 2 °C above pre-industrial level. If the global temperature increases beyond 3°C above the pre-industrial level, what can be its possible impact/impacts on the world?

- Terrestrial biosphere tends toward a net carbon source
- Widespread coral mortality will occur.
- All the global wetlands will permanently disappear.
- Cultivation of cereals will not be possible anywhere in the world.

Select the correct answer using the code given below.

1. 1 only
2. 1 and 2 only
3. 2, 3 and 4 only
4. 1, 2, 3 and 4

Explanation:

Taiga and temperate forests act as an important carbon sink. Global warming by 3°C will turn these forests into carbon source.

Corals are very sensitive to temperature changes. 3°C rise in global temperature will lead to widespread coral mortality.

3°C rise in global temperature will lead to submergence of many low lying coastal wetlands (not all) due to rise in sea levels. Inland wetlands like Keoladeo Ghana National Park will not be effected (it is 100s of kilometers away from coast).

Because of the slow inertia, long response time for parts of the climate system, it has been estimated that we are already committed to a sea-level rise of approximately 2.3 metres (7.5 ft) for each degree Celsius of temperature rise within the next 2,000 years.

<http://goo.gl/LH3RMl>

Cultivation of cereals in tropics will take a hit. But in temperate regions their production increases in the short run.

Earth Summit 1992 - UNCED | CBD | UNCCD

Major International Conventions to Protect Environment

Nature conservation

- United Nations Conference On Environment And Development (UNCED) [This post]
- Convention on Biological Diversity (CBD) [This post]
- [Ramsar Convention on Wetlands \[Given in "Wetland Ecosystem"\]](#)
- [Convention on International Trade in Endangered Species of Fauna and Flora \(CITES\) \[Given in "IUCN Red List"\]](#)
- [The Wildlife Trade Monitoring Network \(TRAFFIC\) \[Given in "IUCN Red List"\]](#)
- Convention on the Conservation of Migratory Species (CMS)
- Global Tiger Forum (GTF)

Hazardous material [Given in "Solid Waste"]

- Stockholm Convention
- Basel Convention
- Rotterdam Convention

Land

- United Nations Convention to Combat Desertification (UNCCD) [This Post]

Marine environment

- International Whaling Commission (IWC)

Atmosphere

- [Vienna convention and Montreal Protocol \[Explained in "Ozone Depletion"\]](#)
- [United Nations Framework Convention on Climate Change \(UNFCCC\) \[Next Post\]](#)
- [Kyoto Protocol \[Next Post\]](#)

UNCED - Earth Summit 1992, Rio De janeiro Brazil

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- Earth Summit 1992 is also known as The **United Nations Conference on Environment and Development (UNCED)**.
- As a follow-up, the **World Summit on Sustainable Development (Rio+10)** was held in 2002 in Johannesburg, South Africa.
- 190 countries pledged their commitment to achieve by 2010, a significant reduction in the current rate of biodiversity loss at global, regional and local levels.
- In 2012, the **United Nations Conference on Sustainable Development** was also held in Rio, and is also commonly called **Rio+20** or Rio Earth Summit 2012.

What is Rio+20 Conference, often mentioned in the news?

- It is the United nations Conference on Sustainable Development
- It is a Ministerial Meeting of the World Trade Organization
- It is a Conference of the Inter-governmental Panel on Climate Change
- It is a Conference of the Member Countries of the Convention on Biological Diversity

Answer: a) United nations Conference on Sustainable Development

The issues touched included

- checking production of toxic components, such as lead in gasoline, or poisonous waste including radioactive chemicals,
- alternative sources of energy to replace the use of fossil fuels,
- new reliance on public transportation systems in order to reduce vehicle emissions, congestion in cities,
- the health problems caused by polluted air and smoke, and
- the growing usage and limited supply of water.

Agenda 21

- Agenda 21 is an action plan of the United Nations (UN) related to sustainable development.
- It was an outcome of the United Nations Conference on Environment and Development (UNCED) held in Rio de Janeiro, Brazil, in 1992.
- The number 21 refers to an agenda for the 21st century.

Landmark Agreements

- An important achievement of the summit was an agreement on the Climate Change Convention which in turn led to the Kyoto Protocol and the Paris Agreement.
- Important legally binding agreements (Rio Convention) were opened for signature:

1. **Convention on Biological Diversity [This Post]**
2. **United Nations Convention to Combat Desertification [This Post]**
3. **Framework Convention on Climate Change (UNFCCC) [Next Post]**

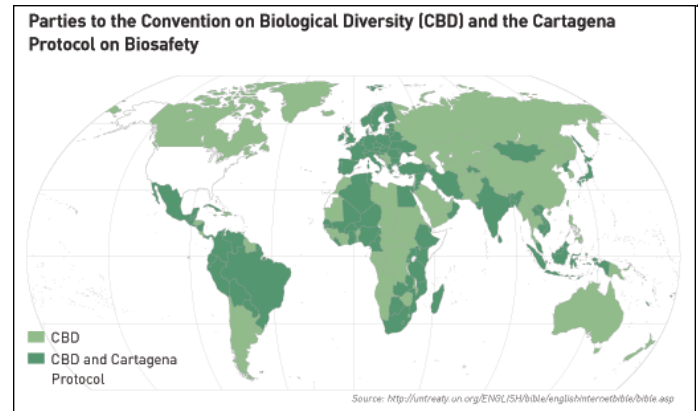
Biodiversity Related:

Convention on Biological Diversity

- Biodiversity knows no political boundaries and its conservation is therefore a collective responsibility of all nations.
- Convention on Biological Diversity (CBD) is a step towards conserving biological diversity or biodiversity with the involvement of the entire world.
- The historic Convention on Biological Diversity (Biodiversity Convention - a multilateral treaty) was opened for signature at the **Earth Summit in Rio de Janeiro** in 1992 and entered into in 1993.
- The convention called upon all nations to take appropriate measures for conservation of biodiversity and sustainable utilisation of its benefits.
- The Convention has three main goals:

1. **conservation of biological diversity (or biodiversity);**
2. **sustainable use of its components; and**
3. **fair and equitable sharing of benefits arising from genetic resources.**

- It is often seen as the key document regarding sustainable development.
- The Convention is **legally binding**; countries that join it ('Parties') are obliged to implement its provisions.
- 195 UN states and the European Union are parties to the convention.
- All UN member states—with the exception of the **United States**—have ratified the treaty.



- At the 2010 10th Conference of Parties (COP) to the Convention on Biological Diversity in October in Nagoya, Japan, the **Nagoya Protocol** was adopted.

Cartagena Protocol

- CBD covers the rapidly expanding field of **biotechnology** through its Cartagena Protocol on **Biosafety**.
- It addresses technology development and transfer, benefit-sharing and biosafety issues.
- The Biosafety Protocol seeks to protect biological diversity from the potential risks posed by living modified organisms resulting from modern biotechnology.

Nagoya Protocol

- The Nagoya Protocol is a 2010 supplementary agreement to the 1992 Convention on Biological Diversity (CBD).
- The Nagoya Protocol is about “**Access to Genetic Resources and the Fair and Equitable Sharing of Benefits Arising from their Utilization**”, one of the three objectives of the CBD.
- It is the second Protocol to the CBD; the first is the 2000 Cartagena Protocol on Biosafety.

International Treaty on Plant Genetic Resources for Food and Agriculture

- Popularly known as the **International Seed Treaty**.
- International agreement in harmony with the Convention on Biological Diversity.
- Aims at guaranteeing food security through the conservation, exchange and sustainable use of the world's plant genetic resources for food and agriculture (PGRFA), as well as the fair and equitable benefit sharing arising from its use.

Consider the following international agreements:

- The International Treaty on Plant Genetic Resources for Food and Agriculture
- The United Nations Convention to Combat Desertification
- The World Heritage Convention

Which of the above has/have a bearing on the biodiversity?

1. 1 and 2 only
2. 3 only,
3. 1 and 3 only
4. **1, 2. and 3**

World Heritage Convention explained in "Biodiversity", Rest two in this section.

United Nations Convention to Combat Desertification (UNCCD)

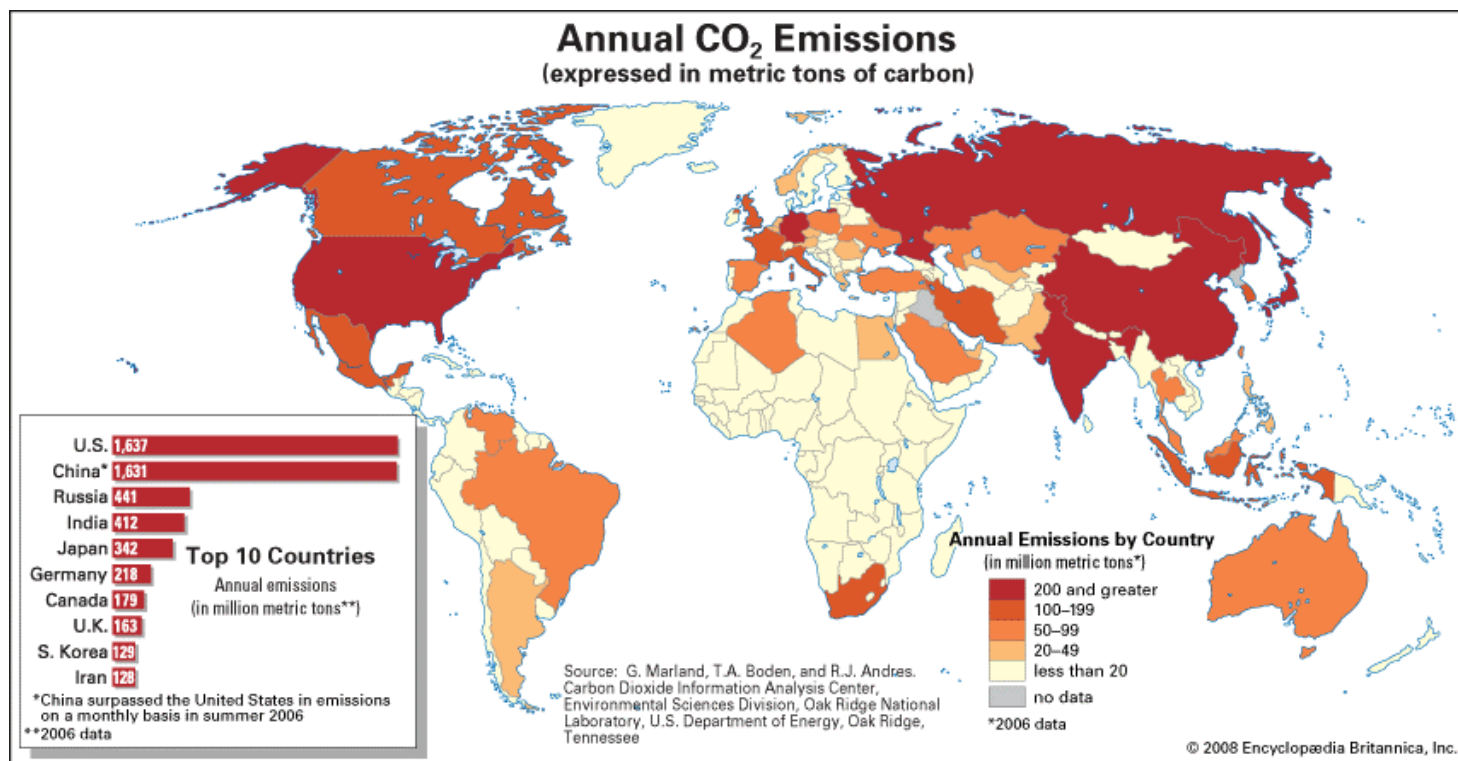
- UNCCD is a Convention to combat desertification and mitigate the effects of drought through national action programs (NAP).
- NAP incorporate long-term strategies supported by international cooperation and partnership arrangements.
- The Convention stemmed from a direct recommendation of the Rio Conference's Agenda 21, was adopted in Paris, France in 1994 and entered into force in 1996.
- It is the only internationally **legally binding** framework set up to address the problem of desertification.

- The Convention is based on the principles of participation, partnership and decentralization—the backbone of Good Governance and Sustainable Development.
- It has 196 parties, making it truly global in reach.
- To help publicize the Convention, 2006 was declared "International Year of Deserts and Desertification".
- The UN Convention to Combat Desertification has established a Committee on Science and Technology (CST).
- CST is composed of government representatives competent in the fields of expertise relevant to combating desertification and mitigating the effects of drought.
- UNCCD collaborates closely with **Convention on Biological Diversity (CBD)**.

UNFCCC - Kyoto Protocol - Carbon Credits Trading

UNFCCC: United Nations Framework Convention on Climate Change

- International environmental treaty that came into existence under the aegis of UN.
- Signed ==> May **1992**.
- Location ==> New York City, USA.
- As of March 2014, UNFCCC has 196 parties (almost all countries).
- UNFCCC is negotiated at the **United Nations Conference on Environment and Development (UNCED)**.
- UNCED is informally known as the **Earth Summit 1992**, held in **Rio de Janeiro, Brazil**.
- Role: UNFCCC provides a framework for negotiating specific international treaties (called "protocols") that aim to set **binding limits** on greenhouse gases.
- Objective of UNFCCC: Stabilize greenhouse gas concentrations in the atmosphere at a level that would prevent dangerous consequences.
- Legal Effect: Treaty is considered **legally non-binding**: The treaty itself set no binding limits on greenhouse gas emissions for individual countries.



Conferences of the Parties (COP) – UNFCCC

The COP is the supreme decision-making body of UNFCCC.

All States that are Parties to the Convention are represented at the COP.

They review the implementation of any legal instruments that the COP adopts.

They promote the effective implementation of the Convention.

- The first COP meeting was held in **Berlin, Germany** in March, **1995**.
- The parties to the convention have met **annually** since 1995.
- In **1997**, the **Kyoto Protocol (3rd COP)** was concluded and established **legally binding** obligations for developed countries to reduce their greenhouse gas emissions.
- The **2010 Cancun agreements** stated that future global warming should be limited to below **0 °C** (3.6 °F) relative to the pre-industrial level.
- The 21st COP (2015) was held in Paris in 2015.
- The 22nd COP (2016) will be held at Marrakesh, Morocco.

List of UNFCCC Summits

- 1995: COP 1, The Berlin Mandate**
- 1996: COP 2, Geneva, Switzerland

3. 1997: COP 3, The Kyoto Protocol on Climate Change

- 1998: COP 4, Buenos Aires, Argentina
- 1999: COP 5, Bonn, Germany
- 2000: COP 6, The Hague, Netherlands
- 2001: COP 6, Bonn, Germany
- 2001: COP 7, Marrakech, Morocco
- 2002: COP 8, New Delhi, India**
- 2003: COP 9, Milan, Italy
- 2004: COP 10, Buenos Aires, Argentina
- 2005: COP 11/CMP 1, Montreal, Canada
- 2006: COP 12/CMP 2, Nairobi, Kenya
- 2007: COP 13/CMP 3, Bali, Indonesia
- 2008: COP 14/CMP 4, Poznan, Poland
- 2009: COP 15/CMP 5, Copenhagen, Denmark
- 2010: COP 16/CMP 6, Cancun, Mexico
- 2011: COP 17/CMP 7, Durban, South Africa
- 2012: COP 18/CMP 8, Doha, Qatar
- 2013: COP 19/CMP 9, Warsaw, Poland
- 2014: COP 20/CMP 10, Lima, Peru**
- 2015: COP 21/CMP 11, Paris, France**
- 2016: COP 22/CMP 12, Marrakech, Morocco**

Criticisms of the UNFCCC

- Nothing except Kyoto Protocol made any binding limits on GHG emissions.
- Never achieved its stated goals of reducing the emission of carbon dioxide.

- Negotiations are governed by consensus and small group of countries often block the negotiations.
- It is easy for the developed countries to escape from their responsibility: **United States**, one of the biggest polluters never ratified Kyoto Protocol. **Canada** pulled out of Kyoto Protocol citing wealth transfers out the country due to binding limits.
- Treaty doesn't cover developing countries who now include the largest CO₂ emitters (India and China).
- **Japan, Russia** didn't sign second Kyoto term because it would impose restrictions on it not faced by its main economic competitors, China, India and Indonesia.

Important UNFCCC (COP) Summits are discussed below.

Kyoto Protocol

- The Kyoto Protocol was adopted in **Kyoto, Japan, in 1997**.
- India ratified Kyoto Protocol in **2002**.
- The Kyoto Protocol came into force in February **2005**.
- There are currently 192 Parties.
- **USA** never ratified Kyoto Protocol.
- **Canada** withdrew in 2012.
- Goal: Fight global warming by reducing greenhouse gas concentrations in the atmosphere to "a level that would prevent dangerous anthropogenic interference with the climate system."
- Kyoto protocol aimed to cut emissions of greenhouse gases across the developed world by about 5 per cent by 2012 compared with 1990 levels.
- The Protocol is based on the principle of **common but differentiated responsibilities**.
- Kyoto Protocol is the **only global treaty with binding limits on GHG emissions**.

What is Common But Differentiated Responsibilities – Kyoto Protocol?

- It puts the obligation to reduce current emissions on **developed countries** on the basis that they are historically responsible for the current levels of greenhouse gases in the atmosphere.
- CBDR divides countries into two categories.
- Historically biggest polluting developed countries like US, UK, France, Japan, Russia etc. (they are polluting the earth since Industrial Revolution).
- Recently polluting developing countries like China, India, Brazil, etc. (polluting since 1950s).
- "Common" = Every country (both developing and developed) must take part in the fight against climate change.
- "But differentiated responsibilities" = Historically biggest polluters should do more compared to the recent polluters = Responsibilities proportional to pollution caused.
- So under CBDR, developed countries like US, UK, Russia etc. must contribute more to reduce GHGs (greenhouse gases).
- They must accept to **certain binding limits** on GHG emissions.
- They must contribute funds towards reducing GHG emissions in developing and least developed countries.
- On the other hand, developing and least developed countries should do everything possible to cut down their GHG emissions. But **nothing is binding** and every initiative is voluntary.

Classification of Parties and their commitments – Kyoto Protocol

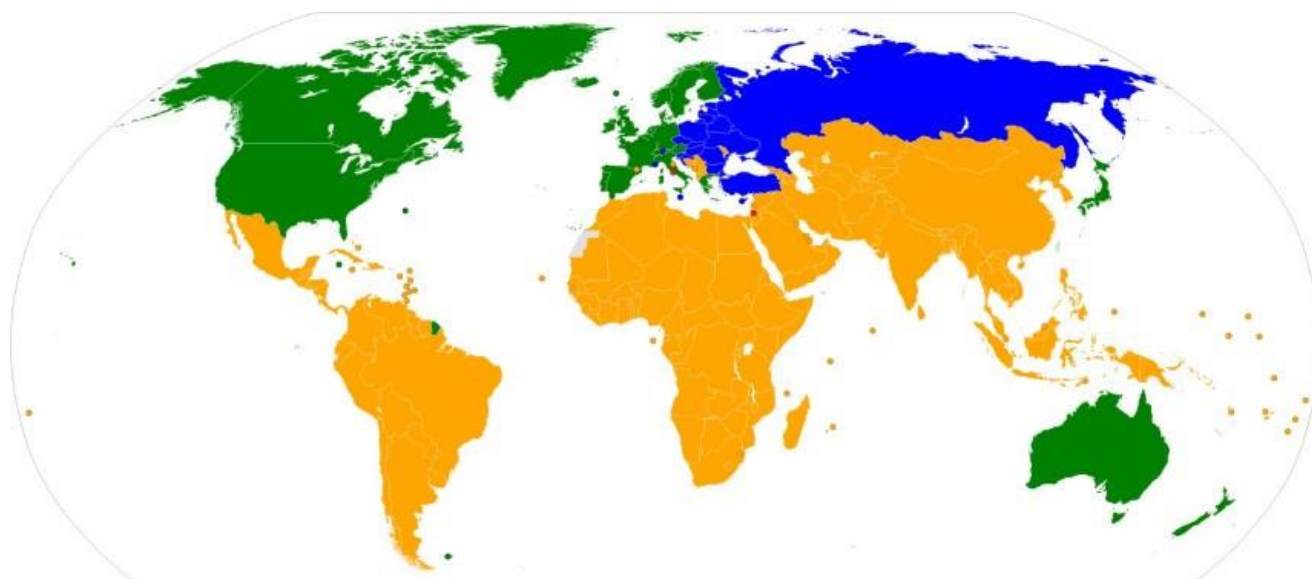
Annex I	Developed countries [US, UK, Russia etc.] + Economies in transition (EIT) [Ukraine, Turkey, some eastern European countries etc.]
Annex II	Developed countries. Annex II is a subset of Annex I. Required to provide financial and technical support to the EITs and developing countries to assist them in reducing their greenhouse gas emissions.
Annex B	Annex I Parties with first or second-round Kyoto greenhouse gas emissions targets. The first-round targets apply over the years 2008–2012 and the second-round

	Kyoto targets, which apply from 2013–2020 . Compulsory binding targets to reduce GHG emissions.
Non-Annex I	Parties to the UNFCCC not listed in Annex I of the Convention are mostly low-income developing countries. No binding targets to reduce GHG emissions.
LDCs	Least-developed countries No binding targets to reduce GHG emissions.

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- Developing countries may volunteer to become Annex I countries when they are sufficiently developed.



Parties to the UNFCCC	
	Annex I and II parties
	Annex I parties
	Non-annex parties
	Observer states

What is commitment period – Kyoto Protocol?

- Under Kyoto Protocol, there are two commitment periods:
 - 2008 – 2012 and**
 - 2013 – 2020.**
- The second commitment period was agreed on in 2012, known as the **Doha Amendment** to the protocol.
- Each commitment period has its own **binding targets** set for developed countries to reduce their GHG emissions.
- Nations that miss their Kyoto target in 2012 will incur a **penalty** of an additional third added to whatever cut they agree under a new treaty in Copenhagen.
- During first commitment period (2008-12), more than 35 countries had binding targets.
- Canada withdrew in 2012** after the first commitment period.
- Japan, New Zealand and Russia** have participated in Kyoto's first-round but have not taken on new targets in the second commitment period.
- As of November 2015, 55 states have accepted the Doha Amendment, while entry into force requires the acceptances of 144 states.
- So second commitment period is an epic failure.
- Negotiations were held in Lima in 2014 to agree on a post-Kyoto legal framework that

would obligate all major polluters to pay for CO₂ emissions.

- China, India, and the United States (three big villains) have all signaled that they will not ratify any treaty that will commit them legally to reduce CO₂ emissions.

The Kyoto Protocol emission target gases include

- Carbon dioxide (CO₂),
- Methane (CH₄),
- Nitrous oxide (N₂O),
- Sulphur hexafluoride (SF₆),
- groups of hydro fluorocarbons (HCFs) and
- groups of Per fluorocarbons (PFCs).

Flexible Market Mechanisms – Kyoto Protocol

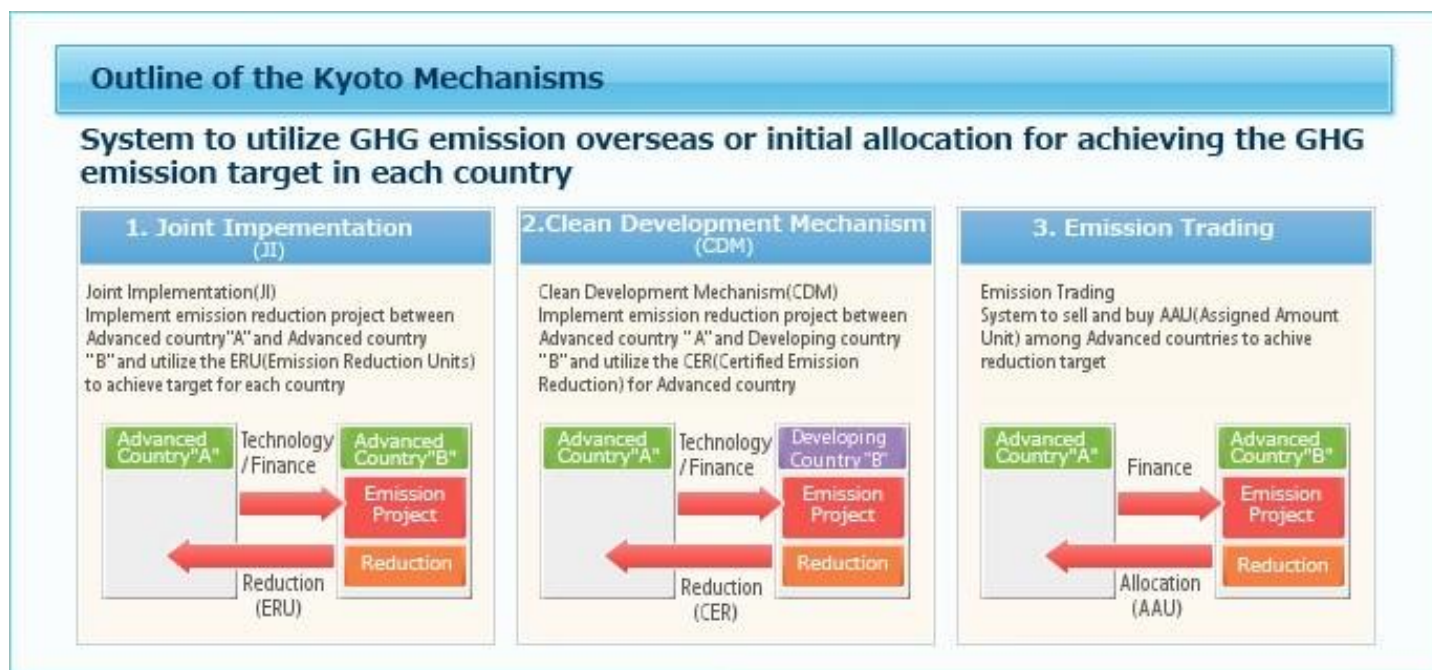
- Countries bound to Kyoto targets have to meet them largely through domestic action— that is, to reduce their emissions onshore.
- But they can meet part of their targets through three “market-based mechanisms”.

The Kyoto Flexible Market Protocol mechanisms:

1. Clean Development Mechanism (CDM)
2. Emission Trading
3. Joint Implementation (JI)

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Clean Development Mechanism (CDM) – Kyoto Protocol

- The Clean Development Mechanism (CDM), defined in the Kyoto Protocol, allows a country with an emission-reduction or emission-limitation commitment under the Kyoto Protocol (Annex B Party) to implement an emission-reduction project in developing countries.

[Hypothetical E.g. of CDM: Australia takes up or finances some environment benefitting project in India (solar power projects, wind power projects, afforestation etc.) and earns some carbon credits (certified emission reduction credits). Now it shows these earned

carbon credits to the world and tells them how it is working towards meeting its Kyoto targets.]

- Such projects can earn saleable certified emission reduction (CER) credits, each equivalent to one tonne of CO₂, which can be counted towards meeting Kyoto targets.
- In simple terms: Developed countries emit more and lose carbon credits. They provide financial assistance to developing and least developed countries to create clean energy (solar, wind energy etc.) and gain some carbon credits = meet their Kyoto Quota (Kyoto units) of emissions without violations.
- Suppose a developed country has a Kyoto Quota of 100 Carbon Credits ==> It can emit 100 tonnes of CO₂.

- Due to negligence it emits 110 tonnes of CO₂ = 10 carbon credits lost = Kyoto Quota violation.
- Now the country has to make up for its lost carbon credits to avoid penalty.
- So it invests some money (equal to 10 carbon credits) in developing and LDCs to build clean energy infrastructure like solar plants, wind farms etc. and will make up for its 10 lost carbon credits and avoid penalty.

Carbon Credits Trading [Carbon Trading] – Kyoto Protocol

Carbon credit – Kyoto Protocol

A carbon credit (often called a carbon offset) is a **tradable certificate or permit**.

One carbon credit is equal to **one tonne** of carbon dioxide.

Carbon credits are a part of attempts to mitigate the growth in concentrations of GHGs.

Carbon credits or carbon offsets can be acquired through afforestation, renewable energy, CO₂ sequestration, methane capture, buying from an exchange (carbon credits trading) etc..

- Carbon trading is the name given to the exchange of emission permits.
- This exchange may take place within the economy or may take the form of international transaction.
- Under Carbon Credits Trading mechanism countries that emit more carbon than the quota allotted to them buy carbon credits from those that emit less.
- In Carbon trading, one credit gives the country or a company **right to emit** one tonne of CO₂.
- A country having more emissions of carbon (less carbon credits) is able to purchase the right to emit more from a country having less emissions (more carbon credits).
- More carbon emitting countries, by this way try to keep the limit of carbon emission specified to them.
- A developing nation such as **India**, turns out to be a **seller** of such credits, which eventually provides them with monetary gains.
- Carbon credits are traded at various exchanges across the world.

- **Multi-Commodity Exchange of India (MCX)** launched futures trading in carbon credits in 2009.

Types of Carbon trading

1. Emission trading and
2. Offset trading.

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Emission trading/'cap-and-trade'

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- Emissions trading allows countries that have emission units to spare - emissions permitted them but not "used" - to sell this excess capacity to countries that are over their targets.
- Carbon is now tracked and traded like any other commodity. This is known as the "carbon market."

Other trading units in the carbon market

1. A removal unit (RMU) by reforestation.
2. An emission reduction unit (ERU) generated by a joint implementation project (explained below).
3. A certified emission reduction (CER) generated from a clean development mechanism project activity.

Offset Trading/Carbon Project/'baseline-and credit' trading

- Another variant of carbon credit is to be earned by a country by investing some amount of money in such projects, known as carbon projects, which will emit lesser amount of greenhouse gas in the atmosphere.
- For example, suppose a thermal plant of 800 megawatt capacity emit 400 carbon-equivalent in the atmosphere. Now a country builds up a 800 megawatt wind energy plant which does not generate any amount of emission as an alternative of the thermal plant. Then by investing in this project the country will earn 400 carbon-equivalent.

Offset Trading is a variant of Emission Trading or Carbon Trading.

Carbon tax (related concept but not related to Kyoto Protocol)

- It is a tax on all fossil fuels in proportion to carbon dioxide emissions.
- Proposed in may developed and developing countries.
- The proposal faced political resistance (politician – corporate nexus, people feared more burden).
- India has a carbon tax of sorts. Budget of 2010-11 introduced a cess of Rs. 50 per tonne of both domestically produced and imported coal. Later it was increased to Rs. 100.
- This cess is used to raise revenues for the **National Clean Energy Fund**.

Non-Compliance of Kyoto And Penalties

- Like most things in life, failure to comply with the Protocol carries penalties.
- If a country does not meet the requirements for measurements and reporting said country loses the privilege of gaining credit through joint implementation projects.
- If a country goes above its emissions cap, and does not try to make up the difference through any of the mechanisms available, then said country must make up the difference plus an additional thirty percent during the next period.
- The country could also be banned from participating in the 'cap and trade' program.

Joint Implementation (JI) – Kyoto Protocol

- The mechanism known as “joint implementation,” allows a country with an emission reduction commitment under the Kyoto Protocol (Annex B Party) to earn emission reduction units (ERUs) from an emission-reduction project in another Annex B Party, each equivalent to one tonne of CO₂, which can be counted towards meeting its Kyoto target.
- Joint implementation offers Parties a flexible and cost-efficient means of fulfilling a part of their Kyoto commitments, while the host Party benefits from foreign investment and technology transfer.

Regarding "carbon credits", which one of the following statements is not correct?

1. The carbon credit system was ratified in conjunction with the Kyoto Protocol

2. Carbon credits are awarded to countries or groups that have reduced greenhouse gases below their emission quota
3. The goal of the carbon credit system is to limit the increase of carbon emission quota
4. Carbon credits are traded at a price fixed from time to time by the United Nations Environment Programme.

Answer d) Carbon credit prices are traded on an exchange. So their prices are never fixed. 237

Benefits of Flexible Market Mechanisms

- Stimulating green investment in developing countries.
- Including the private sector in this endeavor to cut and hold steady GHG emissions at a safe level.
- It also makes "leap-frogging" — possibility to skip older, dirtier technology for newer, cleaner infrastructure and systems, with obvious longer-term benefits.
- Strengthen the Protocol's environmental integrity, support the carbon market's credibility and ensures transparency of accounting by Parties.

Criticism of Kyoto Protocol

- Under Kyoto Protocol, Annex 1 countries can meet their targets by cutting emissions or buying unused allowances (**carbon credits, carbon trading**) from other countries.
- This kind of approach ignores long term social and economic costs. It is like committing only half of what one needs to commit.
- Kyoto Protocol is based on the “common but differentiated responsibility” approach to global warming. Under CBDR, many countries were allowed to increase pollution.
- It excluded most polluting countries like China and India, which have since become the world's largest and fourth largest polluters.

Important Summits Post Kyoto

After the Kyoto Protocol, parties to the Convention have agreed to further commitments. These include the Bali Action Plan (2007), the Copenhagen Accord (2009), the Cancun agreements (2010), and the

Durban Platform for Enhanced Action (2011) etc.

Bali Summit, 2007

- COP 13, CMP 3.
 - Adopted Bali Road Map that included
1. The Bali Action Plan (BAP) [BAP: launch a comprehensive process to enable the implementation of the Convention through long-term cooperative action up to and beyond 2012.]
 2. Launch of the Adaptation Fund,
 3. Decisions on technology transfer and
 4. On reducing emissions from deforestation.
- All developed country Parties have agreed to “quantified emission limitation taking into account differences in their national circumstances.”
 - So they will fix emission limits according to their convenience and try to achieve them.
 - Developed countries stressed developing countries like India and China, which are increasing their emissions as they grow economically, also undertake some kind of emission cuts.

CMP: Conference of the Parties serving as the meeting of the Parties to the Kyoto Protocol.

COP11 / CMP 1 was held in Montreal, Canada in 2005 (Kyoto Protocol was ratified in 2005)

Copenhagen Summit, 2009

- COP 15, CMP 5.
- UNFCCC meet in Copenhagen, capital city of Denmark.
- Produced the **Copenhagen Accord**.
- This accord is an agreement between developing nations block called BASIC (Brazil, South Africa, India and China).
- According to this accord, all countries should pledge **voluntary limits** (no binding obligations) to reduce GHG emissions.
- Binding obligations could not be reached due to discord between developed and developing countries.

- Copenhagen Accord also laid the groundwork for financial commitments from developed countries to developing countries.
- Agrees a "goal" for the world to raise \$100 billion per year by 2020. New multilateral funding for adaptation will be delivered, with a governance structure.
- The Accord states that global warming should be limited to below 2.0 °C (3.6 °F) to the pre-industrial level.

Cancún Summit, 2010

- COP 16, CMP 6.
- An agreement adopted by the COP called for a large “**Green Climate Fund**”, and an “**Adaptation Committee**” at global level to support developing countries in mitigation of GHGs.
- It looked forward to a second commitment period for the Kyoto Protocol.
- As per the Cancun Agreements, all Parties to the Convention (including the developed and developing countries) have agreed to report their voluntary mitigation goals for implementation.

Durban Summit, 2011

- COP 17, CPM 7.
- In 2011, parties adopted the “**Durban Platform for Enhanced Action**”.
- Parties have agreed to “develop a protocol, another legal instrument or an agreed outcome with legal force”. This new treaty is due to be adopted at the 21st COP, and implemented in 2020.
- Second phase of Kyoto Protocol was secured.
- India’s insisted Common but differentiated responsibility retained.

Doha Summit, 2012

- COP 18, CPM 8
- Kyoto Extended: It established a **second commitment period** (2013 – 2020) of the Kyoto Protocol.
- **Japan, Russia** and **Canada** refused to join the second commitment period under the Kyoto Protocol.
- It finalized **Green Climate Fund**.

Green Climate Fund

- The Fund will start operating from **2013**.
- The Green Climate Fund (GCF) is a fund within the framework of the UNFCCC.
- The GCF is based in **Incheon, South Korea**.
- It is a mechanism to redistribute money from the developed to the developing world.
- GCF will help developing countries financially in adapting mitigation practices to counter climate change.
- It is intended to be the centerpiece of efforts to raise Climate Finance of \$100 billion by 2020.

Which of the following statements regarding 'Green Climate Fund' is/are correct?

- It is intended to assist the developing countries in adaptation and mitigation practices to counter climate change.
- It is founded under the aegis of UNEP, OECD, Asian Development Bank and World Bank

Select the correct answer using the code given below.

1. 1 only
2. 2 only
3. Both 1 and 2
4. Neither 1 nor 2

Answer: a) 1 only

Warsaw Summit, 2013

- COP19, CMP9.
- 2013 United Nations Climate Change Conference.
- The G77 and China bloc led 132 poor countries in a walk out during talks about "loss and damage" compensation for the consequences of global warming.
- Poor countries have demanded that the developed world give them \$100 billion annually by 2020.

Paris Summit 2015 - Lima Summit 2014 - INDC

Lima Summit, 2014

- COP20 or CMP10 or Lima Summit was held in Lima, Peru, in December 2014.
- 2014 United Nations Climate Change Conference.

- It is the
- 20th yearly session of the Conference of the Parties (COP 20) to the 1992 United Nations Framework Convention on Climate Change (UNFCCC) and
- the 10th session of the Meeting of the Parties (CMP 10) to the 1997 Kyoto Protocol.
- The overarching goal of the conference is to reduce greenhouse gas emissions (GHGs) to limit the global temperature increase by 2030 to 2 degrees Celsius above 1850 baseline or Pre Industrial era.
- No agreement was reached due to **lack of consensus** between developed and developing countries.
- There was no clarity on the burden that each country has to bear and the summit ended in mere symbolism.

Last-minute deal to end the impasse at Lima Summit

- The last minute deal urged developed countries to provide financial support to developing countries to meet their "ambitious mitigation" goals (Slightly in favor of Common But Differentiated Responsibility).
- The agreement urges parties to take national pledges by finalizing their **Intended Nationally Determined Contributions (INDC)** by November 2015 (Before Paris Summit).
- The agreement was severely criticized for being too shallow in its commitments.

Developed vs. Developing

- Wealthy nations like the U.S, EU argued that emissions from developing countries are consistently rising and they need to commit to more serious emission cuts.
- But India accused them of watering down the CDR principle envisaged in earlier agreements (Kyoto Protocol).
- India stuck to its conventional position that the developed countries should shoulder a bigger burden as they are responsible for the problem in the first place.
- India confirmed that poverty alleviation would continue to be its primary concern and hence it will not compromise with its share of carbon credits.

India's position

- China agreed a deal in 2014 under which its emissions would peak by 2030 (after 2030, it will start reducing its carbon footprint).
- India, the world's third largest carbon emitter, is under pressure since then to make commitments like China did.
- India says carbon emissions will grow as it drives to beat poverty.
- India's emissions are around 1.9 tonnes per person — less than China, which emits around 7.2 tonnes per person and the 5 tonnes world average.
- So India wants to use its carbon credits to alleviate poverty and is not ready to accept anything other than 'common but differentiated responsibilities'.
- India has long defended the principle of "common but differentiated responsibility" (Common but differentiated responsibility = Contribution proportional to pollution: USA, China, Japan and Major European powers are polluting since many years and hence they should contribute more. India and other developing nations along with African nations have stated polluting the environment very recently and hence they will contribute less).

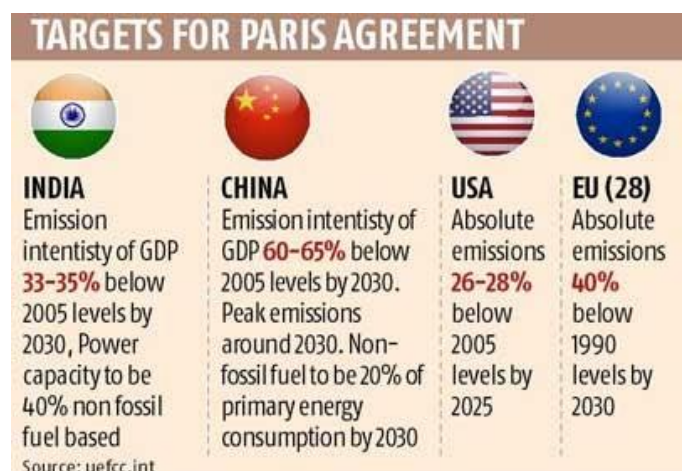
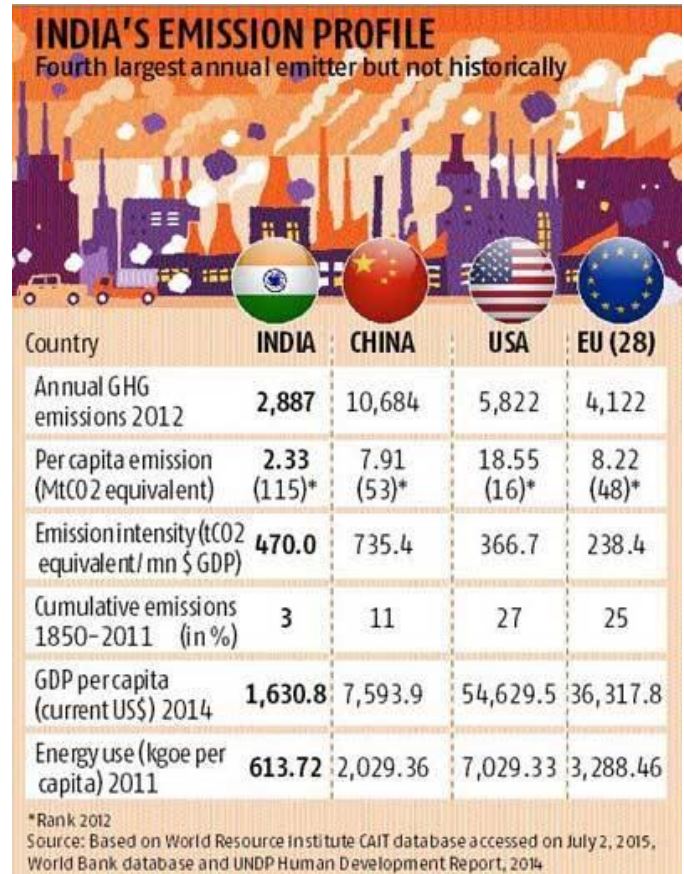
What India must do other than blaming developed nations

- India must demonstrate concerns about climate change issues (2015 Chennai floods, 2013 Uttarakhand floods are all the effects of climate change).
- Emissions in India are rising dangerously and the environmental costs will offset all its economic progress.
- Instead of always passing the burden onto others, it has to take responsibility.
- India had to make a pragmatically determined national pledge before Paris Summit [India announced its INDC in October 2015].

Paris summit, 2015

- 2015 United Nations Climate Change Conference.
- COP 21 or CMP 11.
- November 30 to December 11, 2015.
- Location ==> Paris.
- Previous summit was Held in 2014 in Lima, Peru.
- No agreement was reached in Lima.

- All agreements and decisions were reserved for Paris Summit 2015.
- Paris Summit is one of the most important environmental conference because of the **INDC commitments** made by major polluters.



- The conference objective is to achieve a legally binding and universal agreement on climate to be signed in 2015, and implemented by 2020.
- Prior to the conference, 146 national climate panels publicly presented draft national

climate contributions (so-called Intended Nationally Determined Contributions, INDCs).

China-U.S. deal on emission cuts

- Prior to the summit, China and the United States have agreed on a timetable to limit emission of GHGs.
- It will impose fresh pressure on India to make a voluntary commitment.
- S. agreed to reduce by 2025 its emission of greenhouse gases by 26 per cent to 28 per cent below its 2005 level.
- China stated its intent to peak emissions of carbon dioxide in 2030, if not earlier (from 2030 it will start reducing its emissions). It also agreed to raise the share of non-fossil fuels to 20 per cent in the next 16 years.
- India's per capita emissions are estimated at **one-tenth** of the United States and **one-fourth** of China.
- China – US deal imposed a fresh pressure on India to make a voluntary commitment.
- India announced its INDCs in the end of 2015.



What is an INDC?

- During previous climate negotiations, countries agreed to publicly outline what actions they intend to take under a global agreement well before the Paris Summit 2015.

- These country commitments are known as Intended Nationally Determined Contributions (INDCs).

The main points of contention on INDCs

Inclusion of Adaptation, finance and transfer of technology

- Developed countries are of the view that only actions that help in reducing greenhouse gas emissions should be counted as 'contributions' in INDCs.
- Almost every developing country, including India, however, wants adaptation measures also to be counted.
- Developing countries also want efforts by developed ones on providing money or transferring technology to poorer nations to be included in INDCs.
- This will help in holding the rich countries (biggest culprits that contributed to the increase of GHG emissions since Industrial Revolution) accountable to their promises on ensuring financial and technology flows.

Commitment Period

- India, European Union, China etc. are in favor of a 10-year commitment period.
- The **United States**, however, wants five-year commitment period so that countries can make quicker reviews.

Ex-post Review

- Since the INDCs are 'nationally-determined' and voluntary, the level of ambition in making 'contributions' is likely to be low.
- Some countries want an assessment of each country's INDC to see whether these are in line with the global 2 degree target.
- **India** and the **United States** strongly resent any such provision, saying such an exercise will negate the 'nationally-determined' nature of the 'contributions'.

India's INDC objectives

- Announced in October, 2015 (Lima summit urged every country to announce its INDCs by Nov, 2015)

Reduce emission intensity by 33 to 35%

- Reduce emission intensity by 33 to 35 per cent by 2030 compared to 2005 levels.

How?

- Introduce new, more efficient and cleaner technologies in thermal power generation.
- Reducing emissions from transportation sector.
- Promote energy efficiency, mainly in industry, transportation, buildings and appliances
- Develop climate resilient infrastructure.
- Pursue Zero Effect, Zero Defect policy under Make in India programme.

Produce 40% non-fossil fuel based energy

- Produce 40 per cent of electricity from non-fossil fuel based energy resources by 2030, if international community helps with technology transfer and low cost finance.

How?

- Install 175 GW of solar, wind and biomass electricity by 2022, and scale up further in following years.
- Aggressively pursue development of hydropower.
- Achieve the target of 63 GW of installed nuclear power capacity by 2032.

Create additional carbon sink of 2.5 to 3 billion tonnes

- Create an additional carbon sink of 2.5 to 3 billion tonnes of carbon dioxide equivalent by 2030 through additional forest and tree cover.

How?

- Full implementation of Green India Mission and other programmes of afforestation
- Develop 140,000 km long tree line on both sides of national highways

Sustainable development

- Develop robust adaptation strategies for agriculture, water and health sectors.

How?

- Redesign National Water Mission and National Mission on Sustainable Agriculture.
- Active implementation of ongoing programmes like National Initiative on Climate Resilient Agriculture, setting up of 100 mobile soil-testing laboratories, distribution of soil health cards to farmers.
- Additional impetus on watershed development through Neeranchal scheme.
- Effective implementation of National Mission on Clean Ganga.
- Early formulation and implementation of National Health Mission.
- Complete Integrated Coastal Zone Management plan. Mapping and demarcation of coastal hazard lines.

Money required to meet India's INDC

- At least USD 2.5 trillion (at current prices) required between now and 2030 to implement all planned actions.
- A total of INR 170.84 billion collected through cess on coal production. Being used for funding clean energy projects.
- **National Adaptation Fund** has been created with initial allocation of Rs 3500 million.
- **Tax free infrastructure bonds** of INR 50 billion being introduced for funding renewable energy projects.

National Adaptation Fund on Climate Change (NAFCC)

- Budget 2015 introduced "National Adaptation Fund" for climate change.
- As an initial sum, an amount of Rs 100 crore will be transferred to the Fund.
- Budget provision for the year 2015-16 and 2016-17 is Rs.350 crores.
- Money obtained from coal cess goes into NAF.
- Objective: Assist States and Union Territories that are particularly vulnerable to the adverse effects of climate change in meeting the cost of adaptation.
- The National Bank for Agriculture and Rural Development (NABARD) has been appointed as National Implementing Entity (NIE) responsible for implementation of adaptation projects under the (NAFCC).

Tax free infrastructure bonds

- A bond is an instrument to borrow money.

- Infrastructure bonds are borrowings to be invested in government funded infrastructure projects within a country.
- They are issued by governments or government authorized Infrastructure companies or Non-Banking Financial Companies.
- Infrastructure bonds are good for people who need a fixed income. They offer a decent rate of interest and tax benefits.
- The maturity of these bonds is often between 10 to 15 years with an option to buy-back after a lock-in of 5 years (one can sell their bonds only after 5 years).
- Example: An authority or company wants to raise Rs 5 crore from tax-free bonds. The price of each bond is Rs. 1000. It will issue 50000 units of bonds. The maturity period is ten years. The minimum investment is 5 bonds which is equal to Rs. 5000. You want to invest Rs 10000. If the interest rate which is known as the coupon rate is 10 %, your return per annum is Rs. 1000. So after 10 years, you get a total of Rs. 20000.

What are the tax benefits?

- Investments up to Rs. 20000 are eligible for income tax deduction under Section 80 CCF of the Income Tax Act.

REDD | IPCC | Global Environment Facility | Arctic Council

REDD & REDD+

- REDD (Reducing Emissions from Deforestation and Forest Degradation) is the global endeavor to create an **incentive for developing countries** to protect, better manage and save their forest resources, thus contributing to the global fight against climate change.
- REDD+ goes beyond merely checking deforestation and forest degradation, and includes **incentives for positive elements of conservation**, sustainable management of forests and enhancement of forest carbon stocks.

Forest Carbon Partnership Facility

- It is a global partnership of governments, businesses, civil society, and Indigenous

Peoples focused on **reducing emissions from deforestation and forest degradation**, forest carbon stock conservation, the sustainable management of forests, and the enhancement of forest carbon stocks in developing countries (activities commonly referred to as REDD+).

- The **World Bank** assumes the functions of trustee and secretariat.
- The World Bank, the Inter-American Development Bank and United Nations Development Programme are Delivery Partners under the Readiness Fund and responsible for providing REDD+ readiness support services to distinct countries.

Objectives

- To assist countries in their REDD+ efforts by providing them with financial and technical assistance.
- To pilot a performance-based payment system for REDD+ activities.
- Within the approach to REDD+, to test ways to sustain or enhance livelihoods of local communities and to conserve biodiversity.
- To disseminate broadly the knowledge gained in Emission Reductions Programs (ERPs).

With reference to 'Forest Carbon Partnership Facility', which of the following statements is/are correct?

- it is global partnership of governments, businesses, civil society and indigenous peoples
- it provides financial aid to universities, individual scientists and institutions involved in scientific forestry research to develop eco-friendly and climate adaptation technologies for sustainable forest management
- It assists the countries in their 'REDD+ (Reducing Emission from Deforestation and Forest Degradation+)' efforts by providing them with financial and technical assistance.

Select the correct answer using the code given below

1. 1 only
2. 2 and 3 only
3. 1 and 3 only
4. 1, 2 and 3

It provide financial incentives to countries in their REDD+ efforts. There is no mention of assistance to universities, scientists...

Answer: c) 1 and 3 only

BioCarbon Fund Initiative

- The BioCarbon Fund Initiative for **Sustainable Forest Landscapes (ISFL)** is a multilateral fund, supported by donor governments and managed by the **World Bank**.
- It seeks to promote reduced greenhouse gas emissions from the land sector, from deforestation and forest degradation in **developing countries (REDD+)**, and from sustainable agriculture, as well as smarter land-use planning, policies and practices.

'BioCarbon Fund Initiative for Sustainable Forest Landscapes' is managed by the

1. Asian Development Bank
2. International Monetary Fund
3. United Nations Environment Programme
4. World Bank

Answer: d) World Bank

Intergovernmental Panel on Climate Change

- Scientific intergovernmental body under the auspices of the United Nations.
- Set up at the request of member governments.
- It was first established in 1988 by two United Nations organizations, the World Meteorological Organization (WMO) and the United Nations Environment Programme (UNEP).
- Membership of the IPCC is open to all members of the WMO and UNEP.
- The IPCC produces **reports** that support the United Nations Framework Convention on Climate Change (UNFCCC).
- IPCC reports cover all relevant information to understand the risk of human-induced climate change, its potential impacts and options for adaptation and mitigation.
- The IPCC does not carry out its own original research.
- Thousands of scientists and other experts contribute on a voluntary basis.

- The 2007 Nobel Peace Prize was shared, in two equal parts, between the IPCC and an American Environmentalist.

The aims of the IPCC are to assess scientific information relevant to:

1. Human-induced climate change,
2. The impacts of human-induced climate change,
3. Options for adaptation and mitigation.

IPCC Assessment Reports (AR)

- In accordance with its mandate, the IPCC prepares at regular intervals comprehensive Assessment Reports of scientific, technical and socio-economic information relevant for the understanding of human induced climate change, potential impacts of climate change and options for mitigation and adaptation.

Special Reports

- Special Reports have been prepared on topics such as aviation, regional impacts of climate change, technology transfer, emissions scenarios, land use, land use change and forestry, carbon dioxide capture and storage and on the relationship between safeguarding the ozone layer and the global climate system.

Global Environment Facility

- The Global Environment Facility (GEF) unites 183 countries in partnership with international institutions, civil society organizations (CSOs), and the private sector to address global environmental issues while supporting national sustainable development initiatives.
- Today the GEF is the **largest public funder of projects** to improve the global environment.
- An **independently operating financial organization**, the GEF provides grants for projects related to biodiversity, climate change, international waters, land degradation, the ozone layer, and persistent organic pollutants.
- The GEF also serves as financial mechanism for the following conventions:
 - Convention on Biological Diversity (CBD)
 - United Nations Framework Convention on Climate Change (UNFCCC)

- UN Convention to Combat Desertification (UNCCD)
- Stockholm Convention on Persistent Organic Pollutants (POPs)
- Minamata Convention on Mercury
- The GEF, although not linked formally to the Montreal Protocol on Substances that Deplete the Ozone Layer (MP), supports implementation of the Protocol in countries with economies in transition.

The list below describes 10 GEF agencies that currently operating related to adaptation to climate change:

- United Nations Development Programme
- United Nations Environment Programme
- World Bank Food and Agriculture
- Organization Inter-American Development Bank
- United Nations Industrial Development Organization
- Asian Development Bank
- African Development Bank
- European Bank for Reconstruction and Development
- International Fund for Agricultural Development
- World Wildlife Fund
- United Nations Industrial Development Organization
- Conservation International

Areas of work

- Biodiversity
- Climate change
- International waters
- Land degradation
- Sustainable forest management / REDD+:
- Ozone depletion

With reference to 'Global Environment Facility', which of the following statements is/are correct?

- It serves as financial mechanism for 'Convention on Biological Diversity' and 'United Nations Framework Convention on Climate Change'.
- It undertakes scientific research on environmental issues at global level
- It is an agency under OECD to facilitate the transfer of technology and funds to

underdeveloped countries with specific aim to protect their environment.

- Both (a) and (b)

Explanation:

GEF is an independent financial organization. Page

It may fund scientific research but it is not directly involved in scientific research. | 245

IPCC takes care of the most of the research work.

Answer: a)

Transition to green economy

Three priorities in transition of economy to green economy are

decarbonizes the economy;
commit the environmental community to justice and equity; and
conserve the biosphere.

A key step forward consists in changing our conception of growth and prosperity — achieving more with less and creating real wealth and quality of life.

Measures to adapt green economy

Energy audit can reduce your building's climate footprint and lead to significant savings in energy costs.

Overfishing in many parts of the world threatens to deplete future fish stocks. We can avoid this by working to promote sustainable fishing practices.

Deforestation accounts for close to 20% of the world's greenhouse gas emissions.

Sustainably managed forests can continue to support communities and ecosystems without damaging environment and climate.

Use electronic files to reduce your demand for paper products.

When you support certified sustainable forest products, you support a healthy environment and sustainable livelihoods.

Car-pooling or taking public transport reduces environmental impacts and economic costs while strengthening community.

Walking or riding a bike for short trips is good

for your health - and the environment. Taking small steps towards wise water use can help conserve this precious resource. Resource efficiency is key to a Green Economy and water is one of our most important resources.

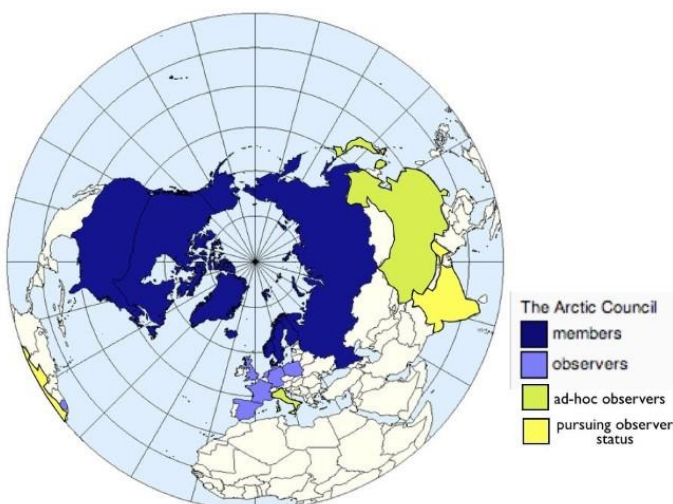
The development of clean, renewable energy by using solar, wind, tidal, etc will contribute to green economy.

Recycling appropriate materials and composting food waste reduces the demand on our natural resources.

Moving towards a green economy has the potential to achieve sustainable development and eradicate poverty on an unprecedented scale, with speed and effectiveness.

Arctic Council

- Intergovernmental forum promoting cooperation, coordination and interaction among the Arctic states, Arctic Indigenous communities and other Arctic inhabitants on common Arctic issues, in particular on issues of sustainable development and environmental protection in the Arctic.
- The Arctic Council consists of the eight Arctic States: **Canada, the Kingdom of Denmark (including Greenland and the Faroe Islands), Finland, Iceland, Norway, Russia, Sweden and the United States.**



- Six international organizations representing Arctic Indigenous Peoples have permanent participant status.

Consider the following countries:

1. Denmark

2. Japan
3. Russian Federation
4. United Kingdom
5. United States of America

Which of the above are the members of the 'Arctic Council'?

- a. 1, 2 and 3
- b. 2, 3 and 4
- c. 1, 4 and 5
- d. **1, 3 and 5**

National Action Plan On Climate Change

India's National Action Plan On Climate Change

- The National Action Plan hinges on the development and use of new technologies.
- The implementation of the Plan include public private partnerships and civil society action.
- The focus will be on promoting **understanding of climate change, adaptation and mitigation, energy efficiency and natural resource conservation.**
- There are Eight National Missions which form the core of the National Action Plan.



1. National Solar Mission
2. National Mission On Sustainable Habitat
3. National Water Mission (NWM)

4. National Mission For Sustaining The Himalayan Ecosystem (NMSHE)
5. National Mission For A Green India
6. National Mission For Sustainable Agriculture (NMSA)
7. National Mission on Strategic Knowledge for Climate Change (NMSKCC)
8. National Bio-Energy Mission

National Solar Mission

- The National Solar Mission is a major initiative to promote ecologically sustainable growth while addressing India's energy security challenge.
- The Mission will adopt a 3-phase approach:
 1. spanning the remaining period of the 11th Plan and first year of the 12th Plan (up to 2012-13) as Phase 1,
 2. the remaining 4 years of the 12th Plan (2013-17) as Phase 2 and
 3. the 13th Plan (2017-22) as Phase 3.
- At the end of each plan there will be an evaluation of progress.

Objectives

- To establish India as a global leader in solar energy, by creating the policy conditions for its diffusion across the country as quickly as possible.
- To create an enabling policy framework for the deployment of 100,000 MW of solar power by 2022.
- To create favorable conditions for solar manufacturing capability, particularly solar thermal for indigenous production and market leadership.

National Mission For Enhanced Energy Efficiency (NMEEE)

- NMEEE seeks to strengthen the market for energy efficiency by creating conducive regulatory and policy regime.
- NMEEE has been envisaged to foster innovative and sustainable business models to the energy efficiency sector.
- The NMEEE seeks to create and sustain markets for energy efficiency in the entire country which will benefit the country and the consumers".

National Mission On Sustainable Habitat

- "National Mission on Sustainable Habitat" seeks to promote sustainability of habitats through improvements in energy efficiency in buildings, urban planning, improved management of solid and liquid waste, modal shift towards public transport and conservation through appropriate changes in legal and regulatory framework.
- It also seeks to improve ability of habitats to adapt to climate change by improving resilience of infrastructure, community based disaster management and measures for improving advance warning systems for extreme weather events.

National Water Mission (NWM) Mission

- Ensuring integrated water resource management for conservation of water, minimization of wastage and equitable distribution both across and within states.
- Developing a framework for optimum water use through increase in water use efficiency by 20% through regulatory mechanisms with differential entitlements and pricing, taking the National Water Policy (NWP) into consideration.
- Ensuring that a considerable share of water needs of urban areas is met through recycling of waste water.
- Meeting water requirements of coastal cities through the adoption of new and appropriate technologies such as low-temperature desalination technologies allowing use of ocean water.
- Revisiting NWP to ensure basin-level management strategies to deal with variability in rainfall and river flows due to climate change.
- Developing new regulatory structures to optimize efficiency of existing irrigation systems.

National Mission For Sustaining The Himalayan Ecosystem (NMSHE)

Primary objectives

- Develop a sustainable National capacity to continuously assess the health status of the Himalayan Ecosystem

- Assist States in the Indian Himalayan Region with their implementation of actions selected for sustainable development.

National Mission For A Green India

Mission Objectives

- Increased forest/tree cover on 5 million hectares (ha) of forest/non-forest lands and improved quality of forest cover on another 5 million ha of non-forest/forest lands (a total of 10 million ha)
- Improved ecosystem services including biodiversity, hydrological services, and carbon sequestration from the 10 million ha of forest/non-forest lands mentioned above
- Increased forest-based livelihood income of about 3 million households, living in and around the forests
- Enhanced annual CO₂ sequestration by 50 to 60 million tons in the year 2020.

National Mission on Seabuckthorn

Seabuckthorn plant is popularly known as **Leh berries**.

The MoEF and DRDO have launched a major national initiative for seabuckthorn cultivation in the **high-altitude, cold desert ecosystems**.

The initiative is one of many conservation measures for fragile high-altitude ecosystems. Seabuckthorn, also called the “Wonder plant” and “Ladakh gold”.

Uses

- It has multi-purpose **medicinal** and **nutritional properties**, and also helps in **soil conservation and nitrogen fixation**.
- Hardy, drought-resistant and tolerant to extreme temperatures from – 43° C to + 40° C, the plant has an extensive root system which can fix atmospheric nitrogen, making it ideal for controlling soil erosion and preventing desertification.
- The initiative is a part of Sub-Mission on Cold Desert Ecosystems under the Green India Mission — which is a part of the National Action Plan on Climate Change.

Government of India encourages the cultivation of 'sea buckthorn'. What is the importance of this plant?

1. It helps in controlling soil erosion and in preventing desertification.
2. It is a rich source of biodiesel.
3. It has nutritional value and is well-adapted to live in cold areas of high altitudes.
4. Its timber is of great commercial value.

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Which of the statements given above is /are correct ?

1. 2, 3 and 4 only
2. **1 and 3 only**
3. 1, 2, 3 and 4
4. 1 only

National Mission For Sustainable Agriculture (NMSA)

The NMSA has identified key dimensions for

- Adaptation and mitigation:
- Improved Crop Seeds, Livestock and Fish Culture
- Water Efficiency
- Pest Management
- Improved Farm Practices
- Nutrient Management
- Agricultural Insurance
- Credit Support
- Markets
- Access to Information
- Livelihood Diversification

The National Mission on Strategic Knowledge for Climate Change (NMSKCC)

Mission Objectives

- Formation of knowledge networks among the existing knowledge institutions engaged in research and development relating to climate science.
- Establishment of global technology watch groups with institutional capacities to carry out research on risk minimized technology selection for developmental choices
- Development of national capacity for modeling the regional impact of climate change on

different ecological zones within the country for different seasons and living standards

- Establishing research networks and encouraging research in the areas of climate change impacts on important socio-economic sectors like agriculture, health, natural ecosystems, biodiversity, coastal zones, etc.

National Bio-Energy Mission

- The government is preparing a national bio-energy mission to boost power generation from biomass, a renewable energy source abundantly available in India.
- The national mission will aim at improving energy efficiency in traditional biomass consuming industries, seek to develop a bio-energy city project and provide logistics support to biomass processing units.
- It will also propose a GIS-based National Biomass Resource Atlas to map potential biomass regions in the country.
- According to estimates, biomass from agro and agro-industrial residue can potentially generate 25,000 MW of power in India.

Indian Network On Climate Change Assessment

- Launched by MoEF in an effort to promote domestic research on climate change.
- Reports prepared by the INCCA will form a part of India's National Communication (Nat Com) to the United Nations Framework Convention on Climate Change (UNFCCC).

National Communication (NATCOM)

- In pursuance of the implementation of the provisions of UNFCCC, India's Initial National Communication (NATCOM) has been initiated in 2002 funded by the Global Environment Facility.

Pollution Environment & Biodiversity Conservation Laws

National Environmental Legislation

- Our constitution, originally, **did not** contain any direct provision regarding the protection of natural environment.
- However, after the **United Nations Conference on Human Environment**, held

in **Stockholm** in 1972, Indian constitution was amended to include protection of the environment as a constitutional mandate.

- Environment related legislation came very late in 1972 with **Wild Life Protection Act 1971**.
- The **forty second amendment** Clause (g) to **Article 51A** of the Indian constitution made it a **fundamental duty** to protect and improve the natural environment.

Clause (g) to Article 51A of the Indian constitution states "It shall be the duty of every citizen of India to protect and improve the natural environment including forests, lakes, rivers and wild life and have compassion for living creatures."

- There is a directive, given to the State as one of the **Directive Principles** of State Policy regarding the protection and improvement of the environment.

Article 48A states "The State shall endeavor to protect and improve the environment and to safeguard the forests and wildlife of the country".

- The Department of Environment was established in India in 1980 to ensure a healthy environment for the country. This later became the Ministry of Environment and Forests (MoEF) in **1985**.
- The **Environment Protection Act of 1986 (EPA)** came into force soon after the Bhopal Gas Tragedy and is considered umbrella legislation as it fills many lacunae in the existing legislations.

Pollution Related Acts

- Among all the components of the environment air and water are necessary to fulfill the basic survival needs of all organisms. So, to protect them from degradation the following acts have been passed.

1. **Water Acts**
2. **Air Acts**
3. **Environment Act**

- A few important legislations of each category with brief description are given below:

The Water (Prevention and Control of Pollution) Act of 1974 and Amendment, 1988

- The main objective of this act is to provide prevention and control of water pollution.

Some important provisions of this Act are given below:

- The Act vests regulatory authority in **State Pollution Control Boards** to establish and enforce effluent standards for factories.
- A **Central Pollution Control Board** performs the same functions for Union Territories and formulate policies and **coordinates** activities of different State Boards.
- The Act grants power to SPCB and CPCB to test equipment and to take the sample for the purpose of analysis.
- Prior to its amendment in 1988, enforcement under the Act was achieved through criminal prosecutions initiated by the Boards.
- The 1988 amendment act empowered SPCB and CPCB to close a defaulting industrial plant.

The Water (Prevention and Control of Pollution) Cess Act of 1977

- The Water Cess Act was passed to generate financial resources to meet expenses of the Central and State Pollution Boards.
- The Act creates **economic incentives** for pollution control and requires local authorities and certain designated industries to pay a cess (tax) for water effluent discharge.
- The **Central Government**, after deducting the expenses of collection, pays the central board and the states such sums, as it seems necessary.
- To encourage capital investment in pollution control, the Act gives a polluter a 70% rebate of the applicable cess upon installing effluent treatment equipment.

The Air (Prevention and Control of Pollution) Act of 1981 and amendment, 1987

- To implement the decisions taken at the **United Nations Conference on the Human Environment** held at Stockholm in

June 1972, Parliament enacted the nationwide Air Act.

- The main objectives of this Act are to improve the quality of air and to prevent, control and abate air pollution in the country.

Important provisions of this Act are given below: | Page

- The Air Act's framework is **similar** to that of the Water Act of 1974. 250
- The Air Act expanded the authority of the central and state boards established under the Water Act, to include air pollution control.
- States not having water pollution boards were required to set up air pollution boards.
- Under the Air Act, all industries operating within designated air pollution control areas must obtain a "consent" (**permit**) from the **State Boards**.
- The states are required to prescribe emission standards for industry and automobiles after consulting the central board and noting its ambient air quality standards.
- The Act grants power to SPCB and to test equipment and to take the sample for the purpose of analysis from any chimney, fly ash or dust or any other.
- Prior to its amendment in 1988, enforcement under the Act was achieved through criminal prosecutions initiated by the Boards.
- The 1988 amendment act empowered SPCB and CPCB to close a defaulting industrial plant.
- Notably, the 1987 amendment introduced a citizen's suit provision into the Air Act and extended the Act to include **noise pollution**.

Environment and Biodiversity Related Acts

- The most important legislation in this category is The Environment (Protection) Act of 1986.
- Through this Act **Central Government** gets full power for the purpose of protecting and improving the quality of the environment.

Environment (Protection) Act of 1986

- In the wake of the **Bhopal tragedy**, the government of India enacted the Environment (Protection) Act of 1986.
- The purpose of the Act is to implement the decisions of the **United Nations Conference**

on the Human Environment of 1972, in so far as they relate to the protection and improvement of the human environment and the prevention of hazards to human beings, other living creatures, plants and property.

- The Act is an “umbrella” for legislations designed to provide a framework for Central Government, coordination of the activities of various central and state authorities established under previous Acts, such as the Water Act and the Air Act.
- In this Act, main emphasis is given to “Environment”, defined to include water, air and land and the inter-relationships which exist among water, air and land and human beings and other living creatures, plants, micro-organisms and property.
- “Environmental pollution” is the presence of pollutant, defined as any solid, liquid or gaseous substance present in such a concentration as may be or may tend to be injurious to the environment.
- “Hazardous substances” include any substance or preparation, which may cause harm to human beings, other living creatures, plants, microorganisms, property or the environment.

The main provisions of this Act are given below

- The Act empowers the **center** to “take all such measures as it deems necessary”.
- By virtue of this Act, **Central Government** has armed itself with considerable powers which include,
 1. coordination of action by state,
 2. planning and execution of nationwide programmes,
 3. laying down environmental quality standards, especially those governing emission or discharge of environmental pollutants,
 4. placing restriction on the location of industries and so on.
 5. authority to issue direct orders, included orders to close, prohibit or regulate any industry.
 6. power of entry for examination, testing of equipment and other purposes and power to analyze the sample of air, water, soil or any other substance from any place.

- The Act explicitly prohibits discharges of environmental pollutants in excess of prescribed regulatory standards.
- There is also a specific prohibition against handling hazardous substances except those in compliance with regulatory procedures and standards.
- The Act provides provision for penalties. For each failure or contravention the punishment included a prison term up to five years or fine up to Rs. 1 lakh, or both.
- The Act imposed an additional fine of up to Rs. 5,000 for every day of continuing violation.
- If a failure or contravention occurs for more than one year, offender may be punished with imprisonment which may be extended to seven years.
- Section 19 provides that any person, in addition to authorized government officials, may file a complaint with a court alleging an offence under the Act.
- This “Citizens’ Suit” provision requires that the person has to give notice of not less than 60 days of the alleged offence of pollution to the Central Government.

The Genetic Engineering Appraisal Committee (GEAC) is the apex body constituted in the Ministry of Environment and Forests under 'Rules for Manufacture, Use, Import, Export and Storage of Hazardous Microorganisms/Genetically Engineered Organisms or Cells 1989', under the Environment Protection Act, 1986. The Rules of 1989 also define five competent authorities i.e. the Institutional Biosafety Committees (IBSC), Review Committee of Genetic Manipulation (RCGM), Genetic Engineering Approval Committee (GEAC), State Biotechnology Coordination Committee (SBCC) and District Level Committee (DLC) for handling of various aspects of the rules

Genetic Engineering Approval Committee (GEAC)

1. **Biosafety concerns** have led to the development of regulatory regime in India.
2. The **MoEFCC** has notified the Rules for Manufacture, Use/Import/ Export & Storage Of Hazardous Micro Organisms/Genetically Engineered Organisms or Cells, 1989 [‘Rules 1989’]
3. Aim of ‘Rules 1989’ is to protect environment, nature and health in connection with

application of gene technology and micro-organisms.

4. These rules cover areas of research as well as large scale applications of GMOs and their products including experimental field trials and seed production.
5. The Rules 1989 also define the competent authorities and composition of such authorities for handling of various aspects of the Rules.

Presently there are six committees

1. **Recombinant DNA Advisory Committee (RDAC):** The functions are of an advisory nature. It recommends safety regulations for India in recombinant research, use and applications.
2. **Review Committee on Genetic Manipulation (RCGM)** established under the Department of Biotechnology, **ministry of Science** and technology is to monitor the safety related aspects in respect of on-going research projects.
3. **Genetic Engineering Appraisal Committee (GEAC)**
4. **State Biotechnology Coordination Committee (SBCC's)** have a major role in monitoring. It also has powers to inspect, investigate and take punitive action in case or violations of statutory provisions.
5. **District Level Committees (DLCs)** have a major role in monitoring the safety regulations in installations engaged in the use of genetically modified organisms/hazardous microorganisms and its applications in the environment.
6. **Institutional Biosafety Committee (IBSC)** is established under the institution engaged in GMO research to oversee such research and to interface with the RCGM in regulating it.

Genetic Engineering Appraisal Committee (GEAC)

- The Genetic Engineering Appraisal Committee (GEAC) is the apex body constituted in the **Ministry of Environment and Forests** under 'Rules for Manufacture, Use, Import, Export and Storage of Hazardous Microorganisms/Genetically Engineered Organisms or Cells 1989', under the **Environment Protection Act, 1986**.

The Genetic Engineering Appraisal Committee is constituted under the

- a. Food Safety and Standards Act, 2006
- b. Geographical Indications of Goods (Registration and Protection) Act, 1999
- c. Environment (Protection) Act, 1986
- d. Wildlife (Protection) Act, 1972

Answer: c)

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The Ozone Depleting Substances Rules

- The rules are framed under the jurisdiction of **Environment (Protection) Act**.
- These Rules set the deadlines for phasing out of various ODSs, besides regulating production, trade import and export of ODSs and the product containing ODS.
- These Rules prohibit the use of CFCs in manufacturing various products beyond 1st January 2003 except in metered dose inhaler and for other medical purposes.
- Similarly, use of halons is prohibited after 1st January 2001 except for essential use.
- Other ODSs such as **carbon tetrachloride** and **methylchloroform** and CFC for metered dose inhalers can be used upto 1st January 2010.
- Further, the use of **methyl bromide** has been allowed upto 1st January 2015.
- Since **HCFCs** are used as interim substitute to replace CFC, these are allowed up to 1st January 2040.

National Ganga River Basin Authority (NGRBA)

- National Ganga River Basin Authority (NGRBA) is a financing, planning, implementing, monitoring and coordinating authority for the Ganges River, functioning under the **Ministry of Water Resources**.
- The mission of the organization is to safeguard the drainage basin which feeds water into the Ganges by protecting it from pollution or overuse.
- In 2014, the NGRBA has been transferred from the Ministry of Environment and Forests to the **Ministry of Water Resources, River Development & Ganga Rejuvenation**.
- It was established by the Central Government of India, in 2009 under Section 3(3) of the **Environment Protection Act, 1986**,

which also declared Ganges as the '**National River**' of India.

Composition of NGRBA

- The **Prime Minister** the chair of the Authority.

Members belonging to the government sector are as follows:

- Prime Minister of India
- Minister of Environment and Forests (Union Minister)
- Minister of Finance
- Minister of Urban Development
- Minister of Water Resources
- Minister of Power
- Minister of Sciences and Technology
- Chief Ministers of **Uttarakhand, Uttar Pradesh, Bihar, Jharkhand and West Bengal**
- Ministry Of Environment and Forests (state minister)
- Ministry Of Environment and Forests, secretary

Wild Life (Protection) Act of 1972 and Amendment, 1982

- In 1972, Parliament enacted the Wild Life Act (Protection) Act.
- The Wild Life Act provides for
 1. state wildlife advisory boards,
 2. regulations for hunting wild animals and birds,
 3. establishment of sanctuaries and national parks,
 4. regulations for trade in wild animals, animal products and trophies, and
 5. judicially imposed penalties for violating the Act.
- Harming **endangered** species listed in **Schedule 1** of the Act is prohibited throughout India.
- Hunting species, like those requiring special protection (Schedule II), big game (Schedule III), and small game (Schedule IV), is regulated through licensing.
- A few species classified as **vermin (Schedule V)**, may be hunted without restrictions.
- Wildlife wardens and their staff administer the act.

- An amendment to the Act in 1982, introduced a provision permitting the capture and transportation of wild animals for the scientific management of animal population.

Forest (Conservation) Act of 1980

- First Forest Act was enacted in 1927.
- Alarmed at India's rapid deforestation and resulting environmental degradation, Centre Government enacted the Forest (Conservation) Act in 1980.
- It was enacted to consolidate the law related to forest, the transit of forest produce and the duty livable on timber and other forest produce.
- Forest officers and their staff administer the Forest Act.
- Under the provisions of this Act, prior approval of the Central Government is required for diversion of forestlands for non-forest purposes.
- An Advisory Committee constituted under the Act advises the Centre on these approvals.
- The Act deals with the four categories of the forests, namely reserved forests, village forests, protected forests and private forests.

Reserved forest

- A state may declare forestlands or waste lands as reserved forest and may sell the produce from these forests.
- Any unauthorized felling of trees quarrying, grazing and hunting in reserved forests is punishable with a fine or imprisonment, or both

Village forests

- Reserved forests assigned to a village community are called village forests.

Protected forests

- The state governments are empowered to designate protected forests and may prohibit the felling of trees, quarrying and the removal of forest produce from these forests.
- The preservation of protected forests is enforces through rules, licenses and criminal prosecutions.

Biodiversity Act 2000

- India's richness in biological resources and indigenous knowledge relating to them is well recognized.
- The legislation aims at **regulating access to biological resources** so as to ensure **equitable sharing of benefits** arising from their use.
- The Biological Diversity Bill was introduced in the Parliament in 2000 and was passed in 2002.

Salient features of the biodiversity legislation

- The main intent of this legislation is to protect India's rich biodiversity and associated knowledge against their use by foreign individuals and organizations without sharing the benefits arising out of such use, and to **check biopiracy**.
- This bill seeks to check biopiracy, protect biological diversity and local growers through a three-tier structure of central and state boards and local committees.
- The Act provides for setting up of a **National Biodiversity Authority (NBA)**, **State Biodiversity Boards (SBBs)** and **Biodiversity Management Committees (BMCs)** in local bodies. The NBA will enjoy the power of a **civil court**.
- BMCs promote conservation, sustainable use and documentation of biodiversity.
- NBA and SBB are required to consult BMCs in decisions relating to use of biological resources.
- All foreign nationals or organizations require prior approval of NBA for obtaining biological resources and associated knowledge for any use.
- Indian individuals/entities require approval of NBA for transferring results of research with respect to any biological resources to foreign nationals/organizations.
- Collaborative research projects and exchange of knowledge and resources are exempted provided they are drawn as per the policy guidelines of the Central Government.
- However, Indian citizens/entities/local people including vaid and hakims have free access to use biological resources within the country for their own use, medicinal purposes and research purposes.
- While granting approvals, NBA will impose terms and conditions to secure equitable sharing of benefits.

- Before applying for any form of IPRs (Intellectual Property Rights) in or outside India for an invention based on research on a biological resource obtained from India, prior approval of NBA will be required.
- The monetary benefits, fees, royalties as a result of approvals by NBA will be deposited in **National Biodiversity Fund**.
- NBF will be used for conservation and development of areas from where resource has been accessed, in consultation with the local self-government concerned.
- There is provision for notifying **National Heritage Sites** important from standpoint of biodiversity by State Governments in consultation with local self-government.
- There also exists provision for notifying items, and areas for exemption provided such exclusion does not violate other provisions. This is to exempt normally traded commodities so as not to adversely affect trade.

How does National Biodiversity Authority (NBA) help in protecting the Indian agriculture?

1. NBA checks the biopiracy and protects the indigenous and traditional genetic resources.
2. NBA directly monitors and supervises the scientific research on genetic modification of crop plants.
3. Application for intellectual Property Rights related to genetic / biological resources cannot be made without the approval of NBA.

Which of the statements given above is/are correct?

- a. 1 only
- b. 2 and 3 only
- c. 1 and 3 only
- d. 1, 2 and 3

The top biotech regulator in India for Genetically Modified Organisms is **Genetic Engineering Appraisal Committee (GEAC)**.

Answer: c) 1 and 3 only

The Scheduled Tribes And Other Traditional Forest Dwellers (Recognition Of Forest Rights) Act, 2006

- Forest Rights Act, 2006 provides for the restitution of deprived forest rights across India.
- The Act provides scope of integrating conservation and livelihood rights of the people.

FRA is tool

- To empower and strengthen the local self-governance
- To address the livelihood security of the people
- To address the issues of Conservation and management of the Natural Resources and conservation governance of India.

For the first time Forest Rights Act recognizes and secures

- Community Rights in addition to their individual rights
- Right to protect, regenerate or conserve or manage any community forest resource which the communities have been traditionally protecting and conserving for sustainable use.
- Right to intellectual property and traditional knowledge related to biodiversity and cultural diversity
- Rights of displaced communities & Rights over developmental activities

Salient Features

- Nodal Agency for the implementation is Ministry of Tribal Affairs (MoTA).
- This Act is applicable for Tribal and Other Traditional Forest Dwelling Communities.
- The Act provides for recognition of forest rights of **other traditional forest dwellers** provided they have for at least three generations prior to 13.12.2005 primarily resided in and have depended on the forests for bonafide livelihood needs.
- The maximum limit of the recognizing rights on forest land is **4 ha**.
- **National Parks** and **Sanctuaries** have been included along with Reserve Forest, Protected Forests for the recognition of Rights.
- The Act recognizes the right of ownership access to collect, use, and dispose of minor forest produce by tribals.
- Minor forest produce includes all non-timber forest produce of plant origin, including bamboo, brush wood, stumps, cane, tussar,

cocoons, honey, wax, lac, leaves, medicinal plants and herbs, roots, tubers and the like.

- The rights conferred under the Act shall be heritable but not alienable or transferable.
- As per the Act, the **Gram Sabha** has been designated as the competent authority for initiating the process of determining the nature and extent of individual or community forest rights.

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National Green Tribunal Act, 2010

- Act of the Parliament of India which enables creation of **NGT** to handle the expeditious disposal of the cases pertaining to environmental issues.
- It was enacted under India's constitutional provision of **Article 21**, which assures the citizens of India the **right to a healthy environment**.
- The specialized architecture of the NGT will facilitate **fast track resolution of environmental cases** and provide a boost to the implementation of many sustainable development measures.
- NGT is mandated to dispose the cases within **six months** of their respective appeals.

Origin

- During the Rio de Janeiro summit of United Nations Conference on Environment and Development in June 1992, India vowed the participating states to provide **judicial and administrative remedies** for the victims of the pollutants and other environmental damage.

Members

- The sanctioned strength of the tribunal is currently 10 expert members and 10 judicial members although the act allows for up to 20 of each.
- The Chairman of the tribunal who is the administrative head of the tribunal also serves as a judicial member.
- Every bench of tribunal must consist of at least one expert member and one judicial member.
- The Chairman of the tribunal is required to be a **serving or retired Chief Justice of a High Court or a judge of the Supreme Court of India**.

Jurisdiction

- The Tribunal has Original Jurisdiction on matters of “substantial question relating to environment” (i.e. a community at large is affected, damage to public health at broader level) & “damage to environment due to specific activity” (such as pollution).
- The term “substantial” is not clearly defined in the act.

Notable orders

- Ban on decade old Diesel vehicles at Delhi NCR
- NGT cleared three-day World Culture Festival (Art of Living) on the floodplains of the Yamuna in March but imposed a fine of Rs 5 crore as an interim compensation.

The National Green Tribunal Act, 2010 was enacted in consonance with which of the following provisions of the Constitution of India?

1. Right to healthy environment, construed as a part of part of Right to life under Article 21.
2. Provision of grants for raising the level of administration in the Scheduled Areas for the welfare of Scheduled Tribes under Article 275(1)

Which of the statements given above is/are correct ?

- a. 1 only
- b. 2 only
- c. Both 1 and 2
- d. Neither 1 nor 2

Answer: a)

Acts for Protecting Coastal Environment and Wetlands

Coastal Regulation Zone (CRZ)

- The coastal stretches of seas, bays, estuaries, creeks, rivers and back waters which are influenced by tidal action are declared "Coastal Regulation Zone" (CRZ) in 1991.
- India has created institutional mechanisms such as **National Coastal Zone Management Authority (NCZMA)** and State Coastal Zone

Management Authority (SCZMA) for enforcement and monitoring of the CRZ Notification.

- These authorities have been delegated powers under Section 5 of the **Environmental (Protection) Act, 1986** to take various measures for protecting and improving the quality of the coastal environment and preventing, abating and controlling environmental pollution in coastal areas.

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Classification Criteria and Regulatory Norms

- The coastal regulation zone has been classified for the purpose of regulation of the permitted activities.

CRZ-I:

- Ecological sensitive area and the area between High Tide Line (HTL) and Low Tide Line (LTL).
- **No new construction** is permitted **except** for a few specified most essential activities like support activities for Atomic Energy Plants and Defense requirements, facilities required for disposal of treated effluents and other port related water front activities.

CRZ-II:

- The area that have been developed up to or close to the shore line which includes the designated urban areas that are substantially built up.
- Buildings permitted only on the landward side of the existing authorized structures as defined in the notification.

CRZ-III:

- The areas that are relatively undisturbed and those which do not belong to either CRZ-I or CRZ-II which includes mainly the rural area and those not substantially built up within designated urban areas.
- The area up to 200 meters from HTL is earmarked as "No Development Zone".
- No construction is permitted within this zone except for repairs to the existing authorized structures.
- Development of vacant plots between 200 and 500 meters of HTL is permitted in CRZ III for the purpose of construction of dwelling units

and **hotels/beach resorts** subject to certain conditions.

CRZ-IV

- The activities impugning on the sea and tidal influenced water bodies will be regulated except for traditional fishing and related activities undertaken by local communities.
- No untreated sewage, effluents, pollution from oil drilling shall be let off or dumped.

Wetlands (Conservation And Management) Rules 2010

- MoEF has notified the rules in order to ensure that there is no further degradation of wetlands.
- The rules specify activities which are harmful to wetlands such as industrialization, construction, dumping of untreated waste and reclamation and prohibit these activities in the wetlands.
- Other activities such as harvesting and dredging may be carried out in the wetlands but only with prior permission from the concerned authorities.
- Under the Rules, wetlands have been classified for better management and easier identification.
- **Central Wetland Regulatory Authority** has been set up to ensure proper implementation of the Rules.

Animal Welfare Board of India

- **Statutory advisory body** advising the Government of India on animal welfare laws, and promotes animal welfare in the country of India.
- It works to ensure that animal welfare laws in the country are followed; provides grants to Animal Welfare Organizations; and considers itself "the face of the animal welfare movement in the country."
- It was established in 1960 under Section 4 of **The Prevention of Cruelty to Animals Act, 1960**.
- Well-known humanitarian **Rukmini Devi Arundale** was instrumental in setting up the board.
- The subject of Prevention of Cruelty to Animals is under **MoEF**.

Functions

- The board is highly concerned about "abuse of animals in research" and "cruelty involved when animals were used in entertainment".
- The Board oversees Animal Welfare Organizations (AWOs) by granting recognition to them if they meet its guidelines.
- The Board provides financial assistance to recognized Animal Welfare Organizations (AWOs)
- The Board suggests changes to laws and rules about animal welfare issues.
- The Board issues publications to raise awareness of various animal welfare issues.

Consider the following statements:

- Animal Welfare Board of India is established under the Environment (Protection) Act, 1986.
- National Tiger Conservation Authority is a statutory body.
- National Ganga River Basin Authority is chaired by the Prime Minister.

Which of the statements given above is/ are correct?

1. 1 only
2. 2 and 3 only
3. 2 only
4. 1, 2 and 3

Animal Welfare Board of India is established under Prevention of Cruelty to Animals Act.

Answer: b) 2 and 3 only

Prominent NGOs

Bombay Natural History Society

- The Bombay Natural History Society, founded in 1883, is one of the largest non-governmental organizations in India engaged in **conservation and biodiversity research**.
- It supports many research efforts through grants and publishes the **Journal of the Bombay Natural History Society**.
- It organizes and conducts nature trails and camps for the general public.
- Many prominent naturalists, including the ornithologists Salim Ali and S. Dillon Ripley, have been associated with it.

With reference to Bombay Natural History Society (BNHS), consider the following statements:

1. It is an autonomous organization under the Ministry of Environment and Forests.
2. It strives to conserve nature through action-based research, education and public awareness.
3. It organizes and conducts nature trails and camps for the general public.

Which of the statements given above is/are correct?

- a. 1 and 3 only
- b. 2 only
- c. **2 and 3 only**
- d. 1, 2 and 3

Indian government has also started some conservation projects for individual endangered species like Hungal (1970), Lion (1972), Tiger (1973), Crocodiles (1974), Brown-antlered Deer (1981) and Elephant (1991-92). They are given in the next post.

Project Tiger - Elephant - Indian Rhino - Snow Leopard

Project Tiger – Tiger Conservation

- Indian tiger population at the end of the 20th century was estimated at 20,000 to 40,000 individuals.
- The first country-wide tiger census conducted in 1972 estimated the population to comprise a little more than 1,800 individuals, an alarming reduction in tiger population.
- In 1973, Project Tiger was launched in the Palamau Tiger Reserve, and various tiger reserves were created in the country based on a 'core-buffer' strategy.
- India has more than 80 national parks and 441 Sanctuaries of which some have been declared as Tiger reserves.
- Tiger reserves are governed by the Project Tiger (1973).
- It is a Centrally Sponsored Scheme of the **Ministry of Environment and Forests**.
- It is administered by the **National Tiger Conservation Authority**.
- Aim: Protect tigers from extinction by ensuring a viable population in their natural habitats.

- Government has set up a **Tiger Protection Force** under PT to combat poachers.
- PT funds relocation of villagers to minimize human-tiger conflicts.

Core and Buffer zones

The Tiger Reserves are constituted on a 'core-buffer strategy'.

Core Zone

- The core area is kept **free of biotic disturbances** and forestry operations, where collection of minor forest produce, grazing, human disturbances are not allowed within.
- These areas are required to be kept for the purposes of tiger conservation, without affecting the rights of the Scheduled Tribes or such other forest dwellers.
- These areas are notified by the **State Government** in consultation with an Expert Committee (constituted for that purpose).

Buffer Zone

- The Act defines buffer zone as the area peripheral to the critical tiger habitat or core area providing **supplementary habitat** for dispersing tigers, besides offering scope for **co-existence of human activity (tribals)**.
- The limits of such areas are determined with the concerned **Gram Sabha** and an Expert Committee constituted for the purpose.

Tiger Task Force

- The implementation of Project Tiger over the years has highlighted the need for a statutory authority with legal backing to ensure tiger conservation.
- On the basis of the recommendations of **National Board for Wild Life**, a Task Force was set up to look into the problems of tiger conservation in the country.
- The recommendations of the **Task Force** include strengthening of Project Tiger by giving it statutory and administrative powers.

National Board for Wild Life ==> Tiger Task Force ==> National Tiger Conservation Authority

National Tiger Conservation Authority (NTCA)

- The National Tiger Conservation Authority was established in December 2005 following a recommendations of the Tiger Task Force.
- National Tiger Conservation Authority administers Project Tiger.
- Administration of the tiger reserves will be in accordance with guidelines of NTCA.
- Tiger reserves in India are administered by field directors as mandated by NTCA.
- No alteration in the boundaries of a tiger reserve shall be made except on a recommendation of the NTCA and the approval of the National Board for Wild Life.
- No State Government shall de-notify a tiger reserve, except in public interest with the approval of the NTCA and the approval of the National Board for Wild Life.

The Wild Life (Protection) Amendment Act, 2006

- The Act provides for creating
 1. National Tiger Conservation Authority and
 2. Tiger and Other Endangered Species Crime Control Bureau (Wildlife Crime Control Bureau).

NTCA members

- The Wildlife Protection Amendment Act, 2006 provides for the constitution of the National Tiger Conservation Authority.
- NTCA was set up under the Chairmanship of the Minister for Environment and Forests.
- The Authority will have
 1. eight experts having qualifications in wildlife conservation and welfare tribals,
 2. 3 MPs,
 3. The Inspector General of Forests, in charge of project Tiger, will be ex-officio Member Secretary
 4. Others

Functions of NTCA

- The Authority lays down standards, guidelines for tiger conservation in the Tiger Reserves, National Parks and Sanctuaries.

- It would provide information on protection measures including future plan for conservation.
- The Authority would facilitate and support tiger reserve management in the States through eco-development and people's participation.
- The Tiger Conservation Authority would be required to prepare an Annual Report, which would be laid in the Parliament along with the Audit Report.
- State level Steering Committees will be set up in the Tiger States under the **Chairmanship of respective Chief Ministers**. This has been done with a view for ensuring coordination, monitoring and protection of tigers in the States.
- A provision has been made for the State Governments to prepare a **Tiger Conservation Plan**.
- Provision will be made for the States to establish a **Tiger Conservation Foundation**, based on the good practices emanating from some tiger reserves.

Steps Taken by the Government

Legal Steps

- Amendment of the Wild Life (Protection) Act, 1972 to Wild Life (Protection) Act, 2006 for providing enabling provisions towards constituting the National Tiger Conservation Authority and the Tiger and Other Endangered Species Crime Control Bureau.
- Enhancement of punishment in cases of offence relating to a tiger reserve or its core area.

Administrative Steps

- Strengthening of anti poaching activities, including special strategy for monsoon patrolling.
- State level Steering Committees under the Chairmanship of Chief Ministers and establishment of Tiger Conservation Foundation.
- Creation of Special Tiger Protection Force (STPF) [Budget 2008].

Financial Steps

- Financial and technical help is provided to the States under various Centrally Sponsored

Schemes, viz. Project Tiger and Integrated Development of Wildlife Habitats.

International Cooperation

- India has a bilateral understanding with Nepal on controlling trans-boundary illegal trade in wildlife.
- India has signed a protocol on tiger conservation with China.
- India has signed a with Bangladesh for conservation of the Royal Bengal Tiger.

- A sub-group on tiger/leopard conservation has been constituted for cooperation with the Russian Federation.
- A Global Tiger Forum of Tiger Range Countries has been created for addressing international issues related to tiger conservation.
- India is a party to **CITES**. CITES's landmark decision states that 'tigers should not be bred for trade in their parts and derivatives'.

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Wiki List of Tiger Reserves in India

	State	Tiger Reserve	Est. No. of Tigers	Total Area (km ²)
1	Andhra Pradesh	Nagarjunsagar-Srisailem Tiger Reserve	29	3568
2	Arunachal Pradesh	Namdapha Tiger Reserve	37	1985
3		Pakhui Tiger Reserve	36	862
4	Assam	Manas Tiger Reserve	35	2840
5		Nameri Tiger Reserve	37	344
6		Kaziranga Tiger Reserve	42	859
7	Bihar	Valmiki Tiger Reserve	42	840
8	Chhattisgarh	Udanti and Sitanadi Tiger Reserve	54	1580
9		Achanakmar Tiger Reserve	42	963
10		Indravati Tiger Reserve	39	2799
11	Jharkhand	Palamau Tiger Reserve	19	1026
12	Karnataka	Bandipur Tiger Reserve	89	866
13	Karnataka	Bhadra Tiger Reserve	36	492
14		Dandeli-Anshi Tiger Reserve	41	875
15		Nagarhole Tiger Reserve	47	643
16		Biligiri Rangaswamy Temple Tiger Reserve	40	540
17	Kerala	Periyar Tiger Reserve	77	925
18		Parambikulam Tiger Reserve	58	391
19	Madhya Pradesh	Kanha Tiger Reserve	87	1945
20		Pench Tiger Reserve	64	758
21		Bandhavgarh Tiger Reserve	79	1162
22		Panna Tiger Reserve	18	542
23		Bori-Satpura Tiger Reserve	44	1486
24		Sanjay Dubri Tiger Reserve	56	831
25	Maharashtra	Melghat Tiger Reserve	45	1677
26		Tadoba-Andhari Tiger Reserve	100	620
27		Bor Tiger Reserve	7-8	138
28		Pench Tiger Reserve	35	257
29		Sahyadri Tiger Reserve	46	569
30		Nawegaon-Nagzira Tiger Reserve		653
31	Mizoram	Dampa Tiger Reserve	48	500
32	Odisha	Simlipal Tiger Reserve	64	2750
33		Satkosia Tiger Reserve[15]	31	523
34	Rajasthan	Ranthambhore Tiger Reserve	68	1334
35		Sariska Tiger Reserve	5	866
36		Mukundara Hills Tiger Reserve		417

37	Tamil Nadu	Kalakad-Mundathurai Tiger Reserve	93	800
38		Mudumalai Tiger Reserve	64	321
39		Sathyamangalam Tiger Reserve	47	524
40		Anamalai Tiger Reserve	49	1019
41	Telangana	Kawal Tiger Reserve	42	893
42	Uttar Pradesh	Dudhwa Tiger Reserve	141	811
43		Amangarh Tiger Reserve		80
44		Pilibhit Tiger Reserve	30	1074
45	Uttarakhand	Jim Corbett Tiger Reserve	160	80
46	West Bengal	Buxa Tiger Reserve	29	759
47		Sunderbans Tiger Reserve	89	2585

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Titbit: There are eight subspecies of tiger: among the eight, at present five subspecies are present in the wild. They are: Bengal, South China, Indochinese, Sumatran, and Siberian. Three subspecies of tiger—Caspian, Bali, and Javan—are extinct.

Reasons for falling number of tigers

- **Pressure on habitat, Habitat fragmentation and Habitat destruction:** Caused due to large-scale development projects such as dams, industry, mines, railway lines etc.
- **Incessant poaching:** Tigers are killed so their body parts can be used for Traditional Chinese Medicine.
- **Invasive species:** Destroy the local producers. This has a cascading effect on the food chain. Tigers are the worst hit as they are at the end of the food chain. [Tigers represent an “Umbrella Species” that indicate the health of the ecosystem.]

Problems involved with Project Tiger

- The Forest Rights Act passed by the Indian government in 2006 recognizes the rights of some forest dwelling communities in forest areas. This has led to controversy over implications of such recognition for tiger conservation.
- The project overlooks the role of abuse of power by authorities in the tiger crisis.

Reasons for slightly increased tiger population recently

- Wireless communication systems and outstation patrol camps have been developed within the tiger reserves, due to which poaching has declined considerably.
- Fire protection is effectively done by suitable preventive and control measures.
- Voluntary Village relocation has been done in many reserves.
- Livestock grazing has been controlled to a great extent in the tiger reserves.
- Various compensatory developmental works have improved the water regime and the ground and field level vegetation.
- Stringent punishments for violators.

- GIS based digitized database development to evaluate tiger population.

Estimation of Tiger Populations

- The process of estimating the number of tigers in a given area is called ‘Tiger census’.
- It is conducted at regular intervals to know the current tiger populations and population trends.
- Besides estimating the number of tigers the method also helps to gather information on the density of the tiger populations and associated prey.
- The most commonly used technique in the past was ‘**Pugmark Census Technique**’.
- In this method the imprints of the pugmark of the tiger were recorded and used as a basis for identification of individuals.
- Now it is largely used as one of the indices of tiger occurrence and relative abundance.
- Recent methods used to estimate the numbers of tigers are **camera trapping** and **DNA fingerprinting**.
- In camera trapping, the photograph of the tiger is taken and individuals are differentiated on the basis of the stripes on the body.
- In the latest technique of DNA fingerprinting, tigers can be identified from their scats.

Project Elephant

Tiger, faces threat of extinction, whereas the elephant faces threat of attrition. The elephant numbers have not increased or decreased drastically but there is an increasing pressure on the elephant habitats and it is a serious concern which has to be addressed by involving people in elephant conservation and welfare.

- Project Elephant was launched in 1992.
- It is a centrally sponsored scheme.

Objectives:

1. to assist states having populations of wild elephants and to ensure long term survival of identified viable populations of elephants in their natural habitats.
2. addressing man-animal conflict.

3. Developing scientific and planned management measures for conservation of elephants.
4. Protecting the elephants from poachers, preventing illegal ivory trade and other unnatural causes of death

- The Project is being mainly implemented in 13 States/UTs, viz. Andhra Pradesh, Arunachal Pradesh, Assam, Jharkhand, Karnataka, Kerala, Meghalaya, Nagaland, Orissa, Tamil Nadu, Uttaranchal, Uttar Pradesh and West Bengal. Small support is also being given to Maharashtra and Chhattisgarh.
- States are being given financial as well as technical assistance in achieving the objectives of the Project.

Elephant Corridor

- An elephant corridor is defined as a stretch/narrow strips of forested (or otherwise) land that connects larger habitats with elephant populations and forms a conduit for animal movement between the habitats. This movement helps enhance species survival and birth rate.
- There are 88 identified elephant corridors in India.
- Out of total 88 corridors, 20 were in south India, 12 in north-western India, 20 in central India, 14 in northern West Bengal and 22 in north-eastern India.

Threats to Elephant Corridors

- Habitat loss leading to fragmentation and destruction caused by developmental activities like construction of buildings, roads, railways, holiday resorts and the fixing solar energized electric fencing, etc.
- Coal mining and iron ore mining is the two "single biggest threats" to elephant corridors in central India.
- Orissa, Jharkhand and Chhattisgarh, are mineral-rich states, but also have the highest number of elephant corridors in the country, which makes them known for elephant-man conflicts.
- There is also a serious poaching problem, as elephant ivory from the tusks is extremely valuable.
- Elephants need extensive grazing grounds and most reserves cannot accommodate them. If protected areas are not large enough,

elephants may search for food elsewhere. This often results in conflicts with humans, due to elephants raiding or destroying crops.

Mitigation

- Fusion of the corridors with nearby protected areas wherever feasible; in other cases, declaration as Ecologically Sensitive Areas or conservation reserves to grant protection.
- During the process of securing a corridor, monitoring for animal movement have to be carried out; depending on the need, habitat restoration work shall also be done.
- Securing the corridors involves sensitizing local communities to the option of voluntarily relocation outside the conflict zones to safer areas.
- Preventing further fragmentation of the continuous forest habitat by encroachment from urban areas.

Monitoring of Illegal Killing of Elephants (MIKE) Programme

Mandated by COP resolution of CITES, MIKE program started in South Asia in the year 2003 with following purpose:

- To provide information needed for elephant range States to make appropriate management and enforcement decisions, and to build institutional capacity within the range States for the long-term management of their elephant populations

Haathi Mere Saathi

- Haathi Mere Saathi is a campaign launched by the Ministry of environment and forest (MoEF) in partnership with the wildlife trust of India (WTI).
- The campaign was launched at the "Elephant-8" Ministerial meeting held in Delhi in 2011.
- The E-8 countries comprise of India, Botswana, the Republic of Congo, Indonesia, Kenya, Srilanka, Tanzania, and Thailand.
- This public initiative was aimed at increasing awareness among people and developing friendship, companionship between people and elephants.

The campaign mascot 'Gaju'

- The campaign focuses on various target audience groups including locals near elephant habitats, youth, policy makers, among others.
- It envisions setting up of Gajah (the elephant) centers in elephant landscapes across the country to spread awareness on their plight and invoke people's participation in addressing the threats to them.

Elephant - 8 Ministerial Meeting

- The E-8 ministerial meeting represented regions with all 3 species of elephants, viz.,
 1. Asian elephant
 2. African Bush Elephant
 3. African Forest Elephant
- Discussions covered a wide range of issues categorised under three basic themes.
 1. Science and conservation
 2. Management and conservation
 3. Cultural and Ethical perspectives of conservation

Indian (One Horn) Rhino Vision 2020

- One horned rhinos are poached for their horns.
- Indian rhino vision 2020 implemented by the department of environment and forests, Assam.
- The programme will be supported by WWF — India, the international rhino foundation (IRF), and a number of local NGOs.
- **Translocations** are the backbone of the IRV 2020 program.
- The goal set was to populate the potential rhino habitat areas identified viz. Manas NP, Dibru Saikhowa WLS, Laokhowa - Bura Chapori WLS with a viable population of rhino through translocations from Kaziranga NP and Pobitora WLS.
- Manas National Park was selected as the first site for translocation of rhinos.
- Ten rhinos have been released into Manas since 2008. Ten more rhinos will be moved from Kaziranga National Park before the end of the year.
- Translocating rhinos will help to create a viable population of this threatened species.

- The vision of this program is to increase the total rhino foundation in Assam from about 2000 to 3000 by the year 2020 and to ensure that these rhinos are distributed over at least 7 protected areas (PA) to provide long term viability of the one-horned rhino population.

Why?

- Concentrating so many rhinos in a single protected area like Kaziranga exposes the species to risks of calamities (epidemics, floods, massive poaching attempts).
- Further, rhinos in Pabitora have exceeded carrying capacity and numbers must be reduced to protect the habitat and to mitigate the increasing rhino-human conflicts.

Project Snow Leopard

- The snow leopard is a globally endangered species.
- Merely 7,500 are estimated to be surviving over two million square kilometers in the Himalaya and Central Asian mountains.
- Most snow leopard occur in China, followed by Mongolia and India.

Threats posed due to

- Human interference,
- competition with livestock – people kill them to save their livestock,
- retreating deeper into mountains due to global warming, and
- poaching.

Why to conserve the high altitude ecosystem?

- The high altitudes of India (> 3000 m) (including the Himalaya and Trans-Himalaya biogeographic zones) support a unique wildlife assemblage of global conservation importance.
- This includes highly endangered populations of species such as the snow leopard, two species of bears, wolf, red panda, mountain ungulates such as the wild yak, chiru, Tibetan gazelle, Tibetan argali, Ladakh urial, two species of musk deer, the hangul, three species of goral, serow, and takin, etc. High altitude lakes and bogs provide breeding grounds for a variety of avifauna including the

black-necked crane, barheaded Geese, brahminy ducks, and brown-headed gulls, etc.

- India has ratified international agreements promoting the conservation of high altitude wildlife species such as the snow leopard.
- In 2003, the **Convention on Migratory Species** included the snow leopard as a Concerted Action Species under its Appendix I.
- Similarly, in 2003, the Convention on International Trade in Endangered Species (CITES) expanded the scope of the CITES Tiger Enforcement Task Force to include all Asian big cat species including the snow leopard.

Sea Turtle Project

- A significant proportion of world's Olive Ridley Turtle population migrates every winter to Indian coastal waters for nesting mainly at eastern coast.
- With the objective of conservation of olive ridley turtles and other endangered marine turtles, MoEF initiated the Sea Turtle Conservation Project in collaboration of UNDP in 1999 with **Wildlife Institute of India, Dehradun** as the Implementing Agency.
- The project is being implemented in 10 coastal States of the country with special emphasis in State of Orissa.
- The project has helped in preparation of inventory map of breeding sites of Sea Turtles, identification of nesting and breeding habitats along the shore line, and migratory routes taken by Sea Turtles, development of guidelines to safeguard and minimize turtle mortality.
- One of the important achievements have been demonstration of use of Satellite Telemetry to locate the migratory route of Olive Ridley Turtles in the sea and sensitizing the fishermen and State Government for the use of Turtle Exclusion Device (TED) in fishing trawlers to check turtle mortality in fishing net.

Indian Crocodile Conservation Project

- The Indian Crocodile Conservation Project has pulled back the once threatened crocodilians from the brink of extinction and place them on a good path of recovery.

Objectives

- To protect the remaining population of crocodilians in their natural habitat by creating sanctuaries.
- To rebuild natural population quickly through 'grow and release' or 'rear and release' technique.
- To promote captive breeding.

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Captive breeding means that members of a wild species are captured, then bred and raised in a special facility under the care of wildlife biologists and other expert.

Bringing an animal into captivity may represent the last chance to preserve a species in the wild.

- To take-up research to improve management.
- To build up a level of trained personnel for better continuity of the project through training imparted at project-sites and through the (erstwhile) Central Crocodile Breeding and Management Training Institute, Hyderabad.
- To involve the local people in the project intimately.

Project Hangul

- The Kashmir stag also called Hangul is a subspecies of Central Asian Red Deer native to northern India.
- It is the state animal of Jammu & Kashmir.
- In Kashmir, it's found in **Dachigam National Park** at elevations of 3,035 meters.
- These deer once numbered from about 5,000 animals in the beginning of the 20th century.
- Unfortunately, they were threatened, due to habitat destruction, over-grazing by domestic livestock and poaching.
- This dwindled to as low as 150 animals by 1970. However, the state of Jammu & Kashmir, along with the IUCN and the WWF prepared a project for the protection of these animals.
- It became known as Project Hangul. This brought great results and the population increased to over 340 by 1980.

Ganges Dolphin

- The Ministry of Environment and Forests notified the Ganges River Dolphin as the **National Aquatic Animal**.

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- The River Dolphin inhabits the Ganges-Brahmaputra-Meghna and Karnaphuli-Sangu river systems of **Nepal, India, and Bangladesh**.
- It is estimated that their total population is around 2,000 and they are listed in Schedule I of the Wildlife Protection Act (1972).
- The Ganges Dolphin is among the four "obligate" freshwater dolphins found in the world — the other three are the 'baiji' found in the Yangtze River (China), the 'bhulan' of the Indus (Pakistan) and the 'boto' of the Amazon River (Latin America).
- Although there are several species of marine dolphins whose ranges include some freshwater habitats, these four species live **only in rivers and lakes**.
- The Chinese River Dolphin was declared functionally extinct by a team of international scientists in 2006.
- In India, the Ganges River Dolphin is **threatened** by river water pollution and siltation, accidental entanglement in fishing nets and poaching for their oil.
- In addition, alterations to the rivers in the form of barrages and dams are separating populations.

Green Revolution - Bringing Green Revolution in Eastern India

Green Revolution – Modern Agriculture

- Modern agriculture has made impressive contribution in increasing food grain production in the country.
- The country could achieve self-sufficiency in food grain production by using better quality of seed, proper irrigation and adequate supply of plant nutrients by using chemical fertilizers and control of pests and diseases in crop plants by using pesticides.
- It has also involved modern cultivation practices using tractors, combine harvesters and tube wells for irrigation.
- Rapid growth in food grain production from using seeds of high yielding variety is termed as Green Revolution.

What is Green Revolution?

- The term “Green revolution” refers to substantial increase in grain yield obtained by

plant breeders by developing new crop varieties.

- The high yielding varieties (HYVs) of wheat and rice have been the key elements in green revolution.
- Specifically the term “green revolution” refers to **wheat and rice** but some agricultural scientists even include **maize, soyabean and sugarcane** where spectacular gains in yield have occurred.

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The factors which have helped to bring about the green revolution are

- Introduction of high yielding varieties of crops.
- Multiple cropping, better irrigation and sufficient supply of fertilizers.
- Use of crop protection measures against disease and pest.
- Transfer of the technology of scientific farming from research farms to village farmers.
- Better arrangements for transporting farm produce from fields to the market.

Green Revolution involves

Introduction of High Yielding Varieties (HYV)

- In 1960s, the average national yield of wheat was very low as compared to the wheat yields of agriculturally advanced countries.
- MS Swaminathan, former Director General of ICAR (Indian Council of Agricultural Research) stressed the need for reorientation of the entire breeding programme of tall varieties.
- On the request of Indian breeders **Norman E. Borlaug** was invited from Mexico in 1963 by the Government of India to assess the possibilities of using dwarf varieties in India.
- **Borlaug** recommended the feasibility of using semi dwarf wheat of Mexican origin as the agro-climatic condition prevailing in India are similar to Mexico.
- On his recommendation two semi dwarf varieties namely Lerma Rajo and Sonora-64 were chosen and were released for cultivation in irrigated fields.
- These varieties gave very high yield and brought in revolution in wheat production.
- In 1970, **Norman E. Borlaug** was awarded Nobel prize for “Green Revolution” which also helped India.
- Through extensive wheat breeding programmes carried out during 1970-80, new

amber seeded, high yielding dwarf wheat varieties were developed.

- The important high yielding varieties include 'Kalyan Sona', 'Sharbati, Sonara', 'Sonalika' were released. These varieties responded favorably to fertilizer and irrigation.

Use of Chemical Fertilizers and Pesticides

- Chemical fertilizers are generally of the following types:
 1. **Nitrogenous fertilizers:** Nitrogen containing fertilizers e.g. ammonium sulphate, ammonium nitrate and urea.
 2. **Phosphate fertilizers:** Phosphate containing fertilizers e.g. ammonium phosphate, calcium dihydrogen phosphate (superphosphate)
 3. **Potassium fertilizers:** Potassium containing fertilizers e.g. potassium sulphate and potassium nitrate.
- Nitrogenous fertilizers promote plant growth and are essential for food production. But they should be used judiciously. Inefficient absorption by crops and wasteful application of fertilizers are the main causes of environment pollution.
- The unused fertilizers then enter surface water (rivers, lakes, ponds) and ground water.
- Fertilizers cause environmental pollution when used indiscriminately and cost us dearly.
- Pesticides are chemicals which have been developed to kill or control organisms called pests which are unwanted in agriculture. They are source of environmental pollution.

Mechanization Of Agriculture

- Increase in productivity on large areas of land brought the idea of farm mechanization.
- To cope up with the shortage of agricultural labour, farm mechanization was the obvious choice for completing agricultural operations.
- The machines which perform various jobs at the farm are water pump, ploughs, combine harvesters, land levelers, cultivators, power operated tractor sprays, reapers, threshers, trolleys and mechanical pickers etc.

Irrigation

- High yielding varieties usually require a lot of water and hence irrigational facilities were a prerequisite for green revolution.
- India has a wide variety of water resources and a highly varied climate. Under such conditions, several different kinds of irrigation systems are adopted to supply water to agricultural lands. These include wells, canals, rivers and tanks.
- **Wells:** There are two types of wells, namely dug wells and tube wells. This kind of irrigation is widely practiced in plain regions of India. Overexploitation of wells is well observed in Punjab-Haryana region.
- **Canals:** This is usually an elaborate and extensive irrigation system. In this system canals receive water from one or more reservoirs or from rivers. Canal irrigation is well suited for regions with clayey soil as clayey soil prevents water percolation. Mostly practiced in south India and Ganga-Yamuna region.
- **River Lift Systems:** In areas where canal flow is insufficient or irregular due to inadequate reservoir release, the lift system is more rational. Water is directly drawn from the rivers for supplementing irrigation in areas close to rivers. Mostly practiced in South India.
- **Tanks:** These are small storage reservoirs, which intercept and store the run-off of smaller catchment areas.

Second Green Revolution For Sustainable Livelihood

- The first Green Revolution was to ensure food security as there was severe scarcity of food in the country.
- The second Green Revolution aims at creating **sustainable agriculture by leveraging advancements in technology**.

Why we need Second Green Revolution

- With the growing population and over-exploitation of land resources, the pressure on food security will continue and rise.
- 65% of the population is still living in the villages and over 70% of the rural people are dependent on agriculture for their livelihood.
- Green Revolution, launched in mid-1960s, were mainly confined to certain areas. He said it was not successful in rain-fed areas, which

contribute about 60% of the country's total food-grain production.

- The Green Revolution has made us self-sufficient in food grains but the environmental consequences and ecological costs are offsetting the progress made.
- The ground water is depleted and polluted. The lakes and ponds are becoming life less due to eutrophication – a direct consequence of Green Revolution.
- Growth in the agricultural sector has been almost stagnant.
- GM Crops are marred in various controversies related to intellectual property, ecological consequences, health consequences etc.
- Global warming is said to engulf productive coastal lands due to rise in sea levels. This creates an urgent need to raise agricultural productivity.

It is necessary to develop a suitable strategy to improve agricultural development in India.

What we want from Second Green Revolution

- Improving agricultural production while generating gainful self-employment for the small farmers and weaker sections of the society.
- Scaling up food production without disturbing the ecological balance.
- Boosting agricultural development, women empowerment and environmental protection. [Women are the major power in agriculture as about 65-70% of the labour in crop production is contributed by women].
- Reclaiming degraded and low fertile lands and lands deprived of irrigation.

Bringing Green Revolution in Eastern India (BGREI)

- Green Revolution that turned India from 'begging bowl' to leading producer of food-grains.
- BGREI is about bringing similar benefits to eastern India that largely remained untouched of the wonder that converted the north-west into a 'grain bowl'.



- BGREI is flagship programme under **Rashtriya Krishi Vikas Yojana (RKVY)**.
- It is intended to address the constraints limiting the productivity of **“rice based cropping systems”**.
- The BGREI program was announced in the Union Budget, 2010-11.
- BGREI focuses on bringing the second Green Revolution in eastern region, which has rich water resources.
- **Assam, Bihar, Chhattisgarh, Jharkhand, Odisha, West Bengal and eastern Uttar Pradesh (Purvanchal)** are the seven states.

Objectives of BGREI

- Harness the water potential for enhancing agriculture production in Eastern India which was hitherto underutilized.
- Yield maximization of rice and wheat per unit area by improving agronomy, Water harvesting and conservation; and Water utilization (recycling of conserved water – surface water as well as groundwater).
- Promotion of recommended agriculture technologies and package of practices by addressing the underlying constraints.

Government Initiatives To Strengthen BGREI

- The ICAR has established IARI, Hazaribagh in Jharkhand and Indian Institute of Agricultural Biotechnology, Ranchi.
- It has also established National Research Centre for Integrated Farming at Motihari in Bihar to further strengthen the agricultural research for the eastern region.
- Central Agricultural University (CAU) has been sanctioned six new Colleges for the North-Eastern States.

How to Make Second Green Revolution a Success

Precision Agriculture

- The wealth of data if harnessed appropriately, can help farmers make the most efficient use of vital inputs such as water and fertilizer by applying them in precise amounts.
- Testing of samples of soil from agricultural fields is vital for achieving nutrient stewardship.
- Mobile-based applications for farmers will form an important part of the data-driven precision agriculture approach.

Efficient Use of Water

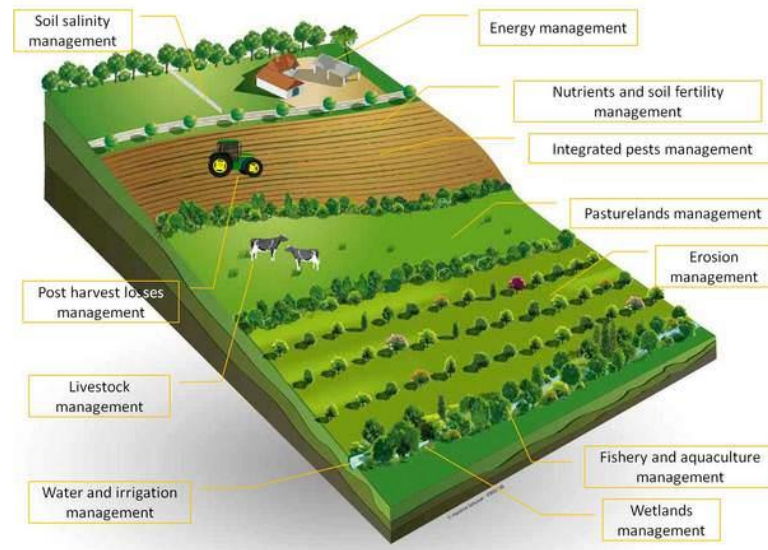
- Laser levelling is a technology that can grade an agricultural field to a flat surface by using a laser-guided scraper.
- Laser levelling has been shown to improve crop yields, reduce labour time spent weeding, and, in particular, reduce water use for irrigation by up to 20-25 per cent.
- Developing additional water sources through tube wells, dug wells and farm ponds.
- Promotion of Flood, Drought, and Salinity tolerant rice varieties.
- Use of Drum seeders for timely planting of direct seeded rice.

Sustainable Agricultural Practices

- Cultivation practices to increase biological and economic stability.
- Selection of improved varieties to suit the need.
- Soil management by proper method of tillage.
- Organic farming.

Sustainable Agriculture - Organic Farming - Biofertilizers

Concept Of Sustainable Agriculture



- Agriculture is a broad term encompassing all aspects of crop production (food and fibre), livestock farming, fisheries, forestry etc.
- Food and fibre productivity has increased by using new technologies, mechanization, increased use of fertilizers and pesticides and expansion of irrigation facilities.
- These changes reduced the labour demand to produce the majority of the food and fibre.
- Although these changes have had the positive effect, they also caused some serious environmental and social problems such as erosion of top soil, depletion and pollution of groundwater and other water resources, unemployment of farm laborers due to their replacement by farm machinery.
- In view of the growing negative consequences of modern agriculture there is growing demand to promote "sustainable agriculture".
- Sustainable agriculture is the production of food, fiber, plant or animal products using farming techniques that protect the environment, public health, human and animal welfare.
- Sustainable agriculture incorporates many environmentally safe agricultural practices and offers innovative and economically viable opportunities for farmers, laborers, consumers, policymakers and many others in the entire food system.
- Sustainable farming systems are those that are least toxic and least energy intensive and yet maintain productivity and profitability. E.g. Organic farming.
- Thus, sustainable agriculture is one that,

1. **supports profitable production;**
2. **protects environmental quality;**

3. **uses natural resources efficiently;**
4. **provides consumers with affordable, high-quality products;**
5. **decreases dependency on non-renewable resources;**
6. **enhances the quality of life for farmers and rural communities; and**
7. **will last for generations to come.**

Methods Of Sustainable Agriculture

- At the planning level one must take into account the local geography (topography), soil condition and nature, local climate, pests, local inputs and the farmer's goals.
- The grower (farmer) must then select appropriate practices.
- Several methods adopted in sustainable agriculture are:

1. **cultivation practices to increase biological and economic stability.**
2. **selection of improved varieties to suit the need.**
3. **soil management by proper method of tillage.**

Mixed cropping or diverse cropping

- In mixed cropping or diverse cropping two or more crops are grown all at the same time in a field.
- If by chance one crop fails, the others crops cover the risk of total crop failure.
- Usually a long duration crop is grown with a short duration one so that both get sufficient nutrition at the time of maturity.
- Generally a leguminous crop is grown along with the main crop.
- Legumes helps to increase soil fertility by fixing atmospheric nitrogen. This saves the cost of chemical fertilizers.
- The various plans followed in diverse or mixed cropping practices are:

1. **polyvarietal cultivation** where several genetic varieties of the same crop are planted.
2. **intercropping** where two or more different crops are grown at the same time on a plot like carbohydrate rich cereal that uses soil nitrogen and nitrogen fixing legume that puts back the nitrogen in the soil.

3. **polyculture**, in which different plants maturing at various times are planted together.

Advantages of Mixed cropping

- This practice has many advantages because fertilizer and water requirement of plants are different so there is less need of these inputs.
- Pests are controlled naturally because their natural predators find multiple habitats to survive.
- It has been found that this practice produces much higher yield per hectare compared to monoculture.

Monoculture

Large scale mechanization lead to the spread of monoculture i.e. **only one crop variety** is sown in the entire area when only one cultivator is planted in a large area.

This system (monoculture) uses lot of fertilizer, pesticide, water. This practice may be productive for some time but causes environmental and economic problems. E.g. Paddy and wheat cultivation.

Crop rotation

- It is practice of growing different crops in regular succession in the same field.
- This practice controls insects and diseases, increases soil fertility and decreases soil erosion.
- Generally soil cannot sustain continuous cropping with high yielding single crop because certain nutrients required by the crop get exhausted totally while others remain unutilized leading to serious **nutrients imbalance** in soil and encouraging certain diseases and pests.
- Sowing a leguminous crops (e.g. green gram) as a rotational crop is very useful because legumes enhance nitrogen level in the soil, reduces the need for chemical nitrogen fertilizer thereby saving the soil from the harmful effects of using high yielding varieties along with the application of large amount of fertilizer, pesticides and water.
- It is possible to grow two or sometimes three different crops in succession on the same land within a year is known as **multiple cropping**. This practice can go on for some time but the

land cannot maintain high yield in the long run.

- Crop rotation takes into account the following factors:

1. **Leguminous crop should be grown after non-leguminous crop.**
2. **Crops require less water (irrigation) should be grown after one – that requires more water.**
3. **Crops requiring less manure should be sown after one that requires more manure.**

- Important crop patterns of crop rotation

1. Green gram - Wheat – Moong
2. Ground nut – Wheat – Moong
3. Arhar – Sugarcane – Wheat – Moong
4. Paddy – Wheat – Moong

Mixed Farming

- Optimum diversity may be obtained by integrating **both crops and livestock** in the same farming operation.

Mixed crop along with livestock operations have several advantages.

1. Growing crops only on more level land and pastures or forages on steeper slopes will reduce soil erosion.
2. Pasture and leguminous forage crops in rotation enhance soil quality and reduce erosion; livestock manure, in turn, contributes to soil fertility.
3. Livestock can buffer the negative impacts of low rainfall periods by consuming crop residue which in “plant only” systems would have been considered crop failure.
4. Feeding and marketing are flexible in animal production systems. This can help cushion farmers against price fluctuations and, make more efficient use of farm labour.

Strip Farming: This method is another method of soil erosion. This involves planting the main crops in widely spaced rows and filling in the spaces with another crop to ensure complete ground cover. The ground is completely covered so it retards water flow which thus soaks down into the soil, consequently reducing erosion problems.

Soil Management

- A healthy soil is a key component of sustainable agriculture. That is healthy soil along with water and nutrients produces healthy crops that are less susceptible to pests and diseases.
- Accordingly, soil must be protected and nurtured to ensure long term productivity and stability.
- Methods of protection include using cover crops, compost, reducing tillage, conserving soil moisture by dead mulches, this increases water hold capacity of the soil.

Nutrient Management

- There are **sixteen nutrients** which are essential for plants.
- Air supplies carbon and oxygen, hydrogen comes from water, and soil supplies the other thirteen nutrients to plants.
- Amongst these thirteen nutrients, six are required in large quantities and are therefore called macronutrients.
- The other seven nutrients are used by plants in small quantities and are therefore called micronutrients
- Deficiency of these nutrients affects physiological processes in plants including reproduction, growth and susceptibility to diseases.

Source	Nutrients
Air	carbon, oxygen
Water	hydrogen, oxygen
Soil	(i) Macronutrients: nitrogen, phosphorus, potassium, calcium, magnesium, sulphur (ii) Micronutrients: iron, manganese, boron, zinc, copper, molybdenum, chlorine

Macronutrients

- Based on the relative abundance in plants, viz., Nitrogen (N); Phosphorous (P), Potassium (K), Sulfur (S), Calcium (Ca) and Magnesium (Mg)

Micronutrients

- Their concentration is very small. They are also referred to as minor elements.
- Iron (Fe); Zinc (Zn); Manganese (Mg), Copper (Cu), Boron (B), Chlorine (Cl) and Molybdenum (Mo).
- In some plants, other than the above, Sodium (Na), Cobalt (Co), Vanadium (Va), Nickel (Ni) and Silicon (Si) are considered as essential micronutrients

Nitrogen (N)

- N is an essential constituent of proteins and is present in many other compounds of great physiological importance in plant metabolism
- N is an integral part of chlorophyll, which is primary observer of light energy needed for photosynthesis.
- N also imparts vigorous vegetative growth and dark green colour to plants.

Phosphorus (P)

- Phosphorus (P) is an essential part of the enzymes which help the crop to fix light energy.
- It forms an integral part of nucleic acids, the carriers of genetic information, and is important in stimulating root growth

Potassium (K)

- Potassium (K) is involved in processes which ensure carbon assimilation and the transportation throughout the plant for growth and the storage of sugars and proteins.
- The potassium ion is also important for water regulation and uptake.
- Furthermore, the presence of potassium in sufficient amounts ensures resistance to frost, drought and certain diseases

Others

- Magnesium occurs in chlorophyll and is also an activator of enzymes,
- Sulphur forms part of two essential amino acids which are among the many building blocks of protein. It is also found in vitamin B1 and in several important enzymes.
- Calcium is required for plant growth, cell division and enlargement. The growth of root

and shoot tips and storage organs is also affected by calcium as it is a component of cell membranes. Calcium is also vital for pollen growth and to prevent leaf fall.

Bio-Fertilizers And Their Use In Agriculture

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- For a sustainable agriculture system, it is essential to use renewable inputs (fertilizer, pesticides, water etc.) which can benefit the plant and cause no or minimal damage to the environment.
- One of the energy efficient and pollution free method is to exploit the ability of certain microorganisms like **bacteria, algae and fungi to fix atmospheric nitrogen, solubilize phosphorus, decompose organic material or oxidize sulphur** in the soil.
- When they are applied in the soil, they enhance growth and yield of crops, improve soil fertility and reduce pollution. They are known as "bio fertilizers".
- Thus bio-fertilizers are living or biologically active products or microbial inoculants of bacteria, algae and fungi (separately or in combination) which are able to enrich the soil with nitrogen, phosphorus, organic matter etc.
- Following are some of the important types of bio fertilizers which can be considered for agro based industries.

Rhizobium biofertiliser

- Rhizobium is a **symbiotic bacteria** forming root nodules in legume plants. These nodules act as miniature nitrogen production factories in the fields.
- The nodule bacteria fix more nitrogen (N₂) than needed by legume plant and the bacteria. The surplus fixed nitrogen is then secreted and fertilizes the soil.
- Rhizobium is **more efficient** than-free living nitrogen-fixing bacteria.

Azotobacter biofertiliser

- Azotobacter are **aerobic free living nitrogen fixers**.
- They grow in the rhizosphere (around the roots) and fix atmospheric nitrogen nonsymbiotically and make it available to the particular cereals.

- These bacteria produce **growth promoting hormones** which helps in enhancing growth and yield of the plant.

Azospirillum biofertiliser

- These are **aerobic free living nitrogen fixers** which live in associative symbiosis.
- In this type of association bacteria live on the root surface of the host plant and do not form any nodule with roots of grasses.
- It increases crop yield and its inoculation benefits crop. They also benefit the host plants by supplying growth hormones and vitamins.
- These bacteria are commonly used for the preparation of commercial inoculants (vaccines, culture medium).

Blue green algae

- Blue green algae (BGA or cyanobacteria) like **Nostoc** and **Anabaena** are free living photosynthetic organisms also capable of fixing atmospheric nitrogen.
- In the flooded rice fields blue green algae serves as a nitrogen biofertiliser.

Consider the following organisms

1. Agaricus
2. Nostoc
3. Spirogyra

Which of the above is/are used as biofertilizer/biofertilizers

1. 1 and 2
2. 2 only
3. 2 and 3
4. 3 only

Answer: b) 2 only [Not sure]

Azolla biofertilisers

- Azolla is a water **fern** inside which grows the nitrogen fixing blue green algae **Anabaena**.
- It contains 2-3% nitrogen when wet and also produces organic matter in the soil.
- The Azolla-Anabaena combination type biofertiliser is used all over the world. This can be grown in cooler regions.

- But there is a need to develop a strain that can tolerant to high temperature, salinity and resistant to pests and diseases.
- Production technology is very easy and can be adopted by rice farmers.
- The only constraint in Azolla is that it is an aquatic plant and **water becomes limiting factor** in growing it particularly in summer.

Phosphorus solubilizing biofertiliser

- Phosphorus is an important element required for plant growth. This element is also **needed for nodulation by rhizobium**.
- Some microorganisms are capable of solubilizing immobilized phosphorus making it available to plants for absorption.

Mycorrhizal fungi biofertiliser

- Mycorrhizal fungi** acts as biofertiliser and are known to occur naturally on roots of forest trees and crop plants.
- Mycorrhizal fungi **resist disease** in plants. The plants also show **drought and salinity resistance**. Plants can **tolerate** adverse soil, pH, high temperature and heavy metal toxicity.
- In soils low in available nutrients there is an increased absorption of nutrients by plants infected with Mycorrhiza.
- The fungus has the ability to **dissolve and absorb phosphorus** that plant roots cannot readily absorb.
- A wise way will be to develop an integrated nutrient supply system involving the combination of chemical fertilizers and biofertilisers.

Mycorrhizal biotechnology has been used in rehabilitating degraded sites because mycorrhiza enables the plants to

1. resist drought and increase absorptive area
2. tolerate extremes of pH
3. Resist disease infestation

Select the correct answer using the codes given below:

- a) 1 only
- b) 2 and 3 only
- c) 1 and 3 only

d) 1, 2 and 3

Answer: d) all

Compost Tea

- Compost Tea is a Liquid fertilizer for flowers, vegetables and houseplants.
- Compost tea is an aerobic (in the presence of oxygen) water solution that has extracted the microbe population from compost (dead and decaying matter) along with the nutrients.
- In simple terms, it is a concentrated liquid created by a process to increase the numbers of beneficial organisms as an organic approach to plant/soil care.
- Air is sent through the water to keep the water oxygenated, as this favors the beneficial bacteria and fungi over the pathogens.
- At the end of the brewing cycle, what you have is a concentrated liquid full of billions of microorganisms (bacteria, fungi, protozoa, nematodes) that can then be sprayed directly onto the leaf surface.
- The liquid fertilizer occupies the infection sites on the leaf surface and is held there by simple sugars that the plant puts out that work as a glue to keep the beneficial microorganisms thriving and protecting the plant.

Bio char

- Bio char is found in soils around the world as a result of vegetation fires and historic soil management practices.
- Intensive study of bio char-rich dark earths in the Amazon (terra preta), has led to a wider appreciation of bio char's unique properties as a soil enhancer.
- Bio char is charcoal that is as soil amendment (minor improvement).
- It created using a pyrolysis process (decomposition brought about by high temperatures), heating biomass in a low oxygen environment.
- Once the pyrolysis reaction has begun, it is self-sustaining, requiring no outside energy input.
- Byproducts of the process include syngas ($H_2 + CO$), minor quantities of methane (CH_4) (greenhouse gas), organic acids and excess heat.
- Once it is produced, bio char is spread on agricultural fields and incorporated into the top layer of soil.

- The syngas and excess heat can be used directly or employed to produce a variety of biofuels.

Agricultural Benefits of Bio char

- It increases crop yields, sometimes substantially if the soil is in poor condition.
- It helps to prevent fertilizer runoff and leeching, allowing the use of less fertilizers.
- It retains moisture, helping plants through periods of drought more easily.
- Most importantly, it replenishes exhausted or marginal soils with organic carbon
- It fosters the growth of soil microbes essential for nutrient absorption, particularly **mycorrhizal fungi**.
- Bio char can increase soil fertility of acidic soils (low pH soils)[$pH < 7$ = Acidic Soils; $pH > 7$ = Basic Soils] [The most fertile soils are slightly acidic (pH just below 7)]
- Bio-char reduces the acidity of the soil, protects the plants from diseases, promotes growth of friendly microorganisms, and reduces the loss of micro nutrients apart from increasing water retain-ability.

Kollam, Kerala, is famous for fishing and fishery-related activities; as an agricultural sector, it does not evoke any enthusiasm. Why?

- With the prevailing soil and climatic conditions which favors leaching and draining of soil nutrients into the Arabian sea and Ashtamudi Lake, the soil has high acidity making it unfit for cultivation of any kind (Why? Because friendly microorganisms don't like acidic medium. They like slightly alkaline or basic medium [having a pH greater than 7]).
- Bio char can make a difference to the agriculture of the region.

Other Environmental Benefits

- Most carbon in the soil is lost as greenhouse gas (carbon dioxide, CO_2) into the atmosphere if natural ecosystems are converted to agricultural land.
- Soils contain 3.3 times more carbon than the atmosphere.
- This makes soils an important source of greenhouse gases but also a potential sink if right management is applied.

- The use of crop residues for bio-energy production reduces the carbon stocks in cropland.
- Further the dedication of cropland to bio-fuel production increases the area of cultivated land and thus carbon loss from soils and vegetation.
- Bio char remains stable for millennia, providing a simple means to sequester carbon emissions.
- If bio char is returned to agricultural land it can increase the soil's carbon content permanently and would establish a carbon sink for atmospheric CO₂.

Organic Farming



- Organic farming is a type of agriculture or farming which avoids the use of synthetic fertilizers, pesticides, growth regulators, and livestock feed additives.
- Organic farming systems rely on **crop rotation, crop residues, animal manures, legumes, green manure, off-farm organic wastes and biofertilisers**, mechanical cultivation, mineral bearing rocks to maintain soil productivity to supply plant nutrients and biological pest control, controlling weeds, insects and other pests.
- All kinds of agricultural products can be produced organically, including grains, meat, dairy, eggs, fibres such as cotton, jute, flowers etc. Thus organic farming creates a sustainable lifestyle for generations to come.

- Organic farmers build healthy soils by nourishing the microbial inhabitants that release, transform, and transfer nutrients.
- Soil organic matter contributes to good soil structure and water-holding capacity.
- Organic farmers feed soil biota and build soil organic matter with cover crops, compost, and biologically based soil amendments. These produce healthy plants that are better able to resist disease and insect predation.
- Organic farmers' primary strategy in controlling pests and diseases is prevention through good plant nutrition and management.
- Organic farmers use cover crops and sophisticated crop rotations to change the field ecology, effectively disrupting habitat for weeds, insects, and disease organisms.
- Weeds are controlled through crop rotation, mechanical tillage, and hand-weeding, as well as through cover crops, mulches, flame weeding, and other management methods.
- Organic farmers rely on a diverse population of soil organisms, beneficial insects, and birds to keep pests in check.
- When pest populations get out of balance, growers implement a variety of strategies such as the use of insect predators, mating disruption, traps and barriers.

Some important benefits of organic farming and organic foods

- Organic farming can be learnt easily by any conventional farmer.
- Switching to organic farming, conventional farmer can actually reduce its production cost by over 25% as.
- Eliminates the use of expensive synthetic fertilizers and pesticides, minimizing soil degradation.
- Organic farms can support substantially higher levels of wildlife especially in low lands and where animals can roam in pastures or graze on grassland.
- Entire ecosystems and ground water are improved by simply following organic farming methods.
- When dairies feed their cows organic feed and graze them on organic fields, the cows experience better health, less sickness, diseases and ultimately produce better tasting milk for consumers.
- Organic farming promotes healthy soils rich in micro nutrients and which can be used for

decades to grow crops without getting exhausted.

- Consumers purchasing organically grown foods are tastier.
- Organically grown products are free from harmful chemicals, artificial flavors and preservatives that ultimately cost consumers more money than non-organically grown products.

Vermicomposting

- Vermicomposting is an appropriate technique for efficient recycling of animal wastes, crop residues and agro-industrial wastes.
- The process of conversion of organic materials into manure is chiefly microbiological.
- Earthworms are important for producing vermicompost from organic wastes.
- Vermicompost can be prepared from all sorts of organic residues – animal waste, sericulture residues, dairy and poultry residues, bagasse from sugarcane factories, sorghum straw, rice straw after feeding cattle, dry leaves, groundnut husk and wheat husk, waste vegetables, weeds (particularly Parthenium hysterophorus or Congress weed before flowering) etc..

Integrated Pest Management (IPM)

- In this approach, each crop and its pests are evaluated as parts of an ecological system.
- Then farmers develop a control programme that includes cultivation, biological and chemical methods applied in proper sequence and with the proper timing.
- The aim of IPM is not to eradicate the pest population completely but to keep the crop damage to economically tolerable level.
- Farmers monitor the field and when they find the pest level to be high enough, they first use biological methods and cultivation practices to control and then use small amounts of insecticides mostly insecticides derived from plants as a last resort.

Biological control includes

- Natural predators, parasites and pathogens of the pests are used.

Examples are:

- Pest on cucumber plant called red spider mite is controlled by using a predatory mite that feed on red spider mite.
- Citrus fruits in California heavily damaged by scale insects which were controlled by Australian ladybird which ate away the insects.

Cultivation practices

- A variety of cultivation practices like crop rotation, polyculture and inter cropping etc. can be used to get rid of the pests.
- Some amounts of insecticides, mostly of plant origin (e.g. Pyrethrum and Rotenone neem product) are applied as a last resort.
- Pest and disease resistant crop plants can be produced by genetic engineering. Example is Bt cotton, insecticidal for bacterial gene (Bacillus thuringiensis) introduced into cotton plant making cotton plant resistant to pest.

Disadvantages of Integrated Pest Management (IPM)

- Farmer should have an expert knowledge about each pest.
- It acts more slowly than conventional pesticides.
- Methods developed for a crop in one area might not apply to areas with even slightest different growing conditions.
- Initial cost may be higher but in the long-term cost become very low.

Genetically Modified Crops - Benefits/Controversies

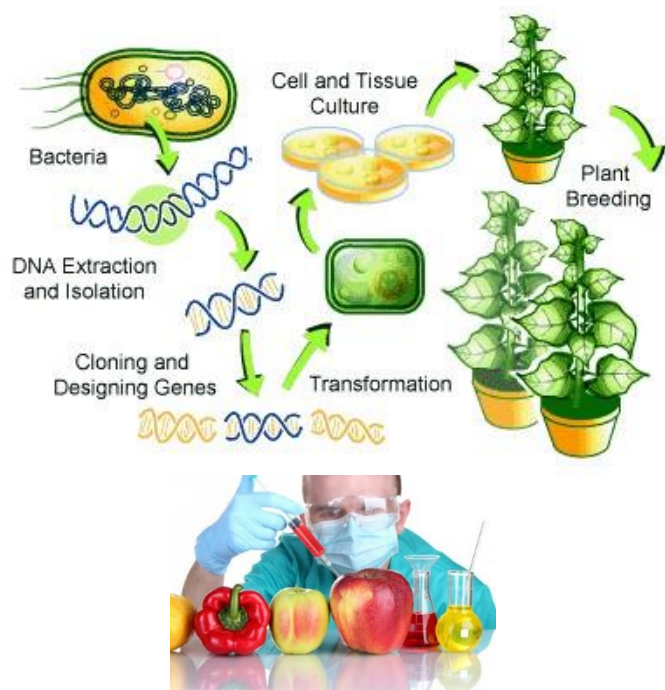
Biotechnology – Genetically Modified (GM)

- With conventional breeding practices reached their saturation point, the “**gene revolution**” seems to hold lot of potential.
- Agricultural biotechnology or gene technology or genetic engineering may act as the second “green revolution” that can be used to create high yielding crop varieties that are:

1. **herbicide tolerant,**
2. **insect resistant,**
3. **resistant to pathogens like virus, bacteria and fungi**

4. have better nutritional value and other commercial properties.

- The crop plants produced by these techniques are called “**transgenic**” or **genetically modified (GM)** plants or genetically modified organisms (GMOs).



- By using the technique of genetic engineering it has been possible to genetically transform large number of agricultural and ornamental crops.

Transgenics have been produced with the following aims:

- Crop resistance to herbicides.
- Crop resistance to insects and diseases.
- Atmospheric nitrogen fixation by cereal crops.
- Tolerance to high salt soils and to flooding in crops.
- Drought resistance in crops.
- Improving nutritional quality of crops.
- Prolonging shelf life of fruits and vegetables.

Some important examples of transgenics or GMOs are:

- Bt cotton** produced by incorporating Bt gene which encodes for **BT toxin** (insecticidal protein in **Bacillus thuringiensis**) in the cotton plant. The plant becomes insect resistant and this gene has been incorporated

in corn, potato, tomato, tobacco etc. making them insect resistant (bio pesticides).

- “**Golden Rice**” a transgenic with enhanced **vitamin A** content producing nutritionally rich rice to save many lives.
- Salt and flood tolerance genes have been incorporated in rice so that Bt rice in China shows higher yield and a huge reduction in pesticide use. Such rice can be grown on saline soil.
- By slowing down and controlling ripening in tomato by introducing a bacterial gene that **prevents ethylene formation** thus delaying ripening. Such tomatoes are easy to handle during transportation and remains on the shelf for a long time.
- Cold damage to crop plants can be minimized by introducing genes for **antifreeze proteins (AFPs)** found in the blood of arctic fishes.
- Frost resistant tomatoes have been produced by introducing gene for antifreeze proteins from polar fish living in ice water.
- Plant biotechnology can help to make intensive agriculture less damaging to the environment as well as help the country to spend less money on fertilizers, pesticides, herbicides etc.

Benefits of GM Crops

Crops

- Enhanced taste and quality.
- Reduced maturation time.
- Increased nutrients, yields, and stress tolerance.
- Improved resistance to disease, pests, and herbicides.
- New products and growing techniques.

Animals

- Increased resistance, productivity, hardness, and feed efficiency.
- Better yields of meat, eggs, and milk.
- Improved animal health and diagnostic methods.

Environment

- “Friendly” bioherbicides and bioinsecticides.
- Conservation of soil, water and energy.
- Bioprocessing for forestry products.
- Better natural waste management.
- More efficient processing.

Society

- Increased food security for growing population.

Controversies Surrounding GM crops

Safety

- Potential human health impact: allergens, transfer of antibiotic resistance markers, unknown effects.
- Potential environmental impact: unintended transfer of transgenes through crosspollination, unknown effects on other organisms (e.g., soil microbes) and loss of flora and fauna biodiversity

Access and intellectual property

- Domination of world food production by a few companies.
- Increasing dependence on industrialized nations by developing countries.
- Biopiracy**—foreign exploitation of natural resources.

Ethics

- Violation of natural organisms' intrinsic values.
- Tampering with nature by mixing genes among species.
- Objections to transferring animal genes in plants and vice versa.
- Stress for animal.

Labeling

- Not mandatory in some countries (e.g. United States).
- Mixing GM crops with non-GM confounds labeling attempts.

Society

- New advances may be skewed to interests of rich countries.

Effectiveness

- The ineffectiveness of genetically modified (GM) cotton against the recent whitefly attack in Punjab and Haryana raised more questions regarding the effectiveness of GMO.
- The whitefly attack in Punjab damaged over 75 per cent crop across the cotton belt.
- This has raised concern among agricultural experts and farmers over the growing dependency on Bt cotton.

Bt cotton

- Strains of the **bacterium *Bacillus thuringiensis*** produce over 200 different Bt toxins, each harmful to different insects.
- Most notably, Bt toxins are insecticidal to the larvae of moths and butterflies, beetles, cotton bollworms but are harmless to other forms of life. (this is why Bt cotton failed against whitefly).

Newer Agricultural Practices & Misc Topics

Newer Agricultural Practices

- Modern agriculture includes animal husbandry, poultry farming, apiculture, fisheries and mushroom culture etc. to provide additional food supplements like milk, meat, fish, egg, mushroom etc.
- In addition to provide nutritional food for the masses, they also reduce load on the consumption of cereals and pulses.

Revolutions in Agriculture

Revolution	Related with
Green	Food grain Production
Golden	Fruit Production
Grey	Fertilizer Production
Blue	Fish Production
Black	Petroleum Production
Pink	Prawn Production
Round	Potato Production
Red	Meat/Tomato Production
Silver	Egg/Poultry Production
White	Milk Production
Yellow	Oil seeds Production

Various Agricultural Activities

Name	Agricultural Activity
SilviCulture	art of cultivating forest trees
Sericulture	rearing of silkworms for the production of raw silk

Apiculture	maintenance of honey bee colonies, commonly in hives, by humans
Olericulture	science of vegetable growing, dealing with the culture of non-woody (herbaceous) plants for food
Viticulture	science, production and study of grapes
Floriculture	discipline of horticulture concerned with the cultivation of flowering and ornamental plants for gardens
Arboriculture	cultivation, management, and study of individual trees, shrubs, vines, and other perennial woody plants
Pomology	branch of horticulture which focuses on the cultivation, production, harvest, and storage of fruit, etc.
Aeroponics	process of growing plants in an air or mist environment without the use of soil or an aggregate medium
Hydroponics	method of growing plants using mineral nutrient solutions, in water, without soil.
Geoponic	farming practice, refers to growing plants in normal soil
Aquaponics	System that combines conventional aquaculture (raising aquatic animals such as snails, fish, crayfish or prawns in tanks) with hydroponics (cultivating plants in water) in a symbiotic environment.

Poultry farming

- Poultry farming is a term used for rearing and keeping of birds such as fowl, duck and hen for egg and meat.
- Poultry farming has become popular because this is comparatively easy to start and maintain.
- It gives quick return within one to six month of investments, is easily manageable and required less space and labour. Poultry birds and their eggs are rich source of nutrients.
- Indian poultry birds provide good quality meat but produces small sized eggs. They have natural immunity against common diseases as compared to exotic varieties bred abroad.

Mushroom culture

- Mushrooms are kind of **fungus** which appear as white tiny balls consisting of a short stem and a cap which opens like an umbrella later.
- They lack chlorophyll and grow on organic matter or waste materials from farms or factories, useless by-products can be recycled as medium to grow mushrooms for human consumption.
- Out of the large number of mushroom species only some are edible.
- Mushrooms are good source of high quality **proteins** and are rich in vitamins and minerals.
- Like fruits and vegetables mushrooms are perishables and require a great deal of attention during storage, marketing and processing.

Apiculture

- Apiculture is also known as **bee-keeping**.
- Apiculture or bee-keeping is the art of controlling colonies of honey bee in large quantity for commercial production of honey.
- There are three major advantages of bee-keeping: (i) provides honey- a valuable food (ii) provides bee wax- which has many uses in industry (iii) honey bees are excellent pollinating agents which increase agricultural yields.

Fish culture and aquaculture

Consequences Of Over fishing in seas

- Over fishing take place to such an extent that very little breeding stock (fish population) is left to maintain the special number.
- Prolonged over fishing leads to commercial extinction when the population of the species becomes so low that it is no longer profitable to hunt them.
- Fishing methods such as trawling and drift nets capture everything in their way indiscriminately. Sometimes 70% of the catch is thrown away.
- Commercial fish catching disturbs non-target marine animals. Dredges and trawls also adversely affect the marine habitats.

Aquaculture offers a potential solution

- Aquaculture offers a potential solution to the depleted ocean fisheries as well as meeting the demand for sea food.

- The current “blue revolution” of aquaculture has taken up the shape of an industry with intensive use of resources and has adverse environmental impacts.
- Ecological aquaculture (Eco-aquaculture) need to be promoted with a focus on developing aquatic farming system that preserve the.
- Areas where fish are reared commercially are known as artificial fisheries. The fishes are bred, reared and later harvested.
- Depending on the nature of water in which fish is reared fisheries can be:

1. Marine fisheries: fishing operations along the coastline, e.g. Mackerels, Sardines, Catfish.
2. Fresh water or inland fisheries: fish found in rivers, irrigation canals, reservoirs, lakes, tanks, ponds etc. e.g. Rohu, Catla, Mystus.
3. Estuarine or brackish water fisheries: estuaries are where river water and sea water get mixed like backwaters, lagoons, coastal lakes, delta channels. They are more common in Bengal and Kerala. E.g. Mullet, Milkfish, Pearlsport.
4. There are several other aquatic resources such as molluscs (oyster, mussels, squids, octopus, cuttlefish etc.) and seaweeds which have been exploited for aquaculture.

- Sea weeds are used for human consumption, as cattle and poultry feed, as manure and for industrial purpose as a source of **agar-agar** and **algin**.

Animal Husbandry

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- The branch of agriculture that deals with breeding, feeding and care of domestic animals is called animal husbandry.
- Animal husbandry is an integral part of modern agriculture as animal sources provide us important food materials like milk, egg, meat etc.
- Cows and buffaloes are our chief sources of milk. Milk producing animals are called milch animals.
- Hens are egg laying animals. Fishes, pigs, hens and goats are our major sources of meat.
- Animal husbandry plays a prominent role in the rural economy in supplementing the income of rural households.

Diseases in Domesticated Animals

Disease	Pathogen	Animal affected	Symptoms
Foot and mouth disease	Virus	Cattle	Blisters on the mouth and foot, excessive production of saliva, loss of appetite, high body temperature, shivering.
Pox		Cow, buffalo, sheep, goat	Appearance of small nodules and fever.
Dermatitis		Goat and sheep	Irritation, blisters and eruptions on the skin.
Tuberculosis	Bacteria	Cattle	Infection of udders, lungs, intestine and other parts, swelling of lungs and fever.
Rinderpest		Cattle	Discharge from eyes, nostrils, loss of appetite, constipation followed by severe diarrhoea.
Anthrax		Cattle, sheep, goat, pigs	Swelling of body, fever, reduction in milk secretion.
Salmonellosis		Cattle	Diarrhoea with blood clots and fever
Mastitis		Cattle	Swallow udders, fever, milk becomes watery .

- The foot and mouth disease of cattle is very common, dangerous and contagious disease. The affected animals are slaughtered and the dead ones are buried deep or burned so as to stop the disease from spreading.
- Another dreaded disease is anthrax which spreads easily. Such animals after death must be burnt and disposed of completely.

Problems of Indian Agriculture

- Fragmentation of land holding.
- Existence of small and marginal farmers.
- Regional variation.
- Dependence of seasonal rainfall.
- Low productivity of land.
- Increasing of disguised unemployment.
- Disorder in marketing of Agricultural products.
- Weak land reformation.
- Environmental concerns due to green revolution.

Crop Classifications

Classification based on climate

- Tropical: Crops grow well in warm & hot climate. E.g. Rice, sugarcane, Jowar etc
- Temperate: Crops grow well in cool climate. E.g. Wheat, Oats, Gram, Potato etc.

Classification Based on growing season

- **Kharif/Rainy/Monsoon crops:** The crops grown in monsoon months from June to Oct-Nov, require warm, wet weather at major period of crop growth, also required short day length for flowering. E.g. **Cotton, Rice, Jowar, bajara.**
- **Rabi/winter/cold seasons crops:** The crops grown in winter season from Oct to March month. Crops grow well in cold and dry weather. Require longer day length for flowering. E.g. **Wheat, gram, sunflower**
- **Summer/Zaid crops:** crops grown in summer month from March to June. Require warm dry weather for major growth period and longer day length for flowering. E.g. Groundnuts, Watermelon, Pumpkins, Gourds.

Agronomic Classification of Crops

Cereals

- Cereals are cultivated grasses grown for their edible starchy grains. Larger grains used as staple food are cereals. Rice, wheat, maize, barley, rye and oats.

Millets

- These are staple food of poor people.

Major millets

- Sorghum or Jowar, Pearl Millet or Bajra and Finger millet or ragi

Minor millets

- Fox tail millet, Little millet, Common millet, Barnyard millet and Kodomillet.

Pulses or Grain Legumes

- Pulses are major source of protein in Indian vegetarian diet.
- Red gram, Black gram, Green gram, Cowpea, Bengalgram, Horsegram, Dewgram, Soyabean, Peas or gardenpea, Garden bean etc.

Oil Seed Crops

Groundnut or peanut, sesamum, sunflower, castor, linseed or flax, rapeseed & mustard.

Sugar Crops

- Sugarcane and sugar beet.

Byproducts of Sugar Industry

- Molasses, bagasse, pressmud
- Molasses used for **alcohol and yeast formation.**
- Bagasse for **paper making and fuel.**
- Pressmud used for soil amendment
- Trash (green leaf + dry foliage) — the waste is used for cattle feed
- Sugar beet — Tuber for extraction of sugar
- Tubers and tops are used as a fodder for cattle feed

With reference to the usefulness of the by-products of sugar industry, which of the following statements is/are correct?

1. Bagasse can be used as biomass fuel for the generation of energy.
2. Molasses can be used as one of the feedstocks for the production of synthetic chemical fertilizers.
3. Molasses can be used for the production of ethanol.

Select the correct answer using the codes given below.

1. 1 only
2. 2 and 3 only
3. 1 and 3 only
4. 1, 2 and 3

Ethanol is alcohol.

Answer: c) 1 and 3 only

Starch Crops or Tuber Crops

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- Potato, cassava, sweet potato etc.

Stem Fibres

- Jute, mesta (pulicha keerai), sun hemp, sisal hemp

Narcotics - Stimulates Nervous System

- Tobacco, betelvine and arecanut.

Plantation Crops

- Tea – leaf, Coffee – seed, rubber, cocoa – seed, palm – oil, sugarcane – sugar etc.

Spices and Condiments

- Products of crop plants are used to flavor taste and sometime color the fresh preserved food. E.g. ginger, garlic, chili, cumin onion, coriander, cardamom, pepper, turmeric etc.

Medicinal & aromatic crops

- Medicinal plants includes cinchona, isabgoli, opium poppy, senna, belladonna, rauwolfra, iycorice and aromatic plants such as lemon grass, citronella grass, palmorsa, Japanese mint, peppermint, rose, jasmine, henna etc.

Classification based on life of crops/duration of crops

- Seasonal crops: A crop completes its life cycle in one season. E.g. rice, Jowar, wheat etc.
- Two seasonal crops: crops complete its life cycle in two seasons. E.g. Cotton, turmeric, ginger.
- Annual crops: Crops require one full year to complete its life cycle. E.g. sugarcane.
- Biennial crops: Crops requires two year to complete its life cycle E.g. Banana, Papaya.
- Perennial crops: crops live for several years. E.g. Fruit crops, mango, guava etc.

Classification based on cultural method/water

- Rain fed: Cultivation of crop mainly based on the availability of rain water. E.g. Jowar, Bajara, Mung etc.

- Irrigated crops: Crops cultivated with the help of irrigation water. E.g. Chili, sugarcane, Banana, papaya etc.

Classification based on root system

- Tap root system: The main root goes deep into the soil. E.g. Tur, Grape, Cotton etc.
- Fiber rooted: The crops whose roots are fibrous shallow & spreading into the soil. E.g. Cereal crops, wheat, rice etc.

Classification based on economic importance

- Cash crop: Grown for earning money. E.g. Sugarcane, cotton.
- Food crops: Grown for raising food grain for the population and & fodder for cattle. E.g. Jowar, wheat, rice etc.

Classification based on No. of cotyledons

- Monocots or monocotyledons: Having one cotyledon in the seed. E.g. all cereals & Millets.
- Dicots or dicotyledonous: Crops having two cotyledons in the seed. E.g. all legumes & pulses and almost all the trees.

Classification based on length of photoperiod required for floral initiation

- Most plants are influenced by relative length of the day & night, especially for floral initiation, the effect on plant is known as photoperiodism.
- Short-day plants: Flower initiation takes place when days are short less than ten hours. E.g. rice, Jowar, green gram, black gram etc.
- Long day's plants: require long days are more than ten hours for floral initiation. E.g. Wheat, Barley, etc.
- Day neutral plants: Photoperiod does not have much influence for phase change for these plants. E.g. Cotton, sunflower, etc.

Cropping pattern

- The yearly sequence and spatial arrangement of crops and fallow on a given area is called cropping pattern

Cropping system

- Multiple cropping: Growing more than two crops in a piece of land in a year in orderly succession. It is also called as intensive cropping. It is used to intensify the production. It is possible only when assured resources are available (land, labour, capital and water)
- Double cropping: Growing two crops a year in sequence. Example: Rice - Pulse
- Triple cropping: Growing three crops a year in sequence. Example: Rice - Rice - Pulse
- Monoculture: Repetitive growing of the same sole crop in the same land.
- Mono cropping: Continuous production of one and the same crop year after year or season after season is called mono cropping.
- Sole cropping: One crop variety grown alone in a pure stand at normal density.

Intercropping

- Growing two or more crops simultaneously with distinct row arrangement on the same field at the same time.
- Base crop: primary crop which is planted/sown at its optimum sole crop population in an intercropping situation.
- Intercrop : This is a second crop planted in between rows of base crop with a view to

obtain extra yields with intercrop without compromise in the main crop yields

Disadvantages

- Higher amount of nitrogen has to be applied for mineralization of organic matter in zero tillage
- Perennial weeds may be a problem
- High number of volunteer plants and buildup of pests

Advantages

- Better use of growth resources including light, nutrients and water
- Suppression of weeds
- Yield & stability - even if one crop fails due to unforeseen situations, another crop will yield and provides some secured income
- Successful intercropping gives higher equivalent yields (yield of base crop + yield of intercrop), higher cropping intensity
- Reduced pest and disease incidences
- Improvement of soil health and agro-eco system

1. Maize + Cowpea
2. Sorghum + Redgram
3. Groundnut + Redgram
4. Potato + Mustard
5. Wheat + Mustard

Inter cropping	Mixed cropping
The main objective of inter cropping is to utilize the space between rows of main crop and to produce more grain per unit area	The main objective of mixed cropping is insurance against crop failure.
There is no competition between main and inter crop (subsidiary crop)	There is competition between component crops. Here all crops are given equal importance and care. Hence, there is no difference between component crops
In inter cropping, the main crop may be a long duration one and the inter crop may be a short duration/early maturing one	Crops may or may not be of same duration
Main and inter crops are sown in definite row arrangement	There is no specific row arrangement. Generally crop seeds are mixed and broadcasted
The sowing time of both the crops may or may not be the same. Sometimes the main crops is sown earlier than the inter crop	The sowing time of component crops is same.

Permaculture

- It is a contraction of "permanent agriculture" or "permanent culture."
- It is defined as a design system for creating sustainable human environments.

- It uses ecology as the basis for designing integrated systems of food production, housing, appropriate technology, and community development.

- Permaculture is built upon an ethic of caring for the earth and interacting with the environment in mutually beneficial ways.
- A central theme in Permaculture is the design of ecological landscapes that produce food.
- Emphasis is placed on multi-use plants, cultural practices such as sheet mulching and trellising, and the integration of animals to recycle nutrients and graze weeds.

Micro Irrigation

- Micro irrigation is defined as the methods in which low volume of water is applied at low pressure & high frequency. The system has extensive network of pipes at operated at low pressure. At pre-determined spacing outlets are provided for emission water generally known as

Sprinkler Irrigation

- In the sprinkler method of irrigation, water is sprayed into the air and allowed to fall on the ground surface somewhat resembling rainfall. The spray is developed by the flow of water under pressure through small orifices or nozzles.

Drip irrigation

- Drip irrigation is also called trickle irrigation and involves dripping water onto the soil at very low rates from a system of small diameter plastic pipes fitted with outlets called emitters.
- Water is applied close to plants so that only part of the soil in which the roots grow is wetted, unlike surface and sprinkler irrigation, which involves wetting the whole soil profile.

Terracing

- "A terrace is an embankment or ridge of earth constructed across a slope to control runoff and minimize soil erosion".
- It reduces the length of the hill side slope, thereby reducing sheet and rill erosion and prevents formation of gullies.

System of Rice Intensification (SRI)

- SRI is a combination of several practices those include changes in nursery management, time

of transplanting, water and weed management.

- It emphasizes altering of certain agronomic practices of the conventional way of rice cultivation.
- All these new practices are together known as System of Rice Intensification (SRI).

Principle - 'More with Less'

- SRI is not a fixed package of technical specifications, but a system of production with four main components, viz., soil fertility management, planting method, weed control and water (irrigation) management.
- Rice yield increased with less water and with reduction in chemical inputs.

Sustainable Sugarcane Initiative (SSI)

- Sustainable Sugarcane Initiative (SSI) is an innovative set of agronomic practices that involves using less seeds, raising seeds in a nursery, and following new planting methods, with wiser seed spacing, and better water and nutrient management to increase the cane yield significantly.
- SSI methods can increase sugarcane yields b at least 20% with 30% less water and a 25% reduction in chemical inputs.
- The SSI method of sugarcane cultivation was evolved from the principles of 'More with Less' followed in SRI (System of Rice Intensification) and introduced in India in 2009.

Tillage - Advantages

- greater volume of soil may be obtained for cultivation of crops,
- excess water may percolate downward to recharge the permanent water table,
- reduce runoff and soil erosion,
- roots of crop plants can penetrate deeper to extract moisture from the water table.
- Clean tillage: It refers to working of the soil of the entire field in such a way no living plant is left undisturbed. It is practiced to control weeds, soil borne pathogen and pests.
- Blind tillage: It refers to tillage done after seeding or planting the crop (in a sterile soil) either at the pre - emergence stage of the crop plants or while they are in the early stages of growth so that crop plants (cereals, tuber

crops etc.) do not get damaged, but extra plants and broad leaved weeds are uprooted.

- Zero tillage (No tillage): In this, new crop is planted in the residues of the previous crop without any prior soil tillage or seed bed preparation and it is possible when all the weeds are controlled by the use of herbicides.

Advantages of Zero tillage

- Zero tilled soils are homogenous in structure with more number of earthworms.
- Organic matter content increases due to less mineralization.
- Surface runoff is reduced due to presence of mulch.

Disadvantages of Zero tillage

- Higher amount of nitrogen has to be applied for mineralization of organic matter in zero tillage.
- Perennial weeds may be a problem.
- High number of volunteer plants and buildup of pests.

Renewable & Non-Conventional Sources Of Energy

Renewable & Non-Conventional Sources Of Energy

With reference to two non-conventional energy sources called 'coal bed methane' and 'shale gas', consider the following 'statements':

1. Coal bed methane is the pure methane gas extracted from coal seams, while shale gas is a mixture of propane and butane only that can be extracted from fine-grained sedimentary rocks.
2. In India abundant coal bed methane sources exist, but so far no shale gas sources have been found.

Which of the statements given above is/are correct? Page 285

- a. 1 only
- b. 2 only
- c. Both 1 and 2
- d. **Neither 1 nor 2**

CBM = Methane

Shale gas = Lot of **Methane** + Little **Ethane, Propane, & Butane** + very little **carbon dioxide, nitrogen, and hydrogen sulfide**.

Abundant shale reserves occur in India.

Sources of energy

Conventional		Non conventional
Conventional non-renewable energy	Conventional renewable energy	
Mostly fossil fuels found under the ground. Coal, oil, natural gas etc. are the examples.	Mostly non-fossil fuels seen above the ground. Fire wood, cattle dung from vegetable wastes, wood charcoal etc. are the examples.	1. Solar energy 2. Hydro power 3. Wind energy 4. Nuclear energy 5. Hydrogen energy 6. Geothermal energy 7. Bio gas 8. Tidal energy 9. Bio-fuel

Fuel	Production	Advantages	Limitations
Nuclear energy	Nuclear fission (splitting of atom) and Nuclear fusion	No air pollution Fuel efficient	High cost of construction of nuclear plant. Fear of security and nuclear accidents. Problem of safe disposal of nuclear waste.
Hydropower	Dams built on river for electricity generation	World's hydroelectricity capacity high	Ecosystems behind dams disturbed. Human settlements uprooted. Habitat loss and biodiversity loss. Developmental cost high. Fertile farmland lost Amount of nutrient rich silt on down river agricultural fields reduced.
Solar energy	From natural sunlight	Environment friendly Ample or unlimited	Limited capacity for storage of sunlight. Cloud cover. Collecting equipment expensive.

		availability.	
Wind energy	Fans for directing winds in use from long for irrigation crops	No pollution Available for free	Not available everywhere or intermittently available. Fans of wind mills visual hazards for flying birds and aeroplanes (visual pollution).
Tidal energy	Harnessing tidal power by suitable structures	Free and clean	Structures (plant) used for harnessing energy expensive. Plant disrupts natural flow of estuary and concentrate pollutants in the area.
Geothermal energy	Wells drilled to trap steam which powers electrical generators. Steam naturally produced from underground water which gets heated due to very high temperature that region.	Environment friendly	Steam contains Hydrogen Sulphide (H ₂ S) having odour of rotten eggs. Minerals in the steam corrosive to pipe lines and equipment causing maintenance problems. Minerals in the water toxic to fish.
Biomass	Cutting trees for fuel wood and burning them straight away	Cheap so popular in under developed and developing countries	Comparatively low level of energy. Bulky so difficult to transport. Burning wood causes air pollution. Destruction of forests to obtain fuel wood and so desertification. Release lot of fly ash.
Biomass conversion	Obtaining energy from chemical energy. Stored in biomass (or live material). Burned directly for cooking or to produce electricity converted to ethanol or methane (biogas)	Renew able energy	May lead to food shortage because nutrients not returned to soil from biomass. Growing maize for ethanol requires more energy expenditure than the amount of energy in the form of alcohol retrieved. Land for growing food used for growing biomass for conversion into fuel.
Solid waste	Waste sorted and burnable material separated	Decreases cost of fresh disposal Reduces need for land fill sites	Causes air pollution for burning releases CO ₂ and other gases. Waste such as bleached paper and plastics have chlorine containing compounds which form. dioxins which are highly toxic and suspected to be carcinogenic.

Biomass [Conventional Source]

Sources of biomass

- Biomass is a renewable energy resource derived from plant and animal waste.
- The energy from biomass (biomass conversion) is released on burning or breaking the chemical bonds of organic molecules formed during photosynthesis.
- Biomass fuels can be used directly or they can be transformed into more convenient form and then used.
- By-products from the timber industry, agricultural crops and their byproducts, raw material from the forest, major parts of household waste and wood.
- Solid Biomass fuels: Wood logs and wood pellets, charcoal, agricultural waste (stalks and other plant debris), animal waste (dung), aquatic plants (kelp and water hyacinths)

urban waste (paper, cardboard and other combustible materials).

Conversion to gaseous and liquid biofuels

- Biomass can be converted into alcohol (liquid biofuels) by distillation.
- Liquid Biofuels: Ethanol, Methanol, Gasohol, Biodiesel.
- Gaseous Biofuels: Synthetic natural gas (biogas), Wood gas: Methane – 70% and CO₂ – 30%.
- Instead of burning loose biomass directly, it is more practical to compress it into briquettes (compressing them into blocks of a chosen shape) improve its utility and convenience of use.
- Such biomass in the biomass briquettes can be used as fuel in place of coal in traditional furnaces or in a gasifier.
- A gasifier converts solid fuels into a more convenient-to-use gaseous fuel called producer gas.

Uses of biomass

- In the developed world biomass is becoming important for applications such as combined heat and power generation.
- Biomass energy is gaining significance as a source of clean heat for domestic heating and community heating applications.

Advantages of biomass energy

- Burning of biomass does not increase atmospheric carbon dioxide because to begin with biomass was formed by atmospheric carbon dioxide and the same amount of carbon dioxide is released on burning.
- Biomass is an important source of energy and the most important fuel worldwide after coal, oil and natural gas.
- Biomass is renewable and is abundantly available on the earth in the form of firewood, agricultural residues, cattle dung, city garbage etc.
- Bio-energy, in the form of biogas, which is derived from biomass, is expected to become one of the key energy resources for global sustainable development.

Bagasse as biofuel

- Indian sugar mills are rapidly turning to bagasse, the leftover of cane after it is crushed and its juice extracted, to generate electricity.
- This is mainly being done to clean up the environment, cut down power costs and earn additional revenue.

Biogas plant

- The biogas plant consists of two components: a digester (or fermentation tank) and a gas holder.
- The gas holder cuts off air to the digester (anaerobiosis) and collects the gas generated.
- Any biodegradable (that which can be decomposed by bacteria) substance can be fermented anaerobically (in absence of oxygen) by methane-producing (methanogenic) bacteria.
- Cowdung or faeces are collected and put in a biogas digester or fermenter (a large vessel in which fermentation can take place).
- A series of chemical reactions occur in the presence of methanogenic bacteria (CH₄ generating bacteria) leading to the production of CH₄ and CO₂.

Petro crops (Plants)

- Recent researches suggest that hydrocarbon producing plants can become alternative energy sources, which can be inexhaustible and ideal for liquid fuel.
- These plants called **petroplants/ petrocrops** can be grown on land which are unfit for agriculture and not covered with forests. **Jatropha curcas** is an important petro plant.
- Biocrude can be obtained by tapping the latex of *Jatropha curcas*.
- Biocrude is a complex mixture of liquids, terpenoids, triglycerides, phytosterols waxes, and other modified isoprenoid compounds.
- Hydro cracking of biocrude can convert it into several useful products like gasoline (automobile fuel), gas oil and kerosene.
- Some potential Petro-crop species belong to family **Asclepiadaceae** and **Euphorbiaceae**.

Geothermal Energy

- Geothermal energy is natural heat from the interior of the earth that can be used to

generate electricity as well as to heat up buildings.

- The core of the earth is very hot and it is possible to make use of this geothermal energy.
- These are areas where there are volcanoes, hot springs, and geysers, and methane under the water in the oceans and seas.
- In some countries, such as in the USA water is pumped from underground hot water deposits and used for heating of houses.
- Geothermal resource falls into three major categories: i) Geopressurized zones, ii) hot-rock zones and iii) Hydrothermal convection zones. Of these three only the first is currently being exploited on a commercial basis.

Geothermal energy in India

- In India, Northwestern Himalayas and the western coast are considered geothermal areas.
- The Geological Survey of India has already identified more than 350 hot spring sites, which can be explored as areas to tap geothermal energy.
- The **Puga valley** in the Ladakh region has the most promising geothermal field.

Environmental impact of geothermal energy

- Geothermal energy can pose several environmental problems which includes on-site noise, emissions of gas and disturbance at drilling sites.
- The steam contains hydrogen sulphide gas, which has the odour of rotten eggs, and cause air pollution.
- The minerals in the steam are also toxic to fish and they are corrosive to pipes, and equipment, requiring constant maintenance.

Hydrogen Energy

- Many scientists believe that the fuel for the future is hydrogen gas.
- When hydrogen gas burns in the air or in fuel cells, it combines with oxygen gas to produce non-polluting **water vapour** and fuel cells directly convert hydrogen into electricity.
- Widespread use of hydrogen as fuel would greatly reduce the problem of air pollution and

danger of global warming because there will not be any CO₂ emission.

- Hydrogen may be a clean source of energy but getting large amount of pure hydrogen for commercial purposes is a problem because hydrogen is present in combination with other elements such as oxygen, carbon and nitrogen thus hydrogen has to be produced from either water or organic compounds like methane etc. requiring large amounts of energy. This is a very costly proposition.
- Producing hydrogen from algae in large scale cultures is possible. It may be possible to control photosynthesis so that green algae are able to produce hydrogen through the process of photosynthesis.
- Hydrogen is a pollution free, cost effective manner and if technologies such as fuel cells can be made cost effective, then hydrogen has the potential to provide clean, alternative energy for diverse uses, including lighting, power, heating, cooling, transportation and many more.

Fuel Cell Technology

- Fuel cells are highly efficient power-generating systems that produce electricity by combining fuel (hydrogen) and oxygen in an electrochemical reaction.
- Fuel cells are electrochemical devices that convert the chemical energy of a fuel directly and very efficiently into electricity (DC) and heat, thus doing away with combustion.
- **Hydrogen** and **phosphoric acid** are the most common type of fuel cells, although fuel cells that run on **methanol**, **ethanol**, and **natural gas** are also available.
- The most suitable fuel for such cells is hydrogen or a mixture of compounds containing hydrogen.
- A fuel cell consists of an electrolyte sandwiched between two electrodes. Oxygen passes over one electrode and hydrogen over the other, and they react electrochemically to generate electricity, water, and heat.
- Though rapid progress has been made; high initial cost is still the biggest hurdle in the widespread commercialization of fuel cells.
- The rapidly depleting fossil fuel sources of energy and escalating demand of energy have made it necessary to look for alternative sources of energy that are known as renewable or inexhaustible. We can define inexhaustible energy resources as 'those

resources which can be harnessed without depletion'. Most of these resources are free from pollution and some of them can be used at all places. These renewable energy resources are also known as non-conventional or inexhaustible or alternate energy sources. These energy sources are solar, flowing water, wind, hydrogen and geothermal. We get renewable solar energy directly from the sun and indirectly from moving water, wind and biomass. Like fossil fuels and nuclear power, each of these alternatives renewable sources of energy has their own advantages and disadvantages. We are going to discuss some of them in detail.

Solar Energy

- Direct solar energy can be used as heat, light, and electricity through the use of solar cells.
- Direct use of solar energy can be used through various devices broadly directed into three types of systems a) passive, b) active c) photovoltaic.

Passive solar energy

As you know some of the earliest uses of solar energy were passive in nature such as to evaporate sea water for producing salt and to dry food and clothes.

In fact solar energy is still being used for these purposes. The more recent passive uses of solar energy is for cooking, heating, cooling and for the day lighting of homes and buildings.

Active use of solar energy

- Active solar heating and cooling systems rely on solar collectors which are usually mounted on roofs.
- Such systems also requires pumps and motors to move the fluids or blow air by fan in order to deliver the captured heat.
- A number of different active solar heating systems are available. The main application of these systems is to provide hot water, primarily for domestic use.

Solar cells or photovoltaic technology

- Solar energy can be converted directly into electrical energy (direct current, DC) by

photovoltaic (PV) cells commonly called solar cells.

- Photovoltaic cells are made of silicon and other materials. When sunlight strikes the silicon atoms it causes electrons to eject. This principle is called as '**photoelectric effect**'.
- A typical solar cell is a transparent wafer that contains a very thin semiconductor.
- Sunlight energizes and causes electrons in the semiconductor to flow, creating an electrical current.

With reference to technologies for solar power production, consider the following statements:

1. 'Photovoltaics' is a technology that generates electricity by direct conversion of light into electricity, while 'Solar Thermal' is a technology that utilizes the Sun's rays to generate heat which is further used in electricity generation process.
2. Photovoltaics generates Alternating Current (AC), while Solar Thermal generates Direct Current (DC).
3. India has manufacturing base for Solar Thermal technology, but not for Photovoltaics.

Which of the statements given above is / are correct?

- a. 1 only
- b. 2 and 3 only
- c. 1, 2 and 3
- d. None

Explanation:

Photoelectric effect = When light strikes on a material, electrons are dislodged [photons dislodge electrons].

Photovoltaic = The dislodged electrons if channeled through a conductor will create electric current (voltage Or potential difference) = **Solar Panels**. [Electric current is nothing but movement of electrons from high potential to low potential area (more electrons to less electrons region)]

Solar thermal = converting light into heat = **solar cooker, solar water heater**.

Photovoltaics generate direct current (DC). [Rotating = AC, Stationary = DC. Electric

generator, wind turbine generate AC while solar panels generate DC]

Solar thermal is mostly used for water heating purposes. Electricity can be generated by using hot water steam to rotate turbine = AC current.

In India both solar panels and solar cookers are manufactured. [Remember Indian – USA WTO ‘domestic content’ dispute?]

Answer: a) 1 only

Tidal energy

- Tidal power projects attempt to harness the energy of tides as they flow in and out.
- The main criteria for a tidal power generation site are that the mean tidal range must be greater than 5 metres.
- The tidal power is harnessed by building a dam across the entrance to a bay or estuary creating a reservoir.
- As the tide rises, water is initially prevented from entering the bay. Then when tides are high and water is sufficient to run the turbines, the dam is opened and water flows through it into the reservoir (the bay), turning the blades of turbines and generating electricity.
- Again when the reservoir (the bay) is filled, the dam is closed, stopping the flow and holding the water in reservoir when the tide falls (ebb tide), the water level in the reservoir is higher than that in the ocean.
- The dam is then opened to run the turbines (which are reversible), electricity is produced as the water is let out of the reservoir.
- The dams built to harness the tidal power adversely affect the vegetation and wildlife.

Hydropower Energy

- Hydroelectric power uses the kinetic energy of moving water to make electricity.
- Generation of electricity by using the force of falling water is called hydroelectricity or hydel power. It is cheaper than thermal or nuclear power.
- Dams are built to store water at a higher level; which is made to fall to rotate turbines that generate electricity.

- One of the greatest advantages of hydropower is that once the dam is built and turbines become operative, it is relatively cheap and clean source of energy.
- Hydropower also has some disadvantages, building of dam seriously disturbs and damages the natural habitats and some of them are lost forever.

Ministry of New and Renewable Energy (MNRE)

- The ministry was established as the Ministry of Non-Conventional Energy Sources in 1992. It adopted its current name in October 2006.
- The Ministry is mainly responsible for
 1. research and development,
 2. intellectual property protection, and
 3. international cooperation, promotion, and coordination in renewable energy sources such as wind power, small hydro, biogas, and solar power.

Aim

- To develop and deploy new and renewable energy for supplementing the energy requirements of India.

Mission

- Bring in Energy Security;
- Increase the share of clean power;
- Increase Energy Availability and Access;
- Improve Energy Affordability; and
- Maximise Energy Equity.

Initiatives

- Jawaharlal Nehru National Solar Mission (JNNSM)
- Remote Village Lighting Programme
- National Biogas and Manure Management Programme (NBMMP)
- Solar Lantern Programme LALA
- Solar thermal energy Demonstration Programme
- National Biomass Cookstoves Initiative (NBCI)[8]
- National Offshore Wind Energy Authority

Key functional area

- Indian Renewable Energy Development Agency (IREDA)
- Integrated Rural Energy Programme (IREP);
- Commission for Additional Sources of Energy (CASE);

Jawaharlal Nehru National Solar Mission (JNNSM)

- Also known as the National Solar Mission
 - Objective
1. To establish India as a global leader in solar energy, by creating the policy conditions for its diffusion across the country as quickly as possible.
 2. To promote ecologically sustainable growth while addressing India's energy security challenges.
- Major contribution by India to the global effort to meet the challenges of climate change.
 - One of the several initiatives that are part of **National Action Plan on Climate Change**.
 - The program was inaugurated in 2010.
 - Initial target was 20GW by 2022 and it was increased to 100 GW in 2015 Union budget.
 - Long term goal: Global leader in solar energy; maximum in energy production.
 - Immediate goal: Setting up an enabling environment for solar technology penetration in the country.

Targets are set for three phases

1. **First phase 2010-13**
 2. **Second phase 2013-17**
 3. **Third Phase 2017-22**
- At each stage progress will be reviewed and roadmap for future targets will be adopted.
 - Total target of **100,000 MW by 2022**.
 - MNRE has proposed to achieve it through 40,000 MW through Rooftop Solar Projects and 60,000 MW through Large and Medium Scale solar projects.

Domestic content controversy

- Guidelines for the solar mission mandated cells and modules for solar PV projects based on crystalline silicon to be manufactured in India.

- This accounts to over 60% of total system costs.
- For solar thermal, guidelines mandated 30% project to have domestic content.
- A vigorous controversy emerged between power project developers and solar PV equipment manufacturers.
- The former camp prefers to source modules by accessing highly competitive global market to attain flexible pricing, better quality, predictable delivery and use of latest technologies.
- The latter camp prefers a controlled/planned environment to force developers to purchase modules from a small, albeit growing, group of module manufacturers in India.
- Manufacturers want to avoid competition with global players and are lobbying the government to incentivize growth of local industry.
- US Trade Representative has filed a complaint at World Trade Organization challenging India's domestic content requirements citing discrimination against US exports.
- WTO ruled in favor of USA.

Indian Renewable Energy Development Agency (IREDA)

- IREDA is a Mini Ratna (Category - I) Government of India Enterprise.
- It is under the administrative control of MNRE.
- IREDA is Public Limited Government Company established as a **Non-Banking Financial Institution** in 1987 engaged in promoting, developing and extending financial assistance for setting up projects relating to new and renewable sources of energy and energy efficiency/conservation with the motto: **"Energy For Ever"**.

Objectives

- To give financial support to specific projects and schemes for generating electricity and / or energy through new and renewable sources and conserving energy through energy efficiency.
- To increase IREDA's share in the renewable energy sector by way of innovative financing.

With reference to the Indian Renewable Energy Development Agency Limited

(IREDA), which of the following statements is/are correct?

1. It is a Public Limited Government Company.
2. It is a Non – Banking Financial Company.

Select the correct answer using the code given below.

- a. 1 only
- b. 2 only
- c. Both 1 and 2
- d. Neither 1 or 2

Answer: c) Both 1 and 2

Non-Renewable Sources Of Energy

Non-Renewable Sources Of Energy

Fossil fuels represent stored solar energy captured by plants in the past geological times.

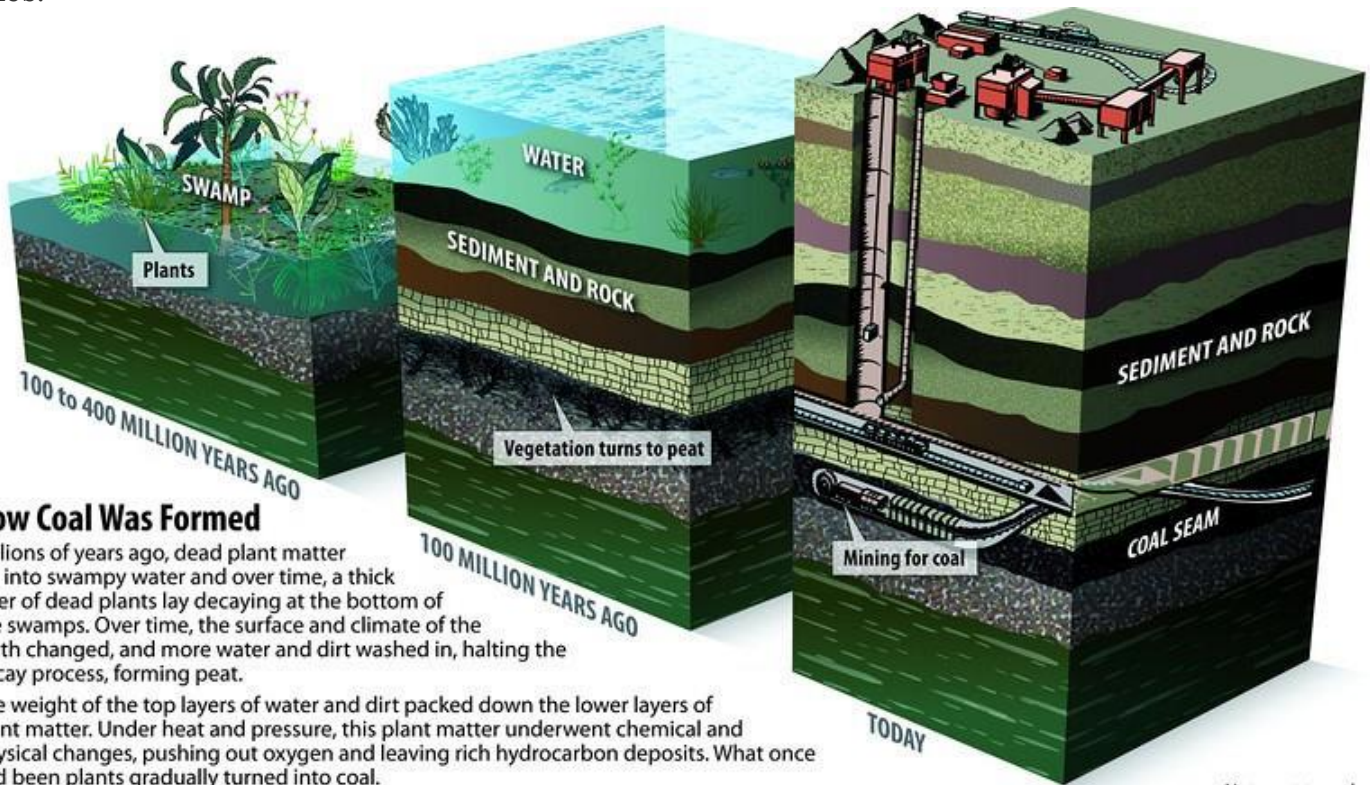
- Coal, petroleum and natural gas are called fossil fuels, as they are the remains of prehistoric plants, animals and microscopic organisms that lived millions of year ago.
- During the **Carboniferous period 275-350 million years ago**, conditions in the world were suitable for formation of large deposits of fossil fuels.

Coal

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- Coal is formed from plants and vegetation buried, 'in situ' or drifted in from outside to a place, which got covered by deposits of sediments.
- Coal is a solid fossil fuel and a sedimentary rock composed primarily of carbon. There are three basic grades of coal: i) **lignite (brown coal)**, ii) **bituminous (soft coal)** and iii) **anthracite (hard coal)**.

Formation of coal

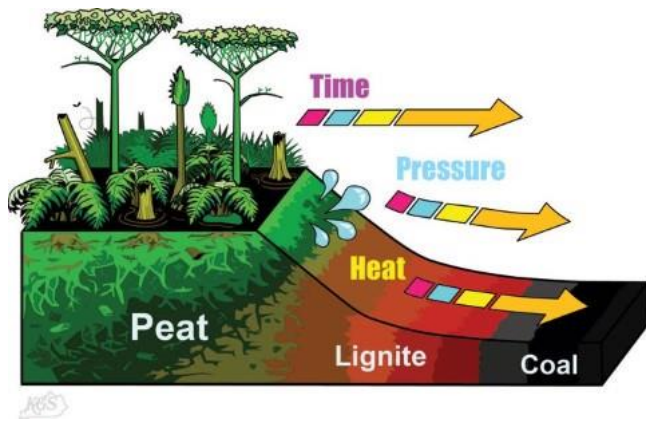


How Coal Was Formed

Millions of years ago, dead plant matter fell into swampy water and over time, a thick layer of dead plants lay decaying at the bottom of the swamps. Over time, the surface and climate of the Earth changed, and more water and dirt washed in, halting the decay process, forming peat.

The weight of the top layers of water and dirt packed down the lower layers of plant matter. Under heat and pressure, this plant matter underwent chemical and physical changes, pushing out oxygen and leaving rich hydrocarbon deposits. What once had been plants gradually turned into coal.

Coal can be found deep underground (as shown in this graphic), or it can be found near the surface.



- Coal is the result of plant material that grew in fresh water swamps approximately three hundred million years ago.
- As this plant material died and accumulated, peat also called peat bog was formed.
- Since the plant material accumulated under water, in the swamps decay was inhibited due to **lack of oxygen**.
- Oceans inundated many of the areas of peat and sediments from the sea were deposited, over the peat.
- The weight of these sediments and the heat of the earth gradually changed the composition of the peat bog and coal was formed.
- Today peat also is used as source of fuel in some parts of the world though its **high water content** makes it a low-grade fuel.
- Peat is changed into coal after many centuries of being compressed by the weight of sediments. It first changes into a low-grade coal known as **lignite (brown coal)**.
- The percentage of carbon in the lignite is higher than in peat. Continued pressure and heat from the earth changes lignite into **bituminous soft coal**.
- If the heat and pressure were great enough then **anthracite coal (hard coal)** would be formed which has the highest heat and carbon content.
- Accordingly energy content is greatest in anthracite coal and lowest in lignite.
- The **sulphur** content of coal is important because on burning low sulphur coal emits less sulphur dioxide (SO₂) so more desirable as a fuel for power plants.

Problems with Coal Mining

- Coal is most abundant fossil fuel on earth, but there are problems associated with its mining, transportation and use.
- Coal is mined from both (i) surface mines, and (ii) underground mines.

Surface mining

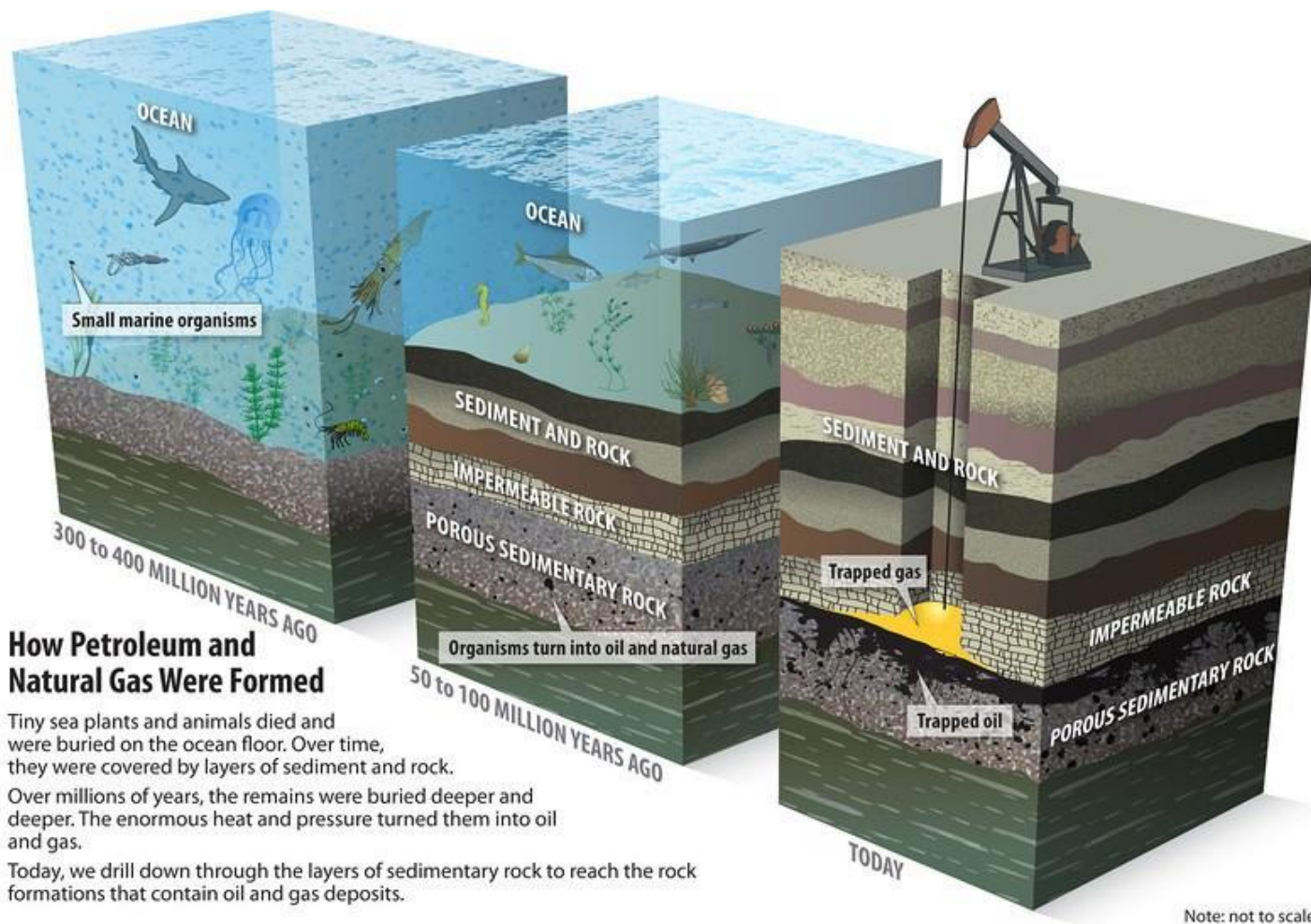
- Surface mining disrupts and drastically changes the natural landscape and destroys the natural vegetation and the habitat of many species, some of which may already be endangered.
- Mining operations, involving digging, blasting, removal of rocks and soil lying over the coal seam, cause serious problems of air and noise pollution.
- Surface mining may also cause soil erosion and silt loading (the discharge of silts into streams) and canals that disrupt and pollute the aquatic ecosystems as well as ground water in places where aquifers are located near or associated with coal seams.

Underground mining

- Underground mining may cause collapse or land subsidence in the mining areas during or after mining operations are over.
- In case of some mines acid mine drainage from the mine waste pollutes long stretches of streams.
- Coal bed methane in underground mines causes fires.

Apart from these problems, burning of coal in thermal power plants for generation of electricity and in industry is the prime source of air pollution.

Petroleum Or Mineral Oil



How Petroleum and Natural Gas Were Formed

Tiny sea plants and animals died and were buried on the ocean floor. Over time, they were covered by layers of sediment and rock.

Over millions of years, the remains were buried deeper and deeper. The enormous heat and pressure turned them into oil and gas.

Today, we drill down through the layers of sedimentary rock to reach the rock formations that contain oil and gas deposits.

- Oil and gas were formed from the remains of plants and animals that once lived in the sea.
- For over millions of years these remains remained buried under mud and rock under great pressure and at high temperatures.
- Under these conditions marine biomass gradually changed into oil and gas.
- Oil and gas are primarily found along **geologically young** tectonic belt at plate boundaries, where large depositional basins are more likely to occur.
- Petroleum or crude oil (oil as it comes out of the ground), is a thick dark liquid consisting of a mixture hundreds of combustible hydrocarbons along with small amounts of **sulphur, oxygen** and **nitrogen impurities**. It is also known as conventional oil or light oil.
- Deposits of crude oil and natural gas are usually trapped together under the sea floor or earth's crust on land.
- After it is extracted, crude oil is transported to a refinery by pipelines, trucks or ships (oil tanker).
- In refineries oil is heated and distilled to separate it into components with **different**

boiling points. The important components are gases, gasoline, aviation fuel, kerosene, diesel oil, naphtha, grease and wax and asphalt.

- Some of the products of oil distillation are called petro-chemicals which are used as raw material for the manufacture of pesticides, plastics, synthetic fibers, paints and medicines etc.

Natural Gas

- Natural gas, primarily consist of **methane**, is often found above reservoirs of crude oil.
- The natural gas is a mixture of 50 to 90% by volume of methane (CH_4), the simplest hydrocarbon.
- It also contains small amounts of heavier gaseous hydrocarbons such as **ethane (C_2H_6)**, **propane (C_3H_8)** and **butane (C_4H_{10})** and also small amounts of highly toxic **hydrogen sulphide (H_2S)**.
- Natural gas is formed through geological processes similar to the processes of crude oil formation described earlier except the organic

material gets changed to more volatile hydrocarbons than those found in oil.

- Almost every oil well produces liquid petroleum along varying amounts of natural gas. However, there are large gas deposits without any liquid petroleum being associated with them.

Conventional natural gas

- It lies above most reservoirs of crude oil. These deposits can be tapped/used only through pipeline.
- But the natural gas that comes out along with oil is often looked as unwanted by product and is burned off.

Unconventional natural gas

- It is found by itself in other underground reservoirs. So far it is very expensive to get natural gas from such unconventional sources but technology is being developed to extract the gases economically.
- When a natural gas field is tapped, **propane** and **butane** gases, present in natural gas are liquefied and removed as **liquefied petroleum gas** (LPG).
- LPG is stored in pressurized tanks or cylinders for use as cooking gas. At a very low temperature natural gas can be converted to **liquefied natural gas** (LNG).
- Natural gas is used as a source of carbon used in **tyre industry**. When natural gas is strongly heated, then methane gets in it decomposed to form carbon and hydrogen.
- The carbon thus formed is called **carbon black** and used as filler in the manufacture of tyres.

Problems associated with oil and gas

- Methane being major component of natural gas, happens to be a greenhouse gas and its leakage contributes to global warming.
- Extraction of oil and gas may cause sinking of land or subsidence.
- Another major problem in the past with onshore oil wells has been brine (salt water). Typically, for every barrel of oil production ten barrels of brine are also extracted.
- In early days the brine was simply discarded into nearby streams or on the soil. Today most brine is reinjected into the well. However, brine can contaminate fresh water aquifers if

the casing lining the well is missing or corroded.

- About half of the oil that contaminates the ocean comes from natural seepage from offshore deposits.
- 20% of the oil contaminating the ocean comes from oil well, blowouts, pipeline breaks and tankers.

Nuclear Energy Sources

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- Radioactive minerals are used to generate nuclear energy through high technological methods.
- There are two methods which can be used to release energy from radioactive minerals:

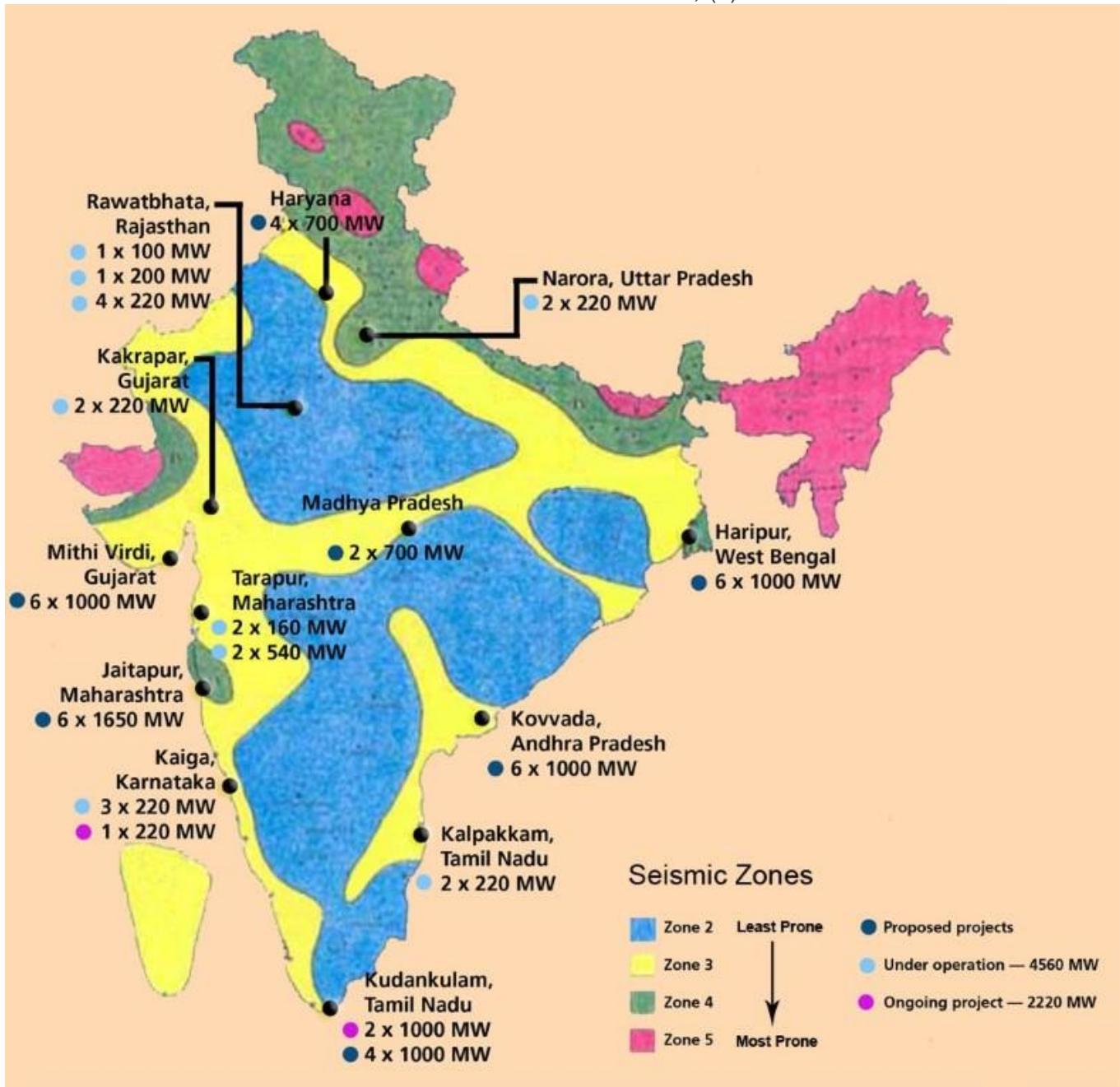
1. **Nuclear fission** – In this process, the nucleus of heavy atom namely of **uranium (U 235)** or **plutonium (P239)** breaks apart into smaller fragments, releasing an enormous amount of energy.
2. **Nuclear fusion** – In this process, small nucleus like those of **isotopes of hydrogen**, namely **deuterium** and **tritium** fuse or join together to form heavier nuclei, releasing vast amounts of energy.

Nuclear fission

- Radioactive mineral, which generates nuclear energy through fission, may be considered a non-renewable alternative source of energy as it is an ore and is found in limited quantities.
- Nuclear fission occurs because the atom of radioactive minerals contains nuclei that are unstable and break or split apart releasing energy.
- Whenever a neutron strikes a nucleus of U-235, energy is released, **krypton** and **barium** are produced, and several neutrons are released.
- These new neutrons may strike other atoms of U-235 to produce a **chain reaction**.
- When this nuclear disintegration takes place particles from the nucleus including neutrons fly out.
- The neutron may cause other atomic nuclei to split releasing more neutrons and more energy.
- Once begun this chain reaction continuous to release energy until the fuel is spent or the neutrons are prevented from striking other nuclei.

- In the reactor of a nuclear power plant, the rate of nuclear chain reaction is controlled and the heat generated is used to produce high pressure steam, which spins turbine that generate electricity.
- Heat produced here is carried away by water coolant and transferred by way of heat exchanger to the water in a steam-generating unit.

- The steam produced powers a turbine that produces electricity. Cooling water is used to condense the steam after it has gone through the turbine
- Two other nuclear technologies for generating electricity from nuclear fuel in a safe and economic way have also been proposed, but so far they have not proved operationally successful. These are: (i) **nuclear breeder reactor**, (ii) **fusion reactor**.



Nuclear breeder reactor

- The nuclear reactors operating today use uranium very inefficiently. About 1% uranium

is actually used to produce steam for generating electricity.

- A nuclear reactor that can utilize between 40% and 70% of its nuclear fuel is called a **breeder reactor**.

- Breeder reactors convert more abundant **uranium -238** or **thorium -232** fissionable isotopes, **Plutonium-239** or **Uranium -233** respectively, that can sustain a nuclear chain reaction.

Nuclear fusion reactor

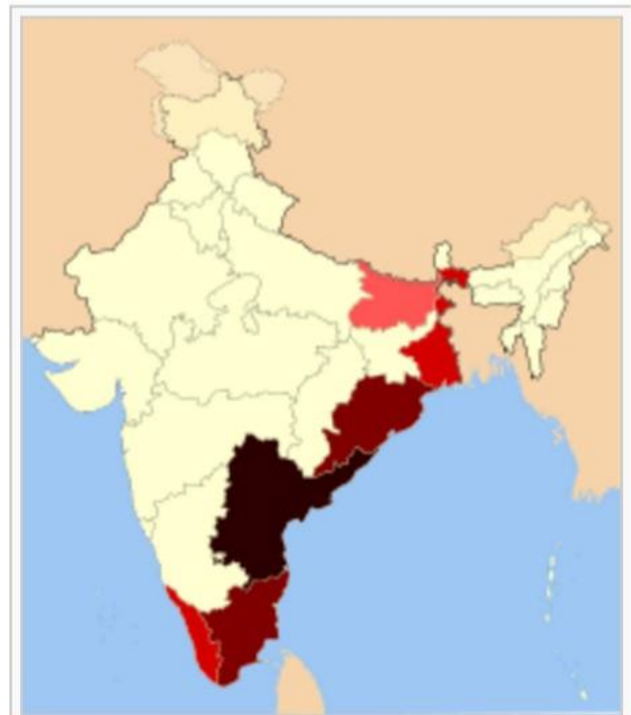
- The principle for nuclear fusion involves, as you are aware, uniting two small atoms to form a large atom with the release of an enormous amount of energy.
- The energy produced by stars and the sun is the result of nuclear fusion.
- Generation of energy by this method so far, however, has not been possible though lot of research has focused on the fusion reaction of deuterium (D) and tritium (T) (two isotopes of hydrogen) which fuse at about **100 million degrees**.

Problems related to nuclear energy generation

The major problems associated with the generation of nuclear power are

- disposal of nuclear waste,
- contamination of environment with long lasting radioactive materials (radioactive pollution),
- thermal pollution,
- health effects from exposure to low levels of radiation,
- limited supplies of uranium ore,
- high construction and maintenance costs,
- questionable reactor safety,
- human or technical error that could result in a major accidents and vulnerability to sabotage,
- developing nuclear weapons by processing reactor waste.
- Problems of dismantling of a nuclear plant's, after their useful life of 30-40 years is over.

Location of radioactive mineral ore in India



India's thorium is mostly found in a **contiguous** belt formed by its eastern coastal states.

2012 reserve estimates:^[9]

- 35% (**Andhra Pradesh**, excluding **Telangana**)
- 15–20% (**Tamil Nadu**, **Odisha**)
- 10–15% (**Kerala**, **West Bengal**)
- 0–5% (**Bihar**)

- In India, **monazite** that is the main source of **thorium**, is found in commercial quantities on the Travancore coast between Kanya Kumari and Quilon, while **uranite** or **pitchblende mineral of uranium** is found in Gaya (Bihar), Ajmer (Rajasthan) and Nellore (Andhra Pradesh).

Energy conservation

- CFL (compact fluorescent lamps) should be replaced by LEDs as they are much more efficient.
- Replacing aging old appliances with energy efficient models.

- Alternative resources i.e. renewable energy sources should be used in place of nonrenewable energy sources e.g. solar energy, biogas, wind energy etc.
- Energy audits of homes, buildings, hotels and factories should be done at regular interval.
- Demonstration of projects involving the introduction of appropriate, renewable solar, wind and biogas energy technologies at the community level.
- Collaborative community/academic research and development in order to produce lowcost, sustainable energy options should be given priority.
- Environment friendly public transport system should be promoted to reduce the use of individual motorized transport.
- Installation of photoelectric controls or timers should be used to make sure that outdoor lighting is sufficient during the day.
- Elevators/lifts should be used for going up beyond three floors and for coming down the usage of lifts may be reduced.
- Whenever two elevators/lifts are provided in a building only single should be operated during “non-peak” hours.
- Conservation and sustainable use of water bodies, including watersheds, rivers barriers and coastal zones will be helpful in the energy conservation at community level.
- Training programme about energy efficient repairs should be organized to conserve energy at community level.
- Advocacy to remove subsidies to inefficient and polluting sources of energy should become essential.
- Locally manufactured, improved cook stoves should be introduced to reduce charcoal/fuel consumption.
- Auditing Regular monitoring and audit of energy consumption in industries results in energy conservation.

- Process modification: replacement of old and more energy consuming processes by the new energy efficient processes. Old factories should now employ process modification.
- Use public transportation as much as possible instead of using own vehicles.
- Avoid free frequent starts and stops of vehicles to reduce fuel consumption.
- Appliances and office equipments should be replaced with energy star rated units.
- Building designs and construction practices should promote energy conservation.

Impact of energy use on the environment

- Depletion of energy resources.
- Pollution of environment from emission of green house gases.

Energy And Economic Development

- Energy development is an integral part of economic development.
- Cheap, efficient and environment friendly energy resources and technologies are must to stay globally competitive.

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